

INTRODUCTION TO SOUND

Acoustics for the Hearing and Speech Sciences

FOURTH EDITION

CHARLES E. SPEAKS





Introduction to Sound

Acoustics for the Hearing
and Speech Sciences

Fourth Edition

Editor-in-Chief for Audiology
Brad A. Stach, PhD



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and Speech Sciences

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Charles E. Speaks, PhD

Distinguished Professor Emeritus
Department of Speech-Language-Hearing Sciences
University of Minnesota
Minneapolis, Minnesota





5521 Ruffin Road
San Diego, CA 92123

e-mail: info@pluralpublishing.com
Website: <http://www.pluralpublishing.com>

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Preface to the First Edition

This book was written to *teach* the fundamental concepts of acoustics, particularly to those who are interested in the discipline of the speech-language-hearing sciences. Readers who are thoroughly grounded in mathematics and physics should be able to move through the various topics quickly. Those who are less comfortable with basic concepts of physics, or with mathematics beyond elementary algebra, will require more careful study of some of the concepts, but ultimately the concepts should be understood.

Students of the speech-language-hearing sciences must have a thorough understanding of the elements of acoustics before they can successfully embark on more advanced study of both normal and disordered human communication. At the University of Minnesota, for example, students in the Department of Communication Disorders who pursue an undergraduate degree must complete a five-credit course in acoustics, which is *prerequisite* to registration in more advanced courses such as Speech Science, Hearing Science, Hearing Loss and Audiometry, Noise and Humankind, Cleft Palate: Oral-Facial Anomalies, and Speech, and Voice Disorders. Treatment of the fundamental concepts of acoustics with two or three weeks of lectures in the context of a broader course such as Speech and Hearing Science or Introduction to Audiology cannot, in our opinion, do justice to the topic or serve students well.

There are many aspects of sound that might interest readers other than students in the speech-language-hearing sciences. Why is a “sonic boom” created when an airplane exceeds the speed of sound? Why is a foghorn designed to emit a low-pitched sound instead of a high-pitched whistle? If you are hunting in the woods, why is your distant prey more likely to hear you if it is downwind from you? How do “whispering galleries” work? In what ways do echoes off a canyon wall behave like billiard balls bouncing off rails of the billiard table? When you contemplate purchasing a stereo system, what does the salesperson mean by terms such as frequency response, noise floor, dynamic range, signal-to-noise ratio, decibels, percentage harmonic distortion, and so on? The answers to these and other questions are sprinkled throughout the text.

In the opening sentence, the word “teach” was emphasized because the fundamental goal is to *teach* the important elements of acoustics, not just present the topics. Two examples should suffice. First, some readers will not know, or will have forgotten, what is meant by “ $\text{antilog}_{10} 2$.” However, everyone will certainly know that $10^2 = 100$. To understand the *concept* of antilogarithms, then,

one needs only to realize that “ $\text{antilog}_{10} 2 = ?$ ” is exactly the same as asking, “what is 10^2 ?” Once the concept is understood, all that remains is to learn the simple steps for solving antilog problems that are computationally, but not conceptually, more difficult. Second, learning of several concepts in acoustics, the decibel for example, can be enhanced by solving problems. For that reason, the accompanying *Course Notes and Workbook for Introduction to Sound* includes nearly 400 practice problems that are followed by *answers and explanations* of how the correct answers were obtained.

The organization of the topics in the book reflects a combination of both logic and personal preference. For example, the concepts of antilogarithms and logarithms must be understood before one can study decibels, and it is difficult to imagine how one can understand complex sound waves without first mastering the concept of sinusoidal wave motion. The location of other topics within the book reflects the author’s preference for teaching. Some might prefer, for example, to begin by reading about “fundamental and derived physical quantities” and “proportionality” from Chapter 1 and “scientific notation” from Chapter 3. Those topics, and some others, should be treated as free-standing modules to be addressed when the reader or teacher elects.

Preface to the Second Edition

The second edition of *Introduction to Sound* retains the singular purpose to *teach* the fundamental concepts of acoustics to students in the speech-language-hearing sciences. To help achieve that objective, this edition differs from its predecessor in three principal ways. **Practice Problems** and **Answers to Practice Problems** have been added for Chapters 1, 5, 7, and 8, and a new set of problems and answers has been added to Chapter 2.

A new section entitled **Frequently Misunderstood Concepts** has been appended to Chapters 1, 2, 4, 5, 6, and 8. An analysis of answers to examination questions by approximately 275 students over the past 3 years led to a distressing realization. Although the mean score was a satisfactory 80%, a few questions were missed consistently by more than half of the students. For example, in response to the question, “An increase in sound pressure by a factor of 4:1 corresponds to an increase by how many decibels?” many students responded with 12 dB SPL or 12 dB IL rather than just 12 dB. The purpose of these new sections, therefore, is to call specific attention to the kinds of mistakes that previous students have made and to attempt further clarification of the basic concepts that had been misunderstood.

Finally, several faculty and student users identified a few errors that appeared in the first edition, and every effort has been made to correct them. I thank them all, and particular appreciation is expressed to Sid Bacon at Arizona State University and Peter Narins at UCLA for their helpful suggestions.

Preface to the Third Edition

The third edition of *Introduction to Sound* continues to retain the singular purpose to *teach* the fundamental concepts of acoustics to students in speech-language-hearing sciences. This edition differs from its predecessors, however, both substantively and cosmetically.

The principal changes are:

- Point-by-point construction of a sine wave is demonstrated by calculating $\sin \theta$ at 11° intervals from 0° to 360° and then plotting the results (Chapter 2);
- Wave equations are used to explain why *particle displacement* must be 180° out of phase with *particle acceleration*, a relation that is seldom intuitively obvious (Chapter 2);
- A table of common logs has been added (Chapter 3);
- The *preferred center frequencies* for constant percentage bandwidth filters are listed, and the reasons for their selection are explained (Chapter 6);
- The “Summary of an Experiment on the Cat” has been deleted (Chapter 7);
- The universality of the inverse square law to physical phenomena other than sound is described (Chapter 8);
- The concepts of *constructive* and *destructive interference* are introduced in a discussion of standing waves (Chapter 8);
- The relation between *displacement* nodes and antinodes and *pressure* nodes and antinodes in consideration of standing waves in acoustic tubes is explained (Chapter 8);
- The higher modes of vibration of strings, and the commonalities between vibration of a string and the standing waves in tubes closed at both ends, are explained (Chapter 8);
- The discussion of sound *absorption* has been expanded to include differences in absorption coefficients as a function of frequency and among common absorbing materials (Chapter 8);
- The discussion of *total absorption*, including the *sabin* and *metric sabin*, has been expanded (Chapter 8);
- The relation between *absorption* and *reverberation time* has been elaborated to show their practical application to optimal and substandard room acoustics (Chapter 8);
- Fourteen equations have been added to the *Alphabetical Listing of Selected Equations*;
- *All illustrations* have been newly prepared with common formatting, several illustrations have been recomposed to achieve greater clarity, some new illustrations have been

prepared, and careful attention has been paid to ensure that all figures are “mathematically correct” (all chapters);

- *Practice Problems* and *Answers to Practice Problems* have been moved to a newly created *Course Notes and Workbook for Introduction to Sound*, which will be accompanied by *printed copies of several hundred PowerPoint slides* that some faculty might choose to use in their classes; and
- A concerted effort has been made to ensure that “nuisance errors” have been virtually eliminated.

In all instances, the modifications, additions, and deletions were driven by a single force: the change must promise to enhance teaching and improve comprehension of the fundamental concepts of acoustics.

Preface to the Fourth Edition

The fourth edition of *Introduction to Sound* is written to *teach* fundamental concepts of acoustics to students in the speech-language-hearing sciences and related behavioral science disciplines.

Chapter 10, *Room Acoustics*, is an important addition to the book. Although noisy and excessively reverberant rooms are annoying (think of the ubiquitous “sports bar”), our principle concern with the acoustical conditions in a room is: “*Can people, some with normal hearing and others with a moderate-to-severe hearing loss, understand speech easily?*”

Design/redesign plans for acoustical treatments of rooms should be formulated to optimize speech understanding. Hearing and speech scientists, audiologists, or speech-language pathologists should participate with acoustical engineers in that endeavor. To prepare for their role on the team, hearing and speech professionals must understand the fundamental principles of room acoustics. We must be thoroughly familiar with techniques for assessing the intelligibility of speech. It is our responsibility to be forceful and effective advocates for the special acoustical requirements of listeners who have moderate-to-severe hearing loss and those who wear hearing aids or other sound enhancement devices.

Hearing and speech professionals must be able to engage in meaningful dialogue with acoustical consultants in the development of plans for optimization of acoustical design/redesign treatments. Chapter 10 is devoted to fundamental concepts of room acoustics (e.g., reflection, “reverberant tails,” absorption, signal-to-noise ratio, etc.). It also includes a chronicle of the research literature on intelligibility testing by focusing on two distinctly different approaches for determining how well speech is understood. One approach is the conventional psychometric test that uses monosyllabic words, sentences, or connected discourse. A second is an engineering-based model in which relevant acoustic parameters of speech and selected elements of the acoustic environment are joined in an algorithm to *predict* rather than *measure* speech intelligibility.

I emphasized the word “*teach*” in the opening sentence. The fourth edition reflects my belief that acoustical concepts can be understood by all of our students, including those who are not thoroughly grounded in mathematics and physics. I hope readers get the sense that as they *read the book* they are *listening to a lecture*.

An effective teacher calls on all the prosodic elements of our language by using many forms of “vocal emphasis.” That does more than just enliven the lecture; it directs attention toward important facts

and ideas. I have tried to deploy the writing parallels of “vocal emphasis” with a liberal use of *italics*, **boldface**, underlining, and sometimes all three when I think ***a concept deserves special emphasis.*** I also make deliberate use of redundancy—some acoustical concepts are repeated in several different contexts to drive home the point that “*this concept is important to know and understand.*”

There also some structural changes in the fourth edition. Some of the paragraphs in the earlier editions were too long. It often was a challenge for students to discern the really important messages embedded in the passage. Many of those lengthy paragraphs are now formatted differently. Key concepts have been extracted and presented in bulleted or numbered fashion. For example:

- The original eight chapters have been restructured into ten, more focused chapters.
- *Practice Problems* and the *Answers to Practice Problems* have been moved from *Course Notes and Workbook for Introduction to Sound* to the textbook and are also available on the PluralPlus companion website. See the inside front cover of your textbook for access instructions.
- A *Glossary* of important terms has been added as a complement to an *Alphabetical Listing of Selected Equations*.
- Finally, *Animations for Introduction to Sound* is now readily available to faculty and students. Visit <https://sites.google.com/a/umn.edu/soundanimations> to gain access to 17 dynamic representations of acoustic events. The animations show—in slow motion—events in acoustics that normally change too rapidly over time to be visualized. The topics include: vibrating tuning forks; pendular motion; simple harmonic motion; transverse and longitudinal wave motion; reflection of sound from plane, concave, and convex surfaces; standing waves; and so forth.

I hope the combination of new material and structural changes will accomplish my objective to *teach* the fundamental concepts of acoustics.

Acknowledgments

I pay a special tribute to my sons, Brandon and Jeffrey, and my daughters-in-law, Liz, and Kelsey. They encouraged me to emerge briefly from retirement and return to writing and teaching. This fourth edition is my “classroom,” and I hope it will serve its purpose of *teaching* concepts in acoustics that are essential for students in the speech-language-hearing sciences.

Edward Carney has done a masterful job with updating and improving *Animations for Introduction to Sound*. His expertise has enhanced the quality of the animations as a valuable teaching tool. I am most grateful for his assistance and collegueship.

I also acknowledge the special role of Jim Jerger in my professional life. His leadership, mentorship, and friendship were invaluable in shaping my career. Jim led by example. He stressed excellence in research and clarity of thought. Jim also advocated for simplicity in writing. I believe his article, “*Scientific writing can be readable*” (1962), should be required reading for all of our graduate students. Jim pointed me in new directions, and his support knew no boundaries. Jim, I cannot repay you, but I can say, “thank you for everything.”

*In Memory of Nancy E. Niccum, PhD
loyal and supportive wife,
Devoted and loving mother, valued colleague, and best friend*

*And to our Sons and Daughters-in-Law
Charles Brandon Speaks (Elizabeth Jensen-Speaks)
Jeffrey Bryant Speaks (Kelsey Lynn Speaks)*



The Nature of Sound Waves

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“If a tree falls in a forest and no one is around to hear it, is there sound?” The answer depends on the distinction between two different perspectives for defining sound: *physical* and *psychological*.

Albers (1970) wrote that sound “in the strict sense, is a compressional wave that produces a sensation in the human ear” (p. 36). When “sensation of hearing” is included in the definition of sound, the *psychological* attributes of sound are invoked: pitch, loudness, and timbre. In other words, from a psychological point of view, “sound is what we hear.”

We certainly are aware of the many “sounds” around us—sounds such as human speech, the barking of a dog, the crying of an infant, the cooing of a dove or of a “significant other,” music of all forms, thunder, traffic noises, and the exhilarating roar of water cascading down the side of a mountain. A psychological approach to defining sound is tempting. It might seem that it would be easier to understand the *physical events* that characterize sound by reference to the psychological sensations or feelings that are associated with the many sounds that we experience daily. But the reverse is more correct; it is easier to explore the nature of the psychological sensations to sound if we thoroughly understand the physical characteristics.

An alternative is to define sound from a *physical* perspective. Sound is defined by reference to properties of the *source* of the event called “sound” and to properties of a *medium* in which, or along which, sound is transmitted. When *physical* properties of sound are emphasized, sound does exist *even if the receiver is absent or is not functional*. In other words, sound exists even if no one is in the forest.

Many objects can serve as a source of sound: vocal folds; the strings of a piano, guitar, or violin; the membrane of a drum; the bars of a xylophone; the metal plates of cymbals; and so on. In each case there is one essential prerequisite for a body to be a source of sound—it *must be able to vibrate*. That requires two physical properties: **mass** and **elasticity**. All bodies in nature possess both of those two properties to some degree.

When a potential source of sound is set into *vibratory motion*, or *oscillation*, sound occurs, and the sound that is created can then be transmitted from the source through, or along, some medium. Air is probably the most familiar medium that we encounter. But, as we shall see, other molecular structures, such as, for example, water, wires, strings, glass, wood panels, steel rails, and so forth can also transmit sound. Because all molecular structures have some finite **mass** and **elasticity**, each is capable of being both a source of sound and a medium for its transmission. Of course, some structures will be more effective sources or more effective transmitters than others.

Although the properties that permit a structure to be a source of sound are essentially the same as the properties that permit a medium to transmit sound, it is convenient to describe the properties of the transmitting medium and the properties of the source separately.

PROPERTIES OF THE TRANSMITTING MEDIUM

Consider air as a medium for transmitting sound. Air consists of approximately 400 billion billion (4×10^{20}) molecules per cubic inch (**in.**). In the quiescent state (before a source of sound begins to vibrate), the air molecules move randomly at speeds that average nearly 940 miles per hour (**mph**), or 1,500 kilometers per hour (**kph**). Although molecular motion is random, the molecules maintain some *average distance* from one another. Thus, we can envision the molecules as being distributed fairly evenly throughout the air space.

The billions upon billions of molecules exert a pressure on whatever they come in contact with. For example, when the randomly moving air molecules impinge on the human eardrum (or any other structure), pressure is exerted on the drum. Interestingly, as we shall see later, *that does not yet produce a sensation of “hearing” sound*. At sea level that pressure, which is called “atmospheric pressure,” amounts to about 14.7 pounds (**lb.**) per square in. (**lb./in.²**), and 14.7 lb./in.² in the English measurement system is equivalent to approximately 100,000 newtons (**N**) per square meter (**N/m²**) or 1,000,000 dynes per square centimeter (**dynes/cm²**) in the metric systems. The **N/m²** and **dyne/cm²** will be defined later when the concepts of both force and pressure are developed more fully.

To conceptualize the pressure in air, consider the cylindrical tube shown in Figure 1–1, which has a cross-sectional area of 1 in.² and extends from sea level to a height of more than 25 miles. At sea level, in the quiescent state, there is a pressure of approximately 14.7 lb./in.² acting downward. At 10 miles above sea level, the pressure is reduced to about 1.57 lb./in.², and at a height of 25 miles, it is only a negligible 0.039 lb./in.²

Air, and all other bodies that can serve to transmit sound, is characterized by two essential physical properties: **mass** and **elasticity**.

Mass

Mass is *the amount of matter present*. Air is a gaseous matter, but the definition of mass also holds for liquids and solids.

Mass Contrasted with Weight

Mass sometimes is confused with **weight**. **Mass** refers to the quantity of matter present, whereas **weight** refers to *the attractive gravitational force exerted on a mass by the earth*. For example, a person is said to weigh 160 lb. because the earth attracts the person with a force of 160 lb. If that person is flown to the moon, the same amount of matter is present, but because of the lessened gravitational pull, the weight amounts to about 27 lb. because the force of gravity is only about one-sixth as great on the moon as it is on earth. The

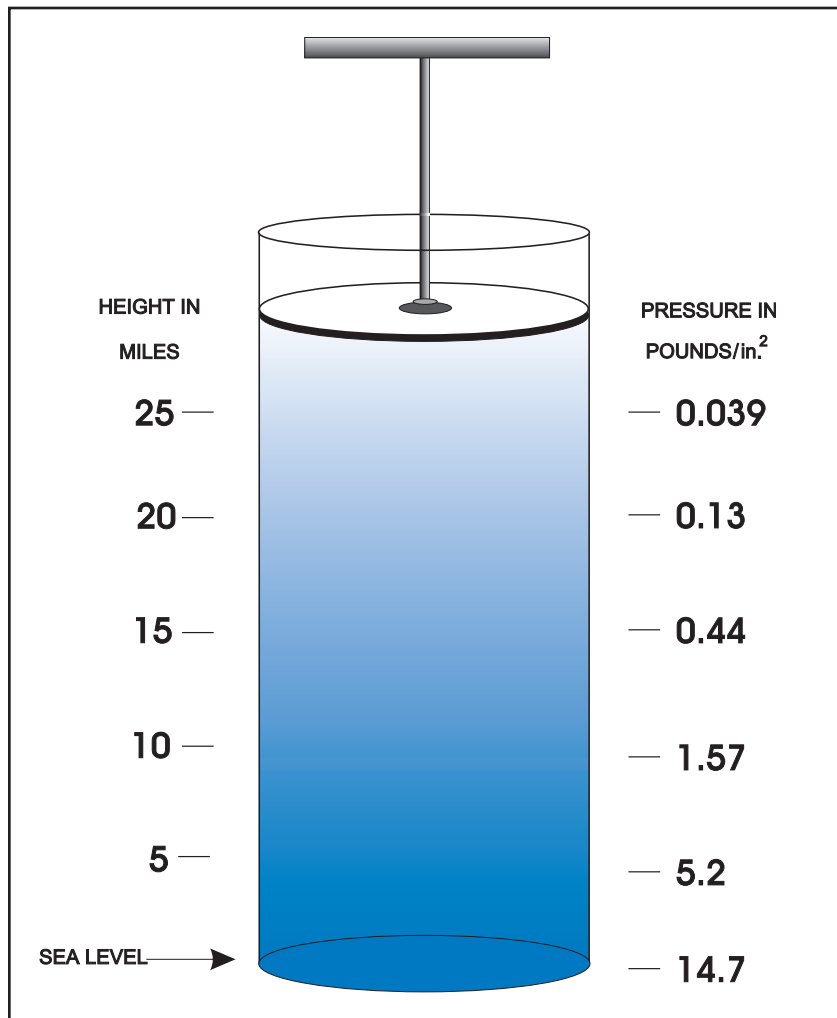


Figure 1-1. A cylindrical tube with a cross-sectional area of 1 in.² that reflects how **pressure** and **density** in an air medium vary with height above sea level.

weight of an object *is directly proportional to its mass*, but weight and mass are simply different concepts. **Weight** is a force, whereas **mass** is the quantity of matter present.

Air has weight as well as mass. A cubic meter of air weighs about 1.3 kilograms (**kg**), and the air in a classroom with the dimensions of $9 \times 12 \times 4$ m weighs about 560 kg. For those who are not yet comfortable with meters and kilograms, a cubic yard (**yd**) of air weighs 35.1 ounces (**oz**), and the air in a classroom with the dimensions of $30 \times 40 \times 12$ feet (**ft.**) weighs about 1,170 lb. From that we might conclude that professors who deliver long lectures in a classroom of that size are “throwing a lot of weight around.”

Mass and Density

It also is important to distinguish between **mass** and **density**. Look again at the cylindrical tube filled with air in Figure 1–1. The air molecules are crowded closely together (darkened regions) near the bottom of the tube, whereas they are rather far apart (lighter regions) in the higher portions of the tube. This occurs because of the pull of gravity.

Because of gravity, the molecules of the atmosphere accumulate near the surface of the earth. A downward force causes the molecules to be compressed into a smaller volume. The volume near the bottom of the tube is more densely packed, and when a greater number of molecules is compressed into a volume of a certain size, the **density** is increased.

Density (ρ) is *the amount of mass per unit volume*. For example, if we exert a force that causes a volume of 1 cubic in. of air to contain 800 billion billion (8×10^{20}) molecules instead of 400 billion billion (4×10^{20}), the density—the mass per unit volume—is doubled. It is easy to see in Figure 1–1 that *the amount of mass per unit volume in the cylinder decreases with increasing height above sea level*.

It might be difficult to imagine the different densities associated with the invisible molecules in volumes of air, but there are more visible examples that might make the distinction between mass and density clear. Imagine a grocery bag with a volume of 0.06 cubic meter that is filled with 50 loosely crumpled sheets of newspaper. If you pack the paper more tightly until the same amount of paper (50 sheets) occupies only half of the bag's volume (0.03 cubic meter), the same amount of matter is present—the **mass**—but the matter is packed into a smaller volume. After compression, the amount of mass per cubic meter—the **density**—has doubled.

With respect to the first property of a transmitting medium, it is useful to refer to both the **mass** of a medium and to the **density** of a medium, a quantity derived from mass. We shall subsequently explain what is meant by “a quantity *derived* from another quantity.”

Elasticity (E)

Elasticity is the second property of a transmitting medium. All matter, whether gaseous, liquid, or solid, undergoes distortion of either shape or volume or both when a force is applied to it. Moreover, all matter is characterized by the tendency to “recover” from that distortion. The property that enables recovery from distortion to either shape or volume is **elasticity**. We shall see subsequently that elasticity is more properly defined as *the ability to resist changes in shape, volume, or position* rather than the ability to recover from such changes.

Imagine a weight attached to a spring suspended from the ceiling. When the spring is stretched and then released, it returns to its

original position (and beyond) unless it has been “overloaded.” By “overloaded” we mean that the original stretching of the spring is sufficient to exceed its **elastic limit**. If the applied force exceeds the elastic limit, deformation is permanent. If the applied force exceeds the elastic limit by a sufficient amount, the object breaks.

A portable radio has a spring that holds the battery in place. If you remove the spring, you can verify that it is relatively easy to stretch it so far that it will not “spring back” when released. Its elastic limit was exceeded. In some forms of matter, the elastic limit is very small. In other forms, such as tempered steel, the elastic limit is very large. The elastic limit of air is so large that it need not concern us.

With air, the concept of elasticity means *the tendency of a volume of air to return to its former volume after compression*. Return again to the air-filled cylinder in Figure 1–1. We know that air molecules are present, that they are in random motion, that—on average—they are equidistant from each other, and that the density of the air is greater near the bottom of the tube.

Suppose we now insert a plunger into the cylinder and push downward. All molecules that were present in the full length of the tube are crowded (compressed) into a smaller space; the **density** is increased. When the plunger is removed, the air molecules return to their former “position,” or more appropriately, *the air volume resumes the density* that existed before compression. The density of the air is *restored*, and the restoring force is **elasticity**.

■ PROPERTIES OF THE SOUND SOURCE

Let us now consider bodies that can serve as a *source of sound*. We will consider the same two properties that characterized the transmitting medium: **mass** (or density) and **elasticity**.

Vibratory Motion of a Tuning Fork

A tuning fork, as shown in Figure 1–2, is one source of sound. The tuning fork is a U-shaped metal bar. The prongs, or tines, of the fork have **mass** (a quantity of matter is present) and they also possess the restoring force of **elasticity**. Because of their elasticity, the tines of the fork return to their former position after they have been displaced. This is illustrated by striking the fork gently with a soft hammer. The tines are set into vibration, which takes the form of each tine moving back and forth.¹

View Animation 1_2. The Vibrating Tuning Fork at <https://sites.google.com/a/umn.edu/soundanimations>

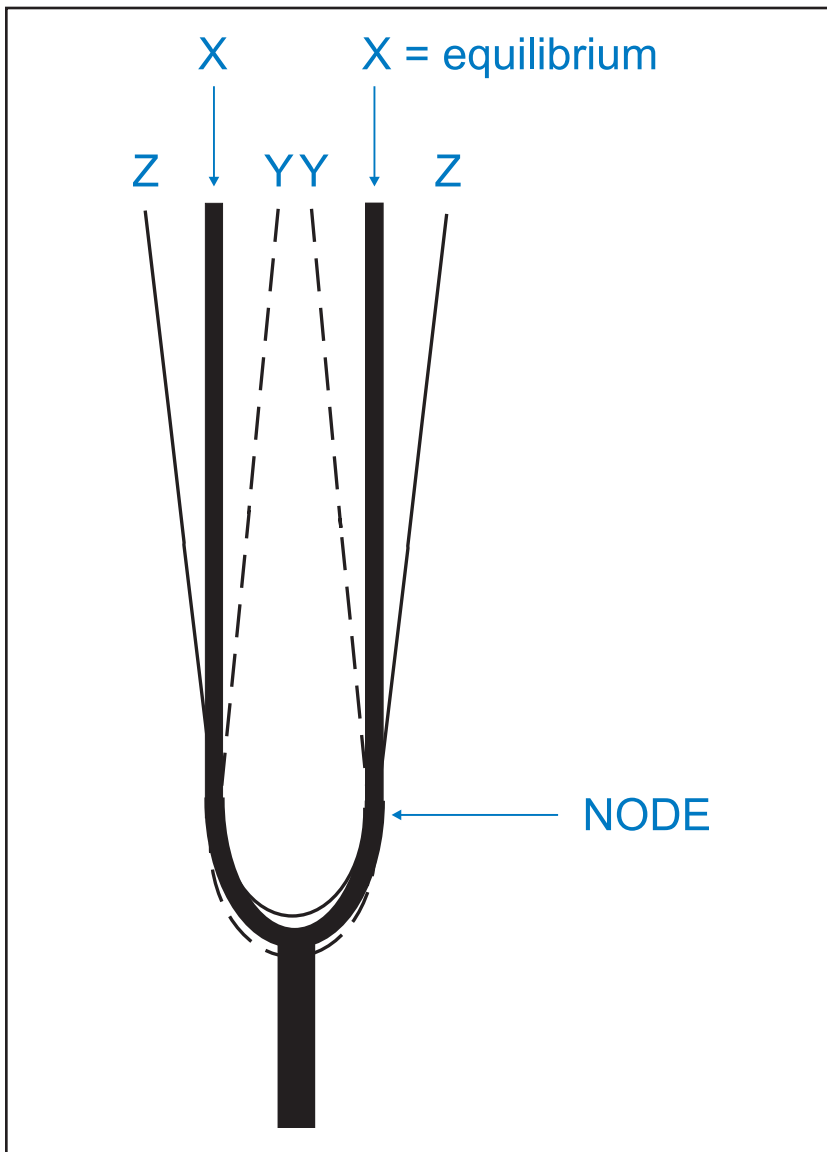


Figure 1–2. The vibratory pattern of a tuning fork, a U-shaped metal bar with the properties of **mass** and **elasticity**. Once struck, the tines move from **X** (equilibrium), to **Y** (maximum displacement in one direction), back to **X**, to **Z** (maximum displacement in the other direction), and back to **X** to complete one **cycle** of vibration.

Displacement from Equilibrium

Imagine that we can “zoom in” and observe the pattern of vibration of the two tines. The position of the fork before a force is applied is its *equilibrium position*, and the heavy solid lines labeled **X** in

Figure 1–2 show equilibrium for the fork. When the hammer strikes the fork, a force is applied that causes the tines to be displaced from equilibrium. Both tines move inward toward positions **Y**, which are shown with the dashed lines.

- *The amplitude of displacement of the tines is proportional to the magnitude of force applied.* In other words, the harder the fork is struck with the hammer, the greater the amplitude of displacement from equilibrium.
- Upon reaching maximum displacement inward, *the tines' motion is momentarily halted*, and then, because of the restoring force of elasticity, the tines return toward equilibrium, **X**.
- Motion does not cease after the tines have returned to positions **X**. Instead, the tines move through equilibrium toward maximum displacement in the opposite direction, which is shown by the thin solid lines labeled **Z**.

Newton's First Law of Motion: Inertia

Movement of the tines beyond equilibrium to maximum outward displacement (**Z**) is explained by the first of Sir Isaac Newton's three laws of motion: *All bodies remain at rest, or in a state of uniform motion, unless another force acts in opposition.* In other words, the tines (and stem) move beyond equilibrium until a reaction, or restoring, force *stops* such motion. The property that Newton's first law addressed is **inertia**, and his first law can be stated as *the tendency of a body in motion to remain in motion, and the tendency of a body at rest to stay at rest.*

The amount of **inertia** that an object has *is directly proportional to the mass of the object.* In fact, we may think of **mass**, the quantity of matter present, as a *measure of inertia*. Thus, a massive object such as a steel ball with a circumference of 1 meter has considerable inertia in comparison with the more negligible mass and inertia of a ping-pong ball. Because of the large inertia of the steel ball, a large force is required to start it in motion, to stop its motion, or to change the direction of its motion. In contrast, a ping-pong ball can easily be set in motion, its motion can easily be stopped, and the direction of its motion can easily be changed.

Vibratory Motion

We have seen that the tines of the tuning fork are displaced (toward **Y**) by application of an external force, and that the magnitude of displacement is proportional to the force applied. Next, we saw that the restoring force of **elasticity** (F_r) causes the tines to return toward equilibrium (**X**), and the tines then move *through equilibrium* toward maximum displacement in the opposite direction (**Z**). The process continues as the restoring force of elasticity returns the tines

back toward equilibrium (**X**), inertia causes the tines to be displaced beyond equilibrium (**Y**), and so on.

Because of the interaction of **elasticity** (an elastic restoring force, F_r) and **inertia**, the tines of the fork continue to move to and fro, and the fork is set into **vibration**. It is convenient to refer to displacement in one direction as positive (+) and displacement in the opposite direction as negative (-); the designation of which direction is positive and which is negative is arbitrary.

Vibratory motion consists of back and forth movement of a body that has **mass** and **elasticity**. Because all bodies are characterized by some finite amount of mass and elasticity, each is capable of being set into vibratory motion.

Newton's Third Law of Motion: Reaction Forces

The fact that two forces are operating rather than just one is consistent with Newton's third law of motion, which states: *with every force there must be associated an equal reaction force of opposite direction*. A hammer, on striking a nail, exerts a force on the nail. The nail exerts an equal, but opposite, force on the hammer, which causes a rapid deceleration of the hammer's movement.

Another example of a reaction force can be observed by pushing your hand against the edge of a desk. You are exerting a force on the desk. Notice, though, that the shape of your hand is distorted, which occurs because of the reaction force exerted on your hand by the desk because the desk is elastic. All solid materials possess some degree of elasticity, and because of their elasticity, they can exert forces on other objects.

Newton's third law tells us that *a force cannot exist alone*, which is why it is impossible to have a one-person tug of war. A more familiar analogy is to consider why a car gets stuck in the snow. Because of the slippery snow, we cannot develop sufficient frictional force to "push the car out." It is stuck, and even a powerful engine might not alter that unhappy state. Instead, we must develop sufficient traction by spreading a substance, such as sand, or by digging down to the earth so that the "push of the ground" causes the car to move.

In terms of the tuning fork, after the original force is applied to cause the tines to be displaced, the tines move in one direction due to inertia, and the movement continued in that direction until **inertia** is overcome by the **elastic restoring force** (F_r). In other words, *elasticity is the "reaction force" to inertia*.

An important feature of the vibratory motion of a tuning fork (and of other sources of sound) is that *we do not need a constant reapplication of the external force* that produces the initial displacement from equilibrium. Inertia and elasticity sustain the movement of the fork much as one push of a child's swing in the park is sufficient to keep the child swinging for some time.

Movement of the tines from equilibrium to maximum displacement in one direction, back to equilibrium, to maximum displacement

in the opposite direction, and finally back to equilibrium constitutes *one cycle of vibration*. Many such cycles will occur before the vibratory motion is finally stopped, and then we must strike the fork again to achieve additional vibration.

■ SOUND SOURCE ACTING ON A MEDIUM

Mass and **elasticity** are the two essential properties of both the source of sound and of the medium that transmits sound. Next, let us place the tuning fork in a medium such as air, apply an external force to cause the tines to be displaced, and see how the moving tines affect the surrounding air medium.

The effects on the air medium are illustrated in Figure 1–3. The rows, from top to bottom, represent discrete positions of the tines of the tuning fork and of individual air molecules at various moments in time during vibratory movement: equilibrium, maximum displacement (+), equilibrium, maximum displacement in the opposite direction (–), equilibrium, and so on. *Each column, from left to right, shows the position of a single air molecule.*

View Animation 1_3. Movement of Air Particles at <https://sites.google.com/a/umn.edu/soundanimations>

Movement of Air Particles

In the top row the molecules are equidistant from one another before they are energized by the moving tines. Imagine, then, that the tuning fork has been vibrating to and fro for some time and that it is possible to visualize the effects of the moving tines on the air molecules, one by one, over time.

The vibratory motion of the fork exerts a force on molecule **A**. In row 2, molecule **A** is displaced from equilibrium to some maximum displacement that is proportional to the magnitude of force applied to it. The molecule also is set into vibratory motion and moves back and forth (from row to row) about its position of equilibrium—because of **inertia** and the **elastic restoring force** (F_r)—in the same way that was described for the tuning fork tines.

Movement of molecule **A** exerts a force on molecule **B**, causing it to commence vibratory motion, and this application of forces is spread from molecule to molecule (from column to column) much as you see when you flick the first of a long row of standing dominoes. We can think of the air medium as a series of masses (the air molecules) that are connected by “springs,” in which the springs reflect the intermolecular forces between the masses.

Figure 1–3 also enables us to see the **difference** between **particle movement** and **wave propagation**. Look at the molecule, or particle,

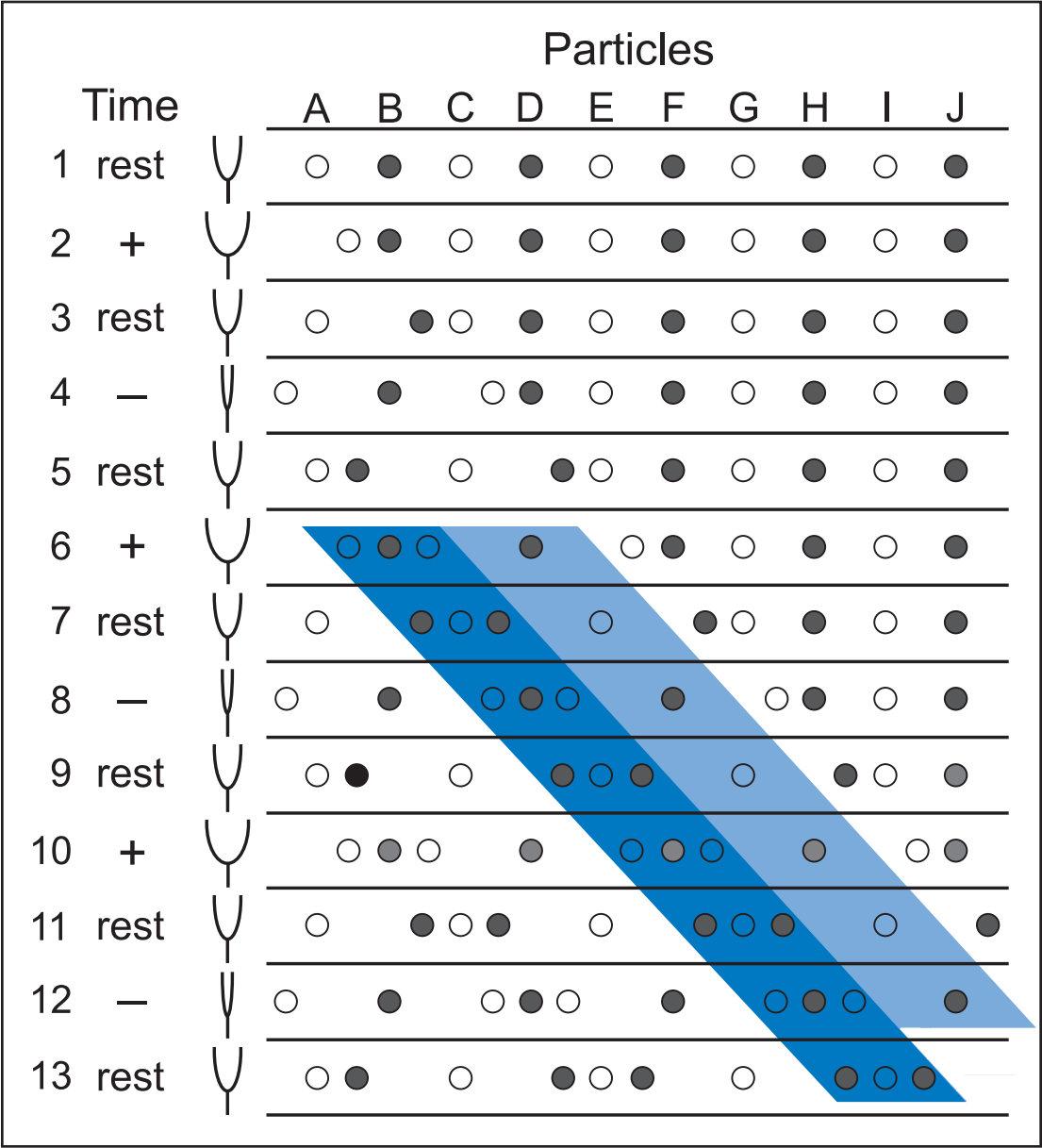


Figure 1-3. Displacement of individual air particles due to the vibratory motion of a tuning fork. Each particle moves back and forth over an infinitesimally small distance about its average position. The darker shaded area traces a **compression** moving to the right over time, and the lighter shaded area traces a **rarefaction**, also moving to the right over time.

in Column A as it moves through three **cycles** of vibration from the top row to the bottom row. Particle A and, in turn, each of the other particles, simply oscillate to and fro about their average equilibrium positions. *The medium is not displaced over an appreciable distance.*

Next, look at the darker of the two shaded areas of the figure.

- In row 6, molecule **A** is maximally displaced to the right, **B** is at equilibrium, and **C** is maximally displaced to the left. Molecules **A**, **B**, and **C** are crowded closely together; *that region in space is in a state of increased density, a state called **compression** because the molecules are “compressed” closely together.*
- One-quarter cycle later (row 7), the region of **increased density**, or **compression**, has moved to the right to involve particles **B**, **C**, and **D**. Over time, the **compression** continues to move rightward through space and involves particles **C**, **D**, and **E**, then particles **D**, **E**, and **F**, and so on.

In summary, the darker of the two shaded areas in Figure 1–3 traces the movement of a compression from left to right away from the vibrating source. Now look at the lighter of the two shaded areas of the figure, commencing with column **D** and row 6.

- Particle **C** has moved maximally to the left, particle **D** is at equilibrium, and particle **E** has moved maximally to the right. The region in space associated with particle **D** is now thinned, or rarefied; *that region in space is in a state of decreased density, a state called **rarefaction**.*
- One-quarter cycle later (row 7), the region of decreased **density**, or **rarefaction**, has moved to the right to particle **E**. Over time, the **rarefaction** continues to move rightward through space to locations involving particles **F**, **G**, **H**, and so on.

In summary, **particle movement** consists of an oscillation to and fro about the average equilibrium position; *the oscillating particles in the medium are not displaced over an appreciable distance*. In contrast, **wave propagation** consists of alternate regions of increased **density** (**compressions**) and decreased **density** (**rarefactions**) that move through the medium away from the source. Some specified region in space that is in a state of **rarefaction** at one moment in time (e.g., column **D**, row 6) is in a state of **compression** one-half cycle later (e.g., column **D**, row 8).

Movement of the Air Mass

Recall that in air there are approximately 400 billion billion (4×10^{20}) molecules per cubic in. It therefore makes more sense to think about the effects of tine movement on the surrounding **air mass** rather than on just a single molecule. This is illustrated in Figure 1–4. The outward movement of the tines toward a positive (+) displacement causes the surrounding air molecules to be crowded, or compressed, as the force of displacement is passed from molecule to molecule. In other words, there is an increase in **density** of the air because the

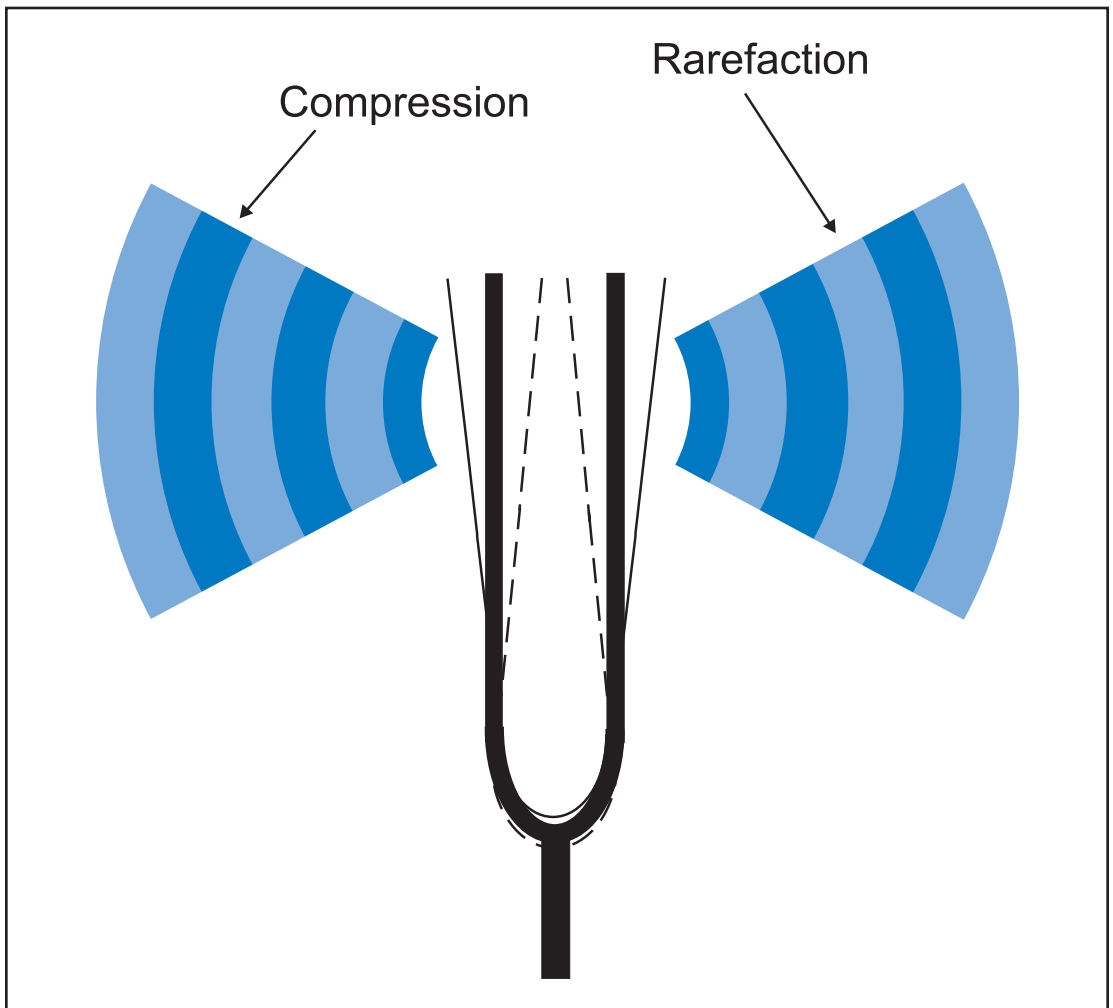


Figure 1–4. Alternate regions of **compression** (darker shading) and **rarefaction** (lighter shading) move outward through an air mass because of the vibratory motion of a tuning fork.

force of the moving tines causes the molecules to be compressed into a smaller volume. This **compression** phase of sound is shown by the darkened bands in Figure 1–4.

View Animation 1_4. Disturbance of the Air Mass at
<https://sites.google.com/a/umn.edu/soundanimations>

As the tines return toward equilibrium because of the elasticity of the fork, the force on the surrounding medium is relieved, and the air molecules return toward their positions of equilibrium. This results in a thinning of the molecules in the sense that they are

now less crowded and the medium is therefore characterized by less density. This **rarefaction** phase of sound is shown in Figure 1–4 by the lightened bands.

As we look at the air mass at points farther and farther from the tuning fork, there are alternate regions in which the molecules are crowded together with an increase in density of the air space (**compression**) and regions in which the molecules are thinned with a decrease in density of the air space (**rarefaction**). These alternate regions of compression and rarefaction are caused by the effects of the moving tines being passed from region to region of the air.

Displacement of the Air Medium and Wave Motion

The medium is not displaced over an appreciable distance during this process. An individual molecule is displaced only over a very small distance. What is moving through the air space is a “wave of disturbance.” Alternate regions of increased and decreased density move through the air medium. If we could take a microscopic look at just one region of the air to see how its density changes over time, we would see that its volume is characterized by alternately increased and decreased density. The increased density of one region, in turn, is followed by an increased density of the next region, and so on. Thus, a region that is in a state of compression [in Figure 1–4] at one moment in time is in a state of rarefaction a brief moment later in time.

In a very elastic medium, such as air, sound is characterized as the *propagation of density changes* through the elastic medium. Although changes in density are propagated through the medium, the individual molecules are not permanently displaced. Think again of the domino analogy. A single domino does not move far—it just falls down. But a wave of disturbance moving along the row of dominoes is clearly visible.

An easy way to see wave motion is to observe “the wave” that is popular at many sporting events. The individual spectators move back and forth and, if sober, are not displaced from their seats. If the movements of the spectators are properly synchronized, however, a “wave” can be seen to move around the stadium.

The properly synchronized flashing of light bulbs can create the same image of wave movement. The entrance and marquee of some theaters contain long rows of light bulbs that can be alternately illuminated and turned off. If all the lights flash on and off simultaneously, we do not see a “wave”; we simply observe that the lights are blinking on and off. If the electrician times their flashing carefully, however, a “wave” is seen to move along the row of lights.

In this discussion of properties of both the sound source and the transmitting medium, we have referred to certain physical quantities such as **mass**, **density**, **force**, **pressure**, **displacement**, and so on. Before we explore the characteristics of sound, it is useful to review

those and other physical quantities in greater detail and separate the physical quantities into two categories: **fundamental** and **derived**.

FUNDAMENTAL PHYSICAL QUANTITIES

There are three **fundamental quantities** in physics: **length**, **mass**, and **time**. All other quantities are derived from those fundamental ones, which is to say that quantities other than length, mass, and time are **derived quantities**. To specify the magnitude of both fundamental and derived quantities, we use either of two measurement systems: the English or the metric. There are two versions of the metric system: the **MKS** and **cgs**.

- With the **MKS** metric system, length is measured in meters (**m**), mass in kilograms (**kg**), and time in seconds (**s**).
- With the **cgs** metric system, length is measured in centimeters (**cm**), mass is measured in grams (**g**), and time in seconds (**s**).

The English **fps** system is more cumbersome. Length is measured in feet (**ft.**), mass in pounds (**p**) (although the pound actually is a measure of gravitational force rather than mass) and time in seconds (**s**).

We will describe the three fundamental quantities and their derivatives primarily by reference to the metric system. The **MKS** metric system is the system of choice in physics. The **cgs** metric system, however, appears in much of the literature that has been written over the decades, and we must therefore be familiar with both systems and be able to move back and forth between them. Both metric systems will be used throughout this text, but the emphasis will be on the **MKS** system. Only on rare occasions will we persist in referring to the **fps** system.

Length

Length is a measure of distance, which is *the extent of spatial separation between any two points*. To determine the magnitude of distance, we select a unit of measure such as the meter (the **MKS** system), and through the process of measurement, we determine how many times this unit (the meter) is contained in a given distance. To measure the length of a board, we select a meter stick (or a yardstick or tape measure in the **fps** system) and determine how many times the meter stick can be contained over the length of the board.

What is “a meter?” For many years the **meter** was defined arbitrarily. In the late 1700s, the meter was defined as 1/10,000,000 of the distance from the equator to the North Pole along the surface of the earth. Confirmation of that distance, however, proved to be virtually impossible as well as hazardous. New definitions emerged

in 1889 and again in 1960. Finally, in 1983 the 46-member General Conference on Weights and Measures met in Paris and agreed to define the meter relative to another of the three fundamental physical quantities—**time**. The speed of light is 299,792,458 meters per second (**m/s**), and *one meter now is defined as the distance traveled by light in a vacuum during $1/299,792,458$ of a second.*

That change in definition of the meter is not really as trivial as it might seem. For example, 1 **m** used to equal *about* 39.37 in., but as of 1983, 1 **m** equals *exactly* 39.37007874 in.; all additional decimal values beyond the eighth are zero. This seemingly small difference is fairly important to the physicist. Previously, 1 in. equaled *approximately* 2.54 **cm**, but now 1 in. corresponds to *exactly* 2.54 **cm** and 1 ft now equals *exactly* 0.3048 **m**. More importantly, the new definition accomplished a goal that physicists had sought for a long time; the physical quantity **time**, which is the most accurately measured of the three fundamental quantities, now is used to define another physical quantity, **length**.

The metric systems, either **MKS** or **cgs**, are convenient for describing multiples or submultiples of the standard unit of length. All are expressed as multiples of 10 (.001, .01, .1, 1, 10, 100, 1,000, and so on). One centimeter (**cm**) equals one one-hundredth (.01) of a meter (**m**), one millimeter (**mm**) equals one one-thousandth (.001) of a meter and one-tenth (0.1) of a centimeter, and one kilometer (**km**) equals 1,000 **m**. That is why the metric system is less cumbersome than the English system, where you must divide by 12 to convert inches to feet; by 3 to convert feet to yards; by 5,280 to convert feet to miles; by 1,760 to convert yards to miles; and so on. Only the submultiples of the inch ($1/2$, $1/4$, $1/8$, $1/16$, $1/32$, $1/64$) possess reasonable regularity.

Mass

Mass is a property of all matter and it refers to the quantity of matter present. By “quantity” we mean that there is an observable property that can be measured or specified numerically. Recall, also, that it is important to distinguish between **mass** and **weight**. The *unit of measure of mass* is the kilogram (**kg**) in the **MKS** system and the gram (**g**) in the **cgs** system. *One kilogram is the mass of a cylinder of platinum housed at the International Bureau of Weights and Measures, and a gram is $1/1,000$ of a kilogram.*

The concept of **mass** is tied inextricably to the concept of **inertia**, and the word “mass” is sometimes used to refer to how much inertia a body has. The greater the mass of an object, the greater its inertia.

Time

Time might be the physical quantity that is most easily and accurately measured, but it also is a quantity that is difficult to define in

a way that has intuitive appeal. Daniloff, Schuckers, and Feth (1980), for example, wrote:

It is difficult to define *time* without saying that it is that quantity we measure in seconds, minutes, or hours. Any physical event that recurs on a regular basis can be used to mark off equal intervals of time. Thus, the period required for one complete revolution of the earth is identified as one day. (p. 75)

Because a solar day is divided into 24 hours (**hr**), an hour into 60 **min**, and a minute into 60 **s**, each day has 86,400 **s**. That led to the concept that *one second is 1/86,400 of a solar day*.

DERIVED PHYSICAL QUANTITIES

A **derived quantity** is *either a quotient, or a product, of the fundamental quantities or of fundamental and other derived quantities*. For example, length is a fundamental physical quantity. The *area* of a rectangle is given by the product of two measures of length ("length" times "width"); area, therefore, is a derived physical quantity. Derived quantities of interest in the study of sound include **displacement (x)**, **velocity (c)**, **acceleration (a)**, **force (F)**, and **pressure (p)**.

Displacement (x)

Displacement (x) is *a change in position*, and is specified by calculating the distance from a reference, or starting position, to a new, or ending, position. Displacement involves two concepts: **distance** and **direction**. A body moves over a certain **distance**, but, for a given distance, the body can move in several different **directions**. Thus, *to specify displacement we must account for both the direction moved and the distance moved*.

Quantities that are specified by both *magnitude and direction* are **vector quantities**, or just **vectors**. Consider the examples shown in Figure 1–5. Starting points are shown by an "**x**" and ending points by an "**o**." The length of the line represents the distance moved, and the direction the arrow points represents the direction moved. Thus, **A** and **B** have *different magnitudes of displacement* because their movements are in different directions, even though both **A** and **B** moved the same distance. **A** and **C** have *different magnitudes of displacement* because they have moved different distances, even though their movements are in the same direction. Finally, **B** and **C** have *different magnitudes of displacement* because their movements are characterized by both different directions and different distances.

Quantities such as **mass**, **time**, and **energy** are specified only by reference to magnitude; they have no direction associated with them. They are **scalar quantities**, or just **scalars**. The distinction between scalars and vectors is particularly important when we want to add

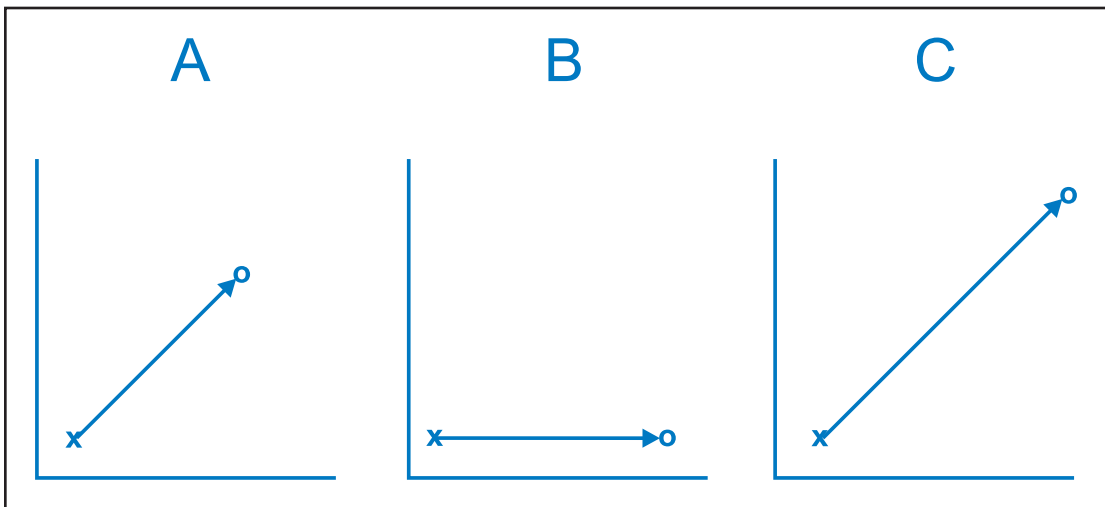


Figure 1-5. Explanation of vector quantities in which the *length* of each line represents the **magnitude of displacement** (distance moved) and the *direction* of the arrow represents the **direction of displacement**.

or subtract the values of a quantity. When the quantities are scalars, simple algebraic addition or subtraction is sufficient. **Time** is a scalar. If the time required to travel from point **a** to point **b** is 1 s, and the time required to travel from point **b** to point **c** is 1.5 s, then algebraic addition tells us that the time required to travel from **a** to **c** must be 2.5 s. When the quantities are vectors, we must use methods of vector analysis rather than algebraic addition or subtraction.

Velocity (c)

Velocity (c), probably from the Latin word *celeritas* for quickness, swiftness, or speed, is *the amount of displacement per unit time*. Velocity is a vector quantity because specification of displacement per unit time incorporates both the distance and the direction moved. Thus, displacement is accomplished during some time period. If we divide displacement by the time required to complete the displacement, we calculate the **average velocity** of the body that was displaced.

If a motorcycle travels in a northwesterly direction for a distance of 180 kilometers (**km**), or 112 miles, in 2 **hr**, the average velocity of travel is 90 **kph** (56 **mph**). We are accustomed to thinking of **speed** rather than velocity in such cases, but speed and velocity are not always equal. Speed corresponds to velocity *when direction is not considered*. Speed is a **scalar** that reflects only magnitude, whereas velocity is a **vector** and reflects both magnitude and direction.

It is easy to understand the distinction between speed and velocity if you imagine you are paddling a canoe across a lake. As you cross

the lake, suppose there is no wind and the surface of the water is glassy. You simply aim toward your desired destination and paddle. By the time you are ready to return, suppose a strong wind is blowing at right angles to your intended path. It is necessary to compensate for the wind by aiming your canoe in a direction different from where you want to end up.

Figure 1–6 illustrates the difference between the speed and velocity of the canoe during the windy condition. Let “x” be the starting point. The length of vector S_1 shows the *speed* with which the canoe is moving in one direction, and the length of vector S_2 shows the *speed* with which the canoe is moving at right angles to

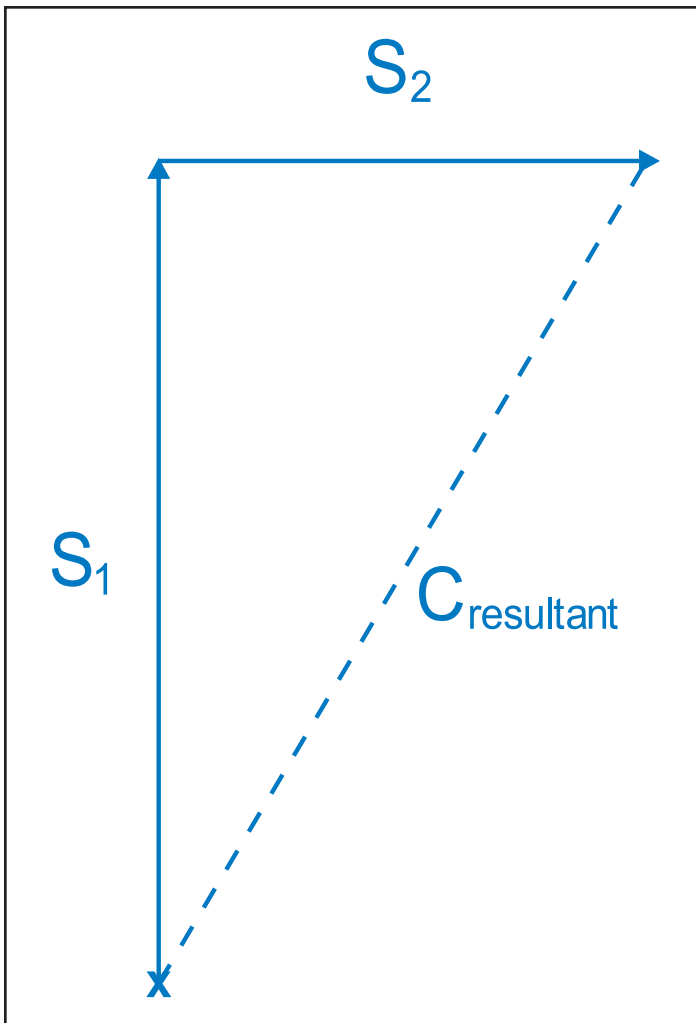


Figure 1–6. Velocity (c), the amount of displacement per unit time, is shown as a vector quantity in relation to the speed of movement in two directions (S_1 and S_2) at right angles to each other.

the first direction because of the wind. Each of those speeds is given by the simple formula:

Equation 1–1

$$s = d/t,$$

where **s** = average speed, **d** = distance, and **t** = time.

The dashed vector $C_{\text{resultant}}$ represents the **resultant velocity**, and its length is greater than the lengths of either S_1 or S_2 that represent only speed. We can calculate the value of the resultant velocity by use of the Pythagorean theorem, which states that the square of the hypotenuse in a right triangle is equal to the sum of the squares of the two sides of the triangle. Thus,

Equation 1–2

$$C_{\text{resultant}} = \sqrt{S_1^2 + S_2^2}.$$

Average and Instantaneous Velocity

We mentioned previously the concept of **average velocity**. The motorcycle moved with an average velocity of 90 **kph** (56 **mph**), but the rate of movement must have varied during the 2-hr trip. To incorporate variability in rate of movement, we distinguish between average velocity and **instantaneous velocity**. *Instantaneous velocity is the velocity measured over an infinitesimally small time interval*, perhaps an interval as small as a millionth or billionth of a second.

The distinction between **average** and **instantaneous** applies also to **speed**. For example, if a car is driven 80 **km** (approximately 50 miles) in 1 **hr**, its **average speed** is 80 **kph**. During the period of travel, the **instantaneous speed**, the speed measured for such a small interval of time that the speed is treated as a constant during the interval, might vary over a considerable range from moment to moment.

Summary of Velocity

Velocity (either average or instantaneous) is displacement per unit time and it can be expressed in many ways: meters per second (**MKS** metric system), centimeters per second (**cgs** metric system), miles per hour, feet per day, or whatever (**fps** system). We simply calculate *the ratio of the measure of displacement to the measure of time* with Equation 1–1.

Acceleration (a)

Velocity is *the amount of displacement per unit time*. That is the same as saying that velocity is *the time-rate of displacement*. Just as velocity is the time-rate of displacement, **acceleration (a)** is *the*

time-rate change in velocity. Acceleration is a vector quantity; hence a change of velocity in one direction is not the same as a change of velocity in another direction. The magnitude might change, the direction might change, or both might change.

Although it might not be intuitively obvious, an automobile traveling at a constant **speed** around a circular path *is undergoing acceleration*. If we think about it for a moment, though, it makes sense. The speed might not vary, but the *direction* of motion is constantly changing. If the direction changes, velocity also must change because the concept of velocity incorporates both speed and direction. Therefore, *if the velocity changes, acceleration must have occurred, because acceleration is the time-rate change in velocity.*

If velocity increases, acceleration is said to be “positive,” and if velocity decreases, acceleration is said to be “negative.” Negative acceleration, of course, commonly is called *deceleration*. Acceleration (**a**) is given by:

$$a = \frac{\Delta c}{t}$$

Equation 1–3

where $\Delta c = c_2 - c_1$.

If a car travels at a velocity of 10 m/s (**c**₁) and then is accelerated to a velocity of 30 m/s (**c**₂), and if that change is accomplished in 5 s, acceleration equals 4 m/s/s, which usually is written as 4 m/s²:

$$a = \frac{(30 - 10)}{5} = 4\text{m/s/s} = 4 \text{ m/s}^2.$$

Force (F)

Force (F) is a *push* or a *pull*, and is given by the product of **mass (m)** and **acceleration (a)**:

$$F = ma.$$

Equation 1–4

Equation 1–4 is Newton’s second law of motion, although his law originally was stated as **a = F/m**: *The acceleration of an object is directly proportional to the net **force** applied to the object and inversely proportional to the object’s **mass**.* When both sides of Newton’s original equation are multiplied by the mass (**m**), we obtain Equation 1–4, which is the more familiar form of his second law.

An equally correct way to state Newton’s second law of motion is that *the net **force** applied to an object is equal to the **mass** of the object multiplied by its **acceleration**.* A force is required to accelerate an object because the object has mass, hence **inertia**. The inertia of the object serves to oppose a change in motion, and therefore a **force** must be applied to overcome the **inertia**.

Consequences of Applying a Force

Force produces two consequences that are particularly important for gaining an understanding of both sound sources and sound transmission.

- First, a force can cause matter to be *distorted*. If a spring is stretched by a pulling force or compressed by a pushing force, the spring is distorted. When the force that produced distortion is removed, the spring returns to its original condition because of elasticity.
- Second, a force can cause matter to be *accelerated*. Therefore, we can define force not only as a *push* or a *pull*, but also as *that which imparts acceleration to a mass*.

Force as a Vector Quantity

Force is a vector quantity. Therefore, we must specify not only the *magnitude* of force but also the *direction* of force. In other words, it *does* matter whether the force is acting to the left or the right or upward or downward in space. Almost without exception, a force of some magnitude pushing on a body from a southerly direction produces no effect in an easterly direction; the exception will be addressed in the section “The Pendulum: An Example of Slow-Motion Vibration.”

Because force is a vector quantity, we must be careful when we speak of the “addition of forces.” If we have two forces of the same magnitude acting on the same body, the resultant force will not necessarily have twice the magnitude of one of the forces alone. The resultant force could, in fact, be zero if the equal forces acted from exactly opposite directions. In that case the body remains at equilibrium. For example, if two people simultaneously apply equal forces to a swinging door, but from opposite directions, the door will not move.

With respect to hearing, you might have reasoned that there is a pressure (the distinction between pressure and force will be made shortly) of $100,000 \text{ N/m}^2$ ($1,000,000 \text{ dynes/cm}^2$) impinging on your eardrums. Fortunately, your eardrums (tympanic membranes) do not “cave in” under this pressure because there is an equal air pressure in your middle ear cavity behind the eardrum that produces a force on the eardrums from the opposite direction. That works well, of course, until something happens to reduce the pressure on the outside of the drum, such as when you are in an airplane that climbs to 40,000 ft or so. When you feel the discomfort or pain that results from the differential pressure on the two sides of the eardrum, you might yawn or chew gum. That opens a tube (Eustachian tube) that leads from the back of your mouth to the middle ear cavity and restores the pressure to the same value that exists outside the drum.

It is important to remember that a resultant force is not simply the algebraic sum of the two forces. *Force is a vector quantity, and vector summation is required to calculate the resultant force.* If the two forces operate at right angles to one another, the resultant force can be solved by application of the Pythagorean theorem in the same way that it was used for determining the resultant velocity in Figure 1–6.

Units of Measure: Force

The unit of measure of force in the **MKS** system is the **newton (N)**, in honor of Sir Isaac Newton, who first defined force in his second law of motion (Equation 1–4). One **newton** is *the force required to accelerate a mass of 1 kg from a velocity of 0 m/s to a velocity of 1 m/s in 1 s*. In other words, when a mass of 1 kg is stationary, its velocity is 0 m/s. To accelerate the mass to move with a velocity of 1 m/s, and to accomplish that change in velocity in 1 s, a force must be applied. The value of that force is defined as 1 N.

The unit of measure of force in the **cgs** system is the **dyne**. The definition of force in the **cgs** system is equivalent to that for the **MKS** system, but the units are changed from kilograms to grams and from meters to centimeters. Thus, one **dyne** is *the force required to accelerate a mass of 1 gram (g) from a velocity of 0 cm/s to a velocity of 1 cm/s in 1 s*. One newton (**MKS**) equals 100,000 dynes (**cgs**).

Pressure (p)

In the study of sound, we are not concerned with measuring the force exerted by one molecule on another. Instead, we want to determine the sum of many such small forces that act collectively on some surface—the eardrum, for example. Thus, we will focus on *the amount of force per unit area*, which is **pressure (p)**, as shown in Equation 1–5:

$$p = F/A,$$

Equation 1–5

where **F** = the force applied and **A** = area.

Figure 1–7 illustrates a **force** of 1 N applied uniformly to a surface area of 100 m² (100 square meters). We can calculate with Equation 1–5 that the **pressure**, the amount of force per unit area, is 0.01 N/m².

Units of Measure: Pressure

Because force is measured in **newtons (MKS)** or **dynes (cgs)**, and because *pressure is the amount of force per unit area*, the unit of measure of pressure is the **newton per square meter (N/m²)** in the

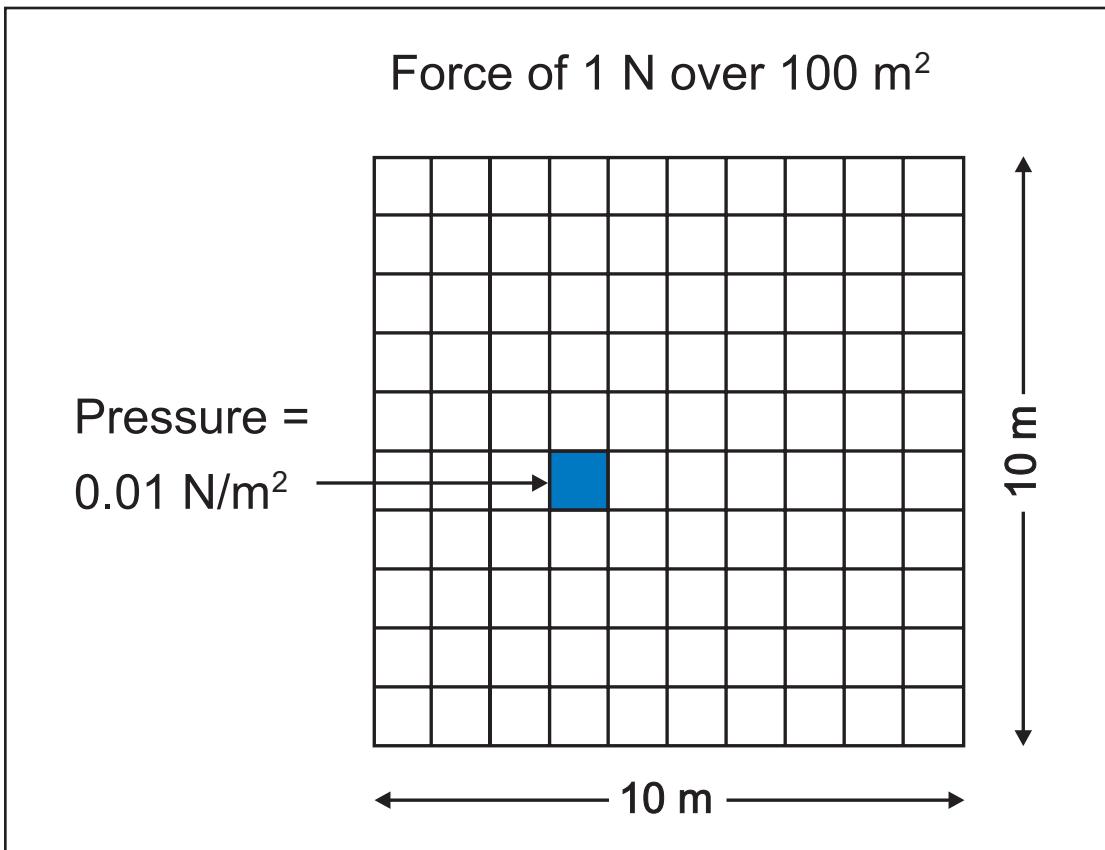


Figure 1–7. A **force** of 1 newton (N) distributed uniformly over an area of 100 m² produces a **pressure**, the amount of force per unit area, of 0.01 N/m².

MKS system and the **dyne per square centimeter (dyne/cm²)** in the **cgs** system. A pressure of 1 N/m² in the **MKS** system corresponds to how many dynes/cm² in the **cgs** system?

$$1 \text{ N} = 100,000 \text{ dynes, and}$$

$$1 \text{ m} = 100 \text{ cm.}$$

$$\text{Therefore, } 1 \text{ N/m}^2 = 100,000 \text{ dynes/m}^2$$

$$= \frac{100,000 \text{ dynes}}{10,000 \text{ cm}^2}, \text{ and}$$

Equation 1–6

$$1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2.$$

An alternative to the newton per square meter in the **MKS** system is the **pascal (Pa)**. The relations among these three units of measure for pressure are:

Equation 1–7

$$1 \text{ Pa} = 1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2.$$

VIBRATORY MOTION OF A SPRING-MASS SYSTEM

Now that the fundamental and derived physical quantities have been defined, we can examine the nature of vibratory motion in a different, but more complete, way by reference to a spring-mass system shown in Figure 1–8. One end of a spring is attached to a rigid structure, and the other end is attached to a mass that lies on a nearly frictionless surface. An arrow points toward a displacement scale. In panel **A** of the figure, the arrow points to “0” (equilibrium) on the scale. The mass is at equilibrium because the net force acting on the spring-mass system is zero; hence, there is no motion.

View Animation 1_8. A Spring-Mass System at <https://sites.google.com/a/umn.edu/soundanimations>

Characteristics of a Spring

A spring is very elastic. It can be displaced from equilibrium either by moving the mass to the left, which *compresses* the spring, or by moving the mass to the right, which *extends*, or *stretches*, the spring. When the spring is *compressed*, as in panel **B** of Figure 1–8, the mass is displaced from equilibrium and the arrow in panel **B** points toward

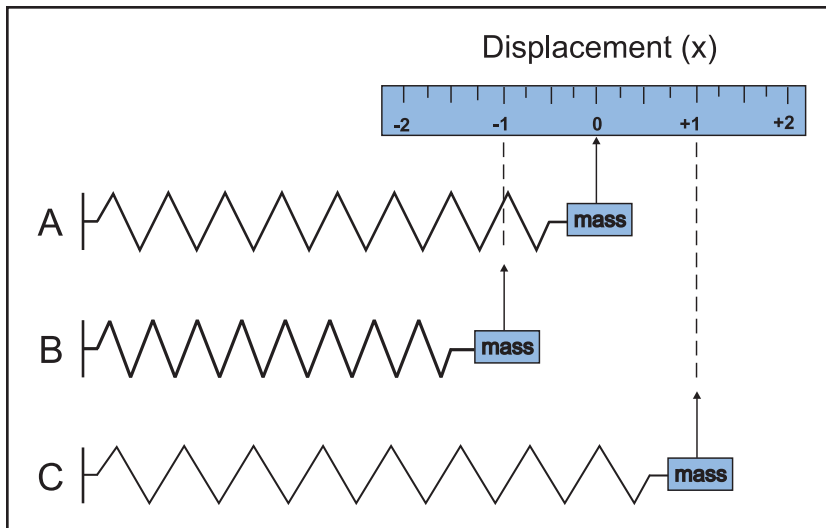


Figure 1–8. A spring-mass system with one end of the spring attached to a rigid structure. In panel A, the spring is in equilibrium, which corresponds to “0” on the displacement scale. In panel B, the spring is compressed, and the mass has moved to “–1” on the displacement scale. In panel C, the spring is stretched (extended), and the mass has moved to “+1” on the displacement scale.

a “–” value on the displacement scale. When the spring is *stretched*, the mass is displaced from equilibrium in the opposite direction, and the arrow in panel C points toward a “+” value on the displacement scale.

Regardless of whether the spring is compressed or extended, the restoring force of elasticity *opposes* the force that acts to deform the spring. As the spring is compressed, an even greater force is required to accomplish additional compression because of the restoring force of elasticity. *The magnitude of the restoring force (F_r) of elasticity is directly proportional to the magnitude of the spring’s displacement (x).* That is Hooke’s law, in honor of Robert Hooke, who published his observation in the late 1600s. Hooke’s law is expressed as:

Equation 1–8

$$F_r = -kx,$$

where F_r is the restoring force of elasticity,
 x is the magnitude of displacement of the spring, and
 k is a spring constant.

The minus sign in Equation 1–8 reminds us that the direction of the restoring force is opposite the direction of displacement.

Stiffness

What is “ k ” in Equation 1–8? All springs are elastic, but some springs require a greater force than others to be either compressed or extended because of differences among springs in **stiffness**. **Stiffness** is the spring constant (k) in the equation. In other words, the stiffer the spring, the greater the value of k , and the greater the force that is required to compress or stretch the spring some given distance.

Compliance

The elasticity of a spring also is defined by a spring’s **compliance**. **Compliance** is *inversely related to stiffness*, which means that springs with relatively little stiffness have a relatively large compliance. As compliance increases, the force required to compress or stretch the spring decreases. Hence, compliant springs can be displaced from equilibrium more easily than stiff springs.

Vibratory Motion of the System

If the mass in Figure 1–8 is moved to the left and then released, *the system is set into vibration, or oscillation, because of the elasticity of the spring and the inertia of the mass*. The vibratory motion is similar in most respects to the motion of the tuning fork that was illustrated in Figure 1–2. The arrow passes through equilibrium

toward maximum displacement to the left ("−" on the displacement scale) as the spring becomes progressively more compressed. When the restoring force of elasticity (F_r) *overcomes the force applied*, *motion is momentarily halted*, and the direction of motion of the system is reversed. The system then passes through equilibrium toward maximum displacement to the right ("+" on the displacement scale) as the spring becomes progressively more stretched, and so on.

We shall see in Chapter 2 that the spring-mass system, the tuning fork, and any vibrating system for which the magnitude of the restoring force is directly proportional to the displacement ($F_r \propto x$), are engaged in what is called **simple harmonic**, or **sinusoidal**, **motion**.

THE PENDULUM: AN EXAMPLE OF SLOW-MOTION VIBRATION

Because movement of the tuning fork is too rapid to see—and, of course, we certainly cannot see movement of the air molecules—the concept of vibratory motion might be difficult to grasp. We can, however, achieve a more intuitive understanding of vibratory motion by examining the characteristics of a slow-moving vibratory system such as the pendulum shown in Figure 1–9.

View Animation 1_9. The Pendulum at
<https://sites.google.com/a/umn.edu/soundanimations>

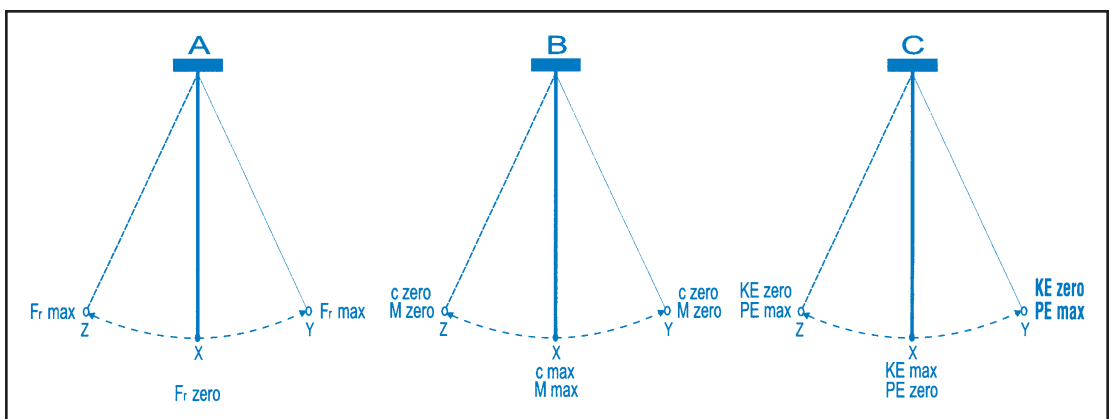


Figure 1–9. Vibratory motion of a pendulum moving from maximum displacement in one direction, through equilibrium, to maximum displacement in the opposite direction, and back to equilibrium. Panel A shows that the elastic restoring force (F_r) is maximal at maximum displacement and zero at equilibrium. Panel B shows that velocity (c) and momentum (M) are zero at maximum displacement and are maximal at equilibrium. Panel C shows that potential energy (PE) is maximal at maximum displacement and zero at equilibrium, whereas kinetic energy (KE) is maximal at equilibrium and zero at maximum displacement.

The pendulum is at equilibrium when the bob is located at point **X**. Next, displace the pendulum to point **Y** and release it. The pendulum returns toward equilibrium **X**. It does not stop when it reaches equilibrium; it continues on toward maximal displacement in the opposite direction **Z**. Upon reaching maximal displacement, it again returns toward and beyond equilibrium **X**, and so on. In other words, the motion of the pendulum appears to be similar to what was seen previously for the tuning fork. Each engages in *vibratory motion*.

When the pendulum or tuning fork tines are maximally displaced in either direction, they return toward equilibrium because of a **restoring force** (F_r). With the tuning fork in Figure 1–2 and the spring-mass system in Figure 1–8, *the restoring force is elasticity*. With the pendulum in Figure 1–9, *the restoring force is gravity*. With this difference in mind, we see that the pendulum is restored to equilibrium from maximal displacement because of the restoring force of gravity, and it moves beyond its position of equilibrium toward maximal displacement because of **inertia**.

The pendulum continues to move toward maximal displacement until the restoring force of gravity (F_r) overcomes inertia. At the point of maximal displacement, the restoring force of gravity is maximal, *and motion is momentarily halted*. As the pendulum then moves back toward equilibrium, the restoring force of gravity diminishes. When the pendulum reaches equilibrium, the net force of gravity on motion is zero and inertia is maximal, which results in continuation of motion toward a new displacement in the opposite direction.

When the concept of **force** as a vector quantity was introduced, we stated that “it *does* matter whether the force is acting to the left or the right or upward or downward in space. Almost without exception, a force of some magnitude pushing on a body from a southerly direction produces no effect in an easterly direction.” The vibratory motion of the pendulum is an exception. Consider the downward-acting gravitational force to be in a “southerly” direction, and the path of excursion of the pendulum to be in an “easterly-westerly” direction. In this circumstance, the force of gravity in a southerly direction *does affect* the vibratory motion that takes place in an easterly-westerly direction.

Momentum (M)

As we consider the opposing forces that result in vibratory motion, it is preferable, and more accurate, to invoke the role of **momentum** instead of inertia. **Momentum (M)** is *the product of mass (m) and velocity (c)*, as shown in Equation 1–9:

Equation 1–9

$$M = mc.$$

Momentum is directly proportional to both **mass** (*the amount of matter present*) and **velocity** (*the time-rate of displacement*); as either mass or velocity increases, momentum increases. A large car traveling with the same velocity as a smaller car has greater momentum

because of its greater **mass**. If two cars have identical mass, momentum is greater for the car traveling with the greater **velocity**. With respect to the moving pendulum, we see in panel B of Figure 1–9 that momentum is maximal, and the net force acting on the pendulum is zero, when the pendulum is passing through equilibrium. At maximal displacement in either direction, momentum is zero, and the restoring force of gravity is maximal; motion is momentarily halted.

Why does the pendulum gain momentum as it moves toward equilibrium? Note the two quantities that determine momentum in Equation 1–9, and then recall the discussion of fundamental and derived physical quantities.

- Momentum of the pendulum does not change because of a change in **mass**—the quantity of matter present remains the same.
- It follows, therefore, that momentum must change because of a change in **velocity**.
- In other words, as the pendulum moves from maximum displacement *toward* equilibrium, momentum increases because the velocity of motion increases.

In summary, at maximum displacement, *where motion is momentarily halted*, both **M** and **c** are zero. As the pendulum passes through equilibrium, both **M** and **c** are maximal.

The Energy Principle

Movement of the pendulum, as well as movement of other matter engaged in vibratory motion, also is described by the **energy principle**. According to the **law of conservation of energy**, any system that can do work (such as the pendulum, the tuning fork, the spring-mass system, or any other body that can be set into vibratory motion) must receive a supply of energy from somewhere.

What is energy? **Energy** is an abstract concept that refers to *something that can produce a change in matter*: displacement of a mass; distortion of the shape of solid matter; expansion of gases; and so forth. When a change in matter occurs, **work** is done, and **energy** is *a measure of the capacity to do work*. **Energy** is something a body *possesses*, whereas **work** is something a body *does*.

Work

Work is accomplished only when a **force** *succeeds in moving the body that the force acts upon*, and the quantity of work accomplished is given by *the product of the force applied and the distance moved*:

$$W = Fd,$$

Equation 1–10

where **F** = the force applied and **d** = the distance moved.

If a force does not give rise to motion, no work is done. The act of pushing against a cement wall with a force of 50 N accomplishes no work unless the wall is moved through some distance. It does not matter how much force is applied or for how long, if the wall does not move through some distance, no work is done.

In the **MKS** system, *force is expressed in newtons* and *distance is expressed in meters*. The unit of measure of **work** in the **MKS** system is the **joule**. One joule of work is accomplished if a force of 1 N acts through a distance of 1 m (1 joule = 1 N × 1 m). In the **cgs** system, the unit of measure of **work** is the **erg** (1 erg = 1 dyne × 1 cm). To relate the two units of measure of **work**, 1 joule = 10,000,000 ergs:

$$1 \text{ N} = 100,000 (10^5) \text{ dynes, and}$$

$$1 \text{ m} = 100 (10^2) \text{ cm.}$$

$$\begin{aligned} \text{Therefore, } 1 \text{ joule} &= 100,000 (10^5) \text{ dynes} \times 100 (10^2) \text{ cm} \\ &= 10,000,000 (10^7) \text{ ergs} \end{aligned}$$

Transformation of Energy

Energy cannot be depleted—it is *transformed* from one form to another, or it is *transferred* from one place to another. If you wind the spring on an old-fashioned clock, the spring gains energy as it is wound. In turn, it loses energy as it does work on the clock. If you pull the trigger on a rifle, exploding gases do work on the bullet, which gives energy to the bullet. When the bullet strikes an object, it does work on the object and loses energy.

Return now to the pendulum shown in panel C of Figure 1–9. When the pendulum is raised from equilibrium to maximum displacement, the pendulum gains **potential energy (PE)**, which is a form of *stored energy*. In the case of the pendulum, potential energy is *an energy of position*, and it sometimes is called **gravitational potential energy**; it has potential energy because of the gravitational attraction of the earth.

The potential energy associated with the pendulum in its raised position is transferred, or converted, to *an energy of motion* when released, which is **kinetic energy (KE)**. When the pendulum reaches the equilibrium position, potential energy is zero and kinetic energy is maximal. As the pendulum moves beyond equilibrium toward maximum displacement, kinetic energy is converted to potential energy, and at maximum displacement, kinetic energy is zero and potential energy is maximal.

At all points during the excursion, *the sum of PE and KE remains constant*. That defines the **total energy** in accordance with the **law of conservation of energy**. According to that law, energy can be transformed from one kind to another, but energy is never created or destroyed—the *total energy remains a constant*.

Frictional Resistance

In the case of the pendulum, the tuning fork, and the spring-mass system, we have proceeded as if vibratory motion continues indefinitely. That does not happen, however, because of *opposition to motion* by a **frictional force**. Because **friction** opposes, or resists, motion, it limits velocity of motion. As the pendulum swings along its path, friction at the point of suspension and friction on the bob of the pendulum due to air resistance cause energy to be *transformed* from **kinetic energy** to **thermal energy** (heat). Because of the opposing force of friction, the amplitude of each excursion (cycle) of the pendulum is less than the amplitude of the previous excursion.

In Chapter 2 we will refer to this diminishing amplitude over time as **damping**, or **damped vibration**. Resistance to motion attributable to friction, which sometimes is called **frictional resistance**, or just **resistance**, is analogous to electrical resistance, which serves to oppose, or impede, the flow of current in an electrical circuit. We will return to the concept of **transfer of energy** when we discuss sound wave propagation.

Characteristics of Pendular Motion

There are two characteristics of the vibratory motion of the pendulum: **amplitude** and **frequency**.

Amplitude

The first characteristic is **amplitude**. Amplitude of displacement, a derived physical quantity, is *a vector quantity that incorporates both magnitude and direction of displacement*. The amplitude of displacement of the pendulum is proportional to the force applied.

Frequency (f)

The second characteristic is **frequency**. When the tines of a tuning fork move from equilibrium to maximum displacement (+), back through equilibrium (0), to maximum displacement in the opposite direction (-), and finally back to equilibrium (0), one **cycle** of vibration is completed. The same is true with the pendulum. Movement from **X** to **Y** to **X** to **Z** and back to **X** in Figure 1-9 defines one cycle of pendular vibration. **Frequency (f)** is *the rate of vibratory motion, that is, the number of cycles completed per second (cps)*. Thus, if the pendulum completes four cycles each second, $f = 4$ cps. If the rate of motion is increased so that 25 cycles are completed each second, $f = 25$ cps.

The unit of measure for frequency has been changed from **cycles per second (cps)** to **hertz (Hz)**, to honor Heinrich Hertz (1857–1894), a

German physicist. Conversion between **cps** and **Hz** is simple: 1 **Hz** = 1 **cps**. Thus, for the two examples mentioned above, $f = 4$ Hz (4 cps) and $f = 25$ Hz (25 cps).

Frequency is the number of *cycles per second*. We also can calculate the number of *seconds per cycle*, that is, *the amount of time required to complete one cycle*. That is the **period (T)** of vibration. Thus, if the pendulum completes 4 cycles of vibration in 1 s ($f = 4$ Hz), then 1 cycle is completed in $1/4$ s. Similarly, if $f = 25$ Hz, one cycle is completed in $1/25$ s. For those two examples, we say that the period (**T**) is $1/4$ (.25) s and $1/25$ (.04) s.

You might have noticed that conversion from frequency to period or from period to frequency is fairly simple:

Equation 1-11

$$T = 1/f$$

and

Equation 1-12

$$f = 1/T$$

Thus, if $f = 4$, $T = 1/4$ (.25); and if $T = .25$, $f = 1/.25 = 4$. When we divide 1 by some number, we *calculate the reciprocal of that number*. Thus, we say that *period is the reciprocal of frequency and frequency is the reciprocal of period*.

Frequency and **amplitude** are independent characteristics of pendular motion (the value of one does not depend on the value of the other), but **frequency** and **period** are not independent.² The value of one completely determines the value of the other.

Determinants of Frequency of Vibration of a Pendulum

What properties of the pendulum determine the **period** and **frequency** of vibration? It is tempting to think that you might increase the frequency (and thereby decrease the period) if a greater force were applied. Recall, however, that *the magnitude of force applied determines the magnitude of displacement, hence, the amplitude of vibration*.

If you think about a child's swing, you should realize that the swing moves at the same rate no matter how hard you push it. It can be displaced farther in response to a greater force, but it does not move with greater frequency. You can make the swing move with greater frequency, however, by looping the rope around the supporting structure a few times. That makes the swing shorter, and the *length* of the swing (or pendulum) is the principal determinant of frequency of vibration.

The equation for the **period (T)** of the pendulum is:

Equation 1-13

$$T = 2\pi\sqrt{\frac{L}{G}},$$

where **L** = length and **G** = gravitational force.

Thus, the **period** of pendular vibration is *directly proportional to the square root of the length*, which means that as length increases, period increases, and as length decreases, period decreases. Because frequency and period are inversely related, **frequency is inversely related to the square root of the length**. That, of course, is what we learned from the child who shortened the length of the rope to “swing” more rapidly.

The dependence of frequency of vibration of a pendulum on length was first discovered by Galileo (1564–1642), who observed the swinging of lamps that hung in the cathedral at Pisa. The frequency of the swinging lamps never changed. Gravity is a constant in the equation with a value of 9.8 m/s^2 or 980 (more precisely, 981.456) cm/s^2 . Those values are the metric equivalents of 32.2 ft per s per s (32.2 ft/s^2) in the **fps** system of measurement.

We have discussed two principal characteristics of pendular vibration:

- **amplitude**, which is directly proportional to the force applied; and
- **frequency**, which is inversely proportional to the length of the pendulum, and frequency is the reciprocal of the **period**.

The concepts of amplitude and frequency will be used subsequently to help define characteristics of vibratory motion in general, not just the movement of the pendulum.

■ PROPORTIONALITY

What do **inversely proportional** and **directly proportional** mean? When **A** is *inversely proportional to B*, as **B** decreases, **A** increases. Conversely, as **B** increases, **A** decreases. An inverse relation is stated as:

$$A \propto \frac{1}{B},$$

Equation 1–14

which means that **A** is inversely proportional to **B**.

Equations 1–11 and 1–12 that specified the relation of frequency and period were of the form shown in Equation 1–14, which means that frequency and period are inversely related.

When **A** is *directly proportional to B*, as **B** increases, **A** increases, and as **B** decreases, **A** decreases. Direct proportionality is described in the form:

$$A \propto B,$$

Equation 1–15

which means that **A** is directly proportional to **B**.

Consider Equation 1–13 for the period of a pendulum. **L** appeared in the numerator, which means that period is directly proportional to the square root of length.

It is easy to inspect an equation and know if you are dealing with an inverse or direct relation. To determine if **A** is directly or inversely related to **B**, look at **B** in the equation.

- If **B** appears in the denominator of the ratio, **A** is *inversely proportional* to **B**.
- If **B** appears in the numerator of the ratio, **A** is *directly proportional* to **B**.
- What if there is no ratio? If there *appears* to be no ratio because there is no denominator (actually, the denominator has a value of 1), the proportional relation takes the form shown in Equation 1–15, $A \propto B$. An example appeared in Equation 1–9, which expresses that momentum (**M**) is directly proportional to both mass (**m**) and velocity (**c**).
- Caution! Look again at Equation 1–13 that described the period of a pendulum. Length (**L**), is in the numerator, but it is incorrect to state that $T \propto L$. Instead, $T \propto \sqrt{L}$. Thus, the period of pendular oscillation is *directly proportional to the square root of the length of the pendulum*.

■ SOUND WAVE PROPAGATION

We return now to a sound wave that results from the vibratory motion of some mass. Suppose a source of sound is energized—a tuning fork, for example, is struck. As shown in Figure 1–4, the action produces a wave of compression that moves through the air medium. The *individual air molecules move back and forth* about their average positions of equilibrium, *but they do not move over a great distance*. In fact, particle displacement is on the order of 0.0000000768 (7.68×10^{-8}) m. Because of the force applied, movement of the air particles produces alternate regions of increased density (**compression**) and decreased density (**rarefaction**). That is the disturbance that is propagated through the medium.

In our previous discussion of vibratory motion, we considered the pendulum (an example of slow-motion vibration) because the vibratory movements of the tuning fork are too rapid to see. It might also be helpful to think of waves of compression that can be seen. Imagine that you are sitting on a dock over a lake. The wind is so calm that the surface of the water is glassy—the water medium is “in equilibrium.” Now, dip a stick in and out of the water one time. What happens? A “disturbance” (a wave of disturbance) moves outward from the point where the stick entered the water. Do it again, but this time dip the stick in and out of the water repeatedly at regular intervals, say once every 2 s. Now the wave of disturbance is repeated, and a series of concentric circles of larger and larger diameter move away from the center. The water wave is reasonably

analogous to the propagation of a sound wave in air, except that the water wave that we see is moving mostly along a two-dimensional surface.

A wave of disturbance moves through the water. But are the water molecules displaced over any great distance? No. Place a small cork on the surface of the water and imagine it represents a “water particle.” As the wave of disturbance moves outward, the cork bobs (approximately) up and down. The cork moves over a relatively small distance, but a *wave of disturbance* is propagated through the water medium for a considerable distance. The same is true for propagation of sound in air—the *propagation of density changes through an elastic medium*.

Two properties are affected at a certain *rate* during sound wave propagation:

1. *the rate of vibratory back and forth movements of air particles*, which is the **frequency** (**f**) of vibratory motion; and
2. *the rate with which alternate regions of compression (high density) and rarefaction (low density) are propagated through the medium*, which is the **speed** (**s**) of sound.

Frequency of Vibratory Motion (f)

The rate at which the source of sound vibrates, expressed in hertz (**Hz**), is called the **frequency** (**f**). What determines the frequency with which the tuning fork, or any other sound source, vibrates? *The frequency of vibration of the source of sound is determined by properties of the source!* It does not matter whether the sound wave that results is propagated through air or water or along a steel rail, and it does not matter what force is applied to the source.

In the case of the tuning fork, the principal factors that determine the frequency of vibration are the *density of the metal* and the *length of the bar*. Thus, a single tuning fork that is always characterized by some constant density and length always vibrates with the same frequency. With the pendulum, length is the principal factor that determines frequency of vibration. We shall see later that the frequency of vibrating strings such as on a piano or guitar depends on the length, mass, and tension of the strings. But it is important to remember that, as long as nothing happens to alter those characteristics of the source, *the frequency of vibration is the same*.

What about the frequency of vibration, to and fro, of air molecules in response to the vibration of the source? *The frequency of air molecule vibration depends on, and is the same as, the frequency of vibration of the source.*

- If the tuning fork vibrates with a frequency of 125 Hz, the air molecules in the medium vibrate with a frequency of 125 Hz.
- If a different fork vibrates with a frequency of 250 Hz, the air molecules in the medium vibrate with a frequency of 250 Hz.

Speed of Sound (s)

The vibration of either tuning fork mentioned above, one with $f = 125$ Hz or one with $f = 250$ Hz, results in a wave of compression that is propagated through the medium. What determines the **speed** of sound wave propagation?

Whereas the **frequency**, or rate, of vibration of the source depends on characteristics of the source, *the **speed** of sound wave propagation depends on characteristics of the medium*. Thus, the waves of compression that result from the two forks vibrating with different frequencies move through the medium with the same speed.

Common experience tells us that some finite amount of time is required for a sound wave to reach our ears. We see the ball leave the tee from a distant golfer before we hear the “crack” of the ball; we see the conductor’s hands signal the celebrity to commence singing the national anthem at the World Series before we hear the first words; and we see the flash of lightning long before (we hope) we hear the crash of thunder.

These differences between how soon we “see something” and how soon we “hear something” occur because light and sound travel at different speeds. The speed of light is 299,792,458 meters per sec (186,282.397 miles per sec; a slight change from the value of 186,282.423 that was accepted before the General Conference on Weights and Measures met in Paris in 1983). *The speed of sound in air at sea level with a temperature of 0° Celsius (C) is only 331 meters per sec (1,085.96 ft. per sec)*. Thus, the speed of light is almost 1 million times faster than the speed of sound.

The speed of sound in air is given by:

Equation 1-16

$$s = \sqrt{\frac{E}{\rho}},$$

where **E** refers to elasticity and **ρ** refers to density.

Thus, speed is *directly proportional* to the square root of **elasticity (E)** of the medium, which is measured in N/m² (**MKS**) or dynes/cm² (**cgs**), and *inversely proportional* to the square root of density (**ρ**) of the medium, which normally is measured in kilograms per cubic meter (kg/m³), but sometimes in grams per cubic centimeter (g/cm³). The relation between those two alternative expressions is given by:

$$1 \text{ g/cm}^3 = \frac{0.001 \text{ kg}}{(0.01 \text{ m})^3} = 1,000 \text{ kg/m}^3.$$

We generally are interested in the speed of sound waves traveling in air that is warmer than 0° C, and *changes in air temperature do affect the speed of sound wave propagation*. As the medium warms, the ambient pressure of the medium remains constant, but the expanding gases cause the density of the medium to decrease. The speed of sound in air also is affected by the humidity of the

medium; speed of propagation is directly proportional to the amount of humidity.

We can reason from Equation 1–16 that the speed of sound increases as temperature increases because an increase in temperature produces a decrease in density, and $s \propto 1/\sqrt{\rho}$. More specifically, the speed of sound increases approximately 0.61 m/s (61 cm/s or 2 ft./s) for each 1°C increase in temperature. For example, if the speed of sound is 331 m/s (33,101 cm/s; 1,086 ft./s) at 0°C, it increases to 343 m/s (34,320 cm/s; 1,126 ft./s) if the temperature rises to 20°C (68° Fahrenheit).

From Equation 1–16 it is obvious that *the speed of sound in air is independent of the frequency of vibration of the source*, because only characteristics of the medium are included in the equation: **elasticity** and **density**. Thus, the speeds of two sound waves, one with $f = 125$ Hz and one with $f = 250$ Hz, are identical when the waves travel in the same medium.

It also follows that the sound wave that results from a tuning fork vibrating with a frequency of, for example, 500 Hz is propagated at different speeds in different sound transmission media. In air, the disturbance travels with a speed of about 331 m/s (1,086 ft./s). In water with a temperature of 0° C (if it has not yet frozen), the disturbance from the same vibrating fork travels much faster, about 1,433 m/s (4,702 ft/s). On a steel rail the speed is even greater, about 4,704 m/s (15,434 ft./s). Wood is an interesting medium for transmitting sound. In certain kinds of wood, the speed of sound is approximately four times greater when traveling with the grain of the wood than when traveling across the grain.

Does the contrast of speed along steel and speed through air make sense from Equation 1–16? Steel is a little more than 6,000 times as dense as air. If we ignore elasticity, we might conclude, incorrectly, that the speed of sound therefore is less in steel because speed is inversely proportional to the square root of density. We cannot, however, ignore elasticity. In fact, steel is about 1,230,000 times more elastic than air. Thus, counter to our sometimes common intuition, **elasticity** can be thought of as *the ability to resist deformation*, and steel certainly offers a greater resistance to deformation than air. The contribution of the difference in elasticity between air and steel far outweighs the difference in density. As a result, the speed of sound is approximately 14 times greater along steel than in air ($\sqrt{1,230,000/6000} = 14.3$).

TYPES OF WAVE MOTION

We have described the nature of wave motion that occurs with a sound wave, and we then drew the analogy of wave motion on the surface of water. Next we will describe the characteristics of *all wave motion* in more detail. We say “all wave motion” because wave motion is not confined to sound. The vibrations, or tremors, of buildings or bridges also are examples of wave motion. Another is

the pattern that you see if you flick one end of a stretched rope that is fastened at the other end to a wall.

The types of waves that result from vibratory motion are classified according to *the direction of vibration of the medium relative to the direction in which the wave is propagated*. There are two types of wave motion: **transverse** and **longitudinal**.

Transverse Wave Motion

One type of wave motion is a transverse wave. Consider the stretched rope in Figure 1–10 with its two ends extending to infinity so that no “reflections” occur. Several dots, representing rope “particles,” are located at equal intervals. Next, flick the rope and, with the aid of a fast camera, periodically “stop the action” and examine the positions of each of the rope particles (dots) at different moments in time. If we take a series of such pictures at regular intervals, we can see the course of movement of each of the rope particles from moment to moment. Time progresses from t_0 at the top of the figure to t_5 at the bottom.

View Animation 1_10. Transverse Wave Motion on a String at <https://sites.google.com/a/umn.edu/soundanimations>

- The first picture, at time zero (t_0), shows the rope stretched—it is in equilibrium.
- In the second picture (t_1), taken a fraction of a second (one-quarter **cycle**) later, particle **A** has moved upward, **B** stays in place, **C** moves downward, **D** stays in place, **E** moves upward, **F** stays in place, and so on.
- As you move down the rows of the figure, you see that, over time, each particle moves up and down. Furthermore, each particle moves up and down *with the same frequency* as every other particle.
- Next, consider one cycle of vibration for particle **A**: movement from equilibrium at time zero (t_0) to maximal displacement upward (t_1), to equilibrium (t_2), to maximal displacement downward (t_3), and back to equilibrium (t_4).
- Every other particle completes one cycle of vibration in the same amount of time. Particle **B**, for example, completes one cycle between t_1 and t_5 . Particle **I** behaves like particle **A** and completes one cycle of vibration between t_0 and t_4 . The upward and downward displacements of particle **I** are shown by the darker shaded region in the figure.

Look again at the positions of the particles along the rope and you see peaks occurring at regular intervals from left to right.

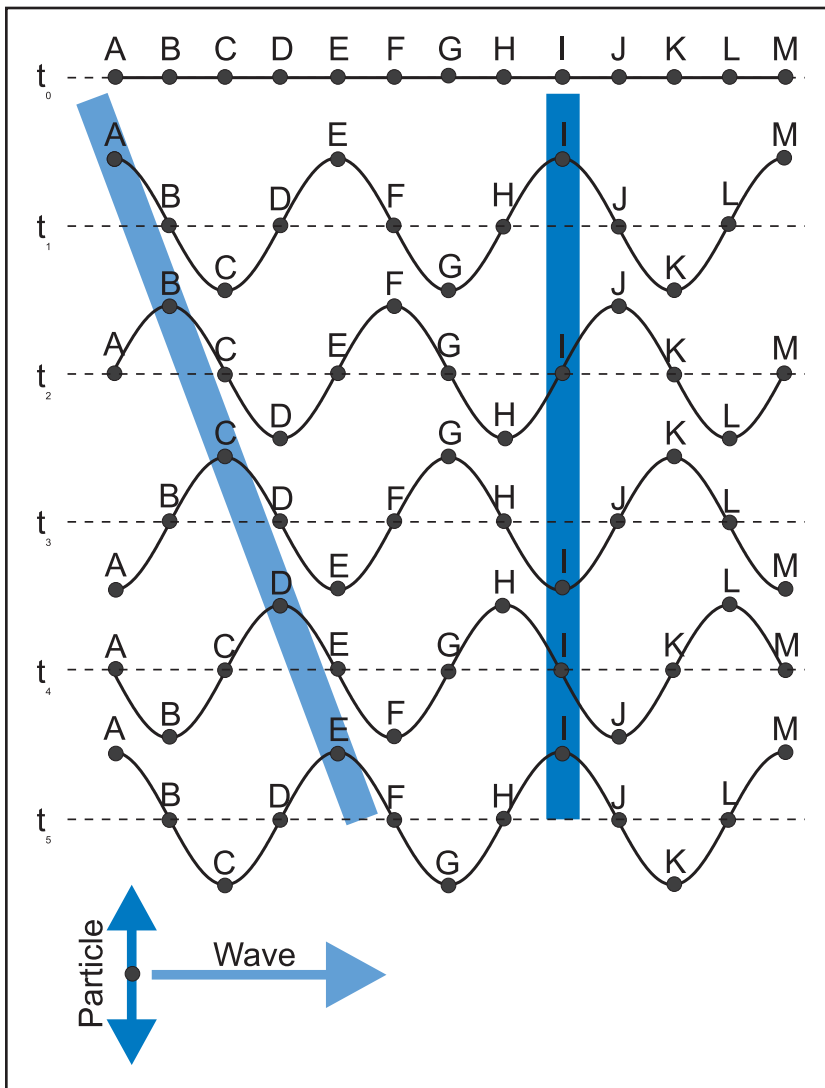


Figure 1–10. Transverse wave motion of a stretched rope. Particles (A through M) painted on the rope move alternately up and down over time (from t_0 through t_5) at right angles to the direction of wave motion from left to right.

- At t_1 , the peaks correspond to particles A, E, I, and M. Those peaks are analogous to regions of **compression**, or **high density**, for sound transmitted through air.
- We also see valleys from left to right that correspond to points C, G, and K. Those valleys are analogous to regions of **rarefaction**, or **low density**, for sound in air.

As we move through time by looking down the rows, *the locations of the peaks move from left to right* (shown by the lighter shaded region in the figure).

- At t_1 , the first peak appears at **A**, but at t_2 particle **A** has moved downward and particle **B** has moved upward.
- As a result, the peak that formerly was at **A** (t_1) now has moved toward the right to correspond to point **B**.
- The same is true for all other peaks, and so at t_2 the peaks are located at **B**, **F**, and **J**.
- Over time, as you move down through time t_5 , the peaks move from left to right.

For now, we won't concern ourselves with what happens when a peak finally moves far enough to the right to reach the imaginary surface to which the rope is attached (at infinity), but that concept will be introduced in Chapter 9.

With the type of wave motion that occurs with the rope in Figure 1–10, the direction of wave movement is from left to right, in contrast to the direction of particle movement, which is up and down. That is transverse wave movement. A **transverse wave** is one in which *the direction of vibration of the medium is at right angles (90°) to the direction of the wave that is propagated through the medium*. Transverse wave motion happens when you pluck the string of a guitar. Another example of (almost) transverse wave motion occurs when you throw a pebble into a pond near to where a cork lies motionless (in equilibrium). A wave of disturbance with peaks and valleys moves outward from where the pebble struck the water. As the wave moves outward, the cork (and the water medium) bobs approximately up and down at right angles to the direction of wave movement.

Longitudinal Wave Motion

The second type of wave motion is a **longitudinal wave**, which is illustrated in Figure 1–11 by a spring-mass system lying on a frictionless surface. “Individual particles” of the spring are identified with dots and labeled **A** through **H**. Time progresses from t_0 at the top of the figure to t_8 at the bottom.

View Animation 1_11. Longitudinal Wave Motion
on a Spring-Mass System at
<https://sites.google.com/a/umn.edu/soundanimations>

- When the spring is in equilibrium (t_0), the dots are spaced at equal intervals.
- At t_1 , a short segment of the spring is compressed and then released. Particle **A** moves to the left.
- From t_0 through t_8 , particle **A**—and the mass attached to the spring—moves leftward and rightward (back and forth) through two complete **cycles** of vibration.

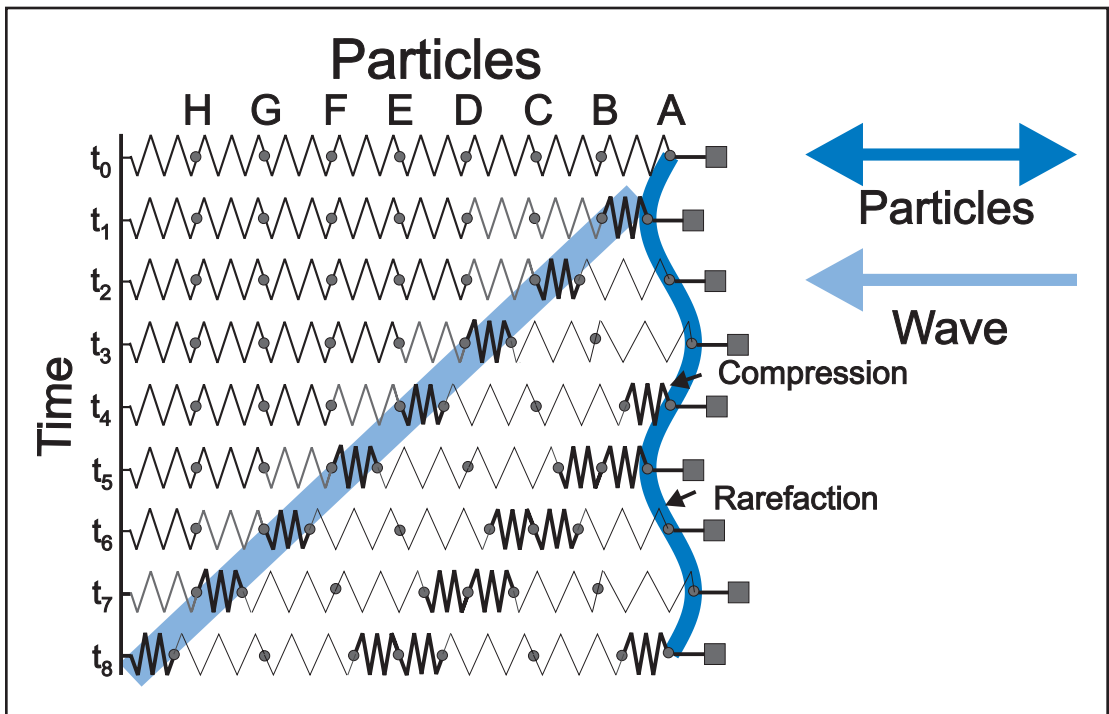


Figure 1-11. Longitudinal wave motion of a spring-mass system lying on a frictionless surface. “Particles” (A through H) painted on the spring move alternately back and forth from left to right over time (from t_0 through t_8) in the same plane as the wave movement from left to right.

- Each new picture shows the positions of the spring particles at progressively later moments in time, and ultimately each of the particles B through H undergoes the same motion that was seen for particle A. That is to say, each particle vibrates back and forth with exactly the same amplitude and frequency as particle A.

Compressions, or **condensations**, are shown by the dots when crowded closely together (high density), and **rarefactions** are shown when the dots are far apart (low density). From picture to picture, the particles of the spring move back and forth, and the compressions and rarefactions move in the same plane as the particles. This leads to the definition of a **longitudinal wave**. A **longitudinal wave** is one in which *the direction of particle movement is parallel to the direction of wave propagation*.

SOUND WAVES

Are sound waves in air transverse or longitudinal? Sound waves in air are **longitudinal** waves; *the direction of air particle movement is parallel to the direction of wave movement*. We can think of the

air molecules as small elements of matter attached to each other by springs. This was illustrated in Figure 1–3 in which 10 individual particles are in the columns from left to right and “frozen moments” in time are in the rows from top to bottom. The source of sound is a tuning fork. The first row shows the fork and the medium in equilibrium. Three complete cycles of vibration are shown in the remaining rows. By way of review, we should be able to identify the following aspects of vibratory motion.

1. The tuning fork in Figure 1–3 moves through three cycles of vibration after it is struck by moving from equilibrium, to maximal displacement (+), through equilibrium, to maximal displacement in the opposite direction (–), and back to equilibrium during each **cycle**.
2. Each particle vibrates back and forth relative to its average position of equilibrium, which was shown in the top row of Figure 1–3. The **amplitude** and **period** of vibration of each particle are the same as they are for every other particle.
3. The individual particles do not vibrate together. That means, not all of them simultaneously move to the left or to the right. Rather, they move in a progressive fashion, just as we observed in Figure 1–10 for the dots on the stretched rope and in Figure 1–11 for the dots on the spring-mass system. If the dots were members of a marching band, we would say they were “marching out of step with one another.” In Chapter 2 we will learn to say that they are “out of phase” with one another.
4. The back and forth movements of the air particles in Figure 1–3 produce a movement, or propagation, of the **compressions** (high density) and **rarefactions** (low density) from left to right. During the compression phase, the particles move to the right, and during the rarefaction phase, the particles move to the left.
5. The individual particles are envisioned to be connected by a spring, which is equivalent to an **elastic bond**, and each particle *exerts a force on the one to the right of it*. By that we mean, during **compression**, each spring *pushes the particle to its right forward*. During **rarefaction**, the springs or elastic bonds are stretched, which *pulls the particle backward*.
6. Each particle does **work** on the particle in front (to the right) of it. This results in a **transfer of energy** *through the medium in the direction in which the wave is traveling*.

■ TRANSFER OF ENERGY

We learned previously that the medium itself is not transferred during sound wave propagation; each particle is displaced from equilibrium over only a very small distance. **Energy**, however, is transferred

through the medium in the direction in which the wave is propagated as each particle does work on the one next in line.

Transfer of energy during transmission of sound is the same as it was in the discussion of the oscillating pendulum. In other words, *potential energy is greatest, and kinetic energy is least, when the molecule is at maximal displacement.* As the particle returns toward equilibrium, potential energy is transformed to kinetic energy. At equilibrium, *kinetic energy is maximal and potential energy is zero.*

So, “What is sound?” The answer can be approached in different ways.

Previously, sound in air was said to be *characterized as the propagation of density changes through an elastic medium.* We also can describe sound as *a mechanical disturbance of a gaseous, liquid, or solid medium.* Both of those statements are true, but it is preferable to *define* sound as *the transfer of energy through an elastic medium.*

In our discussion of pendular vibration, we learned that **friction** limits the velocity of pendular motion. The same thing occurs when sound energy is transferred through a medium such as air. The air mass offers **resistance**, and some of the kinetic energy is transformed into **thermal energy** (heat). When that transformation is complete, the sound energy is dissipated, which means that no more useful work can be accomplished.

The wave motions that we have discussed to this point have come from the vibratory motion of tuning forks, the pendulum, a stretched rope or string, and a spring-mass system. Each of those examples produces the same pattern of wave motion, either transverse or longitudinal, which is called **simple harmonic motion**. The characteristics of simple harmonic motion will be described in Chapter 2.

NOTES

1. The stem of the fork also is set into vibration, but it moves up and down as the tines move back and forth. The fork also has two points, located near to where the metal bars are bent, at which no vibration occurs. Points of no vibration are called **nodes**.
2. Strictly speaking, the **frequency** of pendular vibration is independent of **amplitude** only if the amplitude is small. For our purposes, however, the influence of amplitude on frequency will be disregarded.

FREQUENTLY MISUNDERSTOOD CONCEPTS

1. What happens to the speed of sound wave propagation in air as the period of a sine wave that is propagated through the medium increases?

Nothing. The speed of sound wave propagation in air *is unaffected* by changes in the period of the sine wave because

frequency (and **period**) and **speed** are independent quantities. Speed of sound is determined by the elasticity and density of the air medium and is given by Equation 1–16:

$$s = \sqrt{\frac{E}{\rho}},$$

where **E** refers to elasticity and **ρ** refers to density.

When the air is characterized by *standard conditions* (0° Celsius and a pressure of 1.013×10^5 N/m²), the speed of sound is 331 m/s regardless of the **frequency**, hence the **period**, of the sound wave that is propagated through the medium. As the temperature of the medium increases, the pressure in the medium remains constant, the gases expand, and the **density** decreases. Because the speed of sound is inversely proportional to the square root of the density, the speed of sound increases by 0.61 m/s for each 1° C rise in temperature. Thus, if the temperature is increased to 25° C, the speed of sound wave propagation increases by 15 m/s (0.61×25) from 331 m/s to 346 m/s ($331 + 15$).

2. What happens to potential energy (PE) as a pendulum swings away from equilibrium?

At the precise moment that the pendulum swings *through* equilibrium, **PE** is zero and **KE** is maximal. As the pendulum swings *away from* equilibrium in either direction, **KE**, which is an *energy of motion*, is converted to **PE**. Thus, **PE**, which is a form of *stored energy*, increases.

3. One newton in the MKS system corresponds to how many dynes in the cgs system?

The **newton** and **dyne** are units of measure of **force**, and $1 \text{ N} = 100,000 (10^5) \text{ dynes}$. See Equation 1–6 and review the section on “Units of Measure for Force.”

■ PRACTICE PROBLEMS

1. You have two pendulums of different lengths. Why does the shorter pendulum vibrate with a higher frequency than the longer one?
2. Why does an object traveling with a velocity of 80 **kph** have greater momentum than the same object traveling with a velocity of 50 **kph**?
3. Describe the relation between the direction of vibration of the medium and the direction of wave propagation for a transverse wave.
4. Express 1.5 pascals (**Pa**) in equivalent newtons per square meter (**N/m²**) and equivalent dynes per square centimeter (**dynes/cm²**).
5. What derived physical quantity defines the time-rate change in displacement?
6. What derived physical quantity defines the time-rate change in velocity?
7. List the three fundamental physical quantities and the unit of measure for each in the **MKS** system.
8. What acts as the restoring force in a vibrating spring-mass system?
9. Describe how the magnitudes of potential energy and kinetic energy vary with the magnitudes of displacement (**x**) of a mass from equilibrium during vibratory motion.
10. Why can you ordinarily see a flash of lightning before you hear the crash of thunder?
11. The frequency of a sound wave traveling in air is 100 Hz. If the sound wave then travels through a water medium, what is the frequency of vibration?



Simple Harmonic Motion

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When the concept of transfer of energy was discussed in Chapter 1, vibratory motion of air particles was simulated in Figure 1–11 by a spring-mass system. When the mass is pushed to the left to compress the spring, a rightward reaction force results. When the mass is pulled to the right to extend the spring, a leftward reaction force results. In other words, there is a restoring force (F_r) that returns the mass to equilibrium, and the magnitude of the restoring force is proportional to the change in length of the spring: **Hooke's Law** (Equation 1–8, $F_r = -kx$). *The restoring force of elasticity is proportional to the distance that the mass is displaced from equilibrium.*

During oscillatory motion of the spring-mass system, the magnitude of the restoring force varies over time because the magnitude of displacement is constantly changing. As the mass moves toward equilibrium, the restoring force diminishes, and as the mass moves away from equilibrium toward maximal displacement, the restoring force increases. Because of this buildup and decay of the restoring force, *the mass accelerates as it moves toward equilibrium and decelerates as it moves away from equilibrium.* That means that the velocity of the mass is constantly changing.

The result is **vibration**, or **oscillation**, of the mass backward and forward that simulates the to-and-fro vibratory, or oscillatory, motion of the air molecules. One **cycle** of vibration is movement from equilibrium to maximal displacement (+), back through equilibrium, to maximal displacement in the opposite direction (–), and back to equilibrium.

THE WAVEFORM

It is useful to examine a graph that illustrates the nature of this simple vibratory motion by plotting changes in the magnitude, or amplitude, of displacement over time. We will assume that the system has already been set into oscillation and we plot two cycles of oscillation that commence at the exact moment that the mass in Figure 1–11 passes through equilibrium and heads toward maximal displacement. That is analogous to the air molecules in Figure 1–3 being displaced from equilibrium to the right.

Such a plot is shown in Figure 2–1, which displays the **time-domain waveform**, or just **waveform**. This is the result that would be obtained if, for example, the path of point A for the spring-mass system in Figure 1–11 were traced over time through two cycles of oscillation from t_0 to t_8 . It is important to avoid a serious misinterpretation: *The air mass does not literally undergo an excursion with the shape that is shown in Figure 2–1.* The air molecules simply move back and forth just as the spring-mass system moves back and forth, and the path of the wave of disturbance that moves through the medium is essentially a straight line. The curved line in the figure is simply a *graphic representation* (i.e., the **waveform**) of what transpires. We will see later why straight-line, back and forth movement is represented with the shape shown in Figure 2–1.

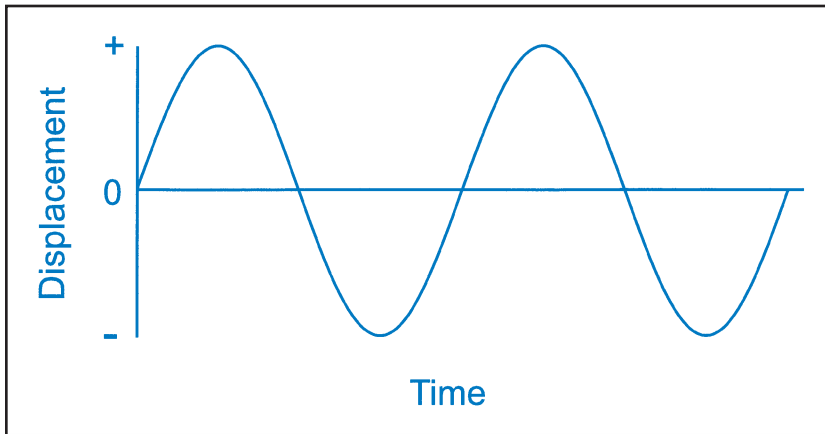


Figure 2–1. Graphic representation of the movement of a spring-mass system over time that was presented in Figure 1–11. The pattern of movement reflects **simple harmonic motion**, which also is called **sinusoidal motion**. This kind of representation is called the **waveform**.

The waveform in Figure 2–1 shows the changing amplitude of *displacement* as a function of time, but the concept of a waveform is not restricted to a display of changing displacement. Many other quantities (e.g., velocity, acceleration, force, pressure, momentum, and so on) can be shown as a function of time, and they also are waveforms.

THE CONCEPT OF SIMPLE HARMONIC MOTION

The type of motion that the spring-mass system (and the air molecules) undergoes is called **simple harmonic motion**, and we shall see subsequently why simple harmonic motion also can be called **sinusoidal motion**. Importantly, simple harmonic, or sinusoidal, motion can be represented as *projected uniform circular motion*.

Uniform Circular Motion

Uniform circular motion occurs when a body moves around the circumference of a circle at a constant number of degrees of rotation per second. Figure 2–2 shows a wheel that is turning counter-clockwise. The dot on the wheel moves at a *uniform rate* around the circumference. One complete revolution of the wheel, and therefore one complete revolution of the dot, is accomplished in 1 s. Because the dot moves through 360° during one revolution, the wheel turns at a uniform rate of 1° per $1/360$ s.

The back and forth movement of the mass, and the back and forth movement of the air molecules, is analogous to a point moving

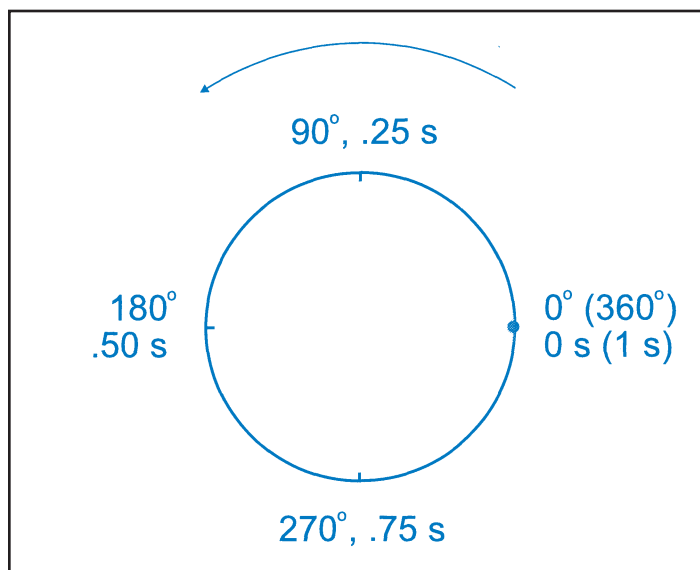


Figure 2–2. A dot moving at a uniform rate around the circumference of a wheel through 360° of rotation in 1 s.

around the circumference of a circle at a constant rate. When that is accomplished, we can say that movements of the mass and of the air molecules are examples of **simple harmonic motion**. We also will be able to explain why straight-line, back and forth motion, which is called **rectilinear motion**, is pictured to have the same pattern shown in Figure 2–1.

A good explanation of simple harmonic motion was provided by Hirsch (1952), and Figure 2–3 is taken from his textbook. A person who is seated on a Ferris wheel points a flashlight at a motorized wall coated with a light-sensitive material. The Ferris wheel starts to move at a constant rate, which means that both the person and the flashlight move around the circumference of the circle at a constant number of degrees of rotation per second. Thus, the rider undergoes **simple harmonic motion**.

View Animation 2_3. A Ferris Wheel Illustrates
Simple Harmonic Motion at
<https://sites.google.com/a/umn.edu/soundanimations>

What will the projected image look like when the wall is stationary? It is a straight vertical line running upward from the middle of the wall, then downward to near the bottom of the wall, and finally back to the middle. The upper limit of the traced line corresponds to the moment in time when the person is at the top of the path of the wheel, the lower limit corresponds to the moment in time

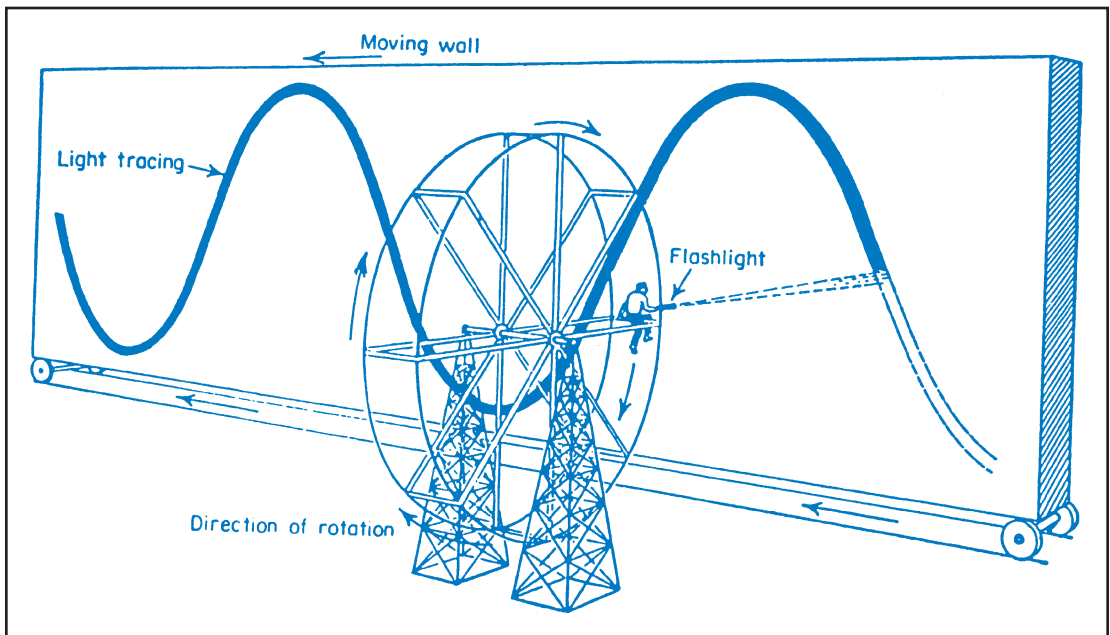


Figure 2-3. An illustration of how a sinusoid is created by the projection of uniform circular motion. From *The Measurement of Hearing* (p. 20) by Ira Hirsch, 1952, New York: McGraw-Hill Book Company, Inc. Copyright by McGraw-Hill Education. Reproduced with permission.

when the person is at the bottom of the path, and the midpoint of the line corresponds to the starting (equilibrium) position. Thus, *circular motion can be **projected** as rectilinear (straight-line) motion*. You can demonstrate this by pointing a flashlight in a circular motion on a path perpendicular to the wall. The beam of light moves upward and downward in a straight line.

Next, suppose that the wall moves from right to left. The straight-line image that was observed when the wall was stationary now takes the form of the curved line shown in Figure 2-3. You can demonstrate this effect for yourself by pointing a flashlight toward a wall and attempting to walk along the wall as you rotate the flashlight in a circular fashion as before. The curved image will result.

Projection of Uniform Circular Motion

In Figure 2-4 the projected circular motion—**simple harmonic motion**—is shown in greater detail. The circle at the left of the figure represents the Ferris wheel with the passenger aboard. P_0 defines the equilibrium, or starting position, and equilibrium also is labeled as 0° . The person then is set into counterclockwise motion around the circumference of the circle through P_1 , P_2 , and so on until the wheel has moved completely around the circumference. The excu-

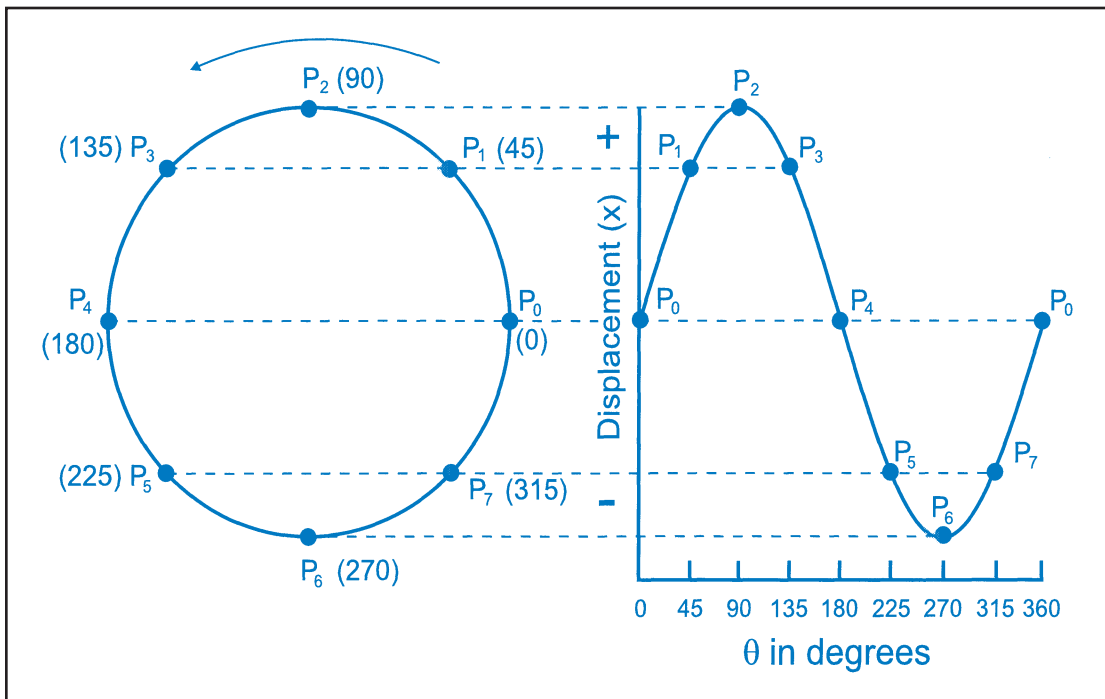


Figure 2–4. Projection of uniform circular motion. The points (P_0 through P_7) on the turning wheel, which is moving at a uniform rate, hence undergoing **simple harmonic motion**, are projected as corresponding points (also P_0 through P_7) on a sinusoidal function. Each point on the sinusoidal function shows the **magnitude of displacement** from the baseline corresponding to that angle of rotation.

sion around the circumference is marked at 45° intervals, and at the end of one complete rotation the passenger has passed through 360° .

View Animation 2_4. Projection of
Simple Harmonic Motion at
<https://sites.google.com/a/umn.edu/soundanimations>

The projection of the circular motion is shown at the right of Figure 2–4, just as the image of the flashlight beam was projected in Figure 2–3. In this case, the projection displays the **displacement waveform** (x). The only difference from what was shown in Figure 2–3 is that the “coated wall” is rotated 90° relative to its position in Figure 2–3 so that the circle and the projection appear in the same two-dimensional plane. The projection (the **waveform**) assumes the same shape that it did in Figure 2–3, where it resulted from a flashlight moving with the Ferris wheel.

The waveform in Figure 2–4 also is marked with dots at regular intervals labeled from P_0 through P_7 . The precise location of each

of those points on the projection corresponds to the location of the corresponding points on the turning wheel. It is worthwhile to examine the nature of the correspondence in detail, because the relation between circular motion and the form of its projection is fundamental to understanding both simple harmonic motion and sound.

- At P_0 , the passenger is located on the “baseline,” or reference line, and the projection of the beam also lies along the baseline.
- As the wheel begins to rotate counterclockwise, the passenger—and the projected beam—are displaced upward from the baseline. At 45° (P_1) of rotation, the horizontal (dashed) line indicates exactly where the projection is at that point, which also is labeled P_1 , and therefore corresponds to 45° .
- Between 45° and 90° , the passenger travels through the same number of degrees as between 0° and 45° , *but the magnitude of linear displacement above the baseline is less between 45° and 90° than it was between 0° and 45° .*
- At 90° , the height of the projected curve is maximal, which corresponds to the passenger being at the highest point above the baseline.
- Between P_2 and P_6 the passenger moves downward, and the projection also moves in a downward, or negative, direction.
- At P_3 , the magnitude of displacement is the same as at P_1 , and in a similar vein, P_4 corresponds to P_0 , and P_5 corresponds to P_7 .
- The relation between the magnitudes of displacement for P_2 and P_6 also should be apparent. P_2 corresponds to maximal displacement above baseline (90°) and P_6 corresponds to maximal (and equal) magnitude of displacement below baseline (270°).
- From P_6 through P_0 the passenger climbs upward toward the baseline (and so does the projection) to complete *one cycle of rotation*, which corresponds to 360° .

Notice that the horizontal axis (abscissa) of Figure 2–5 is labeled in degrees. That allows us to specify the exact location of the person (P) in the excursion around the wheel at any given moment. Thus, the starting point, P_0 , corresponds to 0° , P_1 to 45° , P_2 to 90° , and so on until the passenger returns to the starting point, 360° .

When the passenger reaches the top of the circle, 90° of rotation ($1/4$ of a cycle), the projection reaches its maximal height in a *positive* (+) direction. When the passenger completes exactly half of the circle, 180° of rotation, the projected beam is in the same position that it was at equilibrium, or starting position. Maximal displacement of the beam in a *negative* (–) direction occurs at 270° of rotation. Finally, when movement around the circumference of the circle is complete, the person has returned to the starting point, 360° of rotation, and the projection is back to zero displacement.

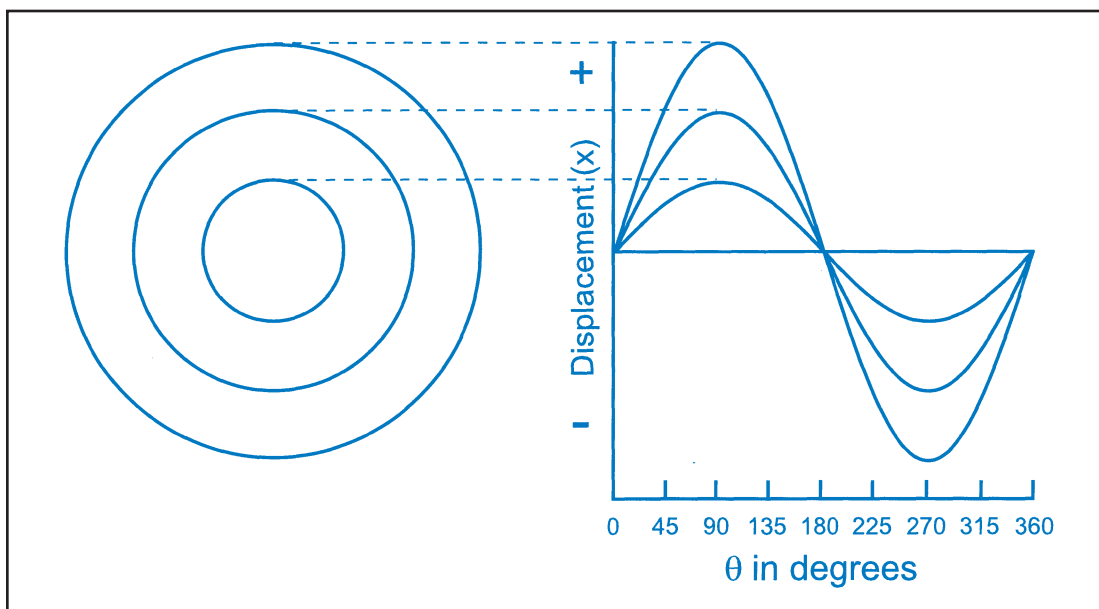


Figure 2-5. Projections of a rotating point on each of three wheels undergoing simple harmonic motion. Because the radii of the wheels are different, the magnitudes of displacement (x) of the projections of the three wheels also are different.

One rotation of 360° defines one **cycle**, and that is identical to our previous definition: one **cycle** corresponds to displacement from equilibrium to maximum, back to equilibrium, to minimum (maximum displacement in the opposite direction), and back to equilibrium. Equilibrium corresponds to 0° , 180° , and 360° ; maximum displacement corresponds to 90° (+) and 270° (-).

The projected graph of uniform circular, or simple harmonic, motion shows the magnitude of **displacement** (x) as a function of degrees of rotation. The abscissa also can be labeled in units of time. Suppose the wheel rotates with a **frequency** (f) of one rotation per second. That means the **period** of rotation—the time required to complete one cycle of rotation—is 1 s because $T = 1/f$, and $1/1 = 1$. Therefore, 90° (1/4 of the cycle) corresponds to 0.25 s (1/4 of 1 s), 180° to 0.50 s (1/2 of 1 s), and so on. If frequency increases to two rotations per second, the period—the time required to complete one cycle of rotation—is 0.5 s. Now, 90° still corresponds to 1/4 of the excursion and to maximal amplitude of displacement, but because of the increased frequency, it corresponds to 0.125 s, which is 1/4 of 0.5 s (the time required to complete one cycle).

The Sine Wave

Figure 2-5 is similar to Figure 2-4, but now the projections of three wheels of different sizes are shown. Look at both the similarities and differences among the three projections.

View Animation 2_5. A Sine Wave at
<https://sites.google.com/a/umn.edu/soundanimations>

- For each of the three wheels, one cycle corresponds to 360° , maximal displacement corresponds to 90° and 270° , and so on.
- Although maximal positive displacement corresponds to 90° for all three wheels, *the largest wheel results in the largest absolute displacement of the projection.*
- Even though the magnitude of displacement (x) varies with the size of the wheel, *the general form, or shape, of the three projections (the three waveforms) is the same.*

The next step is to identify the constant feature that characterizes the three projections. That requires determining the **sine of an angle**. Consider the triangle in Figure 2–6. The angle (θ) is 45° , the

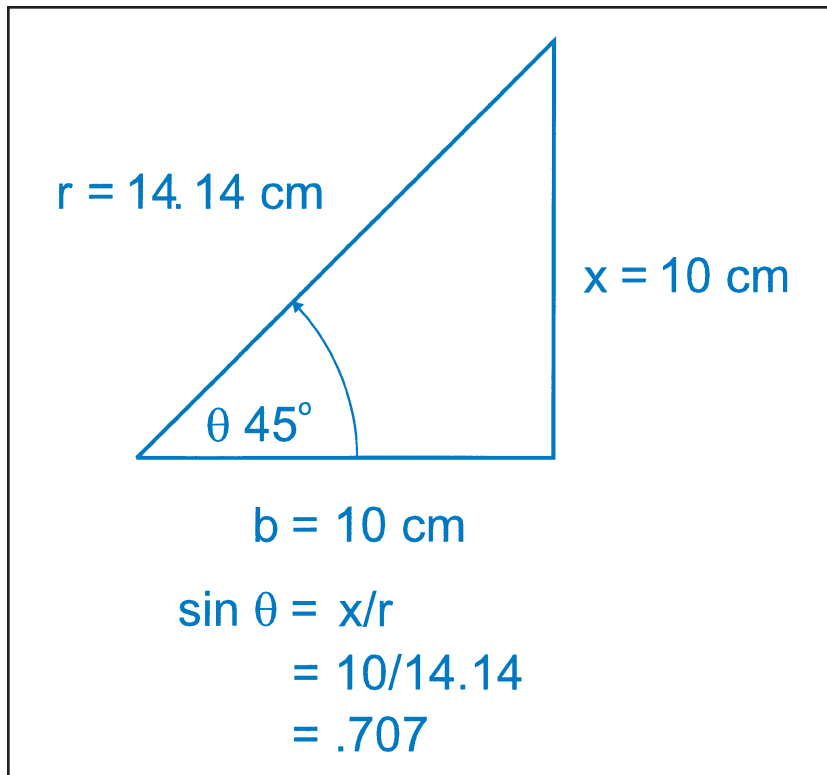


Figure 2–6. A triangle with angle of 45° , hypotenuse r , base b , and side x . The ratio x/r is the **sine of the angle** ($\sin \theta$). The sine of any constant angle such as 45° is a constant, regardless of the absolute magnitudes of x and r . Thus, for a 45° angle, if $x = 1.00$, r equals 1.414, and the sine of the angle is 0.707. If, for the same angle, $x = 2.0$, r equals 2.828, but the sine of the angle still is 0.707.

side opposite the angle is labeled **x**, the base is labeled **b**, and the hypotenuse is labeled **r**. *The sine of the angle, $\sin \theta$, equals x/r .* In other words, if we know the lengths of **x** and **r**, we divide **x** by **r** and the result is the **sine of the angle**. With the example in the figure, the length of **x** is 10 cm, the length of **b** is 10 cm, and the length of **r** is 14.14 cm. The ratio of **x** to **r**, x/r , is $10/14.14 = 0.707$, and *0.707 is the sine of a 45° angle*. That is the same as saying that the length of **x** is 70.7% of the length of **r**.

For any given angle, regardless of the lengths of **x** and **r**, the ratio x/r (the sine of the angle) is a constant, but *for other angles* the ratio x/r (the sine of the angle) will not necessarily be 0.707. You will see later that $\sin \theta = 0.707$ only when the angle is 45° or 135° and -0.707 when the angle is 225° or 315°. The sine of other angles is different from 0.707. For example, you can verify with a pocket calculator that the sine of 90° is 1.00, the sine of 270° is -1.00 , the sine of 180° and 360° is 0.00, and so forth. In all cases, however, the sine of the angle is given by the ratio x/r .

We could have formed other ratios for the triangle as well. For example, the ratio b/r is the **cosine** of the angle, and the ratio x/b is the **tangent** of the angle. Cosines and tangents are constants as well *for a given angle*. In summary, the trigonometric terms “sine,” “cosine,” and “tangent” simply are the ratios of lengths of two specific sides of the right triangle.

Figure 2–7 contains the same three circles that appeared in Figure 2–5, but in Figure 2–7 a triangle with an angle of 45° is placed within each circle. The radius (**r**) of each circle forms the hypotenuse for the triangle of each circle and, of course, the radius of each circle intersects the circumference of that circle. The length of **r** is different for each of the three circles.

The side of the triangle opposite the angle corresponds to *the magnitude of displacement (x)* just as it did in Figure 2–3, where a beam of light was projected from the Ferris wheel. Finally, the lengths of **x** and **b** also are different for each of the three circles.

- *The ratio x/r , the **sine of the angle**, is identical for all three triangles because the angle is constant.*
- *For each of the three triangles, $\sin \theta = 0.707$. In other words, it does not matter what the three lengths are; if the angle is 45°, the ratio of the length of **x** to the length of **r** is 0.707.*
- *Put differently, the length of **x** is 70.7% of the length of **r** when the angle is 45°. In other words, at 45°, displacement always is 70.7% of the maximal displacement that will be achieved at 90°, regardless of the absolute magnitude of maximal displacement.*
- *For a constant angle, $\sin \theta$ is a constant.*

Fortunately, we can use a pocket calculator or published trigonometric tables to find the sine (and cosine and tangent) of any angle. For example:

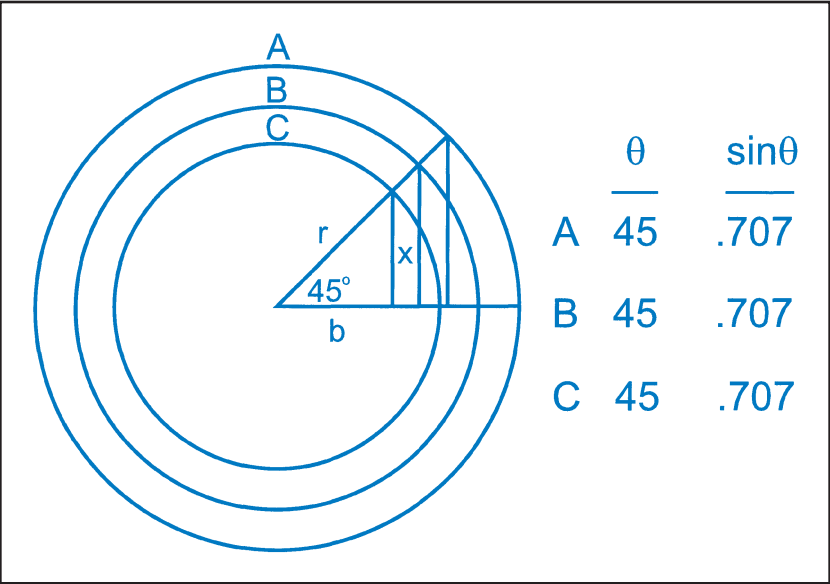


Figure 2-7. Three triangles inserted within the three wheels shown previously in Figure 2-5. Because the circumferences of the three circles are different, the absolute magnitudes of x , b , and r vary, but the sine of the 45° angle, x/r , is a constant 0.707 for all three triangles.

- At 90° , $\sin \theta = +1.00$;
- at 135° , $\sin \theta = 0.707$, and it projects to the same height as at 45° ;
- at 180° , $\sin \theta = 0.00$; and
- at 270° , which projects to the same height (maximal displacement) as at 90° , but in the opposite direction, $\sin \theta = -1.00$.

Displacement (x) Represented by $\sin \theta$

Figure 2-8 displays the three wheels from Figure 2-5, but in this case the heights of the projections at the right are shown on the ordinate by $\sin \theta$ instead of x . Notice that the three projections are superimposed on one another. That happens because for each angle throughout the cycle, the sine of the angle is a constant. In Figure 2-5 the same three circles produced projections that reached three different maximal heights of displacement. That was because we plotted “absolute displacement” above baseline in Figure 2-5, which was different for the three wheels of different sizes. In Figure 2-8 displacement is represented by $\sin \theta$, which is a constant for any constant angle, instead of x . The result is three superimposed curves. That reveals what is constant about uniform circular motion even though that motion can occur along circumferences of different sizes.

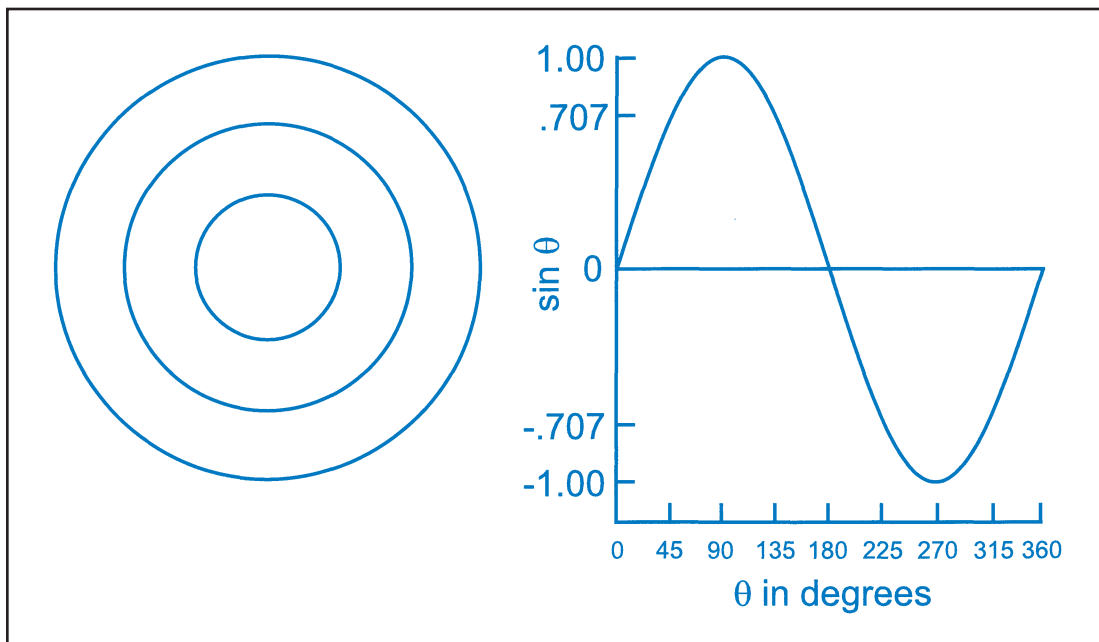


Figure 2–8. Projections of the same three rotating wheels shown previously in Figure 2–5. In Figure 2–5, magnitude of displacement was given by x . The ordinate in this figure, however, is the sine of the angle, $\sin \theta$, as the wheel moves through 360° of rotation. Because the sine of any constant angle is a constant regardless of the absolute magnitude of x , a single projection is common to all three wheels.

Construction of a Sine Wave

The simplicity of understanding a sine wave is illustrated by Table 2–1 and Figure 2–9. Table 2–1 lists the sine of angles ($\sin \theta$) selected at *equal intervals* of 11.25° from 0° to 360° , and the function in Figure 2–9 is formed by plotting $\sin \theta$ as a function of θ in degrees from the entries in Table 2–1: For example:

$$\begin{aligned} \theta = 0^\circ, \sin \theta = 0; & \quad \theta = 11.25^\circ, \sin \theta = .195; & \quad \theta = 45^\circ, \sin \theta = .707; \\ \theta = 90^\circ, \sin \theta = 1.0; & \quad \theta = 135^\circ, \sin \theta = -.707; & \quad \theta = 180^\circ, \sin \theta = 0; \\ \theta = 270^\circ, \sin \theta = -1.0; & \text{and so on.} \end{aligned}$$

It should be apparent that the 33 *discrete* points in Figure 2–9 are forming a sinusoidal function. If an infinite number of such points were displayed, a *continuous* sinusoidal waveform would be constructed.

Summary of Sinusoidal Motion

The *common element* in the projections of simple harmonic, or uniform circular, motion is *the sine of the angle*. For any constant angle,

Table 2–1. Sine of Selected Angles at 11.25° Intervals

θ	$\sin \theta$	θ	$\sin \theta$	θ	$\sin \theta$	θ	$\sin \theta$
0.00	0.000						
11.25	.195	101.25	.981	191.25	–.195	281.25	–.981
22.50	.383	112.50	.924	202.50	–.383	292.50	–.924
33.75	.556	123.75	.831	213.75	–.556	303.75	–.831
45.00	.707	135.00	.707	225.00	–.707	315.00	–.707
56.25	.831	146.25	.556	236.25	–.831	326.25	–.556
67.50	.924	157.50	.383	247.50	–.924	337.50	–.383
78.75	.981	168.75	.195	258.75	–.981	348.75	–.195
90.00	1.000	180.00	.000	270.00	–1.000	360.00	0.000

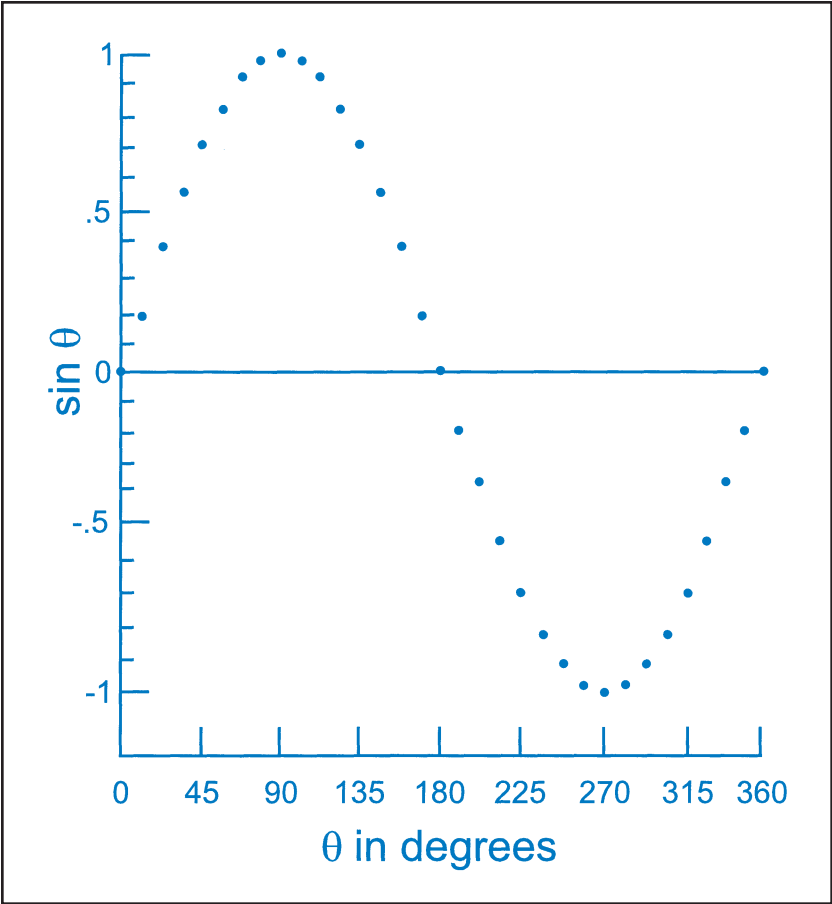


Figure 2–9. Construction of a sine wave by plotting $\sin \theta$ for 33 angles selected at equal intervals of 11.25° from 0° to 360° from Table 2–1.

regardless of the size of the triangle, the sine of that angle is a constant. In other words, each of the three wheels in Figure 2–8 achieved 38.3% of maximum displacement at 22.5° ($\sin 22.5^\circ = 0.383$), 70.7% at 45° ($\sin 45^\circ = 0.707$), 100% at 90° ($\sin 90^\circ = 1.00$), and so on. That is why **simple harmonic motion** also is called **sinusoidal motion**, and the projection, or graph, of simple harmonic, or sinusoidal, motion is called a **sine wave**, a **sinusoidal wave**, or a **sinusoid**.

Movement around the circumference of a circle at a constant number of degrees of rotation per second (uniform circular motion) is called **simple harmonic**, or **sinusoidal, motion** and the projection, or graph, of sinusoidal motion is a **sinusoidal wave**. You should notice, however, that movement of the spring-mass system in Figure 1–11 was *rectilinear* (back and forth in a straight line), not circular, and that movement of air particles in Figure 1–3 also was back and forth, not circular. In spite of that seemingly significant difference, we still represented the back and forth motions as being sinusoidal in Figure 2–1, even though it was not labeled a sine wave at that time.

Rectilinear Motion Represented as Uniform Circular Motion

The challenge that remains is to explain how *back and forth motion of air particles* can be *represented as uniform circular motion*. To do this we rely on another explanation provided by Hirsh (1952), which is illustrated in Figure 2–10. At the left of the figure is a wheel to which a rod is attached. The rod is connected to a piston housed in a piston chamber in a way that is reminiscent of an old-fashioned steam engine.

View Animation 2_9. Simple Harmonic Motion and Sound Waves at
<https://sites.google.com/a/umn.edu/soundanimations>

The point where the piston is attached to the wheel is labeled 90° . The wheel now is set into clockwise rotation at a *uniform rate*, which also is called a **uniform angular velocity**. The point on the wheel where the piston rod is attached to the rim undergoes *uniform circular, or simple harmonic, motion*. What happens to the piston in the chamber? As the wheel turns, the piston rod causes the piston in the chamber to move *back and forth in straight-line, or rectilinear, motion*.

- The position of the piston at the point of equilibrium is labeled 0° .
- When the wheel rotates to 90° , the piston moves maximally rightward and that position also is labeled 90° .

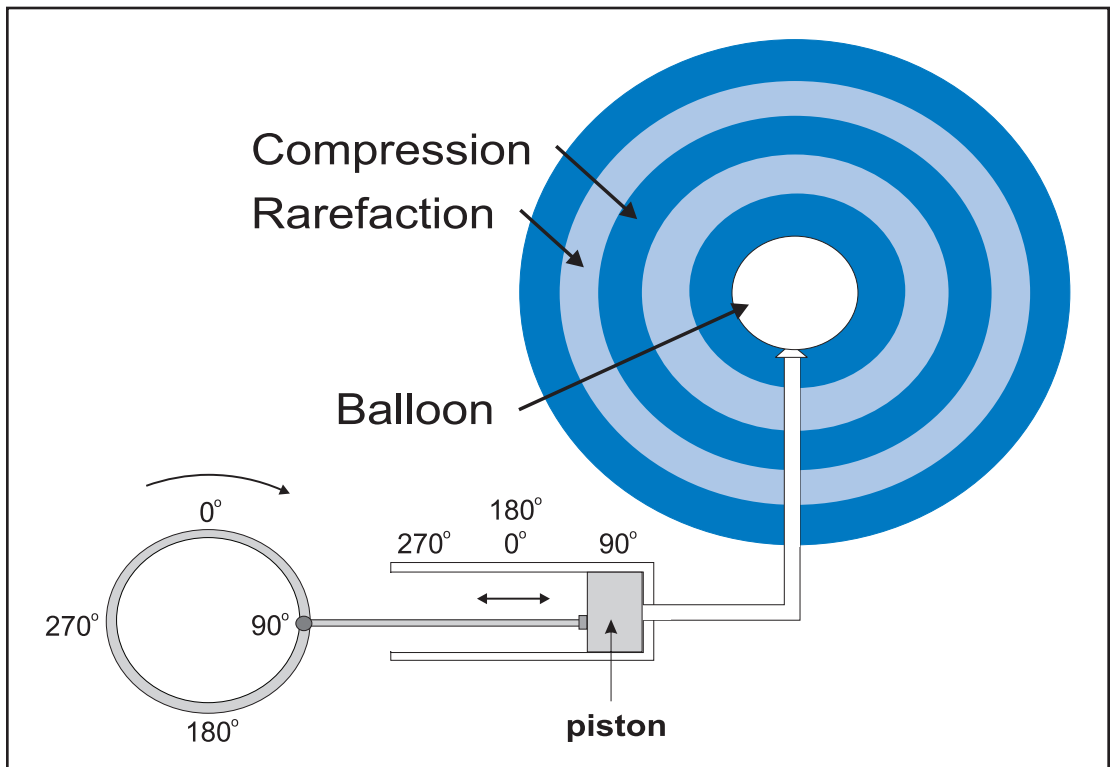


Figure 2–10. The relation between linear back and forth movement and uniform circular motion in the understanding of sinusoidal motion. The wheel, turning clockwise, causes the piston in the chamber to move *back and forth*. Each position of the piston is identified in degrees that correspond to the angle of rotation of the wheel. The chamber is connected to a balloon by a tube. Because of the changing air pressure in the chamber, tube, and balloon, the balloon acts as a pulsating sphere and produces a **spherical sound wave**. **Compressions** and **rarefactions** move outward through the medium in all directions as an increasingly larger sphere that, in a two-dimensional representation, appears as increasingly larger concentric circles. Adapted from *The Measurement of Hearing* (p. 22) by Ira Hirsch, 1952, New York: McGraw-Hill Book Company, Inc. Copyright by McGraw-Hill Education. Reproduced with permission.

- When the wheel reaches 180°, the piston returns to equilibrium, which also is labeled 180°.
- The piston moves maximally leftward when the wheel reaches 270°, and therefore maximal displacement of the piston to the left is labeled 270°.
- Finally, when the wheel completes one full cycle of rotation, the piston returns to equilibrium (360°).

The important point is that *the wheel engages in uniform circular motion*, but *the piston engages in rectilinear (straight-line) motion*. Thus, with a simple “mechanical translation,” we can label points along a rectilinear excursion in degrees in a way that corresponds to points along a circular excursion.

Simple Harmonic Motion and Sound Waves

To achieve sound from the motion observed in Figure 2–10, we rely on the rest of the apparatus shown in the figure. At the right of the piston chamber there is an opening to a tube that leads to a partially inflated balloon. The balloon acts as a pulsating sphere and produces a spherical wave—one in which the compressions and rarefactions move outward from the center of the source in all directions.

When the wheel and piston are motionless, the balloon remains partially inflated, which we shall call equilibrium. When the wheel moves clockwise to 90° , the piston is displaced maximally to the right, which forces air up the tube and causes the balloon to be inflated. As the wheel continues to turn, the piston moves back and forth, and alternately it causes the balloon to move from inflated, to equilibrium, to deflated, to equilibrium, and so on. One *cycle* of the wheel's movement corresponds to one cycle of the piston's movement, which also corresponds to one cycle of movement of the balloon. Each cycle comprises 360° .

When the balloon surface expands outward because air is forced up the tube, *a force is applied to the air molecules* surrounding the surface of the balloon. Thus, the circular motion of the wheel, and the straight-line movements of the piston, cause the inward and outward movements of the balloon surface, which affect the air medium in the same way that occurred with the vibratory motion of the tuning fork. The result is *increases (**compression**) and decreases (**rarefaction**) in density that are propagated through the elastic air medium*. Sound has been created.

If the back and forth movements of the piston are simple translations of the circular motion of the wheel, then we also should be able to label the sound wave in degrees:

- 90° corresponds to the wheel having moved $1/4$ of the way through its cycle and to the piston having moved $1/4$ of the way through its cycle;
- when the piston completes $1/4$ of a cycle, it is displaced maximally, and air is forced up the tube to inflate the balloon;
- thus, maximal outward displacement of the balloon also corresponds to 90° ; and finally
- maximal outward displacement of the balloon produces a compression in the air medium, and maximal compression therefore occurs at 90° . Similarly, maximal rarefaction also corresponds to 270° .

■ DIMENSIONS OF THE SINE WAVE

When the vibratory motions of the tuning fork and pendulum were introduced, concepts such as amplitude, frequency, and period were mentioned. Those are “dimensions” of the sound wave. In this section, those and other dimensions will be considered in greater detail.

Amplitude

The first dimension is **amplitude**, which is a measure of the strength, or magnitude, of the sound wave. In most of the examples that will be cited, use of the term “amplitude” will refer specifically to the **sound pressure**. The top waveform in Figure 2–11, which is adapted from Hirsch (1952), represents the *inward and outward movements*

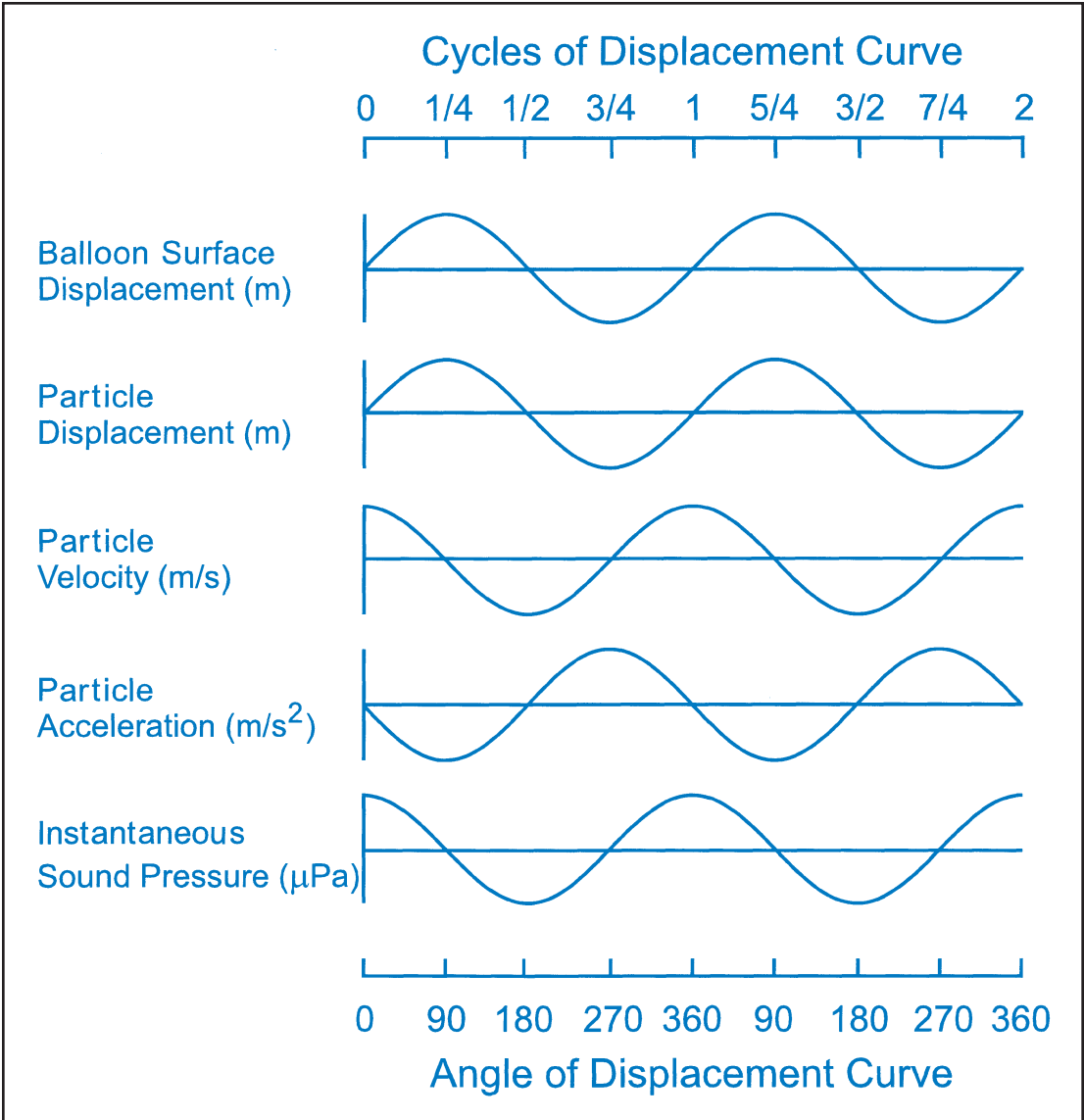


Figure 2–11. Sinusoidal functions that show the relations among **displacement**, **velocity**, **acceleration**, and **sound pressure** as measures of one dimension of sine waves: **amplitude**. Adapted from *The Measurement of Hearing* (p. 24) by Ira Hirsch, 1952, New York: McGraw-Hill Book Company, Inc. Copyright by McGraw-Hill Education. Reproduced with permission.

of the balloon surface that were caused by the turning wheel and moving piston shown in Figure 2–10. The balloon surface is displaced sinusoidally over time, but of course our interest is in the sinusoidal sound wave that results, not in the magnitude of displacement of the balloon surface.

As the balloon is inflated and the surface moves outward, air molecules immediately adjacent to the balloon surface are pushed outward from their average resting position. As more and more molecules are pushed outward, the number of molecules per unit space increases—they are crowded together—which results in *increased density* (**compression**). As the balloon deflates, the molecules move back toward their original positions, but there are fewer molecules in the immediate surface area of the balloon. That is the region of *decreased density* (**rarefaction**).

The second waveform in Figure 2–11 shows the displacement of a single air molecule (particle displacement as a function of time). The individual particle is displaced sinusoidally over time. Stated differently, the **amplitude of displacement** varies sinusoidally over time; maximum amplitude of displacement corresponds to 90° (+) and 270° (–). The magnitude of particle displacement, however, is difficult to measure.

The difficulty of measurement can be overcome by considering some of the derived quantities that were described in Chapter 1. The particles move at some speed, but we also know that because the *direction* of movement is changing, the **velocity** of particle displacement must also change because the concept of velocity incorporates both magnitude and direction of displacement.

The third curve in Figure 2–11 shows the **particle velocity waveform**, and we see that the magnitude of particle velocity also changes sinusoidally over time. It is important to note that *the particle velocity waveform is shifted relative to the particle displacement waveform*. This can be understood by recalling the previous discussion on pendular oscillation.

- When displacement is maximal, velocity is zero and motion is momentarily halted.
- When displacement is zero (when the body is moving through equilibrium), velocity is maximal.
- The relation between displacement and velocity is preserved in Figure 2–11. Particle velocity is maximal when particle displacement is zero, and particle velocity is zero when displacement is maximal (when displacement corresponds to 90° and 270°).¹ Thus, the amplitude of simple harmonic motion can be described by reference to either **particle displacement** or **particle velocity**.²

If particle velocity changes, the particle must be accelerating, either positively or negatively. The fourth curve in the figure shows the *particle acceleration waveform*. Acceleration is greatest when

displacement is maximal. As was noted in the introduction to Chapter 2, a particle is accelerated as it moves from maximal displacement toward equilibrium, and it is decelerated as it moves away from equilibrium toward maximal displacement.

The particle acceleration waveform in Figure 2–11 is shifted relative to both the particle displacement waveform and the particle velocity waveform.³ The relations among **particle displacement**, **particle velocity**, and **particle acceleration** are:

- Particle **velocity** *leads* particle **displacement** by 90° (when $\mathbf{x} = 0^\circ$, $\mathbf{c} = 90^\circ$);
- Particle **acceleration** *leads* particle **velocity** by 90° (when $\mathbf{c} = 90^\circ$, $\mathbf{a} = 180^\circ$); and
- Particle **acceleration**, therefore, *leads* particle **displacement** by 180° (when $\mathbf{x} = 0^\circ$, $\mathbf{a} = 180^\circ$).

Why must acceleration lead displacement by 180° ? First, Equation 1–4 (Newton’s second law of motion) states that *force equals the product of mass and acceleration* (force is directly proportional to both mass and acceleration):

$$\mathbf{F} = m\mathbf{a}.$$

Second, Equation 1–8 (Hooke’s law) states that *the elastic restoring force is directly proportional to the magnitude of displacement*:

$$\mathbf{F}_r = -k\mathbf{x}.$$

In sinusoidal motion, therefore, we have two opposing forces, a force to accelerate a mass ($\mathbf{F}$) and an elastic restoring force ($\mathbf{F}_r$). Third, Newton’s third law of motion states that *with every force there must be an equal reaction force of opposite direction*.

Newton’s third law, expressed as a wave equation, is:

$$\mathbf{F} = \mathbf{F}_r, \text{ and therefore}$$

$$m\mathbf{a} = -k\mathbf{x}.$$

Equation 2–1

Equation 2–2

Equation 2–2 contains two constants, $\mathbf{m}$ (**mass**) and $-\mathbf{k}$ (**stiffness**), and two variables, $\mathbf{a}$ (**acceleration**) and $\mathbf{x}$ (**displacement**). As we saw in Figure 2–11, the two variables, $\mathbf{a}$ and $\mathbf{x}$, vary sinusoidally over time. Finally, to satisfy Newton’s third law of motion ($\mathbf{F} = \mathbf{F}_r$), the two variables, $\mathbf{a}$ and $\mathbf{x}$, *must* be opposite one another.

Recall that atmospheric pressure amounts to $100,000 \text{ Pa (N/m}^2\text{)}$ at sea level. As the balloon surface in Figure 2–10 moves outward and inward, the number of molecules per unit volume (*density of the medium*) changes over time, and **pressure** in the medium is related to the density of the medium. Thus, changes in air pressure also occur sinusoidally over time. Those changes in pressure take the form of

alternate increases and decreases relative to the resting pressure of 100,000 Pa.

The **pressure waveform** is shown at the bottom of Figure 2–11. It reveals that *the instantaneous sound pressures also vary sinusoidally over time*. Moreover, *the pressure waveform “mirrors” the particle velocity waveform*, and both are shifted by 90° relative to the particle displacement waveform. Thus, the pressure waveform *leads* the particle displacement waveform by 90° (when $\mathbf{x} = 0^\circ$, $\mathbf{p} = 90^\circ$).

Figure 2–12 shows two sinusoidal waveforms that display sound pressure as a function of time. The amplitude of wave **A** at 90° is **2 Pa** and the amplitude of wave **B** at 90° is **4 Pa**. Each sine wave’s “pressure” is really many different values of instantaneous pressure that vary sinusoidally over time. In fact, the pressure is only **2 Pa** for wave **A** when particle displacement is zero. (See Figure 2–11 for a review of that relation.)

There are, however, alternative metrics that can be used to express the amplitude of the sine wave. Although the same metrics will be used in Chapter 6 to quantify the amplitude of sound waves that are not sinusoidal, such as speech or music, the equations that are associated with each metric in this chapter *apply only to the sine wave*. Different equations are required for “complex” waves—that is, waves that are not sinusoidal.

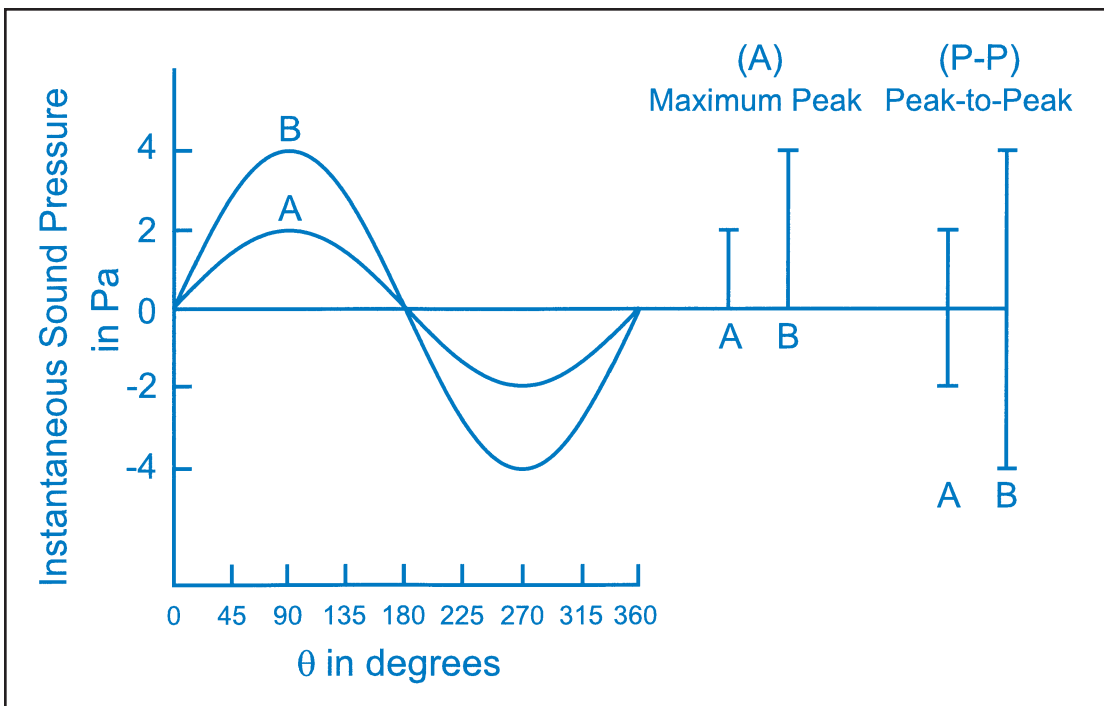


Figure 2–12. Sinusoidal distribution of instantaneous sound pressures for two sound waves of identical frequency that differ in **amplitude**.

Instantaneous Amplitude (a)

By **instantaneous amplitude (a)** we mean *the amplitude of the waveform at any specified instant in time, or at any specified angle of rotation*. Thus, in Figure 2–12, the amplitude of wave **A** is 0 **Pa** at 0°, 2 **Pa** at 90°, and as we shall see later, the amplitude is 1.414 **Pa** at 45° because at 45° the displacement always is 70.7% of maximum ($\sin 45^\circ = 0.707$), and 70.7% of 2 = 1.414.

There are instances in which **instantaneous amplitude** is useful, but not generally so. It never makes sense to specify the instantaneous amplitude without also specifying the corresponding angle of rotation, or moment in time, at which the measurement was taken. Of course, nothing is accomplished by calculating the **mean instantaneous amplitude** of a sine wave, because for every positive instantaneous amplitude there is an equal negative instantaneous amplitude. Therefore, *the mean instantaneous amplitude of sine waves normally is zero* (unless what is called a “direct current [dc] bias” exists).

Maximum Amplitude (A)

The **maximum amplitude (A)**, which also is called the “peak amplitude” (**P**), is *the instantaneous amplitude corresponding to 90° (or 270°)*. Thus, wave **A** in Figure 2–12 has a maximum, or peak, amplitude of 2 **Pa** and wave **B** has a maximum amplitude of 4 **Pa**.

Peak-Peak Amplitude

The **peak-peak** amplitude is *the absolute (not algebraic) difference between the maximum amplitude at 90° and 270°*. Thus,

$$\text{Peak-to-Peak} = 2P$$

Equation 2–3

For the two sine waves in Figure 2–12, the peak-to-peak values are 4 **Pa** for Wave **A** and 8 **Pa** for Wave **B**.

Root-Mean-Square Amplitude (rms)

The **root-mean-square** amplitude (**rms**) is the most common metric for expressing the amplitude of sine waves, but it often seems difficult to achieve an intuitive grasp of what **rms** means. We can achieve an *approximate* understanding of **rms amplitude** by considering it to be *the standard deviation of all the instantaneous amplitudes in a sine wave*.

If we calculate the mean of all instantaneous amplitudes or pressures, the result is zero (unless there is a dc bias). The standard deviation, however, is *the square root of the mean of the squared*

deviations of the instantaneous pressures about their mean of zero. When we square the deviations, we remove the algebraic sign (all values become positive). The standard deviation is a common measure of variability in statistics.

Figure 2–13 shows one cycle of a sinusoidal waveform. For illustrative purposes we will sample only six of the infinite number of instantaneous amplitudes in that single cycle: +2.0, +4.0, and +2.0 during the positive half of the cycle, and –2.0, –4.0, and –2.0 during the negative half of the cycle. Table 2–2 summarizes the steps that are taken to calculate the standard deviation of any set of numbers ($\mathbf{X}$). In this case, the set of numbers comprises the six instantaneous amplitudes.

The sum ($\Sigma \mathbf{X}$) and the mean ($\Sigma \mathbf{X}/N$) of the instantaneous amplitudes are 0.0 (left column). In the second column, the mean of 0.0 is subtracted from each instantaneous amplitude in the first column. Thus, the numbers in the second column are *deviations of the instantaneous amplitudes from the mean of 0.0*. In the third column, each deviation is squared. At the bottom of the third column we see that the sum of the squared deviations ($\Sigma [\mathbf{X} - \bar{\mathbf{X}}]^2$) is 48.0.

The next computation shows that the mean of the squared deviations ($\Sigma [\mathbf{X} - \bar{\mathbf{X}}]^2/N$) is 8.0. In statistics, that is the variance (σ^2) of the distribution of the six numbers ($\mathbf{X}$). The final computation yields the *square root of the mean of the squared deviations*, $\sqrt{\Sigma [\mathbf{X} - \bar{\mathbf{X}}]^2/N}$, which in statistics is the standard deviation (σ). Thus, for the sine wave in Figure 2–13, the value is 2.828. This is one origin of the name

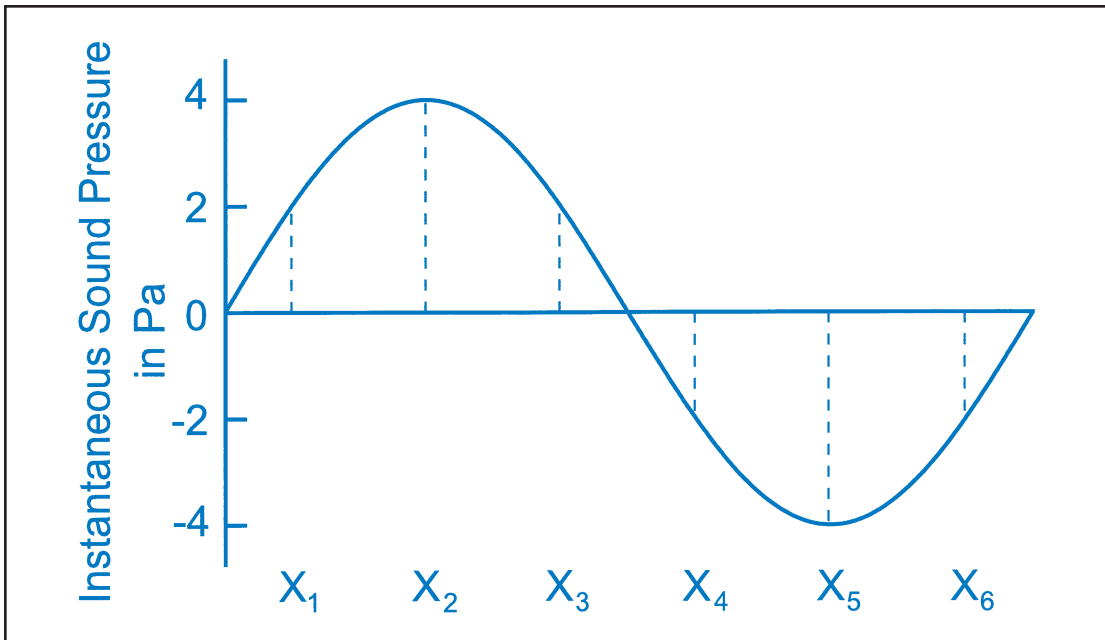


Figure 2–13. Sampling of instantaneous amplitudes to illustrate concept of **root-mean-square (rms)** amplitude.

Table 2–2. Procedures for Calculating the Standard Deviation

X	X – $\bar{X}$	(X – $\bar{X}$)²
2.0	2.0	4.0
4.0	4.0	16.0
2.0	2.0	4.0
–2.0	–2.0	4.0
–4.0	–4.0	16.0
–2.0	–2.0	4.0
$\Sigma X = 0.0$		$\Sigma(X - \bar{X})^2 = 48.0$
$\frac{\Sigma X}{N} = 0.0$	$\sigma^2 = \frac{\Sigma(X - \bar{X})^2}{N} = 8.0 \text{ (ms)}$	
	$\sigma = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}} = 2.828 \text{ (rms)}$	

root mean square (rms). It is the square *root* (**r**) of the *mean* (**m**) of the *squared* (**s**) deviations of the instantaneous pressures; hence, it is the **rms pressure**.

In practice we do not actually calculate the standard deviation of all the instantaneous amplitudes. Instead, Equation 2–4 is used to yield the **rms amplitude** of a *sine wave* from knowledge of the maximum amplitude (**A**):

$$\text{rms} = \frac{A}{\sqrt{2}}.$$

Equation 2–4

Because the square root of 2 equals 1.414, Equation 2–4 can be rewritten as:

$$\text{rms} = \frac{A}{(1.414)}.$$

Equation 2–5

Finally, instead of dividing A by 1.414, you can *multiply by the reciprocal of 1.414*. Thus, Equation 2–5 is changed to read:

$$\begin{aligned} \text{rms} &= A \left(\frac{1}{1.414} \right) \\ &= A (.707), \end{aligned}$$

because the reciprocal of 1.414 is 0.707.

Equation 2–6

To calculate the **rms amplitude** of a *sine wave*, either divide the maximum amplitude (**A**) by 1.414 (Equation 2–5) or multiply the maximum amplitude by 0.707 (Equation 2–6).⁴ What is the unit of measure? The unit of measure of **instantaneous sound pressure** is

the **pascal (Pa)**. The unit of measure of **rms sound pressure** also is the **pascal**. Thus, the **rms pressures** of the two sine waves in Figure 2–12 are **1.414 Pa** for wave **A** and **2.828 Pa** for wave **B**.

Mean Square Amplitude

Mean square sound pressure relates to **rms** sound pressure as the variance (σ^2) in statistics relates to the standard deviation (σ). Thus, variance equals standard deviation squared and *mean square sound pressure equals rms pressure squared*.

$$\text{Because rms} = \frac{A}{\sqrt{2}}$$

$$\text{and because mean square} = \text{rms}^2,$$

$$\text{mean square} = \frac{A^2}{2}.$$

Equation 2–7

Mean square amplitude is used principally for measurements in mechanical dynamics, but it is not likely to be encountered for acoustics in the speech and hearing literature. It was included here only to complete the picture of statistical-acoustical equivalents.

Full-Wave Rectified Average Amplitude (FW_{avg})

What is meant by “rectification”? When you connect an electronic device into an electrical wall outlet, the device receives a sinusoidally alternating electrical current that looks just like (i.e., has the same shape of waveform) the acoustical sine waves we have described. It is called an *alternating* electrical current because during half of its cycle the current flows in one direction (+), and during the other half of its cycle the current flows in the opposite direction (–). The frequency of the electrical current is exactly 60 Hz (in the United States), which means it has a period of 0.0167 s ($T = 1/f$).

The waveform at the top of Figure 2–14 shows the sinusoidally alternating electrical current. The instrument, however, requires a direct current (current that flows in only one direction such as what you obtain with a battery), not the alternating current that the power company provides. The alternating electrical current shown at the top of Figure 2–14 must be converted into direct current such as is shown by the second function in Figure 2–14.

We need not explain all the steps that are required to accomplish the conversion from alternating to direct current, but the first step in the conversion process is important. In the third function of the figure, the negative half of each cycle has been “flipped” to become positive. That is called **full-wave rectification**.

Recall that it makes no sense to calculate the arithmetic mean of the instantaneous amplitudes of the sinusoid because we obtain the same answer, zero, for all sinusoids regardless of differences in

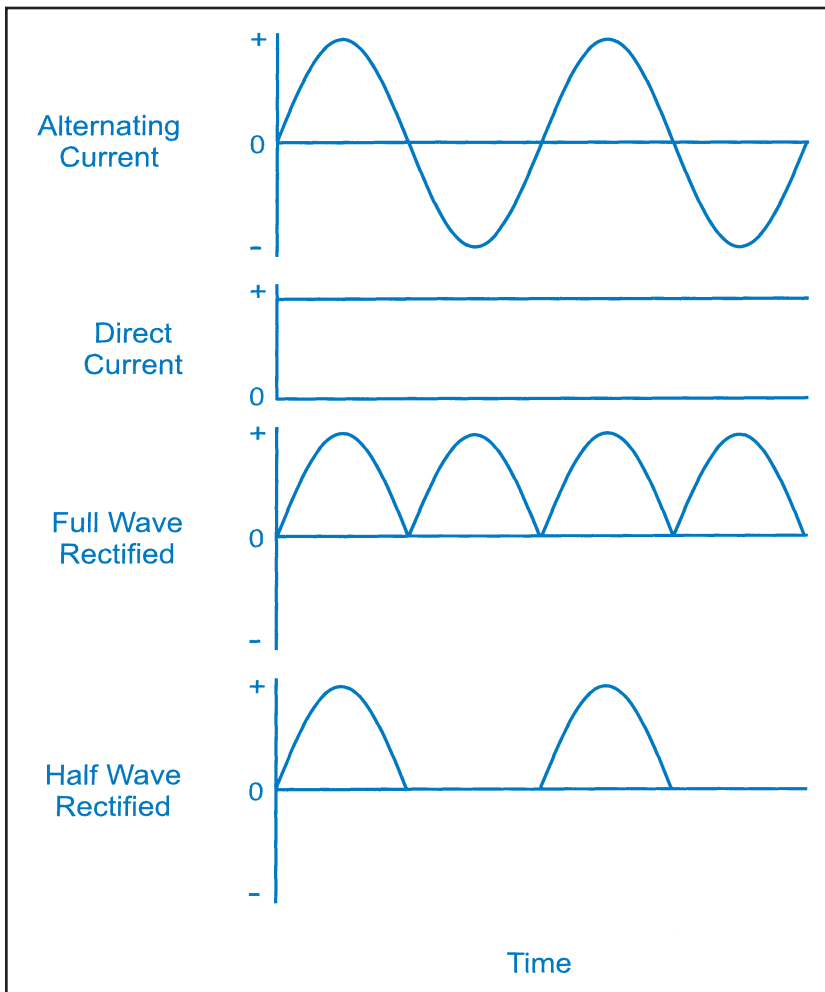


Figure 2-14. Full-wave and half-wave rectification of a sine wave.

their maximum sound pressure (unless a dc bias exists). That circumstance no longer exists with the full-wave rectified sinusoid shown in the figure; all the instantaneous amplitudes are positive. Suppose, therefore, that we *calculate the arithmetic mean of all the instantaneous amplitudes in the full-wave rectified wave*. That is the **full-wave rectified average** (FW_{avg}).

Fortunately, we do not have to perform such a computation. Instead we can use Equation 2-8:

$$\begin{aligned}
 FW_{avg} &= \frac{2A}{\pi} \\
 &= \frac{2A}{3.1416} \quad (\text{substituting the value of } \pi) \\
 &= A (.636) \quad (\text{dividing 2 by 3.1416}).
 \end{aligned}$$

Equation 2-8

Thus, we see that the FW_{avg} is almost two-thirds as large as the maximum sound pressure, whereas the rms is approximately 70% as large as the maximum pressure.

Half-Wave Rectified Average Amplitude (HW_{avg})

The last metric to consider is the **half-wave rectified average** (HW_{avg}). In **half-wave rectification**, *the negative half of each cycle is eliminated rather than flipped* to become positive. The bottom function of Figure 2–14 shows the result. The average of the instantaneous amplitudes that remain is half of what was obtained when we calculated the full-wave rectified average. Thus,

Equation 2–9

$$\begin{aligned}\text{HW}_{\text{avg}} &= \frac{A}{\pi} && \text{(rather than } \frac{2A}{\pi} \text{)} \\ &= \frac{A}{3.1416} \\ &= A (.318) && \text{(taking the reciprocal of 3.1416).}\end{aligned}$$

The FW_{avg} is almost two-thirds (63.6%) as large as the maximum sound pressure, and because $\text{HW}_{\text{avg}} = 0.5 \text{ FW}_{\text{avg}}$, the HW_{avg} is almost one-third (31.8%) as large as the maximum sound pressure.

Comparisons Among Metrics for Measuring Amplitude

The waveform at the top of Figure 2–15 is sinusoidal with a maximum amplitude of 4 Pa. Thus, the instantaneous amplitudes vary sinusoidally from a minimum value of –4 Pa to a maximum value of +4 Pa. The lower portion of the figure shows the rms (Equation 2–6), **mean square** (Equation 2–7), FW_{avg} (Equation 2–8), and HW_{avg} (Equation 2–9).

The *instantaneous amplitudes vary sinusoidally over time*. The other metrics are described by straight lines horizontal to the baseline. They do not vary. Suppose we direct the sinusoid to a measuring instrument that has an indicating meter, or needle, like you might have seen on the VU meter of some older tape recorders. If the measuring instrument is designed to respond to the instantaneous pressure, it would move back and forth about some average value at a rate corresponding to the frequency of the sinusoid. If the instrument is designed to respond to any of the other measures of amplitude, the needle would remain perfectly stationary—*those values remain constant over time because each is time-averaged*.

Frequency (f) and Period (T)

The second dimension of the sine wave is **frequency (f)** and its reciprocal the **period (T)**. **Frequency** is *the rate, in hertz, at which a sinu-*

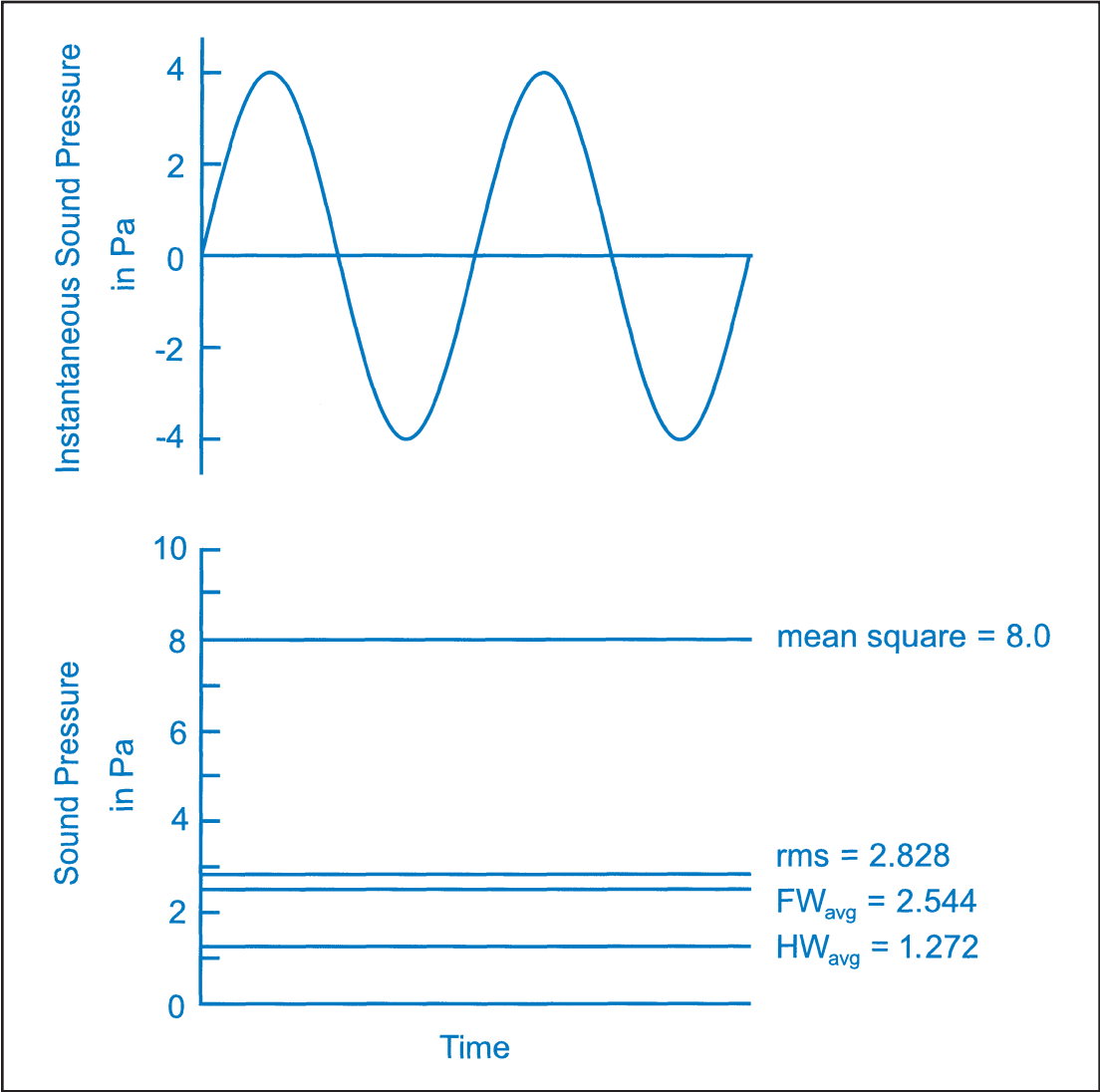


Figure 2–15. Comparison of **mean square**, **root-mean-square**, **full-wave rectified average**, and **half-wave rectified average** measures of amplitude of a sine wave.

*soid repeats itself. **Period** is the amount of time, in seconds, required for a sinusoid to complete one cycle.*

Consider the two sinusoids in Figure 2–16. The period of wave **X** is .001 s, which means that the frequency is 1000 Hz (see Equation 1–12). The period of wave **Y** is .0005 s and its frequency therefore is 2000 Hz. The inverse relation between frequency and period should be apparent. As period is halved (from .001 s to .0005 s), frequency is doubled (from 1000 Hz to 2000 Hz).

We must resort to rather cumbersome numbers when frequency is very large or period is very small. Suppose, for example, that the frequency is 2,345,129 Hz. If that seems cumbersome, think about

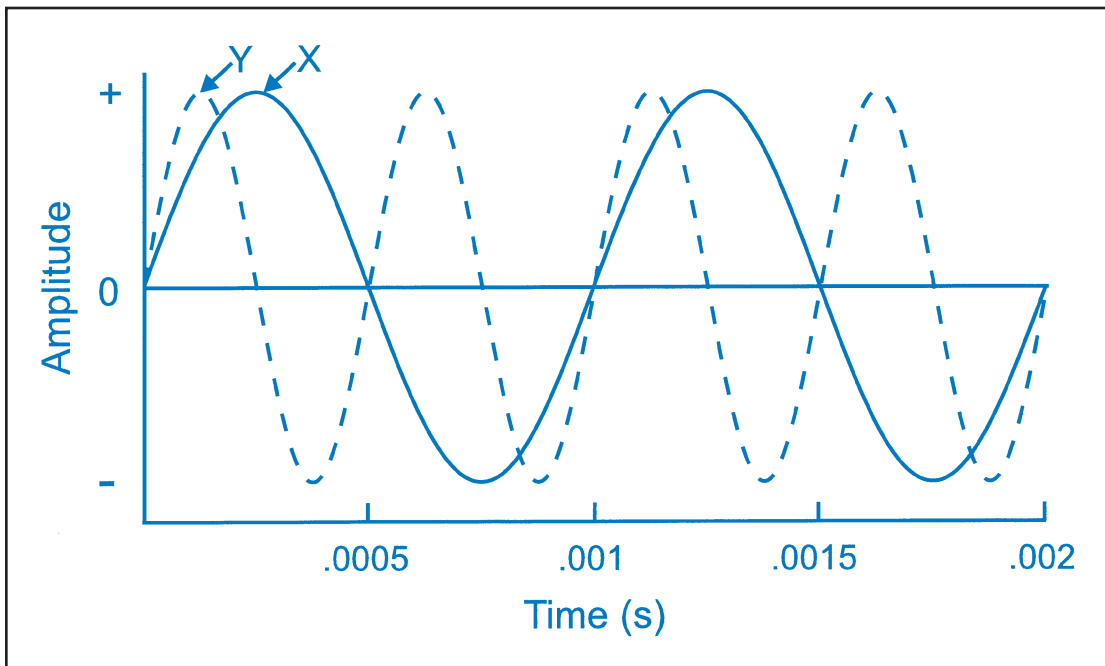


Figure 2-16. Sinusoidal distribution of sound pressure for two sound waves of identical amplitude that differ in **frequency**.

its period (0.0000004264 s). We can partially avoid such cumbersome expressions by using multiples of **hertz** for frequency and sub-multiples of **second** for period. The alternatives commonly used to describe frequency and period are shown in Table 2-3.

We can see from Table 2-3 that

- 1 kilohertz (1 kHz) = 1000 Hz
- 1 megahertz (1 MHz) = 1,000,000 Hz
- 1 MHz also = 1000 kHz, and so on.

For period,

- 1 ms is 0.001 (one-thousandth) s
- 1 μ s is 0.000001 (one-millionth) s, and so on.

Those multipliers are encountered frequently. Consider your AM radio dial. The AM tuner is designed to receive radio signals with frequencies that range from 550 kHz to 1600 kHz. Thus, if you set your dial to 1560, the tuner is adjusted to receive a signal with a frequency of 1560 kHz. If you happen to be driving where the Walhonding and Tuscarawas rivers join to form the Muskingum river, you will be tuned to station WTNS in Coshocton, OH. As a further aside, in the 1950s, Coshocton had three movie theaters: The Pastime, the Sixth Street (because it was located on 6th St.), and the

Table 2–3. Standard Units of Measure for Frequency and Period

Frequency	Multiplier	Period	Multiplier
Hertz (Hz)	1	second (s)	1
Kilohertz (kHz)	1,000	millisecond (ms)	.001
Megahertz (MHz)	1,000,000	microsecond (μ s)	.000001
Gigahertz (GHz)	1,000,000,000	nanosecond (ns)	.000000001

Mu-Wa-Tu. The latter, of course, was named after the three rivers on the west side of this small river town.

Suppose you change your radio from AM to FM and set your FM tuner to 91.2. That doesn't mean 91.2 Hz, but rather 91.2 MHz, which is equivalent to 91,200,000 Hz. The FM tuner covers the range from 88 MHz to 108 MHz, and it is more convenient to label the dial in megahertz than hertz.

The tuner on your television set receives frequencies below and above the range of frequencies for FM radio stations. VHF (very high frequency) tuners range from 54 MHz to 88 MHz (below the FM frequencies) for channels 2 through 6, but from 174 MHz to 216 MHz (above the FM frequencies) for channels 7 through 13.

We know from Equations 1–11 and 1–12 that frequency and period are reciprocally related. Thus, to determine period from knowledge of frequency, calculate the reciprocal of frequency ($T = 1/f$), and to determine frequency from knowledge of period, calculate the reciprocal of period ($f = 1/T$):

- If frequency is expressed in *hertz* and you calculate the reciprocal, *the result for period is expressed in seconds*;
- if frequency is expressed in *kilohertz*, and you calculate the reciprocal, *the result for period is in milliseconds*; and
- if frequency is expressed in *megahertz*, and you calculate the reciprocal, *the result for period is in microseconds*.

Suppose the frequency of a sinusoid is 2 kHz, but you want to express the corresponding period in *seconds rather than milliseconds*. You can proceed in either of two ways. Convert 2 kHz to hertz by multiplying by 1000 (see Table 2–3); the reciprocal of 2000 Hz is .0005 s. Alternatively, calculate the reciprocal of 2 kHz, which is 0.5 ms, and then convert 0.5 ms to seconds by multiplying by 0.001 (see Table 2–3). The answer is the same—0.0005 s.

Determinants of Frequency

The frequency with which a source of sound vibrates is governed by *properties of the source*. Specifically, the frequency of vibration of some simple harmonic oscillator, such as a spring-mass system, is

called the **natural frequency** (f_{nat}) and depends on the **mass** and **stiffness** of the system. The natural frequency is given by Equation 2–10:

Equation 2–10

$$f_{\text{nat}} = (\sqrt{s} / \sqrt{m})$$

The **natural frequency** is

- *directly proportional to the square root of the **stiffness*** (frequency increases as stiffness increases), and
- *inversely proportional to the square root of the **mass*** (frequency decreases as mass increases).

For different sources of sound, different properties are considered, but in all cases, they ultimately relate to **stiffness** and **mass**. We learned previously that the vibratory frequency of a pendulum depends almost entirely on its length (Equation 1–13). What about the strings or wires on a guitar, violin, or piano? In those cases, frequency of vibration depends on the **length**, **cross-sectional mass**, and **tension** of the string, and the relation of each of those three factors to the frequency of a vibrating string is shown in Equation 2–11:

Equation 2–11

$$f = \frac{1}{2L} \sqrt{\frac{t}{m}}$$

where **L** = length, **t** = tension, and **m** = cross-sectional mass.

We can see from Equation 2–11 that the frequency of vibration of a string increases

- as length decreases (*inversely proportional to twice the length*),
- as tension increases (*directly proportional to the square root of the tension*), or
- as mass decreases (*inversely proportional to the square root of the mass*).

Think about the wires on a piano. Tones of lower frequency are produced by longer and more massive wires (inverse relations). To increase the frequency of vibration of an individual wire, you increase the tension—the same process that you go through when you “tune” any stringed instrument such as a guitar. We will learn subsequently that the vibratory pattern of the string or wire is more complex than has just been described. The string vibrates at many frequencies simultaneously. Thus, Equation 2–10 applies to only the lowest frequency of vibration, which is called the **fundamental frequency**.

The equation for the lowest frequency of vibration of a string is particularly important for those who are interested in voice production, because the frequency of vibration of the vocal folds depends mainly on the length, cross-sectional mass, and tension of the folds.

The nature of the proportionalities (direct or inverse) for vocal fold vibration is the same as it is for a string.

The natural frequency for other sources of sound depends on different characteristics. The frequency of a drum's membrane, for example, is directly related to the *tension* of the membrane, but inversely related to its *radius*, *thickness*, and *density*. The equations for calculating frequency of vibration of sources such as stretched membranes, bars, and the like, however, are beyond the scope of this book. In all cases, however, the unifying concept is that f_{nat} is *directly proportional to the square root of the stiffness and indirectly proportional to the square root of the mass*.

Angular Velocity (ω)

Frequency occasionally is expressed by **angular velocity** (ω) instead of hertz. Before we attempt to define angular velocity, let us consider some alternative ways that *could* be used to express frequency, *although they have not been adopted*. We learned earlier that back and forth vibration is analogous to a point on a turning wheel moving about the circumference of a circle. Consider the circle shown at the left of Figure 2–17. When a reference point moves completely around the circumference of a circle, it has completed 360° of rotation. If the 360° rotation is accomplished in 1 s, $f = 1$ Hz, and if the rotation is accomplished in 1 ms, $f = 1000$ Hz, or 1 kHz.

Suppose, instead, that the unit of measure for frequency were *degrees of rotation per second* rather than hertz. Then we would say that $f = 360^\circ$ per s in the first case and $f = 360,000^\circ$ per s in the second case. Frequency has *not* been described that way, but it would

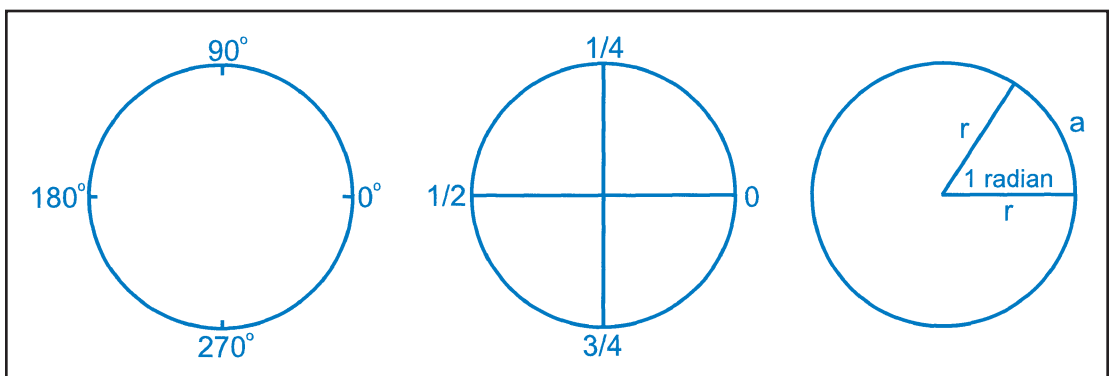


Figure 2–17. An explanation of **angular velocity** in which a point moves around the circumference of a circle. At the left, the frequency of rotation of the point is represented in “**degrees per s**” because the circle is divided into 360 equal parts. At the middle, frequency of rotation is represented in “**quarter cycles per s**” because the same circle is divided into only 4 equal parts. At the right, frequency of rotation is represented in “ **2π radians per s**” because the same circle is divided into 6.2832 equal parts, where 6.2832 equals 2π . See text for an explanation of a **radian**.

be perfectly valid. Of course, it would often result in frequency being described by very large numbers.

We could try a different approach that is illustrated in the middle circle in the figure. Because one cycle consists of *four quarter cycles* (each of which corresponds to 90°), $f = 360^\circ$ per s could be expressed as $f = \text{"4 quarter cycles per s"}$ (dividing 360 by 90), and $f = 360,000^\circ$ per s could be expressed as $f = \text{"4000 quarter cycles per s."}$ In either case, the full circle of 360° is divided into some number of equal segments—360 equal segments to arrive at “degrees per second,” and four equal segments to arrive at “quarter cycles per second.”

It turns out that no one has described frequency by dividing the circle into either 360 equal parts, or four equal parts. Instead, *the circle is divided into 6.2832 parts of equal size*. Why 6.2832 parts? One cycle equals 360° , and 360° corresponds to **2π radians**, which equals 6.2832 radians (2×3.1416). For the circle shown at the right of Figure 2–17, we show one of the 6.2832 possible segments, and it has been drawn to equal 1 radian.

What is a “radian”? You might read in a geometry textbook that *a radian is equal to the angle subtended by an arc whose length is equal to the radius*. Unless you are a “fan of geometry,” that definition might not make a whole lot of sense. So, stated more simply, *an angle equals a radian when the intersection of the two sides of the angle with the circumference yields an arc whose length is exactly equal to the length of the radius*. For the angle shown in Figure 2–17, the two sides of the angle are both of equal length and both are equal to the radius (**r**). An arc (**a**) is formed by the intersection of those two radii with the circumference, and the length of the arc **a** equals the length of the radius **r**.

Every circle comprises 360° , and for every circle there are exactly **2π radians**. Because π equals 3.1416, every circle therefore has 6.2832 (2×3.1416) radians, and 1 radian corresponds to 57.3° (360° divided by 2π). Another way to conceptualize what is meant by radians was suggested to me by Professor Ted Glatke. Measure the diameter of *any* circle and then “snip” the circle and “unroll” it. *Regardless of the size of the circle that is unrolled, the length of the line that is left will be exactly (to four decimal places) 6.2832 times the length of the radius of the circle, or 3.1416 times the length of the diameter of the circle.*

How can radians be used to express frequency? When degrees are used to express frequency, one cycle = 360° , and therefore 1 Hz (cps) corresponds to 360° per s. Because one cycle = **2π radians**, 360° per s corresponds to **2π radians per s**. *When frequency is expressed in radians per s, the unit of measure is called the **angular velocity** (ω).*

Equation 2–12 shows the relation between angular velocity and frequency:

Equation 2–12

$$\omega = 2\pi f.$$

To calculate angular velocity in radians per second, multiply frequency in hertz by **2π** . Thus, if $f = 1$ Hz, $\omega = 6.28$ **rps** (radians per s),

if $f = 100$ Hz, $\omega = 628$ rps, and if $f = 1$ kHz, $\omega = 6280$ rps. In practice, angular velocity is not often used this way. It will be useful to keep angular velocity in mind, however, when the third dimension of the sine wave—**phase**—is introduced.

Phase

The wheel in Figure 2–18 has four reference points (A, B, C, and D). It turns in a counterclockwise direction and the sinusoidal projections from the four reference points are shown at the right. *Degrees of rotation that are used to label the abscissa refer to the course of travel of reference point A*, where 0° is the starting point, 90° is $1/4$ of the way around the circle, and so on.

The four curves are not superimposed on one another because the four reference points on the wheel are at four different locations around the circumference. We can describe their relative positions by reference to *degrees of separation*. Thus, **B leads A** (B ahead of A) by 90° , **C leads B** by 90° , **C leads A** by 180° , and so on. In the same way, the differences among the four sinusoidal projections can be described by reference to *degrees of separation*. Curve **B leads A** by 90° , or we can say, curve **A lags curve B** by 90° .

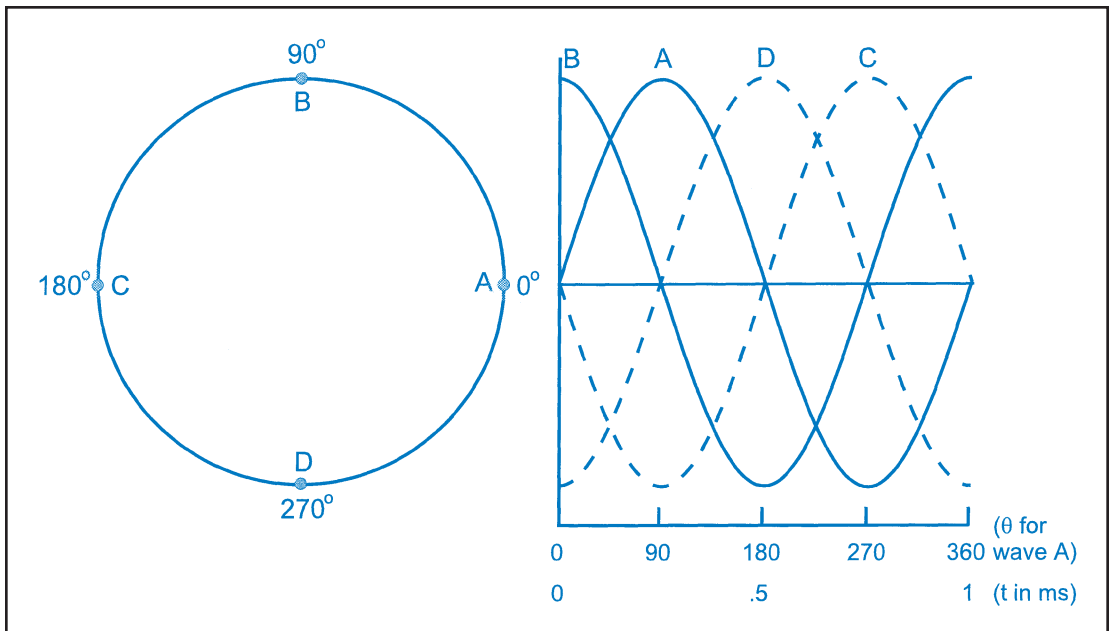


Figure 2–18. An explanation of **starting phase**. At the left, four reference points (A, B, C, and D) are located at different angles: 0° , 90° , 180° , and 270° . When the wheel commences to turn, each point rotates around the circumference with the same amplitude and frequency as the other points. Projections of the sinusoidal motions of the four different points are shown at the right. The four functions differ from one another because of the different angles of rotation at t_0 , which is the **starting phase**.

It is important to realize that all four sinusoids in Figure 2–18 *rotate with the same frequency and have the same amplitude*. That must be the case in this example because there is only one wheel with a fixed circumference (determinant of amplitude), and the rate at which the wheel turns (determinant of frequency) results in all four reference points rotating with the same frequency.

Starting Phase

Even though the four curves in the figure have the same amplitude and frequency, they are different from one another, and the difference is in the **starting phase**. By **starting phase**, we mean *the displacement, in degrees, at the instant the vibration, or rotation, begins*. Thus, the four curves in the figure have different starting phases:

- Wave **A** starts at 0° ,
- wave **B** starts at 90° ,
- wave **C** starts at 180° , and
- wave **D** starts at 270° .

Phase Angle: Instantaneous Phase

With **starting phase**, we referred to **time zero**, a reference time corresponding to the angle when displacement begins. The **phase angle**, also called the **instantaneous phase**, is different from the starting phase; *the phase angle refers to the angle of rotation at any specified moment in time, not just t_0* . Thus, at $t = 0.5$ ms, the **phase angles** are 180° for **A**, 270° for **B**, 360° for **C**, and 90° for **D**. **D** has a starting phase of 270° and therefore leads **A** by 270° , and that separation always remains because **D** and **A** rotate with the same frequency. This contrasts with what we saw in Figure 2–16, which displayed two sine waves with different frequencies.

When the frequencies are different, the two waves move in and out of phase rather than maintain a constant phase separation, even though the starting phases of the two waves happen to be the same. We will return to this concept in Chapter 9 when the phenomenon of **beats** is described.

Phase angles also are described in **radians** rather than **degrees**. Recall that a circle comprises 360° *and* 2π **radians**. The relation between **degrees** and **radians** to describe phase angle is shown in Figure 2–19. If 360° corresponds to 2π **radians**, $0^\circ = 0$ **radians**, $90^\circ = \pi/2$ **radians** ($1/4$ of 2π), $180^\circ = \pi$ **radians** ($1/2$ of 2π), and $270^\circ = 3\pi/2$ **radians** ($3/4$ of 2π).

Another Application of Radians

Students who elect to study **psychoacoustics**, a science concerned with **psychological correlates** of the physical dimensions of sound

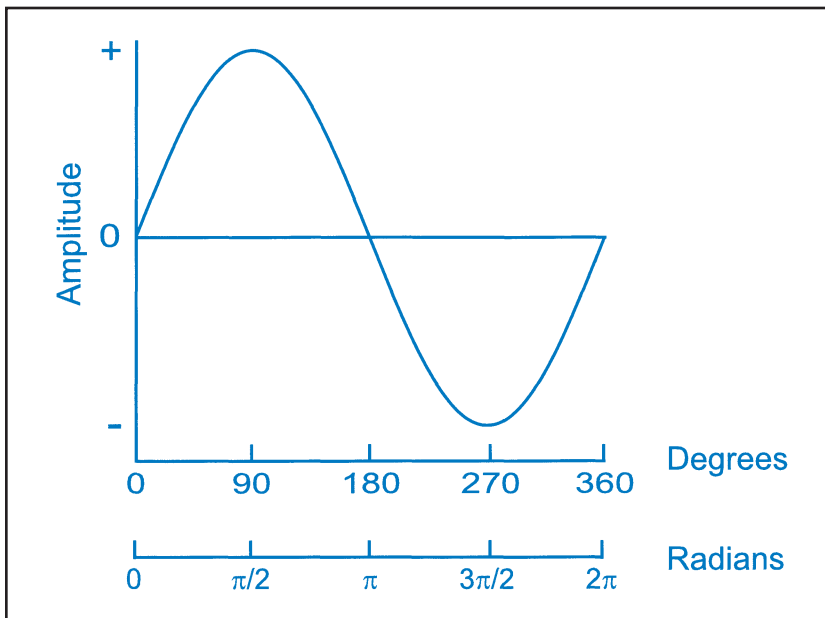


Figure 2-19. The relation between **instantaneous phase** expressed in **degrees** and in **radians**.

waves, will encounter instances in which phase angle is described in radians. Imagine the following experiment. A single sinusoid is presented to a listener who is wearing a pair of earphones. The signal is delivered simultaneously to both ears, and the frequency and the amplitude of the signal are the same in both ears.

Suppose that in one portion of the experiment the *starting phase* of the signal is the same in the left ear as in the right ear. The two signals are said to be *in phase* because there is no difference in phase angle. That condition is called S_0 , which means that the *difference* in phase angle of the signals (S) between the two ears is **0 radians**.

In a second portion of the experiment, suppose that the starting phase of the signal in the left ear is 0° , but the starting phase of the signal in the right ear is 180° . Curves **A** and **C** in Figure 2-18 are examples of this condition. The two signals are obviously out of phase by 180° . Because 180° corresponds to π **radians**, that circumstance is called S_π (see Figure 2-19).

Wavelength (λ)

The last dimension of the sine wave is **wavelength (λ)**. We have described *two quantities that are measured by reference to time*. One is **frequency (f)**, which is the rate at which the sinusoid repeats itself and is measured in hertz. *Frequency is determined by characteristics of the sound source.*

The second is **speed of sound (s)**, which is the rate at which a disturbance is propagated through the medium and is measured in feet per second (**fps**), meters per second (**MKS**), or centimeters per second (**cgs**). *Propagation speed is determined by characteristics (elasticity and density) of the transmitting medium.*

The fourth dimension of the sine wave, **wavelength (λ)**, defines the relation between **frequency** and **propagation speed**. **Wavelength is the distance traveled by a sine wave during one period of vibration, or as the distance between two identical points (identical phase angles) on two adjacent cycles.** Wavelength is directly proportional to the speed of sound and inversely proportional to frequency, as is shown by Equation 2-13:

Equation 2-13

$$\lambda = \frac{s}{f},$$

where λ is wavelength in meters,

f is frequency in hertz, and

s is the speed of sound in meters per second.

Consider the examples in Figure 2-20. Two sinusoids ($f = 1100$ Hz and $f = 550$ Hz) travel in two different media, air and water. Look first at the wave with $f = 1100$ Hz traveling in air. Its period ($T = 1/f$) is 0.9 ms. After 1 s, the wave travels 340 m (assuming the speed of sound is 340 m/s). During just *one period* it travels only 1/1,100 of that distance, which is 0.3 m. That is the distance traveled during one period, which is the definition of wavelength, and that is the answer we obtain with Equation 2-13: $\lambda = 340/1,100 = 0.3$ m.

Compare that outcome with what happens if $f = 550$ Hz in air. The speed of sound still is 340 m/s, which means that the sinusoid still travels 340 m in 1 s. The period now, however, is 1.8 ms. During that 1.8 ms (one period), the wave travels 0.6 m.

What happens when the same two sine waves are propagated through water? For convenience, we will treat the speed of sound in water as if it were four times the speed in air (1,360 m/s), which is a reasonable approximation. When $f = 1100$ Hz, the wave travels 1,360 meters in 1 s, the period still is 0.9 ms, and the distance traveled during 1 period is four times as great (1.2 meters). If $f = 550$ Hz, the wavelength is increased from 0.6 m in air to 2.4 m in water. Thus, we see that *wavelength is directly proportional to the speed of sound and inversely proportional to its frequency*.

The period of most sound waves that are likely to be encountered is typically very small. For example, the period of the sound waves produced by vibrating vocal folds during vowel production by an average adult male is only about 10 ms and the period for the average adult female is just a little over 4 ms. The wavelengths of those sounds in air, however, are relatively large—about 1.5 m (5 ft) for the female and about 3.4 m (11 ft) for the male. In other words, one com-

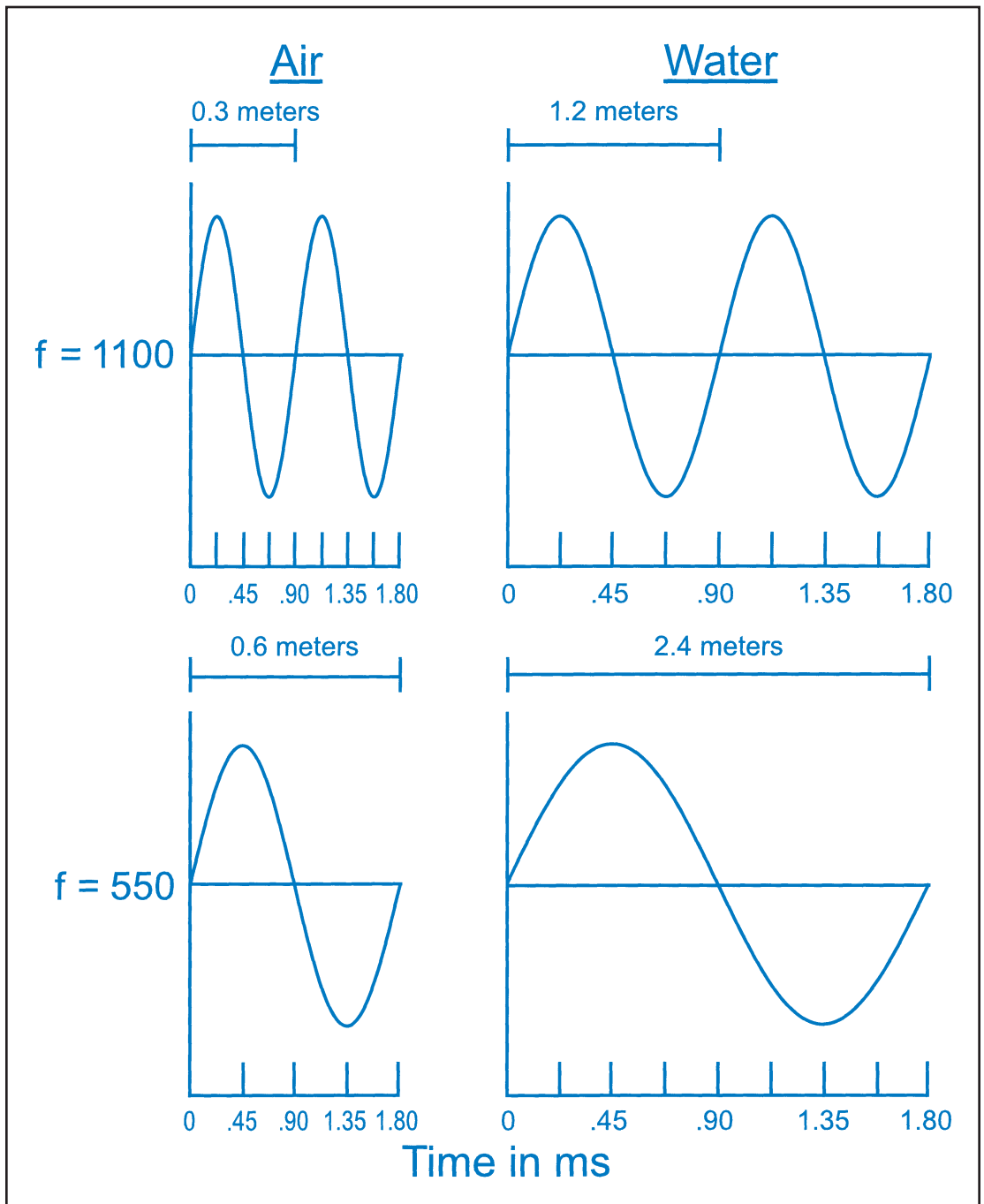


Figure 2-20. An illustration of the dependence of **wavelength** on both **frequency** (rows) and **speed** of sound wave propagation (columns). Comparison of the two functions within a column (left for air; right for water) shows how wavelength varies with frequency when speed of propagation is held constant. Comparison of the two functions within a row (top for 1100 Hz; bottom for 550 Hz) shows how wavelength varies with speed of propagation when frequency is held constant.

plete cycle of vibration spans a distance of about 1.5 m for the vowel produced by the female and 3.4 m for the vowel produced by the male. Will a sound of lower frequency (and longer wavelength) travel through the air medium faster than a sound of higher frequency (and shorter wavelength)? No! Speed of sound is governed by properties of the medium, and frequency of vibration is governed by properties of the source. Wavelength simply shows the relation between those two quantities: s and f .

■ NOTES

1. Particle velocity is the first derivative of particle displacement.
2. Particle displacement is a sine function, and both particle velocity and pressure are cosine functions.
3. Particle acceleration is the first derivative of particle velocity and the second derivative of particle displacement.
4. Equation 2–6 provides a perfectly acceptable *estimate* of the **rms** sound pressure of sine waves. Precise calculations of the **rms** pressure require knowledge of calculus, which is beyond the scope of this book. For the interested reader, however, the defining equation for **rms** is

$$\text{rms} = \sqrt{\frac{1}{T} \int_0^T x(t)^2 dt},$$

where $x(t)$ is the time-domain waveform, T is the period, and $\int$ indicates summation of an infinite number of instantaneous amplitudes during one period (one cycle) of the wave.

■ FREQUENTLY MISUNDERSTOOD CONCEPTS

1. **What is the phasic relation between particle velocity and particle displacement during simple harmonic motion?**

Particle velocity *leads* particle displacement by 90° . Hence, particle velocity is *maximally positive* when the instantaneous phase angle of particle displacement is 0° , and particle velocity is *maximally negative* when the instantaneous phase angle of particle displacement is 180° .

2. **Mean square sound pressure = rms^2 . Should we therefore conclude that mean square pressure will always be greater than rms pressure?**

No. If rms is less than 1.0, mean square less than rms, and if rms equals 1.0, mean square equals rms. For example, if rms = 0.9, mean square = 0.81 (0.9^2), but if rms = 1.0, mean square also = 1.0 (1.0^2). Thus, the numerical value

of mean square may be less than, equal to, or greater than rms, depending on the value of rms. Note, however, that the unit of measure for rms (Pa) is different from the unit of measure for mean square (Pa^2).

3. The frequency of sine wave $X = 1000 \text{ Hz}$ and the starting phase is 0° . Sine wave Y has a frequency of 2000 Hz and a starting phase also of 0° . Describe the relation between the instantaneous phase angles of the two sine waves over the course of one period for sine wave X .

The phasic relation between the two sine waves varies from moment to moment as they move in and out of phase. See Figure 2-16. At $t = 0$, the two waves are *in phase*; at $t = 0.25 \text{ msec}$, Y *leads* X by 90° ; at $t = 0.5 \text{ msec}$, Y *leads* X by 180° ; and so forth. They are *in phase* once again when $t = 1 \text{ msec}$.

4. How does the magnitude of the elastic restoring force of a spring-mass system change over time when the system is engaged in simple harmonic motion?

Elasticity is a restoring force that *opposes* the force that acts to deform the spring. Hooke's law informs us that the magnitude of the restoring force (F_r) is directly proportional to the magnitude of the spring's displacement (x). In other words, as displacement increases, the restoring force of elasticity also increases, and vice versa. The magnitude of the spring's displacement changes sinusoidally over time, and the magnitude of restoring force also varies sinusoidally over time. The sinusoidally changing restoring force is *in phase* with the sinusoidally changing displacement.

■ PRACTICE PROBLEMS

Set 1

1. If $rms = A/\sqrt{2}$, $A =$
2. If $A = rms (1.414)$, $rms =$
3. Given rms , write the equation for peak-to-peak amplitude.
4. Below are values of peak sound pressure. Solve each for rms sound pressure.
1.0 1.5 0.6 3.8
5. Below are values of rms sound pressure. Solve each for peak-to-peak sound pressure.
0.707 1.0 3.5
6. Below are values of peak-to-peak sound pressure. Solve each for rms sound pressure.
1.0 1.414 10.0 2.0 18.0 x
7. Given a sinusoidal wave with a maximum amplitude of 5 V (where voltage is the electrical analog of acoustic pressure), solve for:
 $rms =$ $mean\ square =$ $HW_{avg} =$ $P-P =$
 $FW_{avg} =$ $A =$
8. For each tenfold increase in maximum amplitude, rms increases by a factor of?
9. For each twofold increase in maximum amplitude, the full-wave rectified average increases by?
10. If rms increases by a factor of two,
a. $mean\ square$ increases by a factor of?,
b. the full-wave rectified average increases by a factor of?
11. If rms voltage increases from 10 V to 100 V, $mean\ square$ voltage increases by a factor of?
12. Under what condition will the numerical value of rms equal the value of $mean\ square$?
13. If $FW_{avg} = 2A/\pi$, $HW_{avg} = ?$
14. If $HW_{avg} = 1.3$, $FW_{avg} = ?$

15. If $rms = 1.5 v$, mean square = ?
16. If rms increases by a factor of N , mean square increases by a factor of?

Set 2

1. If the period of a sinusoid is 150 ms, $f = ?$
2. If 1.5 cycles of a sinusoid are completed in 5 ms, $f = ?$
3. Calculate f , in Hz, for each value of period below:
0.002 s 3 ms 1000 μ s
4. Calculate T , in ms, for each value of frequency below:
400 Hz 800 Hz 100 Hz 500 Hz 8000 Hz
5. Calculate f , in kHz, for each of the following values:
100 Hz 0.002 s 1 MHz 5 ms

Set 3

1. Describe the relation between **particle displacement** (x) and **particle velocity** (c) during simple harmonic motion.
2. Describe the relation between **period** and **angular velocity**.
3. Describe the proportional relations between the **frequency** of a vibrating string and the **mass**, **length**, and **tension** of the string.
4. Describe the proportional relations between the **period** of a vibrating string and the **mass**, **length**, and **tension** of the string.
5. Sine wave A has a **frequency** of 100 Hz and a **starting phase** of 0° . Sine wave B has a **frequency** of 110 Hz and a **starting phase** of 90° . Describe the phasic relation between the two waves during one period of vibration for sine wave A.

Set 4

1. Table 2-1 of the textbook listed the sine of selected angles from 0° through 360° at 11.25° intervals. In Figure 2-9 of the textbook, **$\sin \theta$** was plotted as a function of **θ** in degrees. Assume you have a wave with a **maximum amplitude** (A)

of 2 Pa. Calculate the **instantaneous pressure (p)** for each of the angles listed below for the first half of one cycle (three decimal places).

θ	p	θ	p	θ	p	θ	p
0.00	_____						
11.25	_____	67.50	_____	123.75	_____	180.00	_____
22.50	_____	78.75	_____	135.00	_____		
33.75	_____	90.00	_____	146.25	_____		
45.00	_____	101.25	_____	157.50	_____		
56.25	_____	112.50	_____	168.75	_____		

- For the same sine wave in #1 ($P_{\max} = 2 \text{ Pa}$), how would the result differ if you were to continue the computations at 11.25° intervals for the second half of the cycle?
- For the same sine wave in #1, calculate each of the following in Pa (two decimal places).

rms = ? mean square = ? $HW_{\text{avg}} = ?$ $FW_{\text{avg}} = ?$



Acoustic Impedance

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■ REVIEW OF SIMPLE HARMONIC MOTION

When the vibratory motion of a spring-mass system was presented in Chapter 1, and when sine waves were described in Chapter 2, each cycle of sinusoidal motion was represented as being identical in every respect to every other cycle. That representation is appropriate when the sine wave is conceptualized from a strictly mathematical perspective, *and when there is no opposition to motion*. Oscillating systems, however, do encounter opposition to motion in the form of **friction**, or **frictional resistance**, and **friction** limits **velocity**.

■ DAMPING

When amplitude of vibration diminishes over time, the oscillations, or vibrations, are **damped**. To further our understanding of damping, it will be helpful to review and expand on some of the characteristics of vibratory motion described previously.

A spring-mass system, or other simple harmonic oscillators, vibrates sinusoidally. In Figure 3–1, the amplitude of **displacement** (dashed line) of the system varies sinusoidally over time. In accordance with Hooke's law, the restoring force of **elasticity** also varies sinusoidally over time. *The magnitude of the restoring force of elasticity is directly proportional to the magnitude of displacement and is in phase with displacement*; hence, the sinusoidally vary-

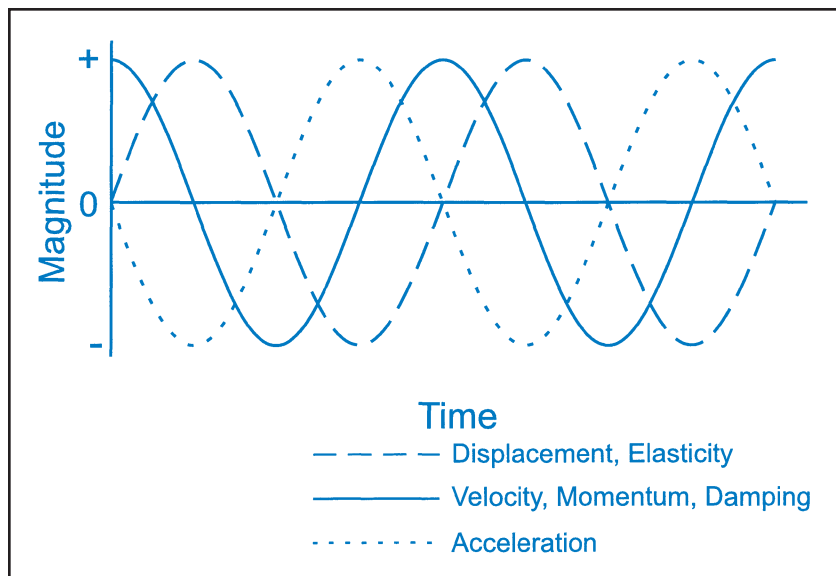


Figure 3–1. The relations among sinusoidal waveforms for **displacement**, **elasticity**, **velocity**, **momentum**, **damping** (frictional resistance), and **acceleration**.

ing elastic force is shown by the same dashed line used to describe displacement.

Velocity (solid line) also changes sinusoidally over time, and velocity *leads* displacement by 90° . Whereas velocity leads displacement by 90° , velocity *lags* **acceleration** (dotted line) by 90° , and acceleration is 180° degrees out of phase with displacement. As the mass of the system moves from maximum displacement toward equilibrium, velocity of motion increases, and the system gains **momentum**.

- Because **momentum** is given by the product of **mass** and **velocity** ($\mathbf{M} = m\mathbf{c}$; Equation 1–9), *momentum is directly proportional to velocity of motion*. Momentum also varies sinusoidally over time, and **momentum** is *in phase* with **velocity**. Thus, momentum is shown by the same solid line that represents velocity.
- Because momentum is in phase with velocity, elasticity is in phase with displacement, and velocity *leads* displacement by 90° , **momentum** also *leads* the restoring force of **elasticity** by 90° .
- Therefore, momentum is greatest when velocity is greatest, velocity is greatest when displacement is zero (i.e., when the vibrating mass is passing through equilibrium), and the restoring force of elasticity is zero when displacement is zero.

Effects of Friction on Vibratory Motion

Once energy is imparted to an object, vibratory motion continues indefinitely with no decrement in amplitude over time *if there is no opposition to motion*—the fictional concept of “perpetual motion.” Because **frictional resistance** limits velocity, vibratory motion does not continue indefinitely.

The amplitude of vibration of the mass diminishes over time, and the vibrations are therefore **damped**. The same is true of the child’s swing in the park, and of all other forms of vibratory motion. One push is sufficient to cause the swing and child to move to and fro, but because of friction, the amplitude of each successive excursion (displacement) is slightly less than the amplitude of the previous excursion; **damping** has occurred.

Damping occurs because of **resistance** of the air medium and of internal friction within an oscillating system. Damping also varies sinusoidally as a function of time, and because frictional resistance is directly proportional to velocity, damping is represented in Figure 3–1 by the same solid line that represents both velocity and momentum. Thus, *as velocity increases, damping increases because kinetic energy is transformed to thermal energy* (heat). Because of the loss of energy due to damping, free vibrations do not continue indefinitely, and the amplitude of vibration diminishes over time.

The Magnitude of Damping

How long will the vibrations of a damped system continue? The magnitude of the *initial* displacement of the mass is directly proportional to the magnitude of force applied. It should make sense, therefore, that the duration of vibratory motion depends on the amount of damping present relative to the magnitude of force applied. Thus, *the duration of vibrations is directly proportional to the force applied and inversely proportional to damping*.

Different vibratory systems have different amounts of damping. In **low-damped** systems, vibrations last for a relatively long time. A tuning fork is a good example of a low-damped system. Energy is imparted to the fork by striking it, and once struck, it continues to vibrate freely for several seconds, depending, of course, on the force with which it was struck. In contrast, in **high-damped** systems, vibrations are very brief and there is a steep decrement in the amplitude of vibration over time.

Figure 3–2 displays examples of the vibrations of a “lossless” system (panel A) where the vibrations continue to infinity, a theoretical concept in which no damping exists. The figure also displays examples of **low-damped** (panel B) and **high-damped** (panel C) systems. In low-damped systems, the amplitude diminishes gradually over time, whereas in high-damped systems, the vibrations last for only a brief period of time, and the decrement in amplitude is steep.

The Damping Factor

Regardless of whether a system is low-damped or high-damped, *the ratio of the amplitudes (A) of any two consecutive cycles of vibration is a constant*. For example, in the low-damped system in panel

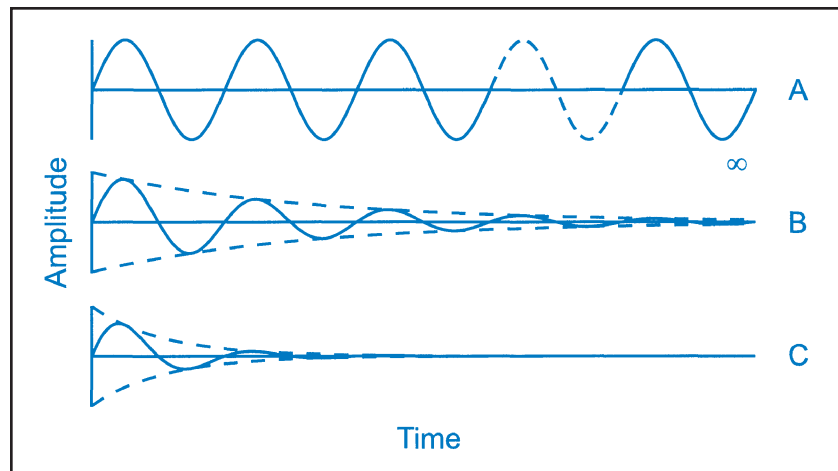


Figure 3–2. Illustrations of **damping** that occurs because of resistance to motion.

B, $A_1/A_2 = A_2/A_3 = A_3/A_4$, and so forth. We will learn in Chapter 4 that *constant ratios* can be expressed logarithmically. For that reason, the *magnitude of damping* (the **damping factor**), symbolized d_f , is given by the natural log ($\log_e$, or $\ln$) of the ratio of the amplitudes of any two consecutive cycles:

$$d_f = \ln \left(\frac{A_1}{A_2} \right).$$

Equation 3-1

Figure 3-3 shows examples of vibrations resulting from five oscillating systems with different damping factors, d_f . The amplitudes of A_1 decrease progressively from .841 unit to .5 unit in the first four examples, and the amplitudes of A_2 decrease from .42 to .031 unit. As the amplitude decreases from A_1 to A_2 in each panel, the damping factor increases from 0.69 (a relatively low-damped system) to 2.77 (a relatively high-damped system).

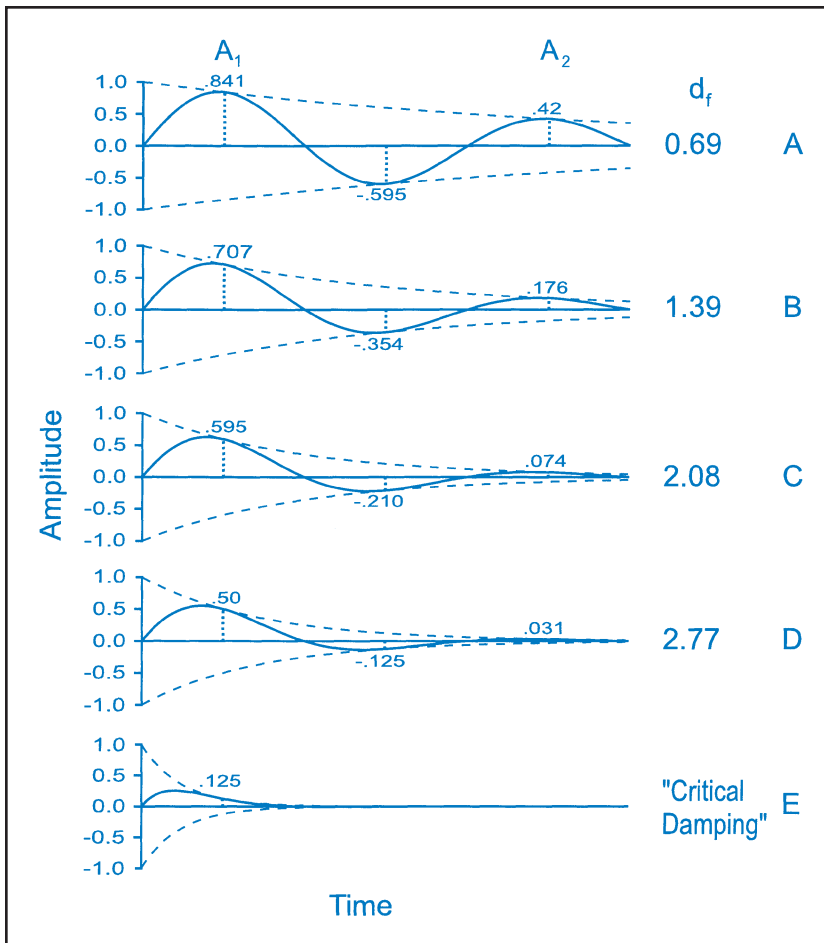


Figure 3-3. Examples of waveforms with different damping (d_f) factors that range from 0.69, to 2.77, to **critical damping**.

After completing a review of logarithms and antilogarithms in Chapter 4, the interested reader might return to this discussion of damping. It should then be apparent that because a constant ratio of 2:1 is preserved between the amplitudes of any two consecutive cycles of the damped system in panel A of Figure 3–3, the damping factor increases linearly in equal (if you carry the calculation to sufficient decimal places) log units of 0.69315 because a constant ratio of 2:1 is preserved and the natural log of 2 is 0.69315.

In a theoretically lossless system, $d_t = 0$, because the ratio of any two amplitudes equals 1, and the log of 1 (regardless of the base) equals 0. The lowest curve in panel E of Figure 3–3 displays **critical damping**. A system that is critically damped is displaced and then slowly returns to equilibrium, but not beyond.

Examples of Damped Systems

There are many familiar examples of damped systems. Shock absorbers on cars, for example, usually are designed to have critical damping (or nearly so) so that the car will not bounce up and down excessively. Of course, as the shock absorbers age, they become underdamped, and the car tends to bounce up and down several times after it hits a bump.

Another example is the level indicator, or needle, on a VU meter of a tape recorder or other electronic device. It typically is designed to be nearly critically damped. If underdamped, the needle would swing back and forth on either side of the “correct value” before finally settling down. On the other hand, if the needle is overdamped, too much time would be required for the needle to reach the correct value and the rapid changes in the level of signals such as speech or music could not be detected. Critical damping represents the boundary between overdamped (non-oscillatory) and underdamped (oscillatory) conditions.

ACOUSTIC IMPEDANCE

The vibrating tuning fork, pendulum, and simple spring-mass system introduced in Chapter 1 engage in simple harmonic motion. Moreover, they engage in **free vibration**, and *each system vibrates freely at its own natural frequency (f_{nat})*.

By **free vibrations** we mean that once energy is imparted to a body that has a low **damping factor**, the body vibrates *freely* for some considerable period of time; no additional outside force is required to urge the continued vibrations.

What is the frequency of vibration? The **natural frequency** (f_{nat}) of a simple harmonic oscillator depends on the **mass** and the **stiffness** of the system (see Equation 3–2).

Equation 3–2

$$f_{nat} = (\sqrt{s} / \sqrt{m}).$$

Specifically, the natural frequency of an oscillating system is *directly proportional to the square root of its stiffness and inversely*

proportional to the square root of its mass. Thus, if the **natural frequency** of a system is 200 Hz and its stiffness is increased by a factor of 4:1, the **natural frequency** *increases* by a factor of 2:1 (square root of 4) to a frequency of 400 Hz. If, instead, the mass of that same system is increased by a factor of 4:1, the natural frequency *decreases* by a factor of 2:1 to a frequency of 100 Hz.

We know, of course, that **free vibrations** do not continue indefinitely. The oscillations are damped because of **resistance**. There is no such thing as perpetual motion because forces exist that oppose, or **impede**, all motion.

Resistance (R)

All motion is opposed by **resistance (R)**, and resistance contributes to, but is not synonymous with, **acoustic impedance**. Acoustic impedance is *the total opposition to motion*. In other words, there are forces in addition to resistance that determine the total opposition to motion.

Because **friction** opposes, or resists, motion, friction serves to limit the velocity of motion. Energy is transformed from **kinetic** to **thermal energy** (heat). Importantly, **resistance**, which is measured in **ohms**, *is independent of frequency*. The independence of resistance and frequency of vibration is important to keep in mind when we discuss the other factors that contribute to impedance. Thus, *a given magnitude of resistance provides a specified opposition to motion of a system, regardless of the frequency with which that system oscillates*. There are, however, additional forces that oppose motion of an elastic system *in a frequency selective way*. Those forces are called **reactance forces** or, simply, **reactances**.

Reactance (X)

When a force is applied to a simple harmonic oscillator, some of the energy associated with the applied force is converted to thermal energy because of **resistance**. A portion of the energy that is applied to a system, however, *is stored in the elastic system as potential energy (PE)*. The component of a system that is responsible for storage of energy is called **reactance (X)**.

Reactance also opposes motion, but in a way that is different from the opposition to motion attributable to **resistance**. In the case of **resistance**, *energy is dissipated*. With **reactance**, *energy is stored in the system*. Thus, the **impedance** of a system has two components:

- an *energy-dissipating component*—**resistance**; and
- an *energy-storage component*—**reactance**.

Resistance, the energy-dissipating component of impedance, *is independent of frequency*. **Reactance**, the energy-storage component of impedance, *is frequency dependent*. The total opposition

to motion, **acoustic impedance**, is the “complex sum” of frequency-independent resistance and frequency-dependent reactance.

Opposition to motion in frequency-selective ways, **reactance**, depends on the **mass** and **compliance** of the system. *Although a system’s mass and compliance each serve to oppose, or impede, motion, they do so in opposite ways.* To explain this, it will be helpful to return to Figure 3–1 and review the phasic relations among displacement, velocity, acceleration, and so forth. The following phasic relations are important for understanding impedance.

1. Opposition to motion due to **resistance**, which is independent of frequency, *is in phase with velocity.*
2. Opposition to motion due to **compliance**, which is frequency dependent, *is in phase with displacement and elasticity, and lags resistance by 90°.*
3. Opposition to motion due to **mass**, which also is frequency dependent, *is in phase with acceleration, and leads resistance by 90°.*
4. It therefore follows that opposition to motion due to **mass** *is 180° out of phase with compliance.*

Reactance has two components: **mass reactance** (X_m) and **compliant reactance** (X_c). The two components of reactance, mass reactance and compliant reactance, *always act in opposition to one another. When one is storing energy, the other is giving it up.*

Mass Reactance (X_m)

The magnitude of **mass reactance** (X_m), measured in **ohms**, is given by:

Equation 3–3

$$X_m = 2\pi f m,$$

where **f** is frequency, **m** is the effective mass, and

2π is a constant.

Recall that **2πf** is **angular velocity** (ω). Therefore, we can say that **mass reactance** is the product of angular velocity and mass. **Mass reactance** (X_m) *is directly proportional to frequency* and is essentially negligible at very low frequencies. For each octave increase (a doubling of frequency, or a frequency ratio of 2:1) in frequency ($f_2 = 2f_1$), mass reactance *increases* by a factor of 2:1. Hence, *mass reactance contributes to the frequency-dependent nature of impedance.*

Compliant Reactance (X_c)

The magnitude of **compliant reactance** (X_c), also measured in **ohms**, is given by:

$$X_c = \frac{1}{2\pi f c},$$

Equation 3-4

where c is compliance.

Compliant reactance also is defined as *the reciprocal of the product of compliance and angular velocity* (because $2\pi f$ equals angular velocity), and as we see in Equation 3-4, **compliant reactance** is *inversely proportional to frequency*. Thus, compliant reactance also contributes to the frequency-dependent nature of impedance.

- Whereas **mass reactance** is negligible at low frequencies, **compliant reactance** is quite large at low frequencies. For each octave increase in frequency, **compliant reactance** *decreases* by a factor of 2:1.
- When $X_m = X_c$, the two opposing reactance forces cancel one another (because they are 180° out of phase), and the only opposition to motion is **resistance**.
- The frequency at which the two opposing forces cancel one another *defines the **natural frequency** (f_{nat}) of the free vibrating system*.

Figure 3-4 shows how the magnitudes of X_m and X_c vary with frequency: $X_m \propto f$ and $X_c \propto 1/f$.

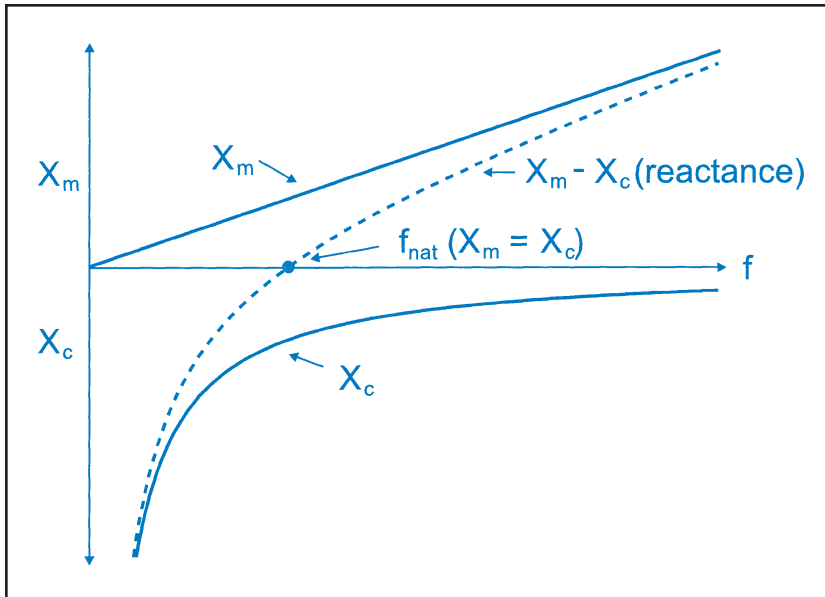


Figure 3-4. Mass reactance (X_m) and compliant reactance (X_c) plotted as a function of frequency. The **net reactance** (X) is shown by the curve labeled $X_m - X_c$. The frequency at which $X_m = X_c$ is called the **natural frequency** (f_{nat}).

- For frequencies below f_{nat} , most of the opposition to motion comes from X_c , and the system is described as *compliance dominant*.
- For frequencies above f_{nat} , most of the opposition to motion comes from X_m , and the system is described as *mass dominant*.

We learned previously that X_m and X_c are 180° out of phase with each other. Therefore, $X_m - X_c$, the net reactance, forms the dashed line in Figure 3–4. The dashed line serves to emphasize the fact that the net reactance ($X_m - X_c$) is largely attributable to X_c at low frequencies and to X_m at high frequencies.

What is the significance of the “dot” on the dashed line? At that point, $X_m = X_c$. When the two reactance components are equal, but opposite, the net reactance must be zero and opposition to motion comes only from R (which is not reflected in Figure 3–4). That defines the **natural frequency**, f_{nat} . In other words, that is the frequency at which *the system vibrates freely with maximum amplitude of vibration*.

Impedance (Z)

Resistance causes energy to be *dissipated*. **Mass reactance** and **compliant reactance** cause energy to be *stored*. All three factors serve to oppose motion and oppose the transfer of energy, but their respective magnitudes cannot be added algebraically to determine the total opposition (**impedance [Z]**) because the three components are not in phase:

- X_m (**mass reactance**) leads R (**resistance**) by 90° ;
- X_c (**compliant reactance**) lags R by 90° ; and
- X_m leads X_c by 180° .

Because the three components of impedance are not in phase, they must be treated as vector-like quantities called **phasor quantities**, or just **phasors**, but the mathematics are the same as for vector solution. Consider the phase diagrams in Figure 3–5. Panel A shows only **mass reactance** (X_m) and **resistance** (R). Mass reactance leads resistance by 90° , and the two quantities are at right angles to one another. The resultant **impedance (Z) vector**, which is shown by the dashed line, is determined by use of Pythagoras’ theorem:

Equation 3–5

$$Z = \sqrt{R^2 + X_m^2},$$

where R is the resistance, and

X_m is the mass reactance.

Panel B shows only **compliant reactance** (X_c) and **resistance** (R). Compliant reactance lags resistance by 90° . Again, the impedance

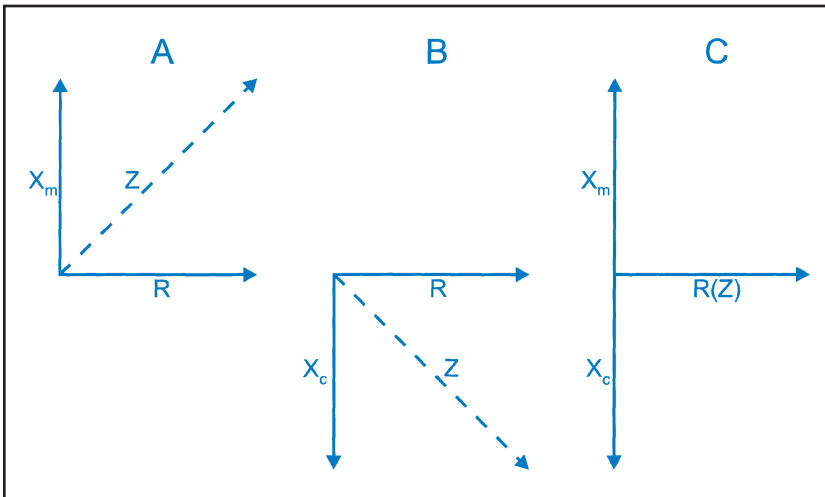


Figure 3–5. Phase diagrams that show **impedance vectors** for **mass reactance** and **resistance** in panel A, **compliant reactance** and **resistance** in panel B, and **mass reactance**, **compliant reactance**, and **resistance** in panel C.

(**Z**) vector is shown by the dashed line, and the resultant impedance is given by:

$$Z = \sqrt{R^2 + X_c^2},$$

Equation 3–6

where **X_c** is the compliant reactance.

Panel C shows all three components of **impedance** (**R**, **X_m**, and **X_c**) with the appropriate phasic relations. The impedance vector now is given by:

$$Z = \sqrt{R^2 + (X_m - X_c)^2}.$$

Equation 3–7

Acoustic impedance (**Z**), expressed in **ohms**, is *the square root of the sum of the square of the **resistance** (**R**) and the square of the **reactances** (**X**)*. In the case of the specific example shown panel C in Figure 3–5, **X_m** = **X_c**, the two reactance components cancel, and **Z** = **R**. As we learned previously, the frequency at which **X_m** = **X_c** defines the **natural frequency** (**f_{nat}**) of the system. In Chapter 7, we will learn that when the **X_m** = **X_c** (the two reactance components equal one another), *the system is “in resonance.”*

What happens when the two reactance components are not equal? Panel A of Figure 3–6 shows a phase diagram in which **X_m** > **X_c**. The contribution of mass reactance to impedance is greater than the contribution of compliant reactance. The opposite relation, **X_c** > **X_m** is shown in panel B of Figure 3–6.¹ You should also note that in those two instances, **impedance** is not determined exclusively by

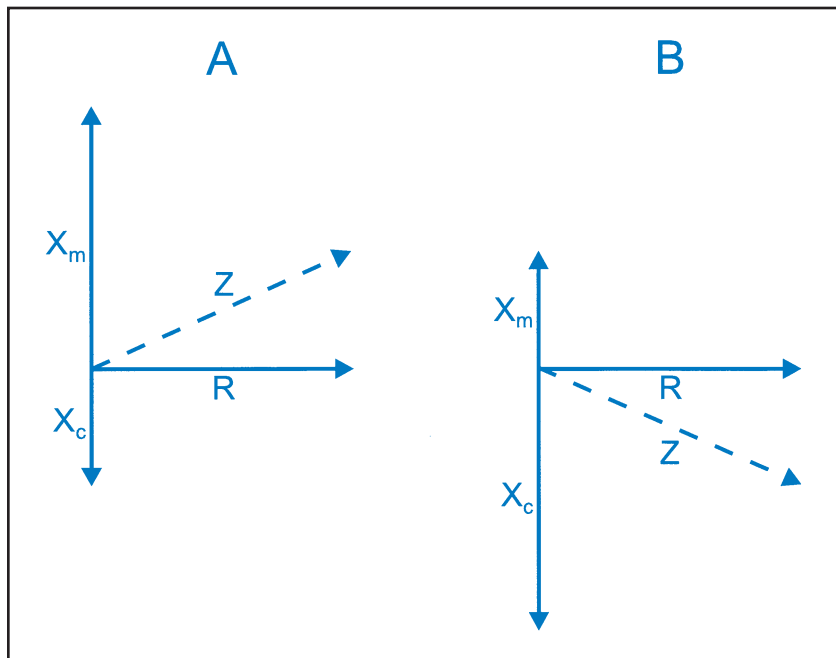


Figure 3–6. Phase diagrams show the **impedance vectors** for all three components of impedance. In panel A, **mass reactance** is dominant, and in panel B, **compliant reactance** is dominant.

resistance. The **reactance** components also contribute to the total opposition to motion, the **impedance**. When $X_m = X_c$, impedance (Z) is minimal.

SUMMARY

Simple harmonic oscillators engage in sinusoidal motion. When a force is applied to such an oscillator, it vibrates freely. The waveform has four dimensions of interest: amplitude, frequency (and period), phase, and wavelength. Finally, free vibrations do not last indefinitely because of opposition to motion from friction. The amplitudes decay over time, and are called damped vibrations. The total opposition to motion or energy transfer is called impedance, and impedance has two components, resistance and reactance (mass reactance and compliant reactance).

Chapter 5 will concern the use of **decibels** to express the magnitude, or specifically the “level,” of a sound wave. To understand the decibel requires knowledge of logs and antilogs. Readers who feel comfortable with those concepts might wish to move directly to Chapter 5. For those who would profit from a review, Chapter 4 is devoted to a detailed description of logs and antilogs.

■ NOTES

1. Impedance is a vector-like quantity called a phasor. Solution of Equation 3-7 reveals the *length* of the impedance vectors in Figure 3-6. The impedance vector in each panel of the figure, however, also must reflect the appropriate phase angle, θ . The phase angle of impedance is given by

$$\theta = \arctan [(X_m - X_c)/R],$$

which is the same as

$$\theta = \tan^{-1} (X/R).$$

Tan⁻¹ is the same as **arctan**, and **tan**⁻¹ likely is the label you will find on your pocket calculator.

■ PRACTICE PROBLEM

1. If $R = 600 \, \Omega$, $X_m = 500 \, \Omega$, and $X_c = 500 \, \Omega$.
 - a. $Z = ?$
 - b. Is the system in resonance?
 - c. If X_c is increased to $600 \, \Omega$, will $Z = R$?



Logarithms and Antilogarithms

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- Scales of Measurement 105
- More on Exponents 111
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- Procedures for Solving Log and Antilog Problems 116
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The magnitude, or amplitude, of a sine wave can be expressed by reference to its sound pressure in newtons per square meter (N/m^2) or pascals (**Pa**). More frequently, however, the magnitude of the sine wave is described by the **decibel (dB)**. It is apparent from reading popular literature such as newspapers and magazine articles and, unfortunately, even an occasional professional paper, that the decibel sometimes is used incorrectly and often is misunderstood. We will consider the decibel (actually the decibels) in detail in Chapter 5, but because decibel notation requires a facility with **logarithms (logs)** and **antilogarithms (antilogs)**, those concepts will be reviewed here.

■ THE CONCEPT OF LOGARITHMS AND ANTILOGARITHMS

Do you know the answers to the two antilog problems below?

$$\text{antilog}_2 2 = ___ \text{ or } \text{antilog}_{10} 2 = ___$$

Some might think that they don't know the correct answer, but they probably do. Almost everyone knows the answers to the next two problems, however, which might *appear* to be easier to understand and solve:

$$2^2 = ___ \text{ (What is 2 squared?) and } 10^2 = ___ \text{ (What is 10 squared?)}$$

The two sets of problems are equal in difficulty because the second set is identical to the first. If you know that $2^2 = 4$ and that $10^2 = 100$, the *concept* of antilogs will be understood easily. They are simply two different ways of asking the same question:

$$\text{antilog}_2 2 = 2^2 = 2 \times 2 = 4,$$

and

$$\text{antilog}_{10} 2 = 10^2 = 10 \times 10 = 100.$$

In other words, asking, " $\text{antilog}_{10} 2 = ___$ " is the same as asking, "what is 10 raised to the second power", or "what is 10 squared?"

The concept of *all* antilog problems is no more difficult than that. All that remains, then, is to learn some simple steps that make it possible to solve problems that are computationally, but not conceptually, more difficult. For example, it will not be immediately obvious that

$$\text{antilog}_{10} 1.7 = 50.12,$$

but we can reframe the question by asking, "what is 10 raised to the 1.7 power?" Thus, the concept underlying that example should be clear, even if the solution to the problem seems more obscure.

We shall see subsequently that the *concept of logarithms* is as easy to grasp as the *concept of antilogs*. Before we describe the computational methods for solving any antilog or log problem, it will be helpful to review “scales of measurement” briefly.

SCALES OF MEASUREMENT

The word “measurement” describes a process whereby *numerals* are assigned to objects according to specific rules. The word “numerals” is emphasized, because, as we shall see, it is important to distinguish between “numerals” and “numbers.”

- A **numeral** is a **symbol** (in effect, a “name,” or “designator”) such as 8, 25, III, A, Red, and so forth.
- A **number** is a **symbol** that bears some fixed relation to other symbols.

For example, the *numbers* 12 and 13 always sum to the *number* 25, whereas the *numerals* 12 and 13 are nothing but names or designators—they *cannot be added or manipulated in other mathematical ways any more than we can add $A + III + \text{orange} = ?$* The importance of distinguishing between numerals and numbers should become more obvious as we review the four scales of measurement: **nominal**, **ordinal**, **interval**, and **ratio** (Stevens, 1951).¹ Before any measurement is undertaken, we must first determine which type of scale is appropriate.

Nominal Scale

With the **nominal** scale, we can sort things into different categories by observing that *one object is the same as, or different from, another*.

- For example, we can sort items according to two categories of male/female, adult/child, fruit/vegetable, fundamental/derived quantities, I's/O's, and so forth, or according to three categories of beef/pork/poultry, or whatever.
- Once the items are sorted into the proper category, all we can do is treat all items in one category as being *different from* the items in another category, or that an item in one category is the *same as* another item in the same category.

In other words, there are strict limits to what we can do mathematically once the items have been sorted into appropriate categories. Within each category we can *count the number of objects that belong*: 6 male/14 female, 13 adults/7 children, and so on. The nominal scale is not very powerful because, basically, *the only arithmetic operation that can be performed legitimately on the entries in the categories is to count them*.

Ordinal Scale

With the **ordinal** scale, we still can say that two or more things are different or the same, but we also can say that one object has *more or less of (or the same as)* some quantity than another object. Consider the familiar letter grades that often are used to describe academic performance. We are accustomed to saying that

$$A > B > C > D > E.$$

An ordinal scale provides an implicit ordering of the letters, but

- *the letters are not numbers and they cannot (or at least should not) be added or subtracted.*
- We know that A exceeds B ($A > B$) and that B exceeds C ($B > C$), but we do not know by *how much* A exceeds B or by *how much* B exceeds C.
- If we do not know by how much A exceeds B or B exceeds C, we obviously cannot know whether A exceeds B *by the same amount* that B exceeds C.

It is a common practice to assign *numerals* to represent those letter grades (e.g., $A = 4$, $B = 3$, $C = 2$, etc.), and from that another ordinal scale is generated:

$$4 > 3 > 2 > 1 > 0.$$

But, do not be deceived. Just because letters have been replaced by numerals does not justify performing many arithmetic operations on those numerals. In spite of that admonition, Offices of Admissions and Records at most colleges and universities do calculate grade point averages (perhaps only athletic departments realize that the GPA is invalid), which requires an arithmetic operation that is not justified with an ordinal scale.

- When the numerals are used on an ordinal scale, you *do know* that $4 > 3$ and that $3 > 2$, *but you do not know the amounts by which 4 exceeds 3 or 3 exceeds 2.*
- Therefore, you *cannot know* that 4 exceeds 3 by the same amount that 3 exceeds 2 any more than you did for the A's, B's, and C's that the numerals replaced. Addition and subtraction of the numerals on the ordinal scale can lead to misleading results, even though academic institutions persist in the practice.

A story adapted from Senders (1958) illustrates what can happen when improper mathematics are performed on the *numerals* of an ordinal scale.

Four men, nicknamed Boomer, Draw, Slice, and Rake, met for the first time at a golf course and wanted to band together to

achieve a competitive match. But, who should be paired with whom to achieve the best team balance? They decided to settle the question by their performance on a driving range. Boomer slammed his drive 280 yards; Draw hooked his ball to the left but netted 265 yards; Slice faded his ball to the right, but it still landed 250 yards away from the tee. Rake took an awkward swing, topped the ball badly, and watched as it dribbled about 50 yards down the range. Thus, by reference to driving distance,

Boomer > Draw > Slice > Rake.

The numeral 4 was assigned to Boomer, 3 to Draw, 2 to Slice, and 1 to Rake because

$$4 > 3 > 2 > 1.$$

To that point everything was legitimate, but how did they use that information to decide who should be teamed with whom? The golfers reasoned that because $4 + 1 = 2 + 3$, Boomer (1) and Rake (4) would compete against Draw (2) and Slice (3). Even though they thought that the two teams would be evenly matched, the outcome was not what they expected. Draw shot a 72 and Slice a 73 for a team total of 145. For the other team, Boomer scored a 68, but poor Rake struggled in with a 117 for a team total of 185. Draw and Slice were the obvious winners. How did this happen when the numerical averages showed the two teams should be equal? Boomer, Draw, and Slice were experienced golfers and each had won several state and regional tournaments. With respect to ability, they were closely bunched. Rake, on the other hand had just appeared at the clubhouse after having tended to the sand traps and had rarely swung a club before. Rake's golfing ability was clearly distant from that of the other three, even though he was only one *numeral* below the numeral assigned to Slice.

We can apply the same logic to a common grading practice. Suppose a professor announces that average test scores of 90 to 100% will be assigned an A, those from 80 to 89% a B, those from 60 to 79% a C, 50 to 59% a D, and those from 0 to 49% an F. The five categories clearly are not of equal size, and using those results in calculation of a grade point average is a clear violation of the rules for an ordinal scale.

Interval Scale

The **interval** scale, which also is called a **linear** scale, represents a large leap in mathematical power. *Ordering still is preserved* from the ordinal scale, but now *the size of the interval is some constant value*.

- The interval is called the **base**. If the numerals 4, 3, 2, 1, and 0 appear on an interval scale, they are *numbers rather than numerals*. Because the **base** (the size of the interval between adjacent numbers) is a known constant, we know that 4 exceeds 3 by the same amount (1) that 3 exceeds 2 (1).
- The interval, or linear, scale consists of *successive units that are generated by adding (or subtracting) some specified **base** (size of interval) to each successive number along the scale.*

Consider the examples for different bases in Table 4–1. In the first series where the **base** = **1**, the base of **1** is added to each successive number to generate a series 1, 2 (1 + **1**), 3 (2 + **1**), 4 (3 + **1**), and so on.

There is no restriction on the starting number in the series. The only difference between the first and second series is that the first series starts with 1 and the second series starts with 0; *the base is the same (1)*. The third series is formed in an identical way, except that a base of **2** is added to each successive number in the series: 0, 2 (0 + **2**), 4 (2 + **2**), 6 (4 + **2**), and so on. In the fourth series, a base of **2** is *subtracted* (added negatively) from each successive number: 10, 8 (10 – **2**), 6 (8 – **2**), 4 (6 – **2**), and so forth. In the final series in the table, a base of **10** is added to form the successive numbers.

- With each of the five examples in the table, the interval between any two successive units equals the interval between any other two successive units, *and the size of the interval is the **base**.*
- With an interval scale, we can say that one object is different from another (as with the nominal scale), that one object has more or less of some quantity than another (as with the ordinal scale), and that one object or event is *a certain number of units more or less than another.*

With respect to the golfers described earlier, driving distance on the range might have been used to assign *numbers* on an interval scale rather than *numerals* on an ordinal scale. In that case it would have been obvious that Boomer and Rake (280 + 50 = 330) would not have been competitive against Draw and Slice (265 + 250 = 515).

Table 4–1. Examples of Interval Scales

Base	Scale	Interval Size
1	1, 2, 3, 4, 5, 6, . . . , n	1
1	0, 1, 2, 3, 4, 5, . . . , n	1
2	0, 2, 4, 6, 8, 10, . . . , n	2
2	10, 8, 6, 4, 2, 0, . . . , n	2
10	10, 20, 30, 40, 50, . . . , n	10

Ratio Scale

The fourth scale is a **ratio** scale, which also is called an **exponential**, or **logarithmic**, scale. In addition to the requirements of an interval scale, the ratio scale requires having *an absolute zero point*. The principal characteristic of the ratio scale is that *one unit on the scale is so many times greater (or less) than another*.

One easy way to understand the necessity of an absolute zero for a ratio scale is to consider the fundamental physical quantity of **time**.

- *Calendar time has no absolute zero.* The year 1992 A.D. is not twice as late in time as the year 996 A.D. because we do not know when time began.
- On the other hand, *periods of time* can be expressed in terms of ratios. Thus, a person who requires only 5 minutes to run a mile can be said to run twice as fast as a person who requires 10 minutes to finish the same distance.

The ratio scale consists of successive units that are generated by *multiplying (or dividing) each successive number along the scale by the base*. Consider the five examples in Table 4–2. In those examples, a base of 1, 2, 1.5, 10, or 0.1 is multiplied by each successive unit along the scale. Thus, in the first series we obtain: 1; 1 ($1 \times \underline{1}$); 1 ($1 \times \underline{1}$); and so forth. In the second series we obtain: 1; 2 ($1 \times \underline{2}$), 4 ($2 \times \underline{2}$), 8 ($4 \times \underline{2}$), 16 ($8 \times \underline{2}$), and so on. In the fourth series we obtain 1, 10 ($1 \times \underline{10}$); 100 ($10 \times \underline{10}$); 1,000 ($100 \times \underline{10}$); 10,000 ($1,000 \times \underline{10}$); and so forth.

In the scales that result with the five different bases in Table 4–2, successive units always differ by some *constant ratio* (1:1, 2:1, 1.5:1, 10:1, and 0.1:1), and *the constant ratio always equals the base*. Thus, when the base = **2**: $64/32 = \underline{2}$; $32/16 = \underline{2}$; $16/8 = \underline{2}$; and so on. Similarly, when the base = **10**: $10,000/1,000 = \underline{10}$; $1,000/100 = \underline{10}$; $100/10 = \underline{10}$; and $10/1 = \underline{10}$. The same must be true for the seemingly more unusual base of **1.5**: $5.0625/3.375 = \underline{1.5}$; $3.375/2.25 = \underline{1.5}$; and $2.25/1.5 = \underline{1.5}$. In each case, *the ratio between two adjacent numbers in the series is constant and equal to the base*.

The **ratio** scale is an **exponential** scale because the scale of numbers can be represented as a **base** raised to successively higher and

Table 4–2. Examples of Ratio Scales

Base	Scale
1	1, 1, 1, 1, 1, , 1
2	1, 2, 4, 8, 16, 32, 64, , n
1.5	1, 1.5, 2.25, 3.375, 5.0625, , n
10	1, 10, 100, 1,000, 10,000, , n
0.1	1, .1, .01, .001, .0001, , n

higher **powers**, or **exponents**. For example, if the base = **2**, we can describe the series by either of two ways:

1, 2, 4, 8, 16, 32, 64
(or)
 $2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6.$

The general expression that defines the exponential series shown above is:

$X^n = \#,$

which means that the base (**X**) is used *n* times in multiplication. Thus, in the example above, where the base is **2**, the same base was used some specified number of times (0, 1, 2, 3, 4, 5, or 6) in multiplication, and a scale was generated that consisted of 1, 2, 4, 8, 16, 32, and 64. If the base of 2 is used 3, 4, or 5 times in multiplication, we obtain the outcome shown in Table 4–3. Thus, 2^3 means that the base of **2** is used 3 (the power) times in multiplication ($2 \times 2 \times 2$) to yield **8**; 2^4 means the base of **2** is used 4 times in multiplication to yield **16** ($2 \times 2 \times 2 \times 2$); and 2^5 means $2 \times 2 \times 2 \times 2 \times 2 = \underline{32}$.

Table 4–4 shows two exponential (ratio) series generated with different base numbers: base = **2** and base = **10**. Two generalizations can be made from the examples in Table 4–4.

Table 4–3. Base = 2 Raised to Powers of 3, 4, and 5

x^n	Operation	Result
2^3	$2 \times 2 \times 2$	8
2^4	$2 \times 2 \times 2 \times 2$	16
2^5	$2 \times 2 \times 2 \times 2 \times 2$	32

Table 4–4. Exponential Series for Base = 2 and Base = 10

Base = 2		Base = 10	
X^n	Result	X^n	Result
2^0	1	10^0	1
2^1	2	10^1	10
2^2	4	10^2	100
2^3	8	10^3	1,000
2^4	16	10^4	10,000
2^5	32	10^5	100,000

- First, *any base raised to the zero power (X^0) equals 1*. Thus, $2^0 = 1$; $10^0 = 1$; $3.1416^0 = 1$; and so forth.
- Second, *any base raised to the first power (X^1) equals the base*. Thus, $2^1 = 2$; $10^1 = 10$; $3.1416^1 = 3.1416$; and so forth.

MORE ON EXPONENTS

Exponents (or powers) specify how many times a base (X) is used in multiplication or division. When the operation is *multiplication*, the expression is:

$$X^n = \#.$$

Equation 4-1

For example,

$$3^4 = 3 \times 3 \times 3 \times 3 = 81.$$

The exponent (4) indicates that the base (3) is used 4 times in multiplication. When the operation is *division*, the expression is written as:

$$X^{-n},$$

Equation 4-2

$$\text{where } X^{-n} = \frac{1}{X^n}.$$

Thus, $3^{-4} = 1/3^4 = 1/(3 \times 3 \times 3 \times 3) = 1/81 = 0.01234567$.

Equations 4-1 and 4-2 will be encountered frequently in decibel notation. There are other defining equations for exponents, although only one of them appears often:

$$X^0 = 1.$$

Equation 4-3

The importance of remembering the rule given by Equation 4-3 cannot be emphasized too strongly. *Any base (X) raised to the zero power equals 1*. Thus,

$$X^0 = 1,$$

$$1^0 = 1,$$

$$2^0 = 1,$$

$$10^0 = 1,$$

and

$$\left[\frac{(2.36)}{(a + bx)} \right]^0 = 1$$

Laws of Exponents

Three laws govern the use of exponents, and they will be of tremendous value when solving decibel problems.

Law 1 of Exponents

LAW 1

$$X^a \times X^b = X^{a+b}.$$

The product of some *common base (X)* raised to two different powers (*a* and *b*) *equals the base (X) raised to the sum of the two powers (a + b)*. For example,

$$2^3 \times 2^2 = 2^{3+2} = 2^5 = 32.$$

It is not especially useful to use Law 1 to solve such a simple problem, because we certainly know that $2^3 = 8$ and $2^2 = 4$, so it is simpler to just multiply 8×4 to get 32. But, consider the next example.

$$2^{1.5} \times 2^{1.5} = \underline{\hspace{1cm}}$$

It might not be immediately obvious how to proceed to determine the value of $2^{1.5}$, and it also might not be intuitively obvious what is meant by *using the base of 2 one and a half times in multiplication*. The solution is easy, however, if we employ **Law 1**.

$$2^{1.5} \times 2^{1.5} = 2^{1.5 + 1.5} = 2^3 = 8.$$

It is important to remember that to use **Law 1** in solution of exponent problems, *the value of the base (X) must be the same in both terms of the product*. Thus, Law 1 does not apply for a problem such as $X^a \times Y^b$ (e.g., $2^3 \times 10^4$).

Law 2 of Exponents

The second law is the most useful of the three laws when solving decibel problems:

LAW 2

$$\frac{X^a}{X^b} = X^{a-b}$$

The ratio of some base (*X*) raised to two different powers equals the base (*X*) raised to *the difference between the two powers (a - b)*. For example,

$$\frac{2^4}{2^2} = 2^{(4-2)} = 2^2 = 4.$$

That problem also is not difficult to solve without use of Law 2 because we know that $2^4 = 16$, and $2^2 = 4$, and $16/4 = 4$. Nevertheless, the example does illustrate the simplicity of the law. The next example would be more difficult to solve without Law 2:

$$\frac{2^{17.5}}{2^{15.5}} = 2^{(17.5 - 15.5)} = 2^2 = 4.$$

As long as the same base appears in both the numerator and the denominator of the ratio, you simply subtract the exponents and then raise the base to the difference.

As mentioned previously, Law 2 will be encountered frequently in decibel problems, and in almost all instances, the base will equal 10. Consider, therefore, four additional examples of Law 2 with base =10.

$$\frac{10^{-9}}{10^{-12}} = 10^{(-9) - (-12)} = 10^3.$$

Example 1

In this example, one negative exponent is subtracted from another negative exponent. Remember that *minus a minus is a plus*. Thus, the difference between -9 and -12 is +3 because it becomes a -9 and a +12 (minus a minus is a plus). The solution could have been carried one step farther to show that $10^3 = 1,000$, but the additional computational step generally is not necessary. It is perfectly acceptable to show the answer simply as 10^3 .

$$\frac{10^{-12}}{10^{-9}} = 10^{(-12) - (-9)} = 10^{-3}.$$

Example 2

Minus a minus is a plus, but in this case that translates to a -12 and a +9, which is a -3.

$$\frac{10^{-9}}{10^{-16}} = 10^{(-9) - (-16)} = 10^7.$$

Example 3

Example 3 is identical in form to Example 1.

$$\frac{10^{-9}}{10^{-9}} = 10^{(-9) - (-9)} = 10^0 = 1.$$

Example 4

The answer of 1 is consistent with Equation 4-3, which defined $X^0 = 1$ (any base raised to the zero power equals 1). You should see from Example 4 that this makes sense, because any number (e.g., 10^{-9}) divided by itself (10^{-9}) equals 1.

Law 3 of Exponents

The third law of exponents will not be used extensively in decibel problems, but it might be encountered occasionally.

$$(X^a)^b = X^{ab}.$$

LAW 3

For example,

$$(2^2)^3 = 2^{(2 \times 3)} = 2^6 = 64$$

■ ANTILOGS AND LOGS

Antilogs

Table 4–4 contained two exponential series generated with different base numbers, **2** and **10**. The two series are shown again in Table 4–5, but Table 4–5 is organized differently. We can ask either of two questions and then perform either of two arithmetic operations, depending on how the entries in Table 4–5 are arranged. As the table is presently organized, we ask:

$X^n = \underline{\hspace{1cm}}$

For those examples, therefore, we ask, “what is the value of 2 raised to the 0 power, 2 raised to the 1st power, 2 raised to the 2nd power, and so on?” Stated more generally, we ask, “*what is the value of the base (**X**) raised to the **nth** power?*” That question also is called *determining the **antilog** of the number*. Thus, the question can be written in either of two equivalent forms:

$X^n = \underline{\hspace{1cm}}$

is *exactly* the same as

$\text{antilog}_x n = \underline{\hspace{1cm}}$

In other words,

$X^n = \text{antilog}_x n.$

Stated differently, “what is the antilog to the base **X** of **n**?” is the same as asking, “what is the value of **X** raised to the **nth** power?” In the examples below, the same questions are posed both ways.

Example 1

$2^3 = 8$ and $\text{antilog}_2 3 = 8.$

Table 4–5. Two Exponential Series of Numbers for Base = 2 and Base = 10

Base = 2		Base = 10	
X^n	Answer	X^n	Answer
$2^0 = \underline{\hspace{1cm}}$	1	$10^0 = \underline{\hspace{1cm}}$	1
$2^1 = \underline{\hspace{1cm}}$	2	$10^1 = \underline{\hspace{1cm}}$	10
$2^2 = \underline{\hspace{1cm}}$	4	$10^2 = \underline{\hspace{1cm}}$	100
$2^3 = \underline{\hspace{1cm}}$	8	$10^3 = \underline{\hspace{1cm}}$	1,000
$2^4 = \underline{\hspace{1cm}}$	16	$10^4 = \underline{\hspace{1cm}}$	10,000

$10^3 = 1,000$ and $\text{antilog}_{10} 3 = 1,000$.

Example 2

Thus, with a base of 2, we ask, “2 raised to the third power equals what number?” (Answer = 8), or we ask, “what is the antilog to the base 2 of 3?” (Answer = 8).

Logs

Table 4–6 contains the same two series of numbers that appeared in Table 4–5, but they are arranged differently to ask a different question. In this case, *the exponent is not given*, and we ask, “to what power must some specified base be raised to equal some particular number?” In other words, we ask:

$X^? = \#$.

Consider the following two examples.

To what power must the base of 2 be raised to equal 8?
(Answer = 3)

Example 1

To what power must the base of 10 be raised to equal 10,000?
(Answer = 4)

Example 2

When you solve problems of that kind, you are *determining the log of a number*, and the following two expressions are identical.

$X^? = \#$
is exactly the same as
 $\log_x \# = \underline{\hspace{1cm}}$

Thus, the question, “to what power must the base X be raised to equal the number in question?” is equivalent to asking, “what is

Table 4–6. Two Exponential Series for Base = 2 and Base = 10

Base = 2		Base = 10	
X^n	Answer	X^n	Answer
$2^0 = 1$	0	$10^0 = 1$	0
$2^1 = 2$	1	$10^1 = 10$	1
$2^2 = 4$	2	$10^2 = 100$	2
$2^3 = 8$	3	$10^3 = 1,000$	3
$2^4 = 16$	4	$10^4 = 10,000$	4

the log to the base **X** of the number?" In the following examples, the same questions are posed both ways.

Example 1 $2^4 = 16$ (Answer = 4) and $\log_2 16 = \underline{\hspace{1cm}}$ (Answer = 4.)

Example 2 $10^3 = 1,000$ (Answer = 3) and $\log_{10} 1,000 = \underline{\hspace{1cm}}$ (Answer = 3.)

Bases for Logs and Antilogs

When solving **log** or **antilog** problems, *the base must be specified*. Thus, $\log 1,000 = \underline{\hspace{1cm}}$ is a meaningless question that cannot be solved. How can we possibly know whether you are asking for the power to which 2 must be raised to equal 1,000, the power to which 10 must be raised to equal 1,000, or whatever? *Any number can serve as the base*, but in practice there are three common bases: "**2**," "**10**" (which is called the **common**, or **Briggsian**, **log**), and "**e**," or **I_n** (2.718, which is called the **natural**, or **Naperian**, **log**). **Base 10** is used with *decibels*, **base 2** is used with *frequency*, and we have already encountered **base e** (**I_n**) in Equation 3–1 of Chapter 3 for the *damping factor*.

Summary

When you "calculate the log of a number" (e.g., $\log_{10} 100 = 2$), *you are solving for an exponent. An exponent is a log, and a log is an exponent*. That is why the title to this chapter is "Logarithms and Antilogarithms" rather than "Exponents, Logarithms, and Antilogarithms."

You should be able to solve the simple examples of log and antilog problems by inspection. Thus, you undoubtedly know that $\log_{10} 1,000 = 3$ and that $\log_{10} 100 = 2$. You might, for example, have simply counted the number of zeros and accepted that as your answer. That is perfectly acceptable for those problems. It might be less obvious, though, that the $\log_{10} 121 = 2.08279$.

Conceptually, $\log_{10} 121 = \underline{\hspace{1cm}}$ is no different from the problem, $\log_{10} 100 = \underline{\hspace{1cm}}$. You are asked, "to what power must the base of 10 be raised to equal 121 in one case and to equal 100 in the other?" The concept is the same, but counting zeros will not always provide you with the correct solution. All that remains, then, is to identify a series of computational steps that will enable determination of the log of any (positive) number.

■ PROCEDURES FOR SOLVING LOG AND ANTILOG PROBLEMS

Components of a Logarithm

A logarithm consists of one or more **integers** (whole numbers) and an endless string of **decimal values**. Thus,

$$\log_{10} 100 = 2.00000000...$$

and

$$\log_{10} 121 = 2.08278537...$$

When we compute the log of 100, the solution is an integer, or whole number (2) and an endless string (usually limited to four, and sometimes only to one or two) of decimals composed entirely of zeros. When we compute the log of 121, the integer also is 2, but the first four values in the string of decimals are 0.0828. Recall that in the first example we learned that 10 (the base) must be raised to a power of 2.0000 to equal 100, and in the second example that 10 must be raised to a power of 2.0828 to equal 121.

Because a log is an exponent, we can rephrase both problems to conform to the language used previously with exponents. The base of 10 is used 2.0000 times in multiplication to equal 100, and is used 2.0828 times in multiplication to equal 121.

- The **integer** is the **characteristic** of a log.
- The **decimal values** are the **mantissa** of a log.
- Thus, in the two examples above, the **characteristic** (integer) is 2 in both cases, but the **mantissa** (decimal values) is 0.0000 for the log of 100 and 0.0828 for the log of 121.

To find the log of some (positive) number, the pocket calculator is an indispensable aid. The steps to be followed might vary with different brands, but the manual for the calculator should provide a useful guide. Most generally, to find the log of some number such as, for example, 121, you enter the number 121 and depress the key labeled [LOG]. The display, depending on how many decimals your calculator will show, should read "2.08278537," where the integer 2 is the **characteristic** and 0.08278537 is the **mantissa**.

It might be useful to solve a few log problems by using a table of logarithms instead of a calculator so that you have an opportunity to become more familiar with the concept of logarithms. Therefore, we shall specify the few easy steps by which the log, both the characteristic and the mantissa, can be found for any positive number.

Scientific Notation

Before attempting to calculate the log of a number, it is convenient to transform the number from conventional notation to **scientific notation**. In the course of discussing frequency and period in Chapter 2, we encountered examples of very large and very small numbers such as 890,000,000 **Hz** and its reciprocal, 0.001123596 **μs** or 0.000000001123596 **s**. In Chapter 5, when the concept of sound intensity is introduced, we will encounter numbers such as 0.000000000001 **watt/m²**.

It should be obvious that there are at least two problems with conventional notation. The expression is cumbersome, and it would

be quite easy to drop or add a zero or so inadvertently. Expressing the number in scientific notation should minimize those two problems. We will learn that for the example of sound intensity cited above, the number would be written as 1×10^{-12} in scientific notation rather than 0.000000000001 in conventional notation.

Let us begin more simply. We know that 100 can be written as 10^2 because $100 = 10 \times 10$. Similarly, 1,000 can be written as 10^3 because $1,000 = 10 \times 10 \times 10$. It therefore should make sense that if 1000 can be written as 10^3 , and because 2,000 is twice as large as 1,000, we can write 2,000 as $2 \times 10 \times 10 \times 10$ or, more simply, as 2×10^3 .

- When a number such as 2,000 is expressed as 2×10^3 , we are using **scientific notation**.
- With scientific notation, a number is written *as the product* of some “simple number” (e.g., 2), the **coefficient**, and
- the **base of 10 raised to some power**.
- Hence $2,000 = 2 \times 10^3$, $5,000 = 5 \times 10^3$, and $6,000 = 6 \times 10^3$. What about the number 5,500, which is halfway between 5,000 and 6,000? We will see that it is written as 5.5×10^3 .

The “simple number,” the first term in the product, is called the **coefficient**, and in scientific notation, the coefficient always is multiplied by the base 10 raised to some power.

- The **coefficient** is any number greater than or equal to ($\geq$) 1.00, but less than 10 (e.g., 9.99).
- You can add as many additional decimal values as might be required to achieve the desired precision, but two decimal places are sufficient for most applications.

Conversion from Conventional Notation to Scientific Notation

Table 4–7 contains several examples of numbers written conventionally and in equivalent scientific notation. For the first five examples

Table 4–7. Comparison of Conventional and Scientific Notation

Conventional Notation	Scientific Notation
10	1.00×10^1
100	1.00×10^2
1,000	1.00×10^3
121	1.21×10^2
800	8.00×10^2
0.1	1.00×10^{-1}
0.0121	1.21×10^{-2}

in the table, the conventional numbers are converted to scientific notation by *counting the number of places that the decimal point must be moved to the left (successive division by 10) to yield an integer between 1 and 9*. That specifies the *exponent* in scientific notation.

Thus:

- To transform 10, the decimal point is moved one place to the left (divided by 10 once) to arrive at an integer between 1 and 9, and the exponent in scientific notation is **1**;
- to transform 100, the decimal point is moved two places to the left (divided by 10 twice) to obtain the integer, and the exponent therefore is **2**; and
- to transform 121, the decimal point is moved two places to the left, and the coefficient is 1.21—thus the exponent is **2** (because the decimal point was moved two places to the left).

In the last two examples in Table 4–7,

- *the decimal point must be moved to the right (successive multiplication by ten) to yield an integer between 1 and 9.*
- That provides the value of the exponent, *but in these cases the exponent is negative.*
- For example, with 0.0121, the decimal point must be moved two places to the right to yield an integer between 1 and 9 (coefficient = 1.21), and therefore the exponent is **–2**.

A few more examples are shown below.

$$.002 = 2.00 \times 10^{-3}$$

Example 1

$$.0002 = 2.00 \times 10^{-4}$$

Example 2

$$2002 = 2.00 \times 10^3$$

Example 3

More precisely, 2002 equals 2.002×10^3 , but for convenience we are limiting ourselves to two decimal values for the coefficient.

$$222 = 2.22 \times 10^2$$

Example 4

Solution for Logs

When a number is expressed in conventional notation, three steps are required to determine the log of that (positive) number. In stating the steps, we will use the example, $\log_{10} 121 = \underline{\hspace{1cm}}$.

Step One: Express the number in scientific notation.

$$121 = 1.21 \times 10^2$$

This step is omitted if the number already is expressed in scientific notation form in the problem ($\log_{10} 1.21 \times 10^2$).

*Step Two: The exponent in scientific notation is the **characteristic** of the log.*

$$\text{Thus, } \log_{10} 121 = \log_{10} 1.21 \times 10^2 = 2.????$$

Step Three: If the base is 10, as it will be most of the time, use the Common Log Table in Table 4–8 to find the **mantissa**. Use the three-numbered *multiplier* (the **coefficient**) in the scientific notation expression to find the appropriate row and column in the log table; for this purpose, ignore the decimal point and view the multiplier simply as three numbers.

- The first two numbers specify the correct *row* in the table; the third number specifies the correct *column*.
- The *cell entry* corresponding to that row and column is the **mantissa** of the logarithm.
- For our example of 1.21, we locate row 12 and column 1, and the cell entry (the **mantissa** of the logarithm) is .0828. The answer is:

$$\log_{10} 121 = \log_{10} 1.21 \times 10^2 = 2.0828.$$

By way of review, when we say that the log to the base 10 of 121 is 2.0828, that means that the base of 10 must be raised to the 2.0828 power (use the base 2.0828 times in multiplication) to equal 121. We should be able to perform a quick mental check to see if that answer is reasonable. We know without use of a log table that the log of 100 is 2 and that the log of 1,000 is 3. Therefore, the log of 121 (which is larger than 100 and smaller than 1,000) must be a number between 2 and 3, and 2.0828 falls within that range.

That kind of checking (i.e., “bracketing” your answer) to determine if an answer is at least reasonable is a good idea. It won’t necessarily reveal whether the answer that you obtained is precisely correct, but it will tell you if it is approximately correct or at least if it is wildly wrong. For example, suppose you calculated (incorrectly) that the $\log_{10} 163 = 3.2122$. You then realize that the log of 100 is 2 and that the log of 1,000 is 3. Because the number 163 lies between 100 and 1,000, the $\log_{10} 163$ must be 2. ___, not 3. ___.

Following are additional examples that you might wish to solve before reading the answer.

$$\log_{10} 648 = \log_{10} 6.48 \times 10^2 = 2.8116.$$

$$\log_{10} 707 = \log_{10} 7.07 \times 10^2 = 2.8494.$$

$$\log_{10} 1,000 = \log_{10} 1.00 \times 10^3 = 3.0000.$$

$$\log_{10} 1 = \log_{10} 1.00 \times 10^0 = 0.0000.$$

Table 4–8. Common Log Table

N	0	1	2	3	4	5	6	7	8	9
10	0.0000	0.0043	0.0086	0.0128	0.0170	0.0212	0.0253	0.0294	0.0334	0.0374
11	0.0414	0.0453	0.0492	0.0531	0.0569	0.0607	0.0645	0.0682	0.0719	0.0755
12	0.0792	0.0828	0.0864	0.0899	0.0934	0.0969	0.1004	0.1038	0.1072	0.1106
13	0.1139	0.1173	0.1206	0.1239	0.1271	0.1303	0.1335	0.1367	0.1399	0.1430
14	0.1461	0.1492	0.1523	0.1553	0.1584	0.1614	0.1644	0.1673	0.1703	0.1732
15	0.1761	0.1790	0.1818	0.1847	0.1875	0.1903	0.1931	0.1959	0.1987	0.2014
16	0.2041	0.2068	0.2095	0.2122	0.2148	0.2175	0.2201	0.2227	0.2253	0.2279
17	0.2304	0.2330	0.2355	0.2380	0.2405	0.2430	0.2455	0.2480	0.2504	0.2529
18	0.2553	0.2577	0.2601	0.2625	0.2648	0.2672	0.2695	0.2718	0.2742	0.2765
19	0.2788	0.2810	0.2833	0.2856	0.2878	0.2900	0.2923	0.2945	0.2967	0.2989
20	0.3010	0.3032	0.3054	0.3075	0.3096	0.3118	0.3139	0.3160	0.3181	0.3201
21	0.3222	0.3243	0.3263	0.3284	0.3304	0.3324	0.3345	0.3365	0.3385	0.3404
22	0.3424	0.3444	0.3464	0.3483	0.3502	0.3522	0.3541	0.3560	0.3579	0.3598
23	0.3617	0.3636	0.3655	0.3674	0.3692	0.3711	0.3729	0.3747	0.3766	0.3784
24	0.3802	0.3820	0.3838	0.3856	0.3874	0.3892	0.3909	0.3927	0.3945	0.3962
25	0.3979	0.3997	0.4014	0.4031	0.4048	0.4065	0.4082	0.4099	0.4116	0.4133
26	0.4150	0.4166	0.4183	0.4200	0.4216	0.4232	0.4249	0.4265	0.4281	0.4298
27	0.4314	0.4330	0.4346	0.4362	0.4378	0.4393	0.4409	0.4425	0.4440	0.4456
28	0.4472	0.4487	0.4502	0.4518	0.4533	0.4548	0.4564	0.4579	0.4594	0.4609
29	0.4624	0.4639	0.4654	0.4669	0.4683	0.4698	0.4713	0.4728	0.4742	0.4757
30	0.4771	0.4786	0.4800	0.4814	0.4829	0.4843	0.4857	0.4871	0.4886	0.4900
31	0.4914	0.4928	0.4942	0.4955	0.4969	0.4983	0.4997	0.5011	0.5024	0.5038
32	0.5051	0.5065	0.5079	0.5092	0.5105	0.5119	0.5132	0.5145	0.5159	0.5172
33	0.5185	0.5198	0.5211	0.5224	0.5237	0.5250	0.5263	0.5276	0.5289	0.5302
34	0.5315	0.5328	0.5340	0.5353	0.5366	0.5378	0.5391	0.5403	0.5416	0.5428
35	0.5441	0.5453	0.5465	0.5478	0.5490	0.5502	0.5514	0.5527	0.5539	0.5551
36	0.5563	0.5575	0.5587	0.5599	0.5611	0.5623	0.5635	0.5647	0.5658	0.5670
37	0.5682	0.5694	0.5705	0.5717	0.5729	0.5740	0.5752	0.5763	0.5775	0.5786
38	0.5798	0.5809	0.5821	0.5832	0.5843	0.5855	0.5866	0.5877	0.5888	0.5899
39	0.5911	0.5922	0.5933	0.5944	0.5955	0.5966	0.5977	0.5988	0.5999	0.6010
40	0.6021	0.6031	0.6042	0.6053	0.6064	0.6075	0.6085	0.6096	0.6107	0.6117
41	0.6128	0.6138	0.6149	0.6160	0.6170	0.6180	0.6191	0.6201	0.6212	0.6222
42	0.6232	0.6243	0.6253	0.6263	0.6274	0.6284	0.6294	0.6304	0.6314	0.6325
43	0.6335	0.6345	0.6355	0.6365	0.6375	0.6385	0.6395	0.6405	0.6415	0.6425
44	0.6435	0.6444	0.6454	0.6464	0.6474	0.6484	0.6493	0.6503	0.6513	0.6522
45	0.6532	0.6542	0.6551	0.6561	0.6571	0.6580	0.6590	0.6599	0.6609	0.6618
46	0.6628	0.6637	0.6646	0.6656	0.6665	0.6675	0.6684	0.6693	0.6702	0.6712
47	0.6721	0.6730	0.6739	0.6749	0.6758	0.6767	0.6776	0.6785	0.6794	0.6803
48	0.6812	0.6821	0.6830	0.6839	0.6848	0.6857	0.6866	0.6875	0.6884	0.6893
49	0.6902	0.6911	0.6920	0.6928	0.6937	0.6946	0.6955	0.6964	0.6972	0.6981
50	0.6990	0.6998	0.7007	0.7016	0.7024	0.7033	0.7042	0.7050	0.7059	0.7067
51	0.7076	0.7084	0.7093	0.7101	0.7110	0.7118	0.7126	0.7135	0.7143	0.7152
52	0.7160	0.7168	0.7177	0.7185	0.7193	0.7202	0.7210	0.7218	0.7226	0.7235
53	0.7243	0.7251	0.7259	0.7267	0.7275	0.7284	0.7292	0.7300	0.7308	0.7316
54	0.7324	0.7332	0.7340	0.7348	0.7356	0.7364	0.7372	0.7380	0.7388	0.7396
55	0.7404	0.7412	0.7419	0.7427	0.7435	0.7443	0.7451	0.7459	0.7466	0.7474
56	0.7482	0.7490	0.7497	0.7505	0.7513	0.7520	0.7528	0.7536	0.7543	0.7551
57	0.7559	0.7566	0.7574	0.7582	0.7589	0.7597	0.7604	0.7612	0.7619	0.7627
58	0.7634	0.7642	0.7649	0.7657	0.7664	0.7672	0.7679	0.7686	0.7694	0.7701
59	0.7709	0.7716	0.7723	0.7731	0.7738	0.7745	0.7752	0.7760	0.7767	0.7774

(continues)

Table 4–8. (continued)

N	0	1	2	3	4	5	6	7	8	9
60	0.7782	0.7789	0.7796	0.7803	0.7810	0.7818	0.7825	0.7832	0.7839	0.7846
61	0.7853	0.7860	0.7868	0.7875	0.7882	0.7889	0.7896	0.7903	0.7910	0.7917
62	0.7924	0.7931	0.7938	0.7945	0.7952	0.7959	0.7966	0.7973	0.7980	0.7987
63	0.7993	0.8000	0.8007	0.8014	0.8021	0.8028	0.8035	0.8041	0.8048	0.8055
64	0.8062	0.8069	0.8075	0.8082	0.8089	0.8096	0.8102	0.8109	0.8116	0.8122
65	0.8129	0.8136	0.8142	0.8149	0.8156	0.8162	0.8169	0.8176	0.8182	0.8189
66	0.8195	0.8202	0.8209	0.8215	0.8222	0.8228	0.8235	0.8241	0.8248	0.8254
67	0.8261	0.8267	0.8274	0.8280	0.8287	0.8293	0.8299	0.8306	0.8312	0.8319
68	0.8325	0.8331	0.8338	0.8344	0.8351	0.8357	0.8363	0.8370	0.8376	0.8382
69	0.8388	0.8395	0.8401	0.8407	0.8414	0.8420	0.8426	0.8432	0.8439	0.8445
70	0.8451	0.8457	0.8463	0.8470	0.8476	0.8482	0.8488	0.8494	0.8500	0.8506
71	0.8513	0.8519	0.8525	0.8531	0.8537	0.8543	0.8549	0.8555	0.8561	0.8567
72	0.8573	0.8579	0.8585	0.8591	0.8597	0.8603	0.8609	0.8615	0.8621	0.8627
73	0.8633	0.8639	0.8645	0.8651	0.8657	0.8663	0.8669	0.8675	0.8681	0.8686
74	0.8692	0.8698	0.8704	0.8710	0.8716	0.8722	0.8727	0.8733	0.8739	0.8745
75	0.8751	0.8756	0.8762	0.8768	0.8774	0.8779	0.8785	0.8791	0.8797	0.8802
76	0.8808	0.8814	0.8820	0.8825	0.8831	0.8837	0.8842	0.8848	0.8854	0.8859
77	0.8865	0.8871	0.8876	0.8882	0.8887	0.8893	0.8899	0.8904	0.8910	0.8915
78	0.8921	0.8927	0.8932	0.8938	0.8943	0.8949	0.8954	0.8960	0.8965	0.8971
79	0.8976	0.8982	0.8987	0.8993	0.8998	0.9004	0.9009	0.9015	0.9020	0.9025
80	0.9031	0.9036	0.9042	0.9047	0.9053	0.9058	0.9063	0.9069	0.9074	0.9079
81	0.9085	0.9090	0.9096	0.9101	0.9106	0.9112	0.9117	0.9122	0.9128	0.9133
82	0.9138	0.9143	0.9149	0.9154	0.9159	0.9165	0.9170	0.9175	0.9180	0.9186
83	0.9191	0.9196	0.9201	0.9206	0.9212	0.9217	0.9222	0.9227	0.9232	0.9238
84	0.9243	0.9248	0.9253	0.9258	0.9263	0.9269	0.9274	0.9279	0.9284	0.9289
85	0.9294	0.9299	0.9304	0.9309	0.9315	0.9320	0.9325	0.9330	0.9335	0.9340
86	0.9345	0.9350	0.9355	0.9360	0.9365	0.9370	0.9375	0.9380	0.9385	0.9390
87	0.9395	0.9400	0.9405	0.9410	0.9415	0.9420	0.9425	0.9430	0.9435	0.9440
88	0.9445	0.9450	0.9455	0.9460	0.9465	0.9469	0.9474	0.9479	0.9484	0.9489
89	0.9494	0.9499	0.9504	0.9509	0.9513	0.9518	0.9523	0.9528	0.9533	0.9538
90	0.9542	0.9547	0.9552	0.9557	0.9562	0.9566	0.9571	0.9576	0.9581	0.9586
91	0.9590	0.9595	0.9600	0.9605	0.9609	0.9614	0.9619	0.9624	0.9628	0.9633
92	0.9638	0.9643	0.9647	0.9652	0.9657	0.9661	0.9666	0.9671	0.9675	0.9680
93	0.9685	0.9689	0.9694	0.9699	0.9703	0.9708	0.9713	0.9717	0.9722	0.9727
94	0.9731	0.9736	0.9741	0.9745	0.9750	0.9754	0.9759	0.9763	0.9768	0.9773
95	0.9777	0.9782	0.9786	0.9791	0.9795	0.9800	0.9805	0.9809	0.9814	0.9818
96	0.9823	0.9827	0.9832	0.9836	0.9841	0.9845	0.9850	0.9854	0.9859	0.9863
97	0.9868	0.9872	0.9877	0.9881	0.9886	0.9890	0.9894	0.9899	0.9903	0.9908
98	0.9912	0.9917	0.9921	0.9926	0.9930	0.9934	0.9939	0.9943	0.9948	0.9952
99	0.9956	0.9961	0.9965	0.9969	0.9974	0.9978	0.9983	0.9987	0.9991	0.9996

Note the last example above. If $X^\circ = 1$ (Equation 4–3), and if an exponent is a log (which it is), then it must be the case that the log, to any base, of 1 = 0.

$$\log_{10} 2 = \log_{10} 2.00 \times 10^0 = 0.3010.$$

$$\log_{10} 0.0002 = \log_{10} 2.00 \times 10^{-4} = \bar{4}.3010.$$

Solution of the first five of the six problems above should be accomplished easily if the three steps are followed:

1. Convert the number to scientific notation;
2. the exponent in scientific notation is the **characteristic** of the log; and
3. the multiplier in scientific notation is used to find the **mantissa** in the log table.

The sixth example, $\log_{10} 0.0002$, deserves additional explanation. Notice, that in the answer to the problem, a negative sign is placed *above the 4* rather than in front of it. That is because *the characteristic can be either positive or negative*, but *the mantissa can only be positive*. Thus, $\bar{4}.3010$ does not equal -4.3010 . If we were to say “ -4.3010 ,” we imply that the number is 4.3010 units *less than zero*. With “ $\bar{4}.3010$,” however, we have a negative characteristic of 4, but a positive mantissa of .3010. With a very loose analogy, that is like having roughly 30 cents in your pocket (an asset—a positive value) and owing a friend \$4.00 (a liability—a negative value). In that case, your net is a deficit of \$3.70.

The log problem should be approached in the same conceptual manner. Therefore, $\bar{4}.3010 = -4.0000 + 0.3010 = -3.6990$. In other words, $\bar{4}.3010$ is really 3.6990 units below zero, not 4.3010 units below zero. Fortunately, when you solve the problem $\log_{10} 0.0002$ with a calculator that has a log function, the answer automatically will display -3.6990 .

Logs with Bases Other Than 10

There are a few occasions in which you need to solve a log problem that has a base other than 10, and it might be difficult to locate a published log table to help. There is an equation, however, that enables you to solve such problems by using a base of 10. Suppose you want to solve the problem, $\log_2 8$. You might see immediately that the answer is 3, which means that the base of 2 must be raised to the 3rd power ($2 \times 2 \times 2$) equals 8. Now try to get the answer in a different way by use of Equation 4–4.

$$\log_Y X = \frac{\log_{10} X}{\log_{10} Y}.$$

Equation 4–4

Substitute the values from the problem into Equation 4–4.

To calculate $\log_2 8$, first find $\log_{10} 8$ (0.9031) and divide that answer by $\log_{10} 2$ (0.3010). The answer is 3.0 (if you ignore the third and fourth decimal places), which is the answer we knew by inspection.

Solution for Antilogs

If your calculator has a log function, it almost certainly has an anti-log function. The key normally is not labeled “antilog,” but rather

10^x . To determine the antilog to the base 10 of 2.0828, enter 2.0828 on your keypad, press the antilog function key, and the number 121 is displayed.

Solution of antilog problems without a calculator is just as easy as solution of log problems. Simply use the same three steps described for solving log problems, but in reverse order. Consider the following example.

$$\text{antilog}_{10} 2.0828 = ___,$$

which means

$$10^{2.0828} = ___.$$

Regardless of the way in which the problem is posed, we seek the answer to the question “what is the base of 10 raised to the 2.0828 power?”, which means “what is the value of 10 used 2.0828 times in multiplication?”

Step One: The exponent, 2.0828, is a logarithm and consists of a characteristic (2.) and a mantissa (0.0828). Because the **mantissa** is 0.0828, locate the cell entry of that value in the log table. The row (12) and column (1) that correspond to the cell entry of 0.0828 yields the *multiplier*, or *coefficient*, in scientific notation. Because the coefficient always is an integer *between 1 and 9* followed by some number of decimal values, row 12 and column 1 tells you that the multiplier is 1.21; that is the only place within those three digits that the decimal can be located to produce an integer between 1 and 9.

Step Two: Because an exponent in scientific notation is the **characteristic** of a log, then the characteristic of a log must be the exponent in scientific notation. Thus, the characteristic of 2 tells us to raise the base 10 to the 2nd power (10^2). The answer, in scientific notation, is 1.21×10^2 .

Step Three: If you wish to complete the process by expressing the answer in conventional notation, multiply (or divide) the answer obtained in Step Two by 10 by moving the decimal point to the right (for positive exponents) the number of places specified by the exponent, or to the left for negative exponents. For most of the problems that will be encountered in this book, the third step will not be executed, and the answer remains in scientific notation.

$$1.21 \times 10^2 = 121.$$

Because the exponent is 2, the decimal point is moved two places to the right.

To provide an example for the three steps in determining both logs and antilogs, the number 121 was used. From those solutions we can see that:

$$\begin{aligned}\log_{10} 121 &= 2.0828 \\ \text{and} \\ \text{antilog}_{10} 2.0828 &= 121.\end{aligned}$$

Laws of Logarithms

The four laws of logarithms help make solution of many decibel problems easy. Each law applies to logarithms with any base, but only a base of 10 is used in the examples.

Law 1 of Logarithms

$$\log ab = \log a + \log b.$$

LOG LAW 1

The log of a product equals the sum of the logs of the factors. Thus,

$$\begin{aligned}\log_{10} 3 \times 2 &= \log_{10} 3 + \log_{10} 2 \\ &= 0.4771 + 0.3010 \\ &= 0.7781.\end{aligned}$$

The answer, 0.7781, can be verified by calculating, $\log_{10} 6$. As with some of the earlier examples, we have saved little time by using Law 1 to solve such a simple problem; it is just as easy to multiply 3 by 2 and calculate the log of 6. Use of **Law 1**, however, will simplify the solution of more difficult problems.

We have already applied **Law 1** to solve more difficult log problems without having identified it as such. Look back to Step One in solution of log problems. When the number 121 was converted to scientific notation, we formed *the product* 1.21×10^2 . That is the log of the product **ab**, in which **a** (the coefficient) = 1.21 and **b** = 10^2 . We then found the log of 10^2 , which is 2.0000, the **characteristic** of the log. Next, we found the log of 1.21 in the log table, which is 0.0828, the **mantissa** of the log. The log of 1.21×10^2 , then, is *the sum of the characteristic (2.0000) and the mantissa (0.0828), which equals 2.0828*.

Use of **Law 1** is particularly helpful when expressing the log of a number less than 1 as having a negative characteristic and a positive mantissa. *The log is the algebraic sum of the negative characteristic and the positive mantissa, which is an application of Law 1 of logarithms.* Below are three more examples of the application of **Law 1**.

$$\log_{10} 10 \times 10 = \log_{10} 10 + \log_{10} 10 = 1 + 1 = 2.$$

$$\log_{10} 20 \times 20 = \log_{10} 20 + \log_{10} 20 = 1.3010 + 1.3010 = 2.6020.$$

$$\begin{aligned}\log_{10} 648 &= \log_{10} 6.48 \times 10^2 \text{ (log of a product)} \\ &= \log_{10} 10^2 + \log_{10} 6.48 \\ &= 2.0000 + 0.8116 \\ &= 2.8116.\end{aligned}$$

Law 2 of Logarithms

LOG LAW 2

$$\log \frac{a}{b} = \log a - \log b.$$

The log of a ratio equals the difference between the logs of the factors. Thus,

$$\log_{10} \frac{3}{2} = \log_{10} 3 - \log_{10} 2 = 0.4771 - 0.3010 = 0.1761.$$

The answer, 0.1761, can be verified by calculating the log of 1.5 (3/2). The major advantage of **Law 2** is that it allows us to subtract logs rather than perform division. It is not imposing to divide 3 by 2, but use of the law simplifies solutions of the two problems below.

Example 1 $\log_{10} \frac{1.8}{1.2} = (\log_{10} 1.8) - (\log_{10} 1.2) = 0.2553 - 0.0792 = 0.1761.$

Example 2
$$\begin{aligned} \log_{10} \frac{(1 \times 10^0)}{(1 \times 10^{-4})} &= (\log_{10} 1 \times 10^0) - (\log_{10} 1 \times 10^{-4}) \\ &= 0.0000 - (\bar{4}.3010) \\ &= 3.6990. \end{aligned}$$

Law 3 of Logarithms

LOG LAW 3

$$\log a^b = b \log a$$

The log of a number raised to the **bth** power equals **b** times the log of **a**. **Law 3** is particularly helpful in Chapter 5 when we explain the difference between equations for decibels of intensity and decibels of pressure.

The first example below is a simple problem that can be solved easily without applying **Law 3**. In the second example, though, you should see how use of the law simplifies computation greatly.

Example 1 $\log_{10} 2^2 = 2 \log_{10} 2 = 0.6020.$

The answer, 0.6020, should be easy to verify by calculating the log₁₀ of 4 (2²).

Example 2 $\log_{10} 2^{1.65} = 1.65 \times \log_{10} 2 = 0.4967.$

Law 4 of Logarithms

LOG LAW 4

$$\log \frac{1}{a} = -\log a.$$

The log of a reciprocal equals minus the log of the number. Thus,

$$\log_{10} \frac{1}{2} = -\log_{10} 2 = -0.3010.$$

Law 4 is not an independent law, but rather is a special application of **Law 2**:

$$\log \frac{a}{b} = \log a - \log b.$$

When we calculate the log of $1/2$, that is an application of the second law (*the log of a ratio equals the difference between the logs of the factors*). The log of 1 in the numerator is always 0.0000. Therefore, the answer will always be a negative number because a positive number in the denominator is subtracted from zero in the numerator. Hence, **Law 4** is written simply as:

$$-\log a.$$

Logs Without Log Tables

You can obtain an *approximate* answer to many log problems without use of a calculator or log table, and to do so it is necessary to *memorize the logs of only four numbers*: 1, 2, 3, and 7. This enables you to check to see if the answer you obtained by following all the steps and using a log table or a calculator is *approximately* correct. Consider the numbers between 1 and 10, and because we are interested in getting an approximate answer, we will use only one or two decimal places in the computations.

- $\log_{10} 1 = 0.0$ **Memorize**
- $\log_{10} 2 = 0.3$ **Memorize**
- $\log_{10} 3 = 0.48$ **Memorize**
- $\log_{10} 4 =$ **Law 1** stated that **$\log ab = \log a + \log b$** .
Because $4 = 2 \times 2$, $\log_{10} 4 = (\log_{10} 2 + \log_{10} 2)$,
which already has been memorized, and
 $0.3 + 0.3 = 0.6$.
- $\log_{10} 5 =$ **Law 2** stated that **$\log a/b = \log a - \log b$** .
Because $5 = 10/2$, $\log_{10} 5 = \log_{10} 10 - \log_{10} 2$
 $= 1.00 - 0.30 = 0.70$.
- $\log_{10} 6 =$ Because $6 = 3 \times 2$, $\log_{10} 6 = \log_{10} 3 + \log_{10} 2$
 $= 0.48 + .30 = 0.78$.
- $\log_{10} 7 = 0.85$. **Memorize**
- $\log_{10} 8 =$ Because $8 = 4 \times 2$, $\log_{10} 8 = \log_{10} 4 + \log_{10} 2$
 $= .60 + 0.30 = 0.90$.
- $\log_{10} 9 =$ $(\log_{10} 3) + (\log_{10} 3) = 0.96$.
- $\log_{10}$ of **10** = You should see that the answer is 1.00.

It should be helpful to try to use that system to solve the following problems without log tables.

- $\log_{10} 20 = ___$

$20 = 10 \times 2$

Therefore, the answer is 1.3
- $\log_{10} 30 = ___$

$30 = 10 \times 3$

Therefore, the answer is 1.48
- $\log_{10} 100 = ___$

$100 = 10 \times 10$

Therefore, the answer is 2.0
- $\log_{10} 300 = ___$

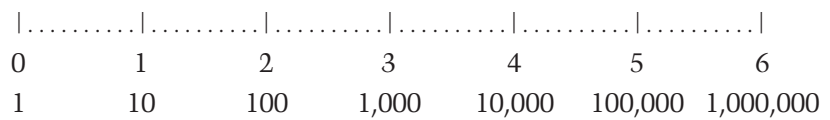
Can you see easily that the answer is 2.48?

A Final Comment About Logarithms

You might have noticed that when you calculate the log of a series of numbers from a ratio scale, the *logs of those numbers are linearly spaced*. That concept is illustrated below by comparing the two columns of numbers. The first column shows numbers that progressively increase by powers of 10 (1, 10, 100, . . . ,1,000,000). The second column shows that the logs of those numbers are linearly spaced (0, 1, 2, . . . ,6).

Column I	Column II
1	0
10	1
100	2
1,000	3
10,000	4
100,000	5
1,000,000	6

Because logs are linearly spaced, linear graph paper can be converted to logarithmic units by calculating the logs of the numbers you wish to display. That operation enables displaying a large range of numbers more economically. In the illustration below, suppose each dot represents a tick mark on your graph paper, and you wish to display information that ranges from 1 to 1,000,000 units. Only 60 lines on the graph paper will be required if the resolution that you desire is restricted to one decimal place.



Chapter 5 concerns the decibel, and in all decibel computations require use of logs and antilogs. The base for all computations involving decibels is 10.

■ NOTES

1. The discussion of the four scales of measurement is largely a condensed version of the extensive treatment of the topic by S. S. Stevens (1951). Readers who wish to explore this topic in greater detail are advised to read his chapter in the *Handbook of Experimental Psychology*.

■ PRACTICE PROBLEMS

Set 1

- Express the following values in scientific notation.
a. 6875 b. 0.0064 c. 109.6
- What is the base of 8^2 ?
- Solve each of the following.
a. 7^3 b. 0.2^4 c. $10^{2.2} \times 10^{1.8}$ d. $2^5 \times 2^{-2}$
e. $2^2 + 2^2$ f. $(5 \times 10^3) \times (2 \times 10^2)$ g. $(xyz)^0$
h. 5.764^1 i. $(1.6/5.2)^0$ j. $(10^2)^3$ k. $(3^2)^2$
l. $(2^{2.5})^2$ m. $(4^2)^{1/2}$ n. $3^{4/2}$ o. 2^{-2}
p. 10^{-3}

Set 2

1. Solve the following.
- | | | | |
|--------------------------|--------------------|--------------------------|---------------------|
| a. $\log_3 27$ | b. $\log_4 16$ | c. $\log_4 16^2$ | d. $\log_4 16^{-2}$ |
| e. $\log_{10} 10$ | f. $\log_2 2$ | g. $\log_{67} 67$ | h. $\log_e e$ |
| i. $\log_{10} 10^2$ | j. $\log_{10} 100$ | k. $\log_2 4$ | l. $\log_3 9$ |
| m. $\log_{12} 144$ | n. $\log_e e^2$ | o. $\log_{4.4} 4.4^{16}$ | p. $\log_{10} 0.1$ |
| q. $\log_{10} (3+4+1+2)$ | | | |
2. Assume a base of 10 and calculate the following.
- | | | |
|-----------------------------|-----------------------------|-----------------------------|
| a. $\log 54$ | b. $\log 3.6$ | c. $\log 0.4$ |
| d. $\text{antilog } 0.3010$ | e. $\text{antilog } 1.5315$ | f. $\text{antilog } 6.7202$ |



Sound Intensity and Sound Pressure: The Decibel

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Sound is a form of wave motion in which a *pattern of pressure*—or a *change in density*—is propagated through an elastic medium. **Energy** is transferred through the medium at some rate. *The rate at which sound energy is transferred through the medium is the acoustic power.*

Power is defined as *the rate at which work is accomplished, or as the rate at which energy is transformed or transferred:*

$$\text{Power} = \frac{\text{work}}{\text{unit time}} = \frac{\text{energy transformed or transferred}}{\text{unit time}}.$$

For example, **power** refers to the amount of work accomplished by an engine per second, or the amount of electrical energy that is transformed into heat energy per second by an electrical heater.

Power and **energy** are not equivalent. Suppose a person engages in a strenuous physical activity such as shoveling snow. The person might shovel for an hour at a rate of two shovelfuls per minute before the amount of energy stored in the muscles is depleted and exhaustion results. If the person is anxious to complete the job more quickly and picks up the pace to four shovelfuls per minute, exhaustion occurs earlier. Thus, the ability to complete such tasks is limited not only by our **energy**, *the capacity to do work*, but also by *the rate at which the energy is expended*, the **power**.

The unit of measure of power, including acoustic power, is the **watt (W)**, which is named in honor of James Watt, developer of the steam engine. If energy is expended at a rate of 1 joule per s (**MKS** system) or 10,000,000 ergs per s (**cgs** system), *one watt of power is expended*. Therefore,

$$1 \text{ watt} = \frac{1 \text{ joule}}{\text{sec}} = \frac{10,000,000 \text{ ergs}}{\text{sec}} \left(\text{or } \frac{10^7 \text{ ergs}}{\text{sec}} \right).$$

■ ABSOLUTE AND RELATIVE MEASURES OF ACOUSTIC POWER

Absolute Measure of Power

When we say that a particular sound wave has an acoustic power of some number of watts, we refer to the **absolute acoustic power** in the sound wave. In the case of sound waves, we should not expect to encounter large values of power such as 60, 100, or 200 watts, and we certainly will find no acoustic rival to the megawatt electrical power station. Instead, in acoustics we commonly deal with very small magnitudes of power, such as, for example, 10^{-8} watt or 2.13×10^{-9} watt. Even though the magnitude of acoustic power is quite small, the absolute measure of acoustic power in watts refers to *the rate at which energy is consumed or transferred*.

Relative Measure of Power

We frequently speak of **relative acoustic power** instead of absolute power. With relative acoustic power, two absolute sound powers are used to form a *product* or a *ratio*, which enables comparison of *the absolute power in one sound wave with the absolute power in another (reference) sound wave*. Thus, we might say that the acoustic power of wave **A** is 10 times greater than the acoustic power of wave **B**:

$$A = 10B. \text{ (a product)}$$

With the expression $A = 10B$, we do not know the absolute power in watts of either wave **A** or wave **B**, but we do know the relation between the two. If the absolute power of **B** is 10^{-6} watt, then the power of **A** is 10^{-5} watt to preserve the ratio of $A = 10B$. If $A = 10B$, then

$$B = A/10. \text{ (a ratio)}$$

Thus, if the absolute power of wave **A** is 6.21×10^{-3} watt, the power of wave **B** must be 6.21×10^{-4} watt:

$$\begin{aligned} B &= \frac{(6.21 \times 10^{-3})}{(1 \times 10^1)} \\ &= 6.21 \times 10^{-4}. \end{aligned}$$

That is the only solution in which the 1:10 ratio is preserved.

Summary

In summary, with relative power we express the **level** of power in a sound wave by forming a ratio (or product) of two absolute powers: the ratio of the absolute acoustic power of the sound wave in question ($\mathbf{W}_x$) to the absolute acoustic power of some reference sound wave ($\mathbf{W}_r$).

$$\text{Level} = \frac{\mathbf{W}_x}{\mathbf{W}_r}.$$

Equation 5-1

Importance of Specifying the Reference Power

Because *level of power* is the ratio (or product) of any two absolute powers, *the measure of level is generally meaningless unless the value of the reference power is specified*. If the level of a sound wave is 1,000, we know only that its power is 1,000 times greater than the power of some unspecified reference sound wave. It could be that the power, $\mathbf{W}_x$, is 10^{-1} relative to (re:) $\mathbf{W}_r$ of 10^{-4} . But, of course, a level of

1,000 would apply equally to $\mathbf{W}_x = 10^{-13}/\mathbf{W}_r = 10^{-16}$. In either case, the ratio of 1,000 is preserved.

A more explicit statement, therefore, is “the level of acoustic power is 1,000 re: 10^{-4} watt.” When the reference power ($\mathbf{W}_r$) is specified, there should be no ambiguity. *If the reference is not known, the concept of level has little meaning.*

■ SOUND INTENSITY

Imagine that we have an “idealized point source of sound” that operates in a “free, unbounded medium.” By an “idealized point source,” we mean a very small sphere (a point) that can pulsate in and out in a manner similar to the pulsation of the balloon in Figure 2–10. By a “free, unbounded medium,” we mean that there are no reflecting surfaces or energy-absorbing materials in the medium to affect sound transmission. In that case, *sound energy is transferred uniformly through the medium in all directions.*

Energy is transferred outward from the point source as an ever-expanding sphere. To measure the acoustic power in such a wave at some distance from the point source, we would have to integrate over the entire surface of the sphere.

- Instead, we generally are more interested in *the amount of power that acts on, is dissipated on, or passes through some much smaller area.*
- The smaller area is the square meter ($\mathbf{m}^2$) in the MKS metric system or the square centimeter ($\mathbf{cm}^2$) in the cgs metric system.
- Thus, instead of referring to the *energy per second* (the **acoustic power**), we speak of the *energy per second per square meter*, which is the **intensity** of the sound wave.

Intensity is *the amount of energy transmitted per second over an area of one square meter.* If the unit of measure of **acoustic power** is the **watt** (energy per second), and if **intensity** is the amount of energy per second per square meter, then it follows that the unit of measure for **intensity** must be the **watt per square meter** ($\mathbf{watt/m}^2$).

Absolute and Relative Measures of Sound Intensity

We have described how sound power is expressed in both absolute and relative terms; the same is true of sound intensity. If we say that a sound has an intensity of 10^{-8} watt/ $\mathbf{m}^2$, we refer to the **absolute intensity** of the sound wave. Alternatively, we can speak of the **relative intensity**, or the level of intensity (not to be confused with “intensity level” because the label “intensity level” implies a specific reference intensity that will be introduced later), by reference to the same kind of ratio used for power:

$$\text{Level} = \frac{I_x}{I_r}$$

Equation 5-2

where I_x is the absolute intensity of wave x , and
 I_r is the absolute reference intensity.

Importance of Specifying the Reference Intensity

It is important to specify the reference for level of sound intensity just as it was to specify the reference for level of sound power. Consider the examples in Table 5-1: six absolute sound intensities in **watt/m²** are referenced to two different reference intensities: $I_r = 10^{-10}$ watt/m² and $I_r = 10^{-12}$ watt/m².

For each absolute intensity (I_x) in the leftmost column, the ratios (I_x/I_r) in the second column are different from the ratios in the third column because those two columns are for two *different reference intensities*, 10^{-10} and 10^{-12} watt/m². For example, the level of intensity in the first entry in the table is 10^2 when referenced to 10^{-10} watt/m², but 10^4 when referenced to 10^{-12} watt/m². It therefore should be apparent that it is unacceptable to say that some sound has a level of intensity of 10^{-1} because, for the examples in Table 5-1, that would apply equally to 10^{-13} re: 10^{-12} and to 10^{-11} re: 10^{-10} . The message is straightforward: the reference must be specified.

THE DECIBEL

When we refer to the level of power or to the level of intensity in a sound wave, we often are faced with rather cumbersome numbers.

For example, a conservative estimate of the range of intensities to which the human auditory system can respond is about 10^{12} :1, and the range of acoustic powers of interest in noise measurements is even greater: about 10^{18} :1. Thus, we are faced with a wide range of intensities, and that presents awkward numerical notations.

Table 5-1. Ratios I_x/I_r for Two Different Reference Intensities

Absolute Intensity in watt/m ²	Relative Intensity, in watt/m ²	
	$I_r = 10^{-10}$	$I_r = 10^{-12}$
I_x		
10^{-8}	10^2	10^4
10^{-9}	10^1	10^3
10^{-10}	10^0	10^2
10^{-11}	10^{-1}	10^1
10^{-12}	10^{-2}	10^0
10^{-13}	10^{-3}	10^{-1}

The problem is minimized by elementary transformations of the numbers. Look again at Table 5–1. All entries in the second and third columns are ratios of the absolute intensities (I_x) to each of two reference intensities ($I_r = 10^{-10}$ and $I_r = 10^{-12}$).

What is the common element about the entries in the second and third columns? *Each is the base 10 raised to some power.* In other words, the base is redundant from one entry to another in the table; *it is always 10.* We can eliminate the base and only list the exponents. How do we justify eliminating the base? Because we know that *an exponent is a log* (Chapter 4). In other words: $\log_{10} 10^2 = 2$; $\log_{10} 10^0 = 0$; $\log_{10} 10^{-3} = -3$; and so on. The numbers are simplified, therefore, by *calculating the logarithm to the base 10 of the ratio of two intensities.*

$$\log_{10} \frac{I_x}{I_r}.$$

The Bel

For the first example in Table 5–1 where the absolute intensity (I_x) equals 10^{-8} watt/m², its level is 10^2 re: 10^{-10} watt/m². Next, we should see that one additional transformation, the log of the ratio 10^2 ($\log_{10} 10^2$), equals 2. This unit of measure of relative intensity is called the **bel** in honor of Alexander Graham Bell, an inventor who patented the first telephone. Thus,

Equation 5–3

$$N \text{ (bels)} = \log_{10} \frac{I_x}{I_r}.$$

where I_x is the absolute intensity of the wave in question in watt/m²,

I_r is the absolute intensity of a *reference* sound wave in watt/m²,

and the **bel** is a unit of level of intensity.

Thus, the relative intensities of the six sound waves in Table 5–1 are 2, 1, 0, –1, –2, and –3 **bels** re: 10^{-10} watt/m², and as 4, 3, 2, 1, 0, and –1 **bels** re: 10^{-12} watt/m².

Summary

Before proceeding, it might be helpful to provide a brief review of the transformations that have been accomplished for specifying the intensity of a sound wave. Consider the first example in Table 5–1.

1. The **absolute intensity** is 10^{-8} watt/m².
2. The **relative intensity**, referenced to 10^{-12} watt/m², is 10,000, and expressed as 10^4 .
3. The *base 10 is redundant* because the level of intensity always is the base 10 raised to some power. The expres-

sion is simplified, therefore, by calculating the log of the ratio. Because an exponent is a log, *the log of 10 raised to some power (exponent) is the exponent*. The level in this example, therefore, is 4 (**bels**).

4. Thus, a sound wave with an absolute intensity (I_x) of 10^{-8} watt/m² has a relative intensity of 4 **bels** re: 10^{-12} watt/m². Because relative intensity varies with the reference that is chosen, that same sound intensity of 10^{-8} has a relative intensity of 2 **bels** re: 10^{-10} watt/m².

From the Bel to the Decibel

The **bel** as a measure of relative intensity is less cumbersome than the ratio of intensities because the wide-range scale of intensities is compressed by transformation to a logarithmic scale. Thus, instead of expressing the relative intensity for various sounds as 0.0001, 0.01, 1,000, 100,000, or 1,000,000,000,000, we say that the relative intensities are -4, -2, 3, 5, and 12 **bels**.

Although the **bel** is less cumbersome than a ratio (a level of 10^{12} is expressed simply as 12 **bels**), the scale is compressed so much that fractional values often are required to reflect the desired precision of measurement. For example, a sound intensity of 2×10^{-8} watt/m² has a relative intensity of 4.3 **bels** re: 10^{-12} watt/m².

$$\begin{aligned}\text{bels} &= \log \frac{(2 \times 10^{-8})}{(1 \times 10^{-12})} \\ &= \log 2 \times 10^4 = 4.3.\end{aligned}$$

Excessive use of decimals is minimized if we use **decibels (dB)** rather than **bels** in the same way that we might use inches rather than feet and centimeters or millimeters rather than meters. A length of 5.5 ft can be expressed as 66 in. because each foot contains 12 in.

The prefix, **deci**, means 1/10. Therefore, a *decibel (dB)* is 0.1 of a *bel*. Because 1 dB = 0.1 bel and, conversely, 1 bel = 10 dB, Equation 5-3 for **bels** is rewritten for **decibels** as:

$$N \text{ (dB)} = 10 \log_{10} \frac{I_x}{I_r} \quad \text{Equation 5-4}$$

Thus, with the preceding example, the relative intensity is 43 dB rather than 4.3 **bels**. What, then, is a decibel? To say that “a **decibel** is one-tenth of a **bel**” is a true statement, but that definition is not sufficient unless the mathematical definition of the **bel** is understood. Instead, the decibel is defined most appropriately by reference to Equation 5-4: *the decibel is ten times the log of an intensity ratio or a power ratio.*¹

A comment about grammar is in order. The singular forms are **bel**, **decibel**, and **dB**, but the plural forms are **bels**, **decibels**, and **dBs** (not *dBs*).

Intensity Level (dB IL)

Use of the word “decibel” is ambiguous unless the reference intensity is specified. How, then, can confusion be avoided?

- First, there should be *no confusion if the reference intensity is specified*. A statement that the relative intensity of a sound is 86 dB re: 10^{-12} watt/m² is explicit: we know the relative intensity (86 dB), we know the reference intensity (10^{-12}) to which some absolute intensity has been compared, and we can calculate (with the use of antilogs) that the absolute intensity of the sound is 3.99×10^{-4} watt/m².
- Second, the potential for confusion has been lessened by adoption of a *conventional reference intensity of 10^{-12} watt/m² (MKS system) for dB IL*.²

Any absolute intensity can be used as the reference for expressing relative intensity, but *when the reference is 10^{-12} watt/m², the result is called **intensity level (dB IL)** rather than just “level of intensity.”* Strictly speaking, the label **intensity level**, and its abbreviation **dB IL**, should only be used when the reference is **10^{-12} watt/m²**. There is, however, still sufficient departure from the convention to warrant one final admonition: *the reference intensity should always be specified when the term **decibel** is used.*

The relation between absolute intensity (I_x) in watt/m² and **dB IL re: 10^{-12} watt/m²** is shown in the following illustration.

dB IL	I_x
70	$= 10^{-5}$
60	$= 10^{-6}$
50	$= 10^{-7}$
40	$= 10^{-8}$
30	$= 10^{-9}$
20	$= 10^{-10}$
10	$= 10^{-11}$
0	$= 10^{-12}$
-10	$= 10^{-13}$
-20	$= 10^{-14}$

When $I_x = 10^{-12}$, **dB IL** = 0 because $I_x = I_r$, the reference intensity. As you move up the scale, I_x increases *multiplicatively* by powers of 10 to the quantities 10^{-11} , 10^{-10} , and so on, to a maximum intensity of 10^{-5} in the illustration.

- For each tenfold increase in sound intensity, **dB IL** increases by 10 dB to a maximum of 70 dB IL when $I_x = 10^{-5}$ watt/m².
- Similarly, if you move down the scale, I_x decreases progressively by powers of 10, and for each power of 10 decrease in sound intensity, **dB IL** decreases by 10 dB to a minimum of -20 dB IL at a sound intensity of 10^{-14} watt/m² in the illustration.

Sample Problems

Two steps should be followed to solve decibel problems for relative intensity (any reference) or **intensity level (dB IL)** (reference = 10^{-12} watt/m²):

1. *Select the proper equation.* If the problem concerns the **intensity** of a sound wave, use Equation 5-4. Later in this chapter we will refer to the **pressure** of a sound wave, and that will necessitate a modification of Equation 5-4.
2. *Form a ratio and solve the problem.*

Problem 1: An increase in intensity by a factor of two (2:1) corresponds to how many decibels?

- Because we are dealing with **intensity**, use Equation 5-4. The ratio is 2:1, which means that an absolute intensity of unspecified value is twice as large as a reference intensity of unspecified value. It might be that the intensity of sound wave **A** is twice as great as the intensity of sound wave **B**. Alternatively, it might be that the intensity of sound wave **A** at one point in time (t_1) is doubled at another point in time (t_2). In either case, *the ratio is 2:1*.
- $$\begin{aligned} \text{dB} &= 10 \log \frac{2}{1} \\ &= 10 \times 0.3010 \\ &= 3.01 \text{ (which normally is rounded to 3 dB).} \end{aligned}$$
- Because the problem could have involved any absolute intensity and any reference intensity that preserved the ratio of 2:1, we see that *as any sound intensity is doubled (the ratio of absolute intensity to reference intensity = 2:1), the level increases by 3 dB*. Correspondingly, if intensity is halved (ratio = 1:2), the level decreases by 3 dB because:

$$\begin{aligned}
 \text{dB} &= 10 \log \frac{1}{2} \\
 &= -10 \log 2 \quad (\text{Log Law 4}) \\
 &= -3.
 \end{aligned}$$

Problem 2: An increase in intensity by a factor of ten corresponds to how many decibels?

$$\begin{aligned}
 \blacksquare \text{ dB} &= 10 \log \frac{10}{1} \\
 &= 10 \times 1 = 10.
 \end{aligned}$$

- Thus, as intensity increases *multiplicatively* by 10, relative intensity increases *additively* by 10 dB. Similarly, if intensity decreases by a factor of 10, relative intensity decreases by 10 dB:

$$\begin{aligned}
 \text{dB} &= 10 \log \frac{1}{10} \\
 &= -10 \log 10 = -10.
 \end{aligned}$$

Problem 3: What is the intensity level re: 10^{-12} watt/m² of a sound with an absolute intensity of 10^{-6} watt/m²?

- Follow the same steps used with the first two problems, but note that the ratio is now different. The intensity is not simply increased (or decreased) from some unknown value by a stated ratio such as 2:1 or 10:1, but rather *a specific absolute intensity* (I_x) is compared with a standard reference intensity 10^{-12} watt/m²:

$$\begin{aligned}
 \text{dB} &= 10 \log \frac{10^{-6}}{10^{-12}} \\
 &= 10 \log 10^6 = 10 \times 6 = 60.
 \end{aligned}$$

Problem 4: What is the intensity level re: 10^{-12} watt/m² of a sound with an intensity of 2×10^{-6} watt/m²?

- There are two approaches that can be taken. First, you can solve the problem in the same way that was used for Problem 3:

$$\begin{aligned}
 \text{dB} &= 10 \log \frac{(2 \times 10^{-6})}{(1 \times 10^{-12})} \\
 &= 10 \log (2 \times 10^6) = 63.
 \end{aligned}$$

Second, a quicker solution is available. We learned in Problem 3 that a sound with an absolute intensity of 10^{-6} has an intensity level of 60 dB. The sound in Problem 4 has an intensity of 2×10^{-6} , which is twice as great as an intensity of 1×10^{-6} . We also know from Problem 1 that when intensity increases by 2:1, *relative intensity increases by 3 dB*. Therefore, if the intensity level of $1 \times 10^{-6} = 60$ dB IL, the intensity level of 2×10^{-6} must = 63 dB. It frequently is useful to inspect a problem to

see if it involves powers of 2 (3 dB) or powers of 10 (10 dB). For example, see if such shortcuts can be used to solve Problem 5.

Problem 5: What is the intensity level re: 10^{-12} watt/m² of a sound with an intensity of 4×10^{-5} watt/m²?

- We know from Problem 4 that 2×10^{-6} corresponds to 63 dB IL. In addition, we should see that 10^{-5} is 10 times greater than 10^{-6} (10 dB), and 4 is 2 times greater than 2 (3 dB). Therefore, we should expect the answer to be 13 dB (10 + 3) greater than for Problem 4, which is 76 dB IL.
- If we solve the problem without taking answers to previous problems into account, we get the same result:

$$\begin{aligned} \text{dB} &= 10 \log \frac{(4 \times 10^{-5})}{10^{-12}} \\ &= 10 \log (4 \times 10^7) = 10 \times 7.6 = 76 \text{ dB IL.} \end{aligned}$$

Problem 6: 3 dB corresponds to what intensity ratio?

- This is an antilog problem, and the solution requires following the steps suggested in Chapter 4.
- $3 \text{ dB} = 10 \log X$
 $0.3 = \log X$ (Divide both sides of the equation by 10)
 $X = 2 \times 10^0$.

(Zero is the **characteristic** of the log as well as the **exponent** in scientific notation; 0.3 is the **mantissa** of the log and, with the aid of a pocket calculator or log table, we determine that the multiplier in scientific notation is 2.)

Problem 7: 13 dB corresponds to what intensity ratio?

- Try to solve the problem initially without paper, pencil, or calculator. 3 dB corresponds to a ratio of 2:1 and 10 dB corresponds to a ratio of 10:1. Because 13 dB consists of 3 + 10, the ratio must consist of $2 \times 10 = 20:1$.
- Check the answer above by solving the problem step-by-step.
 $13 \text{ dB} = 10 \log X$
 $1.3 = \log X$
 $X = 2 \times 10^1$ (which is 20:1).

Problem 8: 14 dB corresponds to what intensity ratio?

- There might not be an obvious combination of 3 dB (2:1) and 10 dB (10:1) that adds to 14 dB, which means that a shortcut that involves powers of 2 and/or powers of 10 does not *seem* to be available.
- You can, however, use knowledge of powers of 2 and powers of 10 to at least “bracket your answer.” We know that 13 dB corresponds to a ratio of 20:1. We should also know that 16 dB corresponds to a ratio of 40:1 because an additional

3 dB represents another doubling. Therefore, the answer must lie between 20:1 and 40:1.

■ $14 \text{ dB} = 10 \log X$

$$1.4 = \log X$$

$$X = 2.51 \times 10^1 \text{ (and 25.1 lies between 20 and 40).}$$

Did you notice that *a shortcut is available*? We know that 20 dB corresponds to a ratio of 100:1. Therefore, 17 dB, which is 3 dB less, corresponds to a halving of sound intensity, or a ratio of 50:1. Finally, 14 dB represents another halving of sound intensity, which corresponds to a ratio of 25:1.

Problem 9: An intensity level of 65 dB re: 10^{-12} watt/m² corresponds to what intensity?

- Combinations of 10 dB and 3 dB will not add to 65 (unless you use a long string such as $10 + 10 + 10 + 10 + 10 + 3 + 3 + 3 + 3 = 65$). In this case it might be easier to solve the problem step-by-step.

- We can, however, determine a range within which the answer must lie by use of powers of 10 and 2, and that enables us to determine if the answer we calculate is at least reasonable. We should know that 63 dB consists of $10 + 10 + 10 + 10 + 10 + 10 + 3$. Each 10 dB involves a tenfold increase in intensity relative to the reference, which therefore corresponds to 10^{-6} (the reference increased by a factor of 10^6). Similarly, each 3 dB involves a twofold increase in intensity relative to the reference. Thus, 63 dB corresponds to an intensity of 2×10^{-6} , and that is a lower boundary of the bracket. To get an upper boundary add another 3 dB. 66 dB represents another doubling of intensity, which is 4×10^{-6} . Because 65 dB lies between 63 and 66, we expect the intensity corresponding to 65 dB to lie between 2×10^{-6} and 4×10^{-6} .

$$65 \text{ dB} = 10 \log \frac{I_x}{10^{-12}}$$

$$6.5 = \log \frac{I_x}{10^{-12}}$$

$$3.16 \times 10^6 = \frac{I_x}{10^{-12}}$$

$$I_x = 3.16 \times 10^6 \times 10^{-12} = 3.16 \times 10^{-6}.$$

■ SOUND PRESSURE

We generally refer to the **pressure** associated with a sound wave rather than its acoustic power or intensity. We saw previously (Equation 1–5) that **pressure** is *the amount of force per unit area*. The unit of measure in the MKS system is the N/m² (newton per square meter)

or **Pa** (pascal), and we learned in Chapter 2 that $1 \text{ N/m}^2 = 1 \text{ Pa}$. It is more common to use the $\mu\text{N/m}^2$ (**microN/m²**) or the μPa (**microPa**) as the unit of measure, and we will use μPa in most computations involving sound pressure.³

When the pressure of a sound wave is, for example, $200 \mu\text{Pa}$ ($2 \times 10^2 \mu\text{Pa}$), we refer to the **absolute pressure** of the wave. As with intensity, we also speak of **relative pressure** in decibels by calculating the log of a pressure ratio. Equation 5-4, however, *cannot be used for decibels of sound pressure*. The equation for decibels of intensity must, therefore, be modified. To understand the reasons for the transformation of Equation 5-4, it is helpful to review the concepts of sound intensity and sound pressure *in relation to impedance*.

Impedance

The **impedance** of a vibratory system, including a volume of air through which sound is transmitted, is determined by the **resistance**, **mass reactance**, and **compliant reactance** of the system. Moreover, *the speed of sound in a medium such as air also is determined by properties of the medium*, namely, **elasticity** and **density**. We might reason, correctly, that **impedance** also is dependent on the speed of sound in a medium.

The **acoustic impedance** for a plane progressive wave is given by the product of the ambient **density** (ρ_0) in kg/m^3 and the **speed of sound** (s)⁴:

$$Z_c = \rho_0 s.$$

Equation 5-5

The subscript (c) is used in the equation because the impedance of the medium with plane progressive waves is called the **characteristic impedance**.

We previously defined intensity as *energy per second per square meter*. Intensity also is defined as *the ratio of the square of rms pressure to the characteristic impedance*:

$$I = \frac{P_{\text{rms}}^2}{\rho_0 s},$$

Equation 5-6

where **I** is intensity,

P_{rms} is root mean square pressure, and

ρ₀s is the characteristic impedance.

Decibels for Sound Pressure

We can see from Equation 5-6 that **intensity** is proportional to the square of **rms pressure**, and conversely, **rms pressure** is proportional to the square root of sound **intensity**:

$$p \propto \sqrt{I},$$

$$I \propto p^2, \text{ and}$$

$$I = \frac{p^2}{Z_c},$$

where **p** refers to rms sound pressure,

I refers to sound intensity, and

Z_c is the characteristic impedance.

Thus, if intensity increases by 16:1, pressure increases by only 4:1 ($\sqrt{16}$); if intensity increases by 10:1, pressure increases by only 3.16 ($\sqrt{10}$); or if sound intensity is doubled, pressure increases by only 1.414 ($\sqrt{2}$). It is important to keep in mind that:

- As sound intensity increases by some factor, rms pressure increases by the *square root* of that factor; and
- as rms pressure increases by some factor, sound intensity increases by the *square* of that factor.

The relation between **pressure** and **intensity** is shown in Figure 5–1. The function is linear because the figure is constructed with log-log

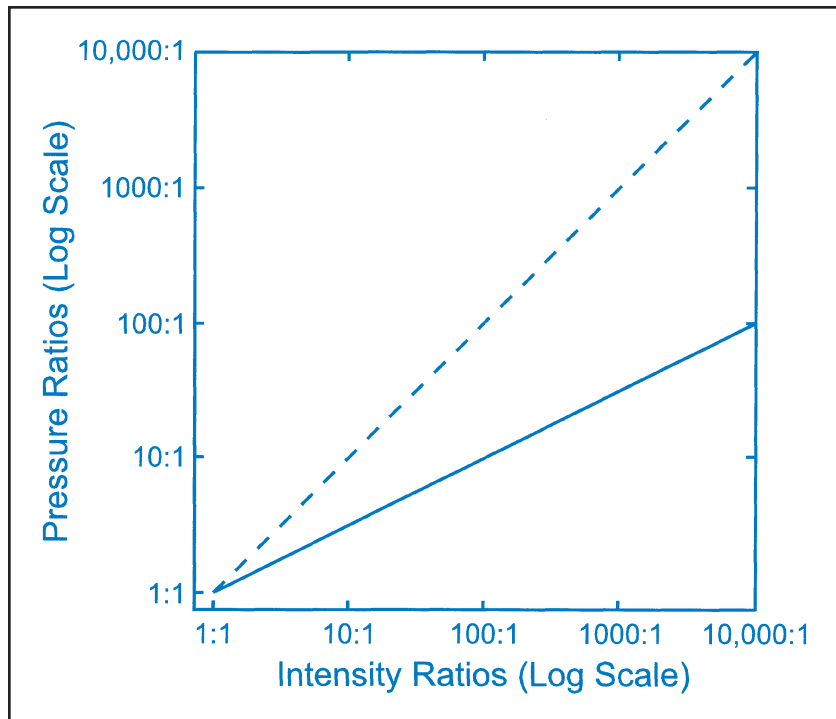


Figure 5–1. Relation between **pressure** and **intensity**. The dashed line reflects what the relation would be *if* pressure were proportional to intensity. The solid line reflects the true relation, namely that pressure is proportional to the *square root* of intensity.

coordinates. The *dashed line* reflects what the relation would be like *if* pressure were proportional to intensity. The *solid line* reflects the true relation, namely that *pressure is proportional to the square root of intensity*.

Equation 5–4 *cannot* be used for decibels of pressure because *rms pressure is proportional to the square root of intensity*, not to intensity. Equation 5–4 can be modified, however, to be appropriate for measures of pressure from our knowledge of the relation between sound intensity and rms pressure:

$$\text{dB} = 10 \log_{10} \frac{I_x}{I_r} \quad \text{Equation 5–4}$$

Because

$$\begin{aligned} I &= \frac{p^2}{Z_c} \\ \text{dB} &= 10 \log \frac{\left(\frac{p_x^2}{Z_c}\right)}{\left(\frac{p_r^2}{Z_c}\right)} && \text{(by substitution)} \\ &= 10 \log \frac{p_x^2}{p_r^2} && \text{(canceling } Z_c) \\ &= 10 \log \left(\frac{p_x}{p_r}\right)^2 \\ &= 10 \times 2 \log \frac{p_x}{p_r} && \text{(Log Law 3: } \log a^b = b \log a) \end{aligned}$$

$$\text{dB} = 20 \log_{10} \frac{p_x}{p_r} \quad \text{Equation 5–7}$$

Equation 5–4 is used to solve problems that involve *sound intensity*, and Equation 5–7 is used to solve problems that involve *sound pressure*. Thus, the decibel now can be defined as *10 times the log of a power or intensity ratio* and as *20 times the log of a pressure ratio*. The only difference between the two equations is that a multiplier of 10 is used for intensity problems and a multiplier of 20 is used for pressure problems.

Sound Pressure Level (dB SPL)

Equation 5–7 applies to all instances in which we represent a pressure ratio by decibels, and the reference pressure always should be specified. We have learned that the standard reference intensity for **dB IL** is $10^{-12} \text{ watt/m}^2$. The standard reference for **sound pressure** in the **MKS** system is **20 μPa** ($2 \times 10^1 \mu\text{Pa}$), *which is the pressure created in air by a sound wave with an intensity of $10^{-12} \text{ watt/m}^2$ under “standard conditions.”*

“Standard conditions” means a temperature of 20° centigrade and a barometric pressure of 760 mm of mercury. When the reference is **20 μPa**, we refer to decibels **sound pressure level (dB SPL)**.⁵ This does not mean that 20 μPa is always the reference for sound pressure level. On the contrary, sound pressure level could be (and has been) referenced to 1 dyne/cm² (cgs) or to other values. The lesson by now, however, should be obvious: *all uses of decibel notation require that the reference be specified.*

In Chapter 1 we emphasized that when an air medium is energized by a vibrating source of sound, *the individual air particles are displaced over a very small distance.* What is meant by a “small displacement”? If the sound wave is a 1000 Hz sinusoid with a sound pressure of 20 μPa (0 dB SPL), the displacement of air particles is approximately 7.68×10^{-8} m, which is about 1/300 of the diameter of a hydrogen molecule (2.34×10^{-6} m).

Think about the vastly different magnitudes that are encountered in the study of sound. For the 1000 Hz sine wave, air particles are displaced a nearly infinitesimal distance (7.68×10^{-8} m), whereas the wavelength of that sound wave is about 0.34 m. If we substitute a 100 Hz sine wave at the same sound pressure level, air particle displacement is unchanged, but the wavelength increases by a factor of 10 to about 3.4 m.

The Relation between Absolute Pressure and Decibels

The relation between absolute pressure (**p_x**) in **μPa** and **dB SPL re: 20 μPa** is shown in the following illustration.

■ When **p_x** = 2×10^1 , dB SPL = 0 because **p_x** = **p_r**.

dB SPL	p _x
100	= 2×10^6
80	= 2×10^5
60	= 2×10^4
40	= 2×10^3
20	= 2×10^2
0	= 2×10^1
-20	= 2×10^0
-40	= 2×10^{-1}

- As you move up the scale, p_x increases *multiplicatively* by powers of 10 to the magnitudes 2×10^2 , 2×10^3 , and so on, to a maximum pressure of 2×10^6 in the illustration.
- For each tenfold increase in sound pressure, dB SPL increases *additively* by 20 dB to a maximum of 100 dB SPL when $p_x = 2 \times 10^6$ μPa .
- Similarly, as you move down the scale, p_x decreases progressively by powers of 10, and for each power of 10 decrease in sound pressure, dB SPL decreases by 20 dB to a minimum of -40 dB SPL in the illustration at a sound pressure of 2×10^{-1} μPa .

Sample Problems

The same two steps used to calculate decibels for intensity should be used to calculate decibels for pressure: (1) Select the proper equation, and (2) form a ratio and solve the problem.

Problem 1: An increase in sound pressure by a factor of two (2:1) corresponds to how many decibels?

- $\text{dB} = 20 \log \frac{2}{1}$
 $= 20 \times 0.3010$
 $= 6.02$ (which normally is rounded to 6 dB).
- As sound *intensity* increases by a factor of 2:1, the level increases by 3 dB. In contrast, as sound *pressure* increases by a factor of 2:1, the *pressure level* increases by 6 dB. That difference occurs because with intensity we multiply the log of 2 (0.3) by 10, whereas with pressure we multiply the log of 2 (0.3) by 20. Correspondingly, if pressure is *halved* (ratio = 1:2), the pressure level *decreases* by 6 dB.

Problem 2: An increase in pressure by a factor of 10 corresponds to how many decibels?

- $\text{dB} = 20 \log \frac{10}{1}$
 $= 20 \times 1 = 20$.
- As pressure increases *multiplicatively* by 10, relative pressure increases *additively* by 20 dB. If the pressure decreases by a factor of 10 (1:10), the level decreases by 20 dB.

Problem 3: What is the sound pressure level re: 2×10^1 μPa of a sound with an absolute pressure of 2×10^4 μPa ?

- $\text{dB} = 20 \log \frac{(2 \times 10^4)}{(2 \times 10^1)}$
 $= 20 \log 10^3 = 60$.

Problem 4: What is the sound pressure level re: $2 \times 10^1 \mu\text{Pa}$ of a sound with an absolute pressure of $4 \times 10^4 \mu\text{Pa}$?

- Note that the pressure in this problem (4×10^4) is twice as great as the pressure in Problem 3 (2×10^4). Because a doubling of pressure corresponds to 6 dB, the answer is 6 dB greater than the answer for Problem 3 ($60 + 6 = 66$),
- $$\text{dB} = 20 \log \frac{(4 \times 10^3)}{(2 \times 10^1)}$$
$$= 20 \log (2 \times 10^3) = 66.$$
- It should not be necessary to use a calculator or log table to solve this problem. The log of 10^3 is 3 (a log is an exponent), and the log of 2 is 0.3. **Law 1** of logarithms stated that the log of a product is equal to the sum of the logs of the factors, so the log of 2×10^3 equals $3 + 0.3$, which multiplied by 20 yields 66.

Problem 5: 26 dB corresponds to what pressure ratio?

- Try to solve this problem by recalling powers of 2 and 10. If not,
- $$26 \text{ dB} = 20 \log X$$
$$1.3 = \log X$$
$$2 \times 10^1 = X$$
$$= 20.$$
- With powers of 2 and 10 in mind, you might have noticed that 26 dB comprises 20 dB (10:1) and 6 dB (2:1). Because 26 dB consists of $20 + 6$, the ratio must be $10 \times 2 = 20:1$.

Problem 6: A sound pressure level of 65 dB re: $2 \times 10^1 \mu\text{Pa}$ corresponds to what sound pressure?

$$65 \text{ dB} = 20 \log \frac{P_x}{(2 \times 10^1)}$$

$$3.25 = \log \frac{P_x}{(2 \times 10^1)}$$

$$1.78 \times 10^3 = \frac{P_x}{(2 \times 10^1)}$$

$$P_x = 1.78 \times 2 \times 10^3 \times 10^1$$

$$P_x = 3.56 \times 10^4.$$

■ THE RELATION BETWEEN dB IL AND dB SPL

In Table 5–2 we see that an *intensity ratio* of 10:1 (row 2) corresponds to 10 dB, whereas a *pressure ratio* of 10:1 (row 3) corresponds to 20 dB. Should you conclude, therefore, that 60 dB IL = 120 dB SPL? *You*

Table 5–2. Relation Between Decibels for Intensity and Decibels for Pressure

Intensity		Pressure	
Ratio I_x/I_r	dB $10 \log_{10} (I_x/I_r)$	Ratio P_x/P_r	dB $20 \log_{10} (P_x/P_r)$
1	0	1.0000	0
10	10	3.1623	10
100	20	10.0000	20
1,000	30	31.6228	30
10,000	40	100.0000	40
100,000	50	316.2278	50
1,000,000	60	1,000.0000	60

should not! If you replied “yes” to the question, you were led down the primrose path.

It is true that an intensity ratio of 10:1 corresponds to 10 dB and a pressure ratio of 10:1 corresponds to 20 dB. If, however, the intensity of a sound wave increases by a factor of 10:1, the pressure of the same sound wave increases *by the square root of 10:1* (3.1623), and the decibel equivalent still is 10 dB ($20 \log 3.1623 = 10$ dB).

Recall that *pressure is proportional to the square root of intensity*, and conversely, *intensity is proportional to the square of pressure*. Other examples of this relation can be seen in Table 5–2. An intensity ratio of 100:1 corresponds to 20 dB and a pressure ratio of 100:1 corresponds to 40 dB. But if the intensity of a sound wave increases by 100:1, the pressure increases by only 10:1 and, in either case, the decibel equivalent is 20 dB.

Thus, 60 dB IL does *not* equal 120 dB SPL. Recall from the derivation of Equation 5–7 that the multiplier of 10 in decibels for intensity was changed to a multiplier of 20 in decibels for pressure. As long as equivalent references are used in the two equations, dB IL = dB SPL, and as we learned previously, $2 \times 10^1 \mu\text{Pa}$ is the pressure equivalent of an intensity of 10^{-12} watt/m².

The equivalence of dB IL and dB SPL is illustrated in Figure 5–2, which shows isodecibel contours. At the left of the figure, intensity values in **watt/m²** increase from 10^{-13} at the bottom to 10^{-4} at the top. At the right of the figure, *corresponding* pressure values in **μPa** increase from 6.32×10^0 to a maximum of 2×10^5 .

Thus, as intensity increases or decreases by a power of 10, pressure changes only by the square root of 10. Each contour is a line that connects a given intensity at the left with a corresponding pressure at the right. The parameter of the figure, then, is decibels, *either dB IL or dB SPL. They are equivalent.*

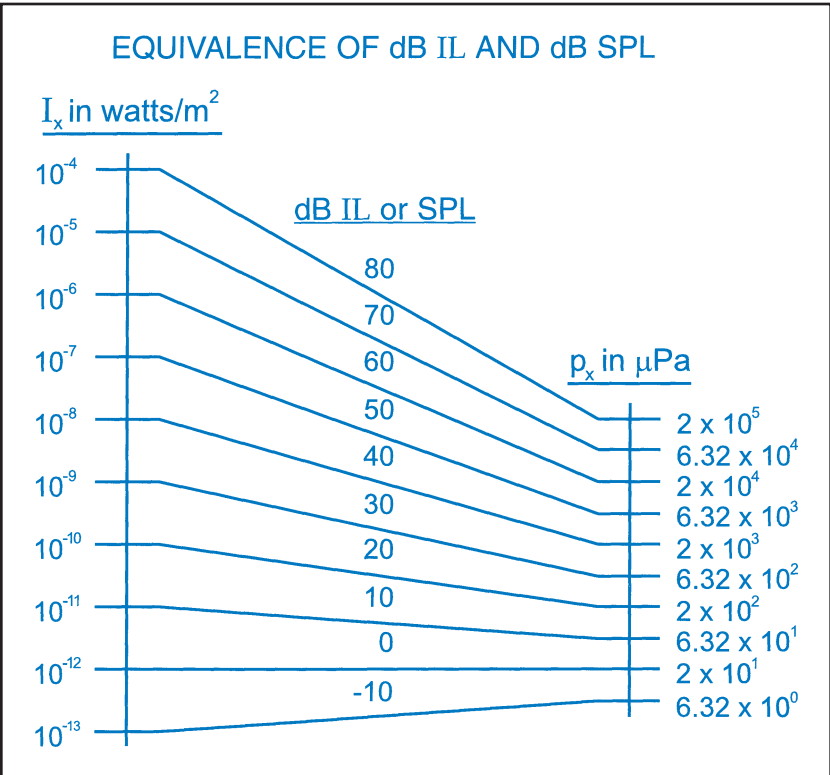


Figure 5–2. Equivalence of **dB IL** and **dB SPL**. The reference intensity for dB IL is 10^{-12} watt/ m^2 . The pressure created in air by a sound wave that has an intensity of 10^{-12} watt/ m^2 is 20 μPa . Thus, the reference pressure for dB SPL is equivalent to the reference intensity for dB IL, and dB IL *always* equals dB SPL.

■ **UNITS OF MEASURE FOR PRESSURE**

In Chapter 1 we indicated that **force** is measured in **newtons (MKS)** or **dynes (cgs)**, and therefore that **pressure** (*the amount of force per unit area*) is measured in **newtons/ m^2 (MKS)** or **dynes/ cm^2 (cgs)**. The popularity of these and other units of measure for pressure has changed over the years. It is necessary to compare and understand the relations among the various systems if we are to read all the scientific literature, not just that written in the past few years.

Table 5–3 compares, in approximate order of their appearance in the literature, the **dyne/ cm^2** (*and its equivalent, the **microbar***) with the **N/ m^2** (*and its equivalent, the **pascal [Pa]***) and with the **$\mu\text{N}/\text{m}^2$** (*and its equivalent, the **μPa***). The first column of Table 5–3 lists various values of dB SPL ranging from 100 to –6. The second column shows the corresponding *absolute* pressure in **dynes/ cm^2** or its equivalent the **microbar**. The third column shows the equivalent absolute pressures in **N/ m^2** and **pascals (Pa)**, which are equal to each

Table 5–3. Comparisons of Various Units of Measure for Pressure

dB SPL	dynes/cm ² or microbar	N/m ² or Pa	μN/m ² or μPa
100	2×10^1	2×10^0	2×10^6
94	10^1	10^0	10^6
80	2×10^0	2×10^{-1}	2×10^5
74	10^0	10^{-1}	10^5
60	2×10^{-1}	2×10^{-2}	2×10^4
54	10^{-1}	10^{-2}	10^4
40	2×10^{-2}	2×10^{-3}	2×10^3
34	10^{-2}	10^{-3}	10^3
20	2×10^{-3}	2×10^{-4}	2×10^2
14	10^{-3}	10^{-4}	10^2
0	2×10^{-4}	2×10^{-5}	2×10^1
–6	10^{-4}	10^{-5}	10^1

other, and the fourth column lists the corresponding absolute pressures in $\mu\text{N/m}^2$ and μPa , which also are equivalent to one another.

We can make several observations from the entries in Table 5–3, some of which have been stressed previously.

- The **reference pressures** for dB SPL appear in the row for 0 dB SPL, and *the three entries are equivalent*. Thus, 2×10^{-4} dyne/cm² (or microbar) = 2×10^{-5} N/m² (or Pa) = 2×10^1 μN/m² (or μPa). In a similar vein, all entries of absolute pressure *in any single row are equivalent*. Thus, for the row corresponding to 74 dB SPL, 10^0 dyne/cm² (or microbar) = 10^{-1} N/m² (or Pa) = 10^5 μN/m² (or μPa).
- Regardless of which unit of measure of sound pressure is selected, any twofold change in pressure corresponds to 6 dB: +6 dB if pressure increases and –6 dB if pressure decreases. For example, 2×10^{-2} Pa = 60 dB SPL; a pressure of 10^{-2} is only half as great as a pressure of 2×10^{-2} , and it corresponds to 54 dB SPL, which is 6 dB less. Similarly, 10^1 μPa = –6 dB SPL, and 2×10^1 μPa, which is twice as great, corresponds to 0 dB SPL, which is 6 dB more.
- A tenfold change in sound pressure corresponds to 20 dB: +20 dB if pressure increases and –20 dB if pressure decreases. Thus, 10^{-3} microbar = 14 dB SPL and 10^{-2} microbar (change in pressure by 10:1) corresponds to 34 dB SPL (+20 dB).
- **What does 0 dB SPL mean?** *It does not mean absence of sound!* It simply means that the pressure in question, p_x ,

equals the reference pressure, p_r . When $p_x = p_r$, the ratio is 1, the log of 1 = 0, and 20 times 0 equals 0 dB. Of course, the same is true for dB IL or any other decibel. A decibel is 10 (intensity) or 20 (pressure) times the log of a ratio, and 0 dB means that $p_x = p_r$ or that $I_x = I_r$.

- **What do negative decibels mean?** It does not matter whether we deal with decibels for pressure or decibels for intensity. Negative decibels mean that the value of the reference pressure (p_r) or reference intensity (I_r) *exceeds* the value of the pressure (p_x) or intensity (I_x) in question, that is, $p_r > p_x$ or $I_r > I_x$. That situation can be seen in the last row of Table 5–3 for all units of measure.

■ CONVERSION FROM ONE REFERENCE TO ANOTHER

The values of absolute pressure listed in Table 5–3 opposite 0 dB SPL are the **standard reference pressures** for decibels sound pressure level. There are, however, occasions on which a different reference might be adopted for some reason. In that circumstance, it might be important to compare decibels corresponding to one reference pressure with decibels referenced to a different pressure to determine whether they are equivalent.

You are likely to encounter older, but still important, literature in which decibels **sound pressure level** were expressed re: **1 dyne/cm²** instead of **2×10^{-4} dyne/cm²** in the **cgs** metric system. Suppose, for example, one author contends that the level of a noise is 74 dB SPL re: 2×10^{-4} dyne/cm². A second investigator describes the level of the same noise as 0 dB re: 1 dyne/cm². Are the results equivalent? We shall see that the answer is “yes,” and only a few calculations are necessary to demonstrate the equivalence.

You might encounter a “conversion rule” that says: to convert from a reference pressure of 2×10^{-4} dyne/cm² to a reference pressure of 1 dyne/cm², *subtract* 74 dB, or, conversely, to convert from 1 dyne/cm² to 2×10^{-4} dyne/cm², *add* 74 dB. That is a correct rule—if you can remember when to add and when to subtract—but, more importantly, it applies only to those two values. What is helpful, then, is a system that will permit conversion between *any* two reference pressures *expressed in the same metric system* (reference pressures that appear in the same column in Table 5–3). Two alternative approaches can be used. One choice, albeit cumbersome, is to calculate p_x (an antilog problem) from knowledge of the original reference pressure and then recalculate decibels (a log problem) using the new reference pressure.

- 74 dB SPL re: 2×10^{-4} dyne/cm² = how many dB SPL re: 1 dyne/cm²? Calculate the pressure, p_x , corresponding to 74 dB re: 2×10^{-4} .

$$\begin{aligned}
 74 &= 20 \log \frac{p_x}{(2 \times 10^{-4})} \\
 3.7 &= \log \frac{p_x}{(2 \times 10^{-4})} \\
 5 \times 10^3 &= \frac{p_x}{(2 \times 10^{-4})} \\
 p_x &= 10 \times 10^{-1} \\
 &= 1 \times 10^0.
 \end{aligned}$$

- From that calculation we know that the sound has an absolute pressure (p_x) = 1 dyne/cm² (1×10^0). The next step is to recalculate dB SPL using Equation 5-7, but with the new reference pressure, 1 dyne/cm², in the denominator.

$$\begin{aligned}
 \text{dB} &= 20 \log \frac{(1 \times 10^0)}{(1 \times 10^0)} \\
 &= 0 \text{ dB SPL}.
 \end{aligned}$$

Consider a second example. 20 dB SPL re: 1 dyne/cm² = ____ dB SPL re: 2 dynes/cm²?

$$\begin{aligned}
 \blacksquare \quad 20 &= 20 \log \frac{p_x}{(1 \times 10^0)} \\
 1 &= \log \frac{p_x}{(1 \times 10^0)} \\
 10^1 &= \frac{p_x}{(1 \times 10^0)} \\
 p_x &= 10^1 \times 1 \times 10^0 \\
 &= 1 \times 10^1. \\
 \blacksquare \quad \text{dB} &= 20 \log \frac{(1 \times 10^1)}{(2 \times 10^0)} \\
 &= 20 \log (.5 \times 10^1) \\
 &= 20 \log (5 \times 10^0) \\
 &= 20 \times 0.7 \\
 &= 14.
 \end{aligned}$$

The approach required two steps: (1) solve an antilog problem to determine the value of p_x , and then (2) solve a log problem to express p_x in decibels re: the new reference. An alternative is to use Equation 5-8, where only one step is necessary.

$$\text{dB}_{p_{r(\text{new})}} = \text{dB}_{p_{r(\text{orig})}} - 20 \log_{10} \frac{p_{r(\text{new})}}{p_{r(\text{orig})}},$$

Equation 5-8

where $p_{r(\text{orig})}$ = the original reference and

$p_{r(\text{new})}$ = the new reference.

To illustrate that the two approaches yield identical results, we will solve the first of the two problems listed above: 74 dB SPL re: 2×10^{-4} dyne/cm² = ? dB re: 1 dyne/cm²:

$$\begin{aligned} \text{dB} &= 74 - 20 \log \frac{(1 \times 10^0)}{(2 \times 10^{-4})} \\ &= 74 - 20 \log (.5 \times 10^4) = 0. \end{aligned}$$

■ COMBINING SOUND INTENSITIES FROM INDEPENDENT SOURCES

If the intensity of a sound is 10^{-5} watt/m², we calculate with Equation 5-4 that the intensity level is 70 dB IL re: 10^{-12} watt/m². Suppose, then, that two independent sound sources operate simultaneously and that each produces a sound with an intensity of 10^{-5} watt/m². We know that each source has an intensity level of 70 dB, but what is the *total intensity level* from the two sources combined? *It is not 140 dB!* Work through the problem and see why 140 dB is not a reasonable answer. To have an intensity level of 140 dB requires an absolute intensity of 10^2 watt/m²:

$$\begin{aligned} 140 &= 10 \log \frac{I_x}{10^{-12}} \\ 14 &= \log \frac{I_x}{10^{-12}} \\ I_x &= 10^{14} \times 10^{-12} = 10^2. \end{aligned}$$

It should be apparent that it is *impossible* to have a total intensity of 10^2 watts/m². Each independent source produces a finite amount of energy. If the intensity (**energy/s/m²**) from each source is 10^{-5} watt/m², then the total intensity from the combined sources can only be twice as great, 2×10^{-5} watt/m², not 10^2 watts/m². If the total **intensity** from the two sources combined is 2×10^{-5} watt/m², what is the **intensity level** from the two combined?

$$\begin{aligned} \text{dB} &= 10 \log \frac{(2 \times 10^{-5})}{10^{-12}} \\ &= 73. \end{aligned}$$

If the **intensity level** is 73 dB, what is the **sound pressure level** from the combined sources? If the reference pressure is 2×10^1 μ Pa, dB SPL also equals 73.

- First, a brief calculation should demonstrate the equivalence. The **intensity** from the combined sources is twice as great as for either source alone.

- A doubling of intensity corresponds to +3 dB, and the total of 73 dB is 3 dB greater than the intensity level of only one source.
- If the intensity increases by a factor of 2:1, what happens to the pressure? *It increases by $\sqrt{2}$ ($p \propto \sqrt{I}$), or 1.414.* From Equation 5-7 we see that sound pressure level also increases by 3 dB from 70 dB SPL: $\text{dB} = 20 \log 1.414 = 3$.
- Second, we learned previously (see Figure 5-2) that dB IL always equals dB SPL *when equivalent reference intensities and pressures are used.* Thus, 70 dB IL (the level for either source alone) equals 70 dB SPL, and if the intensity level increases by only 3 dB, the same must be true for the sound pressure level, which produces a total level of 73 dB SPL.

Unfortunately, the equivalence of dB IL and dB SPL is not always understood. Several years ago a person took a national examination that is required for certification by the American Speech-Language-Hearing Association and encountered the following kind of problem: If two sound sources each produce an identical sound pressure level, by how much will the total sound pressure level of the two combined sources exceed the level for either source alone? By now, you should know that the answer is 3 dB because it is the energies, powers, or intensities that should be added, which requires using the “10-log” equation (Equation 5-4). The person wrote a letter to the editor of the journal *Asha* to complain that the examination was unfair because the alternative answers included 3 dB, but not 6 dB, and the writer was certain that 6 dB was the correct choice.

That letter produced some lively (and correct) responses from Mitchell Kramer, who then was a doctoral student at Northwestern University, and Professors Larry Feth and the late W. Dixon Ward published in *Asha* (March, 1977). Some comments from those three individuals emphasize the main point that has been stressed in this section:

Both sound pressure and power change with a change in energy of an acoustic signal, with power changing as the square of pressure. But, a 3 dB change in pressure is equal, identical, and the very same as a 3 dB change in power or intensity. Of course, a doubling of pressure results in a 6 dB increase (of both pressure and intensity or power) but unless two signals are (1) the same frequency, and (2) added in phase, the intensities add, and a 3 dB increase results. This is equivalent to doubling the sound power, or increasing pressure by a factor of 1.41. But a dB is still a dB. (Kramer, p. 225)

What students often carry away from the lecture on combining sound sources is the rule of thumb: “Double the intensity means a 3 dB increase; double the sound pressure results in a 6 dB increase.” That rule by itself is correct. The errors arise in not understanding how sound pressure waveforms are added and in thinking that there is a difference between intensity level (IL) and sound pressure level (SPL). . . . It must be kept in mind that sound pressure level (SPL) and intensity level (IL) are always synonymous and further are always numerically

equal. When two independent sound sources are combined, one must always add their intensities. Two sound pressure waveforms can only be added if their relative phase is known, and it seldom is. (Feth, pp. 225–226)

What I suspect happened is that the problem went something like this: “If we put two machines side by side, each of which develops an SPL of 80 dB, what will be the SPL when they are both running?” The question is designed to lead the unwary down the primrose path by using SPL instead of IL, so that he or she is hoodwinked into thinking: “Oh yes, doubling the pressure is a 6-dB increase”—which is correct, but, alas, irrelevant, because the pressure simply does not get doubled in this situation. The correct answer is indeed 83 dB SPL—and, assuming that it is still taking place here on earth, also 83 dB IL. In air at standard temperature and pressure, the zeros on the SPL and IL scales have been chosen to represent the same acoustic conditions, so X dB SPL is also X dB IL. . . . To double the pressure (still under optimum conditions) would require four equal machines. (Ward, p. 226)

It is important to remember that when the total intensity level or sound pressure level from uncorrelated (independent) sound sources is desired, it is *the **energies**, or **powers**, or **intensities** that should be summed, not the pressures*. As Feth stated in his letter, we cannot add the pressures unless the relative phases are known, and they seldom are.

There are two approaches that can be taken to solving problems such as these: one for when the source intensities are equal, and another for when the source intensities are unequal.

Equal Source Intensities

Consider first the easier situation in which each of the contributing sources has the same intensity level (therefore, equal sound pressure levels). In that case,

Equation 5–9

$$\text{dB}_N = \text{dB}_i + 10 \log_{10} N,$$

where **i** = dB SPL (or dB IL) from one of the equal sources and
N = the number of sources being combined.

We do not have to calculate the **intensity** of any of the sources or of the total. We simply want to know the total **intensity level** in dB when **N** sound sources, each with equal intensity, are combined.

Example 1

Two snowmobiles each produce 94 dB IL. What is the total intensity level from the two combined?

$$\text{dB}_N = 94 + 10 \log 2 = 97.$$

Each source produces some finite amount of intensity that we do not need to calculate. Because there are two such sources,

the two sources combined must produce twice the intensity as either one alone. If intensity increases by a factor of 2:1, decibels increase by 3: $94 + 3 = 97$. Why does Equation 5–9 “apparently” not contain a ratio? Actually it does. We are asking what happens to the intensity level when we increase from one to two sources. Thus, the reference is 1 and need not be shown in the denominator.

What is the total sound pressure level for eight sound sources combined, each of which produces 92 dB when operating alone?

Example 2

Even though the level in the problem is expressed in **SPL** rather than **IL**, it is the intensities that must be added, not the pressures. Thus, we proceed just as we did with Example 1.

$$\begin{aligned}\text{dB}_N &= 92 + 10 \log 8 \\ &= 92 + 9 \\ &= 101.\end{aligned}$$

You should be able to solve this problem by thinking of powers of 2. Eight (8) is 2^3 , which means that 8 is the result of the base 2 being used 3 times in multiplication. Each doubling corresponds to 3 dB. Thus, eight sources produce a sound with an intensity that is 9 dB ($3 + 3 + 3$) greater than just one of the sources alone.

Unequal Source Intensities

When the contributing sources have different intensities, Equation 5–9 does not apply, but the problem can be solved by following three steps:

1. Calculate the intensity of each source (an antilog problem).
2. Add the intensities, and the sum becomes the numerator of the ratio.
3. Calculate decibels with Equation 5–4 with a reference of 10^{-12} watt/m². The result can be expressed as either dB IL or dB SPL because the two are equivalent.

What is the total SPL that results from combining one source that produces 90 dB SPL with a second source that produces 80 dB SPL?

Example 1

$$90 = 10 \log \frac{I_x}{10^{-12}}, \text{ Therefore, } I_{x(1)} = 10^{-3},$$

and

$$80 = 10 \log \frac{I_x}{10^{-12}}, \text{ Therefore, } I_{x(2)} = 10^{-4},$$

$$\begin{aligned}
 10^{-3} + 10^{-4} &= 1.1 \times 10^{-3}, \\
 \text{dB} &= 10 \log \frac{(1.1 \times 10^{-3})}{10^{-12}} \\
 &= 10 \log (1.1 \times 10^9) \\
 &= 90.4.
 \end{aligned}$$

There are two important observations to make. *First*, when the levels of two sources differ by 10 dB, the total level is only 0.4 dB greater than the level of the source with the higher intensity. That is true of any two levels that differ by 10 dB: 90 and 80 (90.4), 80 and 70 (80.4), 0 and -10 (0.4), and so on. *Second*, one must be careful with addition of the intensities in Step 2 because the exponents are different.

$$\begin{aligned}
 I_{x(1)} &= 1 \times 10^{-3} \\
 + I_{x(2)} &= \frac{1 \times 10^{-4}}{\phantom{10^{-3}}} \\
 &= ?
 \end{aligned}$$

Before the two intensities can be added, the exponent for one must be converted to be the same as the other exponent: *the two exponents must be the same*. We will illustrate the process by converting $I_{x(2)}$ to have an exponent of -3 rather than -4. To convert from -4 to -3 we *multiply* by 10. To preserve equivalence, therefore, we must *divide* the other term in the product, the coefficient, by 10: Thus,

$$1 \times 10^{-4} = 0.1 \times 10^{-3}.$$

Now, when the two intensities are added, we obtain:

$$\begin{aligned}
 &.1 \times 10^{-3} \\
 + &\frac{.1 \times 10^{-3}}{\phantom{10^{-3}}} \\
 = &1.1 \times 10^{-3}.
 \end{aligned}$$

If the validity of that series of steps is not obvious, the two intensities are added below when written in conventional notation rather than scientific notation:

$$\begin{aligned}
 1 \times 10^{-3} &= .001 \\
 + 1 \times 10^{-4} &= \frac{.0001}{} \\
 &= .0011,
 \end{aligned}$$

and .0011 in conventional notation equals 1.1×10^{-3} in scientific notation.

Example 2 A fan operates at an unknown level in a noisy environment. You want to determine the sound pressure level of the fan, but it cannot

be removed from its location and the surrounding noise cannot be turned off.

- Measure the sound pressure level of the combined noise produced by the fan and the surrounding equipment. Suppose this total level is 90 dB SPL.
- Turn off the fan and measure the level produced by the surrounding equipment. Suppose that this new measurement is 86 dB SPL.
- From those two measurements we know that the surrounding equipment and the fan together produce a noise of 90 dB SPL and that the surrounding equipment alone produces a noise of 86 dB SPL. The fan, therefore, must be responsible for the difference of 4 dB, but *do not conclude that its level is 4 dB SPL*.

a. (Intensity of fan *plus* surrounding noise)

$$90 = 10 \log \frac{I_x}{10^{-12}}, \text{ Therefore, } I_x = 10^{-3}.$$

b. (Intensity of surrounding noise *alone*)

$$86 = 10 \log \frac{I_x}{10^{-12}}, \text{ Therefore, } I_x = 3.98 \times 10^{-4}.$$

c. (Intensity of fan alone—by subtraction)

$$(1 \times 10^{-3}) - (3.98 \times 10^{-4}) = 6.02 \times 10^{-4}.$$

d. (Intensity level, or sound pressure level, of fan alone)

$$\begin{aligned} \text{dB} &= 10 \log \frac{(6.02 \times 10^{-4})}{10^{-12}} \\ &= 87.8. \end{aligned}$$

SUMMARY OF DECIBELS FOR SOUND INTENSITY AND SOUND PRESSURE

Two steps have been described for solution of decibel problems.

- First, select the proper equation.
- Second, form a ratio and solve the problem.
- When the problem involves **intensity** or **power**, use Equation 5-4—the “10 log equation.”
- When the problem involves **pressure**, use Equation 5-7—the “20 log equation.”

The distinction between power and pressure applies if decibels are used to express the level of some quantity other than sound. For

example, if you have three dollars, and someone gives you three more dollars, that represents an increase of 3 dB. In contrast, if a committee comprises three people, and three more members are added, that represents an increase of 6 dB. We justify use of the 10 log equation in one case, but the 20 log equation in the other, because “money is power,” but “people are pressure.”⁶

The only confusion that should remain might stem from problems that involve combining sound levels from independent sources. In those instances, it is important to remember that unless the relative phases are known, *it is the **energies** or **powers** or **intensities** that should be added, not the **pressures***. Thus, it does not matter whether the problem is stated by reference to dB IL or dB SPL. *As long as the references are equivalent, dB IL = dB SPL, and you must always use Equation 5-4—the “10 log equation.”*

■ NOTES

1. The proper form for all equations concerning the decibel is “log₁₀ X” preceded by a multiplier of either 10 (**intensity**) or 20 (**pressure**), where “X” is a ratio. Because the base for all calculations of decibels is 10, the base frequently will be omitted for convenience.
2. The equivalent reference intensity in the cgs metric system is 10⁻¹⁶ watt/cm².
3. The unit of measure in the cgs metric system is the dyne/cm² or the microbar.
4. In many reference texts, the symbol for the speed of sound is **c** rather than **s**. Thus, the equation for characteristic impedance ordinarily is expressed as **Z_c = ρ₀c**. We have elected to use **s** so that we can consistently distinguish between speed (**s**), which is a scalar quantity, and velocity (**c**), which is a vector quantity.
5. The equivalent reference pressure in the cgs metric system is 2 × 10⁻⁴ dyne/cm² or 2 × 10⁻⁴ microbar. See Table 5-3 for a more complete description of equivalent measures of pressure in the two metric systems.
6. I trust no reader will believe that this statement was intended to be taken seriously!

■ FREQUENTLY MISUNDERSTOOD CONCEPTS

1. If intensity level increases by 23 dB, the intensity increases by what factor?

200:1. The problem is solved with Equation 5-4:

$$N \text{ (dB)} = 10 \log \frac{I_x}{I_r}$$

$$23 = 10 \log X$$

$$2.3 = \log X$$

$$\begin{aligned} X &= 2 \times 10^2 \\ &= 200. \end{aligned}$$

2. An increase in sound pressure by a factor of 4:1 corresponds to an increase by how many decibels?

12 dB, not 12 dB IL or 12 dB SPL. The problem is solved with Equation 5-7:

$$\begin{aligned} N \text{ (dB)} &= 20 \log \frac{P_x}{P_r} \\ &= 20 \log 4 \\ &= 20 \times .6 \\ &= 12 \text{ dB}. \end{aligned}$$

It cannot be 12 dB SPL or 12 dB IL because those answers require a reference pressure of 20 μPa (SPL) or intensity of 10^{-12} watts/m² (IL). We do not know the value of the reference pressure before the pressure increases. We know only that whatever the value of the pressure originally, it increased by a factor of 4:1. Suppose, for example, that the original pressure was $2 \times 10^4 \mu\text{Pa}$, which corresponds to 60 dB SPL. When the pressure increases by a factor of 4:1, it becomes $8 \times 10^4 \mu\text{Pa}$, which corresponds to 72 dB SPL. The increase, however, is by 12 dB, not 12 dB SPL.

3. An increase in sound pressure by a factor of 6:1 corresponds to an increase in sound intensity by what factor?

36:1. *Intensity is proportional to the square of pressure.* Therefore, if pressure increases by 6:1, intensity increases by a factor of 6^2 , or 36:1.

4. If sound pressure decreases by one half (1:2), sound intensity decreases by how many *decibels*?

6 dB. Because intensity is proportional to the square of pressure, if pressure *decreases* by one half (1:2), intensity *decreases* by $1/2^2$, or a factor of 1:4. Then, $10 \log 1/4 = -6$ dB.

5. 90 dB re: 2×10^{-4} microbar (cgs system) corresponds to how many dB re: 1 microbar?

16 dB. The problem is solved with Equation 5-8:

$$\text{dB}_{P_{r(\text{new})}} = \text{dB}_{P_{r(\text{orig})}} - 20 \log_{10} \frac{P_{r(\text{new})}}{P_{r(\text{orig})}}$$

$$\begin{aligned}
 &= 90 - 20 \log \frac{(1 \times 10^0)}{(2 \times 10^{-4})} \\
 &= 90 - 74 = 16.
 \end{aligned}$$

6. **Four independent sound sources each produce a sound level of 60 dB SPL. What is the intensity (not intensity level) of the four combined?**

4×10^{-6} watt/m². The answer is *not* 66 dB SPL (or 66 dB IL) because the question specifically asked for the **intensity**, which is expressed in watt/m², not intensity level, which is expressed in dB. This problem can be solved easily by breaking it into two parts. First, if each source produces a level of 60 dB SPL, what is the SPL of the four sources combined? Use Equation 5-9:

$$\begin{aligned}
 \text{dB}_N &= \text{dB}_i + 10 \log N \\
 &= 60 + 10 \log 4 \\
 &= 66 \text{ dB SPL}.
 \end{aligned}$$

Second, use Equation 5-4 to solve for intensity, I_x :

$$\begin{aligned}
 \text{dB} &= 10 \log_{10} \frac{I_x}{I_r} \\
 66 &= 10 \log \frac{I_x}{10^{-12}} \\
 &= 4 \times 10^{-6}.
 \end{aligned}$$

7. **Four independent sound sources each produce a sound level of 60 dB SPL. What is the sound pressure (not sound pressure level) of the four combined?**

4×10^4 μPa . The answer is *not* 66 dB SPL, because the question specifically asked for the **pressure**, which is expressed in μPa , not the sound pressure level, which is expressed in dB. We already know the total level for the four sources combined from solving #6: 66 dB SPL. Next, use Equation 5-9 to solve for pressure, p_x :

$$\begin{aligned}
 \text{dB} &= 20 \log \frac{p_x}{p_r} \\
 66 &= 20 \log \frac{p_x}{(2 \times 10^1)} \\
 &= 4 \times 10^4.
 \end{aligned}$$

PRACTICE PROBLEMS

Recall the two-step procedure for solving decibel problems: (1) Select the proper equation and (2) form a ratio and solve the problem. It will be useful to inspect a problem to see if it involves powers of 2 (3 dB for intensity; 6 dB for pressure). Even though some problems cannot be solved that way, you can estimate (“bracket”) the answers by using powers of 2 and powers of 10 to establish upper and lower limits for the “correct” answer. Try to solve as many problems as possible that way, and then use your calculator to check your computations.

Set 1

Convert each of the following **intensity** ratios to decibels.

- a. 1:1 b. 10:1 c. 100:1 d. 1000:1 e. 10^3 :1
f. 2:1 g. 3:1 h. 4:1 i. 5:1 j. 6:1
k. 7:1 l. 8:1 m. 9:1 n. 1:2 o. 1:3
p. 1:4 q. 1:5 r. 1:6 s. 1:7 t. 1:8
u. 1:9 v. 1:10 w. 1:100 x. 1:1000 y. .001:1
z. 100^{-3} :1 aa. 20:1 bb. 200:1 cc. 40:1 dd. 400:1
ee. 60:1 ff. 600:1 gg. 2.45×10^0 :1 hh. 2.45×10^1 :1
ii. $(2.45 \times 10^1) / 10^{-12}$:1 jj. $(2.45 \times 10^{-8}) / 10^{-16}$:1

Set 2

Convert each of the following decibels to intensity ratios.

- a. 0 b. 10 c. 20 d. 30
e. 40 f. 70 g. 3 h. 6
i. 9 j. 12 k. 23 l. 29
m. 46 n. 56 o. 76 p. -10
q. -20 r. -30 s. -23 t. -36
u. 17 v. 62 w. 91 x. 5.4
y. 12.6

Set 3

Calculate dB IL re: 10^{-12} watt/m² for each of the following values of sound intensity (I_x).

- | | | | |
|-------------------------|-------------------------|--------------------------|---------------------------|
| a. 10^{-12} | b. 10^{-11} | c. 10^{-10} | d. 10^{-9} |
| e. 10^{-8} | f. 2×10^{-8} | g. 4×10^{-8} | h. 8×10^{-8} |
| i. 1×10^{-3} | j. 4×10^{-3} | k. 1×10^{-2} | l. 2×10^{-2} |
| m. $.5 \times 10^{-2}$ | n. 0.5×10^{-5} | o. 0.5×10^{-12} | p. 0.25×10^{-12} |
| q. 1.4×10^{-4} | r. 2.8×10^{-4} | s. 1.65×10^{-6} | t. 3×10^{-6} |

Set 4

Calculate sound intensity (I_x) in watt/m² for each of the following values of dB IL re: 10^{-12} watt/m².

- | | | | |
|--------|---------|---------|--------|
| a. 0 | b. 10 | c. 20 | d. 30 |
| e. 40 | f. 60 | g. 13 | h. 23 |
| i. 36 | j. 49 | k. -10 | l. -20 |
| m. -3 | n. -6 | o. -13 | p. -23 |
| q. -26 | r. 41 | s. 62 | t. 73 |
| u. 87 | v. 16.8 | w. 24.2 | x. 38 |
| y. 47 | | | |

Set 5

Convert each of the following pressure ratios to decibels.

- | | | | | |
|---|---|-----------------------|-----------------------|-----------------|
| a. 1:1 | b. 10:1 | c. 100:1 | d. 1,000:1 | e. $10^3:1$ |
| f. 2:1 | g. 3:1 | h. 4:1 | i. 5:1 | j. 6:1 |
| k. 7:1 | l. 8:1 | m. 9:1 | n. 1:2 | o. 1:3 |
| p. 1:4 | q. 1:5 | r. 1:6 | s. 1:7 | t. 1:8 |
| u. 1:9 | v. 1:10 | w. 1:100 | x. 1:1,000 | y. 0.001:1 |
| z. $10^{-3}:1$ | aa. 20:1 | bb. 200:1 | cc. 40:1 | dd. 400:1 |
| ee. 60:1 | ff. 600:1 | gg. $10^{-4}:1$ | hh. $10^{-5}:1$ | ii. $10^{-2}:1$ |
| jj. $10^{-1}:1$ | kk. $10^0:1$ | ll. $10^{-4}/10^{-4}$ | mm. $10^{-3}/10^{-4}$ | |
| nn. $(2 \times 10^{-4})/(2 \times 10^{-4})$ | oo. $(4 \times 10^{-4})/(2 \times 10^{-4})$ | | | |
| pp. $10^{-4}/(2 \times 10^{-4})$ | | | | |

Set 6

Convert each of the following decibels to pressure ratios.

- | | | | |
|--------|--------|-------|--------|
| a. 0 | b. 20 | c. 40 | d. 60 |
| e. 80 | f. 100 | g. 6 | h. 12 |
| i. 18 | j. 24 | k. -6 | l. -12 |
| m. 26 | n. 46 | o. 72 | p. -20 |
| q. -40 | r. 10 | s. 30 | t. 50 |
| u. 44 | v. 17 | w. 62 | x. 5.5 |

Set 7

Calculate dB SPL re: $2 \times 10^1 \mu\text{Pa}$ for each of the following values of sound pressure (p_x) in μPa .

- | | | |
|-----------------------|-----------------------|----------------------|
| a. 2×10^1 | b. 2×10^2 | c. 2×10^3 |
| d. 2×10^4 | e. 2×10^5 | f. 10^5 |
| g. 4×10^1 | h. 8×10^1 | i. 8×10^4 |
| j. 2×10^0 | k. 4×10^0 | l. 4×10^3 |
| m. 1.05×10^6 | n. 1×10^5 | o. 0.5×10^5 |
| p. 4×10^5 | q. 4.25×10^5 | r. 8.5×10^5 |

Set 8

Calculate sound pressure (p_x) in μPa for each of the following values of dB SPL re: $2 \times 10^1 \mu\text{Pa}$.

- | | | | |
|-------|---------|-------|-------|
| a. 0 | b. 6 | c. 12 | d. -6 |
| e. 20 | f. 40 | g. 60 | h. 3 |
| i. 9 | j. 10 | k. 30 | l. 50 |
| m. 43 | n. 46 | o. 36 | p. 34 |
| q. 72 | r. 16.8 | s. -7 | t. -8 |

Set 9

Calculate dB SPL re: 2×10^{-4} dyne/cm² for each of the following values of sound pressure (p_x) in dyne/cm².

- | | | | |
|-----------------------|--------------------|-----------------------|-----------------------|
| a. 0.0002 | b. 0.0004 | c. 8×10^{-4} | d. 2×10^{-4} |
| e. 4×10^{-4} | f. .002 | g. 2×10^{-3} | h. 4×10^{-2} |
| i. 1×10^{-5} | j. 2×10^0 | | |

Set 10

Calculate the total intensity in watt/cm² that results from combining the following intensities from uncorrelated sources.

- | | |
|---|--|
| a. $10^{-8} + 10^{-8}$ | b. $10^{-6} + 10^{-6}$ |
| c. $(2 \times 10^{-6}) + 10^{-6}$ | d. $(2 \times 10^{-6}) + (5 \times 10^{-6})$ |
| e. $(2 \times 10^{-6}) + (5 \times 10^{-6}) + (2 \times 10^{-6})$ | f. $(2 \times 10^{-6}) + (3 \times 10^{-5})$ |

Set 11

Calculate the intensity level re: 10^{-12} watt/m² that results from combining the intensities in Set 10.

- | | | |
|----|----|----|
| a. | b. | c. |
| d. | e. | f. |

Set 12

Calculate the sound pressure level that results from combining the following uncorrelated sound sources whose levels are given in dB SPL.

- | | | |
|-----------------|-----------------|-----------------------|
| a. 20 + 20 | b. 30 + 30 | c. 46.2 + 46.2 |
| d. 20 + 20 + 20 | e. 30 + 30 + 30 | f. 46.2 + 46.2 + 46.2 |
| g. 60 + 70 | h. 60 + 66 | i. 60 + 70 + 80 |



Complex Waves

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All sound waves that have been described to this point were of the **sinusoidal** form that appears in panel A of Figure 6–1. It is important to understand *the sine wave thoroughly because it is the fundamental component of other sound waves that will be encountered.*

The three other sound waves in Figure 6–1 are more complex in form than the simple sine wave, and they are, indeed, called **complex**

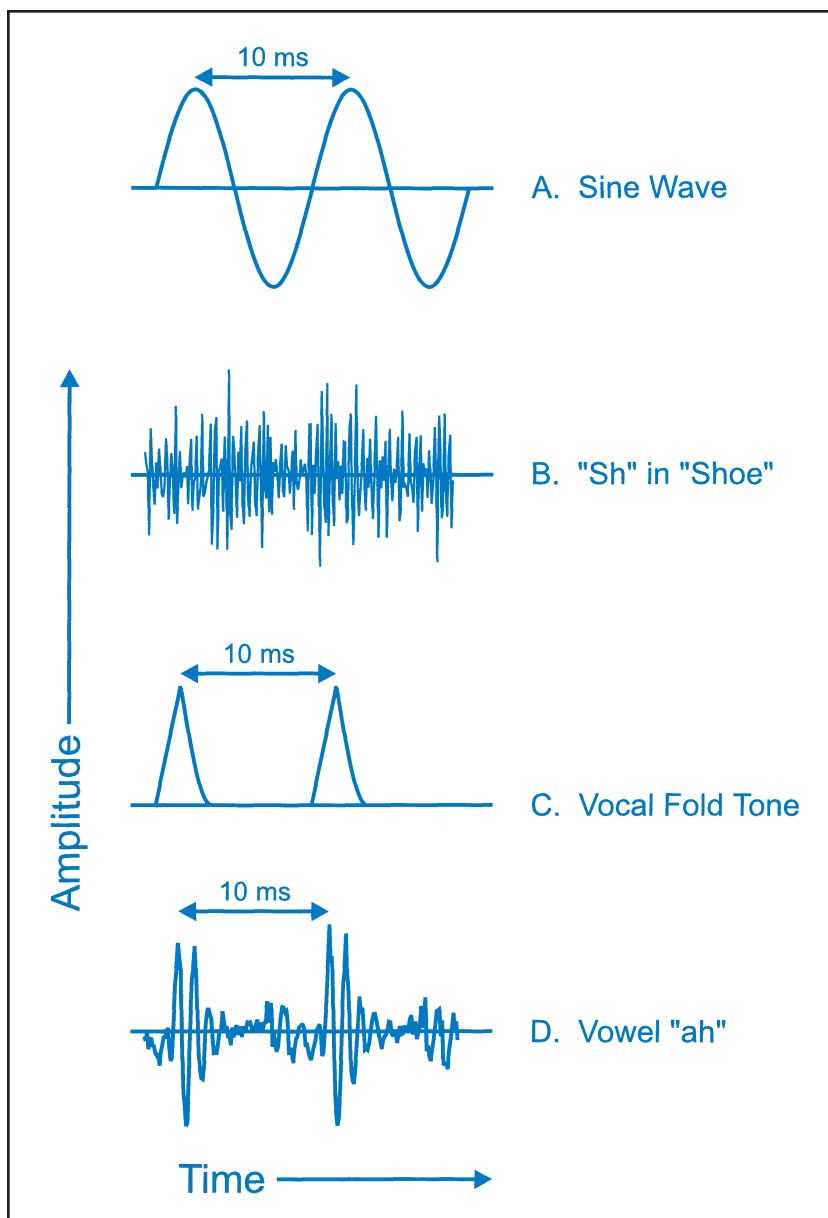


Figure 6–1. Comparison of a sine wave with three complex waves: the “sh” sound from the word “shoe”; the sound wave created by vibratory movement of the vocal folds; and the vowel /a/.

waves. The wave in panel B is similar to the “sh” sound in the word “shoe.” Panel C shows a sound wave produced by the vibratory motion of the vocal folds during production of a vowel. Finally, panel D shows the same vowel as recorded near the mouth opening rather than deep in the throat just above the vocal folds.

FOURIER’S THEOREM

A **complex wave** is *any sound wave that is not sinusoidal*, which although true, does not really explain the nature of complex waves. Fourier’s theorem, however, provides greater insight into the contrast between sinusoidal and complex waves. That theorem states that the complex waves in Figure 6–1—and all other sound waves that are not sinusoidal—are *composed of a series of simple sinusoids that can differ in amplitude, frequency, and phase*. Thus, when two or more sine waves that differ in amplitude, frequency, and/or phase are added together, a complex wave is produced.

The degree of complexity depends on the number of sine waves that are combined and on the specific dimensional values (amplitudes, frequencies, and phases) of the sinusoidal components. The theorem was first stated by Joseph Fourier (1768–1830), a French mathematician who lived at the time of Napoleon I; hence, *the series of sine waves that are combined to compose a complex periodic wave is called a **Fourier series** in his honor.*¹

Fourier’s theorem has two important implications for the study of complex waves.

1. Because a complex wave consists of some number of sinusoids of different amplitudes, frequencies, and/or phases, the nature of any complex wave should be easier to comprehend if we understand the concept of simple harmonic motion associated with each of the sinusoidal components, and if we recall the relevant dimensions of sine waves: **amplitude, frequency, and phase**.
2. The **Fourier series** can be derived by a process called **Fourier analysis**, which means that *a complex waveform can be decomposed, or analyzed, to determine the amplitudes, frequencies, and phases of each of the sinusoidal components*.

All sound waves can be classified by reference to (1) the presence or absence of **periodicity** in the wave and (2) the degree of complexity of the wave.

PERIODIC WAVES

A **periodic wave**, whether sinusoidal or complex, is a wave that *repeats itself at regular intervals over time*. Because the wave repeats itself periodically, it is a **periodic time function**. The sine wave in

Figure 6–1 is an obvious example of periodicity because the characteristics of any one cycle of the wave are duplicated exactly in all other cycles—*each cycle in the wave is repeated regularly over time*.

Sine waves are not the only forms of wave motion that are periodic. The vocal fold wave (panel C) and the vowel (panel D) in Figure 6–1 appear to be reasonably periodic (they are called **quasi-periodic**) because we can verify that all the features within one “cycle” of vibration are duplicated *almost exactly* during the next and all subsequent cycles. Thus, there are two kinds of periodic waves: (1) sinusoidal and (2) complex.

- A **sinusoidal wave** is a wave that results from simple harmonic motion and comes from a mathematical relation that contains a sine function.
- A **complex periodic wave** is a periodic wave, but it is not sinusoidal.

Components of Complex Periodic Waves

According to Fourier’s theorem, *any* complex periodic wave consists of some number of simple sinusoids that are summed, but the sinusoidal components cannot be selected randomly if the resultant sound wave is to be periodic. Instead, they must satisfy a basic mathematical requirement called a **harmonic relation**. The term **harmonic relation** means that *the frequencies of all sinusoids that compose the series must be integer (whole number) multiples of the lowest frequency component in the series*.

- For example, if the lowest frequency sinusoid of a complex wave is 100 Hz, the other sinusoidal components must be selected from the frequencies 200, 300, 400, 500 Hz, and so forth—other frequency values would not satisfy the requirement of being integer multiples of the lowest frequency.
- Similarly, if the lowest frequency sinusoid is 110 Hz, the other sinusoidal components must be selected from the frequencies 220, 330, 440, 550 Hz, and so on.
- We say “must be selected from” because, as we shall see later, *all components of a complex periodic sound wave must be integer multiples of the lowest frequency, but not all integer multiples need be present*.

A Harmonic Series

When a harmonic relation exists among the frequency components, the series of frequencies is called a **harmonic series**, and all sinusoids in the harmonic series are called **harmonics**. The harmonics are numbered consecutively from lowest to highest frequency: 1st harmonic, which also is called the **fundamental frequency** (f_0), 2nd

harmonic, 3rd harmonic, and so on until we reach the n th harmonic, the highest frequency component in the series.

In the first example above, the 1st harmonic (the f_0) = 100 Hz, the 2nd harmonic = 200 Hz, the 3rd harmonic = 300 Hz, and so on. There also is a special circumstance in which the fundamental frequency (1st harmonic) is missing from the series. In that case, all higher frequencies in the harmonic series are integer multiples of what is called the **repetition rate**.

Figure 6–2 shows the waveform of one complex periodic wave that consists of an infinite number of sinusoidal components. Its periodicity is apparent, and because it obviously is not sinusoidal, *it must be a complex periodic wave*. This wave is called a **sawtooth wave** because its form resembles the shape of the teeth of a saw. The sawtooth wave will be of interest to those who become interested in the study of speech acoustics because the waveform of a sawtooth wave resembles the waveform of the sound produced by the vibrating vocal folds.

The period of the fundamental frequency (the 1st harmonic) of the sawtooth wave in Figure 6–2 is 8 ms, which means that f_0 , the fundamental frequency, is 125 Hz. All components—the **harmonics**—of the sawtooth wave are *odd and even whole number (integer) multiples of the fundamental frequency (f_0)*. Thus, if the fundamental frequency is 125 Hz, the harmonic components are: 125 ($f_0 \times 1$), 250 ($f_0 \times 2$), 375 ($f_0 \times 3$), 500 ($f_0 \times 4$), 625 ($f_0 \times 5$) Hz, and so on for an infinite number of odd and even integer multiples.

We see in Table 6–1 that the lowest frequency, 125 Hz, is called the fundamental frequency, f_0 , and the 1st harmonic. The other components are labeled the 2nd harmonic, 3rd harmonic, 4th harmonic, and so on.

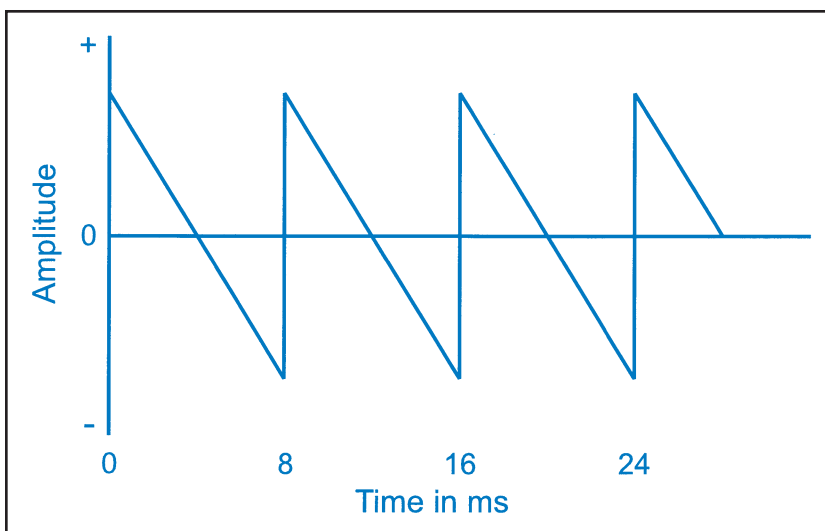


Figure 6–2. A complex periodic sound wave, called a **sawtooth wave**, because its shape resembles the teeth of a saw.

Table 6–1. Fundamental Frequency, Harmonics, Partials, and Overtones in a Complex Periodic Sound Wave

Frequency	Harmonic	Partial	Overtone
125 (f_0)	1	1	
250	2	2	1
375	3	3	2
500	4	4	3
625	5	5	4
750	6	6	5

Partials and Overtones

You occasionally will find that the components in a complex periodic wave are called **partials** or **overtones** instead of **harmonics**. Table 6–1 also shows the relations among those different labels.

The designations of **harmonic** and **partial** are synonymous *if all components are integer multiples of the fundamental frequency*: the 1st harmonic (the fundamental frequency) is the 1st partial, the 2nd harmonic is the 2nd partial, and so on. The word **overtone**, which is encountered in musical literature, derives from the fact that the complex wave can be described as consisting of a fundamental frequency, or fundamental tone, and a series of other tones whose frequencies *lie over* (*are higher than*) the fundamental. Thus, the 2nd harmonic is the 1st overtone, the 3rd harmonic is the 2nd overtone, and so on.

Summary

If a complex wave is periodic, the sinusoidal components *must be integer multiples of the fundamental frequency*. Thus, if $f_0 = 100$ Hz, the other components must be selected from 200 Hz, 300 Hz, 400 Hz, and so forth. When that occurs, the partials are harmonics, and *the sound wave is periodic*. At the end of one cycle of vibration (10 ms), one cycle of the 1st harmonic (100 Hz; $T = 10$ ms), two cycles of the 2nd harmonic (200 Hz; $T = 5$ ms), three cycles of the 3rd harmonic (300 Hz; $T = 3.33$ ms), four cycles of the 4th harmonic (400 Hz; $T = 2.5$ ms), and so on are completed.

Summation of Sine Waves

As more and more sine waves are added (summed) in the harmonic series, the shape of the resultant complex wave progressively changes.

The left side of Figure 6–3 shows four sine waves (S_1 , S_2 , S_3 , and S_4) that have different frequencies and amplitudes, but identical **starting phases** (180°).

Although the exact frequency of each wave to be summed is unimportant, an *appropriate frequency relation* among the four components is maintained; the three higher frequencies are harmonics of the lowest one—the fundamental frequency. Notice, however, in this example, all possible harmonics (integer multiples of the lowest frequency harmonic—the fundamental) are not present. In fact, we have used only *odd integer multiples*: the 1st (f_0), 3rd, 5th, and 7th harmonics. Thus, if the frequency of S_1 is 1000 Hz, the frequencies of the other sinusoids are 3000, 5000, and 7000 Hz; if the frequency of S_1 is 400 Hz, the frequencies of the other sinusoids are 1200, 2000, and 2800 Hz.

At the right of Figure 6–3 the sinusoidal components are summed progressively to form three different complex waves (C_1 , C_2 , and C_3). The complex wave at the top of the figure (C_1) results from $S_1 + S_2$. Wave C_1 is not a sinusoid—it is complex—and comprises two sinusoids that differ in amplitude and frequency, but with the same starting phases (180°).

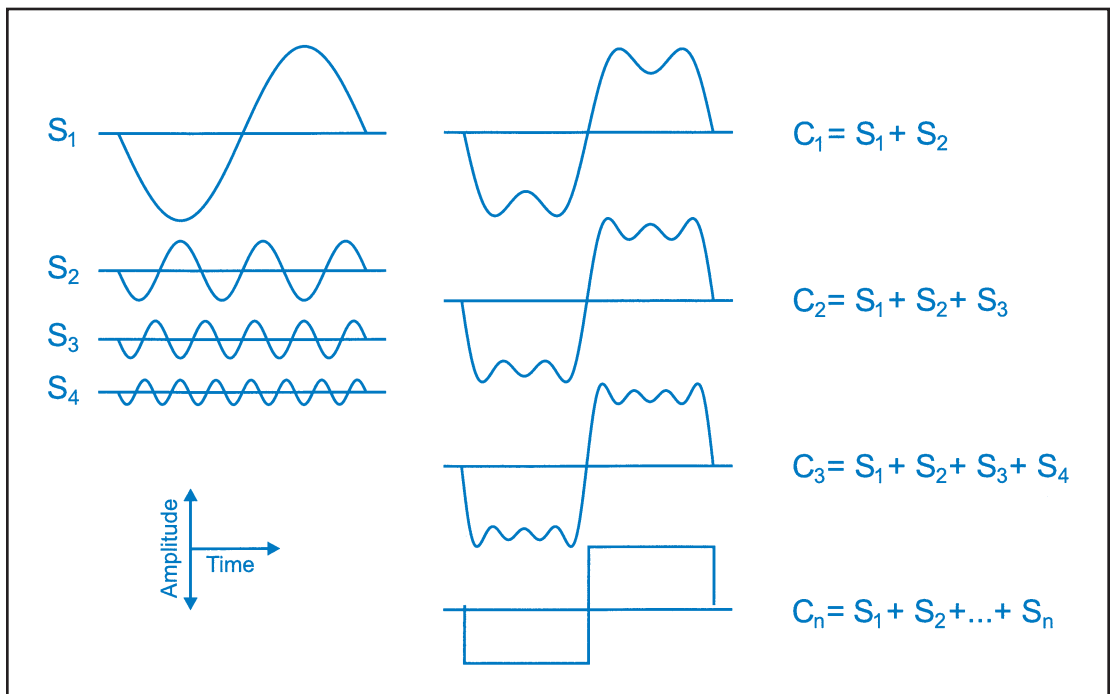


Figure 6–3. Summation of sine waves to form complex waves. The sine waves at the left (S_1 through S_4) are added progressively to form the complex waves (C_1 through C_3) at the right. At the bottom, a **square wave** is created by summation of an infinite number of sine waves of *appropriate amplitudes, frequencies, and starting phases*.

Although the **starting phases** of the two components of C_1 are identical (180°), the **instantaneous phases** of the two sinusoidal components vary from moment to moment because of their different frequencies. If the two sinusoidal components had any other values of frequency and amplitude, the complex wave that results would be different from the one shown as C_1 because the resultant wave depends on all the specific dimensions of the sine waves that compose it.

- Wave C_2 is different from C_1 because its shape results from summation of three sinusoids: $S_1 + S_2 + S_3$.
- Wave C_3 contains all four sinusoidal components. Notice that the complex waves at the right are becoming more and more “square” in shape as additional sine waves are added.
- At the bottom right of Figure 6–3 we see what happens when an *infinite number of odd integer multiples* ($1\times, 3\times, 5\times, 7\times, 9\times, + \dots + n\times$) with appropriate relative amplitudes and starting phases are combined.
- If, for example, the lowest frequency is 100 Hz, the other components are 300, 500, 700 Hz, and so on. A **square wave**, which is a periodic complex wave with a perfectly square shape, is created by *summing an infinite number of sinusoids with frequencies that are odd integer multiples of the fundamental frequency and that have appropriate relative amplitudes and identical starting phases*.

Figure 6–4 shows another example of summation of sine waves to form a different complex periodic wave. The left side of the figure shows three sine waves (S_1 , S_2 , and S_3) that have different frequencies and amplitudes but identical starting phases (0°). S_2 and S_3 are both *odd and even integer multiples of S_1* , not just odd integer multiples such as was shown in Figure 6–3. Thus, if S_1 is 1000 Hz, the two other components are 2000 Hz (S_2) and 3000 Hz (S_3).

At the right of Figure 6–4 we see the results of progressive summation of the sinusoidal components to form the complex waves ($C_1 = S_1 + S_2$ and $C_2 = S_1 + S_2 + S_3$). The two resultant waves, C_1 and C_2 , look different from the resultant waves seen in Figure 6–3 because both odd and even harmonics are included rather than just odd integer multiples of f_0 .

When an infinite number of *odd* harmonics with appropriate relative amplitudes and starting phases is summed, the result is the square wave that was shown in Figure 6–3. When an infinite number of both *odd and even* harmonics with appropriate relative amplitudes and starting phases is summed, the result is a sawtooth wave such as that shown at the bottom right of Figure 6–4. In fact, the complex wave that results from summation of only three components, (C_2), in Figure 6–4 is beginning to assume a sawtooth shape.

In the two examples cited thus far, the **starting phases** were identical for the individual components. Variations in the resultant wave

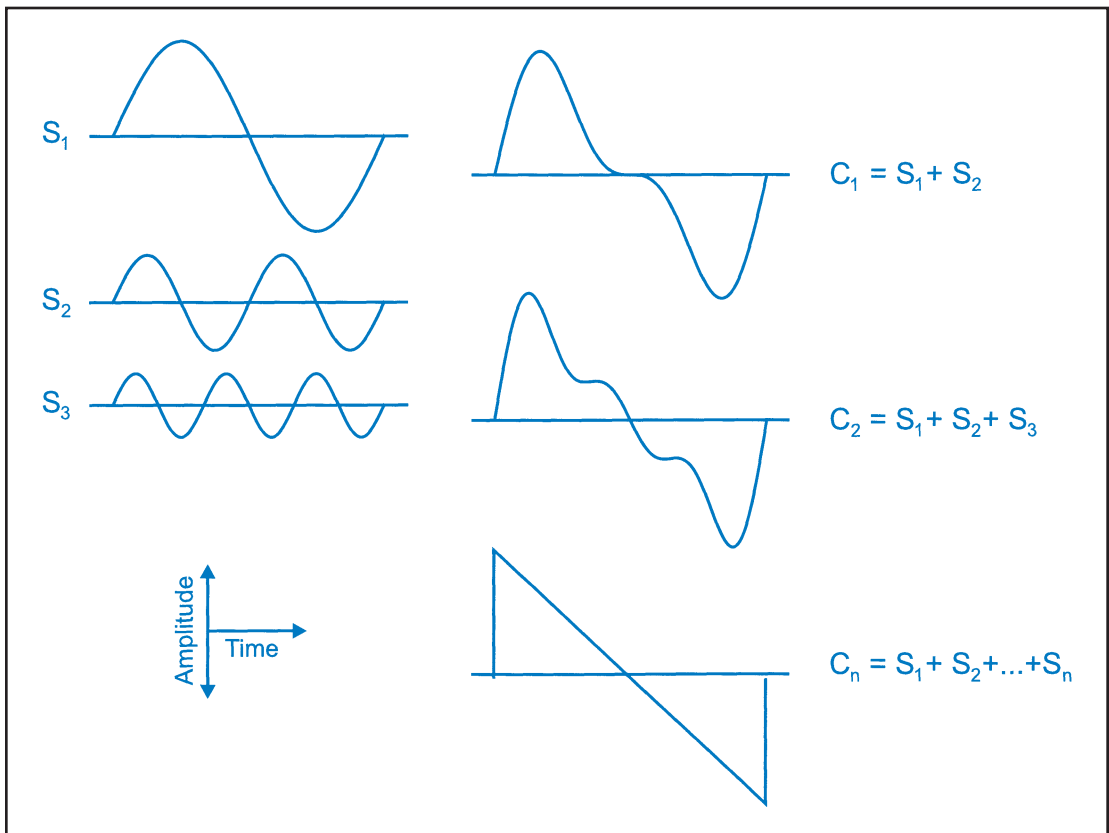


Figure 6-4. Summation of sine waves to form a **complex periodic wave**. Because the shape of this complex wave is different from those shown previously, the parameter values for amplitude, frequency, and starting phase of the sinusoidal components must also be different.

also occur if the starting phases of the components are changed while holding their amplitudes and frequencies constant, as is illustrated in Figure 6-5.

The left side of Figure 6-5 shows the sinusoidal components that are summed to form the complex waves at the right of the figure. The starting phase of S_1 (solid lines) is 0° in all four panels. Four different starting phases are used for S_2 (dashed lines): 0° (panel A); 90° (panel B); 180° (panel C); and 270° (panel D). The four complex waves at the right of Figure 6-5 differ from one another, and the differences are attributable *exclusively* to the differences in the starting phases of S_2 .

APERIODIC WAVES

An important characteristic of **complex periodic waves** is their regularity over time. They repeat themselves indefinitely. A **complex aperiodic wave** is a second category of complex waveform, and its

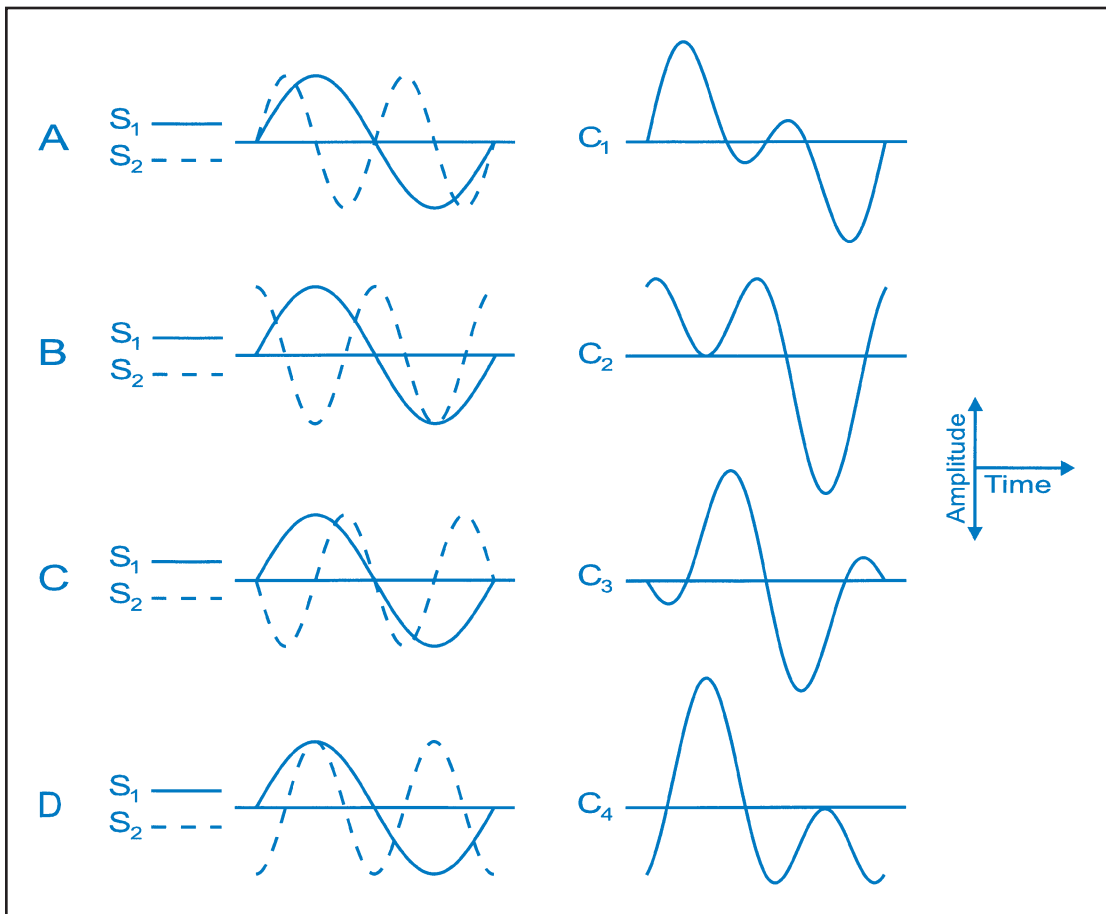


Figure 6–5. Effects of variations in **starting phase** on the shape of a complex wave that results from summation of sine waves.

name derives from an absence of periodicity. Thus, it is very difficult, and ordinarily impossible, to predict what the wave will look like during one interval of time from knowledge of its characteristics during a preceding time interval of equal duration.

The vibratory motion of an aperiodic wave is **random**, and therefore unpredictable, and vibratory motions of this type are called **random time functions**. This is called an **aperiodic sound wave**. Figure 6–6 shows an aperiodic, or random, wave and it is virtually impossible (except by chance) to identify any two time intervals of equal duration in which the characteristics of the vibratory motion are identical in all respects.

We encounter aperiodic sound waves daily. Familiar examples are the noises from aircraft, automobiles, or speedboats. Each of those sounds is characterized by random vibratory motion—aperiodicity—but that does not mean that all aperiodic sounds are unpleasant. The water cascading down the side of a mountain produces an aperiodic sound wave, but under the right circumstances it might pro-

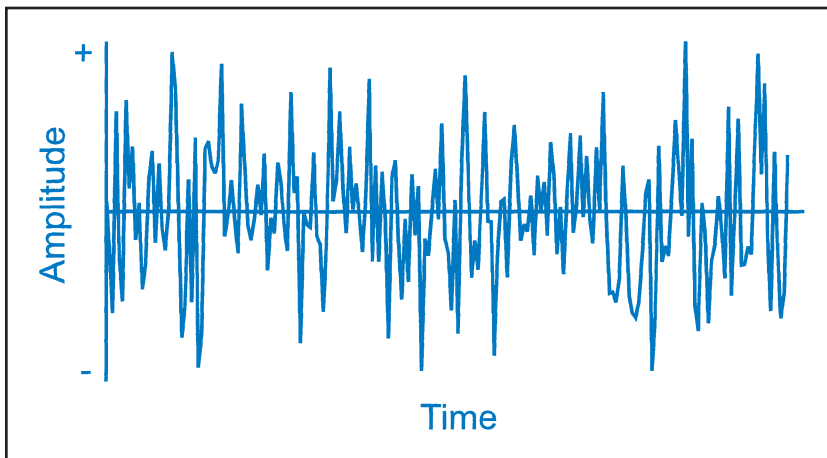


Figure 6-6. An **aperiodic** sound wave.

duce a very satisfying sensation. Many of the sounds of speech (“sh” in “she”; “s” in “see”; “f” in “foolish”; “th” in “three”; and so on) are characterized by random vibratory motion, but we usually don’t think of such sounds as “noise.”

■ WAVEFORM AND SPECTRUM

Graphic representations are used to describe three important characteristics of both periodic and aperiodic sound waves:

1. The **waveform** in the time domain, which also is called the **time domain representation**;
2. The **amplitude spectrum** in the frequency domain, which also is called the **frequency domain representation**;
3. The **phase spectrum** in the frequency domain

Waveform

The graphic representations of periodic (both sinusoidal and complex) and aperiodic waves that have been shown to this point have focused on the **waveform**. By that we mean, we plotted *changes in one variable* (pressure, velocity, acceleration, displacement, and so forth) *as a function of time*. The waveform defines, for example, the distribution of instantaneous amplitudes, or the changing values of pressure, velocity, or acceleration of a sinusoidal or complex wave over time.

Look again at the waveforms for the sawtooth wave in Figure 6-2 and the square wave in Figure 6-3. We can identify the **fundamental period** (T) of each wave, and from that we can calculate the **fundamental frequency** (f_0). Unless we happen to remember, however, that

the *square wave consists of all odd harmonics* and that the *sawtooth wave consists of all odd and even harmonics*, we have no way of knowing what **frequencies** other than the fundamental are present by visual examination of the waveform.

We also cannot determine the **amplitudes** or the **starting phases** of the sinusoidal components by visual inspection of the waveform. We shall see subsequently that both the square wave and the sawtooth wave must satisfy very specific requirements relative to both the amplitudes and starting phases of the components, but the point we wish to emphasize now is that visual inspection of the waveform does not reveal sufficient details about these important dimensions of the sinusoidal components—*even though all that information resides in the waveform*. If the amplitude and phasic information resides in the waveform, we simply need a method for extracting it and displaying it graphically.

Amplitude Spectrum

An alternative to the **waveform** is to display the amplitude spectrum in the frequency domain, or just **amplitude spectrum**. Whereas the waveform shows instantaneous magnitudes such as *amplitude as a function of time*, the **amplitude spectrum** shows **amplitude** (in either absolute or relative values) as a function of **frequency**.

In Figure 6–7, the **waveforms** of the sawtooth and the square wave are shown at the left and their respective **amplitude spectra** are shown at the right.

- The amplitude spectra at the right are shown by plotting *relative amplitude in dB as a function of frequency*.
- The location of each vertical line along the horizontal axis indicates the frequency of that component, and the height of each line is proportional to its relative amplitude.
- Zero dB represents the amplitude of the component with the greatest energy (in this case, the fundamental frequency, f_0), and all other amplitudes therefore are shown as negative because their amplitudes are less than the amplitude of the fundamental; they are shown in *dB relative to the amplitude of the fundamental*.

The **spectral envelopes** in Figure 6–7 are shown by dashed lines that connect the peaks of each of the vertical lines. We see that the **square wave** has energy at all *odd harmonics* ($1/T$, $2/T$, $3/T$, $4/T$ etc.) and the **sawtooth wave** has energy at all *odd and even harmonics*.

Inspection of amplitude spectra reveals information that, although present in the waveform, is not readily apparent by visual inspection. You might have noticed that when the sinusoidal components in Figure 6–3 were summed to create a square wave, the amplitudes of the components decreased with increasing frequency. That also can be seen in the amplitude spectrum for the square wave

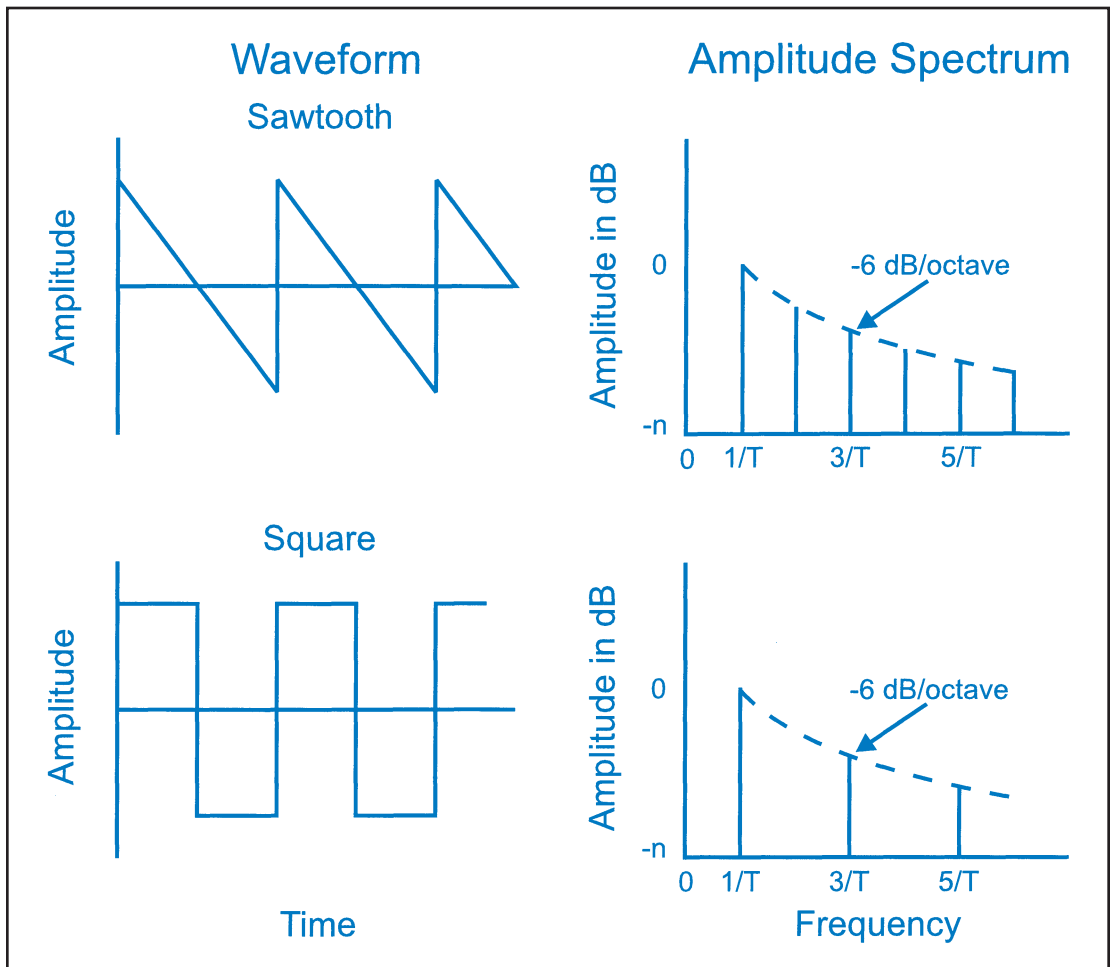


Figure 6-7. A comparison of **waveforms** and **amplitude spectra** for sawtooth and square waves, each of which is a complex periodic wave. Each spectrum is a **line spectrum**.

in Figure 6-7, and now the relation among the amplitudes of the components can be easily seen. For the square wave, the **spectral envelope** in the frequency domain *decreases at a rate of 6 dB per octave; the slope of the spectral envelope is -6 dB per octave.*

An **octave** refers to a doubling or halving of frequency ($2f$ or $1/2f$).²

- Thus, 250 Hz is one octave above 125 Hz; 500 Hz is one octave above 250 Hz and two octaves above 125 Hz.
- An octave always refers to a frequency ratio of 2:1 or 1:2, not to a frequency difference.
- Thus, 2000 Hz is one octave above 1000 Hz and 200 Hz is one octave above 100 Hz because, in both cases, a ratio of 2:1 exists. The fact that the frequency difference is 1000 Hz in one case but 100 Hz in the other is irrelevant.

The white keys on a piano correspond to the musical notes that are designated “A, B, C, D, E, F, G, A.” The lowest note is A_1 , which has a frequency of 27.5 Hz. Seven white keys to the right of A_1 is A_2 , which has a frequency of 55 Hz. Thus, A_2 is one octave above A_1 . At the extreme right of the keyboard is A_8 , which has a frequency of 3520 Hz ($27.5 \times 2^7 = 3520$). Thus, A_8 is one octave above A_7 , which has a frequency of 1760 Hz, and seven octaves above A_1 .

Another example of octave relations can be seen by returning to Table 6–1, where the harmonic components of a complex periodic wave are listed. There we see that the 2nd harmonic is one octave above the 1st harmonic, and conversely, the 1st harmonic is one octave below the 2nd. The 4th harmonic is one octave above the 2nd and two octaves above the 1st, and so forth. In each case, the frequency ratio is either 2:1 or 1:2.

Line Spectra

The amplitude spectra in Figure 6–7 are **line amplitude spectra**, or just **line spectra**; the sinusoidal components of the complex periodic waves are represented by *a set of vertical lines with increasing frequency from left to right*. The location of a line in the frequency domain (horizontal axis) identifies the frequency of that component, and the height of the line along the amplitude scale (vertical axis) identifies the amplitude.

With a line spectrum, *energy is present only at frequencies represented by the vertical lines*. For example, even though the *spectral envelope* for the sawtooth wave at the top of Figure 6–7 is shown by a line that connects the harmonics, *there is no energy at frequencies between adjacent components*.

The line amplitude spectrum for the square wave at the bottom of Figure 6–7 displays the same kind of information. The spectrum for the square wave is different, however, from the spectrum for the sawtooth wave. The amplitude spectrum for a square wave is a line amplitude spectrum, *but energy is present only for odd harmonics of the fundamental*. Notice, though, that the spectral envelope has a slope of –6 dB per octave—the same as the slope for a sawtooth wave.

- How can we say that the slope for a sawtooth wave is –6 dB per octave *when there are no octave relations in the square wave*: the frequency ratios are 3:1; 5:1; 5:3, and so forth?
- We know the amplitude of the lowest frequency is 0 dB. We measure or calculate the amplitudes of the higher odd harmonics and connect their peaks with a line to display the envelope.
- Then, by mathematical interpolation we compute what the amplitude of even harmonics *would have been if energy had been present* at those even harmonics. From that we have, for example, the known amplitude of f_0 and the interpolated

amplitude of the missing 2nd harmonic. That represents an octave, and we now have a means of confirming that the amplitude of the missing 2nd harmonic is indeed -6 dB re: the amplitude of f_0 .

Continuous Spectra

The random, or aperiodic, waveform of the noise in Figure 6–6 is shown again in Figure 6–8 along with its amplitude spectrum. The result is a **continuous amplitude spectrum**, or just **continuous spectrum**, in contrast to the line spectrum shown previously.

- A **continuous spectrum** is one in which *energy is present at all frequencies between specified lower and upper frequency limits.*
- Thus, the complex aperiodic wave does not result from summation of a harmonic series—*energy is present at all frequencies between some lower and upper limits.*

In the case of the noise shown in Figure 6–8, energy is present at all frequencies and the slope of the spectral envelope is 0 dB. In other words, an identical amount of energy is present at all frequencies throughout the range. Equal energy at all frequencies, however, is not a requirement for all aperiodic waveforms, and subsequently we shall describe different types of aperiodic waveforms and their corresponding amplitude spectra.

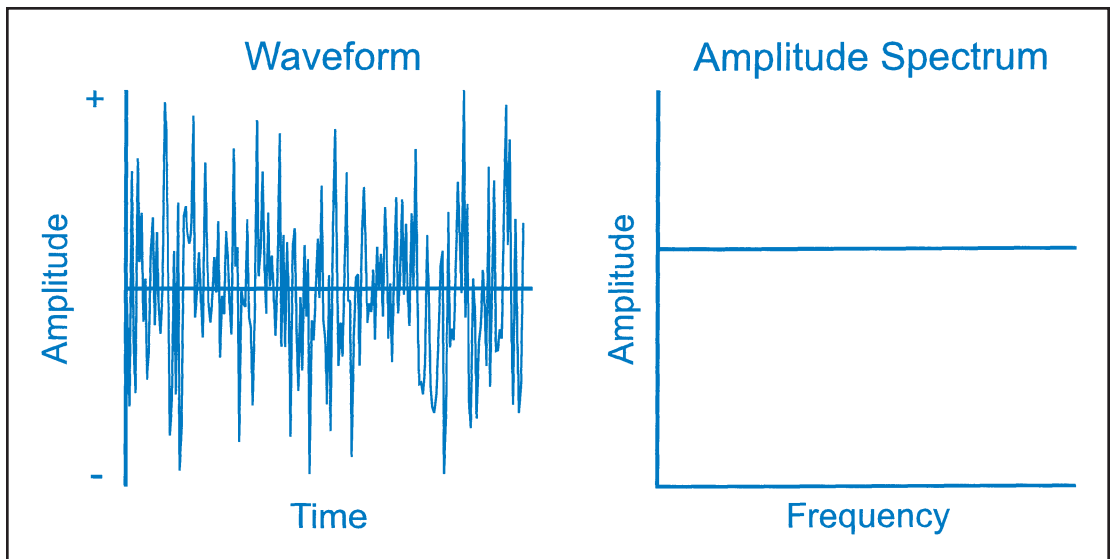


Figure 6–8. Waveform and amplitude spectrum for a complex aperiodic wave. The spectrum is a **continuous spectrum**.

Phase Spectrum

In addition to the **amplitude spectrum** of a sound wave, we also can describe the **phase spectrum in the frequency domain**, or just the **phase spectrum**. Whereas the amplitude spectrum describes relative amplitude as a function of frequency, the **phase spectrum** defines *the starting phase as a function of frequency*. The combination of the amplitude spectrum and the phase spectrum completely defines the waveform in the frequency domain.

■ EXAMPLES OF COMPLEX SOUND WAVES

Examples of four different complex signals, periodic and aperiodic, are shown in Figure 6–9 and are compared with the familiar sine wave at the top of the figure. The panels at the left show the **waveforms**, the middle panels show the **amplitude spectra**, and the panels at the right show the **phase spectra**.

Sawtooth Wave

A **sawtooth** wave is *a complex periodic wave with energy at all odd and even integer multiples of the fundamental frequency*. We can see from Figure 6–9 that the amplitudes of the sinusoidal components decrease with increasing frequency.

- Specifically, the amplitudes decrease as the inverse (the reciprocal) of the harmonic number.
- Thus, the amplitudes, expressed in voltage, of the harmonics are: $\mathbf{h_1(f_0) = 1/1}$; $\mathbf{h_2 = 1/2}$; $\mathbf{h_3 = 1/3}$; $\mathbf{h_4 = 1/4}$. . . $\mathbf{h_n = 1/n}$.
- If the amplitude of $\mathbf{h_1(f_0) = 2\text{ V}}$, the amplitudes of the next three higher harmonics are: $\mathbf{h_2 = 1\text{ V}}$ ($1/2 \times 2$); $\mathbf{h_3 = 0.67\text{ V}}$ ($1/3 \times 2$); and $\mathbf{h_4 = 0.5\text{ V}}$ ($1/4 \times 2$).

The relative amplitude, in decibels, for each component frequency is given by:

$$\text{Equation 6–1} \quad \text{dB} = 20 \log_{10} \frac{1}{h_i},$$

where $\mathbf{h_i}$ is the harmonic number.

By **harmonic number** ($\mathbf{h_i}$) we mean 1st harmonic ($\mathbf{h_1}$), 2nd harmonic ($\mathbf{h_2}$), and so on to the $\mathbf{n}$ th harmonic. Thus, $\mathbf{h}$ always is an integer: 1, 2, 3, and so forth. By application of Law 4 of logarithms ($\log 1/a = -\log a$), Equation 6–1 is simplified to read:

$$\text{Equation 6–2} \quad \text{dB} = -20 \log_{10} h_i.$$

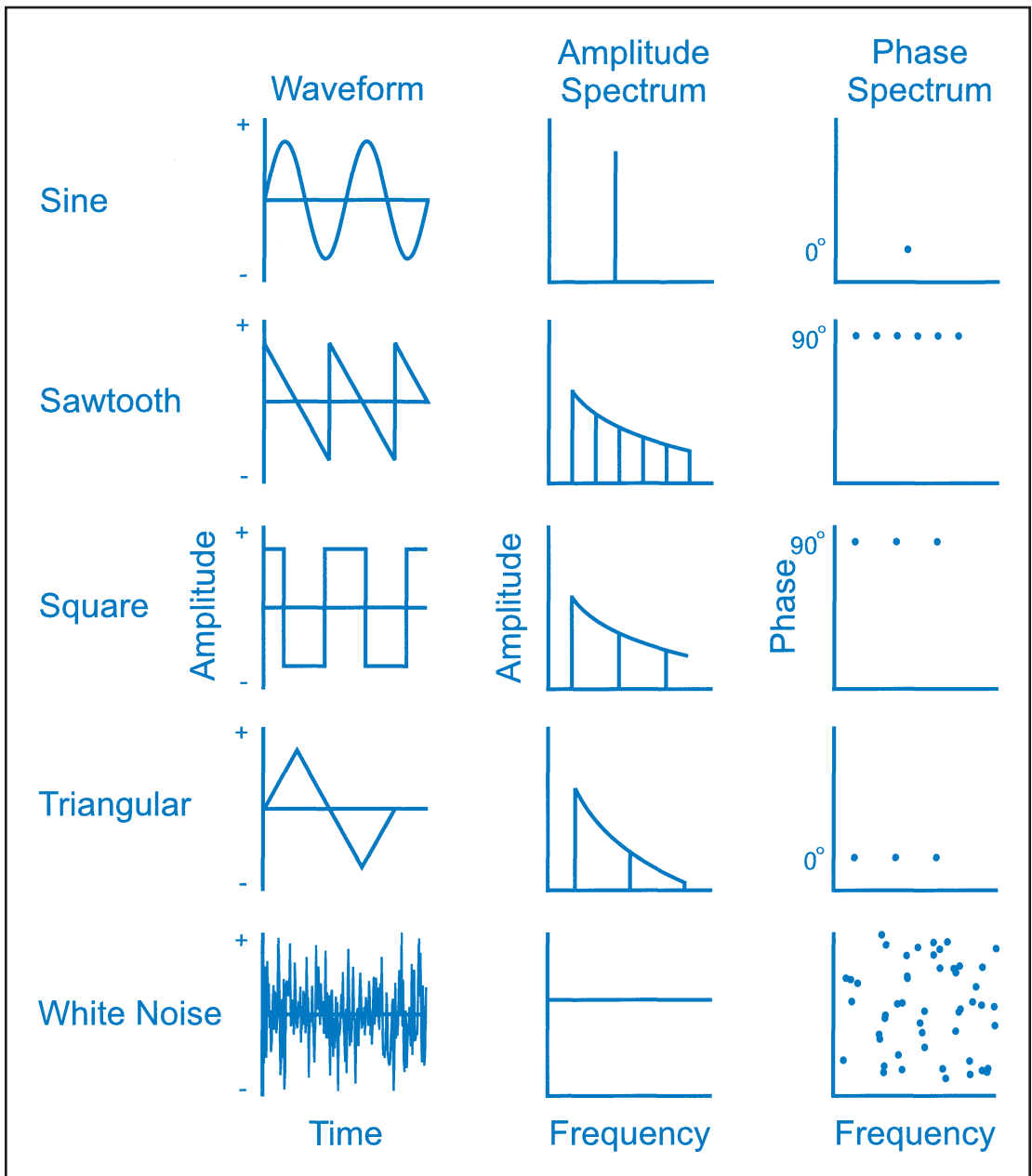


Figure 6-9. A comparison of **waveforms**, **amplitude spectra**, and **phase spectra** for a sine wave, sawtooth wave, square wave, triangular wave, and white noise.

The left-hand column of Table 6-2 lists the harmonic number of the first nine harmonics of a sawtooth wave. The rms voltage (the electrical analog of acoustic pressure) of the fundamental frequency, f_0 , is set arbitrarily to be 2 V. The voltage of each of the harmonics is

Table 6–2. Amplitudes (in Voltage) of Sinusoidal Components of a **Sawtooth** Wave in which the Amplitude of the Fundamental Frequency is 2 V

Harmonic Number	rms Voltage $\left(\frac{1}{h_i} \times 2\right)$	dB = $-20 \log_{10} h_i$
1 (f_0)	$\frac{1}{1} \times 2 = 2$	0
2	$\frac{1}{2} \times 2 = 1$	–6
3	$\frac{1}{3} \times 2 = .67$	–9.5
4	$\frac{1}{4} \times 2 = .50$	–12
5	$\frac{1}{5} \times 2 = .40$	–14
6	$\frac{1}{6} \times 2 = .33$	–15.6
7	$\frac{1}{7} \times 2 = .29$	–16.9
8	$\frac{1}{8} \times 2 = .25$	–18.1
9	$\frac{1}{9} \times 2 = .22$	–19.1

listed in the middle column. Thus, the 2nd harmonic is **1 V** ($1/2 \times 2 = 1$ V), the 3rd harmonic is **0.67 V** ($1/3 \times 2 = 0.67$), the 4th harmonic is **0.5 V** ($1/4 \times 2 = 0.5$), and so on until the 9th harmonic, where the voltage is **0.22 V** ($1/9 \times 2 = 0.22$).

Notice that *the voltage is halved with each doubling of frequency*. Thus, between the 1st and 2nd harmonics, the voltage decreases from 2 V to 1 V. The voltage also is halved between the 2nd and 4th, the 3rd and 6th, and 4th and 8th harmonics. Recall from Chapter 5 that halving of acoustic pressure or electrical voltage corresponds to a change in amplitude of –6 dB:

$$\text{dB} = 20 \log_{10} \frac{1}{2} = -6 \text{ dB}.$$

Therefore, because each doubling of frequency corresponds to an octave, and for each octave increase the amplitude decreases by 6 dB, *the spectral envelope has a slope of –6 dB per octave*. The right-hand

column of Table 6–2 expresses the amplitude of each harmonic in decibels *re: the amplitude of the fundamental frequency*. Thus, for example, with the aid of Equation 6–2 we calculate that the level of the 5th harmonic is –14 dB:

$$\text{dB} = -20 \log_{10} 5 = -14 (-13.98) \text{ dB.}$$

It is important to recognize that the *absolute amplitude* (voltage in this case) for each harmonic listed in column 2 of Table 6–2 *depends on the absolute voltage of the fundamental frequency*. You might wish to try a few calculations and confirm that if the voltage of the fundamental of the sawtooth waveform is 1 V rather than 2 V, the voltages of the eight higher harmonics are: 0.5 V, 0.33 V, 0.25 V, 0.2 V, 0.17 V, 0.14 V, 0.13 V, and 0.11 V.

The absolute voltage of each of the harmonics in the sawtooth wave depends on the absolute voltage of the fundamental. The relative amplitude, in decibels, for each of the harmonics in a sawtooth wave, however, *is independent of the voltage of the fundamental frequency*. In other words, the level of the 2nd harmonic always is –6 dB, the level of the 3rd harmonic always is –9.5 dB, and so on.

If you continue the computations by calculating $-20 \log_{10} h_i$ (Equation 6–2) for each of the calculations that you just made for the case where $f_0 = 1$ V, you should obtain the same answers for a fundamental frequency of 1 V (subject to rounding error) that appear in the third column of Table 6–2 for a fundamental frequency of 2 V. For example, the relative level of the 5th harmonic still is –14 dB.

What does the amplitude spectrum of a sawtooth wave “look like”?

- It is a **line spectrum** because energy exists only at discrete frequencies that are integer multiples of the fundamental, or lowest, frequency.
- The shape of the **spectral envelope** depends on the scales that are adopted for frequency on the horizontal axis (abscissa) and voltage on the vertical axis (ordinate). For example, in panel A of Figure 6–10 the voltage of each harmonic (from column 2 of Table 6–2) is plotted as a function of harmonic number. *The scales for both the y-axis and x-axis are linear. The shape of the spectral envelope that results from this linear-linear plotting is curvilinear.*

In panel B of Figure 6–10, for the same sawtooth wave, both coordinates—the ordinate and the abscissa—are logarithmic. The amplitude scale is logarithmic because we have plotted relative amplitudes in decibels *re: the amplitude of the fundamental frequency* (f_0). Frequency also is plotted on a logarithmic scale.

- *The shape of the spectral envelope that results from log-log plotting is linear.*
- You should see, however, that regardless of whether the amplitude spectrum is plotted with linear-linear (panel A) or

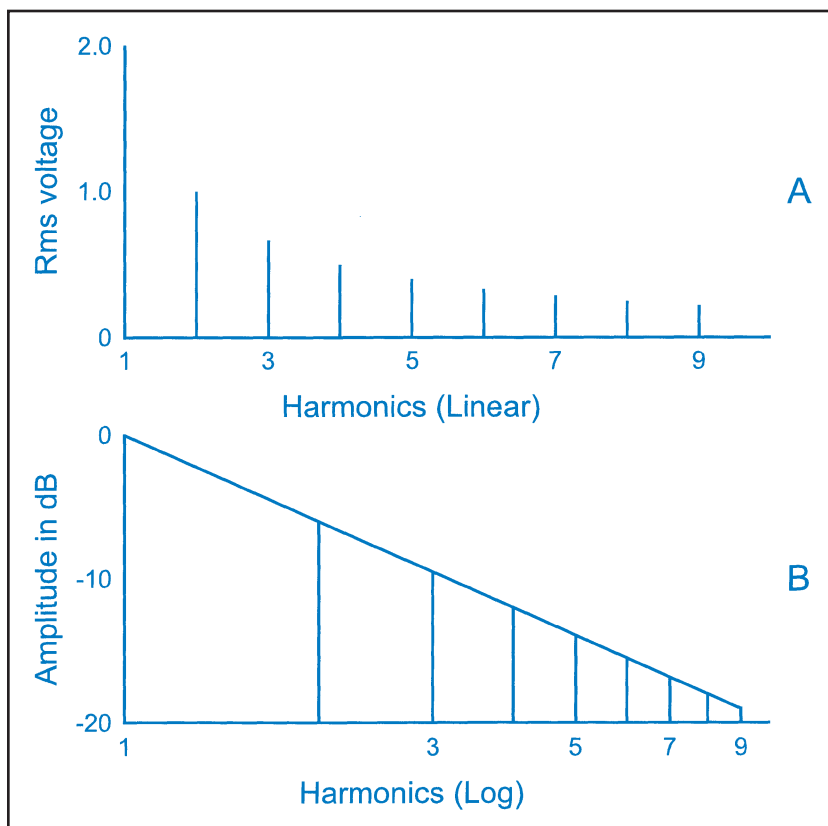


Figure 6-10. Amplitude spectra for a sawtooth wave. In panel A, the measure of amplitude is rms voltage, and the **spectral envelope** is curvilinear. In panel B, the measure of amplitude is decibels, frequency is shown on a logarithmic scale, and the spectral envelope is linear.

log-log (panel B) coordinates, the slope of the spectral envelope is -6 dB per octave; the amplitude decreases by 6 dB with each doubling of frequency.

In summary, a **sawtooth wave** is a complex periodic wave with energy at odd and even integer multiples of the fundamental frequency with a spectral envelope slope of -6 dB per octave. In Figure 6-9, each of the sinusoidal components (harmonics) of the sawtooth wave has a starting phase of 90° . The starting phase could just as well be 180° , 0° , or 270° . It is essential, however, that the starting phases of all frequency components be identical.

Square Wave

A **square wave** also is a complex periodic wave, but it has energy only at odd integer multiples of the fundamental frequency. We can

see from Figure 6–9 that the amplitudes of the sinusoidal components decrease with increasing frequency, just as they did with the sawtooth wave. Moreover, we shall see that the slope of the spectral envelope of a square wave is identical to the slope of the envelope for a sawtooth wave; the amplitudes of the frequency components also *decrease as the reciprocal of the harmonic number*.

The first column of Table 6–3 lists the harmonic number of five odd harmonics (1, 3, 5, 7, and 9) of a square wave. The voltage of the fundamental frequency is 2 V so that the amplitude spectrum for the square wave can easily be compared with the spectrum for the sawtooth wave. We can see that the decrease in voltage for each of the harmonics is identical to the decrease seen for the *same harmonics* in Table 6–2 for the sawtooth wave. For example, the level of the 5th harmonic is –14 dB re: the level of the fundamental frequency for both the square wave and the sawtooth wave.

With the aid of Equation 6–2, try to perform another set of computations with a voltage other than 2 V for the fundamental. Your answers for *absolute voltage* should differ from those in the second column in Table 6–3, but you should obtain the same answers in *decibels* that appear in the third column. For example, if the voltage of the fundamental frequency increases from 2 V to 3 V, the voltage of the 5th harmonic is 0.6 V, but *its relative level in decibels is still –14 dB*. The slope of the envelope of the square wave is still –6 dB per octave even though the amplitude of the fundamental frequency has changed.

In summary, a **square** wave is *a complex periodic wave with energy at odd integer multiples of the fundamental with a spectral envelope slope of –6 dB per octave*. The amplitude spectrum of a

Table 6–3. Amplitudes (in Voltage) of Sinusoidal Components of a **Square** Wave in which the Amplitude of the Fundamental Frequency is 2 V

Harmonic Number	rms Voltage $\left(\frac{1}{h_i} \times 2\right)$	dB = $-20 \log_{10} h_i$
1 (f_0)	$\frac{1}{1} \times 2 = 2$	0
3	$\frac{1}{3} \times 2 = .67$	–9.5
5	$\frac{1}{5} \times 2 = .40$	–14.0
7	$\frac{1}{7} \times 2 = .29$	–16.9
9	$\frac{1}{9} \times 2 = .22$	–19.1

square wave is a **line spectrum**. In the example shown in Figure 6–9, each of the components has a starting phase of 90° , but that is not a requirement. Those who read other introductory reference books or chapters will encounter what, at first glance, might appear to be inconsistencies. Hirsh (1952), for example, shows all components of the square wave to have 0° starting phase, whereas Yost and Nielson (1977) show all starting phases to be 90° .

Figure 6–11 should resolve any confusion. In panel A, the starting phase of each of the harmonics is 0° , and the corresponding waveform begins its first excursion upward (with an infinitely steep slope) from 0° . In panel B, the starting phase of each of the harmonics is 90° , and only half of the first positive-going excursion of the waveform is shown. The result in either instance is still a square wave as long as only harmonics that are odd integer multiples of f_0 are present, all harmonics have *the same* starting phase, and the slope of the spectral envelope is -6 dB per octave.

Triangular Wave

The **triangular wave** shown in Figure 6–9 is a complex periodic wave with energy at *odd integer multiples of the fundamental frequency*. How, then, does a triangular wave differ from a square wave that also has energy at odd integer multiples of the fundamental frequency?

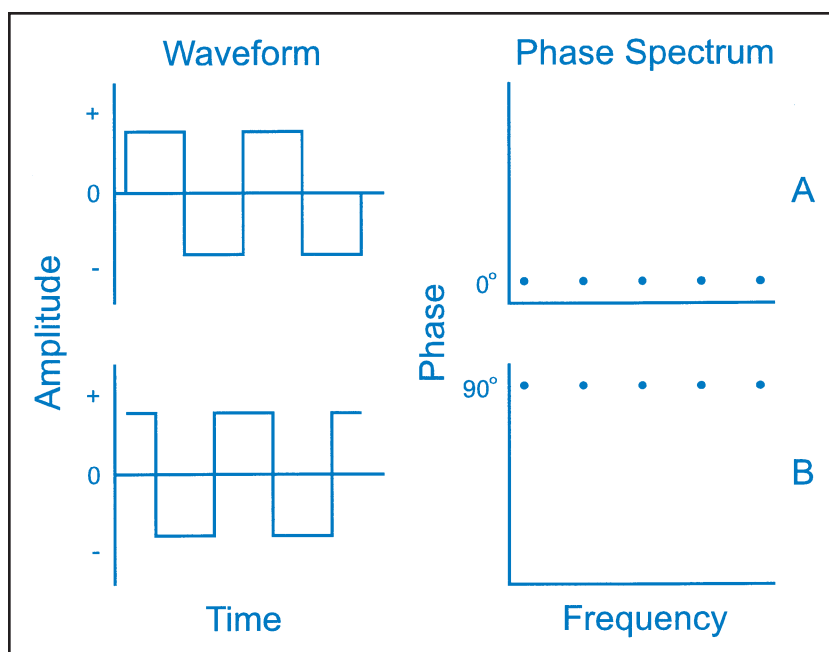


Figure 6–11. A comparison of waveforms and phase spectra for two square waves with different **starting phases**, 0° in panel A and 90° in panel B.

- The first column of Table 6–4 lists the first five harmonics of a triangular wave for which the amplitude of the fundamental frequency has arbitrarily been set at 2 V for easy comparison with calculations for the sawtooth and square waves.
- The amplitudes of the frequency components of the triangular wave decrease at a greater rate than was seen for either the sawtooth wave or the square wave, in which the amplitudes decrease as the reciprocal of the harmonic number.
- The amplitudes of a triangular wave decrease as the *reciprocal of the square of the harmonic number*, and the relative amplitudes in decibels are given by:

$$\text{dB} = 20 \log_{10} \frac{1}{h_i^2}, \quad \text{Equation 6-3}$$

where h_i is the harmonic number.

Equation 6–3 is simplified by applying Law 3 of logarithms ($\log a^b = b \log a$) and Law 4 of logarithms ($\log 1/a = -\log a$):

$$\text{dB} = -40 \log_{10} h_i. \quad \text{Equation 6-4}$$

Consider, for example, the 3rd harmonic. For sawtooth and square waves, the amplitude of the 3rd harmonic is –9.5 dB re: the amplitude of the fundamental frequency because:

$$\text{dB} = -20 \log_{10} 3 = -9.5 \text{ dB}.$$

Table 6–4. Amplitudes (in Voltage) of Sinusoidal Components of a **Triangular** Wave in which the Amplitude of the Fundamental Frequency is 2 V

Harmonic Number	rms Voltage $\left(\frac{1}{h_i^2} \times 2\right)$	$\text{dB} = -40 \log_{10} h_i$
1 (f_0)	$\frac{1}{1^2} \times 2 = 2$	0
3	$\frac{1}{3^2} \times 2 = .22$	–19.5
5	$\frac{1}{5^2} \times 2 = .08$	–29
7	$\frac{1}{7^2} \times 2 = .04$	–33.8
9	$\frac{1}{9^2} \times 2 = .025$	–38.2

In contrast, the level of the 3rd harmonic of a triangular wave is -19.1 dB re: the level of the fundamental frequency because:

$$\text{dB} = -40 \log_{10} 3 = -19.1 \text{ dB}.$$

Thus, the slope of the spectral envelope of a triangular wave is twice as steep, -12 dB per octave, as it is for the sawtooth and square waves.

In summary, a **triangular wave** is a *complex periodic wave with energy at odd integer multiples of the fundamental and a spectral envelope slope of -12 dB per octave*. The amplitude spectrum of a triangular wave is a line spectrum because its waveform is periodic. Triangular and square waves are both characterized by only odd harmonics, but the slope of the envelope is -6 dB per octave for the square wave and -12 dB per octave for the triangular wave. For the example shown in Figure 6–9, all frequency components have a starting phase of 0° .

Pulse Train

Panel A of Figure 6–12 shows a **pulse train**, a repetitious series of rectangular shaped “pulses” of some defined width or duration (P_d). The repetition occurs at a regular rate, which is to say that *it is periodic*. For the example in the figure, the interval between the onset of one pulse and the onset of the next pulse is 10 ms. That interval defines the period (T) of the pulse train.³ By calculating the reciprocal of the period ($1/T$), we obtain the frequency of the pulse train, which for the example in the figure is 100 Hz. This is called the **pulse repetition frequency**.

The pulse train is a *complex periodic waveform*. Therefore, there is energy only at *harmonics* of the pulse repetition frequency: 100 Hz, 200 Hz, 300 Hz, and so on. Panel B of Figure 6–12 shows the amplitude spectrum of the pulse train with frequency plotted on a linear scale.

- First, note that the component with the greatest amplitude corresponds to 0 Hz, which refers to a *dc* (direct current) component of the signal. Recall from Chapter 2 that direct current means that current is flowing only in a single direction, either positive or negative, in contrast to alternating current that alternates back and forth between positive and negative directions.
- Second, notice the irregularly shaped spectral envelope with *lobes* and *valleys* in panel B of Figure 6–12. *Each valley, or “null,” occurs at integer multiples of the reciprocal of the pulse duration, P_d* . Thus, there are nulls at frequencies corresponding to $1/P_d$, $2/P_d$, $3/P_d$, and so on.
- The duration of each pulse in the figure is 2 ms. Therefore, the first null appears at **500 Hz** ($1/0.002 = 500$ Hz), the next null appears at **1000 Hz** ($2/0.002 = 1000$ Hz), and so on.

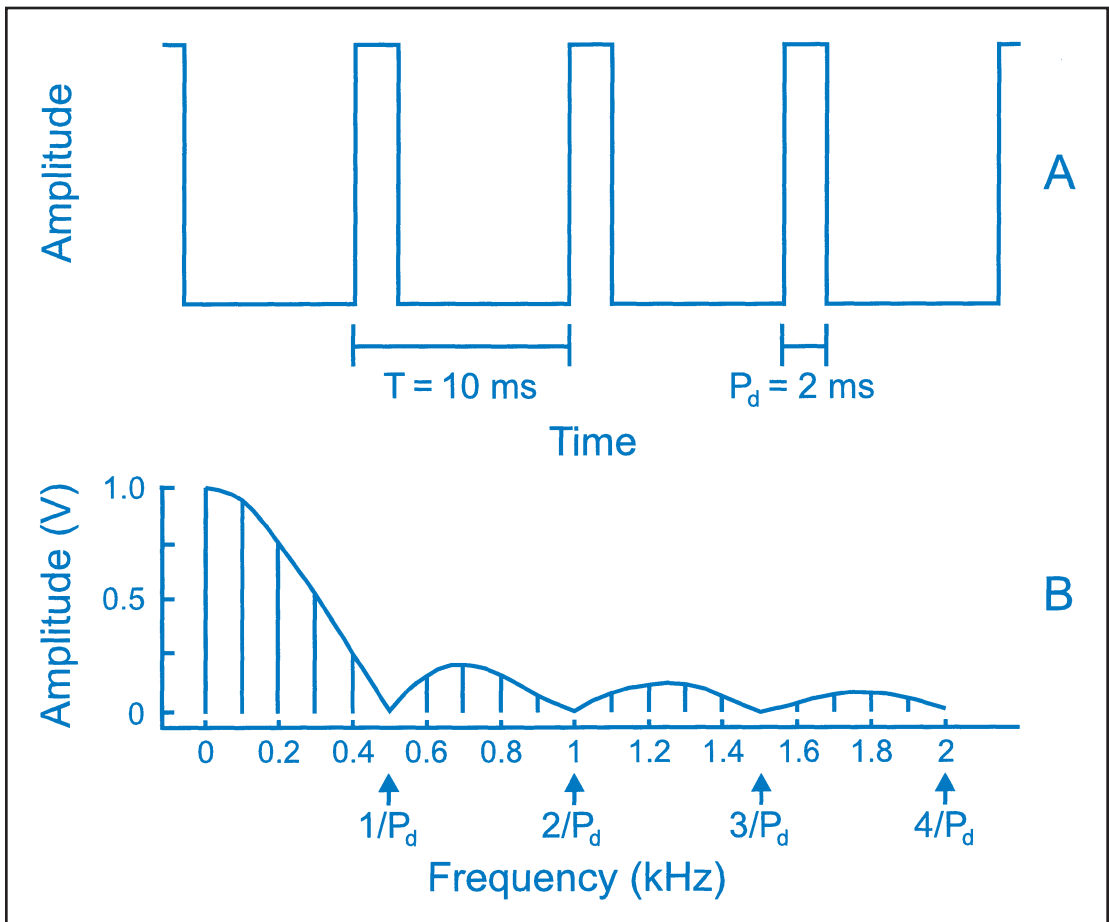


Figure 6-12. Panel A shows a **pulse train** with period = 10 ms and pulse duration (P_d) = 2 ms. Panel B shows the amplitude spectrum corresponding to the waveform in panel A. **Harmonics** are present at integer multiples of the pulse repetition frequency (100 Hz), and **nulls** are present at integer multiples of the reciprocal of pulse duration.

The relation among the starting phases of the frequency components is more complicated than the relation observed for the sawtooth, square, and triangular waves. The components within the first lobe below the first null at 500 Hz have a starting phase of 0° , the components within the second lobe between the first and second nulls (500 Hz and 1000 Hz) have a starting phase of 180° , and the pattern continues to alternate in this fashion from lobe to lobe as frequency increases. Because a pulse train is a *complex periodic wave*, the amplitude spectrum is a **line spectrum**.

White, or Gaussian, Noise

White, or Gaussian, noise, which also was shown in Figure 6-9, has an *aperiodic waveform with equal energy within any frequency*

band 1 Hz wide (from $f - 0.5$ Hz to $f + 0.5$ Hz) and with all phases present in random array. It is called **white noise** to be analogous to white light, which has equal energy in all light wavelengths.

The reason white noise also is called **Gaussian noise** is slightly more complicated. A random time function is described by a **cumulative probability distribution**, which plots the percentage of the total time that any instantaneous value of the waveform's amplitude is less than some specified value. Such a distribution for white noise is shown at the left of Figure 6–13. The **slope** of a cumulative probability distribution is a **probability density function**. For white noise, it takes the form shown at the right of Figure 6–13.

The probability density function at the right of Figure 6–13 has the shape of a normal curve, and the amplitudes (and phases) of white noise are indeed distributed normally. A normal distribution is called **Gaussian** in honor of Karl Friedrich Gauss, a German mathematician, astronomer, and physicist. Therefore, *white noise, which is characterized by a normal probability density function, also is called Gaussian noise.*

White noise is a *complex aperiodic sound wave with a continuous amplitude spectrum*, in contrast to the line spectra that have

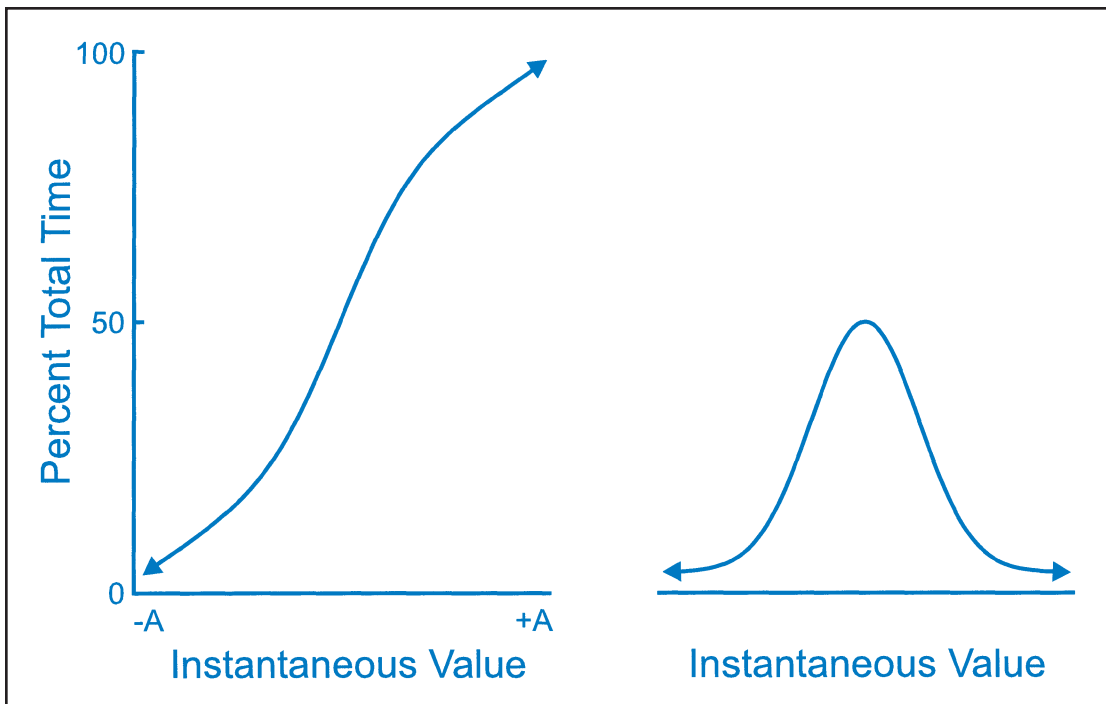


Figure 6–13. At the left is the *cumulative probability distribution* of **white noise**, which shows the percentage of total time that any instantaneous amplitude is less than some specified value. At the right is the *probability density function* for white noise, which is a plot of the slope of the function at the left. Because the probability density function assumes the shape of a normal curve, which also is called a Gaussian curve, **white noise** also is called **Gaussian noise**.

been discussed previously. You can see in Figure 6–9 that the spectral envelope is a line drawn parallel to the baseline, because *white noise has the same amount of energy in every frequency band 1 Hz wide regardless of the value of f* . We will discuss the slope of the envelope of white noise (and introduce “pink” noise) in more detail in Chapter 7 after the concepts of **pressure spectrum level** and **octave band level** have been presented.

A Single Pulse

Panel A of Figure 6–14 shows the waveform of a **single pulse** that has the same width (duration = 2 ms) as each rectangular pulse in the pulse train that was shown in Figure 6–12. *Is the waveform periodic or aperiodic?* Do not be deceived by the fact that the shape of the waveform appears to be “regular” instead of random, as it is for white

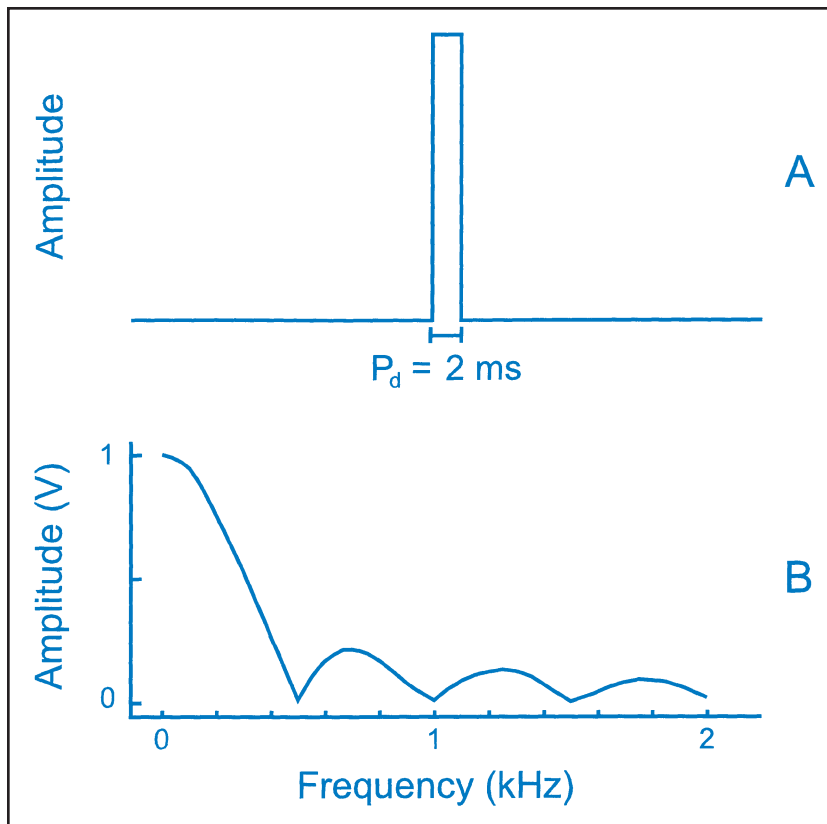


Figure 6–14. The waveform and spectrum of a **single rectangular pulse**. Nulls appear at integer multiples of the reciprocal of pulse duration, just as they did for a pulse train in Figure 6–12. In this case, however, the spectrum is a **continuous spectrum** in contrast to the line spectrum observed previously for a pulse train because a single pulse *cannot* be periodic.

noise; *that is irrelevant!* The concept of *periodicity* means that an event occurs periodically over time. *If there is only a single event (a single pulse), it cannot conceivably occur periodically.*

Recall that the *period of the pulse train* is given by the interval from the onset of one pulse to the onset of the next successive pulse. From that perspective, the “period” of a single pulse is infinity. If a single pulse is not periodic, it must be aperiodic. Therefore, the amplitude spectrum is a **continuous spectrum** rather than a line spectrum.

Look again at the amplitude spectrum of the **pulse train** in Figure 6–12. It is a line spectrum with energy at harmonics of the pulse repetition frequency. For the example in Figure 6–12, the harmonics are spaced at 100 Hz intervals because the period of that pulse train is 10 ms ($f = 1/.01 = 100$ Hz).

Although it might be difficult to imagine what would happen if the period were increased from 10 ms to infinity, we can, with a few examples, progress in that direction. If the period increases from 10 ms to 20 ms, the pulse repetition frequency decreases from 100 Hz to 50 Hz ($f = 1/.02 = 50$ Hz), and the harmonics are spaced twice as closely together at intervals of 50 Hz.

Recall from Chapter 1 that each time the period is doubled, frequency is halved. Therefore, for a complex **periodic** wave,

- *each time the period is doubled, the spacing between harmonics is halved.*
- For example, if we continue to double the period to 40 ms, 80 ms, 160 ms, 320 ms, and so on, the intervals between harmonics in the amplitude spectrum progressively decrease to 25 Hz ($f = 1/.04$), 12.5 Hz ($f = 1/.08$), 6.25 Hz ($f = 1/.16$), 3.125 Hz ($f = 1/.32$), and so on.
- If this process is continued toward infinity, the spacing between harmonics continues to become smaller and smaller. *At infinity, the spacing between harmonics equals 0, and the result is a continuous spectrum of the sort shown in panel B of Figure 6–14.*

The *shape of the spectral envelope* is the same as was shown previously for the pulse train because the width of the single pulse in Figure 6–14 is the same (2 ms) as the width of each of the pulses in the train of Figure 6–12. Thus, the envelope has **nulls** at frequencies that correspond to integer multiples of the reciprocal of the pulse width: $1/P_d$, $2/P_d$, $3/P_d$, and so on.

■ MEASURES OF SOUND PRESSURE FOR COMPLEX WAVES

In Chapter 2 we listed several alternative metrics by which the sound pressure of a **sine wave** is described. We also emphasized that

those equations *apply only to the sine wave*. Table 6–5 contains the sine wave equations shown previously, and the modifications that are required for calculating the **rms**, **mean square**, **FW_{avg}**, and **peak** sound pressure for square waves and typical aperiodic waveforms.

Different equations must be used for different waveforms. This introduces a problem in measurement of sound pressure. One example should suffice to emphasize the importance of knowing the kind of waveform on which a measurement is performed before the measurement is made. An acoustical signal often is converted (transduced) into an electrical signal and then various measurements are performed. Voltage is an electrical correlate of acoustical sound pressure. Thus, a transduced acoustical sine wave is an electrical waveform with sinusoidally fluctuating voltages over time. Measures of voltage are then performed with the aid of a voltmeter, which registers **rms voltage**.

Rms voltage is analogous to rms sound pressure. But one type of voltmeter is called an “average-responding meter” and another type is called a “true rms meter.” The average-responding meter “reads” the peak value of the voltage and then performs a computation to convert the peak reading to rms by dividing the peak value by $\sqrt{2}$. A true rms meter, on the other hand, “reads” the rms directly, thereby avoiding any necessity for conversion.

If the waveform is sinusoidal, you should read the same value with either meter. What happens if an average-responding meter is used to measure the rms voltage of a square wave? Suppose the peak value is 1 V. We can see from Table 6–5 that the rms value is the same as the peak value, 1 V, for a square wave. The average-responding meter, however, does not “know” that it is responding to a square wave. It reads the peak value of 1, divides the reading by $\sqrt{2}$, and registers that the rms voltage is an erroneous 0.707.

Table 6–5. Measures of Sound Pressure for Sine, Square, and Random Waveforms

Metrics	Sine Wave	Types of Waveforms	
		Square Wave	Random Wave
rms	$\frac{A}{\sqrt{2}}$	A	$\sim 0.3 \text{ A}$
mean square	$\frac{A^2}{2}$	A^2	$\sim 0.1 \text{ A}$
FW _{avg}	$\frac{2A}{\pi}$	A	$\sim .25 \text{ A}$
peak	A	A	A

Note. **A** refers to the peak or **maximum amplitude** as defined in Chapter 2.

The same measurement problem occurs with other complex waveforms. One must either know the appropriate conversions or use a more sophisticated and expensive measuring instrument that requires no conversion.

■ SIGNAL-TO-NOISE RATIO IN dB

We almost always listen to signals in the presence of some form of background noise. *The relation between signal level and noise level is quantified by the **signal-to-noise ratio (S/N)** in dB:*

- A positive S/N means that signal level exceeds noise level;
- A negative S/N means that noise level exceeds signal level; and
- An S/N of 0 dB means that signal level and noise level are equal.

Suppose, for example, that a signal with SPL = 70 dB is presented against a background noise with SPL = 66 dB. In that case,

$$\text{dB S/N} = 70 - 66 = +4 \text{ dB.}$$

If the S/N truly is a *ratio*, why do we solve for decibels by *subtracting* noise level from signal level rather than dividing signal level by noise level? Recall Log Law 2 from Chapter 4: *the log of a ratio equals the difference between the logs of the factors*.

A decibel is (10 times) a *log*, and therefore we subtract the denominator from the numerator rather than divide the numerator by the denominator. If, on the other hand, signal intensity and noise intensity were expressed in watts/m², then division would be the appropriate operation. In the example cited above, the *intensity* of the signal is 10⁻⁵ watt/m² (Equation 5-4) and the *intensity* of the noise is 4 × 10⁻⁶ watt/m². In that case,

$$\begin{aligned} \text{dB S/N} &= 10 \log \frac{10^{-5}}{(4 \times 10^{-6})} \\ &= 10 \log (0.25 \times 10^1) \\ &= +4 \text{ dB.} \end{aligned}$$

Obviously, it is easier to subtract decibels.

■ NOTES

1. Although Fourier made an immense contribution to the field of acoustics, his theorem emerged in 1807 during his study of heat diffusion in metal bars (Howard & Angus, 2009).

2. An **octave** always refers to a *frequency ratio of 2:1 or 1:2*. Other frequency ratios are particularly important in musical acoustics. For example, a fifth is a frequency ratio of 3:2, a fourth is 4:3, and so forth. For the octave, the mathematical expression is: $f_2/f_1 = 2^n$, where f_2 is the upper frequency limit of the octave, f_1 is the lower frequency limit, and n is the number of octaves. For one octave, $n = 1$. By way of example, suppose the lower frequency limit of the octave is 125 Hz. What is the upper frequency limit of a frequency band three octaves wide?

$$\begin{aligned} f_2 &= f_1 \times 2^n \\ &= 125 \times 2^3 \\ &= 125 \times 8 \\ &= 1000 \text{ Hz} \end{aligned}$$

3. Some authors, for example, Yost and Nielson (1977), use the symbols **P** for period and **T** for pulse duration. For the sake of consistency, we will continue to use **T** for period and we have adopted the symbol **P_d** for pulse duration.

FREQUENTLY MISUNDERSTOOD CONCEPTS

1. If $f_0 = 150$ Hz for a complex periodic waveform with energy at all integer multiples of f_0 , what is the frequency of the third overtone?

600 Hz. The answer is *not* 450 Hz, which is the frequency of the 3rd harmonic, hence the 2nd overtone. The *3rd overtone* corresponds to the *4th harmonic*, which is given by $f_0 \times 4 = 150 \times 4 = 600$ Hz.

2. A complex periodic waveform comprises two frequencies, each of which has a starting phase of 0° . For one component, $f_1 = 100$ Hz and peak amplitude = 2 V. For the second component, $f_2 = 200$ Hz and peak amplitude = 1 V. Each component has a starting phase of 0° . After 5 ms, what is the instantaneous voltage of the resultant wave?

0 V. The period, **T**, of $f_1 = 10$ ms, which means that the instantaneous phase angle of f_1 is 180° at 5 ms; at that moment, the resultant wave crosses the baseline, and the instantaneous voltage is, therefore, 0 V. The period of $f_2 = 5$ ms, which means that the instantaneous phase angle of f_2 is 0° at 5 ms; at that moment, it also will cross the baseline, and the instantaneous voltage is, therefore, 0 V. Thus, the resultant waveform will cross the baseline at 5 ms, with a resultant instantaneous voltage of 0.

- 3. A sawtooth wave has a fundamental frequency of 123 Hz and a starting phase of 90° . The amplitude of the 5th harmonic is -14 dB relative to the amplitude of f_0 . What is the relative amplitude of the 10th harmonic?**

-20 dB. First, note that the values of f_0 (123 Hz) and starting phase (90°) are irrelevant! Second, the slope of the amplitude spectrum of a sawtooth wave is -6 dB/octave. Because the 5th harmonic has a relative amplitude of -14 dB, and the 10th harmonic is one octave above the 5th harmonic regardless of the value of f_0 , the relative amplitude of the 10th harmonic is -20 dB.

PRACTICE PROBLEMS

1. One of the complex waves described in this chapter has the following characteristics: $f_0 = 125$ Hz, and energy is present at odd and even integer multiples of f_0 . The amplitude of the 4th harmonic is -6 dB re: the amplitude of the 2nd harmonic and -12 dB relative to the amplitude of f_0 . Identify the complex wave and explain the reasons for your choice.
2. A sawtooth wave has a fundamental frequency of 123 Hz and a starting phase of 90° . The relative amplitude of the 5th harmonic is -14 dB re: the amplitude of f_0 . What is the relative amplitude of the 10th harmonic?
3. Describe the important similarities and differences between a square wave and a triangular wave.
4. A complex, periodic waveform with energy at odd and even harmonics has a fundamental period of 5 ms. What is the frequency of the 6th harmonic?
5. If signal level is 70 dB SPL and noise level is 79 dB SPL, what is the signal-to-noise ratio in dB?
6. The lowest note on a tuned piano, A_1 , has a frequency of 27.5 Hz. What is the frequency of A_6 ?
7. If signal intensity is 10^{-5} watt/m² and noise intensity is 8×10^{-5} watt/m², what is the S/N in dB?
8. The waveform of white, or Gaussian, noise is aperiodic with equal energy in every frequency band 1 Hz wide (from $f - 0.5$ Hz to $f + 0.5$ Hz). Suppose the sound pressure level in *any* one band 1 Hz wide is 60 dB SPL, and further suppose that a white noise has 10,000 bands of energy, each of which is 1 Hz wide. Calculate the sound pressure level for the total noise.
9. A pulse train has a pulse repetition frequency of 200 Hz and a pulse duration (P_d) of 3 ms.
 - a. What proportion of each period is occupied by each pulse?
 - b. List the frequencies in Hz of the first five harmonics.
 - c. List the frequencies in Hz of the lowest three nulls in the amplitude spectrum (no decimals).
 - d. If the **pulse train** were replaced by a **single pulse** with $P_d = 3$ ms, list the frequencies in Hz of the lowest three nulls in the amplitude spectrum (no decimals).



Resonance and Filtering

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Sound waves, whether sinusoidal or complex, often are weak, uninteresting, and lacking “richness.” Consider the complex periodic waveform at the left of Figure 7–1 that is produced by the vibrating vocal folds during vowel production. Imagine that a microphone is placed in a location that permits recording the sound produced by the motion of the folds in a way that is not influenced by the supralaryngeal cavities.

The waveform resembles the shape of a sawtooth wave and sounds similar to a “buzz.”¹ The amplitude spectrum at the right of Figure 7–1 shows a fundamental frequency, f_0 , and a series of harmonics that are odd and even integer multiples of the fundamental. An important feature of the vocal fold amplitude spectrum is that the envelope is fairly smooth in the sense that there are no frequency regions with sharp prominences or nulls.

Most of the energy in that vocal fold buzzing sound is located in the lower frequencies, and the amplitudes of the harmonics diminish in a smooth, reasonably regular fashion at a rate of approximately 12 dB per octave. Recall from Chapter 6 that if this were a perfect sawtooth wave, all harmonics would be present and the slope would be -6 dB/octave. If it were a perfect triangular wave, only the odd harmonics would be present, but the slope would be -12 dB/octave.

The vowel component of our language system would be extremely limited if the sound wave shown in Figure 7–1 were all that was available for human vowel production. Basically, the system would have only three principal parameters that could be varied to produce a set of vowels with which to communicate. We could vary the amplitude to produce a series of tokens that differ perceptually in loudness, we could vary the fundamental frequency that would differ in perceived pitch, and we could vary the duration. We would not, however, be able to form the series of vowels that compose an

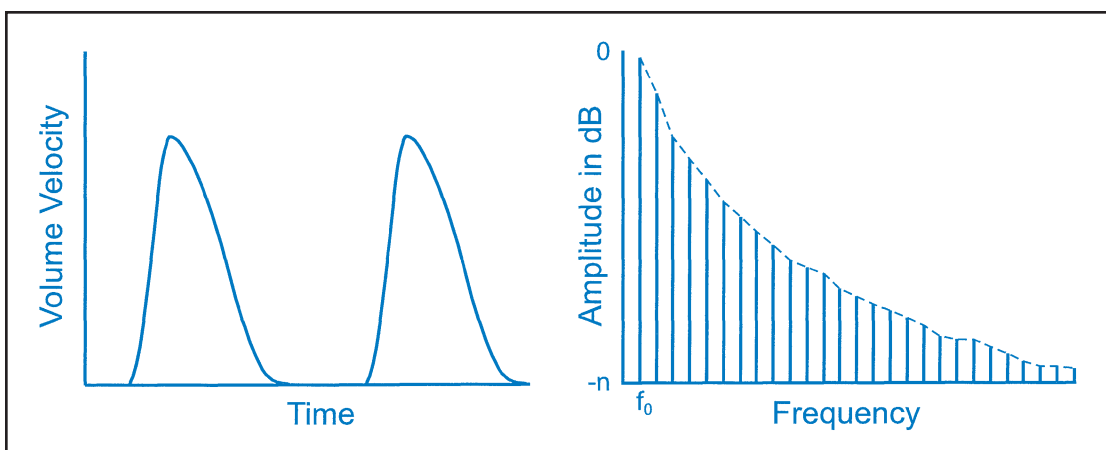


Figure 7–1. The waveform and amplitude spectrum of a sound produced by the vibrating vocal folds. The waveform is approximately sawtooth in shape. The spectrum is a line spectrum with energy at odd and even integer multiples of the fundamental frequency, f_0 , and with a spectral envelope slope of approximately -12 dB/octave.

important part of the phonological system to which we have become accustomed. To compensate for this, the sound wave produced by the vocal fold source—and all other sound waves as well—is “reinforced” by **resonance**, or **filtering**.

The sounds produced by stringed instruments also need to be “enhanced,” rather than rely simply on the vibrating strings alone to produce the desired sound waves. This is accomplished by use of what some call a “mechanical amplifier” known as a sounding board. It is more properly called a **resonator**, because it does not really amplify sound waves. Instead, a resonator such as a sounding board “reinforces” the sound waves emanating from the vibrating strings.

RESONANCE

We begin by attempting to gain an intuitive understanding of what is meant by the word “resonance,” and examine what takes place when a system is “set into resonance.” The next step, then, will be to learn *why and how* resonance occurs.

One way to demonstrate the phenomenon of resonance is to strike a tuning fork and hold it in the air medium. The sound wave that results is approximately sinusoidal with a frequency of vibration that is governed by the mass and stiffness of the fork. Equation 2–10 showed that *the frequency of vibration is (1) directly proportional to the square root of the stiffness and (2) inversely proportional to the square root of the mass.*

$$f_{\text{nat}} = (\sqrt{s} / \sqrt{m})$$

Equation 2–10

The frequency with which a specific fork vibrates is called its **natural frequency**. Because the mass and stiffness of a given fork are constants, it vibrates with the same natural frequency each time it is struck. The natural frequency of the freely vibrating fork gives rise to the perception of some pitch by listeners, and the frequency of vibration does not change as the fork continues to vibrate.

The amplitude of tuning fork vibration depends on the magnitude of force applied when it is struck, and the amplitude of vibration gives rise to the perception of loudness. Finally, whereas frequency of vibration remains constant over time, the amplitude of vibration diminishes because of **frictional resistance**. The decay of amplitude over time is called **damping**.

When the loudness of the tone has become noticeably softer, touch the stem of the fork to a wooden table or some other hard surface. That hard surface also has **mass** and **stiffness**, and therefore it too can be *forced into vibratory motion*. But, unlike the tuning fork, the hard surface does not engage in **free vibration** at its own **natural frequency**. Instead, it is *forced to vibrate* (initially) at the frequency of the vibrating tuning fork.

When the tuning fork is held against the hard surface, observe that the sound immediately becomes louder. Even though the loudness

becomes perceptibly greater when the fork is held against the hard surface, *energy has not been added to the sound wave*. The sound wave that results from touching the stem of the fork to the table does have greater **power** and greater **intensity**, but not greater energy. How is that possible?

- Acoustic power is *the amount of energy per second*, and an increase in power is realized by either (1) an increase in energy or (2) *a decrease in time of vibration*.
- In this case, the greater power that results from touching the vibrating fork to a hard surface is gained at the expense of a shorter time of vibration.
- The duration of vibration is shorter when the fork is held against the hard surface than if the fork were left to vibrate freely in the surrounding air medium instead of holding it against some hard surface.

The Principle of Resonance

As we delve into the meaning of “resonance,” it is helpful to think about the physical properties that a source of sound and a resonator have in common. Recall that a source of sound must be capable of being set into vibration, and that two properties, **mass** and **elasticity** (**stiffness**), enable it to vibrate. Those two properties not only permit an object to be set into vibration, but they also determine the frequency (**natural frequency**, f_{nat}) with which the source vibrates. A “resonator” has the same two properties, **mass** and **stiffness**, and therefore it not only can be forced to vibrate, it also has a **natural frequency** that is determined by those same two properties. What, then, distinguishes a resonator from a source of sound?

The principle of resonance that is demonstrated by applying the vibrating tuning fork to a hard surface comprises two concepts.

1. When a periodically vibrating force is applied to an elastic system, *the elastic system is forced to vibrate initially at the frequency of the applied force, rather than at its own natural frequency*.
2. The nearer the frequency of the applied force to the natural frequency of the elastic system, the greater is the resulting amplitude of vibration.

Suppose the **natural frequency** of the tuning fork, because of its unique mass and stiffness, is 100 Hz, and further suppose that the natural frequency of some hard surface, because of its mass and stiffness, has a natural frequency of 500 Hz.

- When the freely vibrating fork is held against the hard surface, the hard surface is forced to vibrate with a frequency of 100 Hz—well below its own natural frequency—and with some small amplitude of vibration.

- If a different tuning fork with a higher natural frequency of 250 Hz is applied to the hard surface, the hard surface is forced to vibrate at 250 Hz. The amplitude of vibration of the hard surface, though, is greater because 250 Hz is closer than 100 Hz to the **natural frequency** (500 Hz) of the elastic system.
- If the applied frequency increases even more to 400 Hz, the amplitude of vibration is greater than it was at 250 Hz.
- When the frequency applied to the hard surface is 500 Hz, corresponding to the natural frequency of the hard surface, the amplitude of vibration is greatest. The amplitude of vibration then gradually becomes smaller and smaller as higher and higher frequencies are applied.

A Comparison of Two Elastic Systems

Figure 7–2 shows the amplitude spectra that result from driving two elastic systems that have different mass and stiffness characteristics.

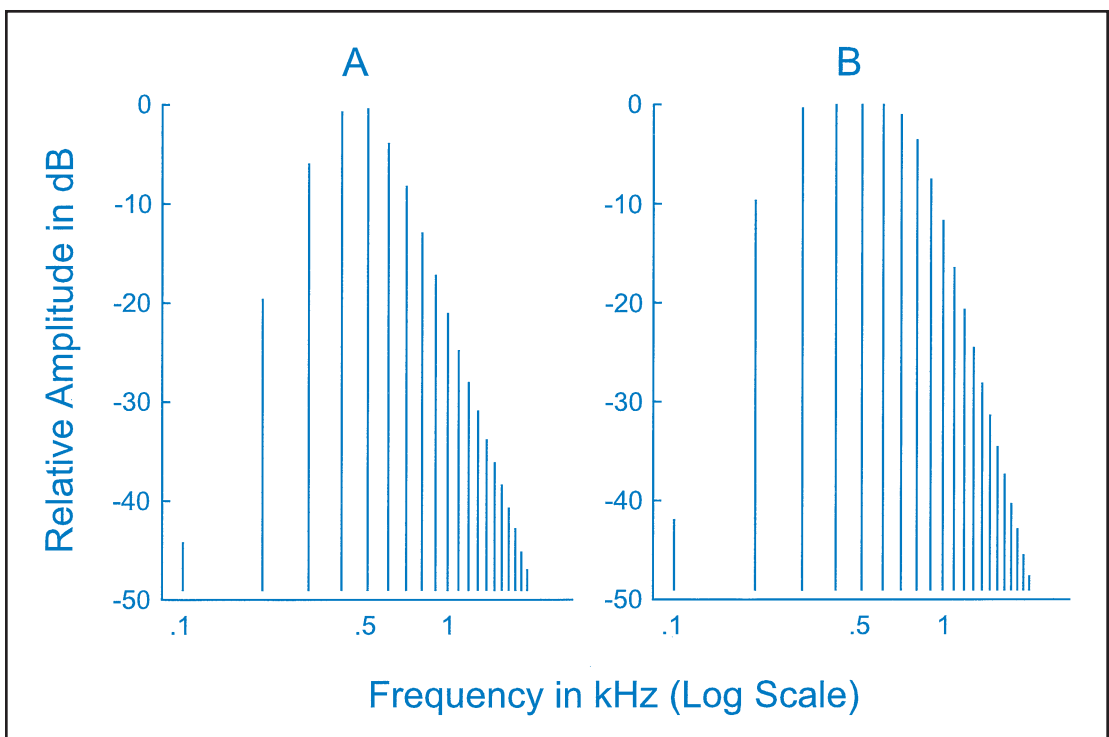


Figure 7–2. An illustration of the **principle of resonance**. Sine waves with different frequencies are applied with equal amplitudes to two resonant systems. The resonant systems are forced to vibrate at the frequency of the applied force, and the nearer *the frequency of the applied force* to the **natural**, or **center**, **frequency**, f_c , of the resonant system, the greater the amplitude of vibration. Consequently, the resulting amplitudes of vibration of the resonant system vary as a function of frequency. The system in panel A is a **narrowly tuned** system with a fairly well-defined f_c , whereas the system in panel B is more **broadly tuned**, and maximum amplitude of vibration occurs over a wider range of frequencies.

Instead of assembling a large array of tuning forks that span a wide frequency range, sine waves of variable frequency are generated by an electronic device called an “audio oscillator,” or “sine wave generator.”

Sine waves of variable frequency *but constant amplitude* are directed to two different systems represented in the figure. *Each system is forced to vibrate at the frequencies of the applied forces*, that is, at the frequencies produced by the sine wave generator. We then can examine, separately for each driving frequency, the amplitudes of vibration *at the outputs of the elastic systems*. Finally, we plot the results shown in Figure 7-2: 0 dB corresponds to the amplitude of the frequency that produces the greatest magnitude of vibration.

First, consider the example in panel A of Figure 7-2.

- When the driving frequency is 100 Hz, the elastic system is forced to vibrate initially with a frequency of 100 Hz; when the driving frequency is 200 Hz, the elastic system vibrates at 200 Hz, and so on.
- This is an example of **forced vibration** because the elastic system is *forced* to vibrate initially at the frequency of the applied force. As we move through higher and higher frequencies, the elastic system always vibrates with the frequency of the force applied. In other words, *resonance does not change the frequency of vibration*. This is consistent with the first part of the principle of resonance that was stated previously: *When a periodically vibrating force is applied to an elastic system, the elastic system is forced to vibrate initially at the frequency of the applied force.*

Return now to the lowest driving frequency, 100 Hz, and look at the amplitude of vibration that results at the output of the elastic system.

- The resulting amplitude is approximately 45 dB lower than the largest amplitude that is achieved at 500 Hz. At 200 Hz, the amplitude of vibration is greater than at 100 Hz, about -20 dB instead of -45 dB, because 200 Hz lies closer to the natural frequency, 500 Hz, of the system.
- As we move to higher and higher frequencies, the amplitude of vibration continues to *increase* until it reaches a maximum, in this case when $f = 500$ Hz. Then, as the frequency of the applied force continues to increase beyond 500 Hz, the amplitude of vibration *diminishes* in a manner that is symmetric (in log frequency) with the increase observed for the lower frequencies.

The driving frequency that produces the greatest amplitude of vibration of the elastic system, 0 dB, corresponds to the **natural frequency** (f_{nat}), or **center frequency** (f_c), of the elastic system, which is 500 Hz in the case of panel A of the figure. That outcome is in agree-

ment with the second part of the principle of resonance: *The nearer the frequency of the applied force to the **natural frequency** of the elastic system, the greater is the resulting amplitude of vibration.*

Panel B of Figure 7–2 shows a different outcome when the same driving frequencies are applied, to a different elastic system. The second system also is forced to vibrate at the driving frequencies. The principal difference between the response of the system at the right and the one at the left of the figure is that maximum amplitude (0 dB) is realized for several driving frequencies, not just one. We shall see later that the elastic system in panel A is a more **narrowly tuned** system, and the one in panel B is a more **broadly tuned** system.

A “Shattering” Example of Resonance

Several years ago there was a television commercial sponsored by a maker of one brand of audio tape. It showed a drinking glass shatter when the recorded voice of Ella Fitzgerald was reproduced near the glass. The glass, as with any system that has **mass** and **stiffness**, has a **natural frequency**, which also is called a **resonant frequency**.

In accordance with the principle of resonance, the glass was forced to vibrate at the frequency of the applied force (the driving frequency), which in this case was determined by the fundamental frequency, f_0 , of Fitzgerald’s singing voice. As Fitzgerald sang higher and higher up the scale, the fundamental frequency of her voice finally reached the resonant, or natural, frequency of the glass. The amplitude of vibration progressively increased until the vibration of the glass became so great that the goblet finally shattered. Did the commercial demonstrate the superiority of that brand of audio tape? Probably not, but it was an excellent, if unintended, demonstration of physics.

■ RESONANCE AND FILTER CURVES

Figure 7–3 shows the same process that was described for Figure 7–2, but in a slightly different fashion. Frequency of applied force again is shown along the abscissa, and amplitude of vibration of the elastic system in dB is shown along the ordinate. In this case, however, there are no sources of sound. Instead, the curve represents *the relative amplitude of forced vibrations as a function of frequency that would be realized if driving forces were applied to the elastic system*. Put simply, it describes properties of the elastic system that reveal how it will respond when forced to vibrate.

From inspection of the curve in Figure 7–3, you can determine the relative amplitude of vibration that can be expected for any frequency of vibration that is applied to the system as long as all of the driving frequencies are applied with exactly the same force.

- If the force of application increases for all frequencies, would the curve be displaced upward?

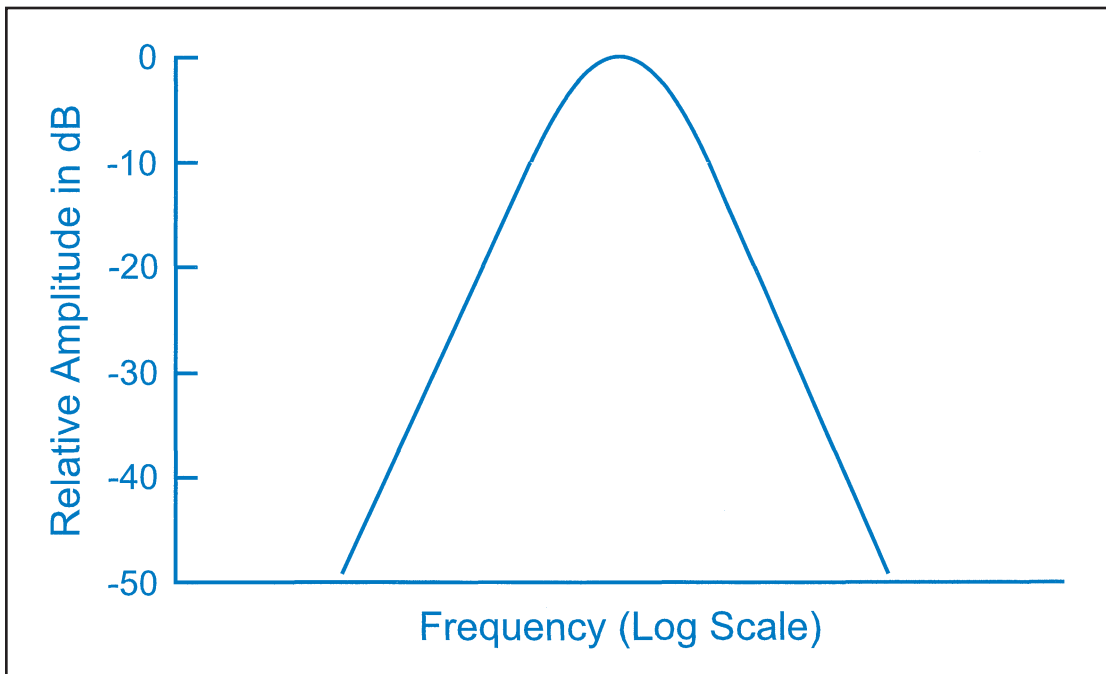


Figure 7–3. A **resonance**, or **filter**, **curve**. The curve shows the relative amplitude of vibration as a function of frequency for a resonator. The curve also is called a **system transfer function**.

- No. It is true that the amplitude of response is increased by an identical amount (to some upper limit) for each frequency, but because the ordinate is scaled in *relative amplitude*, where 0 dB refers to the amplitude of the frequency with the greatest energy, the curve will appear exactly as it does in Figure 7–3.

The curve in Figure 7–3 is a **resonance curve** because it behaves in a manner consistent with the definition of resonance stated earlier: The system vibrates at the frequency of the applied force, and the amplitude of vibration depends on the proximity of the frequency of the applied force to the natural frequency of the elastic system.

The curve in Figure 7–3 also is a **filter curve**, which simply emphasizes a different aspect of the same phenomenon. Look first at the frequency corresponding to the **natural frequency** of the system—the peak of the curve. The amplitude of vibration is greatest at that point, and therefore corresponds to 0 dB.

- As we move upward or downward in frequency from the natural frequency, the system is less responsive, and the corresponding amplitudes of vibration are less than the amplitude at the natural frequency.
- Thus, we can say that *the system responds differentially, or selectively, as a function of frequency*. It therefore is a

frequency-selective system. Frequencies that are remote from the natural frequency are filtered—their amplitudes are diminished.

Summary

When an elastic system is set into resonance, it is forced to vibrate initially at the frequencies of the forces applied to it. The elastic system has **mass** and **stiffness**, and those properties determine the **natural**, or **resonant**, **frequency** of the system. When an elastic system is driven by a frequency corresponding to the **natural frequency** of the system, we obtain the greatest amplitude of vibration. Finally, the amplitudes of vibration are less for frequencies that are lower or higher than the natural frequency.

Why is the amplitude of vibration greatest when the driving frequency equals the natural frequency of the system? To explain this phenomenon, we will review, and expand on, **acoustic impedance**, which was introduced in Chapter 3.

ACOUSTIC IMPEDANCE AND RESONANCE

The **impedance** of an elastic system has two components:

1. One is an *energy-dissipating* component, **resistance**—*the magnitude of resistance is independent of frequency*.
2. The second is an *energy-storage* component, **reactance**—and unlike resistance, *reactance is frequency dependent*.

Recall from Chapter 3 that there are two types of reactance, **mass reactance** (X_m), which *increases with increasing frequency* ($X_m \propto f$), and **compliant reactance** (X_c), which *decreases with increasing frequency* ($X_c \propto 1/f$). **Impedance** (Z) is the complex sum of its components: R , X_m , and X_c .

Effects of Impedance on a Resonance Curve

Figure 7–4 shows five resonance curves. The shape of each curve is *determined by the impedance of the system being driven*. More specifically, the shape of each curve, and its location (from left to right) in the frequency domain, *depends on the relative contribution of the three components of impedance* (**resistance**, **mass reactance**, and **compliant reactance**) to the total opposition to transfer of energy.

If the elastic system offers no resistance (a theoretical, and unachievable condition), maximum transfer of energy occurs, and the response of the system is infinite at the natural frequency. That circumstance is represented by the dashed resonant curve (labeled “Undamped”) with the arrows pointing upward.

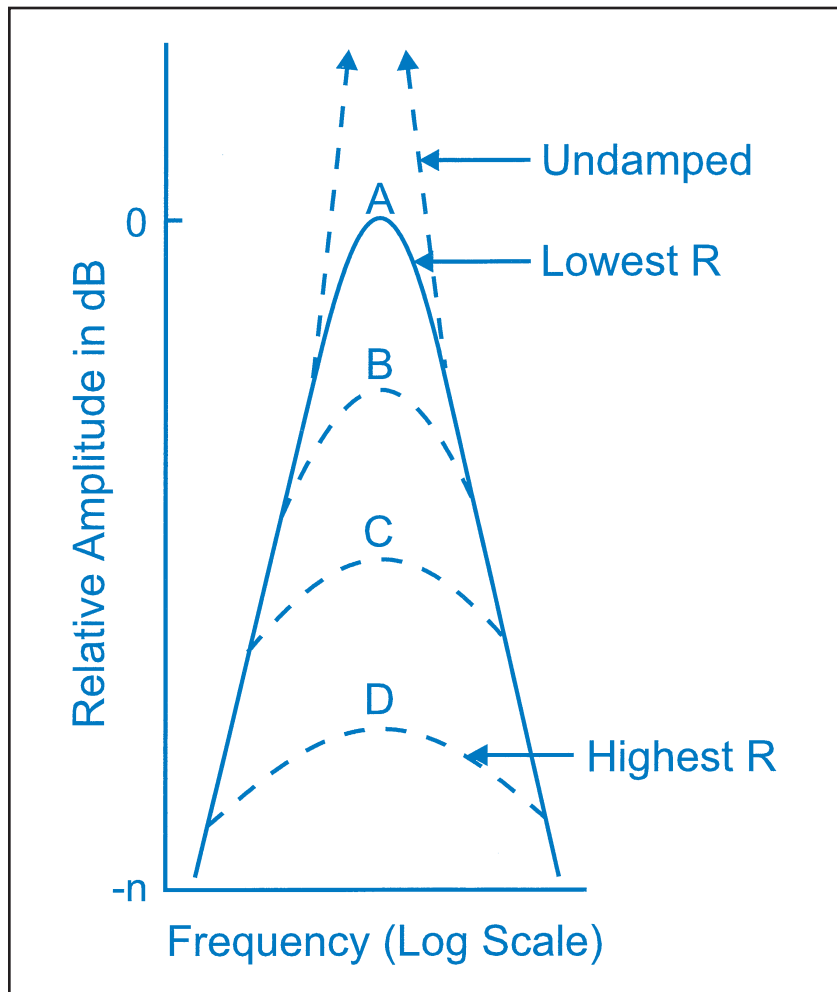


Figure 7-4. A series of resonance curves. The dashed curve labeled “Undamped” reflects the response of a theoretically lossless system with no resistance. From curve A through curve D, resistance increases, which produces two obvious effects on the shape of the curves: (1) the maximum amplitude of response decreases and (2) the system becomes more broadly tuned.

Curve A has a sharp peak. **Resistance** is present, *which limits the maximum amplitude* of vibration relative to the lossless case where there is no dissipation of energy. Curve A is *frequency selective* and the location of the peak of the curve along the frequency axis corresponds to the **natural frequency** of the system. The greatest amplitude of vibration occurs at the **natural frequency** because that is the frequency at which the least **impedance** is encountered.

At the **natural frequency** (f_c), $X_m = X_c$, and the impedance, the total opposition to transfer of energy is determined entirely by the **resistance** because X_m and X_c , though equal in value, are out of phase with each other and therefore their effect cancels.

- When the driving frequency equals the natural frequency of the system, the system is *set into resonance*.
- The amplitude of vibration is less for frequencies below f_c , and for those frequencies the system is called **compliance**, or **stiffness**, **dominant**. By that we mean, the relative contribution of **compliant reactance** to **impedance** is greater than the relative contribution of **mass reactance**. Compliance dominance occurs for lower frequencies because $X_c = 1/2\pi f c$ (Equation 3-4).
- The response (the amplitude of vibration) of the system also diminishes with increasing frequency above f_c , and for those frequencies the system is **mass dominant** because the relative contribution of **mass reactance** to impedance is greater than the relative contribution of **compliant reactance**. **Compliance dominance** occurs for lower frequencies because $X_c = 1/2\pi f c$ (Equation 3-4). **Mass dominance** occurs for higher frequencies because $X_m = 2\pi f m$ (Equation 3-3).

Curves B, C, and D in Figure 7-4 show the effects of progressively *increasing the resistance of a system*. The **natural frequency** is unchanged because there is no change in the **mass reactance** (X_m) or **compliant reactance** (X_c). As **resistance** (which is frequency independent) *increases*, more energy is dissipated, damping increases, and the system becomes more broadly tuned. Stated conversely, as the resistive component of impedance *decreases*, the system becomes less damped and more narrowly tuned.

Admittance

The discussion of impedance has emphasized the “negative.” By that we mean, we have focused on *opposition to motion or transfer of energy*. Thus, we say that the transfer of energy to an elastic system is **impeded** at frequencies remote from the natural frequency, and *the more remote (higher or lower) the driving frequency from the natural frequency of the system, the greater the impedance*. The result is a reduction in the response (amplitude of vibration) of the system as the driving frequency becomes progressively lower or higher than the natural frequency of the system.

We also can emphasize the inverse, or (musically speaking) “accentuate the positive,” by saying, *the closer the driving frequency to the natural frequency of the system, the transfer of energy to the system is greater*. More energy is accepted by, or *admitted* to, the elastic system when the driving frequency equals the natural frequency.

Admittance (Z^{-1}) is inversely proportional to **impedance** (Z), and the symbol for **admittance** therefore is Z^{-1} (recall from Equation 4-2 in Chapter 4 that $1/X^n = X^{-n}$). Thus, the ordinate in Figure 7-4 could be labeled **admittance** (instead of “relative amplitude in dB”). The unit of measure of **impedance** is the **ohm**, whereas the unit of measure of **admittance** is the **mho** (you might notice that “mho” is “ohm” spelled backward).

System Tuning

Figure 7–5 shows two resonance curves. Curve A reflects a **narrowly tuned system** and curve B reflects a **broadly tuned system**. A narrowly tuned system *can be forced to vibrate*, but only with maximum amplitude over a *narrow range of frequencies*. A tuning fork and the wires or strings of stringed musical instruments are examples of narrowly tuned systems, and *they vibrate freely at their own natural frequencies when energized*.

We have, for convenience, discussed sources of sound such as the tuning fork or string separately from resonant systems, which might convey *the erroneous impression that a source of sound cannot be set into resonance*.

- Both the source of sound and the resonant system have some amount of **mass** and **stiffness**, and each therefore is characterized by **resistance** and **reactance**.
- Hence, both have some amount of impedance.
- The tuning fork and the string *are indeed resonant systems* as well as being sources of sound. Thus, when the string is plucked or the fork is struck, each vibrates freely at its own **natural frequency**, which corresponds to the frequency at which the **mass reactance** equals the **compliant reactance**.

The tuning fork also can be forced to vibrate if it is tightly coupled to another tuning fork. But to transfer energy efficiently from one fork to another requires that the two forks have nearly identical

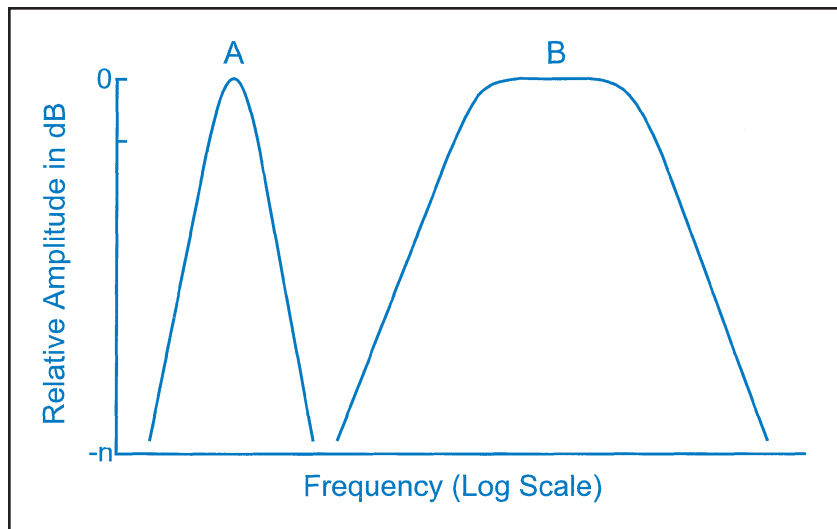


Figure 7–5. Comparison of two resonance systems, one that is narrowly tuned with low damping (panel A) and one that is more broadly tuned with high damping (panel B).

frequency *because the response curve for each reflects very narrow tuning*. That is because the impedance increases (and admittance decreases) rapidly as we descend or ascend in frequency from the natural frequency.

In contrast to a narrowly tuned system, a **broadly tuned system**, such as that shown with curve B in Figure 7–5, can be forced to vibrate with maximum amplitude by external forces over a *wide range of frequencies*. In addition to the table surface we cited previously, other familiar examples of broadly tuned systems include the diaphragms of microphones, earphones, loudspeakers, and our own auditory systems.

Elastic systems that are narrowly tuned *offer minimal resistance*, which means that such systems have little **damping**. That allows free vibrations to continue for some considerable amount of time, and narrowly tuned systems, therefore, are efficient *generators of sound*.

Systems that are **broadly tuned** *offer higher resistance*, which means they have much higher **damping**. Therefore, the free vibrations that occur after the external force is removed are very brief, and broadly tuned systems are efficient *transducers of sound*, systems that convert energy from one form to another. One example of a transducer is a loudspeaker, which converts electrical voltage to acoustic pressure.

Consider a speaker system that you are contemplating purchasing. Transducers such as loudspeakers should be broadly tuned so that they respond efficiently to a wide range of frequencies with little loss in amplitude. The specifications for a particular speaker, for example, might say that it has a flat frequency response from 80 Hz to 10,000 Hz $\pm$ 2 dB. The term “frequency response” is synonymous with the term “resonance curve,” and the specification described above means that the speaker is a broadly tuned system. These transducers are highly damped so that they do not “ring”—that is, so that the transducer does not continue to vibrate freely for any appreciable amount of time after the external force is removed.

Impedance Matching

When a force is applied to an elastic system, *power (energy per second) is transferred from the source (called the “driver”) to the elastic system (called the “load”), and the system is forced to vibrate at the frequency of the applied force*. The maximum amplitude of vibration of the system is achieved when the frequency of the driving force corresponds to the **natural frequency** of the system. When that occurs, **mass reactance** (X_m) equals **compliant reactance** (X_c), and **resistance** (R) is the only component of impedance. **Impedance** (Z) is minimal and **admittance** (Z^{-1}) is maximal at the **natural frequency** of the system.

Transfer of power from the driver to the load *is optimal when the impedance of the source equals the impedance of the load (the*

elastic system). When the wires of a piano are set into vibration, their motion forces the sounding board (a resonator) of the piano to vibrate. The sounding board *does not amplify the sound produced by the vibrating wires*. Instead, it improves the match of the impedance of the driver (the vibrating wires) and the surrounding air mass. Consequently, more sound power is delivered to the air.

The concept of impedance matches or mismatches will be encountered again in Chapter 9 when we discuss sound wave reflection.

■ FREQUENCY-SELECTIVE SYSTEMS: FILTERS

The effects of a **frequency-selective system**—a **filter**—on a driving signal that is directed to its input are illustrated in Figure 7–6.

- Panel A of the figure shows the **amplitude spectrum** of *the input signal*, which is a complex periodic signal *with equal energy in each harmonic*.
- Panel B shows the **filter curve** (also called a “frequency response” or “system transfer function”) of the **frequency-selective system**. The filter curve depicts the amplitude with which the system vibrates at each of the harmonics of the complex periodic input signal.
- The greatest amplitude of response is realized at the **natural frequency** of the system where $X_m = X_c$.

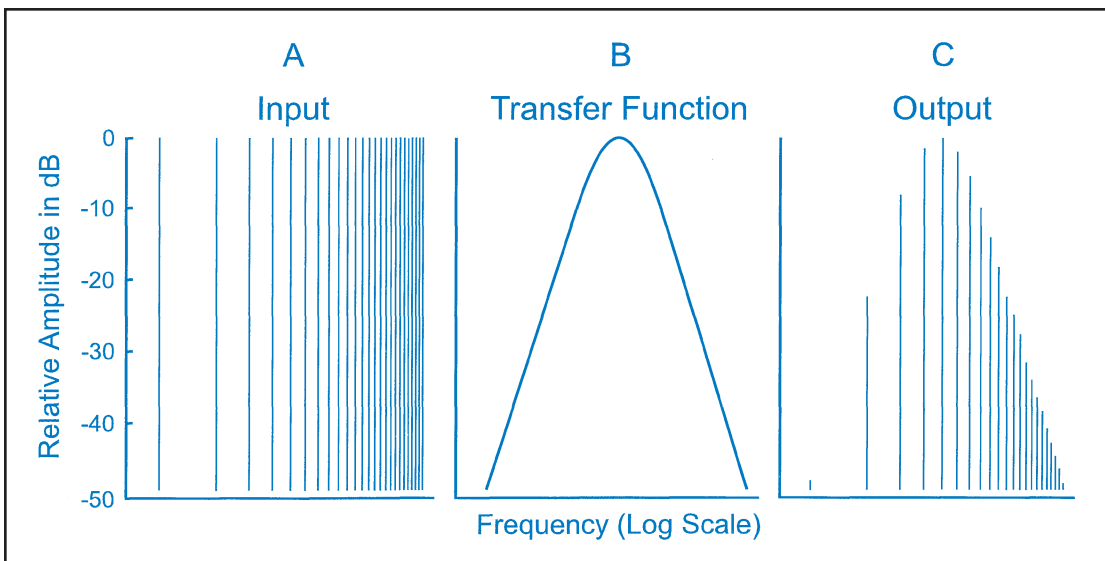


Figure 7–6. The effects of a resonator, or filter, on the amplitude spectrum of sine waves directed to the system. Panel A shows the input signal, panel B shows the transfer function (filter curve), and panel C shows the output signal after filtering. A comparison of the amplitudes of the harmonics in panel C with those in panel A shows the effects of filtering by the transfer function in panel B.

- Frequencies remote from the natural frequency are **filtered**; their amplitudes are diminished relative to the amplitude at the natural frequency because of **mass reactance** and **compliant reactance**, each of which varies with frequency.
- Stated differently, the mismatch of impedances increases as the frequency of the driver departs, upward or downward, from the natural frequency.

The curve in panel B of Figure 7–6 also is called a **system transfer function**. That means that the curve (a resonance, or filter, curve) *describes the way in which the sound wave is altered or transferred (amplitudes changed as a function of frequency) when the input signal is applied to the elastic system*.

The filter curve in panel B is *not a sound wave*. It simply shows the way in which the amplitude spectrum of the sound wave in panel A is altered when the sound wave is applied to the elastic system in panel B and forces the elastic system to vibrate.

Panel C of Figure 7–6 shows the **amplitude spectrum** of the *output signal*. The consequences of filtering are seen by comparing the amplitude spectrum of the *output signal* in panel C with the amplitude spectrum of the *input signal* in panel A. The amplitudes of the harmonics are altered. The greatest amplitude in panel C occurs for the 5th harmonic because its frequency matches the natural frequency of the filter in panel B (the peak of the transfer function). The amplitudes of all other harmonics are attenuated by the filter.

Figure 7–7 is like Figure 7–6, but the input signal (panel A) is **white noise**. White noise, as you recall, *has equal energy in all frequency bands 1 Hz wide and the amplitude spectrum is a continuous*

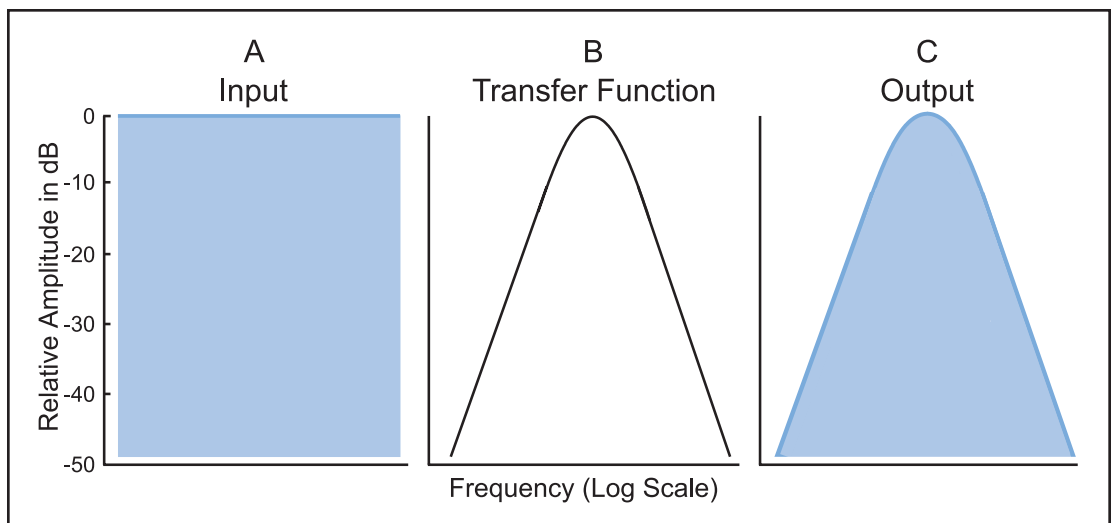


Figure 7–7. This figure is like Figure 7–6 except that the input signal (panel A) is white noise. The result (panel C) is called **band-limited white noise** because the output signal is restricted to some smaller band of frequencies due to the transfer function in panel B.

spectrum. Thus, all frequency bands 1 Hz wide in the white noise signal that are applied (delivered) to the filter have equal energy. The curve in panel B shows the **system transfer function**, or filter characteristics, and it reveals the way in which the system responds when the input signal is applied.

Panel C of Figure 7–7 shows the *amplitude spectrum of the output signal*:

- The input noise in panel A is filtered by the *frequency-selective device* in panel B.
- The sound wave at the output of the filter (panel C) is no longer white noise. Instead, it is a **frequency-limited**, or **band-limited, white noise**.
- The output signal is called a **band-limited white noise** to signify that although a white noise signal was applied to the system, the output is restricted, or limited, to some smaller band of frequencies.

■ PARAMETERS OF A FILTER (SYSTEM TRANSFER FUNCTION)

Two filter curves, or system transfer functions, are shown in Figure 7–8. Filter curves such as those shown in the figure are described by reference to five parameters:

1. the **natural frequency** (f_{nat}), which also is called the **center frequency**, f_c ;
2. the **upper cutoff frequency** (f_U);

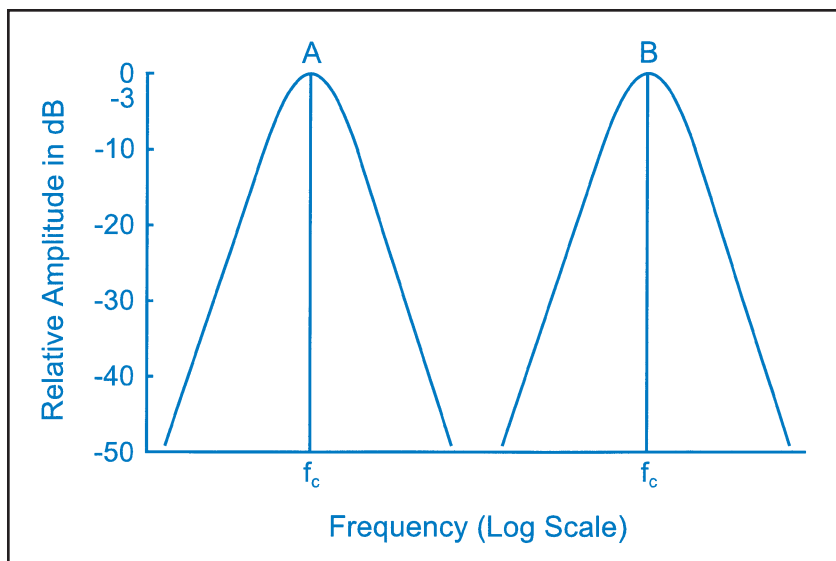


Figure 7–8. Two filter curves with different center frequencies, f_c .

3. the **lower cutoff frequency** (f_L);
4. the **bandwidth** (Δf); and
5. the **attenuation rate**, which also is called the **rejection rate**.

Natural, or Center, Frequency

The **natural frequency** (f_{nat}), or center frequency, f_c , is the frequency that produces a maximum amplitude of vibration. The center frequency depends on the mass and stiffness of an acoustical system² and, as we learned previously, f_{nat} (or f_c) corresponds to the frequency at which $\mathbf{X}_m = \mathbf{X}_c$.

The two curves in Figure 7–8 have different values of f_c . One could easily gain the impression from Figure 7–8 that the obvious difference in f_c for the two curves is the only distinguishing characteristic between them, but after reading a subsequent section on “band-pass filters,” you should return to Figure 7–8 and try to discover another way in which the two curves differ.

Upper Cutoff Frequency (f_U)

A second parameter is the **upper cutoff frequency**, f_U . We have seen that frequencies *above* f_c are attenuated. Higher frequencies are filtered; *their amplitudes are attenuated*.

At what frequency does the filtering begin? Consider the two examples in Figure 7–9, one with solid lines and the other with dashed lines. Both filter curves, **system transfer functions**, have the same f_c , but the response of the two systems is different for frequencies above (and below) f_c . Filtering begins almost immediately for frequencies above f_c and progresses continuously for higher and higher frequencies. A convention has been adopted, however, to describe the “*upper limit*” of the filter by designating a single frequency as the point above f_c where filtering is said to commence.

The upper cutoff frequency, f_U , is defined as the *3-dB down point*, or as the *half-power point*.

- The upper cutoff frequency, f_U , is the frequency *above* f_c for which the amplitude of response is 3 dB less than the amplitude at maximum, f_c .
- It is called the half-power point because it is the frequency at which the power in the resulting sound wave is reduced (attenuated) to one-half the power existing in the wave for f_c .
- It is called the **3-dB down point** because halving of acoustic power corresponds to -3 dB:

$$10 \log_{10} \frac{1}{2} = -3 \text{ dB.}$$

Thus, f_U for the solid filter curve in Figure 7–9 is *higher* in frequency than f_U for the dashed filter curve.

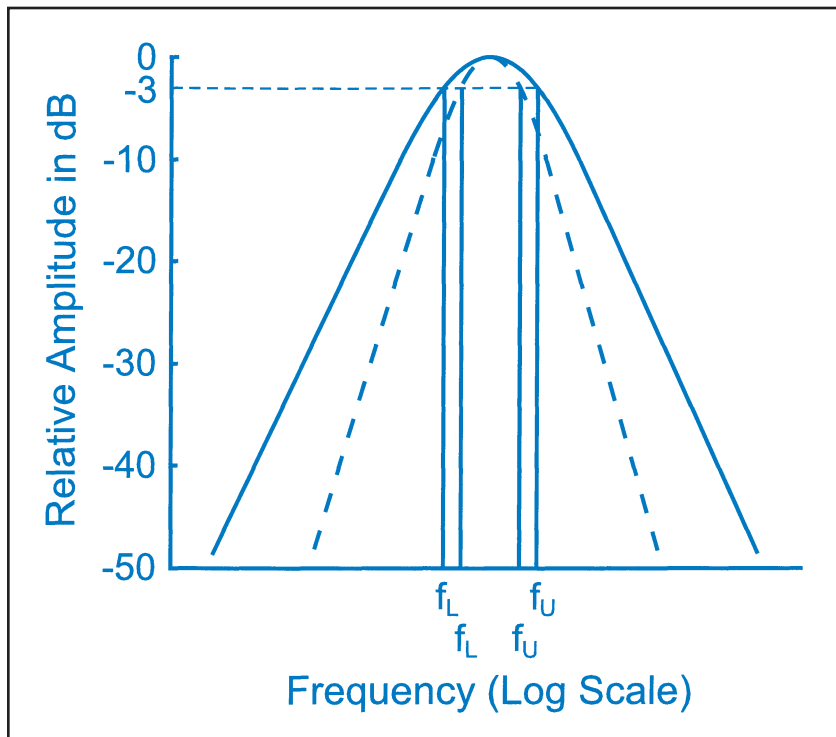


Figure 7-9. Two filter curves with the same center frequency, f_c , but different upper cutoff frequencies, f_U , lower cutoff frequencies, f_L , and bandwidths, Δf .

Lower Cutoff Frequency (f_L)

A third parameter of the filter is the **lower cutoff frequency**, f_L .

- f_L is defined as *the frequency below f_c for which the amplitude of response is 3 dB less than the amplitude of response at maximum, f_c .*
- f_L also is the *half-power point* because it is *the frequency below f_c for which the power in the output wave is reduced (attenuated) to one half the power existing in the wave for f_c .*
- f_L for the solid filter curve in Figure 7-9 is lower in frequency than f_L for the dashed filter curve.

Bandwidth (Δf or BW)

The two filter curves (system transfer functions) in Figure 7-9 have the same f_c , but different upper (f_U) and lower (f_L) cutoff frequencies. They also differ in another important respect; the solid filter curve is *wider* than the dashed filter curve. The width of a filter is quantified by the fourth parameter, the bandwidth, Δf .

- Δf is defined as *the width of the band of frequencies “passed” through the filter*, and

$$\Delta f = f_u - f_L.$$

Equation 7-1

- Thus, *bandwidth refers to the width of the band of frequencies between the lower and upper cutoff frequencies*, and it defines the range of frequencies over which energy is *passed* through the filter.
- It also is called the “passband” of the system to call attention to the band of frequencies that is passed through the filter. The two filter curves in Figure 7-9 have the same f_c , but different upper (f_u) and lower (f_L) cutoff frequencies *and different bandwidths (Δf)*.

Return to Figure 7-4 and examine the resonance curves again. We said previously that as we moved from curve A to curve D, resistance progressively increased and the systems reflected broader and broader tuning. Now we should see that the degree of tuning can be quantified by expressing the bandwidths, Δf . Thus, curve D in Figure 7-4 has a wider bandwidth than curve A.

Attenuation Rate (in dB/Octave)

The fifth parameter describes the *slope* of the “skirts” of a filter curve. It is referred to variously as:

- the **attenuation rate**, *the rate at which the amplitude of response is attenuated as a function of frequency*;
- the **roll-off rate**, *the rate at which the response of the system “rolls off” on either side of f_c* ; and
- the **rejection rate**, *the rate at which energy for frequencies on either side of f_c is “rejected” by the filter*.

The three terms are synonymous. Each is a measure of the slope of the curve on either side of f_c , and *the unit of measure is **dB/octave***.

The two filter curves in Figure 7-10 have the same f_c , but the attenuation rates for the two are different. The response (relative amplitude in dB) at 1000 Hz for filter A is -10 dB re: the response for f_c , and *one octave lower* (500 Hz), the response for filter A is -20 dB. Thus, the **attenuation rate** over this linear portion of the curve for filter A is 10 dB/octave.

For filter B, which has a steeper slope, the response at 1000 Hz is -20 dB, the response at 500 Hz is -35 dB, and the attenuation rate, therefore, is 15 dB/octave. Thus, attenuation rate is one parameter that distinguishes filter A from filter B, and filter B is “more selective” than filter A. Because it is more selective, that is to say, it is more “narrowly tuned,” we know that it offers less resistance to the transfer of energy.

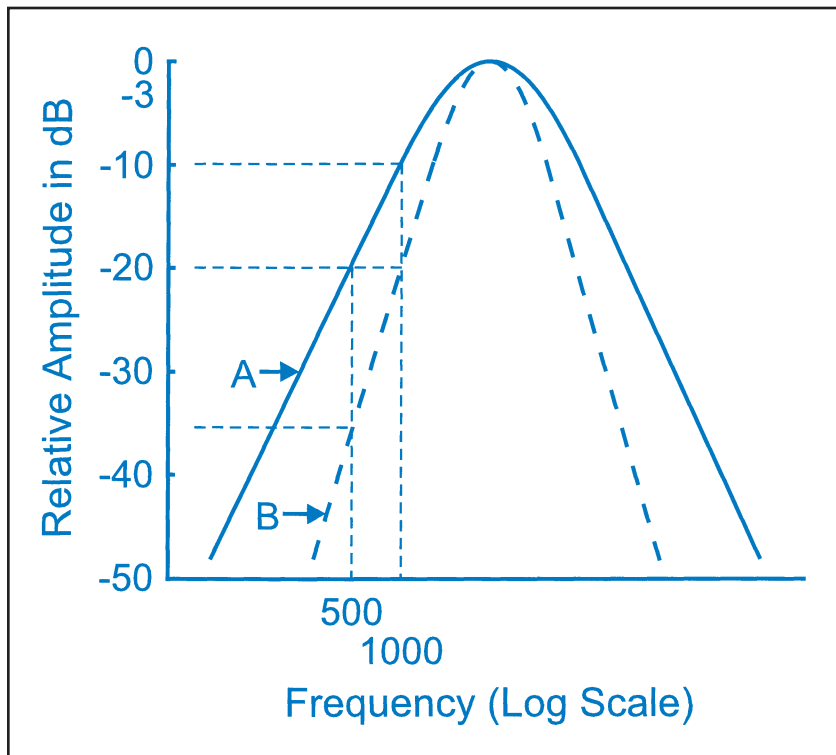


Figure 7-10. Two filter curves with the same center frequency, f_c , but different attenuation rates.

■ IDEALIZED RECTANGULAR FILTER

Consider a filter with the following characteristics: $f_c = 1000$ Hz; $f_L = 800$ Hz; $f_U = 1200$ Hz; $\Delta f = 400$ Hz; and **attenuation rate** = 24 dB/octave. The filter:

- *rejects* energy *below* 800 Hz and *above* 1200 Hz;
- *passes* energy in a frequency band 400 Hz wide centered geometrically on the natural frequency of 1000 Hz; and
- *rejects* energy outside the passband (800–1200 Hz) at a rate of 24 dB/octave.

That filter, which is illustrated in Figure 7-11, has a rectangular shape (transfer function). *All* energy between 800 (f_L) and 1200 Hz (f_U) is *passed* by the idealized filter, *all* energy below 800 Hz and above 1200 Hz is *rejected*, and the filter slopes are infinitely steep. Analog filters are more of the form shown by the curve in Figure 7-11. Because the slopes in the figure are 24 dB/octave rather than infinity, the analog filter in Figure 7-11 falls short of the idealized filter.

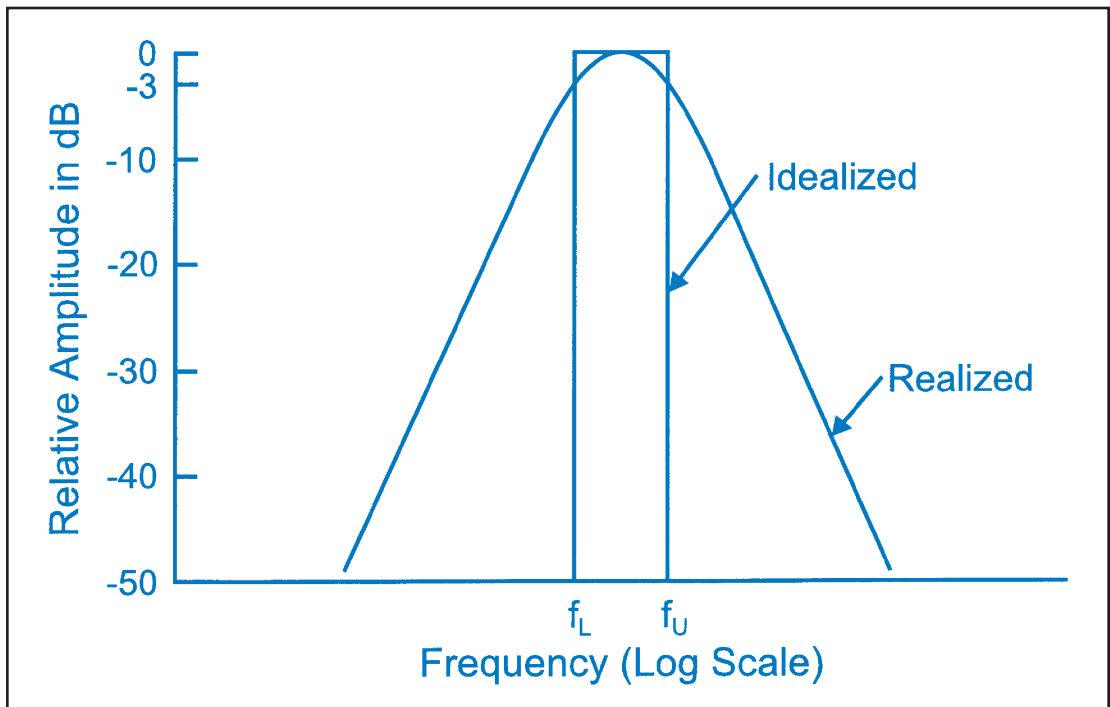


Figure 7-11. A comparison of an analog filter curve (**realized**) with an **idealized rectangular filter** with an infinitely steep attenuation rate.

- When the lower cutoff frequency (f_L), upper cutoff frequency (f_U), and bandwidth (Δf) of the filter are expressed, *the real filter* is described as if it were an **idealized rectangular filter**.
- Because the slope of the real filter is not infinitely steep, specification of the **attenuation rate** provides a measure of the amount by which the real filter *departs* from the idealized filter.
- Filter B in Figure 7-10 is more like the idealized filter than filter A because it has the steeper slope. If another filter is chosen with an attenuation rate of 72 dB/octave rather than 10 or 15 dB/octave, it will be even more like the idealized filter. Thus, *the steeper the attenuation rate, the more the real filter approximates the idealized filter*.

TYPES OF FILTERS

There are four common types of filters: **low-pass**, **high-pass**, **bandpass**, and **band-reject**. The four names correspond closely to the defining properties of each of the filter types and distinguish one from another. We will provide a brief description of each type and specify the parameters that are necessary to describe each type of filter.

Low-Pass Filter

A **low-pass filter** passes energy *below* (lower than) some designated upper cutoff frequency (f_U). The two defining parameters are f_U and **attenuation rate**. Figure 7–12 shows four low-pass filters.

- The two curves (system transfer functions) in panel A have the same attenuation rates, but different upper cutoff frequencies (f_U).
- The two curves in panel B have different upper cutoff frequencies and different attenuation rates.
- In each of the four filter curves, though, energy for frequencies *below* the upper cutoff frequency is passed, whereas energy for frequencies *above* f_U is rejected. Hence, each of the four is a “low-pass” filter.

High-Pass Filter

A **high-pass filter** passes energy *above* (higher than) some designated lower cutoff frequency (f_L). The two relevant parameters are f_L and **attenuation rate**.

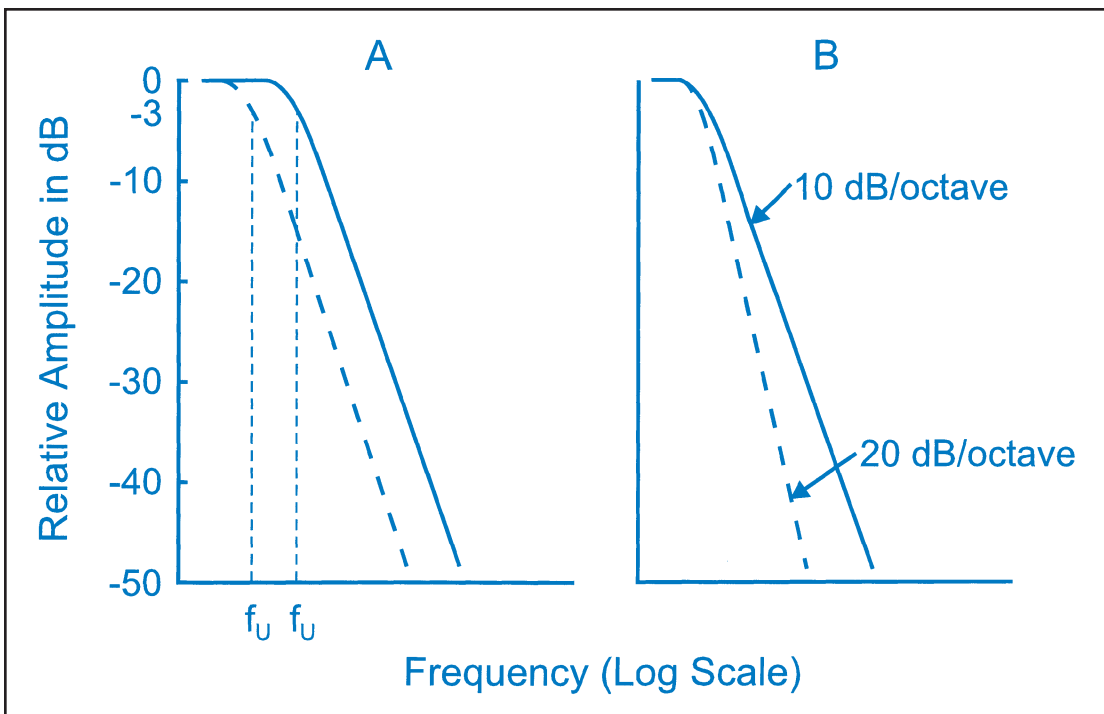


Figure 7–12. Four low-pass filter curves, filters that pass energy *below* some upper cutoff frequency, f_U . The two filter curves in panel A have the same attenuation rates, but different upper cutoff frequencies, f_U . The two filter curves in panel B have different upper cutoff frequencies and different attenuation rates.

- Panels A and B of Figure 7–13 show four high-pass filters that differ in those two parameters.
- In each of the four filter curves, energy for frequencies *above the lower cutoff frequency is passed, whereas energy for frequencies below f_L is rejected*. Thus, each of the four is a “high-pass” filter.

Band-Pass Filter

A **band-pass filter** passes energy in a specified band of frequencies between a lower cutoff frequency, f_L , and an upper cutoff frequency, f_U . All five parameters generally are relevant: f_L , f_U , Δf , and **attenuation rate**. Panels A and B of Figure 7–14 show four band-pass filters that differ in those five parameters.

A band-pass filter reflects the combined effects of low-pass and high-pass filtering as illustrated in Figure 7–15.

- The low-pass filter (solid line) passes energy below a designated **upper cutoff frequency**, which specifies f_U for the band-pass filter.

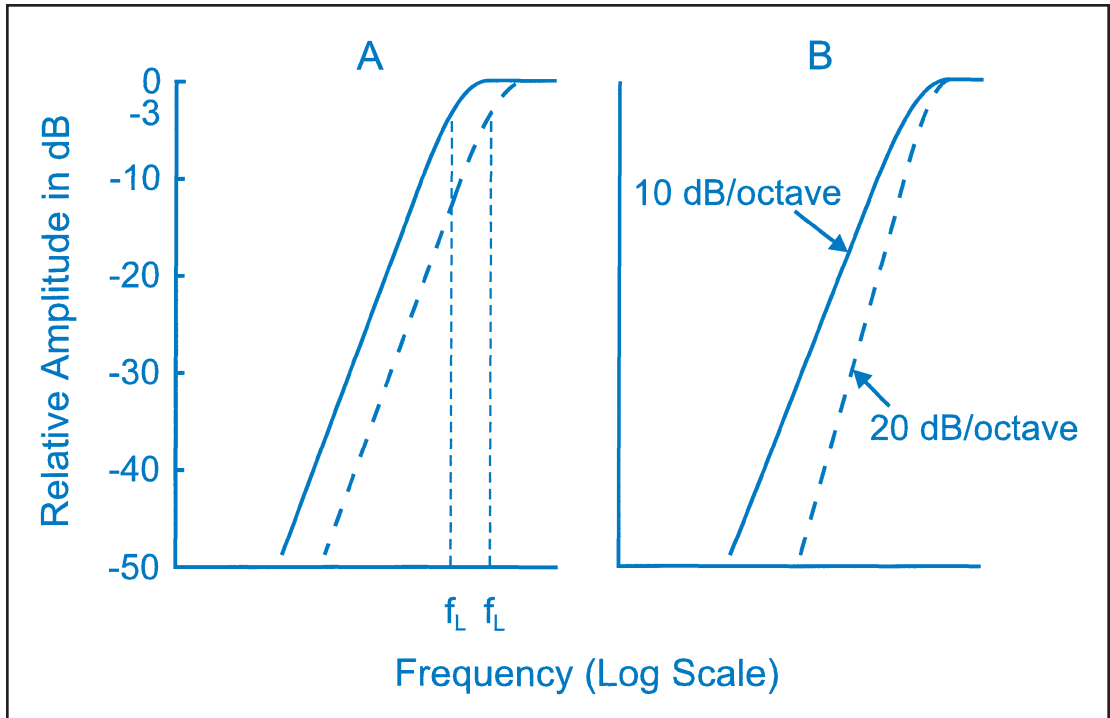


Figure 7–13. Four high-pass filter curves, filters that pass energy above some lower cutoff frequency, f_L . The two filter curves in panel A have the same attenuation rates, but different lower cutoff frequencies, f_L . The two filter curves in panel B have different lower cutoff frequencies and different attenuation rates.

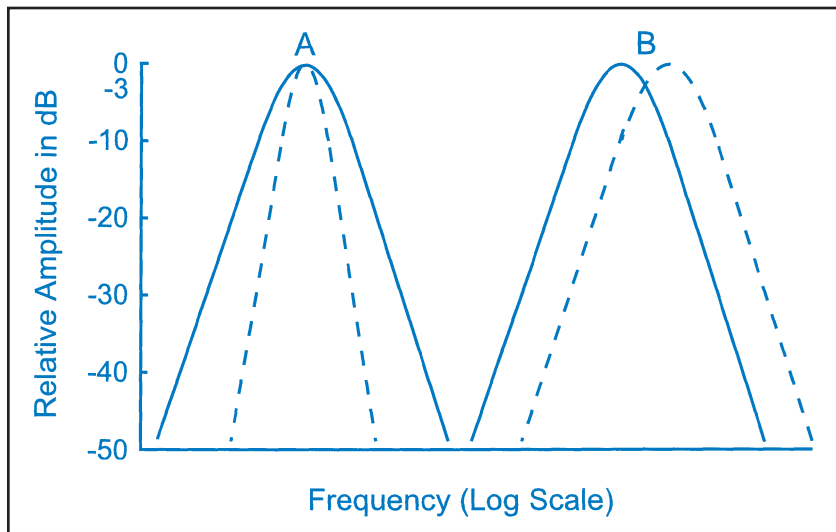


Figure 7-14. Four band-pass filter curves, filters that pass energy in a band of frequencies between f_L and f_U , with different values of f_c , f_L , f_U , Δf , and/or **attenuation rate**.

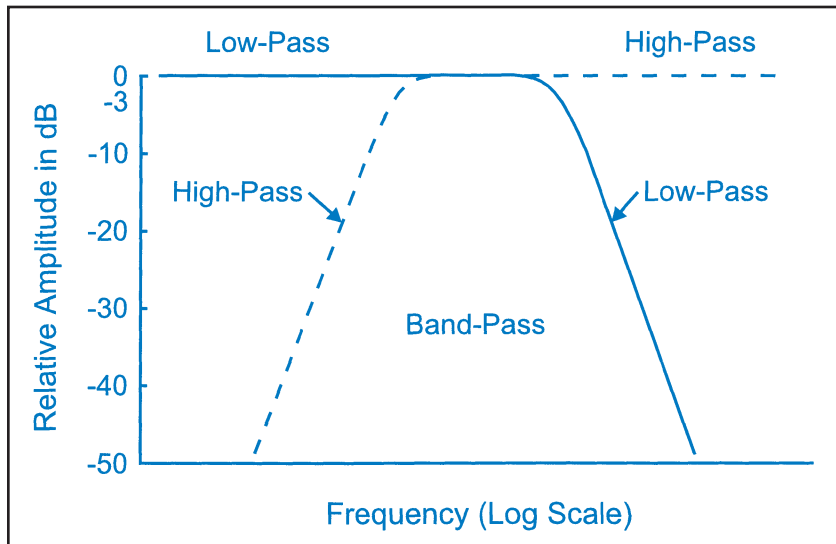


Figure 7-15. An illustration of how a band-pass filter represents the combined effects of a low-pass filter and a high-pass filter connected “in series.”

- The high-pass filter (dashed line) passes energy above a designated **lower cutoff frequency**, which specifies f_L for the band-pass filter.
- The **bandwidth** (Δf) of the band-pass filter is given by the difference between the upper and lower cutoff frequencies, $f_U - f_L$.

- The slope of the low-pass filter curve determines the upper frequency slope of the band-pass filter (the rate of attenuation above f_c), and the slope of the high-pass filter determines the lower frequency slope of the band-pass filter (the rate of attenuation below f_c). Although those two slopes could, in theory, be different from one another, they are almost always identical.

A word of caution is in order when we say that “a band-pass filter is the combined effect of low-pass and high-pass filtering.” That is true *only* if we use a *series, or cascade, system*. By that we mean, a signal is directed to the input of a low-pass filter with an appropriate f_U . The output of the low-pass filter then is directed to the input of a high-pass filter with an appropriate f_L . The signal at the output of the second filter then has been band-passed. The procedure also can be reversed (high-pass followed by low-pass) as long as the output of the first filter is sent to the input of the second filter for additional processing.

For some band-pass filters the *bandwidth, Δf , is independent of the center frequency, f_c* . Thus, Δf and f_c can be manipulated independently of one another. Consider the examples in Figure 7–16.

- Panel A shows the input signal, white noise, and the **amplitude spectrum** of the unfiltered input signal is labeled **X**.
- Panel B shows the transfer functions of two band-pass filters, one with $f_c = 1000$ Hz (labeled **Y**) and a second with $f_c = 4000$ Hz (labeled **Z**).

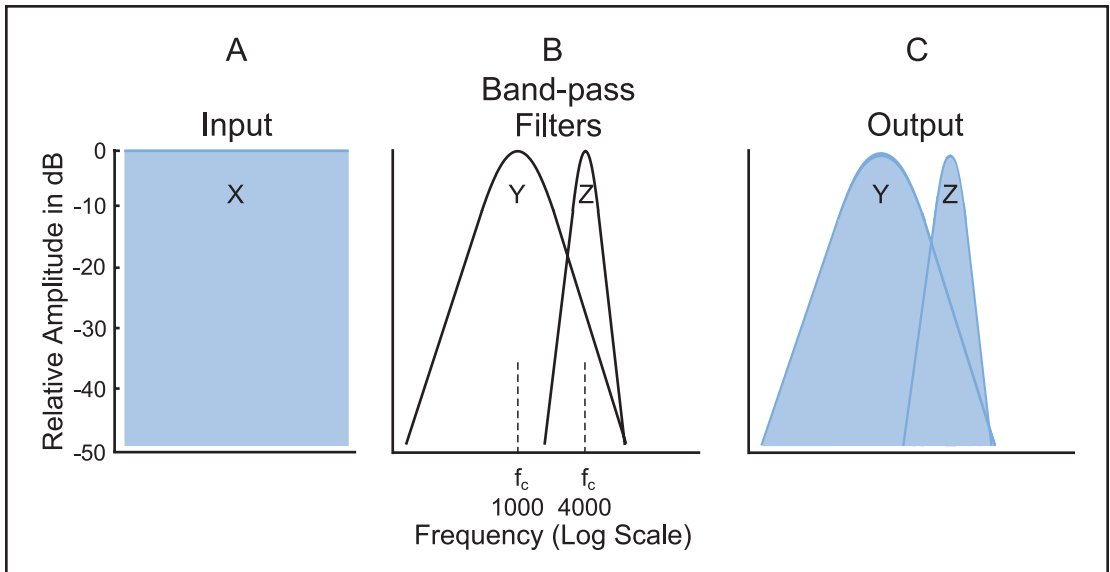


Figure 7–16. A white noise input signal in panel A is filtered by two different band-pass filters in panel B that have identical bandwidths. The different band-limited output noises are shown in panel C. Curve Z at the right in panel B *appears* to have a narrower bandwidth than curve Y because frequency is plotted on a logarithmic scale, but the bandwidths of the two curves, Y and Z, are identical.

- For both filters, $\Delta f = 500$ Hz. Filter **Z** *appears* to have a narrower bandwidth than filter **Y**, but that is because frequency is plotted on a log scale. On a log scale, the linear distance occupied by 500 Hz (the bandwidth) centered on 4000 Hz is only one-fourth the linear distance occupied by 500 Hz (the bandwidth) centered on 1000 Hz.
- Thus, because f_c is raised by a factor of 4, filter **Z** appears to be only one-fourth as wide as filter **Y**, but the two bandwidths are identical. (Return now to Figure 7–8, as suggested previously, and see if f_c was the only parameter that distinguished one filter curve from the other.)

The **amplitude spectra** of the filtered noise in Figure 7–16 are shown in panel C. **Y** shows the amplitude spectrum of the noise modified by the filter with $f_c = 1000$ Hz, and **Z** represents the amplitude spectrum of the noise modified by the filter with $f_c = 4000$ Hz.

- The level of intensity of the white noise signal *at the input to the filter* (that is, the unfiltered white noise), is X_{dB} .
- Only a portion of the original spectrum is preserved at the output of filter **Y** because only a portion of the signal has been passed by the filter; energy at frequencies below f_L and above f_U is rejected.
- The level of intensity at the output of this filter, then, is Y_{dB} , and

$$Y_{dB} < X_{dB}$$

because energy for frequencies *below* f_L and *above* f_U in **X** (the original unfiltered signal) is filtered.

- If some of the energy present at the *input* to the filter is rejected, the output level Y_{dB} must be less than the input level X_{dB} . The signal at the input to the filter has a theoretical bandwidth of infinity because it is white noise, but the output signal is *band limited* by a filter with a center frequency of 1000 Hz and a bandwidth of 500 Hz.

What happens when f_c is raised from 1000 Hz to 4000 Hz? Even though f_c is raised by a factor of 4 (and the bandwidth, when plotted on a log scale in Figure 7–16, appears to be more narrow), the bandwidth is the same (500 Hz) as it was when f_c was 1000 Hz. The bandwidth does not change because *the bandwidth is independent of the center frequency*.

The level of intensity at the output of filter **Z** in panel C, Z_{dB} , is less than the level at the input to the filter, X_{dB} , because the input signal is filtered. Thus,

$$Z_{dB} < X_{dB}.$$

An important factor to understand, however, is that:

$$Y_{dB} = Z_{dB}.$$

Why are the output levels from the two different filters identical? Their bandwidths are identical, and if the input signal is white noise, which has equal energy in all frequency bands 1 Hz wide, the output levels of the two filters remain constant as f_c is changed to any value *as long as the bandwidths do not change*. Of course, a band-pass filter can be adjusted to have many different values of Δf , and the wider Δf , the greater is the energy at the output of the filter. The important feature, though, is that Δf *does not vary automatically* with changes in f_c . The two parameters, f_c and Δf , can be adjusted independently of each other.

Constant Percentage Bandwidth Filter

There is one type of band-pass filter, called a **constant percentage bandwidth filter**, for which *the bandwidth is not independent of f_c* . Instead, Δf *is always some constant percentage of f_c* for a constant percentage bandwidth filter. The concept is like a sales tax. If a state levies a 6% sales tax, the dollar value of tax owed increases as purchase price increases. If the purchase price is \$10, the tax is \$0.60. If the purchase price increases by a factor of 4 to \$40, the tax also increases by a factor of 4 to \$2.40. The purchase price and the sales tax are not independent; the tax is a constant percentage of the purchase price. The same relation exists between Δf and f_c with a constant percentage bandwidth filter. *As f_c changes (increases or decreases) by some factor, Δf changes by the same factor.*

Look at the two filters, **Y** and **Z**, in Figure 7-17. In panel A, a white noise signal is directed to the input of the filters, the amplitude spectrum of the unfiltered noise is represented by **X**, and the level of intensity of the noise at the input is X_{dB} . For filter **Y** in panel B, $f_c = 1000$ Hz, and, for this example,

$$\begin{aligned}\Delta f &= 0.707 (f_c) \\ &= 707 \text{ Hz},\end{aligned}$$

which means that the bandwidth is 70.7% of the value of the center frequency. The level at the output of filter **Y** in panel C is some value Y_{dB} , and as before,

$$Y_{dB} < X_{dB},$$

because the signal is filtered.

Now, insert filter **Z** (panel B) with a center frequency of 4000 Hz to replace filter **Y**. Because the filter is a **constant percentage bandwidth filter** with a bandwidth that always is some constant percentage of the center frequency, and because the center frequency of filter **Z** is four times higher than the center frequency of filter **Y**, the bandwidth of filter **Z** is four times greater than the bandwidth of **Y**:

$$\begin{aligned}\Delta f &= 0.707 (f_c) \\ &= 2828 \text{ Hz}.\end{aligned}$$

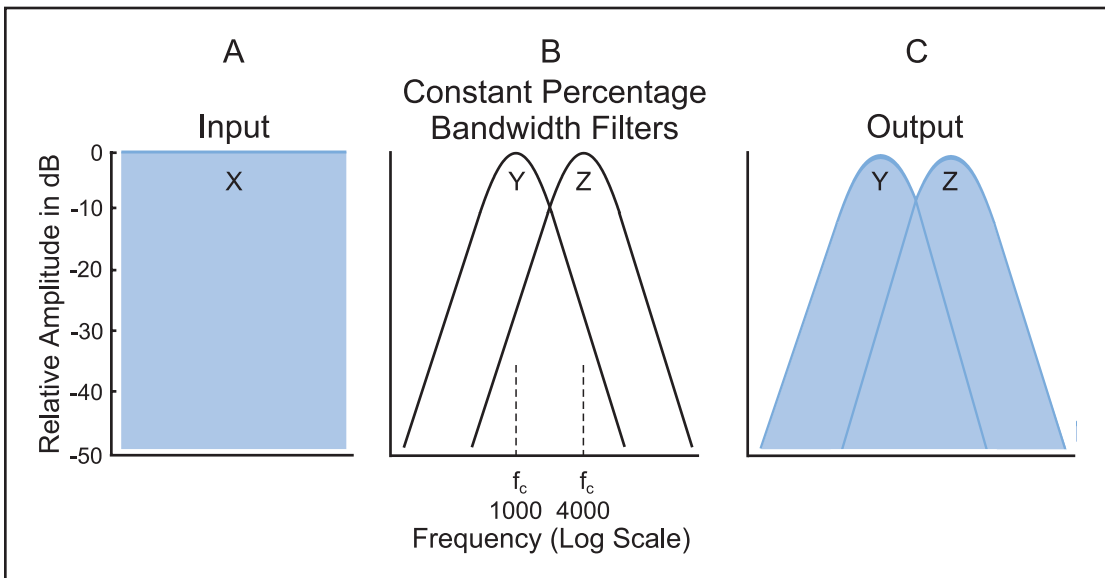


Figure 7–17. This figure is like Figure 7–16 except that the filters in panel B are **constant percentage bandwidth** filters in which the bandwidth is a constant percentage of the center frequency. The two curves, Y and Z, in panel B *appear* to have the same bandwidth because frequency is plotted on a logarithmic scale, but the bandwidth of curve Z is four times greater than the bandwidth of curve Y because its center frequency is four times higher.

- The bandwidths of filters **Z** and **Y** *appear* to be identical in Figure 7–17, but that is because frequency is plotted on a log scale.
- The bandwidth of filter **Z** actually is four times wider than the bandwidth of filter **Y**, but on a log scale they occupy the same linear distance.
- The level at the output of filter **Z** in panel C, Z_{dB} , is of course less than the level at the input, X_{dB} , because the signal is filtered. The important point for these two constant percentage bandwidth filters is that

$$Z_{dB} > Y_{dB}.$$

Why? Because the bandwidth of **Z** is four times wider than the bandwidth of **Y**. The white noise signal has equal energy per cycle at the input to the filter. Filter **Z** passes that equal energy per cycle over a frequency range of 2828 Hz, whereas filter **Y** passes that equal energy per cycle over a frequency range of only 707 Hz. Subsequently, we will describe how much greater the level of Z_{dB} is than the level of Y_{dB} *in decibels*, but the impatient reader might wish to solve the problem in advance by recalling the discussion on “combining sound intensities from independent sources” in Chapter 5.

Common Constant Percentage Bandwidth Filters. In the preceding example, a percentage of 70.7 was used, but any value is acceptable, just as

different local and state governmental bodies levy different amounts of sales tax. There are, however, a few common percentages that will be encountered for constant percentage bandwidth filters. One of the most common is a **1-octave (1/1) filter**. For a **1-octave filter**,

$$\Delta f = 0.707 f_c$$

Equation 7-2

The bandwidth (Δf) of a 1-octave filter is *always 70.7% of the center frequency (f_c)*. Why is it called a **1-octave filter**? An octave is a doubling of frequency ($f_2 = 2f_1$). For a 1-octave filter, and only for a 1-octave filter,

$$f_L = 0.707 f_c$$

Equation 7-3

and

$$f_U = 1.414 f_c$$

Equation 7-4

Therefore, from Equations 7-3 and 7-4, we see that the upper cutoff frequency for a 1-octave filter is *1 octave (2f)* higher than the lower cutoff frequency.

Table 7-1 shows how the parameters of a constant percentage bandwidth filter change with changes in f_c .

- As f_c increases by some factor for a 1-octave filter, f_L , f_U , and Δf increase by the same factor.
- For example, if f_c increases by a factor of 2 from 100 to 200 Hz, the values of the other three parameters also increase by a factor of 2.

Figure 7-18 shows the transfer function of the 1-octave filter from Table 7-1 with a center frequency of 1000 Hz. The center frequency, f_c , appears to be at the “center” of the passband, and in fact it is located at the “geometric center.” The center frequency, however, is not the common arithmetic mean of the upper and lower cutoff frequencies. The *arithmetic mean* of 707 Hz and 1414 Hz is 1060.5 Hz, but we indicate in the table that the center frequency is 1000 Hz. For

Table 7-1. Lower Cutoff Frequency (f_L), Upper Cutoff Frequency (f_U), and Bandwidth (Δf) for Various Values of the Center Frequency (f_c) of a 1-Octave Filter

f_c	f_L	f_U	$\Delta f = f_U - f_L$
100	70.7	141.4	70.7
200	141.4	282.8	141.4
1000	707	1414	707
2000	1414	2828	1414
10,000	7070	14,140	7070

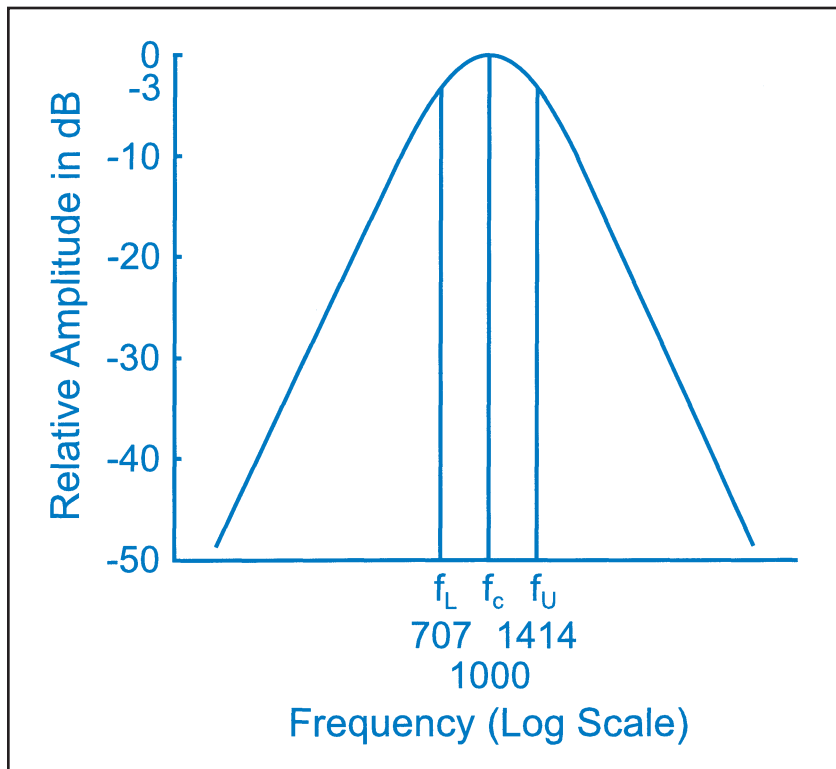


Figure 7-18. A 1-octave (1/1) constant percentage bandwidth filter.

a constant percentage bandwidth filter, the center frequency is *the geometric mean*, not the common arithmetic mean, of the upper and lower cutoff frequencies. Thus,

Equation 7-5

$$f_c = \sqrt{f_L \times f_U},$$

which states that the geometric mean of two numbers is the square root of the product of the two numbers.

There are other common constant percentage bandwidth filters in addition to the 1-octave filter. They include a **1/2-octave filter**, a **1/3-octave filter**, and a **1/10-octave filter**. Because a constant percentage bandwidth filter can, in principle, be any fractional value in addition to unity (1/1), the general expression is: **1/*n*-octave filter**. Each is a constant percentage bandwidth filter, and the common defining characteristic for all such filters is that *the bandwidth is always some constant percentage of the center frequency*. The exact constant percentage varies with the value of *n* that specifies the fraction of an octave that characterizes the filter.

Calculation of Parameters of Constant Percentage Bandwidth Filters. The equations for calculating the important parameters of a constant percentage bandwidth filter are:

$$f_L = \frac{f_c}{2^{1/2n}} = \text{antilog}_{10} (\log_{10} f_c - .3/2n); \quad \text{Equation 7-6}$$

$$f_U = 2^{1/n} f_L = \text{antilog}_{10} (\log_{10} f_c + .3/2n); \quad \text{Equation 7-7}$$

and

$$\Delta f = f_U - f_L. \quad \text{Equation 7-8}$$

Table 7-2 shows how the lower and upper cutoff frequencies and the bandwidths vary as different fraction-octave filters are employed. Each row in the table represents a different constant percentage bandwidth filter (1/1, 1/2, 1/3, and 1/10). In this example, each of the four filters has the same center frequency, 1000 Hz. For the 1-octave, 1/2-octave, 1/3-octave, and 1/10-octave filters in Table 7-2, the upper cutoff frequency is always 1 octave, 1/2 octave, 1/3 octave, and 1/10 octave, respectively, higher than the lower cutoff frequency.

Preferred Center Frequencies for Constant Percentage Bandwidth Filters. Although in theory any percentage can be adopted for a constant percentage bandwidth filter, the center frequencies ordinarily are selected in accordance with national or international standards such as ANSI S1.6-1984, revised 1997 (American National Standards Institute). With that standard, the increment in frequency from one f_c to the next in the series is a constant percentage; hence *a geometric, or exponential, series of center frequencies is formed*. By that we mean that a ratio scale of frequencies is constructed just as we saw in Table 4-2; successive frequency units differ by a constant ratio ($\pm$ a small percentage), and that constant ratio is the base for the ratio, or exponential, scale.³

Table 7-3 lists the **preferred center frequencies** from 100 Hz (Band Number 20, where Band Number 1 is 1 Hz) to 10,000 Hz (Band Number 40). All values in the table are appropriate for 1/3- and 1/10-octave filters. The frequency values shown in **boldface** are the preferred frequencies for 1-octave filters.

We can see from Table 7-3 that the constant ratio (base) for the 1-octave filter is 2 (0.3 log unit). Thus $250/125 = 2$; $500/250 = 2$, $1000/500 = 2$, and so forth. For both the 1/3- and 1/10-octave filters,

Table 7-2. Parameters of Various Constant Percentage Bandwidth Filters. The Center Frequency is a Constant 1000 Hz

1/n	f_c	f_L	f_U	Δf	%
1/1	1000	707	1414	707	70.7
1/2	1000	844	1190	346	34.6
1/3	1000	891	1122	231	23.1
1/10	1000	966	1035	69	6.9

Table 7–3. Frequency Band Numbers (**N**) and Preferred Center Frequencies (**f_c**) for Nominal 1/3- or 1/10-Octave Filters and for 1-Octave Filters (boldface only). Values below band 20 and above band 40 have not been included.

N	f_c	N	f_c
20	100		
21	125	31	1250
22	160	32	1600
23	200	33	2000
24	250	34	2500
25	315	35	3150
26	400	36	4000
27	500	37	5000
28	630	38	6300
29	800	39	8000
30	1000	40	10,000

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the base (constant ratio) is 1.25 (0.1 log unit). Thus, $125/100 = 1.25$, $160/125 = 1.25$, $200/160 = 1.25$, and so on. Finally, the table reveals that the band numbers for successive octave bands are successive integer multiples of 3 (21, 24, 27, 30, , , , 39), whereas the band numbers for successive 1/3- and 1/10-octave filters are successive integers. Table 7–3 also can be used to identify the preferred frequencies for the seldom-used 1/2-octave filters; in that case, successive integer multiples of 3/2 are used.

Band-Reject Filter

A **band-reject filter**, which also is called a **band-stop**, or **notch, filter**, is a filter that *rejects, rather than passes, energy for frequencies between the lower and upper cutoff frequencies*. Such filters usually have a narrow bandwidth, and all five parameters are useful in helping distinguish among various band-reject filters.

Panels A and B of Figure 7–19 contain two band-reject filter curves that differ in **f_c**, **f_U**, and **f_L**. It should be apparent that the meaning of **f_c** for a band-*reject* filter is different from the meaning of **f_c** for a band-*pass* filter. Although the two filter curves in Figure 7–19 appear to have the same bandwidth, we shall see subsequently that the two bandwidths are different; the bandwidth for the curve in panel B is wider than the bandwidth for the curve in panel A. Band-reject

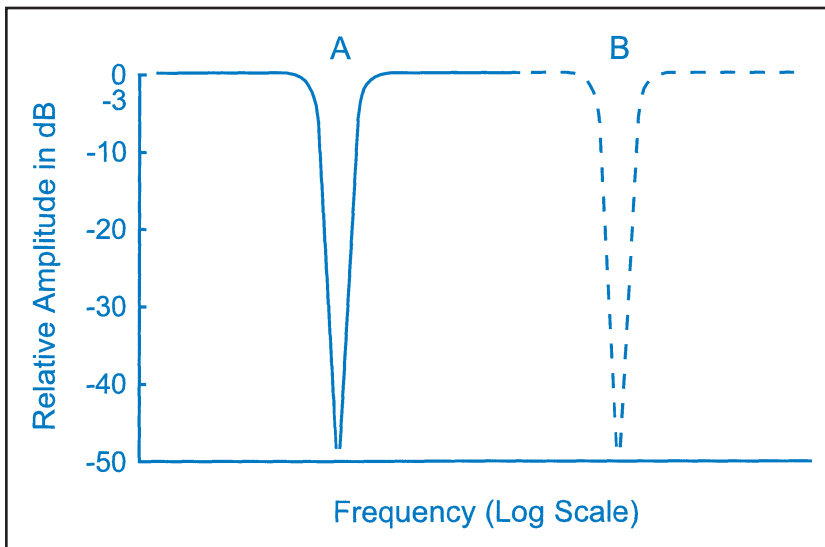


Figure 7-19. A comparison of two band-reject filter curves, filters that reject energy over a band of frequencies.

filters⁴ probably will not be encountered as frequently as the other three types, but they are important for applications such as measuring distortion, which will be discussed in Chapter 8.

SPECIFICATION OF LEVEL AT THE OUTPUT OF FILTERS

We know that the sound pressure level of a signal is reduced at the output of a filter (low-pass, high-pass, band-pass, or band-reject) relative to the level of the signal at the input. The filter *attenuates energy selectively as a function of frequency*. Energy in some frequency ranges is passed by the filter, whereas energy in other frequency ranges is rejected. For example, a **low-pass filter** with an upper cutoff frequency (f_U) of 1000 Hz *passes* energy through the filter for frequencies below 1000 Hz and *rejects* energy for frequencies above 1000 Hz. It therefore stands to reason that the sound pressure level at the output of the filter is less than the sound pressure level at the input.

How can we calculate the actual amount by which signal level is reduced by filtering? Although any sound wave can be filtered, it is convenient to use **white noise** in most of our examples because white noise is characterized by *equal energy per cycle*. That means simply that energy is present at all frequencies of vibration (to reasonable upper and lower frequency limits) and that the level or intensity *within any frequency band 1 Hz wide* (from $f - 0.5$ Hz to $f + 0.5$ Hz) is the same, regardless of the value of f .

The calculations will be much simpler for white noise than for other complex sound waves, but the principles are applicable for specification of the level of other types of sound waves that are filtered.

Before we consider the equations that will be used to accomplish that task, it is useful to introduce a new concept: **pressure spectrum level**, L_{ps} .

Pressure Spectrum Level

Pressure spectrum level, L_{ps} (commonly called just **spectrum level**), is *the sound pressure level in a frequency band of unit width (1 Hz), centered at some frequency, f* . L_{ps} is expressed in decibels. For a white noise signal, we can identify an infinite number of frequency bands that satisfy that definition, and they need not be integers: for example, $f = 300 \text{ Hz}, \pm 0.5 \text{ Hz}$; $f = 1300 \text{ Hz}, \pm 0.5 \text{ Hz}$; $f = 5000 \text{ Hz}, \pm 0.5 \text{ Hz}$; $f = 425.5 \text{ Hz}, \pm 0.5 \text{ Hz}$; and so forth.

When we say that the level or intensity within any frequency band 1 Hz wide is the same as the level or intensity within any other frequency band 1 Hz wide, that is the same as saying that white noise has a **pressure spectrum level slope** of 0 dB; L_{ps} is constant for all values of f .

L_{ps} is given by:

Equation 7-9

$$L_{ps} = \text{SPL}_{wb} - 10 \log \frac{\Delta f_{wb}}{\Delta_o f},$$

where L_{ps} is expressed in dB SPL,

SPL_{wb} refers to the sound pressure level in some wide-band signal,

Δf_{wb} refers to the bandwidth of the wide-band signal, and

$\Delta_o f$ is a reference bandwidth equal to unity (1 Hz).

Because the denominator in the ratio in Equation 7-9 is 1, and because the log of 1 = 0, Equation 7-9 for L_{ps} is simplified as:

Equation 7-10

$$L_{ps} = \text{SPL}_{wb} - 10 \log \Delta f_{wb}.$$

Solution of the following problem should demonstrate the simplicity of the concept of pressure spectrum level. Suppose a white noise signal with a bandwidth of 10,000 Hz has a **sound pressure level** of 80 dB SPL re: 20 μPa . *What is the pressure spectrum level?*

$$L_{ps} = 80 - 10 \log 10,000 = 40 \text{ dB SPL}.$$

The reference pressure for **pressure spectrum level** is the same as the reference pressure used to express the sound pressure level of a sinusoid or of any complex periodic or aperiodic sound wave: 20 μPa (**MKS**) or $2 \times 10^{-4} \text{ dyne/cm}^2$ or microbar (**cgs**). Thus, for the example above, the level of the wide-band (10,000 Hz) signal is 80 dB SPL re: 20 μPa , and the **pressure spectrum level** is 40 dB SPL re: 20 μPa .

The concept of pressure spectrum level is simplified by imagining that the white noise signal is filtered with an **idealized rectangular filter** with a bandwidth of 1 Hz that is centered at any desired frequency (f). The level at the output of the rectangular filter is the **pressure spectrum level**. For the example above, $L_{ps} = 40$ dB SPL, and, because the wide-band signal is white noise, we will measure that exact same pressure spectrum level of 40 dB SPL for any frequency band that is 1 Hz wide regardless of the value of f .

Suppose we do not know the overall level of a signal but are given its bandwidth and **pressure spectrum level**. We can then calculate the overall level of the signal by rearranging Equation 7-10 to read:

$$SPL_{wb} = L_{ps} + 10 \log \Delta f_{wb}. \quad \text{Equation 7-11}$$

For example, if the **pressure spectrum level** of a white noise signal with a bandwidth of 8000 Hz = 45 dB SPL, what is the overall level of the noise?

$$SPL_{wb} = 45 + 10 \log 8000 = 84 \text{ dB SPL}.$$

We have emphasized that with a white noise signal, the level or intensity within any frequency band that is 1 Hz wide (from $f - 0.5$ Hz to $f + 0.5$ Hz) is the same, regardless of the value of f . We can think of each of those frequency bands as being *an independent sound source* and, by definition for white noise, each of those frequency bands, or independent sound sources, has exactly the same level or intensity as any other frequency band.

You might recognize that you have encountered this same kind of problem before. Recall from Equation 5-9 in Chapter 5 that the total intensity level resulting from combining some number of independent sound sources, each of which has equal intensity, is given by:

$$dB_N = dB_i + 10 \log N, \quad \text{Equation 5-9}$$

where dB_i refers to the SPL of one of the sources and

N refers to the number of sources that are combined.

We can rewrite Equation 5-9 and rename the terms in the equation so that it can be applied to the problem of specifying the level of filtered white noise, which results in Equation 7-11, that was presented previously:

$$SPL_{wb} = L_{ps} + 10 \log \Delta f_{wb}, \quad \text{Equation 7-11}$$

where Δf_{wb} is the bandwidth of the signal, and

L_{ps} is the **pressure spectrum level**.

- The bandwidth of the signal (Δf_{wb}) is analogous to “the number of sources being combined” (N).
- The pressure spectrum level (L_{ps}) is analogous to the “sound pressure level of one of the equal sources” (dB_i).
- SPL_{wb} is analogous to “the sound pressure level of the N sources combined.” The result is Equation 7–11.

In summary, if we know the bandwidth and overall level of the white noise signal, we calculate the pressure spectrum level (L_{ps}) with Equation 7–10 by *subtracting* $10 \log \Delta f_{wb}$ from the overall level. If, on the other hand, we know the bandwidth and the pressure spectrum level, we calculate the overall level with Equation 7–11 by *adding* $10 \log \Delta f_{wb}$ to the pressure spectrum level (L_{ps}).

Sample Problems

Problem 1: A white noise signal has a bandwidth of 9000 Hz and a level of 72 dB SPL. What is the pressure spectrum level?

Equation 7–10

$$L_{ps} = 72 - 10 \log 9000 = 32.5 \text{ dB SPL.}$$

Problem 2: A white noise signal has a bandwidth of 10,000 Hz and a level of 34 dB SPL. What is the pressure spectrum level?

Equation 7–10

$$L_{ps} = 34 - 10 \log 10,000 = -6 \text{ dB SPL.}$$

What is meant by saying that the pressure spectrum level in decibels is a *negative* 6 dB? The answer is the same as it was in Chapter 5. Negative decibels simply mean that the sound pressure of the signal of interest is *less than the reference sound pressure*, which for dB SPL is 20 μ Pa.

Problem 3: A white noise signal has a bandwidth of 7000 Hz and a pressure spectrum level of 51 dB SPL. What is the level of the white noise?

Equation 7–11

$$SPL_{wb} = 51 + 10 \log 7000 = 89.5 \text{ dB SPL.}$$

Problem 4: A white noise signal has a bandwidth of 6000 Hz and a pressure spectrum level of -12 dB SPL. What is the sound pressure level of the noise?

Equation 7–11

$$SPL_{wb} = -12 + 10 \log 6000 = 25.8 \text{ dB SPL.}$$

Specification of Level at the Output of Filters: The More General Case

In the previous section, we learned that

- *we can calculate the pressure spectrum level* from our knowledge of (1) the bandwidth and (2) the overall level of a signal (Equation 7–10), or conversely,

- we can calculate the **overall level** of a signal from our knowledge of (1) its bandwidth and (2) pressure spectrum level (Equation 7-11).

Next, we will consider the more general case of how to calculate the level of *any* narrow-band signal (**nb**), not just one that is 1 Hz wide, from our knowledge of the overall level of a wide-band signal (**wb**). Conversely, we can calculate the level of *any* wide-band signal (**wb**) from our knowledge of the level of a narrow-band signal (**nb**).

Figure 7-20 is a schematic representation of a white noise signal with a bandwidth that, because of filtering, becomes narrower as you move from the top of the figure where the bandwidth is 10,000 Hz to the bottom where the bandwidth is 1 Hz. It does not matter what type of filter is used (low-pass, high-pass, or band-pass) because our equations have only two parameters: *level* and *bandwidth*.

With Equation 7-12 we can calculate the level for *any* narrow-band signal (**nb**) from our knowledge of (1) the level of a wide-band signal (**wb**) and (2) the bandwidths (Δf) of both the wide-band and the narrow-band signal:

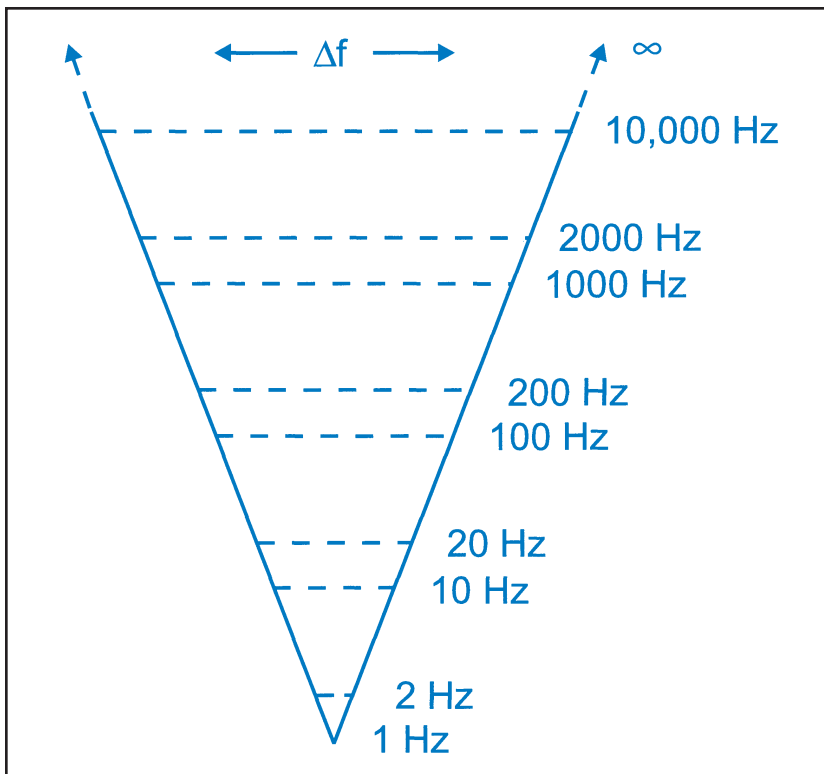


Figure 7-20. Geometric representation of the changing bandwidth of a white noise signal in which the bandwidth becomes progressively more narrow because of filtering. At the top, the unfiltered bandwidth is infinity. At the bottom, the bandwidth is unity (1 Hz).

Equation 7-12

$$\text{SPL}_{\text{nb}} = \text{SPL}_{\text{wb}} - 10 \log \frac{\Delta f_{\text{wb}}}{\Delta f_{\text{nb}}},$$

Note that Equation 7-12 is conceptually the same as Equation 7-10 for pressure spectrum level. The only difference is that now the narrow-band signal is not restricted to a bandwidth of 1 Hz. Because of that, the denominator in the ratio can be any value, not just “1.”

By rearranging Equation 7-12, we obtain Equation 7-13, which enables us to reverse the process and calculate the level for *any* wide-band signal from our knowledge of (1) the level of the narrow-band signal and (2) the bandwidths (Δ) of both the wide-band and the narrow-band signals:

Equation 7-13

$$\text{SPL}_{\text{wb}} = \text{SPL}_{\text{nb}} + 10 \log \frac{\Delta f_{\text{wb}}}{\Delta f_{\text{nb}}},$$

Consider this example. A white noise signal with a bandwidth of 10,000 Hz and a level of 90 dB SPL is filtered with a band-pass filter with $f_c = 2000$ Hz and $\Delta f = 1000$ Hz. *What is the level at the output of the filter?* You can proceed in one of two ways, but first you should recognize that for this particular problem, the value of f_c is irrelevant; f_c is not a term in Equation 7-12.

By use of Equation 7-12, we compute that:

$$\text{SPL}_{1000} = 90 - 10 \log (10,000/1000) = 80 \text{ dB SPL}.$$

Alternatively, we can use a different approach by first computing the **pressure spectrum level** with Equation 7-10 and then using Equation 7-11 instead of Equation 7-12. We must obtain the same answer:

$$L_{\text{ps}} = 90 - 10 \log 10,000 = 50 \text{ dB SPL}$$

and then

$$\text{SPL}_{\text{wb}} = 50 + 10 \log 1000 = 80 \text{ dB SPL}.$$

Sample Problems

For each of the five sample problems, the signal is white noise with a bandwidth of 12,000 Hz and a level of 95 dB SPL.

Problem 1: What is the level at the output of a band-pass filter with $\Delta f = 1200$ Hz?

Equation 7-12

$$\text{SPL}_{1200} = 95 - 10 \log \frac{12,000}{1200} = 85 \text{ dB SPL}.$$

Although it is perfectly acceptable to use Equation 7-12, when there is a series of problems to solve that begin with the same information, it saves time and computational steps to first calculate the **pressure spectrum level** (L_{ps}) with Equation 7-10

and then use Equation 7-11 for the rest of the problems. That approach avoids the necessity of repeated division. For example,

$$L_{ps} = 95 - 10 \log 12,000 = 54.2 \text{ dB SPL},$$

and then

$$\text{SPL}_{1200} = 54.2 + 10 \log 1200 = 85 \text{ dB SPL}.$$

Either approach yields the same answer, but the initial calculation of **pressure spectrum level** saves the step of computing the ratio that is required when using Equation 7-12. It also is advantageous to have alternative approaches to solving a problem because that provides an independent check on the correctness of your solution.

Problem 2: What is the level at the output of a 1-octave filter with $f_c = 2000 \text{ Hz}$? Note that Δf for the filter was not given and, therefore, must be calculated.

$$\Delta f = 0.707 \times 2000 = 1414 \text{ Hz}$$

Equation 7-2

and then

$$\text{SPL}_{1414} = 54.2 + 10 \log 1414 = 85.7 \text{ dB SPL},$$

or

$$\text{SPL}_{1414} = 95 - 10 \log \frac{12,000}{1414} = 85.7 \text{ dB SPL}.$$

Because 85.7 dB SPL is the level at the output of a 1-octave filter, it is called an **octave-band level**. This problem provides a good illustration of why it is easier to solve some of these problems by first computing the **pressure spectrum level** because, in this instance, it saves us from dividing 12,000 by 1414.

Problem 3: What is the level at the output of a low-pass filter with $f_U = 1200 \text{ Hz}$? Note that Δf of the filter is not given. You should know, however, that for a low-pass filter, $\Delta f = f_U$. Therefore,

$$\text{SPL}_{1200} = 54.2 + 10 \log 1200 = 85 \text{ dB SPL},$$

or

$$\text{SPL}_{1200} = 95 - 10 \log (12,000/1200) = 85 \text{ dB SPL}.$$

Problem 4: What is the level at the output of a high-pass filter with $f_L = 1200 \text{ Hz}$?

As with problem 3, Δf of the filter is not given. Unlike problem 3 with a low-pass filter, the value of f_L for a high-pass filter does not equal the bandwidth. The white noise signal at the input to the filter has an upper limit of 12,000 Hz. If that signal

is high-pass filtered with a lower cutoff frequency of 1200 Hz, that means that energy below 1200 is attenuated, and all that remains is the energy between the cutoff frequency of 1200 Hz and the upper limit of the unfiltered noise, 12,000 Hz. Thus, to find the bandwidth of the filtered noise, subtract f_L from the bandwidth of the unfiltered noise ($\Delta f = \Delta f_{\text{unfiltered}} - f_L = 12,000 - 1200 = 10,800$ Hz). Therefore,

$$\text{SPL}_{10,800} = 54.2 + 10 \log 10,800 = 94.5 \text{ dB SPL},$$

or

$$\text{SPL}_{10,800} = 95 - 10 \log \frac{12,000}{10,800} = 94.5 \text{ dB SPL}.$$

Problem 5: Suppose we remove exactly half of the energy in the white noise ($\Delta f = 12,000$ Hz) by some filter. By how much will the level at the input to the filter be attenuated at the output?

Notice that the type of filter is not specified. Does that matter? No! The type of filter is irrelevant if you know the bandwidth, Δf . For example, we might use a low-pass filter with $f_u = 6000$ Hz, a high-pass filter with $f_L = 6000$ Hz, or a band-pass filter with $\Delta f = 6000$ centered on some frequency of our choosing. For each of those examples, the resulting bandwidth is the same, 6000 Hz. Now return to the original question: by how much will signal level be decreased at the output of the filter relative to signal level at the input? The answer must be the same as it was in Chapter 5. If the bandwidth is narrowed to one-half of its original value, the energy or power or intensity of the signal is halved, and if the intensity is reduced to one-half of its original value, the **level of intensity** will decrease by 3 dB because:

$$\text{dB} = 10 \log \frac{1}{2} = -3 \text{ dB}.$$

Thus, signal level is decreased by only 3 dB from 95 dB SPL to 92 dB SPL. We can confirm that answer by using the procedures described for the first four problems.

$$\text{SPL}_{6000} = 54.2 + 10 \log 6000 = 92 \text{ dB SPL},$$

or

$$\text{SPL}_{6000} = 95 - 10 \log \frac{12,000}{6000} = 92 \text{ dB SPL}.$$

Does the Type of Filter Affect the Pressure Spectrum Level?

Think carefully about the definition of **pressure spectrum level** before jumping to the wrong answer. Pressure spectrum level is *the*

sound pressure level in a frequency band 1 Hz wide. If the question concerns white noise, the type of filter (low-pass, high-pass, or band-pass) or the bandwidth of the filter does not affect the value of pressure spectrum level.

To illustrate this, suppose the level of a white-noise signal is 92 dB SPL and the bandwidth is 20,000 Hz. The **pressure spectrum level** of the unfiltered noise is 49 dB SPL:

$$L_{ps} = 92 - 10 \log 20,000 = 49 \text{ dB SPL.}$$

Equation 7-10

Next, suppose the noise is directed to the input of a **1-octave filter** (a **constant percentage bandwidth filter**) with $f_c = 1000$ Hz. What is the **pressure spectrum level** of the noise at the output of the 1-octave filter?

$$\Delta f = 0.707 \times 1000 = 707 \text{ Hz}$$

and then

$$\text{SPL}_{707} = 92 - 10 \log \frac{20,000}{707} = 77.5 \text{ dB SPL.}$$

Finally, calculate L_{ps} from your knowledge of the level of the signal at the *output* of the 1-octave filter. The answer must be the same as computed from knowledge of the bandwidth and level of the unfiltered signal, 49 dB SPL:

$$L_{ps} = 77.5 - 10 \log 707 = 49 \text{ dB SPL.}$$

The answer, 49 dB SPL, is identical to what was calculated before the 1-octave filter was inserted, and that is as it must be. **Pressure spectrum level** is *the sound pressure level in any frequency band 1 Hz wide*. It does not matter which of the infinite number of frequency bands of the noise we had in mind when we calculated the spectrum level. After all, they each had the same amount of energy—hence the same sound pressure level—which for a unit bandwidth is called the **pressure spectrum level**.

In summary, the **pressure spectrum level** of white noise is the **pressure spectrum level**, and neither the type of filter nor the location of the filter along the frequency axis will alter that.

■ ANOTHER LOOK AT SELECTED TYPES OF NOISE

Before introducing new types of noise, we will continue with a **white noise** signal and expand on what was learned in Chapter 6.

White Noise

White noise has energy present at all frequencies (to reasonable lower and upper frequency limits) and the sound pressure level within any frequency band 1 Hz wide from $f - 0.5 \text{ Hz}$ to $f + 0.5 \text{ Hz}$ is the same,

regardless of the value of f . Figure 7–21 shows the “amplitude spectrum” of white noise. The term “amplitude spectrum” is used in a generic sense, and it is important to define precisely what is meant by “amplitude.”

- In panel A of Figure 7–21 the ordinate is **pressure spectrum level** in dB SPL and the abscissa is frequency (log scale). For this example, $L_{ps} = 45$ dB SPL.

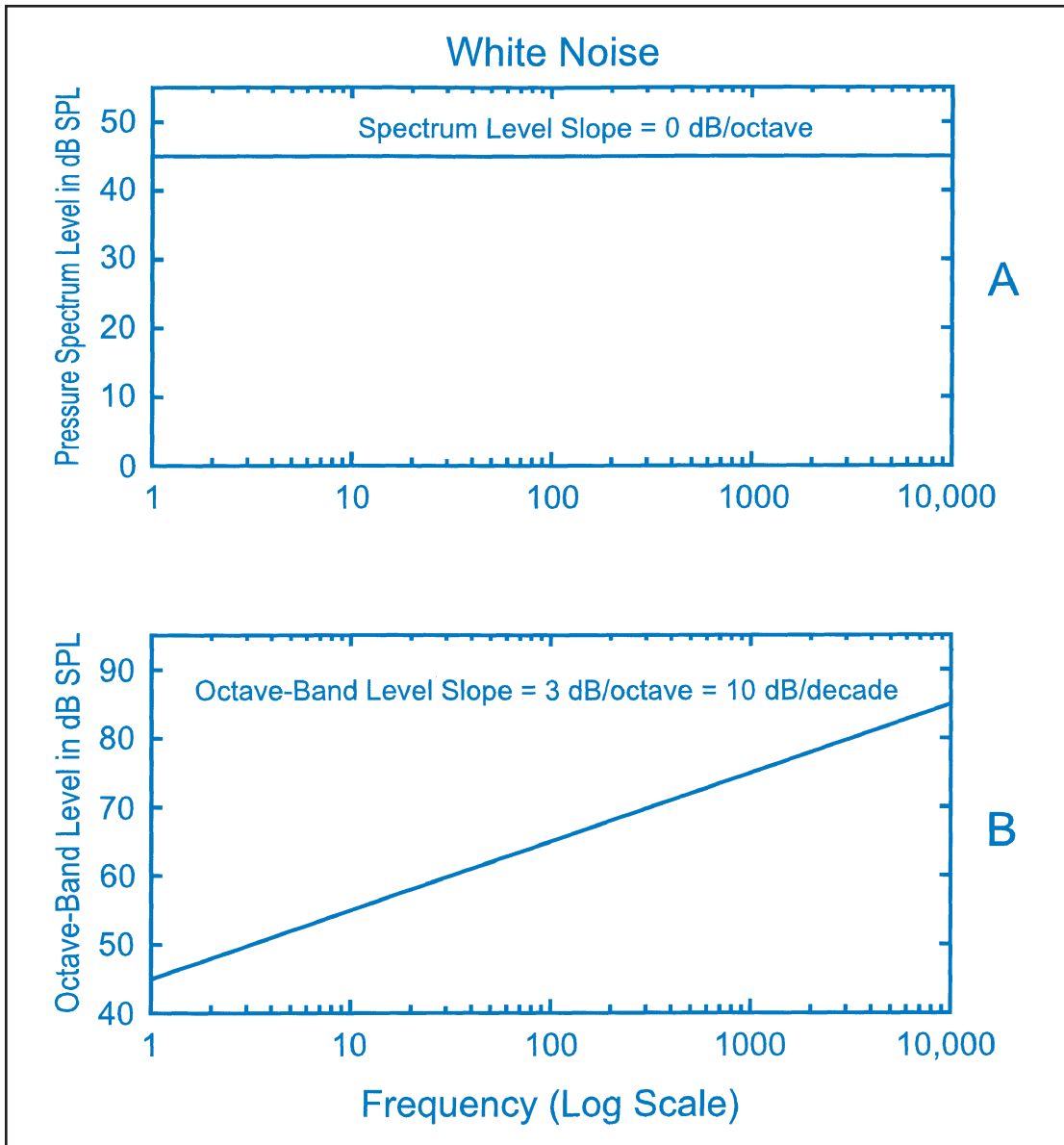


Figure 7–21. Amplitude spectra of white noise. The ordinate in Panel A is **pressure spectrum level**, L_{ps} , and the envelope has a spectrum level slope of 0 dB/octave. In panel B, for the same signal, the ordinate is **octave-band level**, and the envelope has an octave-band level slope of +3 dB/octave or +10 dB/decade.

- Because L_{ps} is the same for all frequency bands 1 Hz wide, the **envelope** of the amplitude spectrum is a straight line parallel to the horizontal axis.
- We emphasize, however, that such a slope only occurs for white noise if the unit of measure on the ordinate is **pressure spectrum level**.
- Thus, **white noise** has a **pressure spectrum level slope** of 0 dB/octave.

Next, that same white noise is directed to the input of a **1-octave filter** and then we calculate the **octave-band levels** at the output of the filter for each of several center frequencies, f_c (100, 200, 400, 800, 1000, 2000, 4000 Hz, and so on). The results are shown in panel B of Figure 7-21. The ordinate now is **octave-band level in dB SPL** instead of pressure spectrum level. **Octave-band level** is used because we are plotting *the level in a frequency band that is one octave wide* ($f_U = 2f_L$).

When $f_c = 100$ Hz, the octave-band level is 63.5 dB SPL because:

$$\Delta f = 70.7 \text{ Hz.}$$

Then,

$$L_{ps} = 45 \text{ dB SPL,}$$

and

$$\text{octave-band level} = 45 + 10 \log 70.7 = 63.5 \text{ dB SPL.}$$

Equation 7-10

Equation 7-11

- With each subsequent increase of f_c by a factor of 2:1, the bandwidth increases by 2:1, twice as much energy is passed by the filter, and the level at the output increases by 3 dB. Thus, the levels are 66.5 for $f_c = 200$ Hz, 69.5 for $f_c = 400$ Hz, and so on.
- With each increase of f_c by a factor of 10:1, the bandwidth increases by 10:1, ten times as much energy is passed by the filter, and the level at the output increases successively by 10 dB. Therefore, if the octave-band level for $f_c = 100$ Hz is 63.5 dB SPL, the octave-band level for $f_c = 1000$ Hz is 73.5 dB SPL.
- We observed in panel A of Figure 7-21 that white noise has a **pressure spectrum level slope** of 0 dB/octave. We also can say that *white noise has an octave-band level slope of +3 dB/octave or +10 dB/decade* (decade = 10:1) because the octave-band level increases by 3 dB with each doubling of f_c (and of bandwidth) and increases by 10 dB with each tenfold increase of f_c (and of bandwidth).

Now, look carefully at panel B of Figure 7-21 where the ordinate is labeled Octave-Band Level in dB SPL. The pressure spectrum level (*the sound pressure level in a frequency band 1 Hz wide*) is 45 dB SPL for the white noise in the figure. Also, observe the values for octave-band level as a function of frequency: $f_c = 1000$ Hz = 73.5 dB; $f_c = 100$ Hz = 63.5 dB; $f_c = 10$ Hz = 53.5 dB; and $f_c = 1$ Hz = 43.5 dB.

If the **pressure spectrum level** *really is the pressure spectrum level*, how can it be that $L_{ps} = 45$ dB, but according to the octave-band level calculations, the level is 43.5 dB when $f_c = 1$ Hz? There is no discrepancy! The pressure spectrum level is 45 dB SPL just as we calculated, and the octave-band level is 43.5 dB SPL when the filter has a center frequency of 1 Hz. But, if $f_c = 1$ Hz, $\Delta f = 0.707$ Hz, and thus the octave-band level is less than the spectrum level, and the octave-band level is:

$$SPL_{0.707} = 45 + 10 \log .707 = 43.5 \text{ dB SPL.}$$

We have learned that white noise has a **pressure spectrum level** slope of 0 dB/octave and an **octave-band level** slope of +3 dB/octave or +10 dB/decade. Suppose the white noise in Figure 7–21 is filtered with a 1/3-octave filter with the same center frequencies used for the 1-octave filter (100, 200, 400, 800, 1000, 2000, 4000 Hz, and so on). The levels at the output of a 1/3 octave-band filter are **1/3 octave-band levels**. Can you state the **1/3 octave-band level** slope for white noise? *It is the same as the octave-band level slope: +3 dB/octave or +10 dB/decade.* If you plot the 1/3 octave-band level as a function of frequency, the result is a line parallel to the octave-band level function in Figure 7–21, but all 1/3 octave-band levels are 4.9 dB less than the 1 octave-band levels. *Why –4.9 dB?* Recall from Table 7–2 that Δf for a 1-octave filter is given by the product of f_c and 0.707, whereas Δf for a 1/3-octave filter is given by the product of f_c and 0.231. Thus,

$$\begin{aligned} \text{dB} &= 10 \log \frac{.231}{.707} \\ &= -4.9. \end{aligned}$$

Therefore, the 1/3 octave-band levels are 58.6 for $f_c = 100$ Hz, 61.6 for $f_c = 200$ Hz, 64.6 for $f_c = 400$ Hz, 67.6 for $f_c = 1000$ Hz, and so on. From those calculations, we should see that the **1/3 octave-band level** slope of white noise also is +3 dB/octave and +10 dB/decade.

What if a 1/10-octave filter with the same values of f_c were used? The slope is still +3 dB/octave and +10 dB/decade, but the **1/10 octave-band levels** are 10.1 dB less than the octave-band levels: $\Delta f = f_c \times 0.069$, and

$$\begin{aligned} \text{dB} &= 10 \log \frac{.069}{.707} \\ &= -10.1. \end{aligned}$$

In practice, we ordinarily do not use center frequencies with values of 100, 200, 400, 800 Hz, and so on for a 1/3-octave filter. *Preferred center frequencies* are listed in Table 7–3.

Regardless of which constant percentage bandwidth filter is used to analyze white noise, will we achieve sufficient *accuracy* by making only one measurement for each value of f_c ? *Almost certainly,*

no! Recall from Figure 6–13 that the probability density function for white noise assumes the shape of a normal curve; the amplitudes (and phases) are distributed normally. *Thus, the value obtained from a single measurement has some finite probability of being less than or greater than the mean instantaneous value.* Because of the time-varying nature of the instantaneous values, *accuracy* will be improved by computing the mean of several hundred, or even several thousand, measurements for each center frequency. If the number of measurements is sufficiently large, the mean for each value of f_c should closely *approximate* the values that we computed, which are the *true values*. With the automated measuring equipment that is available, that task is not as daunting as it might seem.

Accuracy vs. Precision of Measurement

The word *accuracy* was emphasized twice in the paragraph above. The intensity of noise fluctuates over time, which creates a challenge for making *accurate* measurements. We want to achieve “correct” readings, but the concept of “correct” involves distinguishing between **accuracy** and **precision** of measurement.

David Keast, in his book *Measurements in Mechanical Dynamics* (1967), provided an excellent account of what is meant by *accuracy of measurement*, and the difference between **accuracy** and **precision**. It is fitting that he began the discussion by writing, “Probably no term this side of decibel has been more misused and confused than *accuracy*” (p. 53).

Accuracy is defined by the (1) width of the error band about the mean value, and (2) the percentage of observations that lie within the band. The width of the error band is given by σ , which describes the magnitude of dispersion of the multiple readings about the mean value. If we assume that the multiple readings are distributed normally (a Gaussian distribution) about the mean, we know not only the width of the error band, but also the percentage of observations that fall within that band. *Precision*, in contrast to accuracy, refers to *how small a difference can be measured*.

Imagine this hypothetical scenario and see if the requirements for accuracy and precision are met. A technician, having evaluated the acoustic conditions in a classroom, writes that “*the background noise level, as measured with a precision sound level meter, was 72.162 dB SPL.*”

We know that noise level, particularly if conversational speech contributes to the background noise, varies over time. The level also probably is different for different locations in a room. Does the result of 72.162 dB SPL represent a single reading or is it the mean of several readings made at one or multiple locations?

- The report could be improved—but only slightly—if it were amended to read, “*The background noise level in the classroom was measured with a precision sound level meter, and*

ten readings were made at each of three room locations. The mean of those 30 measurements was 72.162 dB SPL.”

- Even though the amended report is better, *it still does not include sufficient information to determine the accuracy of the measurements.*

What about *precision*? Reporting the mean noise level with three decimal places (72.162) implies far greater precision than is possible. It is like a person going to a lumber yard and asking to have a board cut to the dimensions 6 ft. long and 2 ft., 1/256 in. wide.

Perhaps the hypothetical purchase of a board with those dimensions is so far-fetched that it never would happen. But, consider another example that is real. According to Keast (p. 56), in 1930 the National Park Service of the U.S. Department of the Interior posted a sign at the top of a mountain in California proclaiming that the elevation at that point was **14,496.811 ft.** That is equivalent to believing that the *precision of measurement* was 12/1000 of an inch.

How might the technician’s report on sound level measurements be improved? It could read something like, “*The background noise level in the classroom was measured with a precision sound level meter. Ten readings were made at each of three room locations, and the mean of those 30 measurements was 72 dB SPL ($\sigma = 3.0$ dB).*” The revised statement allows us to infer, based on the assumption that the multiple readings are distributed normally, that about 20 of the readings lie between 69 and 75 dB SPL. Moreover, the mean value is reported more realistically by not implying a level of precision that is impossible to achieve.

It is important to keep the distinction between *accuracy* and *precision* in mind when reading Chapter 10 concerning factors such as *background noise level*, *reverberation time*, and *reverberant tails* with respect to the condition of a room’s acoustical environment.

Pink Noise

The noise shown in Figure 7–22 is **pink noise**. The ordinate in panel A is **pressure spectrum level** and the abscissa is frequency (log scale). The spectral envelope of this noise has different characteristics from those for white noise shown in Figure 7–21. We see in panel A that *pink noise has a **pressure spectrum level slope** of -3 dB/octave* and, from panel B, that *pink noise has an **octave-band level slope** of 0 dB/octave*. In other words, the output level from the 1-octave filter for this noise is the same regardless of where in the spectrum the filter is centered.

- If $f_c = 1000$ Hz ($\Delta f = 707$ Hz) and the level at the output is, for example, 45 dB SPL, when f_c is doubled, the bandwidth is doubled to 1414 Hz, but the output level, by definition, remains the same.

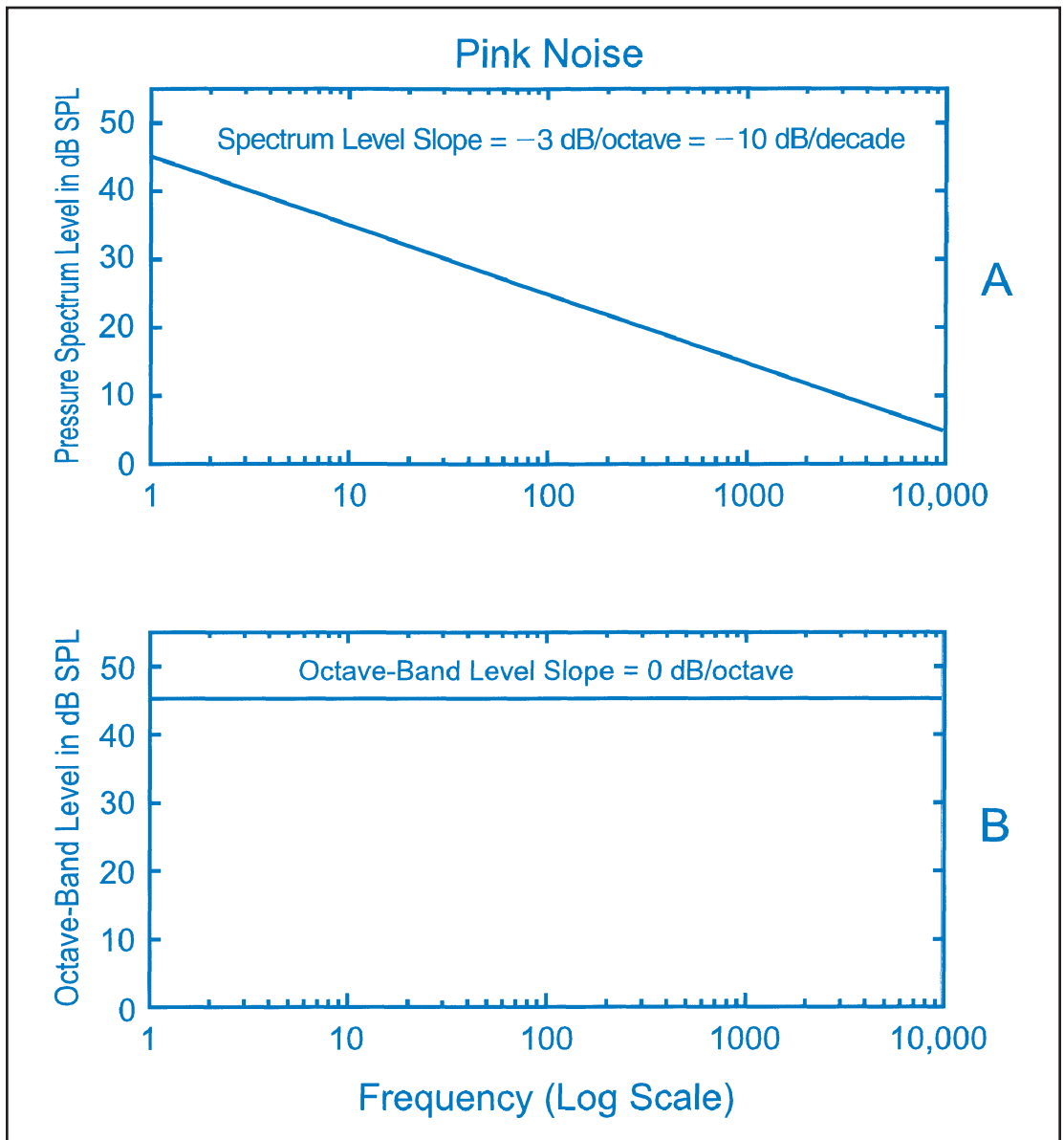


Figure 7-22. Amplitude spectra of pink noise. The ordinate in panel A is **pressure spectrum level**, L_{ps} , and the envelope has a spectrum level slope of -3 dB/octave or -10 dB/decade . In panel B, for the same signal, the ordinate is **octave-band level**, and the envelope has an octave-band level slope of 0 dB/octave .

- If the bandwidth is doubled from 707 Hz to 1414 Hz, energy is passed through the filter at twice as many frequencies.
- If energy is passed at twice as many frequencies, but the octave-band level remains the same, we must conclude that *the energy level per unit bandwidth decreases as frequency increases.*

- That accounts for a negative **pressure spectrum level slope**:
–3 dB/octave or –10 dB/decade.

We can construct a large array of “acoustic colors” such as red, magenta, and so forth by simply manipulating the pressure spectrum level slopes of the acoustic noises in accordance with standards that have been adopted.

Other Signals

Suppose we wish to determine the **amplitude spectrum** of speech spoken by a single talker. In this case, *we must perform measurements* on the signal, followed by calculations, rather than just calculations, as we did with white and pink noises. The measurement process is complicated by the fact that not only does speech intensity vary over time, it also varies among speech sounds (phonemes). Moreover, for any given sound, the amplitude spectrum is not linear over frequency as it was for white and pink noise.

To compensate, we might proceed as follows. First, record a passage with a duration of a minute or so, and ensure that the content of the passage includes all phonemes in the language in approximate proportion to their frequency of occurrence in spoken language. Second, choose a 1/3-octave filter instead of a 1-octave filter because its bandwidth is only 1/3 as wide. The narrower filter increases the accuracy of computation of L_{ps} for each value of f_c . Third, as with measurements of white noise, we *must* make several hundred, or several thousand, measurements for each value of f_c before we compute the mean 1/3 octave-band level and then compute L_{ps} .

Figure 7–23 shows the outcome of measurements made on a recording of a talker (male) reading a passage of nearly 1 minute in duration. The center frequencies, f_c , correspond to the **preferred frequencies** listed in Table 7–3 for a 1/3-octave filter.

The open circles represent the *measured 1/3 octave-band levels*, and the filled circles represent the **pressure spectrum levels**, L_{ps} , that were *calculated* from the *measured 1/3 octave-band levels*. The two curves diverge with increasing frequency from 13.6 dB at $f_c = 100$ Hz, where $\Delta f = 23.1$ Hz, to 33.6 dB at $f_c = 10,000$ Hz, where $\Delta f = 2310$ Hz.

NOTES

1. The ordinate in Figure 7–1 is “volume velocity,” which refers to the particle velocity of air molecules flowing through an area of one cubic meter per second (m^3/s) in the **MKS** system.
2. In an *electrical system*, inductive reactance and capacitive reactance are the relevant determinates of the natural frequency. Inductive reactance is the electrical equivalent of acoustic mass reactance, and capacitive reactance is the elec-

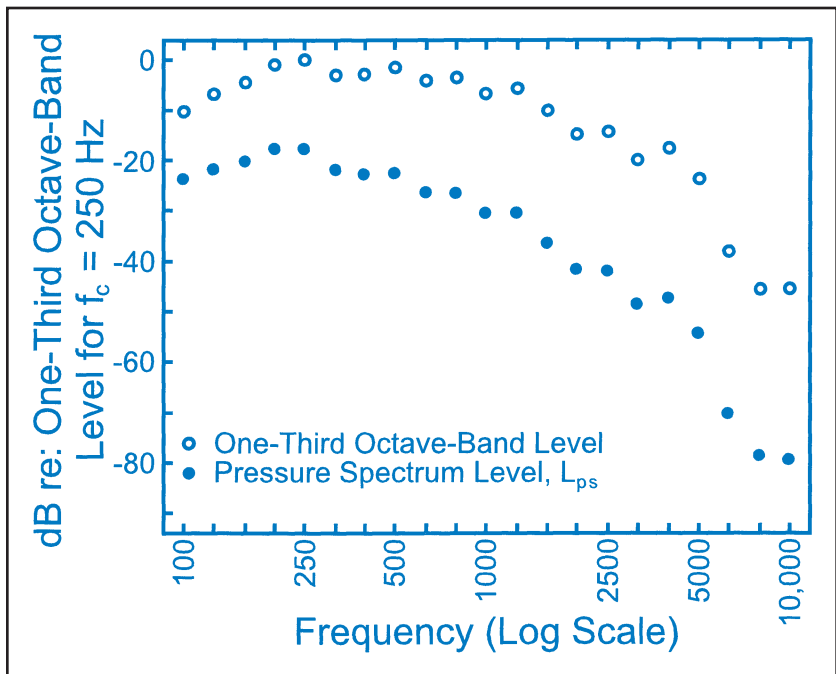


Figure 7-23. Outcome of measurements made on a 1-minute recording of a single (male) talker. The open circles represent the *measured 1/3 octave-band levels*, and the filled circles represent the **pressure spectrum levels, L_{ps}** , that were *calculated* from the 1/3 octave-band levels.

trical equivalent of acoustic compliant reactance. In a volume system, such as the vocal tract—the cavity that extends from the vibrating vocal folds at the lower end to the mouth opening at the upper end—the natural frequency depends on the length of the tube, whether the tube is open at both ends or open at only one end and closed at the other, and on the extent and location of constrictions in the vocal tract.

- As we learned from Table 4-2 in Chapter 4, the lowest, or starting, number (frequency) for the series can, in principle, be any value. The *preferred* reference frequency is 1 Hz. For a 1-octave filter, the constant ratio, or base, is 2. Thus, the **preferred center frequencies** are:

1, 2, 4, 8, 16, . . . , 250, 500, 1000 Hz, . . . , and so on.

For 1/3- and 1/10-octave filters, the reference frequency also is 1 Hz, but the base, or constant ratio, is 1.25. The **preferred center frequencies** for those filters are:

1, 1.25, 1.6, 2, . . . , 100, 125, 160, 200 Hz, . . . , and so on.

The base for a 1/2-octave filter is 1.4. One-half octave filters, however, are not used frequently, and we therefore will not elaborate further on their center frequencies.

ANSI SI.6 lists all **preferred center frequencies** from 1 Hz to 1 MHz for 1-octave, 1/3-octave, and 1/10-octave filters. For most applications that the student of speech and hearing will likely encounter, however, the preferred frequencies will range from about 100 Hz to 10,000 Hz. Lower or higher values can be determined by moving the decimal place appropriately.

4. A **band-reject filter** also represents the combined effects of low-pass and high-pass filtering. In this case, however, the two filters are operated in *parallel*, rather than in *series* as they are for a band-pass filter. By that we mean, a signal is directed simultaneously (in parallel) to the inputs of each of the two filters, one low-pass and one high-pass. When the outputs of the two filters are summed, the result is a signal that is band-rejected.

■ FREQUENTLY MISUNDERSTOOD CONCEPTS

1. A white noise signal has a bandwidth of 5000 Hz and a sound pressure level of 80 dB re: 20 μ Pa. If you insert a high-pass filter with $f_L = 4000$ Hz, what is the SPL re: 20 μ Pa at the output of the filter?

73 dB. First, calculate the pressure spectrum level of the unfiltered noise.

Equation 7-10

$$\begin{aligned} L_{ps} &= \text{SPL}_{wb} - 10 \log \Delta f_{wb} \\ &= 80 - 10 \log (5 \times 10^3) \\ &= 80 - 37 \\ &= 43 \text{ dB SPL.} \end{aligned}$$

Second, calculate the level for the high-pass filtered noise, but the equation requires that you know the bandwidth of the high-pass filtered signal. Energy below 4000 Hz is attenuated by the high-pass filter. Therefore, there is energy from 4000 Hz to the upper limiting frequency of the signal before it was filtered, 5000 Hz. Hence, $\Delta f = 1000$ Hz.

Equation 7-11

$$\begin{aligned} \text{SPL}_{wb} &= L_{ps} + 10 \log \Delta f_{wb} \\ &= 43 + 10 \log 10^3 \\ &= 73 \text{ dB SPL.} \end{aligned}$$

Alternatively, the problem could be solved in one step:

Equation 7-12

$$\text{SPL}_{nb} = \text{SPL}_{wb} - 10 \log \frac{\Delta f_{wb}}{\Delta f_{nb}}$$

$$\begin{aligned}
 &= 80 - 10 \log \frac{5 \times 10^3}{10^3} \\
 &= 80 - 7 \\
 &= 73 \text{ dB SPL.}
 \end{aligned}$$

2. For the same white noise signal ($\Delta f = 5000 \text{ Hz}$; $\text{SPL} = 80 \text{ dB}$), what is the pressure spectrum level of the noise at the output of a 1/3-octave filter with $f_c = 1750 \text{ Hz}$?

43 dB SPL. *Do not make unnecessary calculations!* If the answer is not obvious, review **Does the Type of Filter Affect the Pressure Spectrum Level?** If the signal is white noise, the pressure spectrum level is unaffected by the type of filter or its bandwidth; **pressure spectrum level** is *the sound pressure level in any band 1 Hz wide*.

3. Suppose the bandwidth of the noise in the two problems listed above were increased from 5000 Hz to 10,000 Hz. What is the SPL of the noise before it is filtered?

83 dB SPL. The bandwidth is increased by a factor of two (2:1) from 5000 Hz to 10,000 Hz, and white noise has equal energy in every frequency band 1 Hz wide (from $f - 0.5 \text{ Hz}$ to $f + 0.5 \text{ Hz}$) with all phases present in random array. If the bandwidth increases by 2:1, the intensity of the signal increases by the same factor, 2:1. We learned in Chapter 5 that an increase in intensity by a factor of 2:1 corresponds to 3 dB. Therefore $80 \text{ dB} + 3 \text{ dB} = 83 \text{ dB}$. Alternatively,

$$\begin{aligned}
 \text{SPL}_{\text{wb}} &= \text{SPL}_{\text{nb}} + 10 \log \frac{\Delta f_{\text{wb}}}{\Delta f_{\text{nb}}}, \\
 &= 80 + 10 \log \frac{10^4}{5 \times 10^3} \\
 &= 80 + 3 \\
 &= 83 \text{ dB SPL.}
 \end{aligned}$$

Equation 7-13

4. With the bandwidth of the noise now equal to 10,000 Hz instead of the original 5000 Hz, what is the pressure spectrum level?

43 dB SPL. L_{ps} is unchanged for reasons explained in answer to #2 above. By way of proof:

$$\begin{aligned}
 L_{\text{ps}} &= \text{SPL}_{\text{wb}} - 10 \log \Delta f_{\text{wb}} \\
 &= 83 - 10 \log 10^4 \\
 &= 43 \text{ dB SPL.}
 \end{aligned}$$

5. **One band of filtered white noise with $\Delta f = 100$ Hz and SPL = 90 dB is combined with a second band of filtered white noise with $\Delta f = 4000$ Hz and SPL = 90 dB. What is the total SPL of the two noises combined?**

93 dB SPL. The fact that the two noises have different bandwidths is irrelevant. The *intensity* of each band is 10^{-3} watt/m² (Equation 5–4 from Chapter 5). The intensity of the two combined noises, therefore, is 2×10^{-3} watt/m². Thus,

$$\begin{aligned} \text{dB} &= 10 \log \frac{2 \times 10^{-3}}{10^{-12}} \\ &= 93 \text{ dB SPL.} \end{aligned}$$

Alternatively, the problem can be solved with Equation 5–9 from Chapter 5:

$$\begin{aligned} \text{dB}_N &= \text{dB}_i + 10 \log N \\ &= 90 + 10 \log 2 \\ &= 93 \text{ dB SPL.} \end{aligned}$$

6. **White noise is directed to the input of a 1/3-octave filter with $f_c = 1000$ Hz. The level of the noise at the output of the filter is 61 dB SPL. If f_c is raised to 4000 Hz, what is the SPL at the output of the filter?**

67 dB SPL. A 1/3-octave filter is a *constant percentage bandwidth filter*. Thus, when f_c increases by some factor, 4:1 in this case, Δf increases by the same factor, 4:1. If the bandwidth increases by 4:1, the intensity also increases by 4:1, and as we learned in Chapter 5, an increase in intensity by a factor of 4 corresponds to 6 dB:

$$\begin{aligned} \text{dB} &= 10 \log \frac{I_x}{I_r} \\ &= 10 \log 4 \\ &= 6 \text{ dB.} \end{aligned}$$

The level of the noise therefore increases by 6 dB from 61 to 67 dB SPL. Note that it was not necessary to complicate the solution of the problem by calculating the bandwidths for the two center frequencies of the 1/3-octave filter; we simply understand that if f_c increases by some factor, Δf increases by the same factor. By way of proof, however, we learned from Table 7–2 that the bandwidth of a 1/3-octave filter is given by the product of f_c and 0.231. Thus, with $f_c =$

1000 Hz, $\Delta f = 231 \text{ Hz}$ (0.231×1000). Correspondingly, if f_c increases by a factor of four to 4000 Hz, $\Delta f = 924 \text{ Hz}$ (0.231×4000), which is four times greater than 231 Hz.

■ PRACTICE PROBLEMS

Set 1

For each of the problems in Set 1, you are dealing with a white noise that has the following characteristics: **Bandwidth (Δf) = 10,000 Hz; SPL = 90 dB re: 20 μ Pa.**

1. $L_{ps} = ?$
2. Calculate SPL at the output of each of the following filters:
 - a. Band-pass filter: $\Delta f = 200$ Hz; $f_c = 400$ Hz
 - b. Band-pass filter: $\Delta f = 400$ Hz; $f_c = 400$ Hz
 - c. Band-pass filter: $\Delta f = 1000$ Hz
 - d. Low-pass filter: $f_U = 1000$ Hz
 - e. High-pass filter: $f_L = 9000$ Hz
 - f. 1-octave filter: $f_c = 1000$ Hz
 - g. 1-octave filter: $f_c = 500$ Hz

Set 2

1. When you analyze a white noise with a constant percentage bandwidth filter at several different values of f_c , the SPL at the output of the filter increases at a rate of _____ dB/octave and _____ dB/decade.
2. If $L_{ps} = 40$ dB SPL:
 - a. What is the *sound pressure* in a band 1 Hz wide? _____
 - b. What is the *sound pressure* in a band 10,000 Hz wide?

Set 3

For each of the problems in Set 3, a white noise has the following characteristics:

Bandwidth (Δf) = 5000 Hz; SPL = 80 dB re: 20 μ Pa.

1. $L_{ps} = ?$

2. Calculate the SPL at the output of each of the following filters:

- a. Band-pass filter: $\Delta f = 200 \text{ Hz}$; $f_c = 400 \text{ Hz}$
- b. Band-pass filter: $\Delta f = 400 \text{ Hz}$ $f_c = 400 \text{ Hz}$
- c. Band-pass filter: $\Delta f = 1000 \text{ Hz}$
- d. Low-pass filter: $f_U = 1000 \text{ Hz}$
- e. High-pass filter: $f_L = 4000 \text{ Hz}$
- f. 1-octave filter: $f_c = 500 \text{ Hz}$

3. A white noise is analyzed with a 1-octave filter at seven preferred center frequencies. The octave-band level for 125 Hz is 35 dB SPL re: 20 μPa .

a. Calculate the following octave-band levels.

f_c	dB SPL
125	35
250	—
500	—
1000	—
2000	—
4000	—
8000	—

b. $L_{ps} = \text{—}$

c. If you use a 1/3-octave band filter instead of a 1-octave filter, what is the 1/3 octave-band level slope?



Distortion

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Imagine that a sound wave is delivered to some electronic system such as an amplifier, and we then compare the waveform at the *output* of the amplifier with the waveform at the *input*. If the system reproduces the waveform faithfully in every respect, the signal is *undistorted*. **Distortion**, on the other hand, means that the waveform has been altered. Three types of distortion will be addressed: **frequency distortion**, **amplitude distortion**, and **transient distortion**.

■ FREQUENCY DISTORTION

Amplitude Response of a System

Panel A of Figure 8–1 illustrates a method used to evaluate a component of a system (e.g., amplifier, loudspeaker) to determine its **amplitude response**, which often is called the **frequency response** of the system. A wide array of sinusoidal signals of different frequencies is directed to the input of the system. *The amplitude of the sine waves at the input must be kept constant for all frequencies.*

An appropriate measuring device then is used to assess the amplitudes, or levels, of each of the sine waves at *the output of the system*. When the *output amplitudes* are plotted as a function of frequency, we have defined the **amplitude response** of the system. If the system reproduces all frequencies with the same amplitude, the result is an output amplitude spectrum such as that shown in panel B.

A more realistic *output amplitude spectrum* is shown in panel C. Energy in the low and high frequency regions is attenuated because *the system does not reproduce all frequencies with the same relative amplitude*. In Chapter 7 we said that the input signal has been filtered, and now we say that *if filtering has occurred, the signal has undergone frequency distortion*. A more severe case of **frequency distortion** is shown in panel D. A land-line telephone is one familiar example of frequency distortion, which has an amplitude response (frequency response) that extends from approximately 300 Hz to 3400 Hz ($\Delta f = 3100$ Hz).

System Transfer Function

The three curves shown in panels B, C, and D of Figure 8–1 also describe the **system transfer function**.

- The **system transfer function**, or **amplitude response**, reveals the extent to which a signal undergoes frequency distortion. For example, if you examine the specifications of a moderately priced audio device, you might see that it has an amplitude response (the specifications probably refer to “frequency

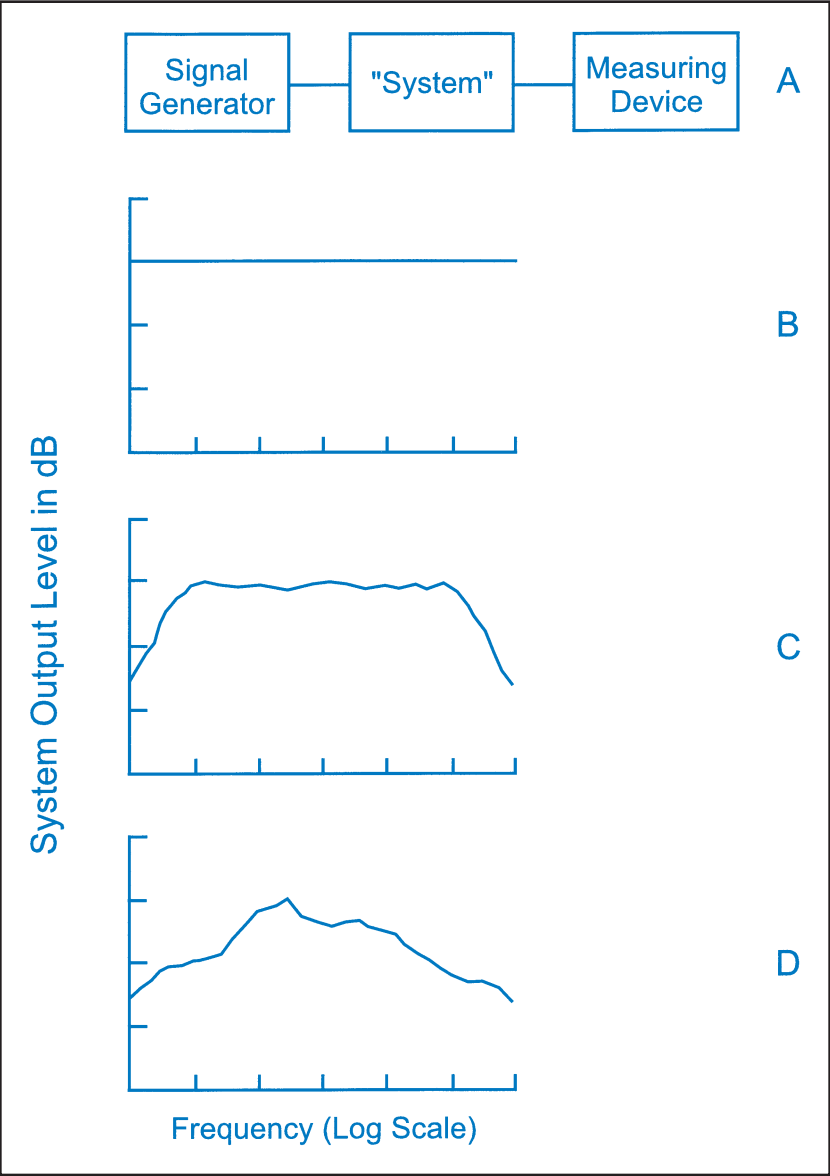


Figure 8-1. Panel A represents the electronic components that can be used for evaluation of a “system.” Sinusoidal signals *of constant amplitude and varying frequency* are directed to some device, the “system,” and the amplitudes of the various sine waves are then measured at the output of the device. Panels B, C, and D show three examples of amplitude response (frequency response) measured for different systems. The response in panel B shows the absence of frequency distortion (a hypothetical construct), whereas the responses in C and D show different magnitudes of **frequency distortion** because the system could not reproduce all frequencies with equal amplitudes.

response”) such as that shown by the curve in panel C of Figure 8–1. The system might have a relatively flat “pass-band” that extends from approximately 100 Hz to 10,000 Hz. Energy for frequencies that lie outside the passband, below 100 Hz and above 10,000 Hz, is attenuated.

The amplitude response of a system is described more completely by also specifying how *flat* the response is *within the pass-band*. For example, even if the width of the passband is the same for two different systems, one that is flat ± 0.5 dB is preferable to one that is flat ± 5 dB because the system with the flatter response within the passband evidences less frequency distortion.

■ AMPLITUDE DISTORTION

A discussion of amplitude distortion requires understanding the distinction between linear and nonlinear systems.

Linear Systems

Linear systems produce only frequency distortion—only the amplitudes (and phases) of the signal are altered by the system. Panel A of Figure 8–2 shows an *input-output function* of a **linear system**.

- *Output amplitude* on the ordinate is plotted as a function of *input amplitude* on the abscissa, and because the system is **linear**, the *input-output function* is a *straight line*. With a linear system, as the amplitude of the input increases (from left to right in the figure), a *proportional increase in output amplitude is achieved* (from bottom to top in the figure).
- Do not conclude that the level of the output signal *equals* the level of the input signal. It might, but it need not. Output level can equal input level, but it also can be greater than the input level (amplification) or less than the input level (attenuation). That is why we emphasize that changes in input level produce *proportional* changes in output level, and proportional changes occur when the input-output function is *linear*.
- In panel B we see that with a linear system, the characteristics of a sine wave at the input of the system (pointing upward) are preserved at the output of the system (pointing rightward); the shape of the output waveform (*sinusoidal*) is the same as the shape of the input waveform—it is not altered by the linear system.

What happens if the device is not linear? Devices that depart from linearity are referred to as **nonlinear systems**.

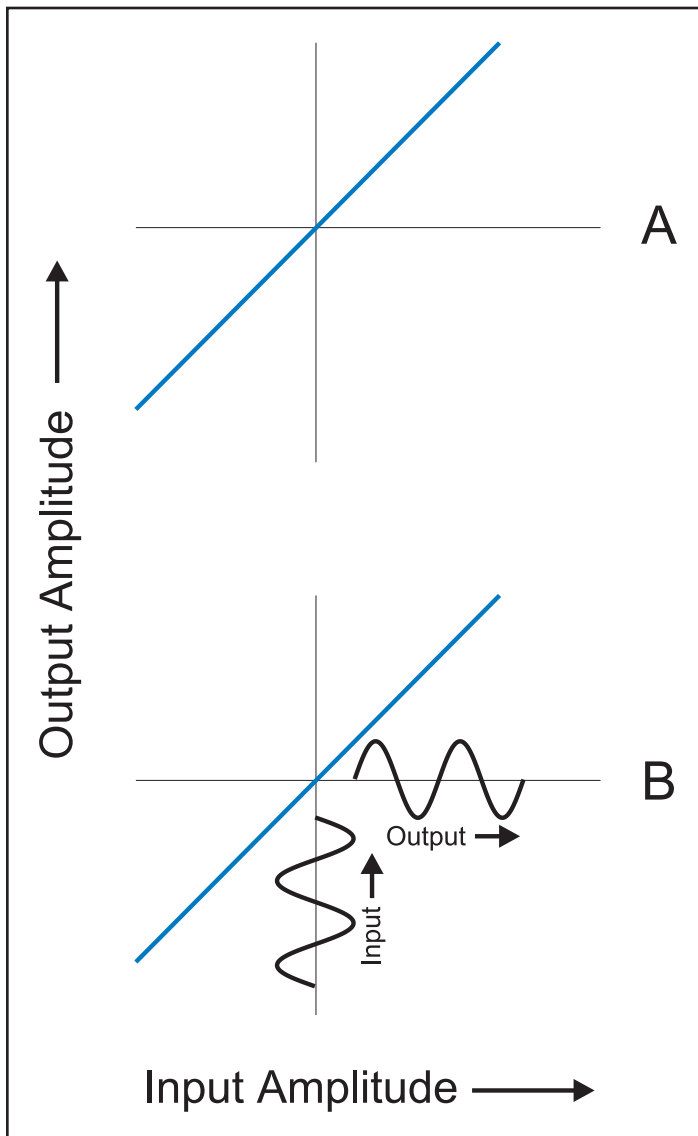


Figure 8–2. Panel A displays an input-output function for a **linear system**, a system for which the amplitude at the output is *proportional* to the amplitude at the input. Panel B shows that the characteristics of the waveform at the input to a **linear system** are faithfully preserved at the output of the system.

Nonlinear Systems

Figure 8–3 compares the input-output functions of a **linear system** in panel A with the input-output function for a **nonlinear system** in panel B. Output amplitude on the ordinate is plotted as a function of input amplitude on the abscissa.

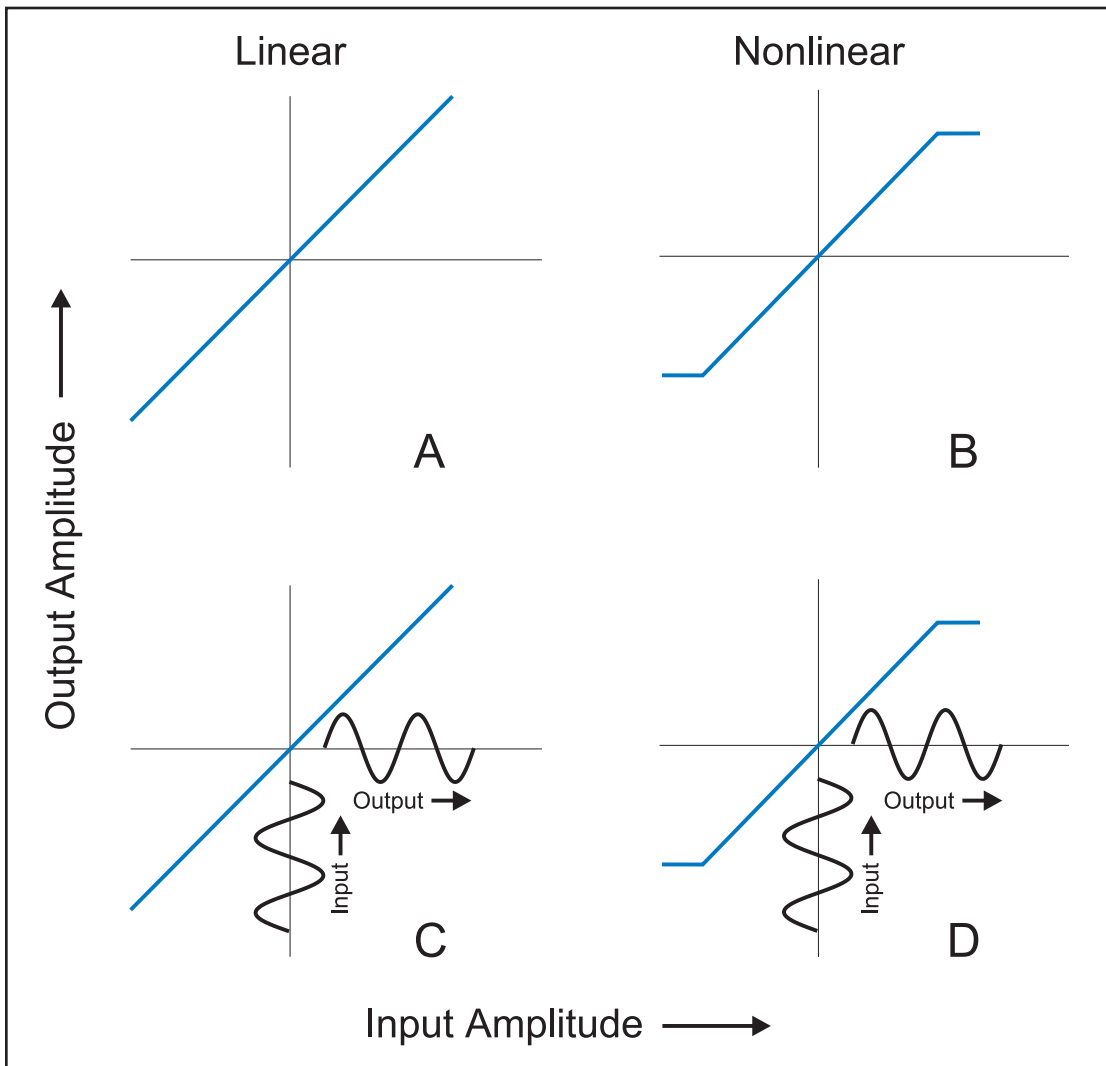


Figure 8–3. Amplitude distortion. Panel A shows an input-output function for the **linear system** that was shown previously in Figure 8–2. Panel B shows an input-output function for a **nonlinear system**. Output amplitude is *proportional* to input amplitude only over the linear portion of the function for the **nonlinear system**. Panel C compares the sinusoidal waveforms at the input and output of the **linear system** in panel A. Panel D compares the sinusoidal waveforms at the input and output of a **nonlinear system**. The waveform characteristics are faithfully preserved at the output of this nonlinear system *because the instantaneous amplitudes of the input signal do not exceed the limits of linearity of the input-output function in panel B*.

- We see in panel C that as the input amplitude to a *linear system* increases, a *proportional* increase in output amplitude is realized.
- In that case, the characteristics of a sine wave at the input in panel C (*pointing upward*) are faithfully preserved at the output of the system (*pointing toward the right*).

The same result happens for a **nonlinear system** (illustrated in panels B and D) *if the instantaneous amplitudes of the input signal do not exceed the limits of linearity of the input-output function of the nonlinear system*. Thus, in panel D, if our attention is confined initially to that portion of the function that is *linear* (a straight line), we observe that as the amplitude of the input signal increases, we again achieve a *proportional* increase in output amplitude. Furthermore, the characteristics of the input waveform are preserved in the output waveform, just as they were in panel C for a linear system. The characteristics of the input sine wave are preserved faithfully because none of the instantaneous amplitudes exceeds the limits of linearity.

Notice, though, that as the input amplitude increases from left to right in panel B of Figure 8–3, a point is reached where the input-output function becomes *nonlinear*, and in that region, *changes in input amplitude do not result in proportional changes in output amplitude*. Thus, for example, when operating on the *nonlinear portion* of the function, a 3-dB increase in input amplitude results in *less* than a 3-dB increase in output amplitude. At the extreme, further increases in input level produce no increase (or even a decrease) in output amplitude. When input level increases so much that we operate on the nonlinear portion of the input-output function, the system is “overdriven,” or “overloaded.”

What happens to a sinusoidal waveform when a nonlinear system is overdriven? The input amplitude to the system is sufficiently large that we operate on the nonlinear portion of the input-output function. The consequences are shown in Figure 8–4. In panel A, a large-amplitude sine wave is directed to the input of the system (pointing upward).

Although many of the *instantaneous amplitudes* of the input sine wave lie within the linear portion of the input-output function, some of the instantaneous amplitudes of the input signal that are *at or near the maximum amplitude* lie on the nonlinear portion of the function. For those, *the output amplitude is not proportional to the input amplitude*. Instead, the instantaneous amplitudes are reduced at the output, and the output waveform (pointing toward the right) is altered: it no longer possesses the sinusoidal shape that was apparent at the input.

The peaks of the sine wave are “clipped,” which is called **peak clipping**. More severe peak clipping is shown in panel B of Figure 8–4. Peak clipping is not the only form of amplitude distortion. For example, peak clipping can be symmetrical, as it is in Figure 8–4. With symmetrical clipping, both the positive and negative peaks of the waveform are clipped, which leaves only the center portions of the waveform unaffected. Peak clipping also can be asymmetrical, in which, for example, the positive peaks are clipped and both the center portions and the negative peaks are preserved faithfully. Furthermore, the signal can undergo **center clipping**, in which the center portions of the wave are eliminated and only the peaks are preserved.

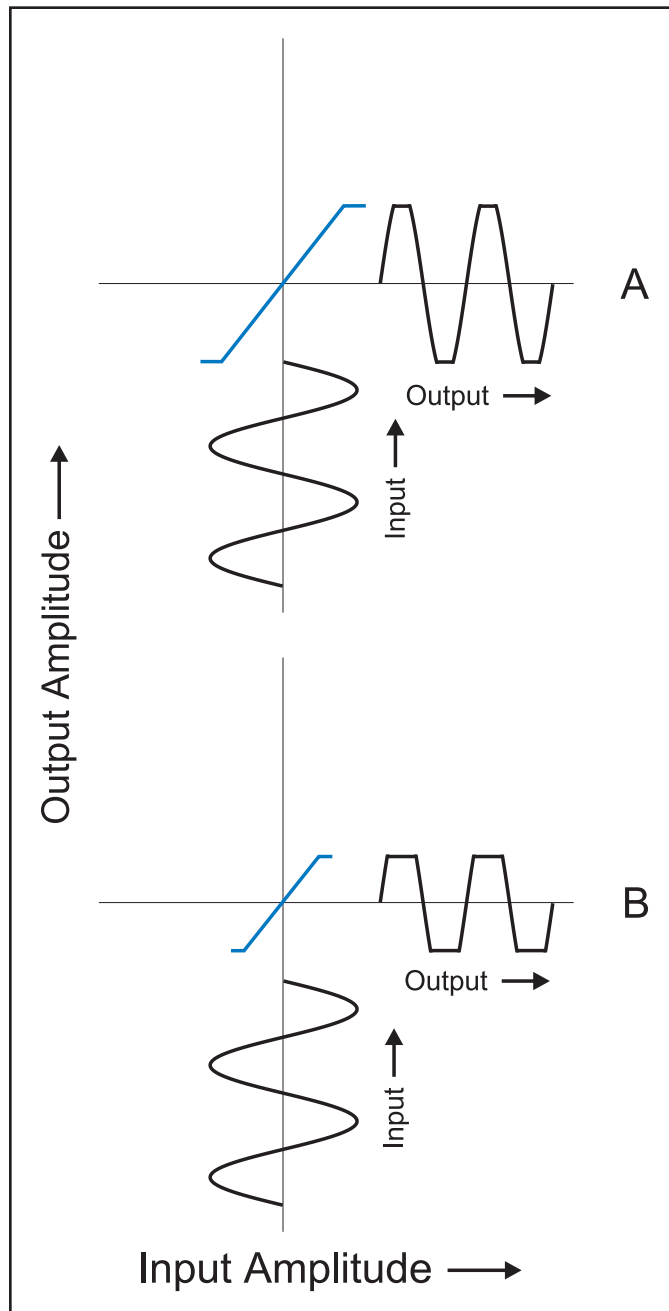


Figure 8–4. Amplitude, or nonlinear, distortion. In panel A, the instantaneous amplitudes at the input to a **nonlinear system** exceed the limits of linearity of the input-output function. The result is that output amplitude is *not* proportional to input amplitude and the signal is **peak clipped**. Panel B displays more severe peak clipping.

The waveform of the signal is altered, and therefore the signal is distorted. Because the *amplitude* of the waveform is the parameter that is altered, this form of distortion is called **amplitude distortion**. Distortion to the waveform occurs because the system is overdriven when operating on the *nonlinear portion* of the input-output function. This type of distortion, therefore, also is called **nonlinear distortion**. **Amplitude distortion** and **nonlinear distortion** are synonymous.

Effects of Amplitude, or Nonlinear, Distortion on the Amplitude Spectrum

We have seen the effects of amplitude, or nonlinear, distortion on the *waveform* of sinusoidal waves. What happens to the **amplitude spectrum**? The input signals in Figure 8–4 are sine waves. The output waveforms in both panels A and B are periodic, *but they are not sinusoidal*. We learned in Chapter 6 that any waveform that is not sinusoidal is a *complex waveform*. Thus, a sine wave that undergoes **nonlinear**, or **amplitude**, **distortion** becomes a *complex periodic waveform*.

Because the output waveform from a nonlinear system is *complex periodic*, we should expect (see Chapter 6) the waveform at the output of the nonlinear system to have energy at **harmonics** of the fundamental frequency—the frequency of the driving sine wave at the input. Therefore, this form of distortion *for a sine wave* also is called **harmonic distortion** because the output signal contains energy at **harmonics** of the fundamental driving signal.

Figure 8–5 shows the **amplitude spectrum** of a 100 Hz sine wave that has undergone harmonic distortion. The fundamental frequency (.1 kHz) of the complex periodic wave is the same as the frequency of the input sine wave, but *there also is energy at harmonics of the fundamental*. The amplitudes of five harmonics ($A_1 - A_5$) are shown in the figure; distortion products (energy at harmonics of the fundamental driving frequency) are present.

Percentage Harmonic Distortion

How do we quantify the amount, or degree, of harmonic distortion? The magnitude of harmonic distortion usually is expressed in *percentage*. If we return to our mythical purchase of an audio system, the specifications might say, for example, that you should expect not more than 1% harmonic distortion. It might not be intuitively obvious, however, what “1% harmonic distortion” means. To gain a better understanding of that concept, we defer to a “logical, or intuitive, derivation” of an equation.

Look again at the amplitude spectrum of the signal that has undergone harmonic distortion in Figure 8–5. A sine wave is delivered to an electronic device. The unit of measure of amplitude is

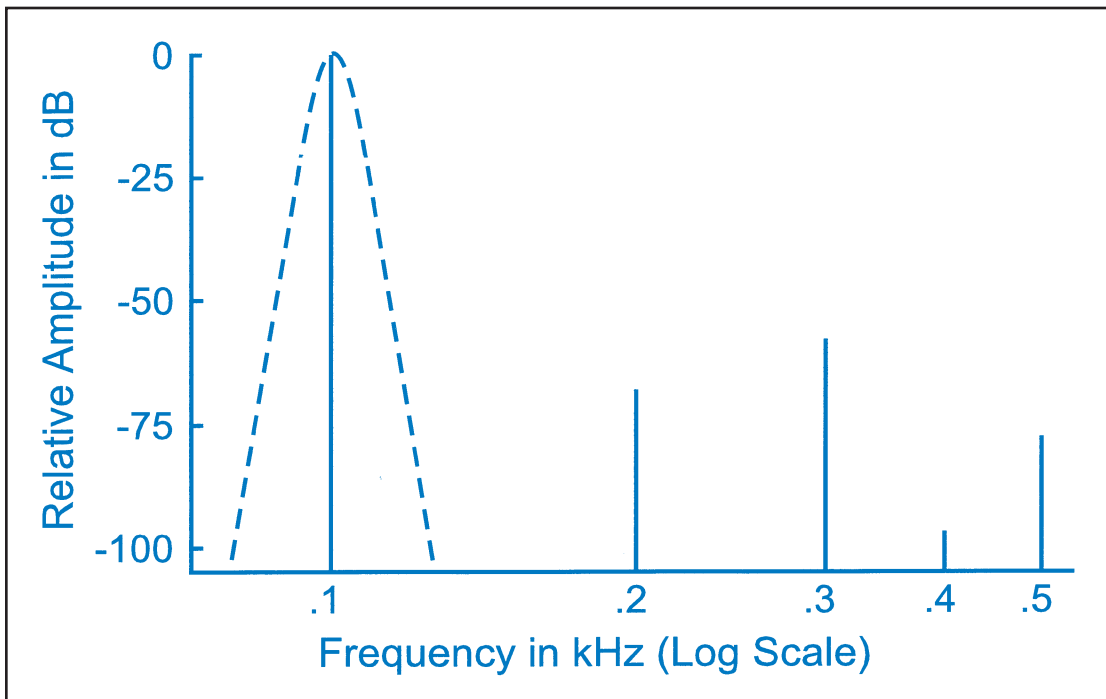


Figure 8–5. The **amplitude spectrum** of a sinusoidal waveform that has undergone **amplitude**, or **non-linear**, **distortion**. As a consequence of operating on the nonlinear portion of the input-output function, the output signal contains energy at **harmonics** of the fundamental driving signal at the input. The dashed curve illustrates the filter curve of a 1/3-octave filter centered on the fundamental frequency of the distorted signal to aid in measuring its voltage independently of the voltages of the higher harmonics.

voltage rather than sound pressure. The voltage (amplitude) of each of the harmonics in the output signal is measured with a wave analyzer. A wave analyzer is an electronic instrument that combines a voltmeter and a narrowly tuned band-pass filter with a variable f_c . Thus, one can “tune” the filter by adjusting the center frequency to a desired value and then measure “only” the voltage of the signal that is passed by the filter.

Suppose, for example, that the wave analyzer contains a 1/3-octave filter such as that shown by the dashed filter function in Figure 8–5. If the frequency of the input sine wave is 100 Hz, we set f_c of the wave analyzer to 100 Hz. The bandwidth of the 1/3-octave band filter is 23.1 Hz (Equations 7–6, 7–7, and 7–8). Thus, virtually all the voltage registered by the voltmeter will be the voltage of only the fundamental frequency of 100 Hz because voltages associated with the higher harmonics are well above the passband of the analyzing filter. The voltage of the fundamental frequency is recorded and the measurement process is repeated several times, each time with the filter tuned to a different center frequency to isolate each of the harmonics of interest: 200 Hz, 300 Hz, 400 Hz, and 500 Hz.

The input signal contains energy only at the fundamental frequency; it is a sine wave. The voltage of the fundamental (V_1) is the *desired energy* in the output signal. The output signal, however, contains energy at the desired frequency and *undesired energy* at higher harmonics (limited to four higher harmonics for convenience) of the fundamental; those voltages are referred to as V_2 , V_3 , V_4 , and V_5 .

- The *total energy* in the output signal equals the *desired energy* in the fundamental frequency and the *undesired energy* in the other four harmonics of the fundamental frequency.
- **Harmonic distortion** is the *proportion of total energy that is undesired energy*, which if multiplied by 100 is expressed as **percentage harmonic distortion**.
- Thus, by reference to the voltages of the fundamental and the higher harmonics in the output signal, we find that:

$$\text{Percentage Harmonic Distortion} = \frac{f(V_2, V_3, V_4, V_5)}{f(V_1, V_2, V_3, V_4, V_5)} \times 100,$$

where f is some undefined function,

and V refers to the voltage of some harmonic, V_1 through V_5 .

Recall from Chapter 5 that we do not ordinarily sum acoustic pressures because we seldom know their relative phases, and the same restriction applies to their electrical analogs, voltages. We can, however, sum the energies or powers or intensities, which is why the “equation” above does not contain “+” signs and therefore is unusable in its present form. We also learned in Chapter 5 that energy, power, or intensity is proportional to the square of pressure ($W \propto p^2$). Electrical energy or power is proportional to the square of voltage ($W \propto V^2$). Therefore, by squaring each of the voltages in the numerator and denominator, we can sum them and compute the ratio. Thus, the equation can be rewritten as:

$$\text{Percentage Harmonic Distortion} = \frac{f'(V_2^2 + V_3^2 + V_4^2 + V_5^2)}{f'(V_1^2 + V_2^2 + V_3^2 + V_4^2 + V_5^2)} \times 100.$$

Finally, the measures can be converted back from “voltage squared” to “voltage” by calculating *the square root of the ratio of the sum of the squared voltages*. The result is Equation 8–1 for harmonic distortion, which is written to accommodate n harmonics.

$$\text{Percentage Harmonic Distortion} = \sqrt{\frac{V_2^2 + V_3^2 + \dots + V_n^2}{V_1^2 + V_2^2 + \dots + V_n^2}} \times 100.$$

Equation 8–1

Equation 8–1 is used to quantify percentage harmonic distortion as *the percentage of total energy in the output waveform that resides in undesired harmonics*.

One technique used to measure percentage harmonic distortion employs a **band-reject filter**. With the filter set to 1000 Hz, the voltage read with an rms (root mean square) voltmeter corresponds to the numerator (voltages of all other harmonics) in Equation 8–1. A separate voltage reading is then made with the band-reject filter removed from the system, which corresponds to the denominator in Equation 8–1. The advantage of this approach is that it avoids the cumbersome computational steps required for measuring the voltages separately for each harmonic and then inserting those values into the equation.

Approximate Percentage Harmonic Distortion

Because most of the total energy in the denominator of Equation 8–1 resides in the fundamental frequency, percentage harmonic distortion sometimes is *approximated* by:

Equation 8–2

$$\text{Percentage Harmonic Distortion} = \sqrt{\frac{V_2^2 + V_3^2 + \dots + V_n^2}{V_1^2}} \times 100.$$

To illustrate, the second column of Table 8–1 lists the voltages, in millivolts, measured in the output signal for each of the five harmonics listed in column 1. The third column lists the squares of the voltages shown in column 2. With the aid of Equation 8–1 we calculate that we have 0.1% harmonic distortion. The same answer is obtained with the simplified Equation 8–2 *for this example*, and the explanation for the equivalence should be obvious. With either equation, the numerator is the same, 0.002626. With Equation 8–1, the denominator for this example is 2,500.002626, whereas with Equation 8–2 the denominator is 2,500, which is only marginally different. Thus, we see for the example in Table 8–1, and for most instances that we are likely to encounter, that Equation 8–2 provides a perfectly acceptable estimate of percentage harmonic distortion.

Table 8–1. Computation of Percentage Harmonic Distortion

Harmonic	Voltage (V) in millivolts	V ²
1	50.0	2,500.0
2	.01	.0001
3	.05	.0025
4	.001	.000001
5	.005	.000025
$\% = \sqrt{\frac{0.0001 + .0025 + .000001 + .000025}{2,500 + .0001 + .0025 + .000001 + .000025}} \times 100 = 0.1\%$		

Amplitude Response and Dynamic Range

Panel A of Figure 8–6 contains input-output functions for some system *for each of several input, or driving, frequencies*, not just one. The upper and lower limits of linearity are constant from about 500 to 5000 Hz, but the range of linearity diminishes for frequencies lower than 500 Hz and higher than 5000 Hz. Next, we choose some *maximum permissible harmonic distortion*, say 0.1%, to accommodate whatever purpose we have in mind for the system.

The point of maximum permissible harmonic distortion is located on each of the functions in the figure as filled dots in panel A. That means that if the input amplitude for that frequency exceeds the value corresponding to that point on the function, the maximum

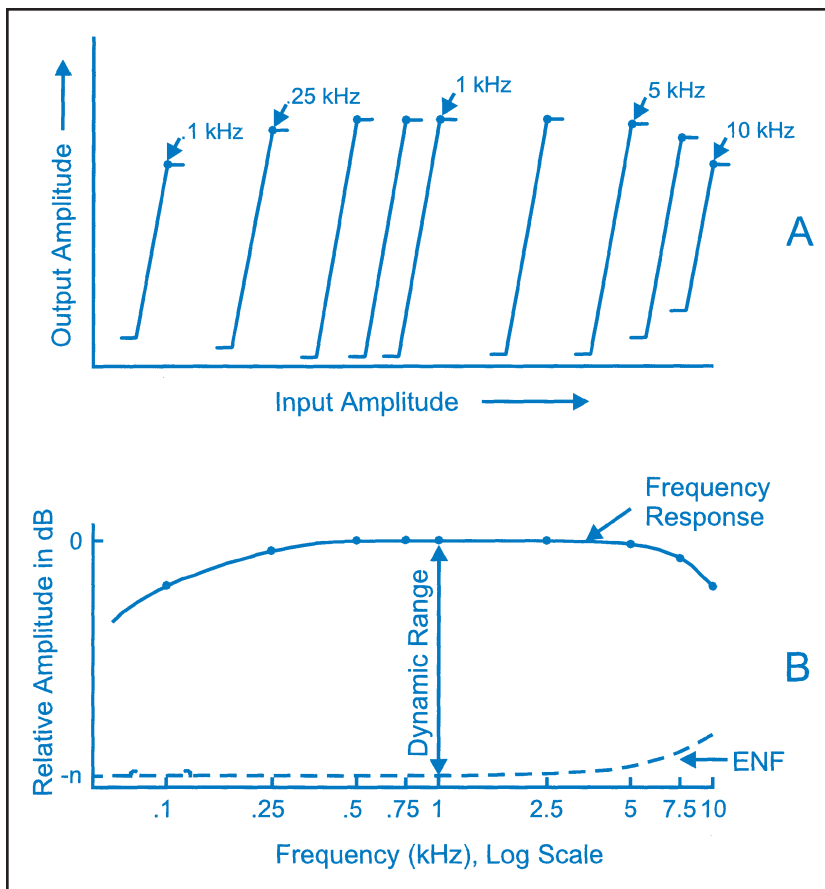


Figure 8–6. Input-output functions for each of nine frequencies are shown in panel A. In panel B, the amplitude response (*solid line*), in relation to the maximum permissible harmonic distortion, is compared with the electrical noise floor (**ENF**) of a system. The distance between the **ENF** and the maximum amplitude response defines the **dynamic range** of a system.

permissible harmonic distortion that was chosen is exceeded. In other words, those points define the maximum driving amplitude as a function of frequency that can be utilized without producing an unacceptable amount of harmonic distortion.

The points of *maximum permissible harmonic distortion* (filled dots) are redrawn in the frequency domain as the upper curve in panel B of Figure 8–6. When the points are connected, the resulting curve defines the **amplitude response** (or **frequency response**) of the system, just as we saw previously in panel C of Figure 8–1. As before, we see that the curve defines how energy is attenuated selectively as a function of frequency *in relation to the maximum permissible harmonic distortion* that we agreed can be tolerated.

Dynamic Range

The specifications for an electronic instrument likely will include a description of the **dynamic range** of the system. The dashed curve in panel B of Figure 8–6 describes *the electrical noise floor (ENF)* of a hypothetical electronic device. All electronic equipment generates a random time function, which as we learned in Chapter 6 is an aperiodic waveform, or *noise*, and the curve depicting the electrical noise floor shows how the level of the noise might vary as a function of frequency.

The **amplitude response curve** in Figure 8–6 reflects *the maximum signal level that the instrument can handle reliably without exceeding the maximum permissible harmonic distortion*. If the input signal level is reduced, percentage harmonic distortion is lowered. We obviously, however, do not want the driving signal level to be so low that the signal is “buried” in the noise. Thus, to achieve an optimal **signal-to-noise ratio**, we want the driving signal to correspond to the highest possible amplitude without exceeding a tolerable amount of harmonic distortion.

Dynamic range is *the distance in decibels between the electrical noise floor and the amplitude response curve*.

- If, for example, the dynamic range is 60 dB, that means that a signal can be delivered to the system with a signal-to-noise ratio of 60 dB or less without experiencing an intolerable amount of harmonic distortion.
- If signal level is reduced, percentage harmonic distortion is reduced, *but at the expense of a less favorable signal-to-noise ratio*.
- It should be obvious that because both the amplitude response curve and the ENF vary with frequency, the dynamic range also is frequency dependent. If you examine the specifications for an instrument, however, the dynamic range probably will be expressed as a single value—perhaps even the most favorable one.

Intermodulation Distortion

Harmonic distortion is synonymous with **amplitude**, or **nonlinear, distortion** *when the driving signal is a simple sinusoid*. Nonlinear distortion is considerably more complicated, however, when the driving signal is a complex waveform. We still experience amplitude, or nonlinear, distortion, but the output spectrum is more complex than “simple” harmonic distortion. Nonlinear distortion for complex input signals is called **intermodulation distortion**. It arises from the interactions among the various frequency components of the signal *and from the interactions among their harmonics*.

To illustrate the differences between **intermodulation distortion** and **harmonic distortion**, consider the simple case of a complex driving signal that has only two components, $f_1 = 100$ Hz and $f_2 = 110$ Hz. The frequency components of the output signal contain two types of **distortion products**: **harmonics** and **combination tones**.

Harmonics

The output signal contains energy at **harmonics** of *each* of the frequency components of the driving signal.

- Thus, there is energy at f_1 (100 Hz), $2f_1$ (200 Hz), $3f_1$ (300 Hz), and so forth.
- There also is energy at f_2 (110 Hz), $2f_2$ (220 Hz), $3f_2$ (330 Hz), and so on.
- Each of those components of the output signal is harmonically related to one or the other of the two components of the driving signal, f_1 and f_2 .

Combination Tones: Difference Tones and Summation Tones

In addition to the harmonically related components, a second type of distortion product comprises **combination tones**. Combination tones are further subdivided into two types: **difference tones** and **summation tones**.

- Examples of **difference tones** include $|f_1 - f_2|$ (10 Hz); $|2f_1 - f_2|$ (90 Hz); $|3f_1 - f_2|$ (190 Hz); and so on.
- Examples of **summation tones** include $|f_1 + f_2|$ (210 Hz); $|2f_1 + f_2|$ (310 Hz); $|3f_1 + f_2|$ (410 Hz); and so forth.

In theory, the combination tones comprise all possible sums (**summation tones**) and differences (**difference tones**) of the primary frequencies, f_1 and f_2 , *and integer multiples of those primary components*. In practice, the amplitudes of many of the higher combination tones are sufficiently small to render their contribution to the

total energy in the output signal negligible. Equation 8–3 defines the harmonics, summation tones, and difference tones produced by a nonlinear system:

Equation 8–3

$$mf_1 \pm nf_2,$$

where **m** and **n** are assigned all integer values: 0, 1, 2, 3, 4, . . . *n*.

If we apply Equation 8–3 to the example in which $f_1 = 100$ Hz and $f_2 = 110$ Hz, we generate the harmonics and combination tones listed in Table 8–2.

■ TRANSIENT DISTORTION

When sine waves were introduced in Chapter 2, their durations were treated as being infinitely long. *Under that circumstance the amplitude spectrum of any sine wave is a line spectrum.* Because duration of sine waves is finite, however, the amplitude spectrum is not a line spectrum.

Effects of Sine Wave Duration on the Amplitude Spectrum

Figure 8–7 shows the effects of sine wave duration on the amplitude spectrum. In panel A, the duration of the sine wave is infinite. It is a periodic signal, and a plot of *its amplitude spectrum, therefore, is a line spectrum with a single vertical line located at a position along the abscissa corresponding to the frequency of the sine wave, 1000 Hz.*

- Panel B shows what happens to the spectrum when the duration is only 100 ms and becomes a “tone burst,” or “sinusoidal pulse.” *Energy is spread to other frequencies, both above and below the driving frequency of the sine wave.* The tone

Table 8–2. Examples of Harmonics and Combination Tones Produced for a Complex Wave with Two Frequency Components, $f_1 = 100$ Hz and $f_2 = 110$ Hz

Harmonics of		Combination Tones	
f_1	f_2	Difference Tones	Summation Tones
$ 1f_1 + 0f_2 = 100$	$ 0f_1 + 1f_2 = 110$	$ 1f_1 - 1f_2 = 10$	$ 1f_1 + 1f_2 = 210$
$ 2f_1 + 0f_2 = 200$	$ 0f_1 + 2f_2 = 220$	$ 2f_1 - 1f_2 = 90$	$ 2f_1 + 1f_2 = 310$
$ 3f_1 + 0f_2 = 300$	$ 0f_1 + 3f_2 = 330$	$ 3f_1 - 1f_2 = 190$	$ 3f_1 + 1f_2 = 410$
etc.	etc.	etc.	etc.

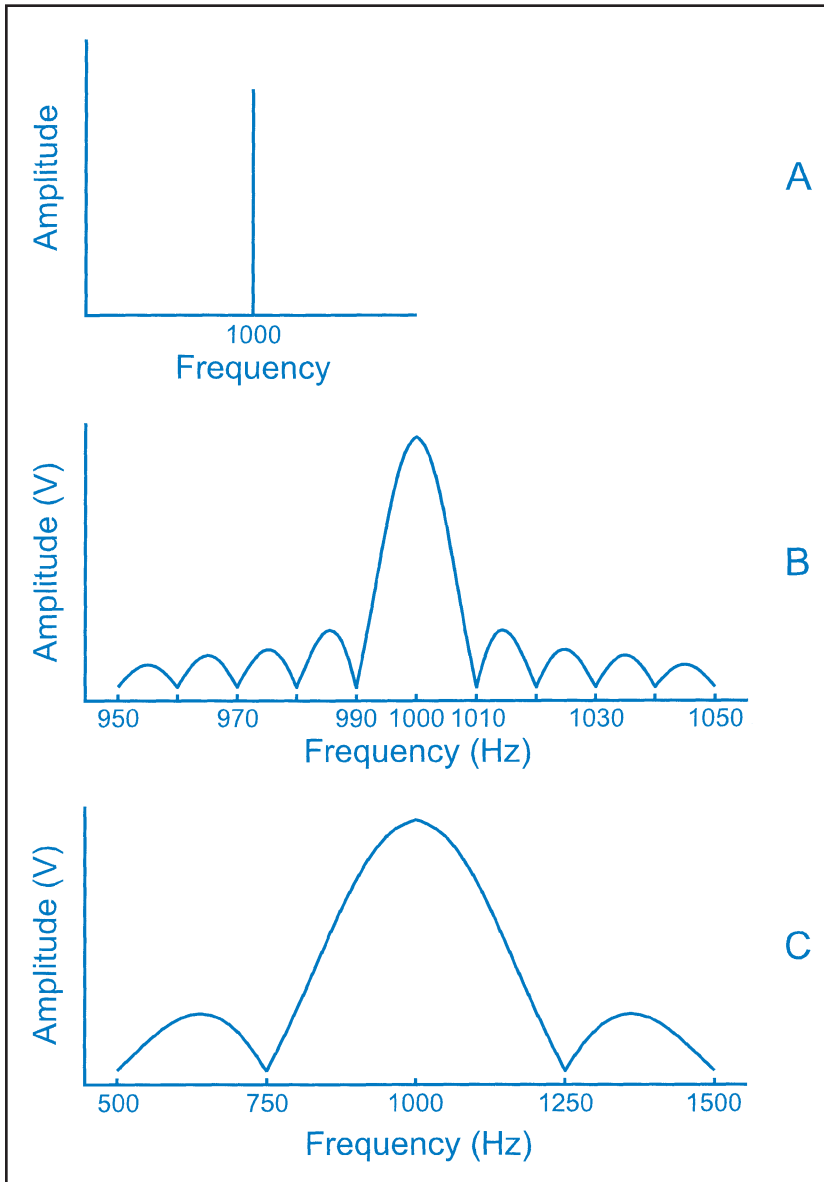


Figure 8-7. Effects of signal duration on the amplitude spectrum of a sinusoid. In panel A, duration is infinite and the amplitude spectrum is a line spectrum with energy at a single frequency. In panel B, duration is 100 ms, and energy is spread to surrounding frequencies to produce a continuous spectrum. In panel C, duration is shortened to 4 ms, which also produces a continuous spectrum. The nulls in panels B and C occur at different frequencies because their locations in the frequency domain are integer multiples of the reciprocal of duration. Adapted from *Signals and Systems for Speech and Hearing* (pp. 188–189) by Rosen, S. & Howell, P. (1991), Academic Press Inc., San Diego, CA. Copyright by Academic Press Limited. Reproduced with permission.

burst is **aperiodic**, and *the amplitude spectrum therefore is a continuous, rather than line, spectrum*. **Lobes** (peaks) and **nulls** (valleys) appear in the spectrum with a pattern that is similar, but not identical, to what was seen in Chapter 6 for a single rectangular pulse.

- The **nulls** occur at very specific frequencies: *Integer multiples of the reciprocal of the duration of the sine wave*. For the example in panel B of Figure 8–7 where duration is 100 ms, **nulls** occur at 10-Hz intervals ($1/0.1 = \pm 10$ Hz; $2/0.1 = \pm 20$ Hz; $3/0.1 = \pm 30$ Hz; etc.). Thus, because the frequency of the sine wave in Figure 8–7 is 1000 Hz, **nulls** occur at 990 Hz, 980 Hz, 970 Hz, and so forth on the lower side of 1000 Hz, and at 1010 Hz, 1020 Hz, 1030 Hz, and so forth on the upper side of 1000 Hz.
- Unlike the amplitude spectrum for the single rectangular pulse, the nulls occur on both sides (above and below) of the frequency of the sine wave (1000 Hz).

Panel C of Figure 8–7 shows what happens if the duration of the sine wave is shortened from 100 ms to 4 ms. In both panels B and C, nulls appear at frequencies corresponding to integer multiples of the reciprocal of duration. Because duration is shortened to 4 ms, the frequency of the first null increases to ± 250 Hz ($1/.004 = 250$ Hz) rather than ± 10 Hz.

- *The width of each band of energy, or lobe (between adjacent nulls), is inversely proportional to pulse duration.*
- Shortening the duration from 100 ms to 4 ms is a factor of 1:25, and consequently each band of energy is widened by a factor of 25:1 (from 10 Hz at 100 ms to 250 Hz at 4 ms).

Effects of Rise/Decay, or Rise/Fall, Time on the Amplitude Spectrum

When sine waves of fairly long duration, say a second or two, are turned on, they are not allowed to reach their maximum amplitude instantly. Instead, the amplitude is made to rise over time from zero to a maximum steady-state value, as is shown in panel A of Figure 8–8. They then remain steady-state for a duration of our choosing. Similarly, when tone bursts are turned off, the amplitudes do not instantly drop to zero; they decay over time from maximum to zero. The dashed curve in panel A defines the **amplitude envelope** in the time domain of the waveform.

Because the signal must be turned on and off, or gated, an amplitude spectrum is created that is more complex than that of a theoretical sine wave as shown in panel B. *Energy is spread to other frequencies, and the amplitude spectrum is a **continuous spectrum** rather than a line spectrum.*

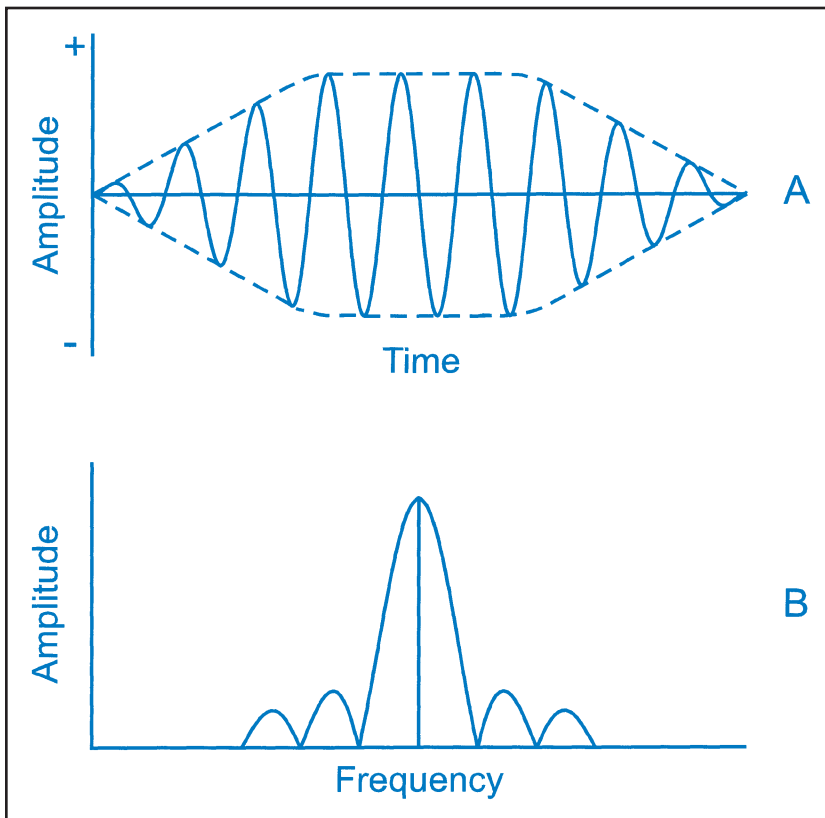


Figure 8–8. Transient distortion. Effects of **rise/decay time** of the waveform (panel A) on the amplitude spectrum (panel B) of a sine wave. The result is a continuous spectrum and reflects what is called **transient distortion**.

Initiating and terminating a signal produces *transients*, which are called “*on-transients*” when the signal is turned on and the amplitude rises, and “*off-transients*” when the signal is turned off and the amplitude decays. This is a form of **transient distortion**.

- The magnitude of transient distortion depends on the time required for the amplitude to *rise* from zero toward maximum or to *decay* from maximum toward zero.
- The longer the duration over which the signal is rising or decaying, the **rise-decay time**, the narrower the frequency band over which energy is spread in both directions from the frequency of the driving signal. Thus, slowly rising and slowly decaying signals result in less transient distortion than signals for which the amplitude rises and decays quickly.
- If the rise or decay is very short, an audible “clicking” sound is heard.

Audiologists test a person’s hearing with an *audiometer*, which is a device that generates sine waves at various frequencies with

variable sound pressure levels. By manipulating those two variables, the audiologist determines the lowest sound pressure levels at which an individual can just detect sine waves of various frequencies. But the sine waves must be turned on and off, and we want to be certain that we determine how well the person hears the sine waves, not the audible clicks that are produced by gating the signal too rapidly.

To ensure that transient distortion is minimized, the American National Standards Institute (ANSI) adopted standards for manufacturers to meet by specifying minimum “rise times” and “decay times” for audiometers. The 1989 standard for rise/decay times (ANSI S3.6-1989), for example, specifies that the time required for the signal amplitude to rise from -20 dB (1%) to -1 dB (79.4%) re: maximum amplitude and to decay from -1 dB to -20 dB shall not be *less* than 20 ms.

In this chapter, we have learned how “ideal” sound waves are altered in various ways that, collectively, are called **distortion**. In Chapters 9 and 10 we will learn other ways in which sound waves are changed from the original state when they are transmitted through a medium.

PRACTICE PROBLEMS

1. An electrical sinusoid is directed to an amplifier in a way that the level of the input signal does *not* exceed the linear portion of the input-output function of the amplifier. When the *input* signal level is $200\ \mu\text{V}$, the output level is $3\ \text{mV}$. If the input level is increased from $200\ \mu\text{V}$ to $2\ \text{mV}$, what should the output level be? Explain your answer.
2. For the same amplifier, you *lower* the input level substantially in an effort to ensure a lower percentage harmonic distortion. What will you sacrifice and why?
3. When making a recording of a person reading a passage of text, you decide to *increase* the input gain so that most of the VU deflections occur at about +2 VU rather than, say, -5 VU in an effort to achieve an improved signal-to-noise ratio. What will you sacrifice and why?
4. The upper limit of the linear portion of an input-output function is approximately $1\ \text{dyne/cm}^2$ (cgs system), which corresponds to 74 dB SPL re: $2 \times 10^{-4}\ \text{dynes/cm}^2$. Express the output level in dB SPL re: $20\ \mu\text{Pa}$ (MKS system).
5. An electrical sinusoid is directed to an amplifier with an input level sufficiently high that you exceed the linear portion of the input-output function of the amplifier. With the aid of an appropriate filter and voltmeter, you measure the output voltages for five harmonics of the distorted output signal: $V_1 = 400\ \text{mV}$; $V_2 = 25\ \text{mV}$; $V_3 = 40\ \text{mV}$; $V_4 = 2\ \text{mV}$; and $V_5 = 4\ \text{mV}$. Calculate the *approximate* percentage harmonic distortion.



Sound Transmission

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When sinusoidal waves were discussed in Chapter 2, each cycle of sinusoidal motion *initially* was represented as identical to every other cycle. That is appropriate when the sine wave is conceptualized from a strictly mathematical perspective.

- We then learned that sound waves produced and transmitted in the real world encounter **frictional forces** that cause *the amplitude of the sound wave to diminish over time*. The sound waves are **damped**, and they ultimately fade away.
- We also know from experience that the intensity of sound *diminishes over distance traveled* as well as over time. All other factors being equal, if you are positioned too far from even an intense source of sound, the sound might be quite faint, if indeed it is even audible or measurable.

It also has been convenient to treat sound waves as if they are transmitted in a free, unbounded medium that has no obstacles for the sound waves to encounter. When obstacles are present, or when certain changes in conditions exist in the medium, important characteristics of sound waves and their transmission are altered.

Chapters 9 and 10 will describe how the characteristics of sound waves are altered in various ways during transmission through a medium. Although the factors that produce change usually act in combination to affect sound waves in fairly complicated ways, for the sake of simplicity they will be presented one at a time.

■ A FREE, UNBOUNDED MEDIUM

Imagine that a sound wave is propagated in a *free, unbounded medium*. By that we mean a medium that contains no obstacles or conditions to affect the propagation or to alter the characteristics of the sound wave in any way as it is transmitted through space. How far might the sound wave travel in such a medium?

Suppose a weapon is fired and the resultant sound wave has a peak sound pressure level of 110 dB at a distance of 0.5 miles from the weapon. As the sound wave is propagated through the free, unbounded medium, *the intensity of the sound wave diminishes in a lawful way*. The decrease in intensity in such an idealized medium behaves in accordance with the **inverse square law**. Could the shot be heard at a distance of 8,192 miles from the source if the listener's threshold of hearing (in the appropriate frequency range) is 20 dB SPL? We will subsequently demonstrate that the answer, perhaps surprisingly, is "yes."

How can that possibly be true? Think of a point source of sound, a spherically shaped source of vibration, or pulsation, with a very small diameter. Figure 9–1 shows that when the point source pulsates by application of an external force, alternate regions of *increased density* (**compressions**) and *decreased density* (**rarefactions**) are cre-

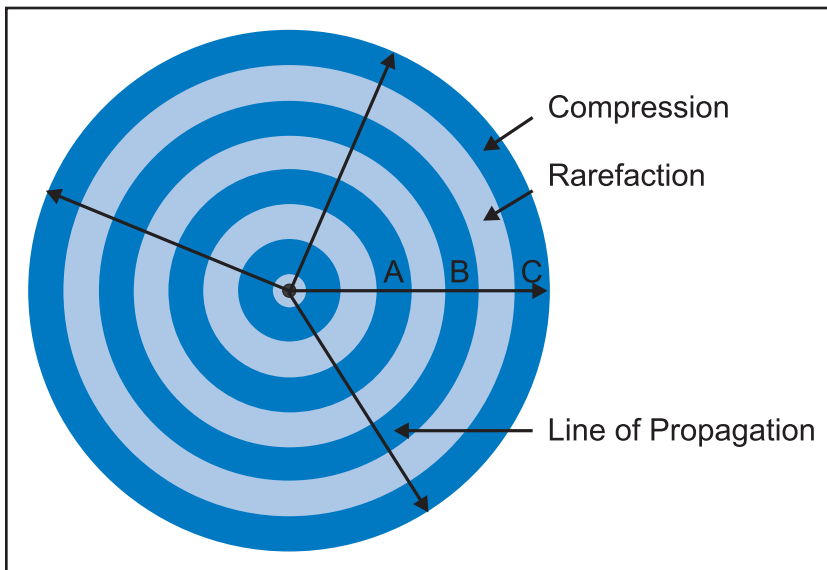


Figure 9–1. Alternate regions of compression and rarefaction in an air mass produced by pulsation of a point source of sound. Four lines of propagation from the source are shown. The compressions form a spherical shell around the spherical point source, and the shell is called a **wave front** or, more precisely, a **spherical wave front**.

ated. A “disturbance” is propagated through the medium, and sound, which is a form of energy, is transferred from the source through the medium. *Sound energy radiates away from the source in all directions*, and four such “lines of propagation” are shown in the figure.

Spherical Waves

We continue to assume that the medium has uniform properties and is unlimited in all directions; the medium is “free and unbounded.”

- The **compressions**, or crests, form a **spherical shell** around the spherical point source.
- The spherical shell is called a **wave front**.
- The disturbance propagated through the medium is an ever increasingly larger **spherical wave front**.

Plane Waves

A spherical wave is similar to a soap bubble that becomes larger and larger as it is blown at a constant rate; the surface of the soap bubble represents a **wave front**. When the wave front is a considerable

distance from the point source, the radius of the sphere is sufficiently large that the spherical wave front becomes a **plane wave front** *because the curvature of the sphere is negligible over some small distance. Thus, sine waves also are **plane progressive waves** that are propagated through a medium.*

■ THE INVERSE SQUARE LAW

As the wave fronts move outward from the source, sound energy is spread over a larger and larger area (think again of the soap bubble). As we move from point A to points B and C in Figure 9–1, energy is dissipated over a progressively larger and larger area; the surface area of the sphere becomes progressively larger. Although a minuscule amount of the sound energy is absorbed by impurities, humidity, and so forth in the air medium, the decrease in sound intensity over distance occurs principally because energy is spread over a wider and wider area.

How does the **inverse square law** explain the decrease in sound intensity over distance? Recall from Chapter 5 that **power is energy/unit time (energy/s)**.

- As the wave fronts move away from the point source, the same amount of **power** is dissipated over the surface of the increasingly larger sphere at the distances represented by A, B, and C in Figure 9–1, but the **energy/s/m² (MKS system)** *must diminish. Why? Because the same amount of power is dissipated over a larger and larger area.*
- **Energy/s/m²**, also called the **energy density**, defines the **intensity** of a sound wave. If sound **power** remains constant, but the **power (energy/s)** is dissipated over a larger and larger area as wave fronts move farther and farther from the source, the **intensity, (energy/s/m²)**, must decrease with increasing distance from the point source.

These concepts of *constant power* but *diminishing intensity* are illustrated by the three-dimensional representation in Figure 9–2.

Imagine that we place our eye at the point source and then look toward the increasingly larger area bounded by the four lines of propagation.

- At distance **X**, a finite amount of **power** (energy/s) is dissipated over an area of 1 m². *That defines the **intensity** of the sound wave at distance **X** from the source because intensity is **energy/s/m²**.*
- As we move twice as far away from the source to distance **2X**, the same finite amount of **power** is dissipated over a larger surface area.
- How much larger? The surface area of a sphere is given by Equation 9–1:

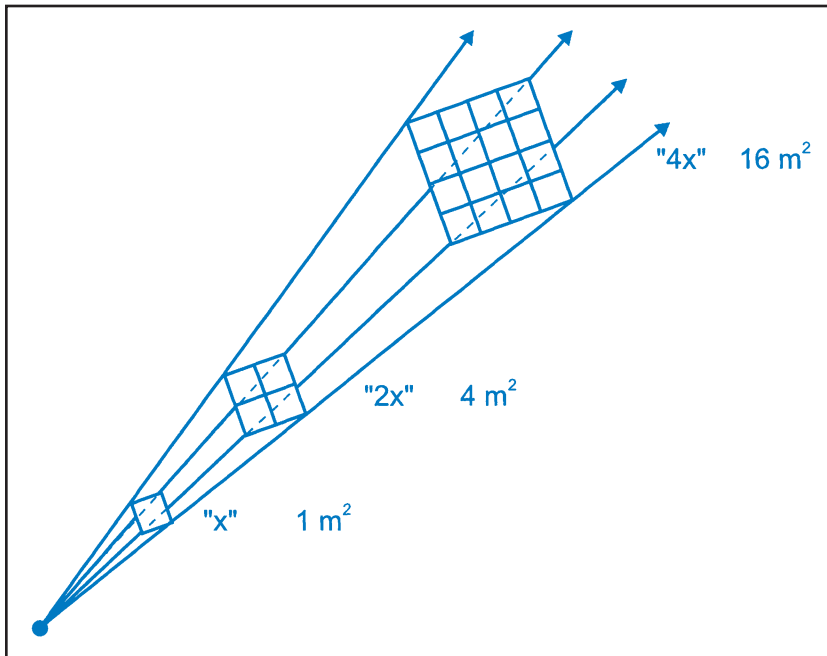


Figure 9-2. A three-dimensional representation of the **inverse square law**. As the distance from the point source of sound increases from **X** to **2X** to **4X**, a finite amount of power is dissipated over a larger and larger area (from 1 m^2 at **X** to 4 m^2 at **2X** to 16 m^2 at **4X**). Hence, *the **intensity** (energy/s/m^2) decreases inversely with the square of the distance from the source.*

$$A = 4\pi r^2,$$

Equation 9-1

where **r** is the radius of the sphere.

- For example, if distance **X** is 1 m from the source, the area (**A**) of the sphere at **X** = 12.6 m^2 ($4\pi 1^2$). At distance **2X**, **A** = 50.3 m^2 ($4\pi 2^2$), and at distance **4X**, **A** = 201.1 m^2 ($4\pi 4^2$).
- *Each time the radius increases by some factor, the surface area of the sphere increases by the square of that factor.*

If the surface area of the sphere increases with increasing distance from the source, *the energy/s/m^2 (the **intensity**) must decrease*. The area (4 m^2) bounded by lines of propagation at point **2X** is four times larger than the area at point **X** (1 m^2). Because intensity is energy/s/m^2 , the intensity is diminished as the sound wave travels from **X** to **2X**. More precisely, the intensity at distance **2X** is only *one-fourth* the intensity at **X** because the power is dissipated over an area that is four times larger. Importantly, *the decrease in intensity in a free, unbounded medium is not attributable to a loss in sound power; it simply is a matter of geometry.*

We have seen that when the distance from the source *increases* by a factor of 2:1, sound intensity *decreases*, and the ratio of the

intensity at point **2X** to the intensity at point **X** is 1:4. *Because $4 = 2^2$, and because the relation is an inverse one, this phenomenon is called the **inverse square law**.*

As we proceed farther and farther from the point source in the *free, unbounded medium*, the **inverse square law** continues to hold. At distance **4X**, which is twice as far from the source as distance **2X**, **power** is dissipated over an area of 16 m^2 in comparison with an area of 4 m^2 at distance **2X**. Thus, the distance has increased by another factor of 2:1, the surface area of the wave front has increased by a factor of $2^2:1$, and the sound intensity, therefore, is changed once again by a ratio of 1:4. We subsequently will express the inverse square law in decibels. But if you recall how intensity is expressed in decibels from Chapter 5, you might be able to answer the following question before reading the explanation: “By how many dB does sound intensity decrease with each doubling of the distance from the source?”

The **inverse square law** (intensity varies inversely with the square of the distance) is expressed as:

Equation 9–2

$$I \propto \frac{1}{D^2},$$

where **D** refers to the ratio of two distances

from the sound source (d_i/d_r),

d_i is the distance of interest, and

d_r is a reference distance.

For example, compare the intensities at two distances: 200 m (d_i) and 100 m (d_r). The intensity at d_i is inversely proportional to the square of the ratio of the two distances, (d_i/d_r). Because $D = (d_i/d_r)$, Equation 9–2 is rewritten as:

Equation 9–3

$$I \propto \frac{1}{\left(\frac{d_i}{d_r}\right)^2},$$

which states that *intensity is inversely proportional to the square of the ratio of two distances*, the ratio of the distance of interest (d_i) to some reference distance (d_r).

Generality of the Inverse Square Law

The **inverse square law** applies to physical phenomena other than transmission of sound waves. For example, the force of gravity decreases over distance in accordance with the inverse square law. Near the surface of the earth, a falling mass is accelerated at a rate of 9.8 m/s^2 because of *the attractive gravitational force exerted on the mass by the earth*. As we move farther and farther from the surface

of the earth, the gravitational force varies inversely as the square of the distance from the earth's core. The radius of the earth, the distance from its center core to the surface, is approximately 6,400 km. That defines the reference distance (d_r). At 6,400 km from the earth's surface (12,800 km from the core, d_i ; a ratio of 1:2), the gravitational force is only one-fourth of what it is at the surface of the earth.

The moon is a mass. How does the inverse square law inform us about the magnitude of acceleration on the moon? The distance from the surface of the earth to the surface of the moon is about 384,000 km (d_i). Thus, the ratio, d_i/d_r , is $384,000/6,400 = 60$. In accordance with the inverse square law, F_a (the force of acceleration) is given by:

$$F_a = \frac{1}{60^2} = \frac{1}{3,600} = 0.00028 \text{ m/s}^2,$$

in contrast to 9.8 m/s^2 on earth. In other words, as Newton calculated, the gravitational attraction of the earth on any mass varies inversely with the square of the distance from the earth's center core.

Another example also illustrates the generality of the inverse square law. Recall from a high school science class that “*like*” *electrostatic charges repel* with some force, whereas “*unlike*” *charges attract* with some force. Charles-Augustin de Coulomb (1736–1806) demonstrated experimentally that the magnitudes of either repulsive or attractive forces vary directly with the product of the individual charges and *inversely with the square of the distance between the two charges*. With respect to generality of the law, the **inverse square law** applies to virtually all fields of force.

The Inverse Square Law Expressed in Decibels

We learned in Chapter 5 that we generally refer to the *level* of a sound in decibels rather than to its absolute intensity in watts/m². To accomplish the transformation to decibels for the **inverse square law**, calculate the log of the distance ratio and multiply by –20 to obtain Equation 9–4. Why multiply by –20 when we are dealing with *intensity*, not *pressure*? Recall the laws of logarithms, specifically Laws 3 and 4:

$$\begin{aligned} \text{dB} &= 10 \log_{10} \frac{1}{\left(\frac{d_i}{d_r}\right)^2} \\ &= -10 \log_{10} \left(\frac{d_i}{d_r}\right)^2 && \text{(Log Law 4)} \\ &= -20 \log_{10} \frac{d_i}{d_r} && \text{(Log Law 3)} \end{aligned} \quad \text{Equation 9–4}$$

Figure 9–2 should provide an intuitive understanding of Equation 9–4.

- The area over which power is dissipated at distance $\mathbf{X}$ is 1 m^2 .
- If we use Equation 9–4 to compare the intensity at distance $\mathbf{X}$ ($\mathbf{d_i}$) with the intensity *at the same distance* $\mathbf{X}$ ($\mathbf{d_r}$), we obtain 0 dB, because the ratio is 1:1 and the log of 1 is 0.
- If we compare the intensity at distance $2\mathbf{X}$ ($\mathbf{d_i}$) with the intensity at reference distance $\mathbf{X}$ ($\mathbf{d_r}$), we calculate that:

$$\begin{aligned} \text{dB} &= -20 \log \frac{2}{1} \\ &= -6 \text{ dB.} \end{aligned}$$

That answer agrees with what we discovered previously. The intensity at $2\mathbf{X}$ is one-fourth the intensity at $\mathbf{X}$, and as we learned in Chapter 5, if intensity decreases by a factor of four, the *level* decreases by 6 dB (Equation 5–4). Obviously, therefore, between $2\mathbf{X}$ and $4\mathbf{X}$ the intensity decreases by another 6 dB because:

$$\begin{aligned} \text{dB} &= -20 \log \frac{4}{2} \\ &= -6 \text{ dB.} \end{aligned}$$

Finally, between $\mathbf{X}$ and $4\mathbf{X}$ the intensity decreases by 12 dB because:

$$\begin{aligned} \text{dB} &= -20 \log \frac{4}{1} \\ &= -12 \text{ dB.} \end{aligned}$$

The **inverse square law**, therefore, sometimes is defined (correctly, but incompletely) by saying that “the intensity decreases by 6 dB for each doubling of the distance.” Although the statement is correct, it is an insufficient definition of the law because it does not describe the more general effect of distance ratios other than 2:1 on intensity. For that reason, the previous definition—*the intensity varies inversely with the square of the distance*—is preferable.

Sample Problems

Problem 1: If the sound pressure level is 80 dB at a distance of 100 m from the source, *by how much is the SPL decreased* at a distance of 200 m?

$$\text{dB} = -20 \log \frac{200}{100} = -6 \text{ dB.}$$

Note that the SPL at $\mathbf{d_r}$ (100 m) is irrelevant. The intensity at $\mathbf{d_i}$ is attenuated by 6 dB re: the intensity at $\mathbf{d_r}$, regardless of the SPL at $\mathbf{d_r}$.

Problem 2: If the sound pressure level is 80 dB SPL at a distance of 100 m from the source, what is the SPL at a distance of 200 m?

$$\text{dB} = 80 - 20 \log \frac{200}{100} = 74 \text{ dB SPL.}$$

It stands to reason that if the SPL decreases by 6 dB between the two distances, the SPL at the second distance is given by the SPL at the first, or reference, distance (80 dB) minus the amount by which the intensity is attenuated (6 dB).

Problem 3: If the SPL at distance 100 m is 80 dB, what is the SPL at distance 850 m?

$$\text{dB} = 80 - 20 \log \frac{850}{100} = 61.4 \text{ dB SPL.}$$

Problem 4: Finally, solve the problem that was posed at the beginning of the chapter. If the SPL of a gunshot is 110 dB at a distance of 0.5 miles from the weapon, could the shot be heard at a distance of 8,192 miles (*in a free, unbounded medium*) from the source if the observer's threshold of hearing in the appropriate frequency range is 20 dB SPL?

$$\text{dB} = 110 - 20 \log \frac{8,192}{0.5} = 110 - 84.2 = 25.8 \text{ dB SPL.}$$

If the intensity of the sound is 25.8 dB SPL and the observer's threshold of hearing is 20 dB SPL, the shot should be heard. You also should note that if the distance from the source of the sound wave is doubled again to 16,384 miles, the SPL decreases by only another 6 dB (to 0.2 dB below the hypothetical threshold) because *intensity varies inversely with the square of the distance*. Finally, you might wish to test your ability to still work problems such as those in Problem Set 8 of Chapter 5 by attempting to calculate just how far from the reference distance of 0.5 miles the observer would have to be positioned for the sound to just equal the observer's threshold of 20 dB SPL. The only difference is that now you are solving for d_i , a distance of interest, rather than p_x , a pressure of interest. You should calculate that the sound should just reach the observer's threshold of 20 dB SPL at a distance of approximately 15,811 miles, if the medium is free and unbounded. Is it realistic to believe that the gunshot can be heard at a distance of 8,192 miles from the source? No! The calculation assumes that the sound wave travels through a "free, unbounded medium" that offered no obstacles to affect sound wave propagation. Hence, there are limitations to the inverse square law.

Limitations to the Inverse Square Law

The inverse square law *only holds strictly* when sound waves are propagated through a free, unbounded medium. Such a medium, of

course, is a theoretical construct. In reality, the transmitting media of interest contain obstacles and present other conditions that interfere with sound transmission in important ways.

For example, suppose you attend an outdoor concert in a park. The grass-covered ground *absorbs* a considerable amount of the sound energy. Moreover, the skin and clothing of people also *absorb* some of the sound energy. Under those circumstances, the decrease in sound intensity with distance is greater than predicted by the inverse square law. More generally, when sound waves encounter obstacles or a change in conditions in the transmitting medium, depending on the nature of the obstacles or conditions, the sound waves are **reflected** (they “bounce” back from the obstacle), **diffracted** (they scatter about the obstacle), **refracted** (they “bend”), or **absorbed** (they penetrate the obstacle).

REFLECTION

What happens when a rubber ball is thrown against a cement wall? The ball does not penetrate the obstacle—it bounces off the wall because the density of the wall greatly exceeds the density of the air. Many young children also quickly learn the approximate path that the ball will take on its return so that they can get into position to catch it.

At the right of Figure 9–3, a ball is thrown toward a wall along a path marked by the dashed line (labeled “Perpendicular Path”) that is perpendicular to the wall (90° angle). If, *and only if*, the ball is thrown toward the wall on a path that is precisely along the perpendicular, and if there is no rotation to the ball, the ball returns to the thrower along the same perpendicular (dashed) line. If the ball is thrown more to the left along the path marked P_i , it returns along the path marked P_r . Why?

View Animation 8_3. Reflection of a Ball
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The path of the thrown ball is the “*incident path*” (P_i) and the path of the bounced (reflected) ball is the “*reflected path*” (P_r). From the point where the thrown ball strikes the wall, a dashed line is shown that is perpendicular to the wall. *The angle of the reflected path (P_r) to the perpendicular equals the angle of the incident path (P_i) to the perpendicular.*

That rule applies to any angle of the incident path to the perpendicular, regardless of whether the ball is thrown to the left, the right, or straight ahead. The young child probably cannot recite that rule of reflection, but some children seem to “know” the rule or else learn it rather quickly. Unless the rule is mastered, attempts to participate in any sporting event in which reflection is part of the game (billiards, handball, etc.) are futile.

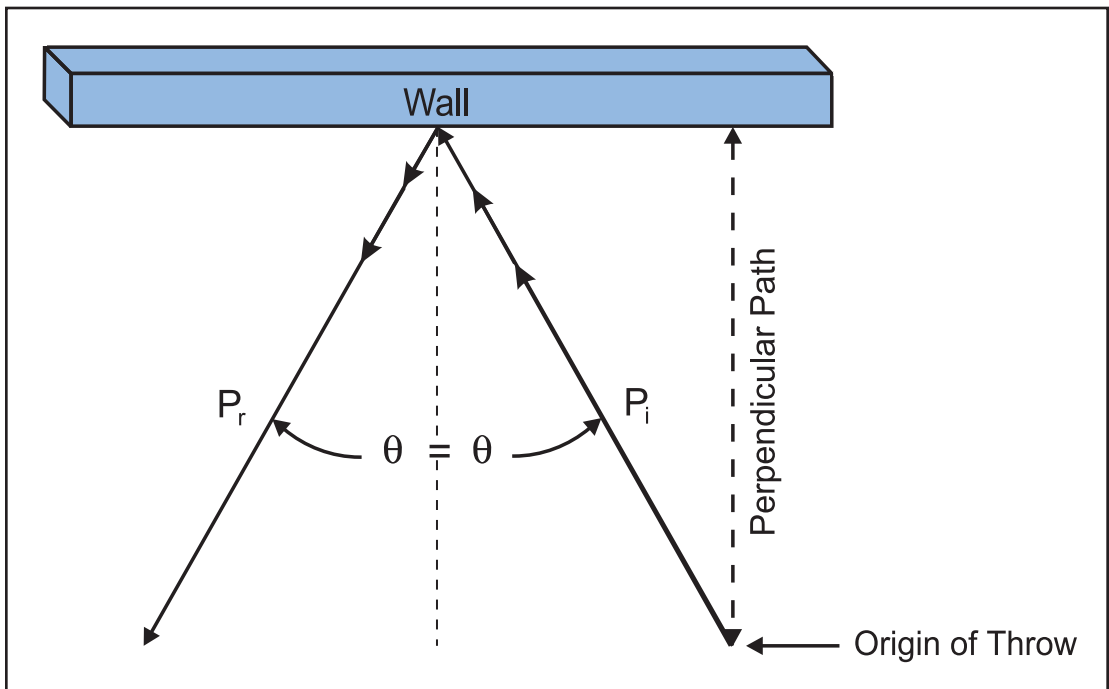


Figure 9–3. The paths of a ball thrown toward a wall and then reflected from the wall. When the ball is thrown straight at the wall along the dashed line, it bounces straight back to the thrower. When the ball is thrown at an angle along the path marked P_i , the ball bounces off the wall and away from the thrower along the path marked P_r . In both cases, and all others, the angle of the reflected path, P_r , to the perpendicular equals the angle of the incident path, P_i , to the perpendicular.

Reflection of Sound Waves

Sound waves are affected by obstacles in the same way that the wall affects the path of the thrown ball. In Figure 9–4, spherical wave fronts move from right to left and encounter a high-density plane obstacle such as a thick steel wall. The sound wave that moves from right to left *toward the wall* is the **incident wave**, and one *ray* (a line perpendicular to the wave front) of the incident wave (**i**) points leftward toward the plane obstacle. The point source of the sound wave is identified by **s**.

View Animation 8_4. Reflection of Sound by a Wall
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- If the obstacle were not present, the sound wave would continue to move leftward toward point s' , which is shown to be as far to the left of the obstacle as point s is to the right of the obstacle.

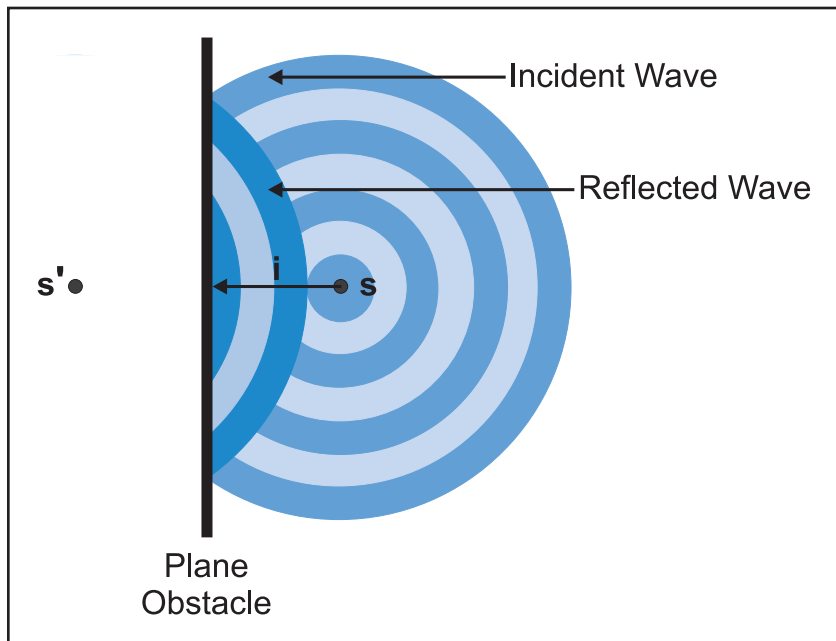


Figure 9–4. Sound wave **reflection**. A spherical wave front moving from right to left from source **s** encounters a plane obstacle. The obstacle causes the sound wave to be **reflected** toward the source *with no change in speed of propagation*. An observer hears the reflected wave as having originated from point **s'**, which is as far to the left of the obstacle as point **s** is to the right of the obstacle.

- The **density** of the obstacle, however, is much greater than the density of the air, and it offers a large **acoustic impedance** to sound transmission. Very little sound energy penetrates the wall. Instead, the sound wave is **reflected** back toward the source (**s**) *with no change in speed of propagation*. The reflected wave moving from left to right toward the original point source also is shown in the figure. This is sometimes called “specular reflection,” and behaves in the same manner as when light waves are reflected from a mirror in accordance with Snell’s law. We will return to Snell’s law later in this chapter in the discussion of refraction.
- *Because the speed of the **reflected wave** is identical to the speed of the **incident wave**, the reflected wave, at any given moment, will be just as far to the right of the plane surface as it would have been to the left of the plane surface if the obstacle had not been present.*

The precise relation of reflected rays to incident rays obeys the same rule that governed the path of the thrown ball in Figure 9–3; *the angles of reflected rays (**r**) to the perpendicular equal the angles of incident rays (**i**) to the perpendicular*. An observer located in the

medium perceives the reflected sound wave as having originated from point s' at the left of the obstacle, not point s . The sound wave is reflected and sound energy is retained in the medium. *Because energy is retained in the medium, the **inverse square law** does not hold strictly for sound transmission in a medium with reflecting surfaces.*

Reflection from Plane Surfaces

The nature of reflected waves depends on the shape, or configuration, of the obstacle: plane surface, convex surface, concave surface, irregularly shaped surface, and so forth. Figure 9–5 illustrates the principle of sound wave reflection from a *plane surface*. In panel A, an incident ray (i) strikes the plane surface at an angle of 45° to the perpendicular of the surface. The incident ray is reflected, and the angle of the reflected ray (r) to the perpendicular also is 45° . In panel B, the angle of i to the perpendicular is 30° , and the angle of r to the perpendicular also is 30° .

In panel C, the angle of i to the perpendicular is 0° . Therefore, the angle of $r = 0^\circ$ and the ray is reflected back along the original incident path. Anyone who has successfully played billiards or pool in River City¹ must be familiar with these concepts. The principle, therefore, is the same as that illustrated in Figure 9–3 for a ball thrown against a wall: *the angles of reflected rays to the perpendicular equal the angles of incident rays to the perpendicular*. That concept is important because, as is shown in Figure 9–4, there are a myriad of incident rays that strike the obstacle.

Reflection from Convex Surfaces

The rule governing the relation of the angles of the reflected rays to the angles of the incident rays for plane surfaces also holds for other reflective surfaces. In panel A of Figure 9–6, a spherical wave moves from right to left toward an obstacle with a surface that is *convex toward the source*. As the wave fronts approach the obstacle at some distance from the source, the *spherical* waves become *plane* waves. Two incident rays, i_1 and i_2 , of the plane wave fronts are shown by the solid lines that point leftward; the perpendicular is shown by the dashed lines.

View Animation 8_6. Reflection by Convex and Concave Surfaces at animations.cla.umn.edu

- The rule that specifies the angle of reflected rays still applies, and we see that the reflected rays *diverge*.
- *Sound energy is scattered*, just as the reflected paths of thrown balls are scattered if they are thrown at various angles against a wall that is convex toward the thrower.

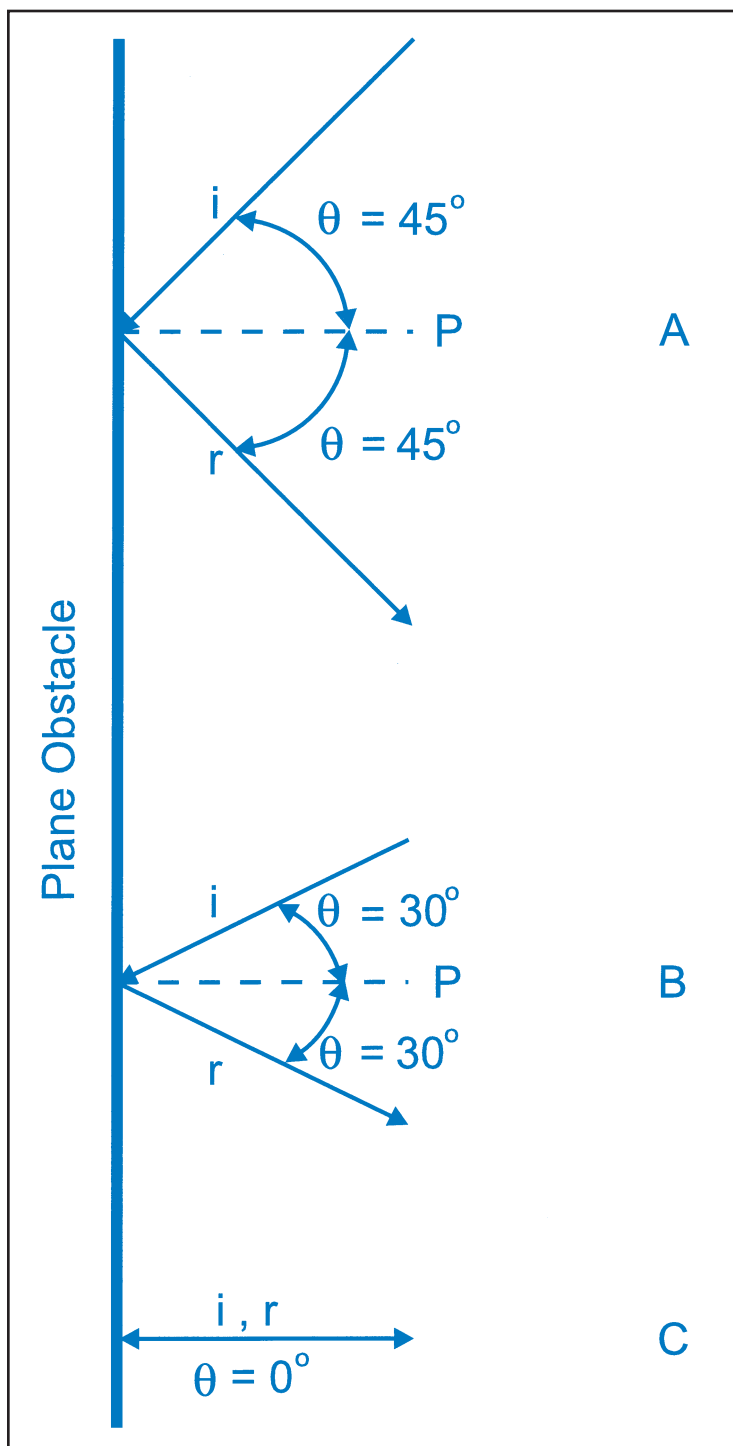


Figure 9–5. Incident rays, i , of a sound wave moving from right to left encounter a plane obstacle. The angles of the reflected rays, r , to the perpendicular equal the angles of the incident rays, i , to the perpendicular.

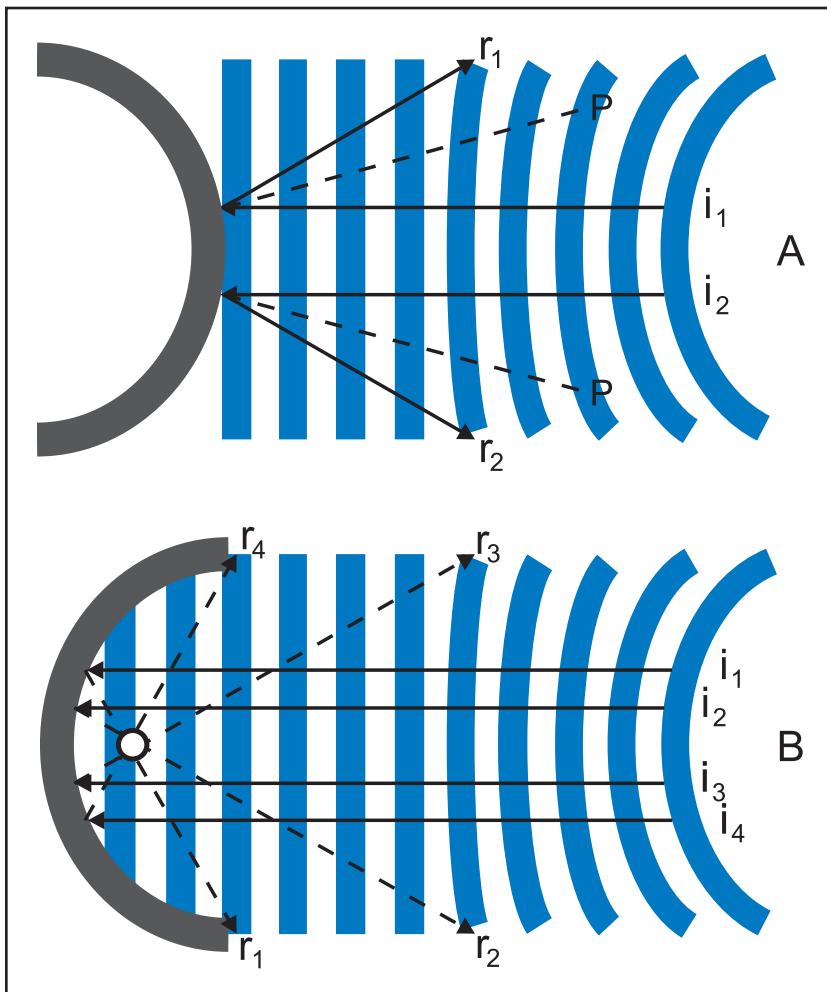


Figure 9-6. Plane progressive sound waves, with wave fronts sufficiently far from the source of sound that their curvature is negligible, are moving from right to left and encounter an obstacle that is **convex** toward the source (panel A) or **concave** toward the source (panel B). The reflected rays *diverge from the convex surface*, but *converge toward a focal point from the concave surface*. In both cases, the angles of reflected rays to the perpendicular equal the angles of incident rays to the perpendicular.

The divergence of reflected rays in panel A of the figure explains what happens to a sound wave in air that strikes the *exterior* surface of a dome-shaped roof of a building. *The rays of the reflected sound wave diverge.* The intensity of the reflected sound wave at some distance from the surface is less than the intensity of the incident sound wave at the same distance, because the sound energy in the reflected wave is scattered.

Reflection from Concave Surfaces

In panel B of Figure 9–6, a plane progressive wave moves from right to left, but this time it encounters a surface that is *concave toward the source*.

- Four incident rays (solid lines), $\mathbf{i}_1$ through $\mathbf{i}_4$, are shown and, once again, the angles of the reflected rays (dashed lines) to the perpendicular equal the angles of the incident rays to the perpendicular.
- When the sound wave encounters the concave surface, the rays of the reflected wave, $\mathbf{r}_1$ through $\mathbf{r}_4$, *converge toward a focal point* (shown by an open circle), the point at which all reflected rays converge.

Convergence of reflected rays of sound explains how “whispering galleries” work in structures such as the U.S. Capitol Building, St. Paul’s Cathedral in London, and several state capitol buildings. Some readers will have had the experience of standing in a room that has a domed ceiling—a surface that is concave toward the floor. You stand in several locations on one side of the room but cannot hear what someone is whispering on the other side of the room. If you then move to just the right point, which probably was marked on the floor, the same whispered speech is audible.

The point marked on the floor identifies the **focal point**, the point at which the rays of the whispered sound *converge* after having been reflected from the concave surface of the ceiling. *The intensity of the reflected wave is greatest at the focal point*, and, indeed, it is even greater than the intensity of the incident wave because the sound energy, in a sense, has been “collected”; *the energy density, or intensity, is maximal*.

Perhaps you have observed a television crew at a football game. The producer wants to enliven the audio portion of the broadcast by picking up the grunts, groans, and occasional profanities uttered by the players. The crew *does not point a microphone toward the playing field* to pick up the *incident* sound waves. Instead, a reflector dish is held with its *concave surface pointed toward the playing field*. The microphone is mounted on the dish with its diaphragm pointed *away from the playing field and toward the reflector*. *The microphone picks up the reflected sound waves from the concave surface of the dish*. The diaphragm of the microphone is located precisely at the focal point of the reflector where the intensity of the reflected sound waves is greatest. The same arrangement also can be used to record animal sounds in nature.

Echoes, Reverberation, and Reverberation Time

Reflected sound waves sometimes are called echoes, or reverberating waves, and rooms with hard surfaces (ceiling, floor, walls,

etc.) are **reverberant** rooms. In contrast, rooms that are designed to nearly eliminate reverberation are **sound treated** or, at the extreme, **anechoic** rooms.

If you have heard your voice bouncing from a canyon wall, you know that you can hear a few echoes, but finally they fade away.

- The time required for the reflected wave to decay is called the **reverberation time (RT)**, which is defined as *the time required for a sound wave to be attenuated by 60 dB relative to its peak level*. **Reverberation time**, therefore, is symbolized RT_{60} .
- Look again at Equation 5–4 from Chapter 5 and see why **reverberation time** also is defined as *the time required for sound intensity to be attenuated to one-millionth of its peak value*.

One practical use of reflected waves was made in 1883 (Stewart, 1924). The explosion of a volcano in Krakatoa in the East Indies produced an intense compressional wave that progressed around the earth in all directions and ultimately converged at a point opposite Krakatoa. At that point, the wave was reflected toward its source. From inspection of barometric records, the *speed of propagation* was measured to be 320 m/s, which closely agrees with modern, more technically accurate measures of the speed of sound.

Standing Waves

When a sound wave is reflected from a surface and “bounces” between two or more reflecting surfaces, *sound energy is retained in the medium*. This gives rise to a rather complex acoustic phenomenon called **standing waves**. Standing waves occur when *two progressive waves, the incident wave and the reflected wave, of the same frequency and amplitude, travel through the same medium in opposite directions*. Compressions and rarefactions always are located at the same fixed positions in the medium between the reflecting surfaces.

The resulting standing waves sometimes are called the “modes” of the elastic system. We shall see subsequently that because the standing waves reside in a resonant system, whether the system is air, a string, or some other elastic system, the lowest resonant frequency mode is called a “first-order mode” and integer multiples of the first-order mode are called “higher-order modes.” Those modes also are called **resonances**, and their **resonant frequencies** are numbered in a similar fashion: **1st resonant frequency**; **2nd resonant frequency**, and so on.

In summary, **standing waves** literally do not move through the medium; *they stand still in a fixed position*. Standing waves for transverse and longitudinal wave motion will be described separately to help bring greater clarity to understanding this phenomenon.

Transverse Wave Motion and Standing Waves

Transverse standing waves are illustrated in Figure 9–7. A string or wire is anchored at both ends. Dots are painted on the string at equal intervals to help identify the pattern of motion of each of those points on the string over time.

View Animation 8_7. Standing Waves
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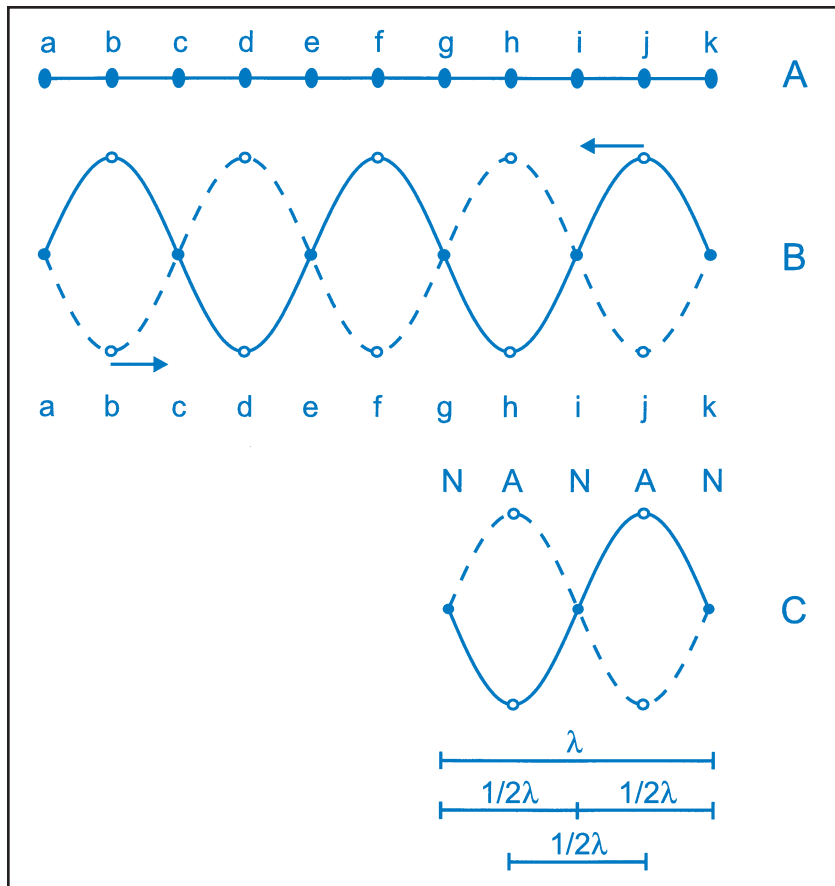


Figure 9–7. Creation of **standing**, or **stationary**, **waves** on a vibrating string that is stretched tautly and anchored at both ends. Two trains of waves of identical amplitude and frequency are moving in opposite directions along the string. The wave shown by the solid line moves from right to left and the wave shown by the dashed line moves from left to right. Points b, d, f, h, and j move alternately upward and downward over time and are called **displacement antinodes**, *which are points of maximum vibratory movement*. Points a, c, e, g, i, and k remain stationary and are called **displacement nodes**, *which are points of no vibration*. The distance between successive antinodes or between successive nodes equals one-half of the wavelength of the two waves moving in opposite directions.

In panel A, the string is stretched with a fixed tension and the identifying points are labeled **a** through **k**. In panel B, a source of sound causes a transverse wave of a certain frequency and amplitude to travel from *left to right* (the dashed line). At the same time, another sound source causes a second transverse wave of *identical amplitude and frequency* to travel in the opposite direction, from *right to left* (the solid line).

In Chapter 1 we learned that, with an unanchored stretched rope and only one transverse wave present, *each individual point on the rope moves upward and downward with identical frequency*. Because the string in Figure 9–7 is anchored, and because we have two identical transverse waves traveling in opposite directions, the pattern of movement of the points along the string is different from what we observed previously.

- Points **b, d, f, h, and j** on the string *move upward and downward*. Those points are **displacement antinodes**, which are *points of maximum displacement* upward and downward. For example, at point **h**, the wave traveling from right to left shows a *maximum downward displacement*, whereas the wave traveling from left to right displays a *maximum upward displacement*.
- Points **a, c, e, g, i, and k** on the string *remain stationary at their equilibrium positions*. The string does not vibrate (is not displaced from equilibrium) at those points; points of “no vibration” are called **displacement nodes**.
- Successive antinodes and nodes are spaced at equal intervals along the string, and *between each pair of nodes, loops are formed*. The center of each loop corresponds to a **displacement antinode**.

Panel C of Figure 9–7 shows one cycle of each of the two transverse waves from panel B. Two loops are formed and the locations of the **displacement antinodes (A)** and **displacement nodes (N)** are labeled.

Recall from Chapter 2 that the distance traveled during one cycle defines the wavelength (λ). Therefore, *the distance between successive antinodes (A to A), or successive nodes (N to N), corresponds to one-half the wavelength of the moving waves*. For example, if the wavelength of each of the two waves is 1 m, both **nodes** and **antinodes** are spaced at 0.5 m intervals. Although the two waves in Figure 9–7 are *moving through each other, the resultant wave is stationary*; it is a **standing wave**.

Longitudinal Wave Motion and Standing Waves

Standing waves also occur with **longitudinal wave motion**. Panel A of Figure 9–8 contains an air-filled tube of uniform cross-sectional area that is *open at one end and closed at the other* (an open-closed tube). Incident waves that travel from left to right in the tube are reflected from the hard-surfaced closed end and travel in the air-filled

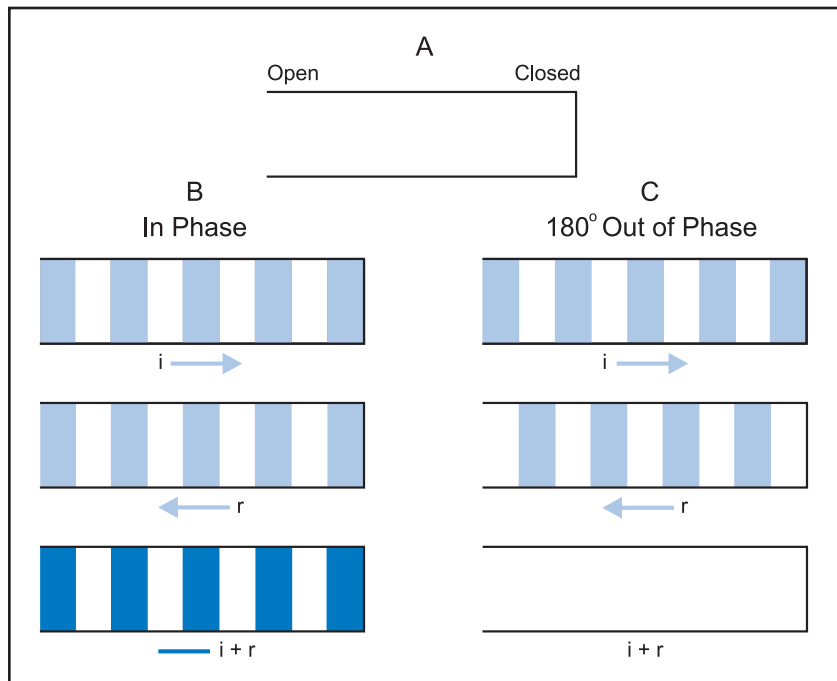


Figure 9–8. Creation of standing waves with longitudinal wave motion in a tube open at one end and closed at the other (panel A). Panel B shows incident waves moving from left to right toward the closed end of the tube and reflected waves moving from right to left. At that moment in time, the incident and reflected waves are *in phase*, which results in **reinforcement**. In panel C (one-fourth of a period later in time) the incident and reflected waves are 180° *out of phase* with one another, and **cancellation** occurs.

tube in a direction opposite (right to left) to the direction of the incident wave. The incident waves and the reflected waves interact all along the length of the tube. Both waves possess alternate regions of **compression** (high density) and **rarefaction** (low density) and *the two waves move through each other along the length of the tube, but in opposite directions*.

Panel B illustrates that, *at a particular moment in time*, the **compressions** (darker regions) of the *incident wave*, i , moving from left to right are precisely aligned with—in phase with—the **compressions** of the *reflected wave*, r , moving from right to left. One wave maximally *reinforces* the other wave because the air particles are displaced maximally; *the result is a maximal increase in intensity*. This is **constructive interference** because, although the incident and reflective waves interfere with each other, they do so in a “constructive” way.

Panel C shows what happens one-fourth of a period later. At that moment, *compressions of the incident wave are 180° out of phase with compressions of the reflected wave all along the length of the tube*. The air medium in the tube cannot simultaneously, at that moment, at that location, be in a state of high density and a state

of low density. Thus, the waves “cancel” one another, and **cancellation** means that *at that moment, at that location, the medium is not vibrating*. This is **destructive interference**. At given points along the length of the tube, the medium alternates over time between reinforcement (**constructive interference**) and cancellation (**destructive interference**) and the intensity of the sound wave at those points alternately increases and decreases.

In summary, there are three defining features of longitudinal standing waves.

1. At certain points in the tube (whose locations depend on the length of the tube and the wavelength of the sound waves), the incident and reflected waves *are always in phase* with one another.
2. At other points along the tube, the incident and reflected waves *are always 180° out of phase*.
3. *The interactions of the incident and reflected waves produce **standing waves***. They are called standing waves because they “stand still” rather than move through the tube. Both the incident and reflected waves move along the tube in opposite directions. *It is the resultant wave—the sum of the incident and reflected waves—that is stationary.*

Figure 9–9 further illustrates the interactions of incident and reflected waves. In each panel, the dotted line represents the incident wave moving from right to left and the dashed line represents the reflected wave moving from left to right.

- *The solid line shows the resultant wave, which is a point-by-point summation over time of the incident and reflected waves.*
- Each panel shows the status of events at a point in time, and each successive panel from top to bottom is separated by an interval equal to one-eighth of a period.
- Thus, if the frequency of the waves is 125 Hz, the period is 8 ms, and each successive group from panel A to panel F is separated by an interval of 1 ms.

Examine what happens at a given location from moment to moment at the locations marked by the three *solid* vertical lines. The solid lines are positioned on **displacement antinodes** in the *resultant wave*, which is similar to what was observed for point **b** in Figure 9–7.

- At those three locations in panel A, *the two waves are partially in phase*. Because the incident and reflected waves are partially in phase, displacement amplitude of the resultant wave (solid line) is greater than the displacement amplitude of either the incident or the reflected wave alone. That is **partial reinforcement**.

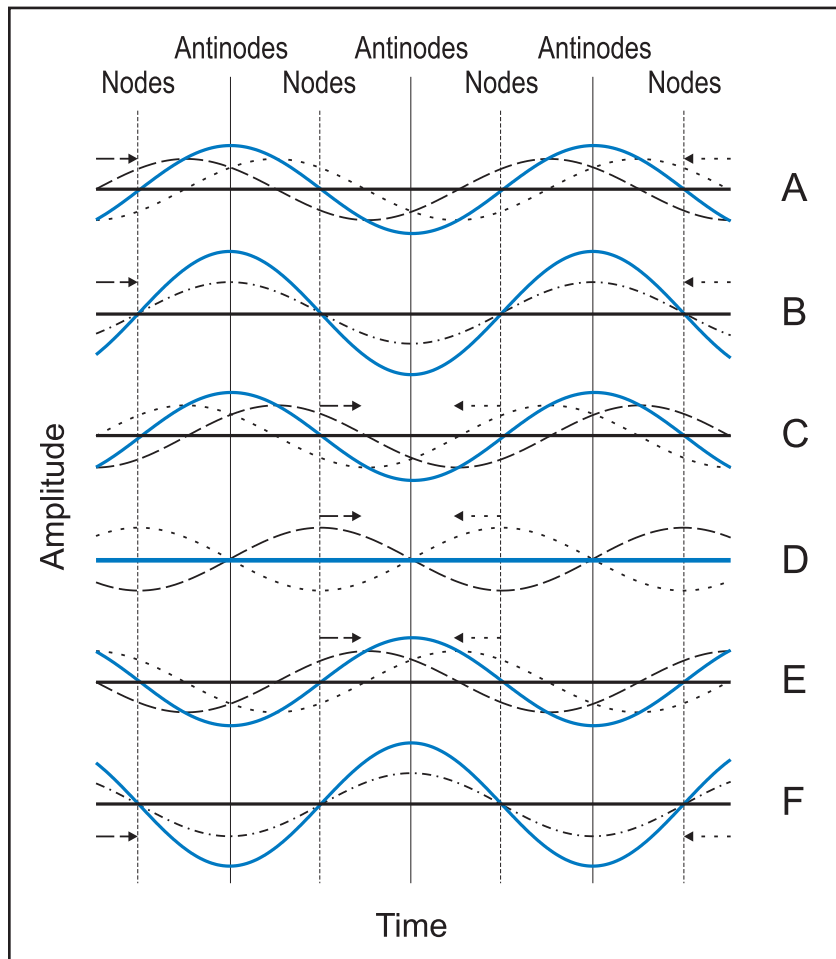


Figure 9-9. Interactions of incident (*dotted lines*) and reflected (*dashed lines*) waves of identical frequency and amplitude moving in opposite directions to produce resultant waves (*solid lines*) that vary in amplitude over time (from panel to panel). At the points indicated by the three solid vertical lines, the resultant wave, over time, ranges from **partial reinforcement**, to **maximal reinforcement**, to **cancellation**, to **partial cancellation**. The points along the four dashed vertical lines represent nodes—points of no vibration. The resultant sound wave is a standing, or stationary, wave.

- In panel B (one-eighth of a period, or 1 ms, later), *the two waves are in phase*. The result is **maximal reinforcement**, and therefore maximal intensity, because the amplitudes of both the incident and reflected waves are maximal and the amplitude of the resultant wave is the sum of those two waves.
- In panel C, the resultant wave is identical to what was observed in panel A: **partial reinforcement**.
- In panel D, *the two waves are 180° out of phase*, and the result is **cancellation** in the resultant wave.

- In panel E, the displacement amplitudes are identical to those in panel C, but in the opposite direction, and in panel F, maximum displacement in the opposite direction occurs.

Note what happens at all locations that fall on the four *dashed* vertical lines in Figure 9–9. In a sense, nothing happens. The resultant wave (solid line) always is located on the time axis. Those are **displacement nodes**. There is no vibratory motion at those locations. Thus, the dashed vertical lines identify **displacement nodes**, whereas the solid vertical lines identify **displacement antinodes**. **Displacement antinodes** are locations at which the amplitude of displacement alternates from maximally positive to maximally negative.

Standing Waves and Resonant Frequency

Finally, let us examine the relation between sound wave reflection in an air-filled circular tube and the **resonant frequency** (actually resonant frequencies) of the tube. Consider first a tube that is *open at one end and closed at the other*, and begin with the simplest case where there is only one **displacement node** and one **displacement antinode**.

Panel A of Figure 9–10 shows the *displacement pattern* of the air mass in an *open-closed* tube.

- The solid horizontal line in the middle of the tube represents equilibrium (0°).
- All along the length of the tube, displacement of the air mass progresses upward in the tube through 45° (70.7% of *maximum upward displacement*) to *maximum displacement at 90°* (dashed line), *downward through equilibrium* (solid line), on through 225° (70.7% of maximum downward displacement), to *maximum displacement in the opposite direction at 270°* (dashed line), and then *upward to equilibrium*.

The two dashed lines (labeled 90° and 270°) display the *maximum displacement pattern* at 90° (upward) and 270° (downward), and they are redrawn in panel B with solid lines, without showing the intermediate displacements of 90° and 270° . Thus, throughout the length of the tube, the air particles are displaced upward and downward over time, and panel B shows only the envelopes (maximum displacements in either direction) of that displacement pattern.

- A **displacement node**, a point of no vibration of air particles, *always is located at the closed end of the tube* where the air is not free to move.
- A **displacement antinode**, a point of maximum displacement of air particles, *always is located at the open end* where the air can move freely.

In addition to the displacement nodes and antinodes, **pressure nodes** and **pressure antinodes** also develop in the tube. Both the

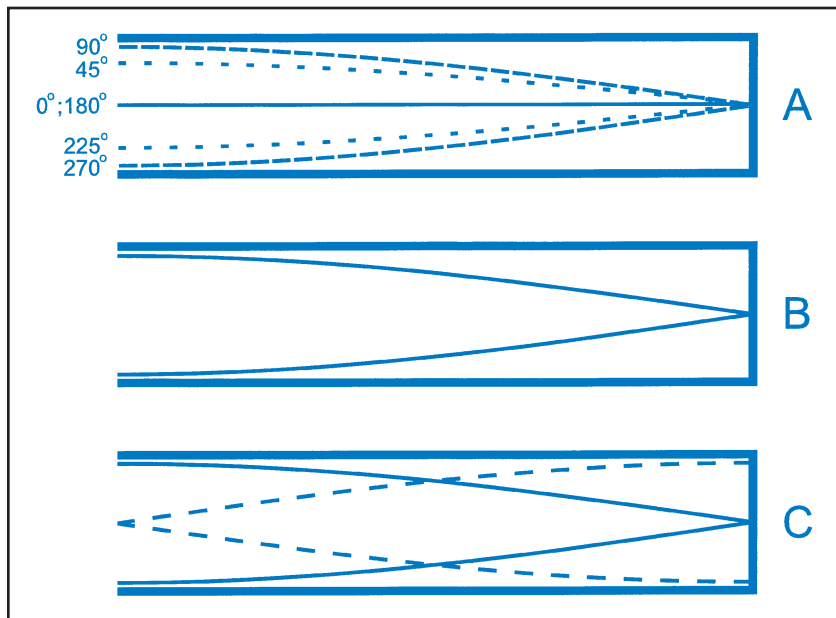


Figure 9–10. A tube open at one end and closed at the other. Panel A shows the changing displacement pattern over time from 0° through 360° . Panel B displays only the maximum displacements at 90° and 270° from panel A. In both panels A and B, a **displacement node** is located at the closed end of the tube, and a **displacement antinode** is located at the open end of the tube. In panel C, the displacement pattern is redrawn from panel B (*solid lines*) and contrasted with the pressure pattern for the tube (*dashed lines*). A **pressure node** is located at the open end of the tube, and a **pressure antinode** is located at the closed end.

displacement pattern (solid lines) and the *pressure pattern* (dashed lines) are shown in panel C. To understand the relation between displacement and pressure nodes and antinodes, recall from Figure 2–11 in Chapter 2 that *pressure and displacement are 90° out of phase with one another; pressure leads displacement by 90° .*

- A **pressure node**, a point of zero pressure, corresponds to a **displacement antinode** and is located *at the open end of the tube*.
- A **pressure antinode**, a point of maximum pressure, corresponds to a **displacement node** and is located *at the closed end of the tube*.

We learned previously (see Figure 9–7) that the distance between two adjacent nodes (or two adjacent antinodes) corresponds to one-half wavelength. Thus, in panel C of Figure 9–7, where three nodes and two antinodes were shown, we observed that *one wavelength corresponds to two loops in the standing wave*.

In panel B of Figure 9–10 there is only one **displacement node** (at the closed end) and one **displacement antinode** (at the open end). Therefore, we have only *one-half of one loop*.

- If two loops in a standing wave correspond to *one* wavelength (1λ), and one loop corresponds to *one-half* wavelength ($1/2\lambda$), then one-half loop must correspond to *one-fourth* wavelength ($1/4\lambda$).
- Thus, in this simple case, where only one node and one antinode are present, a standing wave pattern is created that has a wavelength equal to *one-fourth of the length of the tube*.

Recall from Chapter 2 that **wavelength** (λ) is given by:

$$\lambda = \frac{s}{f}.$$

Equation 2-13

By rearranging Equation 2-13 to solve for **frequency** (f), we obtain:

$$f = \frac{s}{\lambda}.$$

Equation 9-5

In the case of the tube that is open at one end and closed at the other (an *open-closed tube*), a standing wave pattern with *one node and one antinode* is created when the frequency of the sound wave has a *wavelength that is four times the length of the tube*. Therefore, Equation 9-5 can be modified to read:

$$F_1 = \frac{s}{4L}.$$

Equation 9-6

where **L** is the length of the tube and

F₁ refers to the *center frequency of the first of a series of resonances*.

What does Equation 9-6 mean? In an open-closed tube, there is a **resonance** with a *frequency equal to the ratio of the speed of sound in the tube to four times the length of the tube*. That is called the “lowest,” or “first,” resonance, and occasionally it is called the “first mode.”

Why do we say that **F₁** is the lowest, or first, of a “series of resonances”? Because a tube that is open at one end and closed at the other has more than just one node and one antinode. In fact, if the circular tube has a uniform cross-sectional area, there are theoretically an infinite number of nodes and antinodes and therefore an infinite number of resonances. But, we will limit the discussion to only three such resonances. What are their resonant frequencies? They are *odd integer multiples of the lowest resonant frequency, F₁*. Figure 9-11 provides an explanation for the existence of the resonances with frequencies that are higher than **F₁**.

Panel A shows the *lowest resonant frequency* of an open-closed tube; the displacement pattern is identical to what was seen in panel B of Figure 9-10. Panel B shows the next higher resonant frequency, **F₂**. In fitting this displacement pattern in the tube, we still must follow the constraints that:

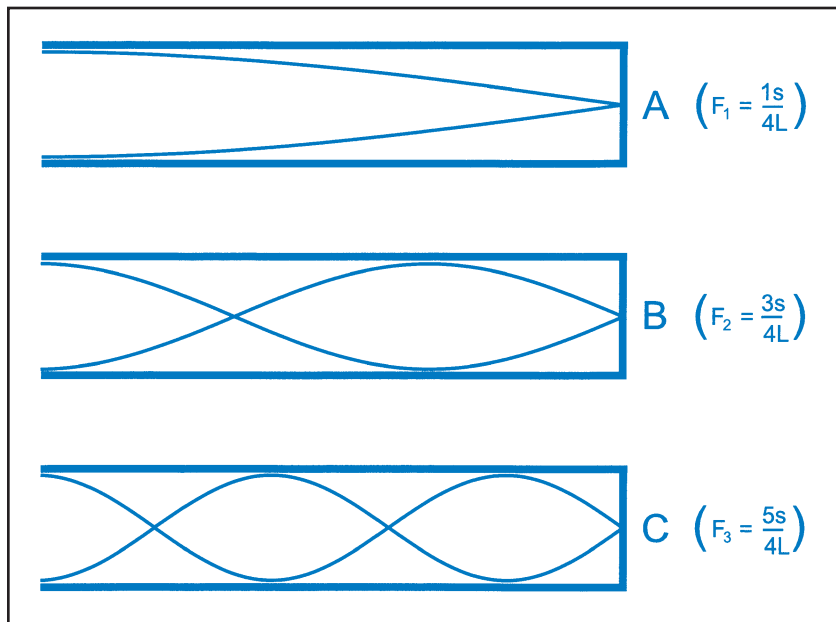


Figure 9-11. Panel A shows the displacement pattern again from **Figure 9-10**. One **displacement node** and one **displacement antinode** form one-half of a **loop**, which corresponds to the lowest resonant frequency of the tube, F_1 . In panel B, two **nodes** and two **antinodes** form one and one-half **loops**, which corresponds to the next higher resonant frequency, F_2 . A third standing wave pattern is shown in panel C; three **nodes** and three **antinodes** form two and one-half loops, which corresponds to F_3 . All resonant frequencies are odd integer multiples of F_1 .

1. a **displacement node** *must be located at the closed end* where the air is not free to move;
2. a **displacement antinode** *must be located at the open end* where the air can move freely; and
3. *the distance between a node and an adjacent antinode still corresponds to $1/4\lambda$.*

Thus, in panel B, the next higher resonant frequency, F_2 , that will fit in the tube and conform to the rules has *two nodes and two antinodes*. There now are one and one-half loops. If one-half loop corresponds to $1/4\lambda$, then one and one-half loops must correspond to $3/4\lambda$. Thus, the equation for this next higher mode of vibration is:

Equation 9-7

$$F_2 = \frac{3s}{4L}.$$

In panel C, a third standing wave pattern still is constrained by the restriction that there must be a displacement node at the closed end and a displacement antinode at the open end. The resulting displacement pattern has three nodes, three antinodes, and two and one-half loops. If one loop corresponds to $1/2\lambda$, then two and one-half

loops corresponds to $5/4\lambda$. The equation for this next higher mode of vibration therefore is:

$$F_3 = \frac{5s}{4L}. \quad \text{Equation 9-8}$$

The two higher modes of vibration, F_2 and F_3 , are odd integer multiples of the lowest resonant frequency, F_1 :

$$\begin{aligned} F_1 &= 1s/4L; \\ F_2 &= 3s/4L; \text{ and} \\ F_3 &= 5s/4L. \end{aligned}$$

A few calculations should help clear up any confusion. Consider an example in which the length of the tube in Figure 9-11 is 0.2 m. If we set the speed of sound to be 340 m/s, with Equation 9-6 we calculate that the lowest resonant frequency, F_1 , is 425 Hz:

$$F_1 = \frac{340}{4 \times .2} = 425 \text{ Hz}.$$

With Equations 9-7 and 9-8, we see that the next two higher resonant frequencies, F_2 and F_3 , are indeed odd-integer multiples of F_1 :

$$F_2 = \frac{3 \times 340}{4 \times .2} = 1275 \text{ Hz, which} = 3 \times 425 \text{ Hz,}$$

and

$$F_3 = \frac{5 \times 340}{4 \times .2} = 2125 \text{ Hz, which} = 5 \times 425 \text{ Hz}.$$

Finally, Equations 9-6, 9-7, and 9-8 can be consolidated to form Equation 9-9 so that it applies to all resonances, F_n , of an open-closed tube:

$$F_n = \frac{[(2n) - 1]s}{4L}, \quad \text{Equation 9-9}$$

where n refers to the n th resonance.

Why are the resonant frequencies of an open-closed tube restricted to odd-integer multiples of F_1 rather than odd and even integer multiples?

- A **displacement node** must be located at the closed end of the tube, and a **displacement antinode** *must* be located at the open end of the tube.
- It is impossible for waves that are *even integer multiples* of F_1 to have a node at the closed end and an antinode at the open end.

- For example, if F_1 corresponds to $1/4\lambda$, then $2F_1$ (an even integer multiple) must correspond to $1/2\lambda$. To fit a wave that is $1/2\lambda$ into the tube, we must either have antinodes at both ends with a node in the middle, or nodes at both ends with an antinode in the middle. Neither circumstance is possible.
- For that reason, standing waves that are even integer multiples of the lowest resonant frequency *cannot exist* in a tube that is open at one end and closed at the other.

Should we expect the same displacement (and pressure) patterns for tubes that are either open at both ends or closed at both ends? No! A tube that is either *open at both ends* (panel A of Figure 9–12) or *closed at both ends* (panel B of Figure 9–12) behaves quite differently from the tube that is open at only one end.

- If a tube is *open-open* (a tube that is open at both ends) as in panel A, a **displacement antinode** is located at each end where the air can move freely. There still must be at least one **displacement node** within the tube if there is to be a standing wave, and in Panel A there is a **node** located midway in the tube.
- If the tube is *closed-closed* (a tube that is closed at both ends) as in panel B, a **displacement node** is located at each end, and a **displacement antinode** is located midway in the tube. *In either case, an open-open or closed-closed tube, a standing wave pattern is created with a wavelength that is $1/2\lambda$ instead of $1/4\lambda$.* Therefore, if the frequency of the incident wave is two times the length of the tube ($2L$), a standing wave pattern is created. For either of these tubes,

Equation 9–10

$$F_1 = \frac{s}{2L}.$$

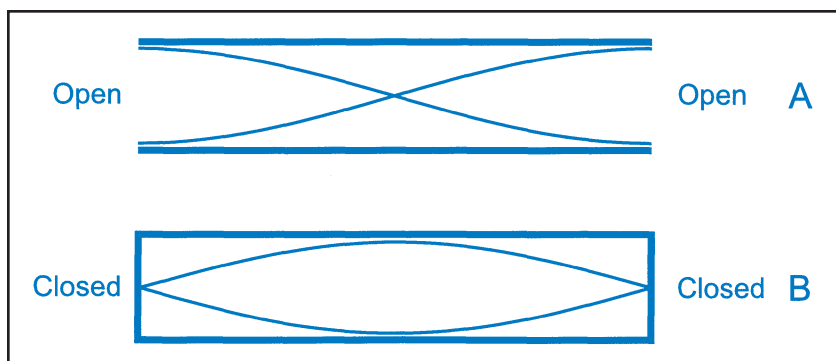


Figure 9–12. Standing wave patterns for the lowest resonant frequency in tubes open at both ends (panel A) or closed at both ends (panel B). In each case, the lowest resonant frequency corresponds to one full loop, $1/2\lambda$, which is formed either by two **antinodes** and one **node** (panel A) or two **nodes** and one **antinode** (panel B).

- If L is set at a fixed length such as 0.2 m,
- F_1 is twice as high in frequency (850 Hz) for *tubes that are either closed at both ends or open at both ends* as it would be for a tube that is open at only one end (425 Hz).
 - Moreover, if the tube is either closed at both ends or open at both ends, successively higher resonant frequencies are *odd and even* integer multiples of F_1 . Thus, in Figure 9–12, the first three resonant frequencies for panels A and B are 850 Hz, 1700 Hz, and 2550 Hz (odd and even integer multiples of F_1).

Equation 9–10 for F_1 is rewritten as Equation 9–11 to apply to all resonant frequencies for *open-open* or *closed-closed* tubes:

$$F_n = \frac{ns}{2L},$$

Equation 9–11

where n is the n th resonance.

The Vocal Tract: An Open-Closed Resonant System. The relation between standing wave patterns and resonant frequencies is important for understanding the acoustics of speech production. The human vocal tract is a prominent example of an *open-closed resonant tube* during vowel production. The vibratory movement of the vocal folds produces a complex quasiperiodic waveform with harmonics that are odd and even integer multiples of the fundamental frequency. The vocal tract, the cavity that extends from the vibrating vocal folds at one end to the mouth opening at the other end, is an air-filled tube, or resonator.

If the opening to the nasal cavity is closed by elevation of the soft palate, the vocal tract can be modeled, as a first approximation, as an *open-closed* circular tube with uniform cross-sectional area. It is *open at one end (the mouth opening) and closed at the other end (the vibrating vocal folds)*. This configuration of the vocal tract is approximated when the talker utters the *schwa* vowel, the first sound in the word “*about*.”

The average length of the vocal tract for an adult male is about .17 m. With the aid of Equations 9–6, 9–7, and 9–8, or with the more general Equation 9–9, we calculate that the frequencies of the first three resonances for the tube with a length of .17 m are:

$$F_1 = 500 \text{ Hz};$$

$$F_2 = 1500 \text{ Hz}; \text{ and}$$

$$F_3 = 2500 \text{ Hz}.$$

Note that each of the resonant frequencies is an odd integer multiple of F_1 . If the vocal tract is shorter than .17 m, the frequencies of the resonances increase and, of course, if the vocal tract is longer, the frequencies of the resonances decrease because *resonant frequency is inversely proportional to the length of the tube*. In speech acoustics, and sometimes in musical acoustics, the resonances are called

formants, and the center frequency of a formant is called the **formant frequency**, F_n .

The vocal tract, of course, does not have a uniform cross-sectional area, and the configuration of the tract as a resonant tube varies from vowel to vowel. For example, when producing the vowel /i/ (as in *see*), a constriction in the tract is located such that the cavity behind the constriction is larger than the one in front of it. For the vowel /u/ (as in *Sue*), the forward cavity increases in size and the back cavity becomes smaller. These variable placements and degrees of constriction result in resonances that are more irregularly spaced in the frequency domain than would be calculated with Equation 9–9.

A seminal paper, *Control Methods Used in a Study of Vowels* (Peterson & Barney, 1952), reported average formant frequencies for adult males, adult females, and children that clearly define the relation among formant frequencies and the locations and degrees of constriction in the oral cavity. Another paper that falls into the “must read” category is *The Elements of an Acoustic Phonetic Theory* (Peterson & Shoup, 1966). Readers who wish to pursue this and related concepts in the acoustics of speech production are advised to consult, among others, *Speech Sciences* (Kent, 1997), *The Acoustic Analysis of Speech* (Kent & Reed, 2002), and *Fundamentals of Sound with Applications to Speech and Hearing* (Mullin et al., 2003).

The Ear Canal: An Open-Closed Resonant System. Another example of an *open-closed* tube is the ear canal (external auditory meatus). The canal is closed at the deep end by the ear drum (tympanic membrane) and open to the surrounding air mass at the entrance to the external ear. Consider a canal with a length of 34 mm. By use of Equation 9–6, and accepting the speed of sound as approximately 340 m/s, we calculate that the canal has a first resonant frequency of about 2.5 kHz:

$$\begin{aligned} F_1 &= s/4L \\ &= 340/(4 \times 34) \\ &= 2.5 \text{ kHz.} \end{aligned}$$

The calculation assumes that the ear canal has uniform cross-sectional area and that the ear drum at the closed end is rigid to provide maximal reflection. In fact, though, the cross-sectional area is not uniform and the tympanic membrane is somewhat flexible. Readers who wish to pursue this and related concepts in hearing are advised to read, among others, *Fundamentals of Hearing: An Introduction* (Yost & Nielsen, 1977) and *Fundamentals of Sound with Applications to Speech and Hearing* (Mullin et al., 2003).

Vibration of Strings

Why should we be interested in learning about the vibratory pattern(s) of strings? For one, though perhaps not very compelling to some, it is

hard to imagine how the study of sound could be complete without examining the interesting way in which strings vibrate. For another, learning about string vibration provides a rough, first-order approximation to learning about the vibratory pattern of human vocal folds during vocal production. Finally, what would our aesthetic world be like if we could not enjoy listening to the beautiful music produced by a wide variety of stringed instruments? Put simply, the violin, viola, harp, piano, guitar, and so forth are examples of stringed instruments, and *the vibration patterns of strings are an example of a resonant system.*

The vast array of vibrating systems, though different in many important details, share certain acoustic properties. We shall see, for example, that the resonant properties of a vibrating string are about the same as the resonant properties of a *closed-closed air-filled tube*.

Equation 2–11 (Chapter 2) informed us that the “frequency” of a vibrating string or wire on, for example, a violin or guitar, is given by:

$$f = \frac{1}{2L} \sqrt{\frac{t}{m}},$$

Equation 2–11

where **L** = length, **t** = tension, and **m** = cross-sectional mass.

Thus, the frequency of vibration of a string is

- *inversely proportional to twice the length,*
- *directly proportional to the square root of the tension, and*
- *inversely proportional to the square root of the mass.*

“Frequency” was placed in quotation marks because the string actually vibrates in many modes simultaneously (with many different frequencies of vibration), and Equation 2–11 provides the solution for only *the lowest of those frequencies*, which is the **fundamental frequency**, f_0 , of the vibrating string.

The string also vibrates at **harmonics** of the fundamental frequency, which are *odd and even integer multiples of the fundamental, or lowest, frequency in the series*. By examining the displacement pattern of the string during vibration, we learn why there also is energy at harmonic frequencies.

The string is a low-damped, narrowly tuned elastic system. When it is displaced from equilibrium, *it continues to vibrate at its natural frequency* (governed by the mass and stiffness of the string) for some considerable amount of time because of its low-damped property. Waves of vibration travel along the string *in both directions*. When the waves reach the ends where the string is fixed, they are reflected in the opposite direction.

- Many of those reflected waves interfere with one another (**destructive interference**), and their energy is dissipated.
- Waves with a frequency corresponding to the **natural, or resonant, frequencies** of the string continue to travel.

In Figure 9–13, the solid line of panel A shows a string attached at both ends to pegs, and stretched to be rather taut. The dashed line in panel A shows the string after it has been displaced from equilibrium.

- The array of dashed lines in panel B shows a series of displacement patterns over time after the string has been displaced and then released. The string is displaced from equilibrium

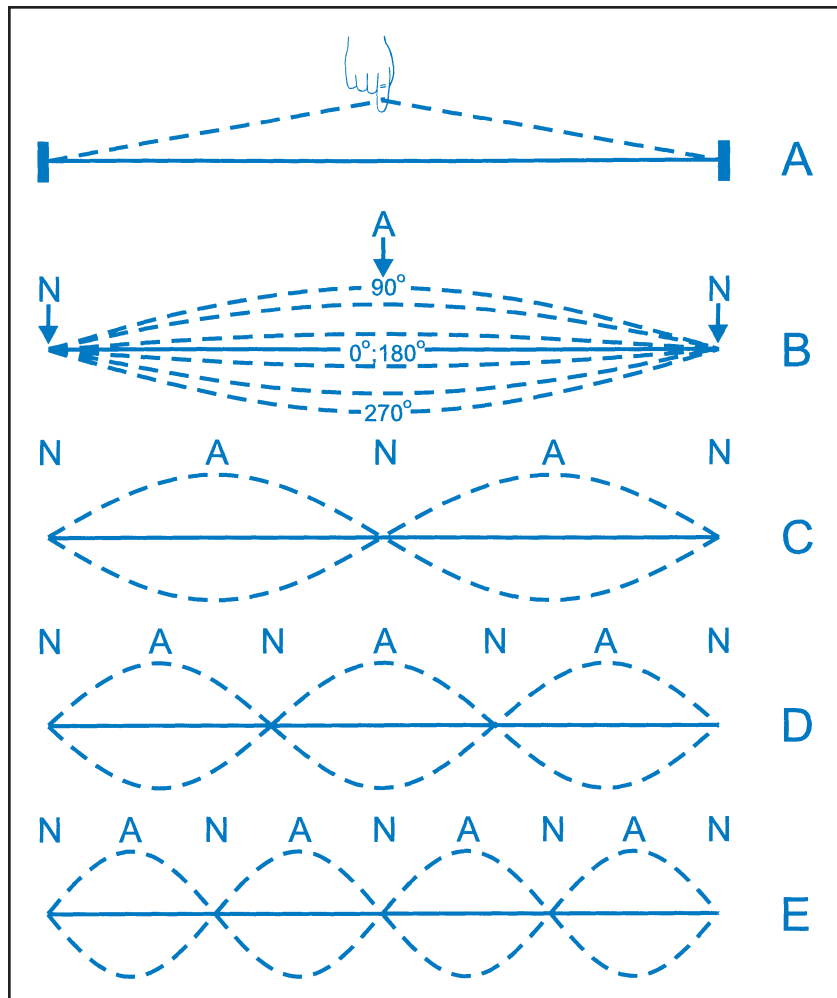


Figure 9–13. Displacement pattern of a vibrating string. In panel A, a stretched string is displaced. Panel B shows the displacement pattern over time from 0° through 360° for the lowest resonant frequency; one **loop** is formed by two **displacement nodes** and one **displacement antinode**. Panels C, D, and E show higher modes of vibration of the string (higher resonant frequencies) as more and more **loops** are formed by an increasing number of **nodes** and **antinodes**. Resonant frequencies of the string are odd and even integer multiples of F_1 , which for a string usually is called f_0 , the **fundamental frequency**.

(0°) upward to maximum displacement (90°), downward through equilibrium (180°) and beyond to maximum displacement in the opposite direction (270°), and then back to equilibrium (360°). That describes *one cycle of vibration*.

- The time required to complete one cycle of vibration defines the **fundamental period** (T). The reciprocal, $1/T$, yields the **fundamental frequency**, f_0 .

Panel B also shows **displacement nodes** (N) at the two fixed ends of the string *where it is not free to vibrate*. There also is a single **displacement antinode** (A) midway between the fixed ends *where the string can vibrate freely*. We learned previously that **standing waves** occur when two progressive waves, an **incident wave** and a **reflected wave**, of the same frequency and amplitude travel through the same medium in opposite directions. Importantly, *those standing waves occur at more than a single frequency*. Panel B shows only the *lowest* frequency at which a standing wave is created: the **fundamental frequency**, f_0 , which physicists call the **first frequency**, f_1 .

In panel B, with two displacement nodes and one displacement antinode, a single **loop** is formed. As we learned previously, one loop corresponds to one-half wavelength. Because the single loop in panel B corresponds to one-half wavelength, the equation for the **fundamental**, or **lowest**, frequency is:

$$f_0 = \frac{s}{2L},$$

Equation 9-12

where f_0 is the fundamental frequency,

s is the speed of sound along the string ($s = \sqrt{\frac{t}{m}}$), and

L is the length of the string.

The **transverse** standing wave pattern for a vibrating string fixed at both ends behaves like the **longitudinal** standing wave pattern for a tube closed at both ends. Therefore, the higher modes of vibration for the vibrating string are *both odd and even integer multiples of the lowest frequency*.

Finally, Equation 9-12 is modified because more than a single loop appears on the string:

$$f_n = \frac{ns}{2L},$$

Equation 9-13

where n is the n th frequency, which corresponds to the number of **loops** that are formed on the string.

If the length (L) of the string is 1.68 m, and the speed (s) of wave transmission along the string is 420 m/s, the frequency of the first harmonic (h_1 , or f_0) is:

$$\begin{aligned}
 F_0 &= \frac{ns}{2L} \\
 &= \frac{(1 \times 420)}{(2 \times 1.68)} \\
 &= 125 \text{ Hz.}
 \end{aligned}$$

Why does the sound wave generated by a string have energy at odd and even harmonics of the fundamental frequency? The string vibrates in many modes, and consequently, additional displacement nodes and antinodes are located along the length of the string. Hence, *more than one single **loop** is formed.*

- In panel C of Figure 9–13, three **displacement nodes** and two **displacement antinodes** combine to form *two loops*.
- In panel D, *three loops* are formed by four **displacement nodes** and three **displacement antinodes**.
- Finally, in panel E, five **displacement nodes** and four **displacement antinodes** result in *four loops*.
- *Each loop corresponds to one-half wavelength*, and with Equation 9–13 we calculate that the frequencies of vibration for the string in panels C, D, and E are, respectively:

$$\begin{aligned}
 f_2 &= \frac{(2 \times 420)}{(2 \times 1.68)} = 250 \text{ Hz,} \\
 f_3 &= \frac{(3 \times 420)}{(2 \times 1.68)} = 375 \text{ Hz,} \\
 &\text{and} \\
 f_4 &= \frac{(4 \times 420)}{(2 \times 1.68)} = 500 \text{ Hz.}
 \end{aligned}$$

The coefficients (*n*) are integers. Because the other terms (*s* and *L*) in Equation 9–13 for a given string are constant, each of the frequencies of vibration is an integer multiple of the lowest frequency. The second harmonic ($2f_0$), the third harmonic ($3f_0$), the fourth harmonic ($4f_0$), and so forth are created.

We have addressed five important concepts regarding the vibratory patterns of strings or wires:

1. The string vibrates in a complex manner.
2. Standing waves are created that correspond to the **natural frequencies** of the string.
3. The natural frequencies (and the corresponding standing waves) depend on the length (*L*) of the string and the speed (*s*) of wave transmission along the string.
4. *When the string is displaced, it vibrates in all its modes.*
5. Once excited, the string moves upward and downward as a series of loops (between adjacent nodes and antinodes) as shown in panel E of Figure 9–13.

The Guitar. Acoustics of the guitar will be discussed briefly to demonstrate the properties of string vibration of just one musical instrument. The common acoustic guitar has six strings. The cross-sectional mass of each string is fixed. The effective length of the string can be altered by pressing a finger against a fret, and because the strings are wound around a peg, the tension on each string can be adjusted independently. When tuned, the fundamental frequencies on what is called the “tempered” scale of the six strings are: **E**₂ (82.5 Hz), **A**₂ (110 Hz), **D**₃ (146.8 Hz), **G**₃ (196 Hz), **B**₃ (246.9 Hz), and **E**₄ (329.6 Hz).

When the strings are plucked with no fretting, each vibrates in its first mode (**fundamental frequency**) at the frequencies listed above. But the string is fixed around pegs at both ends and the waves of vibration therefore are reflected. *Standing waves result from the constructive and destructive interference between incident waves traveling in one direction and reflected waves traveling in the opposite direction.* Thus, the guitar strings function as *closed-closed* resonant systems with resonant frequencies corresponding to both *odd and even integer multiples of their fundamental frequencies (f_o)*. The resonant frequencies are calculated by using Equation 9–13:

$$F_n = \frac{ns}{2L},$$

Equation 9–13

where **n** is the **n^{th}** frequency, which corresponds to the number of **loops** that are formed on the string.

The actual displacement pattern of the string on a musical instrument depends on whether the string is plucked or bowed. Guitar strings ordinarily are plucked, whereas other stringed musical instruments can be either plucked or bowed (or hammered in the case of the piano). Students who are interested in learning more details concerning musical acoustics and the wide variety of musical instruments (string, woodwind, brass, piano, and percussion) are advised to consult *The Acoustical Foundations of Music* by John Backus (1977), among others.

REFRACTION

When an incident wave strikes an obstacle that offers a large acoustic impedance to sound transmission, the wave is **reflected** from the obstacle *with no change in speed of propagation*. Because the *speed of the reflected wave is the same as the speed of the incident wave*, at any given moment, the **reflected wave** is just as far from the obstacle in one direction as the **incident wave** would have been in the other direction if the obstacle had not been present. What happens, though, if the wave fronts encounter a *change in properties of the medium, or a new medium with different properties*, instead of a reflecting barrier?

For example, suppose that sound waves pass from one medium of transmission, **M**₁, to another medium, **M**₂, and suppose that the

speed of sound for M_1 (s_{M1}) is different from the speed of sound for M_2 (s_{M2}). A familiar example is when a sound wave traveling in air (M_1) strikes the surface of water and then travels in the water medium (M_2). The speed of sound in water is about 4.3 times faster than the speed of sound in air. The incident rays are bent and the direction of wave propagation changes because $s_{M2} \neq s_{M1}$.

We can gain insight into the bending of waves by considering the light waves depicted in Figure 9–14. A stick is held in air, M_1 , and pointed toward the surface of water, M_2 , at various angles to the water surface. Some of the light energy penetrates the surface of the water. The speed of light, however, is different in the two media and, for light, speed of propagation is greater in air than in water: $s_{M1} > s_{M2}$. The stick is not really “bent,” but because of the differences in speed of propagation of light waves in air and water, the *image* of the stick appears bent at the surface of the water; the light ray bends toward the perpendicular (dashed line) where the ray enters the water. *That perception is due to a change in speed of wave propagation in accordance with Snell’s law.*

As we move from left to right in Figure 9–14, the angle of the **incident ray** (the stick) to the surface decreases, and the appearance of bending lessens. At the far right, the incident ray is perpendicular to the surface, and now the angle of the **refracted ray** to the surface (the image of the stick in water) equals the angle of the **incident ray** to the surface.

The phenomenon shown in Figure 9–14 for light waves also happens when sound waves penetrate a new medium (for example, air

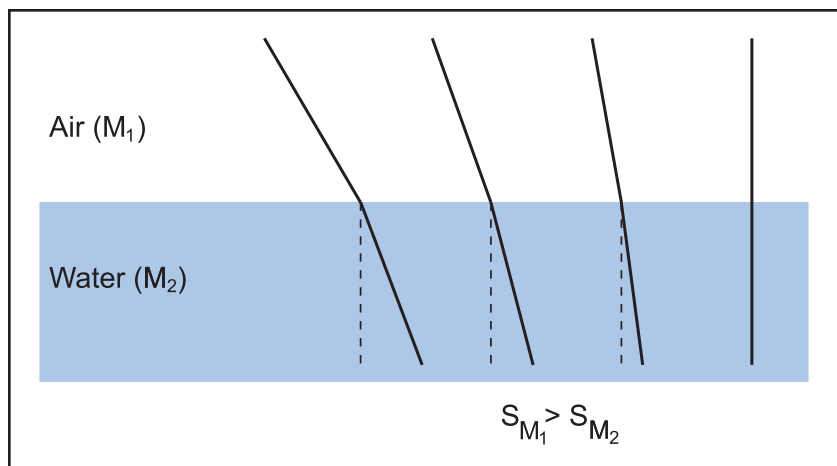


Figure 9–14. Refraction. A stick is pointed at the surface of a body of water at various angles. The ray of light corresponding to the image of the stick in the water appears to be bent as it passes from one medium (air) to a different medium (water) because *the speeds of propagation are different in the two media*. The bending of the light rays is called **refraction**.

to water) or when they encounter any change in the “conditions” of a medium that causes *the speed of wave propagation* to be changed. The rays of the incident wave are bent; they are **refracted**. Thus, **refraction** is a *bending of sound waves, or a change in direction of sound-wave propagation, due to a change in speed of propagation*.

Examples of Refraction and Reflection

An understanding of wave **reflection** and wave **refraction** helps explain some natural occurrences in sound transmission that are due to a change in properties of the medium. For example, why does sound “carry better,” that is, travel *farther*, when it travels *with the wind* rather than *against the wind*? It is tempting to assume that when traveling with the wind, the wind somehow “carries the sound wave along” and that when traveling against the wind, the wind “pushes the sound wave back” toward the source. Actually, the effect of the wind in that sense is negligible.

The correct answer comes from a combination of **refraction** and **reflection** as shown in Figure 9–15. In panel A, **spherical wave fronts** are propagated from left to right toward a receiver, **R**. If there is no wind, and if the sound wave is neither refracted nor reflected, the wave fronts look much like you see in panel A. One ray of the wave is parallel to the surface of the ground and progresses toward the receiver.

View Animation 8_12. Sound Travel With and Against Wind at animations.cla.umn.edu

Now, introduce a windy condition. First, note that *wind speed normally increases with increasing height* above ground. In panel B, wind speed is shown by vectors in which the lengthening of the vectors with increasing height reflects the increasing wind speed blowing from left to right in the direction of the arrows.

In panel C, the sound wave is propagated from left to right *against the wind* that blows from right to left. The greater wind speeds at the higher elevations impede sound wave transmission more than the lesser wind speeds at the lower elevations near the ground. Thus, the higher the elevation above the source of the earth, the less the speed of sound wave propagation. Because of the change in speed with increasing elevation above the surface, the sound wave is **refracted** (bent) upward and sound energy is directed upward into the atmosphere and well above the receiver.

The opposite occurs when sound is propagated *with the wind*. We see in panel D that when traveling with the wind, the greater wind speed at increasingly higher elevations causes the upper reaches of the wave fronts to travel faster than the lower portions. As before, the

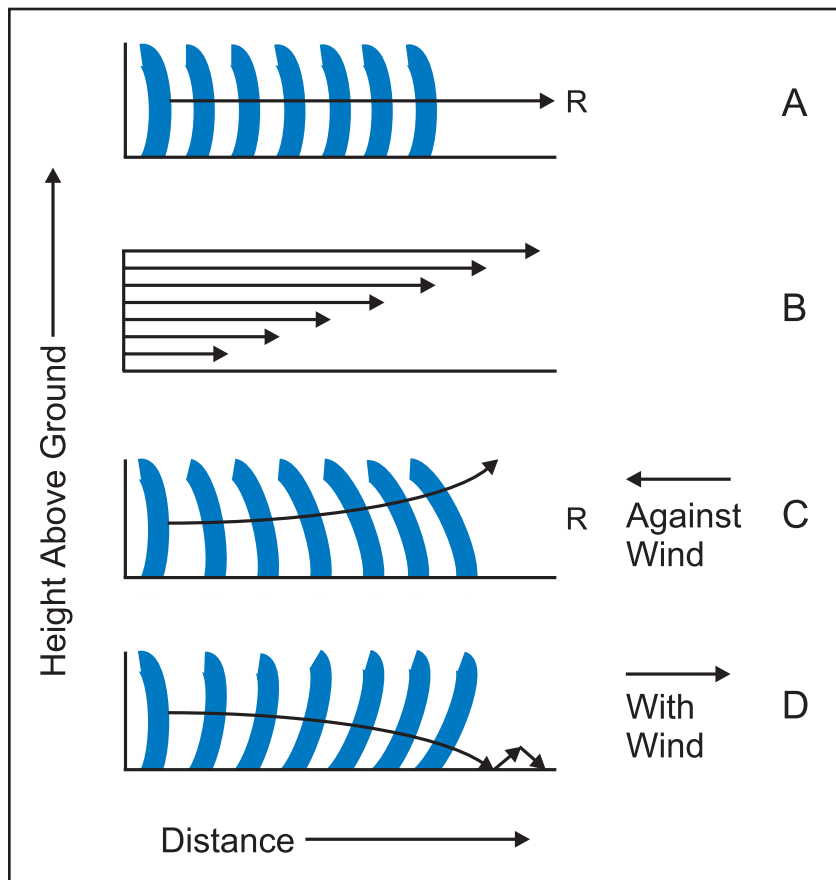


Figure 9–15. Sound wave **refraction**. Differential effects of a sound wave being propagated against the wind and with the wind. In panel A there is no wind, and the wave fronts progress from left to right unaltered. Because wind speed varies as a function of height above ground (panel B), the wind offers different amounts of resistance to different portions of the wave front. *Against the wind* (panel C), the wind offers greater opposition to the upper reaches of the wave fronts, and the sound wave is refracted over the receiver and into the atmosphere. *With the wind* (panel D), the wind causes the upper reaches of the wave fronts to be propagated with greater speed, and the sound wave is refracted toward the surface of the earth, reflected from the earth's surface, refracted downward, reflected upward, and so on. Consequently, sound travels over a greater distance with the wind.

sound wave is **refracted** downward toward the surface of the earth, **reflected** upward because of the **acoustic impedance** offered by the surface of the earth, **refracted** downward, and so on. Consequently, sound travels over a much greater distance with the wind. If you are lost and shout for help, you should hope there will be someone downwind from you who has a chance of hearing your plea.

Consider another example of “environmental influences” on sound transmission. Why does sound (typically) travel farther early in the morning than at midday? Normally, air near the ground cools overnight so that in the early morning, the air is warmer at higher elevations above the surface of the earth. We know from Chapter 1 (Equation 1–16) that *the speed of sound is directly proportional to the square root of elasticity and inversely proportional to the square root of the density of the medium*. As temperature increases, density decreases, and therefore *the speed of sound is directly proportional to the temperature*.

Suppose you are sitting in a canoe on a lake early in the morning. To keep matters simple, also suppose there is virtually no wind so that the surface of the water is glassy. You notice a couple sitting in another canoe at some considerable distance across the lake. To your surprise, you can easily hear them talking, which means that their sound has traveled a long way.

This is another example of the combined effects of refraction and reflection, as illustrated in panel A of Figure 9–16. *Speed of sound increases with increasing distance above the surface because of the presence of warmer air*. The sound wave is **refracted** downward, **reflected** from the surface of the water, **refracted** downward, and so on. The sound wave therefore travels over a long distance.

View Animation 8_13. Sound Travel Early in the Morning and at Midday at animations.cla.umn.edu

You return to the same spot on the lake later in the day and, once again, see a couple across the lake. You think they might be talking, but you cannot hear them. As we see in panel B, by now the sun has warmed the air near the surface of the water so that the air *is cooler with increasing height above ground*. Thus, the speed of sound is greater at the lower elevations, and sound waves created by the distant couple are **refracted** upward into the atmosphere.

When we described the early morning condition in panel A of Figure 9–16, we said that sound travels farther early in the morning because the sound wave *is refracted downward toward the surface of the water, reflected upward from the surface*, and so forth. Why doesn’t the sound that is refracted downward simply penetrate the water and travel as a sound wave in water? It does, but only partially.

The **acoustic impedance** of water is much greater than the impedance of air, which creates an *impedance mismatch* between the two media. About 0.1% of the incident sound energy is absorbed by the water surface and travels as a sound wave in water. Therefore, approximately 99.9% of the sound energy *is reflected from the surface of the water* and retained in the air medium. With Equation 5–4 in Chapter 5, we find that the intensity of sound in air is attenuated

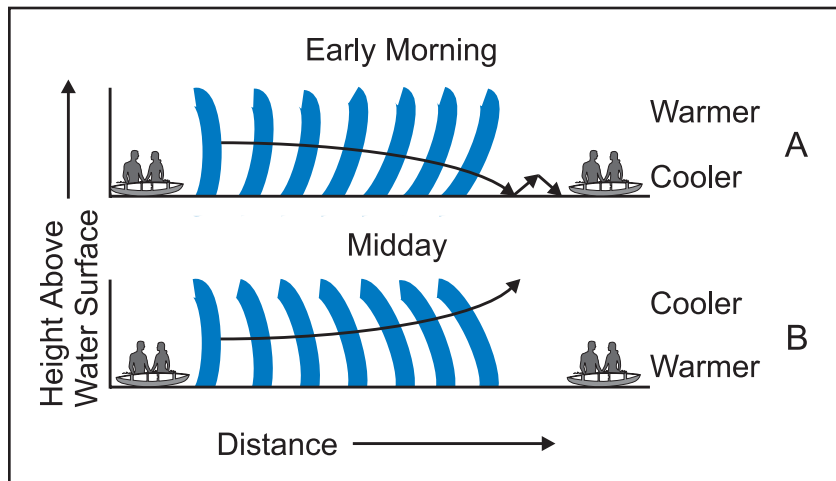


Figure 9-16. Sound wave **refraction**. Differential effects of a sound wave in air propagated in the early morning (panel A) when the surface of the earth has cooled overnight and later in the day (panel B) when the surface of the earth has warmed. Because temperature of air varies with height above ground (warmer in the morning; cooler later in the day) and speed of propagation is directly proportional to temperature of the transmitting medium, the sound wave is refracted. Early in the morning, the sound wave is refracted toward the surface, reflected from the surface, and so on. Hence, sound travels over a great distance. Later in the day, the sound wave is refracted upward into the atmosphere.

by approximately 30 dB when it penetrates the surface of the water because

Equation 5-4

$$\begin{aligned}
 \text{dB} &= 10 \log_{10} \frac{I_x}{I_r} \\
 &= 10 \log \frac{1 \times 10^{-3}}{1 \times 10^0} \\
 &= 10 \log (1 \times 10^{-3}) \\
 &= -30 \text{ dB.}
 \end{aligned}$$

■ DIFFRACTION

When water waves encounter a pier, a dock, or some other obstacle, the water is not “parted” (unless it is the Red Sea). Instead, it “bends” around the obstacle and wave motion continues beyond the obstacle as if it were not present. Sound waves behave the same way. The bending or scattering of a sound wave around an obstacle is called **diffraction**.

- The amount of **diffraction** depends on *the size of the obstacle relative to the wavelength of the sound wave*.
- **Diffraction** is more efficient when the wavelength of the sound wave is much larger than the size of the obstacle.
- Thus, diffraction varies inversely with frequency, because as wavelength increases, frequency decreases.

Consider a water wave. If the wave encounters a small obstacle such as a reed, the wave is diffracted around the reed almost as if the reed were not there. If the wave encounters a larger object such as a floating log or an anchored boat, the wave is diffracted. A “shadow region” forms immediately behind the object where there is no wave motion—the water behind the object is quite still. More distant from the obstacle, the wave reforms and continues.

Panel A of Figure 9–17 shows a plane progressive sound wave *traveling from left to right* that encounters a barrier. If the size of the barrier is large relative to the wavelength of the sound wave, most of the energy is reflected toward the source. In this case, however, the size of the barrier is small relative to the wavelength of the incident wave. Some of the sound energy is reflected from right to left from the barrier as the leftward pointing *dashed* arrows represent.

The wave fronts are diffracted. They then converge at the right side of the barrier. *At some distance from the obstacle, the shape of the wave fronts is nearly identical to the shape that would have*

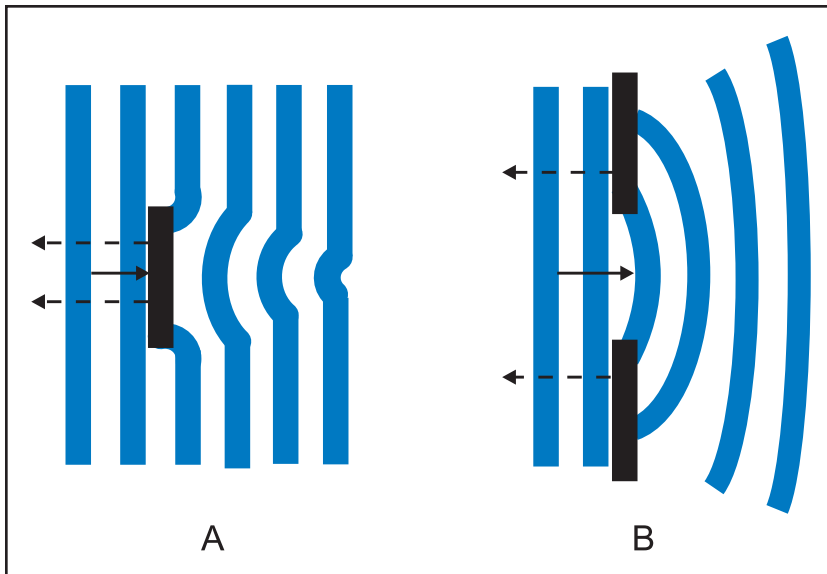


Figure 9–17. Diffraction of waves around an obstacle (panel A) and through an opening (panel B).

been seen had the obstacle not been present. A person sitting some distance to the right of the obstacle hears the sound in a nearly unaltered state.

The sound wave, after bending around the obstacle, continues as a plane progressive wave from left to right until, in panel B of the figure, it encounters an opening. Some of the sound energy is reflected from the walls on both sides of the opening. The portion of the wave front “striking” the opening passes through and reforms on the outside of the room. The implication should be obvious. A person outside the room hears the sound in a nearly unaltered state. The amount of diffraction depends on the wavelength of the sound wave in relation to the size of the opening. The transition between “small” and “large” wavelengths occurs when the size of the opening, in m^2 , is approximately 0.67λ .

Think about another experience you might have had. You relax in a room and enjoy music emanating from a good sound system. You appreciate the full range of the music from the bass to the treble. You then leave the room, walk down a hallway, and enter another room. You still hear the music, but now it “sounds different.” The bass notes are more prominent than the treble because the long wavelengths of the bass notes are more likely to be diffracted (scattered) around the obstacles (walls of the hallway and rooms) than the shorter wavelengths of the treble notes.

■ OTHER PHENOMENA IN SOUND TRANSMISSION

Beats

Suppose two people walk along a path at different rates. If one walks at a rate of 60 steps per minute and the other at a rate of 56 steps per minute, there are moments when they are “in step” (*in phase*) with each other and moments when they are “out of step” (*out of phase*). The one who walks with the faster pace gains four steps each minute on the other person, and they are “in step” four times each minute. Thus, the number of times that the two people are “in step” equals the difference between their two walking rates:

$$(60 \text{ steps/min}) - (56 \text{ steps/min}) = 4 \text{ steps/min.}$$

The same principle applies when two sine waves with *slightly different frequencies coexist in the same medium*. Because their frequencies are different, they move in and out of phase just as the two walkers did. If the frequency of one sine wave is 55 Hz and the frequency of the second one is 50 Hz, at a given place in the medium, the compressions of the two waves are in phase five times each second. When that happens, the two waves maximally *reinforce* (**constructive interference**) each other and the result is an *increase in amplitude*.

Similarly, five times each second the compressions of one sine wave are 180° out of phase with the compressions of the second wave. When that happens, the two waves maximally *interfere* (**destructive interference**) with each other and the result is a *decrease in amplitude*. The consequence of alternating between reinforcement and interference is a *periodic increase and decrease in amplitude of the resultant wave; the changes in amplitude are called beats, and the rate at which the beats occur is called the beat frequency*.

The **beat frequency**, the rate at which the *intensity* increases and decreases, is given by $f_2 - f_1$. In the case of the two waves mentioned above, a listener hears a single sound wave with a *frequency* equal to $(f_2 + f_1)/2$ (57.5 Hz) and with an intensity that changes at a rate given by $f_2 - f_1$ (5 Hz). Thus, even though there are two sine waves with frequencies of 55 Hz and 50 Hz traveling in the medium, the listener hears a single sound wave with a frequency of 57.5 Hz and with an intensity that waxes and wanes at a rate of five times per second.

Panels A and B of Figure 9–18 show two sine waves with equal amplitude, but different frequencies: $f_1 = 50$ Hz ($T = 20$ ms) and $f_2 = 55$ Hz ($T = 18.2$ ms). Panel C shows the resultant wave, which is the point-by-point sum of f_1 and f_2 over time.

- At $t = 0$, f_1 and f_2 are *in phase* and reinforce each other (**constructive interference**).
- After 100 ms, f_1 has completed 5 cycles ($100 \div 20$), f_2 has completed about 5.49 cycles ($100 \div 18.2$), and f_1 and f_2 are 180° out of phase (**destructive interference**); **cancellation** occurs.
- The two sine waves are next in phase at $t = 200$ ms, by which time f_1 has completed 10 cycles and f_2 about 10.99 cycles. Maximum reinforcement occurs.
- Thus, between $t = 0$ and $t = 200$ ms, the resultant wave changes from maximum reinforcement ($t = 0$), to cancellation ($t = 100$ ms), and to maximum reinforcement ($t = 200$ ms). Why are the two maxima separated by an interval of 200 ms? The **beat frequency** ($f_2 - f_1$) = 5 Hz, and the **beat period** ($T = 1/f$) = 200 ms.

A guitarist can use the perception of beats to help tune the instrument. Suppose the lowest (6th) string (E_3) has previously been tuned to 165 Hz and you now wish to tune the 5th string (A_3) to 220 Hz. First, press the 6th string against the 5th fret, which means that it produces A_3 (220 Hz) rather than E_3 (165 Hz). Then, alternately pluck the 6th *fretted string* and the 5th *unfretted string*, and systematically adjust the tension on the 5th string until the two produce the same perception of pitch.

When the tension of the 5th string is *not adjusted properly*, the guitarist hears a *single pitch that alternately increases and decreases in loudness because of a change in intensity: beats*. The rate at which the *intensity* increases and decreases defines the **beat frequency**. By counting the rate at which the beats occur, the beat frequency, one can estimate just how mismatched the two strings are in frequency.

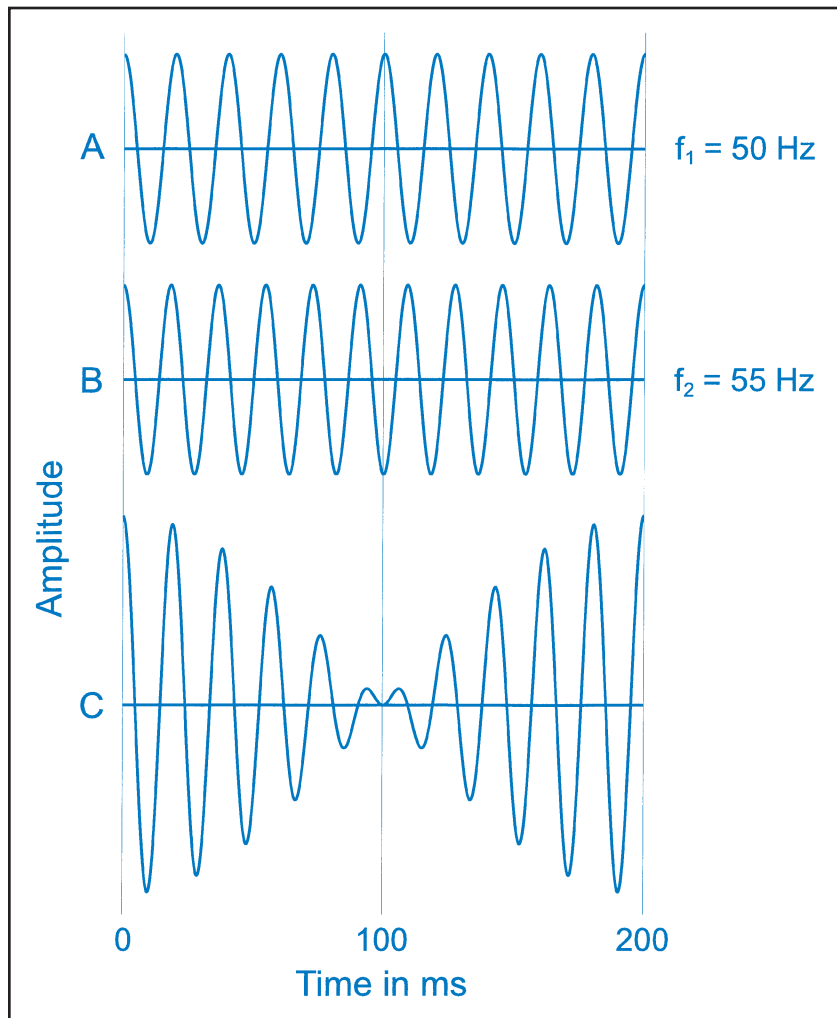


Figure 9-18. Beats. Panels A and B show two sine waves of equal amplitude, but different frequency. Panel C shows the resultant wave, a point-by-point summation of f_1 and f_2 . The *amplitude* of the resultant wave increases and decreases periodically, and the rate at which those periodic changes in *amplitude* occur is called the **beat frequency**.

The Doppler Effect

We learned in Chapter 1 that (1) the *frequency* of a sound wave is governed by *properties of the source* and that (2) the *speed* of sound wave propagation is governed by *properties of the medium*.

- Thus, if the source vibrates with a frequency of 250 Hz, the air particles in the medium are displaced at a rate of 250 Hz (the same frequency as the source), and a disturbance is propagated through an air medium at a rate of about 340 m/s.

- If the resulting sound wave penetrates the surface of water and travels as a sound wave through the water medium, *the frequency of particle displacement is still 250 Hz, but the speed of wave propagation is different* because of the different values of elasticity and density of the two transmitting media.

There is an exception, however. Position yourself at a certain location at the race track during the running of the Indianapolis 500. As the Indy cars approach, the *pitch* of their whining engines increases and reaches a *highest pitch* as the cars pass. As they then speed farther away, the pitch decreases.

This phenomenon is called the **Doppler effect** and is illustrated in Figure 9-19:

- Point **A** marks *the location of a moving source* of a sound wave.
- Point **C** marks *the location of a stationary receiver* positioned 340 meters from the source, **A**.

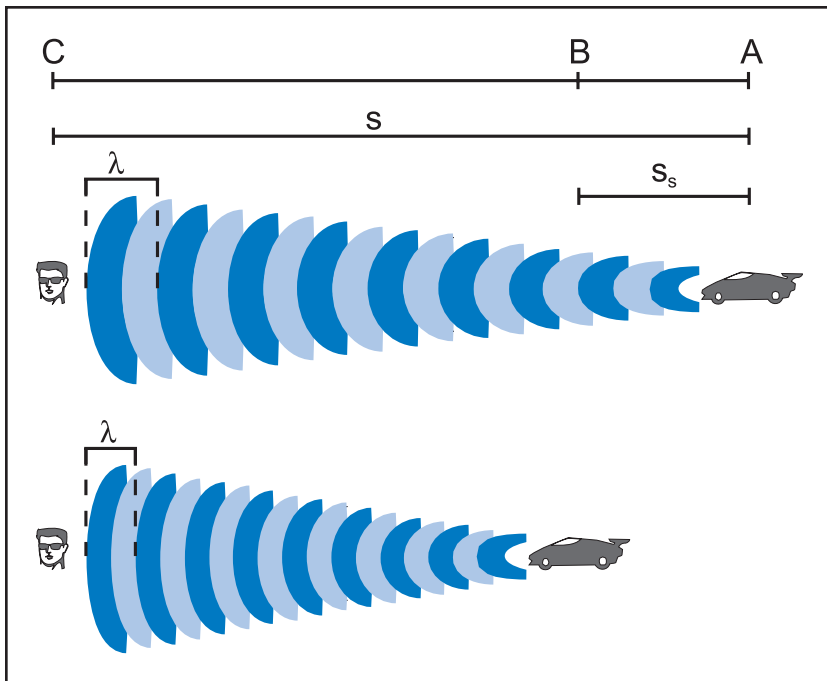


Figure 9-19. The Doppler effect. Point **A** is the location of a moving source of sound, point **C** is the location of a receiver, distance **s** is the speed of sound, and distance **s_s** is the speed of the moving source. If the source is stationary, after 1 second the compressions and rarefactions are spread over distance **AC**. But, the source is moving, and after 1 second it has moved to location **B**. Thus, the same number of compressions and rarefactions now are crowded into distance **BC**. The wavelength, the distance from one compression to another, decreases, and the frequency with which wave fronts move by the receiver **C** increases.

- Vector $\mathbf{s}$ represents the *speed of sound*.
- Vector $\mathbf{s}_s$ represents the *speed of the moving source*.
- Vector $\mathbf{AC}$ corresponds to a *distance* equal to the *speed of sound* (340 m/s).
- Vector $\mathbf{AB}$ corresponds to a *distance* equal to the *speed of the moving source* (85 m/s).

If the source **A** remains stationary, after 1 second the waves of compressions and rarefactions are spread over distance **AC**. The wavelength, λ , is shown as the distance between two compressions.

Suppose that during that 1 second, source **A** moves to location **B** at speed $\mathbf{s}_s$. Now, the same number of compressions and rarefactions are crowded into distance **BC** rather than distance **AC**, as is shown in the bottom row of the figure. If the same number of compressions are crowded into a smaller distance, *wavelength decreases*. Thus, the number of compressions that pass by the receiver **C** each second increases because frequency is inversely proportional to wavelength, $f \propto 1/\lambda$.

By how much will the frequency change? The magnitude of change in frequency is *directly proportional* to both frequency and the speed of the moving source. If the source of sound moves *toward* the observer, the altered frequency (f') is given by Equation 9–14:

Equation 9–14

$$f' = f \left(\frac{s}{s - s_s} \right),$$

where f' is the altered frequency,

f is the frequency produced by the source,

s is the speed of sound wave propagation, and

s_s is the speed of the moving source.

For example, if frequency = 400 Hz, speed of sound = 340 m/s, and the speed of the moving source = 85 m/s, the altered frequency is:

$$f' = 400 \left(\frac{340}{340 - 85} \right) = 533 \text{ Hz.}$$

If *the source moves toward the observer*, the observer hears the pitch rise to a maximum that corresponds to a frequency of 533 Hz instead of 400 Hz. If *the source moves away from the observer*, the pitch of the sound decreases, and the altered frequency is given by:

Equation 9–15

$$f' = f \left(\frac{s}{s + s_s} \right).$$

Therefore, in this case, the frequency decreases from 400 Hz to 320 Hz:

$$f' = 400 \left(\frac{340}{340 + 85} \right) = 320 \text{ Hz.}$$

Sonic Booms

Why is a “sonic boom” created when an airplane exceeds the speed of sound? An airplane traveling with a speed greater than the speed of sound is moving at supersonic speed. Supersonic speeds are rated in **Mach numbers**; a Mach number is the ratio of the speed of the plane to the speed of sound. Thus, if the airplane’s speed is 680 m/s and the speed of sound is 340 m/s, the airplane has achieved Mach 2.

Before the airplane achieves Mach 1, the point at which the speed of the airplane equals the speed of sound, it behaves as a *moving source*. An observer hears a change in pitch in accordance with the Doppler principle. The pitch rises as the plane moves toward the observer and falls as the plane moves away from the observer. The plane with a speed <340 m/s, in effect, “is chasing its own sound waves” because the speed of sound exceeds the speed of the moving source, the airplane.

- When the airplane accelerates until its speed approaches Mach 1, *the sound waves in front impede the flight of the airplane*; additional thrust is required to achieve Mach 1. That is called “breaking the sound barrier.”
- When the barrier is broken and Mach 1 is achieved, and then exceeded, the airplane has moved in front of its own sound wave.
- The successive compressions of the sound wave pile upon one another, rather than be separated by rarefactions, to form a very *large compression* that contains considerable sound energy. The energy may be sufficiently great to break a window.
- The sound we hear is called the “sonic boom.”

NOTES

1. River City refers to the mythical town of River City, Iowa, the setting for the 1962 film, *The Music Man*. The film starred Robert Preston as “Professor” Harold Hill, who, among other activities, chased Marian the Librarian, played by Shirley Jones, and sang the memorable lyrics to “Ya Got Trouble.” Hill warned the townsfolk of the evils of young boys playing pool “right here in River City.”

FREQUENTLY MISUNDERSTOOD CONCEPTS

1. A tube with length = 0.5 m, open at one end and closed at the other, is filled with air at a temperature of 15°C. Calculate the first three resonant frequencies of the tube.

Before using Equations 9–6, 9–7, and 9–8 (or Equation 9–9) to calculate F_1 , F_2 , and F_3 , each of which has speed of sound

(s) in the numerator, we must first consider the effects of temperature on s. We learned in Chapter 1 that the speed of sound at 0°C is 331 m/s, and that the speed of sound increases by 0.61 m/s for each 1°C increase in temperature. Therefore, the speed of sound at 15°C is given by:

$$s = 331^\circ + (0.61 \times 15^\circ) = 340 \text{ m/s.}$$

Next, the frequencies of the three lowest resonant frequencies are given by:

Equation 9-6

$$F_1 = \frac{s}{4L} = \frac{340}{2} = 170 \text{ Hz}$$

Equation 9-7

$$F_2 = \frac{3s}{4L} = \frac{1,020}{2} = 510 \text{ Hz}$$

Equation 9-8

$$F_3 = \frac{5s}{4L} = \frac{1,700}{2} = 850 \text{ Hz}$$

The higher resonant frequencies are *odd integer multiples* of F_1 ($F_2 = 170 \times 3 = 510 \text{ Hz}$; $F_3 = 170 \times 5 = 850 \text{ Hz}$; and so forth).

2. **The air in the tube described in #1 is replaced with helium at a temperature of 20°C. The speed of sound is 927 m/s under those conditions. Calculate the first three resonant frequencies of the tube.**

In this case, we should *not* correct for the fact that the temperature of the helium is 20°C because the speed of sound in helium at 20°C is already given as 927 m/s. Because the resonant frequencies are *directly proportional* to the speed of sound propagation, and because speed in helium at 20°C is 2.73 times faster ($927/340 = 2.73$) than speed in air at 15°C, we simply multiply the three frequencies calculated for #1 by 2.73 to find the correct answers:

$$F_1 = 170 \times 2.73 = 464 \text{ Hz;}$$

$$F_2 = 510 \times 2.73 = 1391 \text{ Hz; and}$$

$$F_3 = 850 \times 2.73 = 2318 \text{ Hz.}$$

Alternatively, we can compute the resonant frequencies and obtain the same results by using Equations 9-6, 9-7, and 9-8 (or Equation 9-9).

3. **The tube described in #2 is replaced with one that is open at both ends. By how much will the *lowest* resonant frequency increase?**

2:1. It will increase by a factor of 2:1 ($464 \times 2 = 928$ Hz), subject to rounding error, because F_1 in a tube open at both ends is given by:

$$F_1 = \frac{s}{2L} = \frac{927}{1} = 927 \text{ Hz}$$

- 4. If intensity level decreases by 6 dB with each doubling of distance from the source, sound pressure level decreases by how many dB?**

6 dB. A common, but *incorrect*, answer is 12 dB. If it is not clear why 12 dB is incorrect, review “The Relation Between dB IL and dB SPL” in Chapter 5. We learned that $2 \times 10^1 \mu\text{Pa}$, the reference pressure for dB SPL, is the pressure equivalent of the reference intensity for dB IL, 10^{-12} watt/m². As long as equivalent references for pressure and intensity are used, *dB IL = dB SPL*. Strictly speaking, of course, this was not really an inverse square law problem.

- 5. If sound intensity level is 85 dB at a distance of 200 m from the source, by how many decibels will the level be *decreased* at a distance of 2,000 m from the source?**

20 dB, not 65 dB SPL. Solution of this problem is straightforward.

$$\begin{aligned} \text{dB} &= -20 \log \frac{d_i}{d_r} \\ &= -20 \log \frac{2,000}{200} \\ &= -20 \text{ dB} \end{aligned}$$

Equation 9-4

It is true that the SPL at 2,000 m will be 65 dB SPL ($85 - 20$), but the question asked is, “by how many decibels will the level be *decreased*?”

- 6. When two sound waves with slightly different frequencies coexist in the same medium, what does the beat frequency reveal?**

The **beat frequency** corresponds to the rate at which the *intensity increases and decreases*. Suppose the frequencies are 256 Hz and 260 Hz. The **beat frequency**, the rate at which the periodic increases and decreases in *intensity* occur, is given by $f_2 - f_1$, which in this case is 4 Hz ($260 - 256 = 4$ Hz). The listener should hear a single sound wave with a pitch corresponding to a frequency of 258 Hz ($f_1 + f_2/2 = 258$ Hz) whose intensity waxes and wanes at a rate of 4 Hz, the **beat frequency**.

■ PRACTICE PROBLEMS

1. The intensity of a sound in a free, unbounded medium is 95 dB SPL at 200 meters from the source. In accordance with the inverse square law, by how many decibels will the SPL be decreased at 2,000 meters from the source?
2. For Problem #1, what is the SPL at a distance of 2,000 meters from the source?
3. Explain why the following statement is correct: "According to the inverse square law, the intensity decreases by 6 dB for each doubling of the distance."
4. Describe the ways in which reflecting surfaces and absorbent materials in a medium should cause the inverse square law not to hold strictly.
5. In the discussion of standing waves and resonant frequency for a tube that is *open at one end and closed at the other*, we stated that a displacement node (a point of no vibration) always is located at the closed end of the tube because the air is not free to move, and a displacement antinode (a point of maximum vibration) always is located at the open end where the air moves freely. We can also talk about a pressure node (a point of zero pressure) and a pressure antinode (a point of maximum pressure), because not only are the air particles moving (being displaced), but also sound pressure increases and decreases in the tube. Describe the relation between displacement and pressure nodes and antinodes.
6. A "fish locator," which should be called a "depth finder," is a popular electronic device that is used to measure water depth at a particular location on a body of water. The device emits a series of pulses from a transducer mounted on a boat just below the surface of the water, and the pulses travel to the bottom of the river or lake. How would you speculate that the device can reveal the water depth?



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■ ABSORPTION

We learned in Chapter 9 that when obstacles in the transmitting medium offer a large acoustic impedance to the propagation of a sound wave, some of the energy is **reflected** back toward the source. The sound wave need not encounter a dense, rigid obstacle such as a steel wall to cause the wave to be reflected. In fact, *opposition to sound transmission exists at any boundary between different transmitting media that have different impedances.*

- If the impedance offered by the obstacle *is infinite*, the intensity of the reflected wave equals the intensity of the incident wave.
- If the impedance offered by the obstacle *is not infinite*, some of the energy penetrates the obstacle; that is **sound absorption**.

When sound energy is absorbed, energy is dissipated in the form of heat and the energy in the reflected wave is reduced—even though the amount of heat that is generated is negligible. The intensity of the reflected wave, therefore, is less than the intensity of the incident wave. Different materials, of course, offer different amounts of impedance. Some materials therefore absorb sound energy more efficiently than others. For example, a steel wall impedes sound transmission more than a wall made of wood, which, in turn, impedes transmission of sound more than a wall covered with cotton batting.

When a sound wave strikes a barrier that offers a large acoustic impedance, the barrier is set into vibration unless it is infinitely rigid. Some of the energy is absorbed due to internal frictional losses in the material. Energy also is lost because the sound wave travels through the barrier if its composition is porous.

Published tables are available that list the proportion of sound energy absorbed by various materials, but those tables should be read carefully. All carpets, for example, are not equally absorbent. The amount of energy absorbed by a particular carpet also depends on the nature of the material on which the carpet is laid—felt pad, rubberized pad, wood, cement, and so forth. Moreover, the absorption efficiency of carpeting—and most other materials—varies, sometimes greatly, with the frequency of incident sound waves. For example, some carpeting might absorb as much as 60% of sound energy at 4 kHz, but as little as 5% at 125 Hz (Everest & Pohlman, 2015).

Absorption by draperies generally is fairly high, and folded draperies absorb sound energy more efficiently than those that are hung flat because of the greater surface area provided by folding. The weight of the drapery also is important. For example, for a frequency of 1 kHz, absorption is nearly 50% for a drapery weight of 18 oz./yd², but only about 18% for a weight of 10 oz./yd² (Beranek, 1954). Once again, there is an interaction with frequency in that the importance of drapery weight on absorption efficiency is negligible for frequencies below 250 Hz and above 2 kHz.

Although porous absorbers generally are quite efficient, all porous absorbers are not equal. For example, at 1000 Hz, absorption is about 40% for carpeting, 80% for cotton curtains, and nearly 100% for fiberglass. In nearly all cases, however, there is an interaction with frequency.

The Absorption Coefficient (α)

The concept of **sound absorption** means that *some* of the incident sound energy penetrates (is absorbed by) an obstacle. The effectiveness of a material to absorb sound energy is expressed by an **absorption coefficient**. A **coefficient** is a dimensionless multiplier used in the measurement of some property. An **absorption coefficient** is *the proportion of energy in the incident wave absorbed by some material*. We will see subsequently that the coefficient is a multiplier in the equation for calculating the “**total absorption**” by a particular material. If the absorption coefficient for a given material is listed as 0.40, that means that 40% of the incident energy is absorbed. The **absorption coefficient** (α), therefore, is *the ratio of the sound energy absorbed (I_a) to the total sound energy in the incident wave (I_i)*,

$$\alpha = \frac{I_a}{I_i}.$$

Equation 10–1

We learned in Chapter 9 that when a sound wave traveling in air strikes the surface of water, approximately 99.9% of the sound energy is reflected from the surface of the water and 0.1% is absorbed by the water. Suppose the sound pressure level of the incident wave in air is 70 dB. That means its intensity is 10^{-5} watt/m² (Equation 5–4 from Chapter 5). If only 0.1% of the sound energy is absorbed by the water medium, then the intensity of the sound wave in water is 40 dB SPL and the absorption coefficient is 0.001 (10^{-3}). You might want to retrieve that fact from your memory bank if you ever have a fishing partner tell you, “Be quiet—you’ll spook the fish!”

We have seen that because 0.1 % of the sound energy is absorbed by the water medium, the intensity of the sound wave traveling in water is attenuated by 30 dB. If 99.9% of the sound energy is reflected back into the air from the surface of the water, by how much is the reflected wave attenuated?

$$\text{dB} = 10 \log 0.999 = 0.0043 \text{ dB}.$$

The answer is “barely!”

The absorption coefficient can, in theory, range between 0.0 (*none of the energy is absorbed; all energy is reflected*) and 1.0 (*none of the energy is reflected; all is absorbed*). In the example above, an absorption coefficient of 0.001 means that very little (0.1 %) of the energy of the sound wave is absorbed by the water.

The absorption coefficient *is independent of the intensity of the incident wave*. Thus, if the intensity of the incident wave is 90 dB SPL instead of 70 SPL,

- the absorption coefficient still is 0.001,
- 99.9% of the sound energy is reflected from the surface of the water and retained in the air medium, and
- the SPL of the sound wave in water is 60 dB SPL.

■ ABSORPTION AND REFLECTION

Sound wave **absorption** *is inversely proportional to sound wave **reflection***. As the absorption coefficient increases, reflection decreases and reverberation time decreases. Porous absorbers (curtains, draperies, fiberglass, etc.) generally absorb better at higher frequencies, whereas panel (resonant) absorbers tend to perform better at lower frequencies.

When one attempts to increase sound absorption in a room, it is important to first measure the *spectral composition* of the ambient noise to help guide in the selection of appropriate absorption materials. Although the absorption coefficient of different materials varies with the frequency of the incident sound wave, I will select a single mid-range frequency of 1000 Hz to illustrate how the proportion of sound energy absorbed varies among common materials in a room (Backus, 1977).

- Walls made of unpainted concrete, marble, or glazed tile have an absorption coefficient of about 0.01. Thus, even though those materials absorb about 10 times as much sound energy as the surface of water (where the absorption coefficient is about 0.001), they still are not efficient absorbers because they only absorb about 1% of the sound energy.
- If heavy carpeting is attached to the walls, the absorption coefficient increases from 0.01 to about 0.37, and approximately one-third of the energy is absorbed by the material.
- There might also be other materials in a room that contribute to the absorption of sound: upholstered seats, clothing, people's skin, and so forth.
- Some sound energy is absorbed by the air medium itself, and in that case, the amount of energy absorbed depends on the temperature and humidity of the air and the frequency of the sound wave.

How are absorption coefficients rated with respect to their effectiveness? Because the coefficients are different for different angles of the incident rays striking the surface, *the coefficient that is reported typically is an average of the coefficients for many different angles for a particular frequency*.

We have emphasized that the *absorption coefficient depends on both the nature of the surface material in a room and the fre-*

quency of the incident sound wave. The relation, however, is not a simple one.

- With some materials, absorption coefficients *increase* with increasing frequency ($\alpha \propto f$). For example, in the frequency range from 125 Hz to 4000 Hz, the coefficient for thick carpeting installed over concrete increases from .02 (2% of the energy is absorbed) at 125 Hz, to .37 (37% of the energy is absorbed) at 1000 Hz, and to .65 (65% of the energy is absorbed) at 4000 Hz.
- If the same carpeting is installed over a felt pad laid on concrete, the coefficient *increases at some frequencies* (e.g., from .06 to .27 at 125 Hz, and from .14 to .39 at 500 Hz), *decreases at some frequencies* (e.g., from .60 to .48 at 2000 Hz), and *remains essentially the same for other frequencies* (e.g., from .37 to .34 at 1000 Hz, and from .65 to .63 at 4000 Hz).
- With other materials, absorption coefficients *decrease* with increasing frequency ($\alpha \propto 1/f$). Plywood ($1/4''$) is an example of this inverse relation. The absorption coefficients are .60 at 125 Hz, .30 at 250 Hz, and range from .09 to .10 for frequencies from 500 Hz to 4000 Hz.
- Finally, *there are some materials for which absorption coefficients are largely independent of frequency*. The coefficients range from .01 to .02 for glazed tile, from .01 to .03 for unpainted concrete, and from .02 to .03 for asphalt tile laid on concrete.

Suspended acoustical tiling is an example of an effective sound-absorbing surface; *absorption coefficients range from a low of .76 at 125 Hz to a high of .99 at 1000 and 2000 Hz*. But it is important to understand that all acoustical tiling is not equally effective; you must know the exact coefficient before installation.

Total Absorption and Reverberation Time (RT_{60})

In the foregoing discussion of absorption, attention was called only to the absorption of sound energy by materials in, for example, a room. *The total sound energy absorbed in a room—any room—depends not only on the absorption coefficients of the materials in the room, but also on the volume of the room.* **Total absorption** is quantified in **absorption units**, and the unit of measure is the **sabin** (fps) or **metric sabin** (MKS), named in honor of Wallace C. Sabine, who performed pioneering empirical studies on architectural acoustics at Harvard University near the turn of the nineteenth century.

Imagine a large, bare room with six hard interior surfaces (ceiling, floor, and four walls) and a 1 m^2 opening (an open window) in one wall. There is no carpeting, upholstery, people, or other sound-absorbing materials in the room except for air. An intense impulse

noise is generated and immediately turned off. The open window effectively “absorbs” all (100%) of the sound energy that strikes it. The 1 m² opening therefore has an **absorption coefficient, α** , of 1.0 (100% of the sound energy is absorbed). That defines **1 absorption unit**, which is expressed as **1 metric sabin (MKS)**. Thus, a **metric sabin (MKS)** is the total absorption by a window (an opening) with an area of 1 square meter (1 m²), a **sabin** is the total absorption by a window opening with an area of 1 square foot (1 ft.²), and

1.0 sabin equals 0.093 metric sabin.

How do **absorption** and **reflection** relate to one another? In the room described above, each of the six surfaces is assumed to absorb the same amount of sound energy. The theoretical reverberation time, **RT₆₀**, is calculated with Equation 10–2. **RT₆₀** is directly proportional to the volume of the room, **V**, and inversely proportional to the area of the opening, **a**:

Equation 10–2

$$RT_{60} = k (V/a)$$

where **k** is a constant.

The value of **k** is 0.049 in the fps system, where volume is expressed in cubic feet and area in square feet, but 0.161 in the MKS system, where volume is expressed in cubic meters and area in square meters.

- If room volume increases by some factor (say 1.5:1), with area of the window held constant, **RT₆₀** increases by the same factor (1.5:1), and sound energy is retained in the room for a longer time. You should keep this relation in mind if you have a home built and decide, for aesthetic reasons, to have a high vaulted ceiling in the living, or “great,” room. With all other factors held constant, the room is more reverberant than it would be without the high ceiling because of the greater room volume.
- If room volume is held constant, but the area of the opening increases by some factor, **RT₆₀** decreases by the same factor, and sound energy escapes from the room more quickly.

Before continuing the discussion of absorption, the absorption coefficient, and reverberation time, it is worthwhile to pause and reflect on Professor Sabine’s career as a scientist. Countless scientific discoveries are serendipitous; others are born from an inquiring mind and diligent, painstaking effort. Sabine’s contributions to architectural acoustics belong to the latter category, and his seminal work remains as influential today as it was more than a century ago.

Graham Chedd (1970) provided a fascinating account of how Sabine arrived at the study of acoustics and the special attention he paid to reverberation and absorption. The vignette that follows is a synopsis of the story written by Chedd.

Sabine was an Assistant Professor of Physics at Harvard University who conducted research in electricity and optics. The President of Harvard approached him to complain about the terrible acoustic environment in the large Hall of the Fogg Art Museum. The President implored Sabine to “do something” about it. Sabine, an untenured Assistant Professor, accepted the challenge. This took place in 1895, and as you can imagine, the technology at that time could best be described as lying somewhere between non-existent and “crude.” He armed himself with a few organ pipes, a stopwatch, and, eventually, an abundant supply of cushions.

After listening in the Hall of the Museum for a bit, Sabine decided that the principal problem was *excessive reverberation*. With the aid of his stopwatch, he attempted to measure “how long a sound remained audible—to him—after the sound source was terminated.” He concluded that the reverberations remained audible for a little more than $5^{1/2}$ seconds. Sabine reasoned that the long reverberation times were likely the principal source of the noise problem and that the duration of the reverberations might be shortened by introducing some absorbing materials to the room—that is where the cushions came into play. He added a few cushions to the room, measured, added a few more cushions, measured again, and so on. Sabine continued this progression until finally the ceiling, walls, floors, and 436 seats were covered with the array of cushions that measured a total of 242 meters in length. Average reverberation time, as measured by the stopwatch and his ability to still hear the sound, dropped from an average of 5.61 seconds to 2.22 seconds. He also found that adding more than 180 meters of cushions yielded virtually no further improvement.

That “simple” start led to many modifications in his experiments such as replacing the cushions with other potentially absorbing materials and by making measurements of surface areas and room volume. Finally, after about three years of study, and with no advances in technology (and no federal funding to support his research), he wrote Equation 10–2 for calculation of reverberation time—an equation that has remained largely unmodified for more than 100 years.

With the close of that historical account, we return to consider additional factors in sound absorption. The room described previously had a 1 m^2 opening with an absorption coefficient of 1.0.

- Suppose the opening (1 m^2) in the hypothetical room is covered with an “ideal” material that also has an **absorption coefficient** of 1.0. Sound energy now dissipates at the same rate that it did with the uncovered opening.
- If another material is substituted that has an **absorption coefficient** of 0.5, what happens to **total absorption**? **Total absorption (A)** of a surface is given by *the product of the surface*

area of the covered opening and the absorption coefficient of the material. Thus,

Equation 10–3

$$A = S\alpha,$$

where **A** is total absorption, **S** is surface area, and α is the absorption coefficient.

For example, if $S = 1 \text{ m}^2$ and $\alpha = 0.5$, the total absorption, **A**, of the surface is 0.5 **metric sabin**. If α remains 0.5, but **S** increases to 2 m^2 , **A** for the larger surface becomes 1 **metric sabin**.

Equation 10–3 enables calculation of the **total absorption, A**, where there is only one absorbing surface, in this case, one covered opening. Equation 10–4 applies to the more general case to calculate the **total absorption, A'**, for a room that has multiple absorbing surfaces:

Equation 10–4

$$A' = S_1\alpha_1 + S_2\alpha_2 + \dots + S_n\alpha_n.$$

Thus, the **total absorption, A'**, for multiple absorbing surfaces *equals the sum of the total absorption of all individual surfaces*. For example, for four absorbing surfaces with different values of **S** and α , with each expressed in metric units (MKS), **A'** is

$$\begin{aligned} A' &= (1.0 \times 0.5) + (2.0 \times 0.5) + (1.0 \times .68) + (2.0 \times .39) \\ &= 2.96 \text{ metric sabin.} \end{aligned}$$

Recall that *sound absorption is inversely proportional to sound reflection*.

- Too much absorption, with correspondingly decreased reverberation, results in a “dead” room, and at the extreme, an **anechoic room** (a room without echoes).
- Too little absorption, with correspondingly increased reverberation, results in a “live” room, and at the extreme, a **reverberant room**.

Optimal Reverberation Time

We have seen that there are two critical elements that determine the reverberation time (**RT₆₀**) of a room: (1) *room volume* and (2) *total absorption*. When designing or redesigning a room to achieve a desired reverberation time, a decision to increase room volume would need to be offset by adding materials that provide greater sound absorption. What is the optimal **reverberation time**? There is no single answer! As a rule, most people seem to prefer “live” rooms over “dead” rooms for listening to live music, whereas “dead” rooms are preferable for recorded music because reverberation ordinarily is incorporated into the recording.

Optimal **reverberation time** for a large room such as an auditorium is a compromise between satisfactory “clarity” (*clarity decreases as RT_{60} increases*) and satisfactory sound intensity (*maximum intensity increases as RT_{60} increases*). Moreover, *optimal reverberation time for speech is different from what is considered optimal for music*. For speech in a conference room, optimal RT_{60} ranges between 400 ms and 800 ms, whereas optimal RT_{60} ranges between 900 ms and 1400 ms for chamber music, and between 1500 ms and 2400 ms for organ music (Backus, 1977). Optimal RT_{60} is mainly a matter of listener preference, status of a listener’s hearing, age of the listener, and whether or not the listener is wearing hearing aids. Reverberation time and optimal reverberation time will be fleshed out further in a subsequent section on room acoustics.

Sample Computations

Suppose a room has the dimensions 8 m wide, 15 m long, and 3 m high. Reverberation time varies with frequency, but we will compute RT_{60} only for 500 Hz. The floor is covered with a heavy carpet ($\alpha = 0.40$), the walls and ceiling are constructed of plaster ($\alpha = 0.10$), and the total absorption, A' , is computed with Equation 10–4:

	Surface Dimensions	Surface Area (S)	α	A
Side wall 1	15 × 3	45	0.10	4.5
Side wall 2	15 × 3	45	0.10	4.5
End wall 1	8 × 3	24	0.10	2.4
End wall 2	8 × 3	24	0.10	2.4
Floor	8 × 15	120	0.40	48.0
Ceiling	8 × 15	120	0.10	12.0

$$A' = 73.8.$$

Room volume (V) = 360 m^3 ($15 \times 3 \times 8$). With Equation 10–2, but with A' instead of A in the denominator, we calculate that RT_{60} is:

$$\begin{aligned}
 RT_{60} &= k \frac{V}{A'} \\
 &= .161 \frac{360}{73.8} \\
 &= 785 \text{ ms.}
 \end{aligned}$$

Thus, with a room volume of 360 m^3 and a **total absorption** of 73.8 **metric sabins**, the **reverberation time**, RT_{60} , is about 785 ms. Suppose 785 ms is longer than desired. If acoustical tiling with $\alpha = 0.76$ is installed on the ceiling and one wall at the end of the

room toward which a talker is expected to face, total absorption, A' , increases from 73.8 to 168.8 metric sabins. The new value of RT_{60} is:

$$\begin{aligned} RT_{60} &= k \frac{V}{A'} \\ &= .161 \frac{360}{168.8} \\ &= 343 \text{ ms.} \end{aligned}$$

Installation of acoustical tiling on the ceiling and one end wall decreases RT_{60} from 785 ms to 343 ms.

Installing carpeting in a classroom or a restaurant is problematic because of the potential for staining from spillage. Now that acoustical tiling has been placed on the ceiling and one end wall, if we install asphalt tile on the floor with an absorption coefficient, α , of .03 instead of a heavy carpeting with an absorption coefficient of .40, what happens to RT_{60} ? We know that the reverberation time increases, but will it lie in an acceptable range?

The **total absorption**, A' , decreases from 168.8 to 124.4 **metric sabins**. Finally, we compute the new value of RT_{60} to be:

$$\begin{aligned} RT_{60} &= k \frac{V}{A'} \\ &= .161 \frac{360}{124.4} \\ &= 466 \text{ ms.} \end{aligned}$$

Thus, when carpeting is replaced with asphalt tile, RT_{60} becomes approximately 466 ms, and that lies within what we believe to be an acceptable range of 400–800 ms.

There are three important caveats.

- The sample computations were made for only a single frequency, 500 Hz, and different results will be obtained for other frequencies. When RT_{60} is computed for more than a single frequency, say 500, 1000, and 2000, the results can be reported in a more informed way as $RT_{60/500}$, $RT_{60/1000}$, and $RT_{60/2000}$, or as $RT_{60/500,1000,2000}$ for the mean of the three frequencies.
- The presence of seats, tables, people, wall hangings, and so forth was not considered. Even wooden chairs absorb sound energy, and each person in the room might add as many as 3 to 5 absorption units, particularly at 500 Hz. Beranek (1960), however, found that total absorption for a half-filled auditorium is the same as for a full auditorium, if the smaller number of people is distributed uniformly throughout the room.
- The equations for calculation of **reverberation time** and **total absorption** are susceptible to some error because of certain assumptions that are made. All carpeting is not the same,

all acoustic tiles are not the same, all wooden floors are not the same, and so forth. To increase the accuracy of the computations, therefore, one needs to know the absorption coefficient for the specific material being used.

The computations do provide, however, a useful starting point for determining the approximate acoustic conditions of a room. Equation 10–2 tends to be more accurate for “live” rooms than for more absorbent rooms.

Because of those caveats, *measurement of reverberation time almost always is more accurate than a complex array of calculations*. Finally, decisions about how best to modify a room to achieve optimal acoustical conditions is not a recommended undertaking for amateurs—seeking the advice of a qualified acoustical engineer/consultant is strongly recommended.

Absorption and Diffraction

All surfaces in a room absorb some amount of sound energy. *The air medium in which sound is transmitted also absorbs sound energy.* That helps explain why the foghorn on a ship emits a very low-frequency sound rather than a high-frequency whistle. When conditions are foggy, air contains water droplets with a small diameter relative to the long wavelength of the sound emitted by the foghorn. Sound energy is **diffracted**; it bends around the small-diameter water droplets, reforms, and is transmitted as an effective warning signal. If a high-frequency whistle with a very short wavelength approaching the diameter of the water droplets is used, too much of the sound energy is **absorbed** by the water droplets, and the effectiveness of the alerting signal is diminished.

ROOM ACOUSTICS

The singular form “room” is used commonly, but can be misleading. A more forthright title is “rooms acoustics,” which surely violates some syntactic rule. The plural form is more accurate because there is a vast array of room types and the acoustical characteristics vary from room to room. Moreover, rooms with similar physical and acoustical characteristics are used in different ways, and both design characteristics and usage matter. For those reasons, the challenges faced by an acoustical engineer to achieve an optimal acoustical environment are complex—there is no single strategy that will suit all situations. For example, optimal acoustical environments are different for classrooms, homes, conference or seminar rooms, auditoria, restaurants, and so forth. Within musical venues, optimal conditions are different for choral groups, string quartets, organs, orchestras, and so on. For those reasons, a discussion of “room acoustics” must be specific to both room type and room usage.

An Overview

The discussion of room acoustics in this chapter focuses on classrooms, with an occasional foray into the needs for other venues. *The principal factor that guides the topics to be covered is to identify the acoustical characteristics that are requisite for maximizing speech understanding (intelligibility).* The topics include: sound-treated rooms, sound isolation, measurement of speech understanding, background noise level, signal-to-noise ratio, reverberation, absorption, and a case study of successful classroom redesign.

Sound-Treated Rooms

The two extremes of sound-treated rooms are **anechoic** and **reverberant**. **Anechoic** rooms are designed to virtually eliminate all reverberations. Nearly all the sound energy transmitted *within* the room is absorbed because the room surfaces have very high **absorption coefficients**. Anechoic rooms can be eerily quiet and are commonly used to test electroacoustic instruments such as hearing aids. The background noise level in one such chamber in Minneapolis, Minnesota is -9 dB SPL. One unintended consequence of an anechoic room is that some people cannot remain in the room very long without experiencing discomfort and even hallucinations.

To create an ideal **anechoic** room, fiberglass wedges are fixed to all six surfaces, though there usually is a metal grating lying over the floor wedges to accommodate a person and equipment. Because fiberglass has a high absorption coefficient (nearly 100%), it acts as an “acoustic sponge.” If the *exterior surfaces* of the room are made of steel or some other material with a high transmission loss, in addition to being **anechoic**, the room is **sound isolated** because very little sound energy originating from *outside* the room penetrates the walls to be transmitted as sound waves *within* the room.

Sound-isolated and **sound-treated** rooms fall between being highly anechoic and highly reverberant. A **sound-treated room** is intended to:

1. have a high degree of absorption of sound energy created within the room;
2. maximize reflection of sound energy generated outside the room, thereby to *maximize* **transmission loss** through the walls of the room; and
3. minimize electromagnetic interference.

Sound-treated rooms commonly are used to test a person’s hearing or to conduct experiments on hearing. The person being tested is isolated from sound generated outside the room to minimize interference with the sound waves generated inside the room. Moreover, if the sound waves in the room are transduced by a loudspeaker instead of earphones, very little sound energy is reflected by the

absorbent walls and retained in the medium to interfere with the incident waves.

A well-designed, and professionally installed, sound-treated room should be able to provide satisfactory acoustic **absorption** and **isolation**.

Sound Isolation

Sound isolation *is not synonymous* with **sound absorption**. With sound absorption, materials are selected to *absorb a finite amount of sound energy that is generated within the room*. With sound isolation, the room is isolated from sound generated *outside* the room.

- *Sound absorption* requires materials that *absorb* much of the sound energy.
- *Sound isolation* requires materials that *reflect* much of the sound energy that strikes them.
- *Materials that provide a desired amount of absorption are not likely to provide good isolation.*

Think about the difference between absorption and isolation when each is expressed with decibels. Material with an absorption coefficient of 0.90 is an excellent absorber (90%) of sound energy. But, that same material allows 10% of the energy to pass through, which is a reduction of sound energy by only 10 dB. That is *unacceptable sound isolation*, in that an external sound of say 80 dB SPL penetrates the barrier between rooms and has an intensity of 70 dB in the second room. *Effective sound isolation requires a minimum of 40 dB of attenuation.*

How is *sound isolation* achieved? Think about **acoustic impedance**. A sound isolator, a barrier intended to minimize sound transmission from one medium to another, has two important properties: **mass** and **stiffness**.

- *An isolator is a resonant system with a **resonant**, or **natural**, frequency (f_{nat}). When the driving frequency of the incident sound wave matches the natural frequency of the isolator, vibration amplitude is maximal.*
- *The resonant system is not a good sound isolator at its resonant frequency. Suppose the resonant frequency of a panel is 1000 Hz, the frequency at which **mass reactance** equals **stiffness reactance**. The amplitude of vibration of the panel at that frequency is maximal, and **sound transmission loss (STL)** therefore approaches 0 dB, which is poor isolation.*
- *For frequencies below 1000 Hz, **stiffness increases**, acoustic impedance increases, and **transmission loss increases**. For frequencies above 1000 Hz, **mass increases**, acoustic impedance increases, and **transmission loss increases**.*

For frequencies above or below the natural frequency (f_{nat}) of an isolator, 1000 Hz in this case, transmission loss can be expected to approach 30 dB or more and a reasonable amount of sound isolation is achieved. Thus, an understanding of acoustic impedance should lead to the realization that *an acceptable amount of sound isolation can only be achieved by focusing on the mass and/or stiffness of the barrier between rooms*. Increasing the **stiffness** of the barrier is effective for isolating low-frequency energy (frequencies *below* f_{nat} of the barrier) by as much as 35 dB or so. Increasing the **mass** of the barrier is effective for isolating high-frequency energy (frequencies *above* f_{nat} of the barrier) by as much as 45 dB.

- It is tempting to think that increasing the mass of a single barrier (a single partition) would be a good way to achieve better sound isolation because each doubling of the mass increases the attenuation by 3 dB. The problem with that solution is that “coincidence resonances” (high frequency resonances formed from the interaction of the thickness of the partition and how sound energy is propagated through the barrier) develop and reduce the attenuation to less than 3 dB.
- *Optimal isolation is achieved by using two independent isolators* (two masses) that are separated by an air cavity. Sound isolation increases by about 3 dB for each doubling of the mass of the isolators. For example, the walls, floor, and ceiling of a sound-treated room might consist of 2” steel panels, a 2” air cavity, and a second set of 2” steel panels. In that case, the effective transmission loss for noise generated *outside* the room is nearly twice as great as it would be if only one wall were used, even if the thickness of one wall were increased from 2” to 4”.

A double-walled, sound-treated room that is satisfactory for testing hearing sensitivity by an audiologist, or for experiments conducted on hearing, should approximate the following specifications:

1. **sound transmission loss (STL)** of ~45 to ~48 dB at low frequencies (e.g., 125 Hz) and >90 dB for frequencies of 1000 Hz and higher; and
2. **sound absorption coefficient (α)** >0.40 at low frequencies (125 Hz, 250 Hz), >0.80 at 500 Hz, and >0.95 from 1000 Hz to 4000 Hz.

How much sound isolation is optimal for a room? The obvious answer is that it depends on how the room is used. A restaurant might require less isolation than a classroom, which requires less isolation than a sound booth that is used to test hearing sensitivity.

The unit of measure of isolation is **transmission loss (TL)**, or **sound transmission loss (STL)**, expressed in decibels. If the intensity of the sound on the outer side of the barrier is 95 dB SPL, but only

50 dB SPL inside the room, **STL** = 45 dB (not 45 dB SPL). The amount of transmission loss varies with frequency. A metric, **sound transmission class (STC)**, was introduced in 1961 to provide a means for describing the influence of frequency on transmission loss and for comparing one type of barrier with another. **STC** is calculated as the mean **STL** for 16 frequencies ranging from 125 Hz to 4000 Hz. The result is a standard set of **STC** reference curves in which sound transmission loss, **STL** in decibels, is shown on the ordinate, frequency in **Hz** is shown on the abscissa, and **STC** is the parameter.

Effectiveness of Sound Transmission Class Values

Once the **STC** is determined, what can be expected with respect to effectiveness of sound isolation?

STC	Unwanted External Noise	Will It Be Heard?
25	Speech spoken with ordinary intensity	Yes, and probably understood
35	Speech spoken loudly	Yes, but probably not understood
45	Speech spoken loudly	No
50+	Musical instruments or stereo	Yes, but faintly

Those values reveal the **STC** that must be met to satisfy a personal criterion. For almost all circumstances, an **STC** ≥ 45 will provide sufficient sound isolation.

STC and Type of Partition

Below is a sampling of the **STCs** that can be expected from different types of sound isolators (Harris, 1993).

STC	Sound Isolator
33	$\frac{1}{2}$ " drywall on each side of wooden studs— <i>no insulation between</i>
39	$\frac{1}{2}$ " drywall on each side of wooden studs— <i>fiberglass insulation between</i>
45	double layer of $\frac{1}{2}$ " drywall on each side of wooden studs— <i>batt insulation</i>
50	10" hollow concrete masonry.

A room is likely to have windows, and single-pane glass windows have an **STC** rating of about 27. Using double-pane glass is likely

to increase the **STC** to only about 32. If there is too much sound transmission through existing windows, acoustical windows can be installed with **STC** ratings that range from 35 dB to about 60 dB. A room also has at least one door (perhaps with as much as 1" air space between the bottom of the door and the floor). Unless the door is properly sealed, exterior sound is transmitted into the room. Acoustical doors are available, however, and one that has an **STC** rating of ≥ 45 provides sufficient sound isolation to satisfy nearly any need.

Challenges of a Multi-Use Room

The challenge for the consulting engineer is compounded when confronted with a multi-use facility that serves different purposes and audiences. The engineer and the client must make a choice: Which usage and audience will be given preference? A good example is U.S. Bank Stadium in Minneapolis. Construction was completed in 2016 at a cost of approximately \$1.1 billion. The primary tenant is the professional football Minnesota Vikings, but the stadium is used for concerts as well. The 240,000-square foot roof, which is 60% ethylene tetrafluorethylene (ETFE), a fluorine-based clear plastic, and 40% glass, is not retractable, and the seating capacity is 66,200, expandable to 70,000. The room is highly reverberant.

Football teams want their facility to be noisy. The Vikings website "proudly" proclaims that the roof is constructed of "acoustically **reflective** materials" and advises that fans attending the football games might wish to wear hearing protection. In other words, the stadium was deliberately designed to *maximize reverberations* rather than to strike a balance between reflection and absorption.

The opening event in the stadium was a country music show, followed the next evening by the heavy-metal band Metallica. The sound level was reported to be 105 dB, but the report did not specify where in the stadium the reading was taken and whether the reading was dBA or dB SPL. A few representative comments by fans who attended the concerts were:

- "The noise tended to just bound around."
- "My ears are still (the next day) ringing."
- "There's a lot of echo."
- "The space is so large, there's so much reverberation."
- "With all this glass, this is an echo chamber."
- "I think I've suffered long-term hearing damage" (*Minneapolis Star Tribune*, August 22, 2016, p. E3).

The story was different following the first Vikings football game. Sound intensity was measured to reach 114 dB SPL. If there was negative fan reaction to the intense noise, none was reported by the local media.

The bottom line should be clear; you can't have it both ways. If a facility is designed to be "highly reflective," it will not be a satisfactory acoustic environment for other venues that require a different balance between reflection and absorption.

Background Noise Level

A sound level meter is used to assess the overall noise level in a room. The meter typically covers a frequency range from about 30 Hz to 8000 Hz, and an intensity range from about 30 to 130 dB, depending on the manufacturer. There ordinarily are two weighting networks (filters), **A** and **C**. The **C** response is flat ± 1.5 dB from about 100 Hz to 4000 Hz. Measurements made with the **C** weighting network typically are reported as dBC.

A-scale readings deemphasize the contribution of low frequency energy. For example, the response at 100 Hz is 20 dB less than the response at 1000 Hz. The attenuation rate of the **A**-weighting network from 1000 Hz to lower frequencies is intended to reflect the 40-phon equal-loudness contour, whereas the **C** network reflects the 100-phon equal loudness curve.¹ Measurements made with the **A** weighting network usually are reported as dBA.

In addition to assessing the overall intensity of background noise, it is important to *measure the spectrum of the noise* by observing noise levels at the outputs of 1-octave or 1/3-octave filters that span the frequency range of interest: typically from ~100 Hz to ~8000 Hz.

- Assessment of the noise spectrum is essential because virtually all design or redesign efforts include options that are frequency dependent.
- HVAC (heating, ventilating, and air-conditioning) systems often are not equipped with silencers and potentially are a primary source of room noise.
- The signal-to-noise (S/N) ratio is an important factor in speech understanding, and calculation of S/N requires knowledge of the level of background noise. For adults with normal hearing, speech understanding is not adversely affected until the S/N is decreased to about 0 dB. But for adults with a hearing loss, and if the room is not appreciably reverberant, the S/N ratio should be at least +5 to +10 dB. If adults with hearing loss are listening in a room with significant reverberation, the S/N needs to be increased by an additional 3 to 6 dB (Crandell & Smaldino, 2000a, 2000b).
- How is S/N affected by reverberation time? Boothroyd (2004) has shown that the effective signal level (**S**) is *the combined energy in the direct signal and the early reverberant energy*, whereas noise level (**N**) is *the combined energy of the background noise and the reverberant tails*. When both **S** and **N** are expressed in decibels,

$$S/N_{\text{dB}} = S_{\text{dB}} - N_{\text{dB}}.$$

- Additional measures of background noise ideally should be made in the room while occupied because the presence of people adds to the level of the noise (scooting chairs, shuffling paper, laughing, talking, coughing, etc.)

Is background noise in classrooms a prevalent problem? Siebein, Gold, Siebein, and Ermann (2000) described the noise characteristics in 56 mostly public classrooms in Florida. They claimed that the rooms were representative of the more than 5,000 classrooms in the state. Floors were carpeted in half the rooms and ceilings were constructed with acoustical ceiling tile with a noise reduction coefficient thought to be between 0.50 and 0.60. That means that approximately 50%–60% of sound energy striking the ceilings is absorbed by the tiling. Walls were mostly painted masonry for ~75% of the rooms and gypsum board for ~25% of the rooms. Most ceilings were flat with heights that ranged from 8.5 ft. to 11 ft.

There were two important findings:

1. HVAC systems were the primary source of background noise; and
2. the high ceilings (because they increase room volume) resulted in rooms that were highly reverberant.

Two acoustical factors have been identified as the primary culprits that adversely affect speech understanding in a room: *reverberant tails* that arrive more than 50 ms after arrival of the direct sound wave; and *the S/N ratio*. According to Bradley, Reich, and Norcross (1999), “the S/N is the much more important determinant of speech intelligibility than C_{50} . . . for the 16 test conditions in this experiment that represent the full range of acoustical conditions likely to occur in rooms for speech.” Their measure of “intelligibility” was a modification of the Rhyme Test in which listeners are instructed to identify the initial consonant of words in the test.

Reverberation and Reverberant Tails

A room with a rectangular shape, high ceiling, painted sheet-rock or masonry walls, several glass windows, hard-surfaced floor, inadequately sealed doors, and a poorly chosen or poorly functioning HVAC system is ideally suited to produce an annoying acoustic environment that adversely affects speech understanding.

Imagine being in such a room: the large living, or “great,” room that sometimes is two stories high in many modern homes; most restaurants; many classrooms; a “sports bar”; and so forth. Many of those

rooms undoubtedly fit the italicized description above. They probably were designed to satisfy someone's concept of functionality, to provide an appealing aesthetic quality, to minimize construction expense, or more likely, because "that is the way it's done these days." One thing is almost certain—there was little or no consultation with a qualified acoustical engineer. Such rooms probably have excessive noise, are highly reverberant, and generally are unpleasant (acoustically).

The concept of sound reverberations (echoes) was discussed in Chapter 9, in the section on "Reflection." That section could easily have left the reader with *two erroneous conclusions*: (1) all sound reflections (reverberations) are "bad;" and (2) RT_{60} is a sufficient measure of reverberation time.

RT_{60} is the time required for the intensity of the reflected signal to decay to 1 millionth of its peak level. Without question, RT_{60} is an important measurement, but it does not reveal the ways in which *reverberations might actually enhance speech understanding*. A room does not have only a single reflection. Sound energy is reflected from each of the room surfaces, but the reflections do not occur simultaneously. Ordinarily, reflections from ceilings are first, with those from walls following a few milliseconds later. Moreover, sound energy might be reflected from a single surface more than once.

Imagine a conventional rectangular room with minimal materials in the room to absorb sound energy. An impulse sound is generated near one end of the room to aid in measurement of sound wave reflections, and a listener (or microphone) is located near the back of the room. An impulse sound is used because of the belief that an impulse excites the air in the room in the same way as each syllable of spoken discourse (Cremer & Muller, 1982). What sound energy arrives at the ear of the listener?

1. *The direct, unreflected wave front arrives first* because it is not impeded by any surfaces in the room and is transmitted over the shortest path to the listener.
2. *The next wave front to arrive at the listener's ear is reflected sound energy from the ceiling*, unless the ceiling is exceptionally high. Reflections from the ceiling arrive with reduced intensity because of the longer path to the listener and because some of the sound energy striking the ceiling is absorbed.
3. *Additional reflected sound waves from the walls arrive a bit later* because they have an even longer path to the listener. They also have reduced intensity.

In what ways do these staggered time-of-arrival reflections affect speech understanding?

- If the sound waves reflected from the ceiling and the walls arrive within about 50 ms or less of the arrival of the direct wave front, the energy of those reflected waves is added to

the energy of the directly transmitted sound and is considered to be “*useful energy*” that can enhance speech understanding (Haas, 1972; Siebein, 2004).

- Sound waves that arrive last after having been reflected from multiple surfaces (ceiling, walls, floor) are called “*reverberant tails*.” This later-arriving sound energy, when combined with the background noise, is deemed to be detrimental to speech understanding (Siebein et al., 2000). *In effect, the speech signal is generating its own masking noise in the form of later-arriving reverberant energy* (Boothroyd, 2004).

Bradley (1986) utilized the “useful vs. detrimental” concepts in formation of a “useful-to-detrimental energy ratio.” The direct- and early-arriving sound energy (numerator) is compared with the deleterious effects of the reverberant tails and background noise (denominator). The point of division between useful and detrimental sound energy is 50 ms (Bradley, 2011) and is designated C_{50} .

Sabine’s equation for reverberation time (RT_{60}) is a useful measure, but it defines only one component of a room’s sound quality; it does not distinguish among direct-arriving, early-arriving, and later-arriving sound energy (Bradley, 2011). Instead, it reflects the combined effect of all three. A comprehensive analysis of room acoustics should include technology that enables the consulting engineer to observe both the relative amplitudes and arrival times of the entire series of reflected sounds in the room.

What is an “ideal reverberation time?” For adults with normal hearing, speech understanding likely will not be adversely affected until RT s exceed 1000 ms, but for adult listeners with sensorineural hearing loss, RT s should not exceed about 400 ms (Crandell, 1991, 1992, 1993; Crandell, Smaldino, & Flexer, 1995; Olsen, 1981).

Another important factor concerning the effects of reverberant sound energy is *the distance between the talker and the listener*. Boothroyd (2004) emphasized that when the listener is “close” to the talker, the direct sound energy dominates the later-arriving reflections by as much as 10 dB. When the listener is “far” from the source, reverberant sound energy dominates, also by as much as 10 dB. Thus, listeners who are seated far from the talker are mainly hearing the reverberant sound. *The critical distance to maximize speech understanding is that distance at which the levels of direct and reverberant sounds are equal.*

Two acoustical factors have been identified as the primary culprits that adversely affect speech understanding in a room: *reverberant tails that arrive more than 50 ms after arrival of the direct sound wave*; and *the S/N ratio*. According to Bradley et al. (1999), “. . . the S/N is the much more important determinant of speech intelligibility than C_{50} for the 16 test conditions in this experiment that represent the full range of acoustical conditions likely to occur in rooms for speech.”

Blending Effective Sound Treatment and Aesthetics: A Case Study

Installation of acoustic tiles on the ceiling and perhaps one wall in a room increases sound absorption and thereby decreases sound reflections from those surfaces. That form of treatment, however, often is not considered to be aesthetically pleasing. This case study describes how effective sound treatment can be blended nicely with the room's pretreatment architecture and decor.

The setting is Room 110, Shevlin Hall on the campus of the University of Minnesota–Twin Cities. Shevlin is a historic building that was constructed in 1904 as a facility where “young ladies could escape from the overly aggressive campus wolves.” For several decades, it has been home for the Department of Speech-Language-Hearing Sciences, and it houses offices, clinics, research laboratories, and two classrooms.

Room 110 in Shevlin is now the Richard P. McDermott classroom. The ceiling is two stories high, the floor is asphalt tile, and one side wall has several windows. From an aesthetic point of view, it is a very attractive room with an abundance of dark stained woodwork. Unfortunately, the room was not a satisfactory classroom because it was excessively reverberant. An acoustical consulting firm, Olson Sound Design, was retained to assess the acoustical characteristics, design and implement an acoustical treatment program, and make a final set of sound measurements after the project had been completed.

The aim of treatment was to decrease reverberation time (**RT**) by increasing total sound absorption, to achieve a higher Speech Transmission Index (see the section “Other Physical/Predictive Indices”) and to preserve the aesthetic qualities afforded by the dark woodwork. *Prior to the redesign project*, the room had been fitted with 17.1 m² of sound absorption panels, which covered 2% of the room's surface area.

No changes were made to the ceiling, floor, or windows. The redesign involved installation of acoustical panels with a Noise Reduction Coefficient (NRC) rating of 0.85 at strategic locations on the walls of the room. Fifty-five square meters (55 m²) of acoustic panels were installed. When combined with the pre-existing paneling, 9% of the surface area of the room was covered. As can be seen by Figure 10–1, the acoustical panels blended well with the existing wood paneling and were not in the least obtrusive.

A color photograph that highlights the blending of colors and dimensions of the sound-absorbing panels with the room's woodwork can be seen on the PluralPlus companion website. See the inside front cover of your textbook for access instructions.

The summary of the project that follows is based on a technical paper presented at the 167th meeting of the Acoustical Society of America (Jaramillo & Olson, 2014).



Figure 10–1. Room 110, Shevlin Hall on the campus of the University of Minnesota–Twin Cities. Photograph courtesy of Professor Robert Schlauch. Used with permission of the University of Minnesota.

	Before Treatment	After Treatment
Speech Transmission Index (STI)	~0.57	~0.70
Total absorption	56.5 metric sabins	112.6 metric sabins
RT_{60} (500 Hz, 1000 Hz)	1160 ms	680 ms

The Shevlin Hall project is testimony that careful measurement and expert redesign concepts can lead to a vastly improved acoustic environment without compromising the aesthetic qualities of a room. The project also provides an excellent example of the importance of close collaboration between acoustical engineers and hearing specialists.

■ SPEECH INTELLIGIBILITY: AN OVERVIEW

Why is a discussion of *speech intelligibility* included in a book entitled *Introduction to Sound*? If we set aside the special issues important in *musical acoustics*, consider what we mostly are concerned about when we think of room acoustics. At the extreme, the noise in a room might be so intense, or the room might be so reverberant, that people are “annoyed.” The ubiquitous “sports bar” and many restaurants are examples that come to mind. For the most part, however, the defining issue is “*can people, some with normal hear-*

ing and others with a moderate-to-severe hearing loss, understand speech easily?"

What does "*understand speech easily*" mean? For about 90 years, that term has meant that listeners obtain high scores (perhaps 95% or higher) on a test of speech intelligibility. The "test," at least in the early years, was a carefully constructed list of nonsense syllables or monosyllabic words. Although the composition of the intelligibility test has evolved into more complex language constructions during the past few decades, *monosyllabic words, in various formats, have prevailed.*

Modern approaches for assessing *speech intelligibility* fall broadly into two categories: (1) **psychophysical/behavioral** and (2) **physical/predictive**. Although there are many variations among *psychophysical/behavioral* approaches, the common theme is to structure a task so that *intelligibility of speech can be not just quantified, but "counted."* A speech message (e.g., monosyllabic words, sentences, etc.) is spoken by a talker and delivered to a listener through ear phones or loudspeakers (use of "live-voice" testing is deliberately excluded from consideration because of its inherent lack of reliability). The listener's task is to record (by one method or another) what message was heard. The "*intelligibility score*" is "*counted*" by the tester and expressed as *the percentage of items recorded correctly*. One attractive feature of the psychophysical approach is that it attempts to assess speech intelligibility directly: a talker, a listener, and a response that can be *counted* as correct or incorrect.

Physical/predictive measures assess speech intelligibility indirectly in that they do not rely on either a talker or a listener. Instead, relevant acoustic parameters of speech and measures of selected elements of the acoustic environment (e.g., signal-to-noise ratio, reverberation characteristics) are joined in an algorithm to *predict* rather than *measure* speech intelligibility. The advantages of the *physical* assessment method are attractive; they do not require use of either a talker or a listener, and the *prediction* of speech intelligibility can be completed in as little as 30 sec with the aid of commercially available instrumentation.

With one such physical predictor equation (which will be described in more detail subsequently), if the calculated value lies between 0.60 and 0.80, "*intelligibility is said to be good*," and if the value lies between 0.80 and 1.0, "*intelligibility is said to be excellent*." It is reasonable to ask, however, "*Does the prediction based on the physical parameters of speech and the acoustic environment provide an accurate assessment of speech intelligibility?*" To address that question, advocates of a *physical* approach have circled back and used psychophysical techniques to determine *the degree of correspondence between the physical prediction and the psychophysical measure*. How else would we be able to say confidently that "a value of 'X' obtained by the physical calculation means that 'Y' percentage intelligibility can be expected? Thus, *the ultimate utility of the physical predictive method requires psychophysical confirmation from behavioral test scores.*

Speech Intelligibility and Speech Understanding

The term “speech intelligibility” has been effectively synonymous with “word intelligibility” because monosyllabic word tests have been the dominant instrument used for intelligibility assessment. I choose to use the term “speech understanding” instead of “speech intelligibility” because, often, “intelligibility” is not clearly defined.

Consider an example from *Acoustics and Psychoacoustics* by Howard and Angus (2013). In their discussion of the effect of noise on “speech intelligibility,” they constructed Figure 7.29 (p. 405) that displayed a typical psychometric function that showed “*Intelligibility (in percent)*” on the ordinate and “*Speech to noise ratio (in dB)*” on the abscissa. They wrote, “a speech to noise ratio of greater than +7.5 dB is required for adequate intelligibility. Ideally, a signal to noise ratio of greater than 10 dB is required for very good intelligibility” (p. 406).

What did Howard and Angus mean by “speech intelligibility”? Was it the percentage of monosyllabic words, disyllabic words, or sentences repeated or recorded correctly by listeners? Were the speech tokens presented in an open- or closed-message format? *The term “speech intelligibility score” is meaningless unless it is defined operationally every time it is used.*

Moreover, the type of noise used as a masker was not described. Was it white noise, band-limited noise, speech-shaped noise, a multi-talker babble, a steady-state or fluctuating masker, or some other noise? Different noises produce different outcomes, which is why it is imperative to specify the noise characteristics explicitly. The questions are important because the “intelligibility” outcomes are different, sometimes dramatically so, depending on the answers.

Other examples of ambiguity appear in conclusions reached by Siebein (2004) and his colleagues (Siebein [Gary], Gold, Siebein, [Glenn], & Ermann, 2000) in reports of the effects of selected room acoustic features on speech understanding:

- “Sound reflections can assist listeners for improved speech understanding” (Siebein et al., 2004; p. 376).
- “The STI and RASTI have both been shown to be very closely related to speech intelligibility scores. A STI or RASTI score for a room of 0.75 to 1.0 suggests that most normal-hearing adult listeners will be able to understand speech easily. Conversely, a score below 0.45 suggests that many normal-hearing adult listeners will experience significant difficulty understanding speech in that room” (Siebein, 2004, p. 142).

Those statements open the door to the same questions that were posed about the claims made by Howard and Angus. Put simply, but emphatically, a “speech intelligibility” score has questionable value unless it is accompanied by a description of:

1. *composition of the message set* (monosyllabic words, disyllabic words, sentences, connected discourse, etc.);

2. *the number of tokens in, or the length of, the test;*
3. *the testing format* (open or closed message set);
4. *a description of the listeners and number of listeners tested;*
5. *the nature of the listener's task* (write down, repeat back, button push, answer questions, etc.);
6. *the testing environment* (listening in quiet or in the presence of some form of defined speech degradation such as a masking noise or filtering);
7. *a specific definition of the dependent variable* (percentage of items repeated, identified, written down correctly, or rated); and
8. *whether the tokens were delivered live-voice or from a recording.*

In summary, I believe the following to be essential requirements for any test of speech intelligibility.

- A test of speech intelligibility is *not* a list of orthographic representations of the items—the test is a *recording* that can be presented multiple times with essentially no acoustic changes over time.
- A test of speech intelligibility has little value until the *reliability of the scores* that it generates has been established and is satisfactorily high.
- As a rule, the test should be administered only to members of the population on which the reliability of the scores was established.

Use of live-voice testing fails to satisfy any of those three requirements, and scores reported from live-voice testing are at best suspect.

■ PSYCHOPHYSICAL/BEHAVIORAL ASSESSMENT OF SPEECH UNDERSTANDING

Intelligibility of Nonsense Syllables, Words, and Sentences

In many ways, the decision of how best to assess the level of speech understanding is akin to trying to assemble a 1,000-piece jigsaw puzzle, only to discover that a few pieces are missing and several others have frayed edges. The task becomes even more challenging when you discover that some of the pieces belong to other puzzles.

It might seem that the issue is straightforward. *How well do people understand speech under a certain acoustic condition?* The question is indeed simple. It is the answer that is complex and elusive. The complexity arises mainly from two main issues: (1) What exactly is meant by “understand speech”? and (2) once defined, how can it be quantified reliably?

Even though the research literature on intelligibility testing spans nearly 90 years, consensus still is lacking regarding many key factors: content of message, format of the test, nature of listeners' task, measure of speech intensity, hence measure of signal-to-noise ratio, and so forth. The early efforts on intelligibility testing at the Harvard Psycho-Acoustic Laboratory and the Bell Telephone Laboratories were summarized by French and Steinberg (1929); Miller, Weiner, and Stevens (1946); Hudgins, Hawkins, Karlin, and Stevens (1947); and Egan (1948).

The World War II years provided the impetus for testing intelligibility because of the special communication problems confronted in the noisy environments of ships, submarines, and aircraft. What test materials and testing format have been used most frequently over the years? Most early tests of speech understanding (intelligibility) comprised lists of nonsense syllables or words, and later, sentences. The test items were arranged in either an open- or a closed-set format. *The intelligibility score was the percentage of items recognized or identified correctly as a function of some physical dimension of the signal, such as, for example, signal intensity or the S/N ratio.*

As intelligibility testing advanced in both laboratories and audiology clinics, several new tests emerged. Although Jim Egan was not the first scientist to explore intelligibility testing in a rigorous manner, his seminal paper (Egan, 1948) on "articulation testing" set the standard for others to follow. The nearly 650 citations of his paper are testament to the impact that he had. Egan wrote:

A quantitative measure of the 'intelligibility of speech' may be obtained by counting the number of discrete speech units correctly recorded by the listener in an articulation test. Typically, an announcer reads aloud lists of syllables, words, or sentences to a group of listeners, and the percentage of items correctly recorded by those listeners is called the articulation score. (p. 955)

Many versions of syllable, word, or sentence tests soon followed. Prominent among them are tests such as *CID W-22*; *CNC*; *NU-6*; *Rhyme*; *Modified Rhyme*; *CID Sentences*; *Speech Perception in Noise (SPIN)*; and so forth. Readers might wish to consult the literature such as Egan (1948); Silverman and Hirsh (1955); Fairbank (1958); Peterson and Lehiste (1962); House, Williams, Hecker, and Bell (1965); Tillman and Carhart (1966); Kalikow, Stevens, and Elliott (1977); and Olsen, Van Tasell, and Speaks (1997), among others.

Although the focus over the years clearly was on word tests, attempts also were made to assess speech intelligibility with *sentences* (Fletcher & Steinberg, 1929; Hudgins, 1947; Davis & Silverman, 1960). In each case, the listener was asked to repeat the sentence after it had been presented.

Speaks and Jerger (1965) developed the **Synthetic Sentence Identification (SSI)** test as an alternative way to quantify speech understanding. Listeners were asked to *identify*, not *repeat*, which

sentence had been spoken from a closed set of 10 alternative synthetic sentences—sentences constructed to represent various levels of approximation to syntactic rules for real English sentences. For example, one first-order approximation was “Do mind instead edge drop quickly till,” one second-order approximation was “Due his fit along sick near nearly,” and one third-order sentence was “Agree with him only to find out.” Speaks and Karmen (1967) extended the development by defining psychometric functions for the SSI test in the presence of a background white noise masking signal. The SSI was moved from the laboratory to the audiology clinic by Jerger and Jerger (1971) who used it as a diagnostic tool for evaluating patients with cochlear and those with retrocochlear disorders, and by Jerger and Hayes (1976), who evaluated the utility of SSI in hearing aid evaluation.

Throughout the evolution of testing the intelligibility of speech with syllables, words, and sentences, the **Central Institute for the Deaf (CID) W-22** and the **Northwestern University Auditory Test No. 6 (NU-6)** word tests remained the “tests of choice,” particularly for audiological assessment. Still, though, the question remains as to whether word tests are the best choice for assessing listeners’ ability to understand speech. As Bilger (1984) emphasized, the clearly articulated utterances (*particularly monosyllabic words*) of the talkers, coupled with other factors, *result in estimates of speech recognition that most likely underestimate an individual’s difficulty in understanding speech*. Research by Pollack, Rubenstein, and Decker (1960), Bilger, Neutzel, Rabinowitz, and Rzczowski (1984), and Bell, Dirks, and Trine (1992) provided support for Bilger’s contention.

A Common Source of Measurement Error

In addition to the problems inherent in the dependent variable (percentage performance), when the independent variable is *speech intensity* or *signal-to-noise (speech-to-noise) ratio*, another source of error emerges. *Specification of speech intensity and the S/N ratio is an “inexact science” and is a potential source of experimental error common to nearly all studies involving speech intelligibility testing*. When a report states that speech was presented at 70 dB SPL, or at an S/N ratio of +7 dB, what degree of accuracy should we expect?

The common approach to specifying speech intensity is to *observe the level of frequent peaks on a VU meter*. Unfortunately, reading VU meter deflections is imprecise for even experienced observers. For example, Brady (1965) reported a standard deviation of 1.5 VU for a single observer and 3 VU for multiple observers. Levitt and Bricker (1970) reported a standard error of measurement of 1.3 VU (range = 5 VU) for 10 readings by inexperienced and experienced observers. This source of error increases as the complexity of the message set increases from syllables to words to sentences to connected discourse. Why is specification of speech intensity so difficult?

- Whereas the rms level of noise is relatively steady-state and can be specified with reasonable accuracy, *the intensity of speech fluctuates dynamically as a function of time* (typically over a 20-dB range). That makes it difficult to specify speech level accurately (Horii, House, & Hughes, 1971).
- The dynamic range of speech—the range in average intensity from the highest intensity vowel to the lowest intensity consonant—is about 28 dB. Within vowels, the range is about 5 dB, and consonants have a range of approximately 23 dB.
- The frequent peaks on a VU meter arise primarily from energy in vowels, even though the lower intensity consonants are the primary information-bearing elements of speech.
- If the message is connected discourse, and the masker is a competing speech passage instead of a steady-state noise, the problem is compounded—the dynamic fluctuations of intensity in the competing message *open “acoustic windows” in the masker when the masking level fluctuates downward and close “acoustic windows” in the masker when the masking level fluctuates upward.*

Those potential sources of error, coupled with different crest factors² between the speech signal and the masker, make it difficult to compare results obtained by different investigators in different laboratories unless they all use a common recording. The problem is exacerbated when the speech message is presented “live-voice” to a listener rather than from a standardized recording. It is astonishing that some investigators persist in using a “live-voice” approach to testing.

One possible solution when the S/N ratio is the independent variable is to generate a signal-correlated masker with an amplitude envelope that mirrors the dynamic amplitude envelope of the speech message (Horii, House, & Hughes, 1971). With this approach, the S/N ratio can be specified quite accurately because it is a constant value, even though specification of speech level itself remains elusive.

Beck and Speaks (1976) followed the lead of Horii et al. (1971). To accomplish synchronization of the speech message and masker, the speech waveform was full-wave rectified, and positive or negative values of the rectified signal were randomly assigned. They concluded: “The outcome underscores the value of incorporating a signal-correlated noise masker that yields scores that are relatively unaffected by small differences in signal level among passages” (p. 1075).

Measurement of Speech Understanding

Why is there such a reliance on word tests? Perhaps it is because they are easily administered, and scoring the listener’s performance is straightforward; we simply *count the number of correct responses*. But perhaps we should be guided by the immortal words frequently attributed to Albert Einstein, among others:

Not everything that can be counted counts, and not everything that counts can be counted.

If we recast those words, we might ask, (1) *when we count the number of monosyllabic words repeated correctly, does that mean we have measured intelligibility?* and (2) *is it possible that the ability to understand speech does not lend itself to counting?* Perhaps a different approach to assessing speech understanding should be explored.

If Bilger (1984) was correct when he asserted that *monosyllabic word tests are prone to underestimating the difficulty that a listener has with understanding speech*, what alternatives are available? First, I believe the term “speech understanding” is more appealing than terms such as “speech intelligibility” or “articulation score” because most people probably have an intuitive sense of what it means to “understand speech.” Second, we know that *frequency and intensity are physical quantities that can be observed and measured (easily counted)*, whereas *pitch and loudness are subjective attributes that are quantified by ratings or judgments*. Maybe we should entertain the possibility that speech understanding also is a form of *subjective attribute* that is more amenable to rating or judgment than to traditional measurement or counting?

People make judgments on an almost daily basis about how well they understand what someone is saying, and they then act on that judgment in a way that they hope will improve speech understanding:

- *I didn't understand you, would you please say it again?*
- *Please turn up the radio, I can only understand about half of what she is saying.*
- *Let's move to a different room; the noise in here makes it impossible to carry on a conversation.*
- And so on.

Even though most people might genuinely believe they know what “speech understanding” means, the trick is to find a way to quantify it in a reliable manner. Although we will see subsequently that acoustical engineers have developed a mechanism for *predicting* speech understanding, *it is premature to predict something that has not yet been clearly defined and quantified.*

Understanding Free-Running Speech: Testing with Connected Discourse

How does one proceed to test how well a listener understands free-running speech such as a passage of connected discourse? One approach (Giolas & Epstein, 1963; Giolas, 1966) is to *present a passage of speech to a listener and then ask a series of questions about the content of the message*. Although that produces a “quantitative measure” obtained by “counting something,” it might not be an exclusive assessment of “intelligibility.” Factors such as memory,

intelligence, or prior knowledge might also influence the answer given by a listener.

Another option is “shadowing,” or “concurrent repetition” (Cherry & Taylor, 1954; Huggins, 1965; Rupf, House, & Hughes, 1971). A passage of connected discourse is presented to a listener who is instructed to *repeat the message as closely as possible as it is heard*. The “score” is a *count* of the percentage of words or syllables repeated correctly. Hopkins (1967), however, maintained that a listener’s performance on the task might be influenced in part by “competitive feedback” because of the interaction between the nearly simultaneous reception of the message and the listener’s verbal shadowing response. In addition, scoring the listener’s performance is inordinately time-consuming and deceptively difficult. For example, the tester must decide how to score “additions to a word (e.g., pluralization or attachment of a prefix), omissions of part of a word, or correct repetition of the word, but in an incorrect word order” (Speaks & Trooen, 1974, p. 640). Because of the problems with question and answer and shadowing formats, most researchers and clinicians have continued to employ word tests to assess speech understanding.

Perhaps the solution to testing the ability to understand free-running speech is not as elusive as we might think. Seventy years ago, Falconer and Davis (1947) employed a novel approach to assessment of understanding *passages of connected discourse*. They used a psychophysical “method of adjustment” procedure, which also was called “Bekesy tracking.” *The listener adjusts the intensity of speech to reach a criterion level of understanding defined by the tester. The measure of speech understanding was quantified in decibels rather than counted as percentage-correct word scores.*

Hawkins and Stevens (1950) used a similar approach to define two criterion levels of understanding speech: the “*threshold of detectability*” (**TD**) and the “*threshold of intelligibility*” (**TI**). Each criterion was defined in quiet and in the presence of a masking noise. **TI** exceeded **TD** by 9 dB. Their results were corroborated by Chaiklin (1959). The 9-dB difference between the intensity at which speech is “just detectable” (**TD**) and when the listener can “just understand the gist of the message” (**TI**) suggests that the underlying, but undefined, psychometric function for understanding speech is quite steep—perhaps on the order of 12%/dB in contrast to 5%–6%/dB typically reported for monosyllabic words.

A steep psychometric function presents a significant challenge, because a small measurement error in decibels (e.g., ± 2 dB) translates to a large error in percentage understanding ($\pm 24\%$). Moreover, it seems preferable to express concepts such as thresholds of detectability and intelligibility (and points in between) in the more conventional “*percentage understanding*” rather than *decibels*.

At first, assessing percentage understanding of connected discourse seemed difficult to achieve, so an attempt was made to see if speech understanding of connected discourse could be *predicted* satisfactorily from scores on more conventional word tests. Giolas

and Epstein (1963) compared scores for word lists with understanding of connected discourse (using the question and answer testing method), and Giolas (1966) compared scores for CID sentences with word scores. They reported that:

1. When speech was subjected to frequency distortion, speech understanding scores for sentences were higher than scores on word tests.
2. Performance on a phonemically balanced (PB) word list *was not a valid predictor* of speech understanding of connected discourse.
3. Performance on a sentence test was a more valid predictor of speech understanding for connected discourse.

Giolas (1966) speculated that sentence scores (on the CID sentence lists) held promise for “estimating a patient’s ability to hear and understand colloquial speech” (p. 38).

Speaks (1969) compared performance-intensity function for the SSI test with the **TI** results reported by Hawkins and Stevens (1950). Hawkins and Stevens had defined **TI** as the “level at which the listener is just able to obtain without perceptible effort the meaning of almost every sentence and phrase” (p. 11) of the connected discourse. **SSI** scores of 99.5% reported by Speaks and Karmen (1967) in the presence of a white noise masker at levels ranging from 35 to 75 dB SPL corresponded to **TI** reported by Hawkins and Stevens. Thus, for listeners with normal hearing, **SSI** scores held promise for predicting the level of speech understanding for connected discourse—to the extent that **TD** is reflective of “speech understanding.” The research to that point provided a mechanism for *translating the threshold of intelligibility from decibels to percentage*, but did not address percentage speech understanding for other points on the still-undefined psychometric function that relates percentage speech understanding to speech intensity or to the S/N ratio.

Speaks, Parker, Harris, and Kuhl (1972) employed a different approach to quantifying the level of speech understanding. They sought to *define, rather than predict, “percentage speech understanding”* directly by using methods that were conceptually similar to the approach used by Falconer and Davis (1947) and Hawkins and Stevens (1950). A series of short stories served as the messages. The passages were spoken by a male talker who read with “normal vocal effort and at a comfortable rate” (pp. 592–593).

Two listener tasks were used:

1. **Tracking to Constant Percentage Criteria:** a method of adjustment procedure in which *the listener adjusted the intensity* of speech in the presence of a white noise masker *to speech understanding levels of 25%, 50%, and 75%*; and
2. **Listener Estimates, or Ratings, of Intelligibility:** *the tester adjusted speech intensity and the listener estimated (rated) the percentage of speech understanding.*

Listeners performed both tasks easily, and the two different methods yielded similar results. In a subsequent report, Gray and Speaks (1978) applied the same technique successfully on listeners with impaired hearing.

In the Speaks et al. (1972) experiments, listeners also were tested with the **CID** sentences (Silverman & Hirsh, 1955). The correlation of percentage of key words repeated correctly from the **CID** sentences with estimated speech understanding of connected discourse was +0.93. Speaks et al. concluded that:

- Both the tracking and rating procedures were a promising method for *assessing, rather than predicting*, the level of speech understanding;
- the rating procedure permitted speech understanding to be *expressed in percentages rather than decibels*; and
- both the tracking and ratings procedures made it possible to define the psychometric function for speech understanding in the presence of a noise masker.

The SIR and RSIR Tests of Speech Understanding

One drawback to the Speaks et al. (1972) study was its failure to ensure equivalence of “scores” among the short-story messages. Cox and McDaniel (1989) refined the method for using connected discourse to measure speech understanding by developing the **Speech Intelligibility Rating (SIR)** test. They created 72 alternative passages of connected discourse and sought to achieve equivalence among passages with respect to vocabulary, sentence structure, and passage length. Speech was masked by a multi-talker babble instead of white noise, and listeners *rated the percentage of speech understanding*. They then selected 20 passages from the 72 that produced mean ratings between 60% and 70% to form the **SIR** test.

Speaks, Trine, Crain, and Niccum (1994) modified the **SIR** test to form the **Revised Speech Intelligibility Rating Test (RSIR)**. They tested 12 listeners with normal hearing with 72 passages of the **SIR** test masked by the same multi-talker babble used by Cox and McDaniel. *Forty-two* of the 72 passages yielded equivalent results (± 0.5 dB; $\pm 7.5\%$). Minor adjustments then were made to speech intensity of selected passages to achieve even greater homogeneity among passages, and the experiment was repeated with 18 new listeners. The number of equivalent passages for the **RSIR** test, after the intensity adjustments, *increased from 42 to 64 of the original 72 passages*.

By way of summary, *the level of speech understanding can be rated* by use of the **SIR** test or the **RSIR** test. Those tests of speech understanding take slightly longer to administer than is required for a 25- or 50-item word test, but if they produce a more valid index of how well a person understands speech, the additional time seems justifiable.

A Methodological Curiosity

Approximately 51% of the population of the United States are female. Therefore, about half of the people we *talk* to are women, and, of course, about half of the people we *listen* to are women. The spectral composition of speech spoken by males is different from the spectral composition for females. Somehow the implications of those two facts seem to have escaped the attention of investigators of speech intelligibility—to the best of my knowledge, all or nearly all readily available recorded tests of speech intelligibility/understanding have used a male talker. Surely we do not believe that the results obtained with female talkers would be the same as those for male talkers!

PHYSICAL/PREDICTIVE ASSESSMENT OF SPEECH UNDERSTANDING

There is an abundance of research demonstrating that the degree of speech understanding can be established by using passages of connected discourse. That approach, however, has not found favor with acoustical engineers who design or redesign rooms to create an acoustical environment that optimizes speech understanding. They chose to devise methods by which selected features of the acoustic environment (e.g., S/N ratio and reverberant tails) in a room are combined with essential parameters of the speech signal to calculate a value that enables them to *predict* how well speech would be understood in that environment.

Speech Transmission Index (STI)

The quest for a satisfactory predictive measure led to development of the **Speech Transmission Index (STI)**, which incorporates modulation transfer functions in the equation. “The reduction in amplitude modulations is assessed for . . . 98 combinations of 7 acoustical octave bands and 14 modulation frequency bands. These are reduced to a single number STI value” (Bradley, Reich, & Norcross, 1999, p. 1821). Thus, **STI** was developed as a measure of the *quality of speech transmission*, and yields a value (an index) that ranges from 0.0 to 1.0.

The **STI** is an appealing *predictive measure* because it is a *purely physical measure* that does not rely on conventional word, sentence, or connected discourse tests of intelligibility (Houtgast, Steeneken, & Plomp, 1980; Houtgast & Steeneken, 1985). Moreover, it can be calculated easily and quickly because it does not require using a talker, a panel of listeners, and lists or passages of speech. Instead, it incorporates important characteristics of a transmission channel (e.g., air in a room, the telephone, electro-acoustic equipment).

In effect, STI estimates the ability of the channel to transmit the relevant features of speech by including several factors in the mathematical derivation: *Reverberation time, room volume, background noise level, vocal output of the talker, and the talker-listener distance* (Houtgast et al., 1980). In a subsequent report, Steeneken and Houtgast (1994) claimed that the index is well correlated with “subjective” measures of speech intelligibility (performance on word tests). But, if conventional word tests are likely to *underestimate* the difficulty that a person has understanding speech (Bilger, 1984), then it seems important to know how well **STI** predicts listeners’ performance with free-running speech (connected discourse) rather than how well it predicts performance on a conventional word intelligibility test.

One additional attractive feature of **STI** is that *the STI value is independent of the language that is spoken* because it reflects the ability of the transmission channel to transmit faithfully the physical parameters of speech elements—not a particular spoken language. Howard and Angus (2009) suggest that **STI** values should be interpreted as follows:

- $0.0 < \text{STI} < 0.4$ “intelligibility is *poor*”
- $0.4 < \text{STI} < 0.6$ “intelligibility is *fair*”
- $0.6 < \text{STI} < 0.8$ “intelligibility is *good*”
- $0.8 < \text{STI} < 1.0$ “intelligibility is *excellent*”

The utility and ease of use of the **STI** is unassailable. Although it is widely accepted by acoustical engineers, and has the virtue of being based on a *physical rather than psychophysical measurement*, two questions remain unanswered: (1) How is “intelligibility” defined; and (2) what are the percentage definitions of *poor, fair, good, and excellent*?

Other Physical/Predictive Indices

Other physical/predictive indices of speech intelligibility focus on what is believed about (1) the *beneficial effects* of **direct-arriving** (not reflected) sound energy and **early-arriving** (<50 ms) reflected energy, and (2) the *detrimental effects* on intelligibility of **reverberant tails** (>50 ms) and **background noise level** that arrive at the ear of the listener.

Lochner and Burger (1964), in their investigation of acoustics in auditoria, devised the “useful-to-detrimental” (**U/D**) ratio, expressed in decibels. In its most simple form, that index is expressed as:

Equation 10–5

$$\text{U/D} = 10 \log [E_c / (E_L + E_n)],$$

where

U is useful energy,

D is detrimental energy,

E_e is the early-arriving reverberations
(direct plus <50 ms) speech energy,

E_L is the later-arriving reverberant tails (>50 ms), and

E_n is the ambient noise energy.

Equation 10–5 is viewed as a “defining equation” that expresses the concept of a “useful-to-detrimental” ratio. Other versions of the ratio have been derived that reflect variants of the parameters in the equation. Those include: C_{50} , in which 50 ms is used to divide early- from later-arriving reverberations; C_{80} , in which 80 ms is the point of division; $STI - N$, which is a special variant of the STI that excludes the effects of noise and speech intensities; and so forth.

Bradley (1998) provided a comprehensive comparison of eight physical/predictive measures on the intelligibility of speech. Correlations between specific pairs of indices ranged from +.90 to +.99. Thus, for all practical purposes, the various indices that he evaluated were interchangeable—they all accounted for nearly the same amount of variance. Importantly, however, Bradley did not include a psychophysical/behavioral measure of intelligibility in his research. Thus, even though the intercorrelations among the eight physical measures were impressively high, *we do not know how well any of the eight predict speech understanding or intelligibility in the traditional sense.*

The array of physical/predictive measures of speech intelligibility are attractive because they are based on physical parameters of the speech signal and the acoustic environment. They are language-independent, the presence of a talker and listener is not required, and the measurements and calculations can be made quickly. *Acceptance of these measures as valid predictors of speech understanding by hearing and speech professionals, however, would be enhanced by empirical investigations of the predictability of performance on tests of connected discourse (e.g., the **SIR** or **RSIR**) by one or more of the physical indices.*

THE DESIGN/REDESIGN TEAM

Objective of Acoustic Treatment: Enhance Speech Understanding

The overarching objective of the design/redesign team is to formulate a plan that, once implemented, will create an acoustical environment that is conducive to improved speech understanding. We have discussed three acoustical features that are known to contribute to an unfavorable acoustical environment: *reverberation time, reverberant tails, and background noise level*. Which is most important to address, or should the design plan address all three with equal weight?

Bradley et al. (1999) investigated the effects of several acoustical conditions on speech intelligibility. Listeners were tested in synthesized

sound fields in which the $S/N(A)$ ratio and room acoustic characteristics could be altered instantaneously. Both speech and noise level were **A-weighted**, which is why they used the symbol $S/N(A)$ instead of the familiar S/N . The test of speech intelligibility was the *Rhyme Test* (Fairbanks, 1958) as modified by Latham (1979). Ten listeners “with no known hearing problems” (p. 1822) received a 50-word test in each of 16 different sound-field configurations. A male talker was used and the words apparently were presented *live-voice* rather than from a recording. Speech was presented at a nominal 55 dBA, and noise levels were adjusted to produce $S/N(A)$ ratios ranging from -5 dB to $+10$ dB in 5-dB intervals.

- Intelligibility scores (Rhyme Test) increased as either C_{50} or $S/N(A)$ increased.
- The $S/N(A)$ seemed more important than C_{50} with respect to intelligibility scores.

With respect to formulating a sound-treatment program, Bradley et al. advocate that it is important to focus on signal-to-noise ratio rather than on attempting to achieve the optimal reverberant conditions, either RT_{60} or the timing of reverberant tails.

An emphasis on signal-to-noise ratio might be appropriate for a classroom, but diminishing the importance of reverberant characteristics might not apply to venues such as a busy restaurant, where the patrons are both listeners *and the dominant source of the background noise*. In that case, greater attention should be paid to the importance of increasing sound absorption, thereby lessening the deleterious effects of late-arriving reverberant tails.

I have three additional concerns about how the Bradley et al. paper should be interpreted.

1. **Specification of Speech Intensity:** If the words on the modified Rhyme Test were in fact presented “live-voice,” accuracy of specification of speech intensity, hence the signal-to-noise ratio, is compromised. Even if the words were presented from a recording, speech intensity undoubtedly reflected mainly vowel intensity, but the intelligibility test that was used was mainly a test of intelligibility of initial consonants, not words.
2. **Measure of Speech Intelligibility:** The finding that $S/N(A)$ and C_{50} were reliable predictors of performance on a modified Rhyme Test *does not mean that those physical indices are reliable measures of “true” speech understanding*.
3. The status of hearing of the listeners was not tested, and at least one of the listeners was 62 years old.

The Acoustical Engineer

The design/redesign plans for creating a favorable acoustic environment require the expertise and experience of a qualified acoustical

engineer/consultant. Put simply, a client should not contemplate going to a big box store and purchasing whatever acoustical ceiling tiles the store offers.

The acoustical engineer has, at minimum, four responsibilities: (1) performing a pretreatment assessment of the acoustical conditions in the room; (2) recommending appropriate sound treatment with respect to both optimal types and locations within the room; (3) modeling the predicted outcome of treatment; and (4) performing posttreatment assessments to determine whether the treatment outcomes are satisfactory.

If the acoustical environment in a room is unsatisfactory, several measurements ordinarily are made to inform the strategy for formulating an optimal design/redesign plan. Foremost among those are:

- background noise level when the room is empty;
- background noise level when the room is occupied;
- spectral composition of the background noise; and
- reverberant characteristics, including RT_{60} and the timing of early- and later-arriving reverberant tails.

With that information in hand, the design/redesign engineer can attempt to answer the question, “*what treatments are likely to achieve an optimal acoustical environment (optimal reverberation time and signal-to-noise ratio) that will ensure that speech understanding is satisfactory while preserving or enhancing existing aesthetic features of the room?*” That goal, though, presupposes that a reliable, valid measure of speech understanding is available.

The Hearing and Speech Professional

Professionals in speech-language-hearing need to understand the fundamental elements of room acoustics. As Siebein (2004) wrote, “It is necessary to include an audiologist and an acoustical consultant on the design team for new and remodeled facilities to produce a satisfactory learning environment” (pp. 152–153). I would rephrase Siebein’s recommendation to include a *hearing and speech professional* (audiologist, speech-language pathologist, or hearing and speech scientist) to the design team. The consulting acoustical engineer should be well equipped to perform and interpret acoustical measurements, to provide computer-aided modeling to help with design or redesign recommendations and decisions, and to provide accurate predictions of treatment outcomes.

The hearing and speech professional must (1) know what outcome he or she wants to achieve, (2) have sufficient understanding of the fundamentals of acoustics to engage in meaningful dialogues with the engineer, (3) be prepared to explain the desired outcome, (4) be able to ask appropriate questions, (5) be prepared to understand answers from the engineer, and (6) be willing to express the importance of achieving an optimal level of *speech understanding*, not just word recognition. If the hearing and speech specialist is told that

speech intelligibility will be improved from, say, 60% to 95% after treatment, the speech and hearing professional should be equipped to question the basis on which that assurance is made.

■ A CLOSING COMMENT

In these 10 chapters, we have gone a full “ 2π radians” (or “full circle” if you prefer). Initially, sound waves were treated as if their durations were infinite, as if the distance traveled had no effect on the properties of the sound waves, as if there were no obstacles in the transmitting medium to interfere with sound wave propagation, and as if the characteristics of the waveform were preserved indefinitely in all respects. Those assumptions were convenient to help build a foundation for understanding sound. Subsequently, we have seen that none of those four assumptions holds strictly, but we should have a more complete understanding of how sound is generated and propagated by taking those factors into account.

■ NOTES

1. Intensity is a *physical property of a sound wave* and the unit of measure is the watt/m². *Loudness is a sensation*. Although the magnitude of loudness is determined primarily by the intensity of a signal, *an increase in intensity by a factor of 2:1 does not result in a doubling of loudness*. How is the magnitude of loudness determined? Loudness is a subjective attribute that cannot be “measured” physically in the same way that intensity or sound pressure is measured. It can, however, be quantified by psychophysical rating or matching techniques.

There are two aspects of loudness: (1) *loudness level*, expressed in **phons**, and (2) *loudness*, expressed in **sones**. The experimental methods for assessing loudness level and loudness are described briefly below.

Loudness Level: The listener receives a 1000 Hz sine wave at a fixed reference intensity such as 40 dB sensation level (SL)—this is the reference signal. The listener also receives another sine wave of different frequency (e.g., 200 Hz)—this is the comparison signal. *The listener’s task is to adjust the intensity of the comparison signal until it sounds equally loud as the reference signal*. When completed, the two sinusoids sound equally loud, but they are not necessarily equal in intensity. The “loudness level” of the comparison signal (e.g., 200 Hz) is 40 phons. Then, while holding the intensity of the reference signal constant, new comparison tones with frequencies ranging from <100 Hz to >10,000 Hz are presented to the listener. The result is a *40-phon loudness contour*.

Additional phon contours are defined by varying the intensity of the reference tone, for example to 50, 60, 70 dB SL and so on. For each new reference intensity, an equal loudness contour is established in the same way as with the 40-phon contour, and the family of contours are called Fletcher-Munson curves (Fletcher & Munson, 1933). The loudness level contour that displays dB SPL as a function of frequency is *not linear*. For a 100 Hz tone, a loudness level of 40 phons requires 62 dB SPL, for 5000 Hz, 57 dB SPL, and for 10,000 Hz, 40 phons requires 52 dB SPL. Thus, even though the intensities were different for the various frequencies, all were judged to be equally loud.

Loudness: The concept of **loudness**, expressed in **sones**, is different from **loudness level**, expressed in **phons**. Loudness is assessed by a magnitude estimation psychophysical procedure. One sone is the perceived loudness of a 1000 Hz sinusoid presented with an intensity of 40 dB SPL. *Therefore, by definition, 1 sone equals 40 phons.* The investigator then presents the listener with two sinusoids of identical frequency. The listener adjusts the intensity of one of the signals until it is just, for example, twice as loud as the other signal. *That defines 2 sones.* The experiment continues by changing both the intensity and frequency of the paired sinusoids. After completion, it is possible to define the relation between loudness level in phons and loudness in sones. By way of brief summary, 1 phon = 1 sone (equally loud, by definition), 60 phons = 4 sones (four times as loud), and 80 phons = 16 sones (16 times as loud). There are two important messages to glean from this research:

- A doubling of loudness (e.g., from 1 to 2 sones) corresponds to a 10-dB increase in intensity; and
 - the A-weighting network on a sound level meter is based on the 40-phon equal loudness contour. In other words, the filter for the A-weighting network has a rejection rate that matches the 40-phon contour.
2. The crest factor for a signal is the ratio of the peak value (A) to the rms value (rms). Because $\text{rms} = A/1.414$, that crest factor for a sinusoidal waveform is 1.414.

■ PRACTICE PROBLEMS

1. An empty classroom has the following dimensions: 10 m wide, 20 m long, and 3 m high. The floor is covered with heavy carpet and the walls and ceiling are constructed with plaster. $f = 500$ Hz, α for the carpeting is 0.4 and α for the plaster is 0.1.

a. Calculate each of the following:

	Surface Dimensions	Surface Area (S)	α	A
Side wall 1	_____	_____	_____	_____
Side wall 2	_____	_____	_____	_____
End wall 1	_____	_____	_____	_____
End wall 2	_____	_____	_____	_____
Floor	_____	_____	_____	_____
Ceiling	_____	_____	_____	_____
				$A' =$ _____

b. $RT_{60} =$ _____

- c. If acoustic tiling with $\alpha = 0.76$ replaces the plaster on the ceiling and one end wall,

$A' =$ _____

$RT_{60} =$ _____

Glossary

Absorption. Sound wave absorption occurs when the wave encounters an obstacle in which the impedance offered is not infinite, a portion of the energy is transferred to the obstacle, and energy is dissipated in the form of heat; inversely proportional to sound wave reflection.

Absorption coefficient. The proportion of energy in the incident wave absorbed by the material of an obstacle; the ratio of the sound energy absorbed to the total sound energy in an incident sound wave; the coefficient reported for a particular frequency typically is an average of the coefficients for many different angles.

Absorption (total). The total absorption of some surface is the product of the surface area and the absorption coefficient of the surface material.

Absorption unit. The unit of measure of total absorption; in the MKS metric system, the unit is the metric sabin—the total absorption by a window (opening) with an area of 1 square meter; in the English fps system, the unit is the sabin—the total absorption by a window (opening) with an area of 1 square foot.

Acceleration (*a*). The amount of displacement per unit time; the time-rate of displacement; in accord with Newton's Second Law of Motion, the acceleration of an object is directly proportional to the net force applied to the object and inversely proportion to the object's mass. The unit of measure is the newton in the MKS metric system and the dyne in the cgs metric system.

Accuracy of Measurement. Defined by the width of the error band and the percentage of observations that fall within that band; see contrast with **precision**.

Acoustic impedance (*Z*). The total opposition to vibratory motion; measured in ohms; the square root of the sum of the square of the resistance and the square of the reactances (mass and compliant); acoustic impedance for a plane progressive wave is given by the product of the ambient density and the speed of sound.

Acoustic power. See Power.

Admittance (Z^{-1}). Energy accepted by, or admitted to, a system; unit of measure is the mho (ohm spelled backward).

Amplitude distortion. Occurs when a system is "overdriven" by the instantaneous amplitudes of the input waveform exceeding the limits of linearity of the system. Synonymous with nonlinear distortion and harmonic distortion for a sinusoidal input signal.

Amplitude response. Display of the output amplitudes of a system as a function of frequency; also called a system transfer function; popularly called the frequency response.

Amplitude spectrum. See Spectrum (amplitude).

Anechoic room. An extreme form of sound-treated room designed to maximize sound absorption and minimize sound reflection; all room surfaces have an absorption coefficient of nearly 100%.

Angular velocity (ω). A unit of measure of frequency expressed in radians per second.

Antilogarithm. Answers the question “what is the value of the base X raised to the n th power?” For example, $\text{antilog}_2 2$ is the same as 2^2 .

Antinode. Displacement antinodes are points in a medium of maximum positive or negative displacement; pressure antinodes are points in a medium of maximum positive or negative pressure; pressure antinodes lead displacement antinodes by 90° .

Atmospheric pressure. 14.7 pounds per square inch; 100,000 newtons per square meter; 1,000,000 dynes per square centimeter.

Attenuation rate (dB/octave). The rate at which the amplitude of response of a filter is attenuated as a function of frequency.

A-weighting network. Filter on a sound level meter with an amplitude response based on the 40-phon equal loudness contour.

Band-pass filter. See Filter (band-pass).

Band-reject filter. See Filter (band-reject).

Bandwidth (Δf). The width of the band of frequencies between the lower cutoff frequency and the upper cutoff frequency.

Base. The size of the interval between adjacent numbers on an interval, or linear, scale of measurement.

Bases for logs and antilogs. 2; 10 (common, or Briggsian); and e (2.718).

Beat frequency. The rate at which beats occur.

Beats. Periodic increases and decreases in amplitude of the resultant wave when two waves that differ slightly in frequency travel through the same medium.

bel. The log of an intensity ratio.

Broadly tuned systems. Elastic systems that vibrate with maximum amplitude over a wide range of frequencies.

Center frequency (f_c). Synonymous with natural frequency (f_{nat}); the frequency that is at the geometric center of a band-pass filter and produces the maximum amplitude of vibration.

Centimeter (cm). One hundredth of a meter; 1 inch equals exactly 2.54 cm.

cgs metric system. Length measured in centimeters, mass measured in grams, and time measured in seconds.

Characteristic. The integer(s) of a logarithm.

Coefficient. A dimensionless multiplier used in the measurement of some property.

Combination tones. Distortion products in amplitude, or nonlinear, distortion for a complex periodic waveform; consists of two forms of distortion products—difference tones and summation tones.

Complex wave. Any sound wave that is not sinusoidal.

Complex aperiodic sound wave. A sound wave that does not repeat itself regularly over time; a random time function.

Complex periodic sound wave. A periodic wave that is not sinusoidal.

Compliance. The inverse of stiffness.

Compliant reactance (X_c). Measured in ohms; given by the reciprocal of the product of compliance and angular velocity.

Compression. A region in air of increased density.

Constant percentage band-pass filter. See Filter (constant percentage).

Constructive interference. When a reflected wave encounters an incident wave at a point in the medium such that they are in phase with one another, causing particles of the medium to be displaced maximally and produce an increase in intensity.

Cosine of an angle. Ratio of the base to the hypotenuse of a right triangle.

Crest factor. Ratio of the peak value to the rms value of a signal.

Critical damping. A system, after displacement, slowly returns to equilibrium, but not beyond.

Cumulative probability distribution. A plot of the percentage of the total time that any instantaneous value of a waveform's amplitude is less than some specified value.

Cycle. Movement from equilibrium to maximal displacement in one direction, back through equilibrium to maximal displacement in the other direction, and back to equilibrium; 360° of uniform circular motion.

Damping. Diminishing amplitude of vibration over time due to frictional resistance; occurs because of resistance in the transmitting medium and within the oscillating system; damping increases as kinetic energy is transformed to thermal energy.

Damping factor. A measure of the magnitude of damping; given by the natural log of the ratio of the amplitudes of two adjacent cycles— A_N/A_{N+1} .

Decibel (dB). 10 times the log of an intensity ratio; 20 times the log of a pressure ratio; one-tenth of a bel.

Density (ρ). The amount of mass per unit volume.

Derived physical quantity. A quotient or a product of the fundamental physical quantities or a fundamental and other derived quantities; displacement, velocity, acceleration, force, pressure, and so forth.

Destructive interference. When a reflected wave encounters an incident wave at a point in the medium such that they are partially or completely out of phase with one another, causing particles of the medium to not vibrate and produce a decrease in intensity, and at the extreme, cancellation.

Diffraction. The bending or scattering of sound waves around an obstacle; the amount of diffraction depends on the size of the obstacle relative to the wavelength of the wave; diffraction is more efficient when wavelength is much larger than the size of the obstacle, hence the magnitude varies inversely with frequency.

Directly proportional. As A increases, B increases; $A \propto B$.

Displacement (x). A change in position from a starting point to an ending point; a vector quantity.

Distortion. Alteration of the waveform at the output of a device in the form of frequency distortion, amplitude distortion, or transient distortion.

Doppler effect. The rise or fall in frequency—and perceived pitch—as a moving source of sound moves toward or away from a stationary observer.

Dynamic range. The distance in decibels between the electrical noise floor and the amplitude response curve.

Ear canal. An open-closed tube closed at the deep end by the tympanic membrane (ear drum) and open to the surrounding air mass at the entrance to the external ear.

Elasticity (E). The ability to resist changes in shape or volume; the tendency of a volume of air to return to its former volume after compression.

Erg. Unit of measure of work in the cgs metric system; 1 erg = 1 dyne × cm.

Energy. Something that can produce a change in matter—displacement of a mass, distortion of shape of a solid, expansion of a gaseous medium, and so forth; something that a body possesses.

Equal loudness contour. See Loudness level.

Exponent. A power that specifies how many times a base (X) is used in multiplication or division; an exponent is a logarithm.

Exponent: law 1. The product of some common base X raised to two different powers equals the base X raised to the sum of the two powers.

Exponent: law 2. The ratio of some base X raised to two different powers equals the base X raised to the difference between the two powers.

Exponent: law 3. $(X^a)^b = X^{ab}$.

Filter (band-pass). A filter that passes energy in a specified band of frequencies between a lower cutoff frequency and an upper cutoff frequency.

Filter (band-reject). A filter that rejects energy between the lower and upper cutoff frequencies; also called a band-stop, or notch, filter.

Filter (constant percentage). A band-pass filter for which Δf (the bandwidth) is always a constant percentage of the center frequency (f_c).

Filter (high-pass). A filter that passes energy above some designated lower cutoff frequency.

Filter (low-pass). A filter that passes energy below some designated upper cutoff frequency.

Filter (one-octave). A constant percentage band-pass filter for which the upper cutoff frequency is one octave higher than the lower cutoff frequency; $\Delta f = 0.707 f_c$.

Filter (one-third octave). A constant percentage band-pass filter for which the upper cutoff frequency is one-third octave higher than the lower cutoff frequency; $\Delta f = 0.231 f_c$.

fps system. Length measured in feet, mass measured in pounds, and time measured in seconds.

Force (F). A push or a pull; in accord with Newton's Second Law of Motion, the product of mass and acceleration.

Fourier series analysis. A complex waveform can be decomposed, or analyzed, to reveal the amplitudes, frequencies, and phases of each of the sinusoidal components.

Fourier's theorem. All complex waves are composed of a series of simple sinusoids that can differ in amplitude, frequency, and phase.

Free, unbounded medium. A medium for transmission of sound that contains no obstacles or conditions to affect the propagation, or to alter the characteristics, of the wave.

Free vibration. A source of sound, after receiving an applied force, continues to vibrate freely at its own natural frequency until acted upon by an opposing force that opposes motion.

Frequency (f). The number of cycles per second; the reciprocal of the period of vibration.

Frequency-selective system. An elastic system that responds differentially, or selectively, as a function of frequency—a filter or resonator.

Friction (frictional resistance). Serves to limit velocity; kinetic energy is transformed into heat.

Full-wave rectified average amplitude. The arithmetic mean of all instantaneous amplitudes in a full-wave rectified signal; given by the product of the maximum amplitude and 0.636.

Fundamental frequency (f_0). The lowest frequency in a complex periodic wave; the reciprocal of the fundamental period.

Fundamental period. The time required to complete one cycle of vibration; the reciprocal of the fundamental frequency.

Fundamental physical quantities. Length, mass, and time.

Gigahertz. A unit of measure of frequency; one gigahertz equals 1,000,000,000 hertz.

Half-wave rectified average amplitude. The arithmetic mean of all instantaneous amplitudes in a half-wave rectified signal; given by the product of the maximum amplitude and 0.318.

Harmonic. A frequency component of a complex periodic wave; an integer multiple of the lowest frequency component in the wave.

Harmonic distortion. Occurs when a system is "overdriven" by the instantaneous amplitudes of the input waveform exceeding the limits of linearity of the system. Synonymous with nonlinear distortion and amplitude distortion for sinusoidal input signals.

Harmonic relation. The frequencies of all sinusoids that compose the series must be integer multiples of the lowest frequency component in the series.

Hertz. A unit of measure of frequency; 1 Hz = 1 cps (cycles per second)

High-pass filter. See Filter (high-pass).

Hooke's Law. The restoring force of elasticity is directly proportional to the distance that the mass is displaced from equilibrium; elastic restoring force is in phase with displacement.

HVAC. Heating, ventilation, and air-conditioning.

Impedance. See Acoustic Impedance.

Indirectly (inversely) proportional. As A increases, B decreases; $A \propto 1/B$.

Inertia. In accord with Newton's First Law of Motion, the tendency of a body in motion to remain in motion, and the tendency of a body at rest to remain at rest; directly proportional to the mass of a body.

Input-output function. A plot of output amplitude on the ordinate as a function of input amplitude on the abscissa.

Instantaneous amplitude. The amplitude of a waveform at any specified instant in time or at any specified angle of rotation.

Instantaneous velocity. The velocity that is measured over an infinitesimally small time interval such as a millionth or billionth of a second.

Integer. A whole number; the characteristic of a logarithm.

Intensity. The amount of energy transmitted per second over an area of one square meter (MKS); the unit of measure is the watt per square meter (MKS) or dyne per square centimeter (cgs); also called the energy density.

Intensity (absolute). The amount of intensity expressed in watts per square meter (MKS).

Intensity (relative). The ratio of the absolute intensity in one sound wave to the absolute intensity in another, or reference, sound wave.

Intensity level (dB IL). Decibels for intensity in which the reference intensity is 10^{-12} watt/m².

Intermodulation distortion. A form of amplitude distortion for a complex wave at the input to a system; contains two forms of distortion products—harmonics and combination tones.

Interval (linear) scale. The size of the interval between two adjacent numbers is a constant value; the interval is the base.

Inverse square law. The intensity of the sound wave varies inversely with the square of the distance from the sound source.

Joule. The unit of measure of work in the MKS metric system; 1 joule = 10,000,000 (10^7) ergs.

Kilohertz. A unit of measure of frequency; 1 kilohertz = 1000 Hz.

Kinetic energy. An energy of motion.

Length. The extent of spatial separation between any two points.

Linear system. A system in which only the amplitudes and phases of an input signal are altered, resulting in frequency distortion; as the amplitude of an input signal to a system is increased, achieve a proportional increase in output amplitude.

Logarithm. Asks, "to what power must some specified base X be raised to equal some particular number ($X^? = N$)?" Consists of a characteristic and a mantissa; a log is an exponent.

Logarithm law 1. The log of a product equals the sum of the logs of the factors.

Logarithm law 2. The log of a ratio equals the difference between the logs of the factors.

Logarithm law 3. The log of a number a raised to the b^{th} power equals b times the log of a .

Logarithm law 4. The log of a reciprocal equals minus the log of the number.

Longitudinal wave motion. The direction of particle movement is parallel to the direction of wave propagation.

Loudness. Expressed in **sones**; the perceived loudness of a 1000 Hz sinusoid presented with an intensity of 40 dB SPL

Loudness level. Expressed in **phons**; the perceived loudness of a sinusoid of some particular frequency that sounds equally loud as a 1000 Hz sinusoid presented at 40 dB sensation level.

Loudness level contour. A contour displaying the intensity of comparison tones of varying frequency that sound equally loud as that of a reference tone.

Low-pass filter. See Filter (low-pass).

Lower cutoff frequency (f_L). That frequency below f_c for which the amplitude of response is 3 dB less than the amplitude of response at maximum, f_c .

Mach number. The ratio of the speed of a moving source of sound (e.g., an airplane) to the speed of sound.

Mantissa. The endless string of decimal values in a logarithm.

Mass. The quantity of matter present, in contrast to weight, which is the attractive gravitational force exerted on a mass by the earth.

Mass reactance (X_m). Measured in ohms and given by the product of angular velocity and mass.

Maximum amplitude. The peak amplitude; the instantaneous amplitude at 90° .

Megahertz. A unit of measure of frequency; 1 megahertz equals 1,000,000 hertz.

Meter. The distance traveled by light in a vacuum during 1,299,792,458 of a second.

Metric sabin. See Absorption units.

Microsecond. A unit of measure of time; 1 microsecond equals 0.000001 second.

Millisecond. A unit of measure of time; 1 millisecond equals 0.001 sec.

MKS metric system. Length measured in meters, mass measured in kilograms, and time measured in seconds.

Momentum. The product of mass and velocity; momentum is in phase with velocity and leads displacement and the elastic restoring force by 90° .

Nanosecond. A unit of measure of time; 1 nanosecond equals 0.000000001 second.

Narrowly tuned system. An elastic system that vibrates with maximum amplitude over a very narrow range of frequencies.

Natural frequency (f_{nat}). The frequency with which a resonant system vibrates with maximum amplitude of vibration; directly proportional to the square root of the stiffness and inversely proportional to the square root of the mass; corresponds to that frequency at which mass reactance equals compliant reactance, leaving resistance as the only force opposing free vibratory movement.

Newton's First Law of Motion. All bodies remain at rest, or in a state of uniform motion, unless another force acts in opposition; the First Law defines **inertia**.

Newton's Second Law of Motion. The net force applied to an object equals the product of the mass and its acceleration.

Newton's Third Law of Motion. With every force, there must be associated an equal reaction force of opposite direction.

Nodes. Displacement nodes are points in a medium of zero displacement; pressure nodes are points in a medium of zero pressure; pressure nodes lead displacement nodes by 90°.

Noise (pink). A complex aperiodic waveform with equal energy in each frequency band one octave wide; has an octave-band level slope of 0 dB/octave and a pressure spectrum level slope of -3 dB/octave and -10 dB/decade.

Noise (white). Also called Gaussian noise; a complex aperiodic waveform with equal energy within any frequency band 1 Hz wide (from $f - 0.5$ Hz to $f + 0.5$ Hz) and with all phases present in random array; has a pressure spectrum level slope of 0 dB/octave, and an octave-band level slope of +3 dB/octave or +10 dB/decade.

Nominal scale. Things or objects are sorted into different categories by observing that one object is the same as, or different from, another; A; B; III; etc.

Nonlinear system. A system in which the instantaneous amplitudes of the input signal exceed the limits of linearity of the input-output function; increases in input amplitudes do not result in proportional increases in output amplitudes.

Number. A symbol that bears a fixed relation to other symbols; see contrast with numeral.

Numeral. A symbol—not a number that can be added, subtracted, multiplied, or divided.

Octave. A frequency ratio of 2:1; a doubling of frequency.

Octave-band level. The sound pressure level at the output of a one-octave filter.

One-octave filter. See Filter (one-octave).

One-third octave filter. See Filter (one-third octave).

One-third octave-band level. The sound pressure level at the output of a one-third octave filter.

Ordinal scale. An object has more of, less of, or the same as, some quantity than another object; $A > B$; $C < D$; $E = F$, etc.

Overdamping. A system, once displaced, returns toward equilibrium, but ceases motion short of equilibrium.

Overtone. Name for a frequency component of a complex periodic sound wave; alternative to the designation of harmonic, in which n th harmonic equals the $n^{\text{th}}-1$ partial—the second harmonic is the first overtone.

Partial. Synonymous with harmonic if all components of the wave are integer multiples of the fundamental frequency.

Particle acceleration. Acceleration of individual particles of a medium; the first derivative of particle velocity and the second

derivative of particle displacement; leads particle velocity by 90° and leads particle displacement by 180° .

Particle velocity. Velocity of movement of individual particles of a medium; the first derivative of particle displacement; leads particle displacement by 90° .

Peak clipping. A form of amplitude distortion in which positive and/or negative peaks of the input signal are “clipped,” leaving only the center portion of the waveform unaffected; can be symmetrical or asymmetrical.

Peak-to-peak amplitude. The absolute (not algebraic) difference between the maximum amplitudes at 90° and 270° .

Percentage harmonic distortion. A measure of harmonic distortion in percent; the proportion of total energy in the output waveform of a system that is undesired energy at harmonics of the sinusoidal input signal multiplied by 100.

Period (T). The number of seconds per cycle; the reciprocal of frequency.

Periodic sound wave. A wave, whether sinusoidal or complex, that repeats itself at regular intervals over time; a periodic time function.

Phase angle (instantaneous phase). The angle of rotation at any specified moment in time.

Phase spectrum. See Spectrum (phase).

Physical/predictive. Measure of speech intelligibility that does not rely on a talker or listener; assesses speech intelligibility indirectly with an algorithm that incorporates relevant acoustic features of speech and selected elements in the acoustic environment; see contrast with psychophysical/behavioral.

Pi. Equals 3.1416.

Pink noise. See Noise (pink).

Potential energy. A form of stored energy.

Power. The rate at which sound energy is transferred through a medium; the rate at which energy is expended; the unit of measure is the watt.

Power (absolute). The amount of power expressed in watts.

Power (relative). The ratio of the absolute power in one sound wave to the absolute power in another, or reference, sound wave.

Precision of measurement. Refers to how small a difference can be measured; see contrast with **accuracy**.

Pressure (p). The amount of force per unit area; unit of measure is the newton per square meter or pascal in the MKS metric system and the dyne per square centimeter in the cgs metric system.

Pressure spectrum level (L_{ps}). The sound pressure level in a frequency band of unit width (1 Hz) centered on some particular frequency (f_c).

Probability density function. The slope of a cumulative probability distribution.

Psychophysical/behavioral. Measure of speech intelligibility with tests comprising nonsense syllables, monosyllabic words,

sentences, or connected discourse. See contrast with physical/predictive.

Pulse repetition frequency. The reciprocal of the period of a pulse train; the repetition rate of a pulse train.

Pulse (single). A single pulse event; a complex aperiodic wave that does not repeat itself.

Pulse train. A complex periodic wave consisting of a repetitious series of rectangular shaped pulses of some defined width or duration.

Radian. An angle equals one radian when the intersection of the two sides of the angle with the circumference yields an arc whose length is exactly equal to the length of the radius; every circle comprises 6.2832 radians; 1 radian equals 57.3°.

Rarefaction. A region in air of decreased density.

Ratio (exponential) scale. One unit on a scale is so many times greater (or less) than another.

Ray. A line perpendicular to the wave front.

Reactance (X). The component of an oscillating system responsible for storage of energy; comprises mass reactance and compliant reactance.

Rectilinear motion. Straight-line back and forth motion.

Reflection. When an incident sound wave strikes an obstacle with a large acoustic impedance, energy that is not absorbed by the object bounces (is reflected) from the obstacle as a reflected wave with no change in speed of propagation. The angle of the reflected path to the perpendicular equals the angle of the incident path to the perpendicular; the results are called echoes or reverberating waves.

Refraction. A bending of sound waves, or a change in the direction of sound-wave propagation, due to a change in speed of propagation.

Resistance (frictional resistance). A force that opposes motion and limits velocity; measured in ohms and is independent of frequency of vibratory motion; with resistance, energy is dissipated in the form of heat, in contrast to reactance in which energy is stored.

Resonance (principle). When a periodically vibrating force is applied to an elastic system, the system is forced to vibrate *initially* at the frequency of the applied force rather than at its own natural frequency; the nearer the frequency of the applied force to the natural frequency of the elastic system, the greater the resulting amplitude of vibration. The greatest vibration amplitude occurs when mass reactance equals compliant reactance, and the only opposition to motion is resistance. At that frequency, the system is "set into resonance."

Reverberant room. A room with surfaces that have a high acoustic impedance resulting in minimal sound absorption and maximal reverberation.

Reverberant tails. Reverberant sound waves that arrive later than the initial reflected wave; reverberant tails arriving more than 50 ms later, when combined with background noise, are thought to be detrimental to speech understanding.

Reverberation time (RT_{60}). The time required for a sound wave to be attenuated by 60 dB (one-millionth) from its peak level; RT_{60}

of a room is directly proportional to room volume and inversely proportional to the total absorption from all surfaces in the room.

Reverberation time (optimal). A compromise between satisfactory clarity and intensity; dependent on room usage.

Rise/decay (rise/fall). The time required for a signal to rise from zero to near maximum amplitude (rise time) and fall from near maximum to zero amplitude (fall time).

Root-mean-square amplitude. The square root of the mean of the squared deviations of the instantaneous amplitudes; given by the ratio of the maximum amplitude to the square root of 2; the product of the maximum amplitude and 0.707.

Root-mean square pressure. Proportional to the square root of sound intensity.

Sabin. See Absorption units.

Sawtooth wave. A complex periodic wave with energy at all odd and even integer multiples of the fundamental frequency with a spectral envelope slope of -6 dB/octave.

Scalar quantity. A quantity specified only by magnitude. Examples include speed and time.

Scales of measurement. A process by which numerals are assigned to objects according to specified rules; the four scales are the nominal, ordinal, interval (linear), and ratio (exponential).

Scientific notation. A common number is written as the product of some simple number, the coefficient, and the base of 10 raised to some power. For example, $200 = 2 \times 10^2$.

Second (sec). $1/86,400$ of a solar day.

Signal-to-noise ratio (S/N). The ratio of signal intensity to noise intensity; when the two intensities are expressed in decibels, S/N in dB = S (dB) – N (dB) in accord with Law 2 of logarithms.

Simple harmonic motion (sinusoidal motion). Represented as the projection of uniform circular motion.

Sine of an angle. Ratio of the length of the side opposite the angle to the hypotenuse of the angle of a right triangle.

Sonic boom. Occurs when a moving source of sound (e.g., an airplane) reaches a speed that exceeds Mach 1; compressions pile upon one another rather than be separated by rarefactions; form a very large compression with considerable sound energy.

Sound-isolated room. A room constructed with walls that are highly reflective to minimize transmission of externally generated sound into the room; inversely related to sound absorption; optimal isolation of a room is achieved by using two independent sound isolators separated by an air cavity.

Sound pressure level (dB SPL). Decibels for sound pressure when the reference pressure is $20 \mu\text{Pa}$ (MKS) or 2×10^{-4} dyne/cm² (cgs).

Sound transmission class (STC). A metric that expresses the amount of transmission loss as a function of frequency; calculated as the mean STL for 16 frequencies ranging from 125 Hz to 4000 Hz.

Sound transmission loss (STL or TL). The amount of attenuation in decibels of sound energy between the exterior and interior of a room; governed by the mass and stiffness of the wall—for frequencies

above the natural frequency of the wall, as mass increases, acoustic impedance increases, and transmission loss increases—for frequencies below the natural frequency of the wall, as stiffness increases, acoustic impedance increases, and transmission loss increases.

Sound-treated room. A room designed to increase sound absorption and decrease sound reflection.

Spectrum (amplitude). Distribution of amplitudes in either absolute or relative values as a function of frequency.

Spectrum level. See Pressure spectrum level.

Spectrum (phase). Distribution of starting phases as a function of frequency.

Speech transmission index (STI). A measure of the quality of speech transmission by incorporating relevant characteristics of a transmission channel and the ability of that channel to transmit the relevant features of speech.

Speed of sound. Given by the square root of the ratio of the elasticity to the density of the medium.

Square wave. A complex periodic wave with energy at only odd integer multiples of the fundamental frequency and a spectral envelope slope of -6 dB/octave.

Standing waves. Occur when two progressive waves, the incident wave and the reflected wave, of the same frequency and amplitude, travel through the same medium in opposite directions.

Starting phase. The angle of rotation at the moment displacement begins.

Stiffness. The inverse of compliance; the spring constant ($-k$) in an elastic system.

String vibration. Transverse vibration of a string or wire medium that is anchored at both ends; example of a closed-closed system that vibrates with a fundamental frequency and its harmonics; a low-damped, narrowly tuned elastic system.

System transfer function. See Amplitude response.

Tangent of an angle. Ratio of the side opposite the angle to the base of a right triangle.

Thermal energy. Heat.

Total absorption. See absorption (total).

Transient distortion. A form of distortion in which energy is spread to frequencies above and below the fundamental driving frequency resulting in a continuous spectrum; attributable to the short duration of the input signal and/or the rapid onset or termination of the input signal.

Transmission loss. See Sound transmission loss.

Transverse wave motion. The direction of vibration of the medium is at right angles to the direction of the wave propagated through the medium.

Triangular wave. A complex period wave with energy at odd integer multiples of the fundamental frequency and a spectral envelope slope of -12 dB/octave.

Uniform circular motion. Occurs when a body moves around the circumference of a circle at a constant number of degrees of rotation of second.

Upper cutoff frequency (f_U). That frequency above f_c for which the amplitude of response is 3 dB less than the amplitude at maximum; the 3-dB down point.

Vector quantity. A quantity that is specified by both magnitude and direction.

Velocity (c). The amount of displacement per unit time; leads displacement and the elastic restoring force by 90° ; lags acceleration by 90° .

Vibratory motion. Back and forth movement of a body that has mass and elasticity.

Vocal tract. The cavity extending from the vibrating vocal folds at one end to the mouth opening at the other end; modeled as an open-closed acoustic tube.

Waveform (time-domain waveform). Plot of changes in the magnitude of some quantity (displacement, pressure, velocity, etc.) as a function of time.

Wave front. Compressions form a spherical shell around a point source of sound—the spherical shell is called a wave front.

Wavelength (λ). The distance traveled during one period of vibration; the distance between two identical phase angles on adjacent cycles.

Wave propagation. Alternate regions in air of increased density and decreased density that move through the air medium.

Weight. The attractive gravitational force exerted on a mass by the earth; contrast with mass, which is the quantity of matter present.

White noise. See Noise (white).

Work. Is accomplished when a force succeeds in moving a body; the product of the force applied and the distance moved.

Answers to Practice Problems

■ CHAPTER 1

1. We learned from Equation 1.13 that the **period** of pendular vibration is *directly proportional* to the square root of the length of the pendulum. Because **frequency** and **period** are inversely proportional, it follows that **frequency** is *inversely proportional* to the square root of the length of the pendulum. Thus, the frequency of vibration of the shorter pendulum is greater than the frequency of vibration of the longer pendulum.
2. Because it is the same object that is traveling at two different velocities, we should assume that **mass** is a constant. We learned from Equation 1.9 that **momentum** equals the product of **mass** and **velocity**. Therefore, momentum is *directly proportional* to velocity. Thus, the greater the velocity, the greater the momentum.
3. The direction of vibration of the medium is *perpendicular* to the direction of wave propagation through the medium for a **transverse wave**. In contrast, for a **longitudinal wave**, the direction of particle movement of the medium is *parallel* to the direction of wave propagation.
4. We learned from Equation 1.7 that:

$$1 \text{ Pa} = 1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2.$$

It follows, therefore, that:

$$1.5 \text{ Pa} = 1.5 \text{ N/m}^2 = 15 \text{ dynes/cm}^2.$$

5. **Velocity**, the amount of displacement per unit time; or, the ratio of the measure of displacement to the measure of time.
6. **Acceleration**, the time-rate change in **velocity**.
7. **Length** (meter), **mass** (kilogram), and **time** (second).
8. **Elasticity**. In contrast, **gravity** is the restoring force with pendular motion.

9. When **displacement** (x) is maximal, motion is momentarily halted, **potential energy** is greatest, and **kinetic energy** = 0. When $x = 0$ (as the mass passes through equilibrium), **potential energy** = 0, and **kinetic energy** is maximal.
10. The speed of light (299,792,458 m per s) is nearly 1 million times faster than the speed of sound (331 m per s at sea level at a temperature of 0° centigrade).
11. **100 Hz. Frequency** of vibration is dependent on properties of the source, which in this case is 100 Hz. The sound wave is transmitted through water with a greater **propagation speed** (about 4.3 times faster), but that is not relevant to answering the question.

CHAPTER 2

Set 1

1. rms ($\sqrt{2}$); or rms (1.414); or (rms/0.707)
2. A/1.414; or A (.707)
3. 2 (rms $\sqrt{2}$); or 2 (rms $\times$ 1.414); or 2 (rms/0.707)
4. a. 0.707 b. 1.06 c. 0.42 d. 2.69
5. a. 2 b. 2.828 c. 9.9
6. a. 0.35 b. 0.50 c. 3.5 d. 0.707 e. 6.36 f. 0.5x(.707)
7. a. 3.53 Given by 5×0.707
b. 10 Given by 5×2
c. 12.5 Given by $A^2/2$
d. 3.18 Given by $2A/\pi$; or A (.636)
e. 1.59 Given by A/π ; or A (.318)
f. 5 Given
8. 10
9. 2
10. a. 4 b. 2
11. 100
12. rms = 1 Because mean square = rms², and $1^2 = 1$
13. A/π
14. 2.6
15. 2.25 V²
16. N²

Set 2

1. 6.67 Hz
2. 300 Hz Note that 1.5 cycles is to 5 ms as 1 cycle is to X ms. This is generally noted in the form $[1.5 / 5 : 1 / X]$. The equation then is solved by cross-multiplication: $1.5X = 5 / 1$; $X = 3.33$ ms. Therefore, if $T = 3.33$ ms, $f = 1 / 3.33 = .3$ kHz = 300 Hz.
3. a. 500 b. 333 c. 1000
4. a. 2.5 b. 1.25 c. 10 d. 2 e. 0.125
5. a. 0.1 b. 0.5 c. 1000 d. 0.2

Set 3

1. **Particle velocity** *leads* **particle displacement** by 90° . For example, when particle displacement is 0° , particle velocity is 90° .
2. **Period** is *inversely proportional* to **frequency** ($T = 1/f$) and **frequency** is *directly proportional* to **angular velocity** ($\omega = 2\pi f$). Therefore, **period** is *inversely proportional* to **angular velocity**.
3. **Frequency** is *inversely proportional* to twice the **length** of the string, *inversely proportional* to the square root of the string's **mass**, and *directly proportional* to the square root of the string's **tension**.
4. Because **period** is *inversely proportional* to **frequency**, it follows that **period** is *directly proportional* to twice the **length**, *directly proportional* to the square root of the **mass**, and *inversely proportional* to the square root of the **tension**.
5. Because the two waves differ in frequency, the phasic relation between the two varies from moment to moment; the difference of 90° in the two starting phases is *not* preserved throughout the period.

Set 4

1. $P_{\max}$ is 2 Pa and is achieved at 90° where $\sin \theta = 1.00$. The sine of a given angle indicates the proportion (percentage if multiplied by 100) of $P_{\max}$ realized at that angle. Thus, to calculate the instantaneous value of p for a given angle, multiply the sine of that angle by $P_{\max}$ (2 Pa in this case).

θ	Pa	θ	Pa
0.00	0.000		
11.25	0.390	101.25	1.962
22.50	0.766	112.50	1.848
33.75	1.112	123.75	1.662
45.00	1.414	135.00	1.414
56.25	1.662	146.25	1.112
67.50	1.848	157.50	0.766
78.75	1.962	168.75	0.390
90.00	2.000	180.00	0.000

2. The values of **p** are the same as in #1, but the sign is negative (-).
For example, if $\theta = 11.25^\circ$, **sin θ** = .195, and **p** = 0.390 **Pa** (.195 $\times$ 2).
Then if $\theta = 191.25^\circ$ ($180^\circ + 11.25^\circ$), **sin θ** = -.195, and **p** = -0.390 **Pa** (-.195 $\times$ 2).
3. a. 1.414 Pa
b. 1.999 Pa² (or 2.0 Pa²)
c. 0.637 Pa
d. 1.272 Pa

■ CHAPTER 3

1. a. $500 \, \Omega$. $Z = \sqrt{R^2 + (X_m - X_c)^2}$. Because $X_m = X_c$, $Z = R$.
b. **Yes.** A system is in resonance when $X_m = X_c$.
c. **No.** When $X_m \neq X_c$, $Z > R$. In this case, where $R = 600 \, \Omega$, $Z = 608 \, \Omega$.

CHAPTER 4

Set 1

1. a. 6.875×10^3 b. 6.4×10^{-3} c. 1.096×10^2
2. 8
3. a. 343; or 3.43×10^2 $(7 \times 7 \times 7)$
 b. 0.0016; or 1.6×10^{-3} $(.2 \times .2 \times .2 \times .2)$
 c. 10,000; or 10^4 $(10^{2.2 + 1.8} = 10^4)$
 d. 8; or 8×10^0 $(2^{5+(-2)} = 2^3)$
 e. 8; or 8×10^0 You are adding, not multiplying;
 Law 1 does not apply
 f. 10^6 $(10 \times 10^{3+2})$
 g. 1 Any base raised to the 0 power = 1
 h. 5.764 Any base raised to the 1st power =
 base
 i. 1 Same rule as for Problem 3-g; there-
 fore, you need not divide 1.6 by 5.2
 j. 10^6 $(10^{2 \times 3})$
 k. 81; or 8.1×10^1 (3^4)
 l. 32; or 3.2×10^1 (2^5)
 m. 4 $(4^{2 \times 0.5} = 4^1 = 4)$
 n. 9 (3^2)
 o. $1/4$; or 2.5×10^{-1} $(2^{-2} = 1 / 2^2 = 1/4)$
 p. 0.001 $(1 / 10^3)$

Set 2

1. a. 3 b. 2 c. 4 d. -4
 e. 1 f. 1 g. 1 h. 1
 i. 2 j. 2 k. 2 l. 2
 m. 2 n. 2 o. 16 p. -1
 q. 1
2. a. 1.7324 b. 0.5563 c. $\bar{1}.6021$; or -0.3979
 d. 2 e. 34 f. 5,250,000; or 5.25×10^6

CHAPTER 5

Set 1

Equation 5-4 should be used to convert intensity ratios to decibels. In each of these problems, the reference is not specified, but the ratio I_x/I_r is known. For example, if the intensity ratio is 12:1,

$$\begin{aligned} N(\text{dB}) &= 10 \log 12 \\ &= 10 \times 1.08 \\ &= 10.8 \text{ dB.} \end{aligned}$$

- | | | | | |
|----------|-----------|----------|---------|---------|
| a. 0 | b. 10 | c. 20 | d. 30 | e. 30 |
| f. 3 | g. 4.8 | h. 6 | i. 7 | j. 7.8 |
| k. 8.5 | l. 9 | m. 9.5 | n. -3 | o. -4.8 |
| p. -6 | q. -7 | r. -7.8 | s. -8.5 | t. -9 |
| u. -9.5 | v. -10 | w. -20 | x. -30 | y. -30 |
| z. -30 | aa. 13 | bb. 23 | cc. 16 | dd. 26 |
| ee. 17.8 | ff. 27.8 | gg. 3.9 | | |
| hh. 13.9 | ii. 133.9 | jj. 83.9 | | |

Notes

(a). 0 dB does not mean “silence.” It means that $I_x = I_r$. Therefore, the ratio $I_x/I_r = 1.0$ regardless of the absolute value of I_x .

(b, c, d). Each of these involves powers of 10, and each power of 10 corresponds to 10 dB. Thus, in 1-c the ratio is 100, which is a power of 10 twice (10^2). Because each power of 10 corresponds to 10 dB, the answer is given by $10 \text{ dB} + 10 \text{ dB} = 20 \text{ dB}$.

(f). A power of 2, which is 3 dB.

(h). Another power of 2. The ratio 4:1 corresponds to a power of 2 twice (2^2), and each power of 2 is 3 dB. Thus, $3 \text{ dB} + 3 \text{ dB} = 6 \text{ dB}$.

(i). Can you see that the ratio 5:1 involves powers of 10 and powers of 2? The ratio 5:1 can be thought of as the ratio 10:1 (+10 dB) divided by the ratio 2:1 (-3 dB), which is 7 dB.

(1). A power of 2 three times (2^3), which therefore involves $3 \text{ dB} + 3 \text{ dB} + 3 \text{ dB} = 9 \text{ dB}$.

(n-z). These are the inverse of the otherwise identical problems that were solved earlier in this set. Because the absolute value of $I_r > I_x$, the answer in decibels is negative. Solution of such problems is simplified by recalling Log Law 4 ($\log 1/a = -\log a$). Thus, for example:

$$10 \log 1/2 = -10 \log 2 = -3 \text{ dB.}$$

(aa-dd). By now you should quickly see that each involves a combination of powers of 10 (10 dB) and powers of 2 (3 dB). Thus, 400 consists of a power of 10 twice (+20 dB) and a power of 2 twice (+6 dB): $10 \times 10 \times 2 \times 2 = 400$. The corresponding quantities in decibels are $10 \text{ dB} + 10 \text{ dB} + 3 \text{ dB} + 3 \text{ dB} = 26 \text{ dB}$.

(ee, ff). These can only be approximated by powers of 2 and 10. To see how the approximation works, consider the ratio 600:1. You know that 400 ($10 \times 10 \times 2 \times 2$) is 26 dB and that 1,000 ($10 \times 10 \times 10$) is 30 dB. So, the answer for 600 must lie between 26 dB and 30 dB. If you think back to Problem 1-i, you should be able to set the limits even closer. You solved that 5:1 is 7 dB ($10/2$). If 5:1 is 7 dB, 500 must be another 20 dB, for a total of 27 dB. Therefore, because 600 lies between 500 and 1,000, the answer must lie between 27 and 30. See if you can find a way, still using powers of 2 and 10, to lower the upper limit. (*Hint*: you should be able to set the upper limit to 29 dB.)

(gg, hh). Solution of these problems is made easy by recalling two concepts from Chapter 4. First, both problems involve the log of a product. We learned from Log Law 1 that “ $\log ab = \log a + \log b$.” Thus, with Problem 1-gg, we need only to add the logs of the two factors: $\log 2.45 + \log 10^0$. You will need a calculator or log table to determine the log of 2.45, but a second concept from Chapter 4 allows you to determine the log of 10^0 without reference to a log table: “An exponent is a log, and a log is an exponent.” Thus, the log of 10 raised to any power is the value of the power. $\log 10^0 = 0$; $\log 10^1 = 1$; $\log 10^2 = 2$; . . . ; $\log 10^n = n$.

(ii, jj). These problems represent application of two laws of logarithms and one law of exponents. Consequently, the problems can be solved in two ways. We will use 1-ii as an example. We can apply Log Law 1, where $a = 2.45$ and $b = (10^1 / 10^{-12})$. The log of 2.45 is determined from a calculator or log table, but there are two approaches available for determining the log of the ratio $10^1 / 10^{-12}$. (1) Apply Log Law 2: $\log a / b = \log a - \log b$. Because we know that an exponent is a log, the log of that ratio is given by the difference between the exponents, $1 - (-12)$, which is 13. From Log Law 1, then, the log of the product is $0.4 + 13 = 13.4$, which multiplied by 10 = 134 dB. (2) Apply Law 2 of exponents, which is a companion to Log Law 2: $X^a / X^b = X^{a-b}$. Therefore,

$$10^1 / 10^{-12} = 10^{13}$$

$$\log 10^{13} = 13$$

and

$$\text{dB} = 10 (0.4 + 13) = 134 \text{ dB.}$$

Why do both approaches give the same answer? With one we use Log Law 2. With the other we use the second law of exponents. *We get the same answer because a log is an exponent.*

Set 2

Equation 5–4 also should be used to convert each of the decibels to intensity ratios. Note that you are not solving for the absolute intensity, I_x , but just the ratio of the two intensities, I_x/I_r . For example, if $\text{dB} = 5$,

$$5 = 10 \log X$$

$$0.5 = \log X \quad (\text{dividing both sides of the equation by } 10)$$

$$x = 3.16 \quad (\text{the ratio of the unknown } I_x \text{ to the unknown } I_r).$$

- | | | | |
|-----------------------|-----------------------|-----------------------|-------------------------|
| a. $10^0:1$ (1:1) | b. 10^1 | c. 10^2 | d. 10^3 |
| e. 10^4 | f. 10^7 | g. 2 | h. 4 |
| i. 8 | j. 16 | k. 2×10^2 | l. 8×10^2 |
| m. 4×10^4 | n. 4×10^5 | o. 4×10^7 | p. 10^{-1} |
| q. 10^{-2} | r. 10^{-3} | s. 5×10^{-3} | t. 2.5×10^{-4} |
| u. 5×10^1 | v. 1.58×10^6 | w. 1.26×10^9 | x. 3.47×10^0 |
| y. 1.82×10^1 | | | |

Notes

(a–f). Each of the decibel values is divisible evenly by 10, and each 10 dB of intensity corresponds to a power of 10. Thus, for these six problems, the solutions are a power of 10: the powers of 0, 1, 2, 3, 4, and 7, respectively. A second, but not independent, approach is to recall that, when you convert decibels to intensity ratios, the first step is to divide by 10. The result is a log, which in these cases is an integer of 0, 1, 2, 3, 4, and 7 followed by .0000. The integers are the characteristics, and they indicate the exponents in scientific notation. Thus, the results are 10^0 , 10^1 , 10^2 , 10^3 , 10^4 , and 10^7 .

(g–j). Each of these is a value evenly divisible by 3, and each 3 dB corresponds to a power of 2. Thus, the answers are 2, 4, 8, and 16.

(k–o). Each of these involves a combination of powers of 10 and powers of 2. For example, 2-k is 23 dB, and 23 dB consists of $10 + 10 + 3$. Each 10 dB is a power of 10, and each 3 dB is a power of 2. Thus, the answer is $10 \times 10 \times 2 = 200 = 2 \times 10^2$.

(p–t). Each of these involves either a power of 10 or a combination of powers of 10 and powers of 2. Because the decibel is negative, we know that *the exponent is negative*. Thus, 10 dB corresponds to 10^1 , whereas -10 dB corresponds to 10^{-1} .

(u). Although you might not see it on first inspection, this problem also can be solved with powers of 10 and 2. The number 17 comprises $10 + 10 - 3$. Thus, 17 dB corresponds to a tenfold increase in intensity twice, and a halving of intensity once: $(10 \times 10) / 2 = 50$.

If the problem had involved 14 dB, could you have used the same approach? Yes. $14 = (10 + 10 + 3 + 3)$. Thus, 14 dB corresponds to: $(10 \times 10) / (2 \times 2) = 25$.

(v, w). These, too, can be worked as combinations of powers of 10 and 2, but it might be quicker to solve them step-by-step with a log table or more quickly with your calculator rather than to spend time seeing if the powers of 10 and 2 rules apply. It is surprising, however, to see how many problems can be solved in that simple way without use of log tables or calculators. Consider 2-v. $62 \text{ dB} = 10 + 10 + 10 + 10 + 10 + 3 + 3 + 3 + 3$. So, we have a power of 10 five times (10^5) and a power of 2 four times (2^4). That corresponds to 16×10^5 , which is 1.6×10^6 in scientific notation. Can you see how Problem 2-w can be approached in the same way? (*Hint*: You need to sum 10s and subtract 3s.)

(x, y). There is no quick solution available for these, but you should be able to determine upper and lower limits to check to see if the answers you calculate are reasonable.

General Comment

We have emphasized that reasonably quick solutions to many problems can be realized by employing powers of 10 (10 dB) and powers of 2 (3 dB). Because the $\log$ of 2 = .30103, not 0.30000, you will sometimes experience a rounding error that might or might not be tolerable, depending on the precision that is required. Thus, 3 dB corresponds to an intensity ratio of 1.9953:1 rather than 2:1, and that probably will not pose any difficulty most of the time. Look at Problem 2-1, however. For 29 dB the answer was listed as 800:1, but the correct answer (two decimals) is 794.33:1. When greater precision is required, you should use the powers of 10 and powers of 2 shortcut only to estimate the answer and to aid in determining if the answer you calculate is reasonable.

Set 3

Equation 5-4 also should be used for these problems. All problems in the set are conceptually identical to those in Set 1. The only difference is that in Set 1 you dealt only with a ratio of intensities (I_x/I_r) where the absolute values of I_x and I_r were unknown. In Set 3, both I_x and I_r are known. The first step, then, is to solve the ratio by reference to Law 2 of exponents ($X^a / X^b = X^{a-b}$), and the exponent in the result is the $\log$ of the ratio. For example,

$$\begin{aligned} \text{dB} &= 10 \log (10^{-7} / 10^{-12}) \\ &= 10 \log 10^{(-7) - (-12)} \\ &= 10 \log 10^5 \\ &= 10 \times 5 \\ &= 50. \end{aligned}$$

- | | | | |
|---------|---------|---------|---------|
| a. 0 | b. 10 | c. 20 | d. 30 |
| e. 40 | f. 43 | g. 46 | h. 49 |
| i. 90 | j. 96 | k. 100 | l. 103 |
| m. 97 | n. 67 | o. -3 | p. -6 |
| q. 81.5 | r. 84.5 | s. 62.2 | t. 64.8 |

Notes

(a–e). Each involves a power of 10, and by now you should be able to solve these quickly. For example, in 3-c, 10^{-10} is two powers of 10 larger than the reference of 10^{-12} , each power of 10 corresponds to 10 dB, and the answer, therefore, is 20 dB.

(f–h). These are combinations of powers of 10 and powers of 2.

(m). This also is a combination of powers of 10 and powers of 2. Therefore, 10^{-2} involves 10 ten-fold increases (100 dB), then 0.5×10^{-2} (which is only half as great) must be 3 dB less, or 97 dB.

(q–r). Problem 3-r cannot be solved by inspection. Having already solved 3-q, the answer to 3-r must be 3 dB greater than the answer to 3-q because 2.8×10^{-4} is twice as great as 1.4×10^{-4} .

Set 4

Equation 5–4 also should be used for these problems, which are conceptually identical to those in Set 2. The only difference is that in Set 2 you solved only for the intensity ratio (X), whereas in Set 4 you must carry the computation one step farther to determine the actual value of I_x .

- | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|
| a. 10^{-12} | b. 10^{-11} | c. 10^{-10} | d. 10^{-9} |
| e. 10^{-8} | f. 10^{-6} | g. 2×10^{-11} | h. 2×10^{-10} |
| i. 4×10^{-9} | j. 8×10^{-8} | k. 1.00×10^{-13} | l. 1.00×10^{-14} |
| m. 0.50×10^{-12} | n. 2.50×10^{-13} | o. 0.50×10^{-13} | p. 0.50×10^{-14} |
| q. 0.25×10^{-14} | r. 1.26×10^{-8} | s. 1.58×10^{-6} | t. 2.00×10^{-5} |
| u. 5.00×10^{-4} | v. 4.79×10^{-11} | w. 2.63×10^{-10} | x. 6.30×10^{-9} |
| y. 5.00×10^{-8} | | | |

Notes

(m). If you followed the step-by-step procedures, you probably obtained 5×10^{-13} for your answer rather than 0.5×10^{-12} , but the

two answers are equivalent. The answer of 0.5×10^{-12} came from inspecting for powers of 2 and 10. We know that 0 dB corresponds to an intensity of 10^{-12} , so -3 dB must correspond to half as much intensity, or 0.5×10^{-12} . The same explanation applies to Problems 4-n, o, p, and q.

(u–y). Did you notice that you could solve these by inspection for powers of 10 and 2?

Set 5

Equation 5–7 should be used to convert the pressure ratios to decibels. Each of these problems is identical in concept to those in Set 1, and Problems 5-a through 5-ff are identical numerically. The only difference is that now you are presented with *pressure ratios* rather than intensity ratios, and therefore the log of the pressure ratio is multiplied by 20 rather than 10. Solution of the two sets of problems otherwise is identical.

Because the two sets of problems are virtually identical, except for the multiplier, there are no explanatory notes to accompany these problems. When in doubt, consult the notes for the corresponding problem in Set 1. As a general reminder, the majority of the problems can be solved by inspection for powers of 10 and powers of 2, where a power of 10 for pressure corresponds to 20 dB ($20 \log 10$) and a power of 2 for pressure corresponds to 6 dB ($20 \log 2$).

- | | | | | |
|----------|----------|----------|----------|---------|
| a. 0 | b. 20 | c. 40 | d. 60 | e. 60 |
| f. 6 | g. 9.5 | h. 12 | i. 14 | j. 15.6 |
| k. 16.9 | l. 18 | m. 19.1 | n. -6 | o. -9.5 |
| p. -12 | q. -14 | r. -15.6 | s. -16.9 | t. -18 |
| u. -19.1 | v. -20 | w. -40 | x. -60 | y. -60 |
| z. -60 | aa. 26 | bb. 46 | cc. 32 | dd. 52 |
| ee. 35.6 | ff. 55.6 | gg. -80 | hh. -100 | ii. -40 |
| jj. -20 | kk. 0 | ll. 0 | mm. 20 | nn. 0 |
| oo. 6 | pp. -6 | | | |

Set 6

Equation 5–7 should be used to convert decibels to pressure ratios. The only difference in solutions of these problems from those encountered with Set 2 is that the first step is to divide by 20 rather than by 10 because the problems involve pressure rather than intensity.

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| a. 10^0 (1:1) | b. 10^1 | c. 10^2 | d. 10^3 |
| e. 10^4 | f. 10^5 | g. 2 | h. 4 |
| i. 8 | j. 16 | k. .5 | l. .25 |
| m. 2×10^1 | n. 2×10^2 | o. 4×10^3 | p. 10^{-1} |
| q. 10^{-2} | r. 3.16×10^0 | s. 3.16×10^1 | t. 3.16×10^2 |
| u. 1.58×10^2 | v. 7×10^0 | w. 1.26×10^3 | x. 1.88×10^0 |

Set 7

Equation 5–7 should be used for these problems, and all are identical in concept to the problems in Set 5. The only difference is that in Set 5 you dealt only with a pressure ratio (p_x / p_r) where the absolute values of p_x and p_r were not specified. In Set 7, both p_x and p_r are known. The first step, then, is to solve the ratio by use of the 2nd Law of exponents ($X^a / X^b = X^{a-b}$). The exponent in the result is the characteristic in the log of the ratio. For example,

$$\begin{aligned}
 \text{dB} &= 20 \log (3 \times 10^3) / (2 \times 10^1) \\
 &= 20 \log (1.5 \times 10^2) \\
 &= 20 \times 2.18 \\
 &= 43.6.
 \end{aligned}$$

- | | | |
|---------|---------|---------|
| a. 0 | b. 20 | c. 40 |
| d. 60 | e. 80 | f. 74 |
| g. 6 | h. 12 | i. 72 |
| j. -20 | k. -14 | l. 46 |
| m. 94.4 | n. 74 | o. 68 |
| p. 86 | q. 86.6 | r. 92.6 |

Notes

Almost all of the problems can be solved by use of powers of 10 (20 dB) and powers of 2 (6 dB).

(a). In this problem $p_x = p_r$, the ratio is therefore 1:1, the log of 1 is 0.0000, and the answer must be 0 dB.

(a–e). As you proceed from 7-a through 7-e, you progressively increase by one power of 10 (10^1), which for pressure corresponds to increases of 20 dB. Thus, the answers are 0, 20, 40, 60, and 80 dB SPL.

(f–h). Problem 7-f is one power of 2 (2^1) *less* than 7-e, which means that the sound pressure level for 7-f must be 6 dB less than the sound pressure level for 7-e. Similarly, you should see powers of 2 relations between 7-g and 7-a, between 7-h and 7-g, between 7-i and 7-d, and so on.

(m). The value of p_x is only fractionally greater than 10^6 . By using powers of 10 and 2 you should see that if p_x were 10^6 , $\text{SPL} = 94$ dB (2×10^6 equals 100 dB, so 1×10^6 , which is half as much pressure, must be 6 dB less). If $\text{SPL} = 94$ dB when $p_x = 1 \times 10^6$, SPL must be only fractionally greater when $p_x = 1.05 \times 10^6$. Thus, the answer of 94.4 seems reasonable.

Set 8

Equation 5–7 also should be used for these problems, which are identical in concept to those in Set 6. The only difference is that in Set 6 you solved only for the pressure ratio (X), whereas in Set 8 you must carry the computation one step farther to determine the value of p_x .

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| a. 2×10^1 | b. 4×10^1 | c. 8×10^1 | d. 10^1 |
| e. 2×10^2 | f. 2×10^3 | g. 2×10^4 | h. 2.82×10^1 |
| i. 5.64×10^1 | j. 6.32×10^1 | k. 6.32×10^2 | l. 6.32×10^3 |
| m. 2.82×10^3 | n. 4×10^3 | o. 1.26×10^3 | p. 10^3 |
| q. 8×10^4 | r. 1.38×10^2 | s. 8.93×10^0 | t. 7.96×10^0 |

Notes

(a–g). As with many decibel problems, laborious, step-by-step calculations can be avoided with Problems 8-a through 8-g by inspecting to determine if powers of 2 (6 dB) or powers of 10 (20 dB) apply. In 8-a, 0 dB *always* means that $p_x = p_r$; therefore, the answer must be 2×10^1 . The answer to 8-b must be a power of 2 *greater* (6 dB) than the answer to 8-a, and therefore is 4×10^1 ; 8-c is one power of 2 *greater* than 8-b; 8-d is one power of 2 *less* than 8-a; 8-e is one power of 10 (20 dB) *greater* than 8-a; 8-f is one power of 10 *greater* than 8-e; and 8-g is one power of 10 *greater* than 8-f.

(h, i). You need to calculate the answer to 8-h, but having done so, the answer to 8-i must be one power of 2 greater than 8-h because the difference between the two is 6 dB.

(j, k, l). You need to calculate the answer to 8-j, but having done so, you should see that 8-k and 8-l are powers of 10 relative to 8-j.

Set 9

Equation 5–7 should be used for all problems in this set. The only difference between these and the problems in Set 7 is that pressure now is expressed in dynes/cm² (**cgs** system) rather than μPa , and the reference pressure (p_r) is 2×10^{-4} dyne/cm². For example,

$$\begin{aligned} \text{dB} &= 20 \log (3 \times 10^{-4}) / (2 \times 10^{-4}) \\ &= 20 \log 1.5 \\ &= 20 \times 0.18 \\ &= 3.6. \end{aligned}$$

- | | | | |
|--------|-------|-------|-------|
| a. 0 | b. 6 | c. 12 | d. 0 |
| e. 6 | f. 20 | g. 20 | h. 46 |
| i. -26 | j. 80 | | |

Notes

(a). The answer to 9-a must be 0 dB because $p_x = p_r$.

(b, c). The values of p_x in 9-b and 9-c are, respectively, one and two powers of 2 (each power of 2 corresponds to 6 dB) greater than the value of p_x in 9-a.

(d). 2×10^{-4} dyne/cm² is the same as 0.0002 dyne/cm² in 9-a.

(e–j). All remaining problems in Set 9 involve powers of 2 (6 dB) and/or powers of 10 (20 dB).

Set 10

You are asked to determine the *total intensity* in watts/m² rather than the *level* in decibels that corresponds to the total intensity. Therefore, execute Steps 1 and 2 of the three-step procedure described in Set 12 that follows.

- | | |
|-----------------------|-------------------------|
| a. 2×10^{-8} | b. 2×10^{-6} |
| c. 3×10^{-6} | d. 7×10^{-6} |
| e. 9×10^{-6} | f. 3.2×10^{-5} |

Notes

(f). The only difficulty, if any, that should be encountered is with 10-f. The key is that you must be certain that both intensities have the same exponent before you add. Thus, for example, $2 \times 10^{-6} = 0.2 \times 10^{-5}$, which when added to $3 \times 10^{-5} = 3.2 \times 10^{-5}$.

Set 11

The procedure used to solve these problems should have been sufficiently mastered that no explanation is necessary.

- | | | |
|---------|---------|---------|
| a. 43 | b. 63 | c. 64.8 |
| d. 68.5 | e. 69.5 | f. 75.1 |

Set 12

Even though the uncorrelated noise levels that are being combined are expressed in dB SPL, it is the energies, powers, or intensities that are added, not the pressures (see “Combining Sound Intensities from Independent Sources,” Chapter 5). If the sources have equal intensity, you can use Equation 5–9:

$$\text{dB}_N = \text{dB}_i + 10 \log N.$$

For example, if five sources each produce a noise level of 72 dB,

$$\begin{aligned}\text{dB}_N &= 72 + 10 \log 5 \\ &= 72 + 10 (0.7) \\ &= 79.\end{aligned}$$

If the source intensities are not equal, you must execute three calculations (Steps 1–3).

1. Calculate the *intensity* in watt/m² for each source (Equation 5–4).
2. Add the intensities to determine the value of I_x to be used in the third step.
3. Calculate decibels with Equation 5–4 where $I_r = 10^{-12}$ watt/m². The result is expressed as dB IL or dB SPL; they are equivalent.

For example, if two sources have noise levels of 80 dB SPL and 83 dB SPL:

Step 1:

$$\begin{aligned}80 &= 10 \log I_x / 10^{-12}; \text{ therefore, } I_x = 10^{-4}. \\ 83 &= 10 \log I_x / 10^{-12}; \text{ therefore, } I_x = 2 \times 10^{-4}.\end{aligned}$$

Step 2:

$$\begin{aligned}I_x + I_x &= (1 \times 10^{-4}) + (2 \times 10^{-4}) \\ &= 3 \times 10^{-4}.\end{aligned}$$

Step 3:

$$\begin{aligned}
 \text{dB} &= 10 \log (3 \times 10^{-4})/10^{-12} \\
 &= 10 \log (3 \times 10^8) \\
 &= 10 \times 8.48 \\
 &= 84.8 \text{ (dB IL or dB SPL)}.
 \end{aligned}$$

- | | | |
|---------|---------|---------|
| a. 23 | b. 33 | c. 49.2 |
| d. 24.8 | e. 34.8 | f. 51 |
| g. 70.4 | h. 67 | i. 80.5 |

Notes

(a–f). Problems 12-a through 12-f involve equal source intensities, which therefore permits you to solve the problems with Equation 5–9. Thus, for 12-a, the answer is $20 + (10 \log 2) = 23$ dB. For 12-f, $46.2 + (10 \log 3) = 51$ dB.

(g, h, i). For each of these problems you should use the three-step procedures described above. The solution for 12-i is shown below.

Step 1:

$$\begin{aligned}
 60 &= 10 \log I_x/10^{-12} = 10^{-6}. \\
 70 &= 10 \log I_x/10^{-12} = 10^{-5}. \\
 80 &= 10 \log I_x/10^{-12} = 10^{-4}.
 \end{aligned}$$

Step 2: (Convert to common exponent of 10^{-4} .)

$$\begin{aligned}
 &0.01 \times 10^{-4} \\
 &0.1 \times 10^{-4} \\
 &+ 1.0 \times 10^{-4} \\
 &= 1.11 \times 10^{-4}.
 \end{aligned}$$

Step 3:

$$\begin{aligned}
 \text{dB} &= 10 \log (1.11 \times 10^{-4})/10^{-12} \\
 &= 10 \log (1.11 \times 10^8) \\
 &= 10 \times 8.05 \\
 &= 80.5
 \end{aligned}$$

CHAPTER 6

1. **Sawtooth wave.** Square waves and triangular waves have energy only at *odd harmonics*. In addition, the wave in question has an envelope slope of -6 dB/octave, which also is consistent with a sawtooth wave.
2. **-20 dB.** The envelope slope of a **sawtooth wave** is -6 dB/octave. The level of the 5th harmonic is -14 dB, the 10th harmonic is one octave above the 5th harmonic; therefore, the level of the 10th harmonic is -20 dB. f_0 and **starting phase** are irrelevant.
3. Each has energy only at odd harmonics, but the envelope slopes are different: -6 dB/octave for a **square wave**, but -12 dB/octave for a **triangular wave**.
4. **1200 Hz.** If $T = 5$ ms, $f_0 = 200$ Hz. The frequency of the 6th harmonic is given by $6 \times 200 = 1200$ Hz.
5. **-9 dB S/N.** dB S/N is given by signal level in decibels *minus* noise level in decibels. Thus, $70 - 79 = -9$.
6. **880 Hz.** A_6 is five octaves above A_1 . An octave is a doubling in frequency. Therefore, multiply $27.5 \times 2^5 = 27.5 \times 32 = 880$.
7. **-9 dB.** Noise level is 8 times signal level, and $10 \log 8 = 9$ dB. Because noise level exceeds signal level, the **S/N ratio** is -9 dB. Or, we can use a longer procedure. With Equation 5-4, calculate **signal level** = 70 dB IL and **noise level** = 79 dB IL. Therefore, **dB S/N** = -9 dB ($70 - 79 = -9$).
8. With **white noise**, the starting phases of the 10,000 bands are in random array. Therefore, treat this as an instance of 10,000 *independent*, or *uncorrelated*, noise sources. Thus, with Equation 5-9, calculate that:

$$\begin{aligned}
 \text{dB}_N &= \text{dB}_i + 10 \log N \\
 &= 60 \text{ dB} + 10 \log 10,000 \\
 &= 100 \text{ dB SPL.}
 \end{aligned}$$

9. a. **0.60 or 60%.** If **pulse repetition frequency** = 200 Hz, $T = 5$ ms. $P_d = 3$ ms, which is 60% of 5 ms.
- b. The harmonics are *odd and even* integer multiples of the pulse repetition frequency: 200 Hz; 400 Hz; 600 Hz; 800 Hz; and 1000 Hz.
- c. Nulls are integer multiples of $1 / P_d$: 333 Hz; 667 Hz; and 1000 Hz.
- d. Nulls are integer multiples of $1 / P_d$, just as they were for the **pulse train**: 333 Hz; 667 Hz; and 1000 Hz.

CHAPTER 7

Set 1

1. 50 dB $L_{ps} = SPL_{wb} - 10 \log \Delta f = 90 - 10 \log 10,000.$
2. a. 73 dB $SPL_{wb} - 10 \log 10,000/200$; or $L_{ps} + 10 \log 200$:
Note: f_c is irrelevant; bandwidth was specified.
- b. 76 dB f_c is irrelevant.
- c. 80 dB
- d. 80 dB The key to solution is bandwidth.
- e. 80 dB The bandwidth is the same as in 2-d.
- f. 78.5 dB With a 1-octave filter you must calculate bandwidth: $\Delta f = f_c \times .707.$
- g. 75.5 dB Either use the same procedure employed in 2-f, or realize that this bandwidth is 1/2 as wide as that in 2-f, which means that the output in 2-g is 3 dB less than that in 2-f.

Set 2

1. 3 dB/octave and 10 dB/decade
2. a. 2×10^3 $40 \text{ dB} = 20 \log p_x / (2 \times 10^1)$; $p_x = 2 \times 10^3.$
- b. 2×10^5 If $L_{ps} = 40 \text{ dB}$, then $SPL_{10,000} = 80$. Now, solve for p_x .

Set 3

1. 43 dB $L_{ps} = SPL_{wb} - 10 \log \Delta f$
2. a. 66
- b. 69
- c. 73
- d. 73
- e. 73
- f. 68.5

3. a. The preferred center frequencies for a 1-octave filter are spaced at octave intervals, and the **octave-band level** slope for white noise is +3 dB/octave. Therefore,

f_c	dB SPL	
125	35	(given)
250	38	
500	41	
1000	44	
2000	47	
4000	50	
8000	53	

- b. **15.5 dB SPL.** $L_{ps} = \text{SPL}_{wb} - 10 \log \Delta f_{wb}$. Your calculation of L_{ps} should be the same regardless of which of the seven outcomes in 3-a you elected to use in the equation.
- c. **+3 dB/octave.** The slope is *not* affected by your choice of constant percentage bandwidth filter.

■ CHAPTER 8

1. **30 mV.** Over the linear portion of the **input-output function**, output level is *proportional* to input level. Input signal level increases by a factor of 10:1 (200 μV to 2 mV). Therefore, output signal level also increases by a factor of 10:1, from 3 mV to 30 mV.
2. If input level is reduced, the effective **signal-to-noise ratio** is reduced because signal level lies closer to the electrical noise floor of the system.
3. This is the reverse of the situation in the second problem. As you increase signal level to achieve a better **signal-to-noise ratio**, **percentage harmonic distortion** increases because you will operate closer to, or on, the nonlinear portion of the input-output function.
4. **74 dB SPL.** A reference pressure of 2×10^{-4} dyne/cm² in the **cgs** metric system is equivalent to a reference pressure of 20 μPa in the **MKS** system.
5. **11.8%.** The problem is solved with Equation 8–2.

$$\begin{aligned} &\text{percentage harmonic distortion} = \\ &\sqrt{(V_2^2 + V_3^2 + V_4^2 + V_5^2 / V_1^2)} \times 100 = 11.8\% \end{aligned}$$

CHAPTER 9

1. 20 dB

$$\begin{aligned} \text{dB} &= -20 \log 2,000/200 \\ &= -20 \log 10 \\ &= -20 \end{aligned}$$

Equation 9-4

2. 75 dB SPL

$$\begin{aligned} \text{dB} &= 95 - 20 \\ &= 75 \end{aligned}$$

3. The distance is doubled, which is a distance ratio of 2:1. Thus,

$$\begin{aligned} \text{dB} &= -20 \log 2/1 \\ &= -20 \times 0.3 \\ &= -6 \end{aligned}$$

4. With sound wave **reflection**, sound energy is retained in the medium. Therefore, attenuation over distance is *less* than the amount predicted by the **inverse square law** where sound energy is *not* retained in the medium. With sound wave **absorption**, sound energy is absorbed by material in the medium and it is attenuated over distance. Thus, total attenuation is greater than what is predicted by the inverse square law.

5. Think back to Chapter 2 where we learned that pressure (and velocity) leads displacement by 90°; pressure and velocity are *not* in phase with displacement. Therefore, a **displacement node** corresponds to a **pressure antinode**, and a **displacement antinode** corresponds to a **pressure node**. Thus, at the *closed end* of the tube we have a **displacement node**, but a **pressure antinode**. At the *open end* of the tube we have a **displacement antinode**, but a **pressure node**. (See panel C of Figure 9-10.)

= 850 Hz.

6. **Reflection** is the key. Pulses that travel from the instrument to the bottom of the river or lake are **incident pulses**. When they strike the bottom, they return to the instrument as **reflected pulses**. Because the reflected pulses travel with the same speed as the incident pulses, the instrument measures the total travel time and converts travel time to distance traveled. It also is possible to infer whether the bottom is hard (rocky) or soft (mucky). Hard surfaces offer a larger impedance, and strong signals are reflected; soft surfaces offer a lesser impedance, more of the energy is absorbed, and the reflected pulses are weaker.

CHAPTER 10

1. a.

	Surface Dimensions	Surface Area (S)	α	$\underline{A}$
Side wall 1	20×3	60	0.10	6.0
Side wall 2	20×3	60	0.10	6.0
End wall 1	10×3	30	0.10	3.0
End wall 2	10×3	30	0.10	3.0
Floor	10×20	200	0.40	80.0
Ceiling	10×20	200	0.10	20.0

$$A' = 118.0.$$

b. 818 ms

$$\begin{aligned}
 RT_{60} &= k (V/A') \\
 &= .161 (20 \times 10 \times 3)/118 \\
 &= .818 \text{ s} \\
 &= 818 \text{ ms}
 \end{aligned}$$

c. $A' = 269.8$ **metric sabins**; $RT_{60} = 358$ ms.

A for the end wall increases from 3.0 to 22.8, A for the ceiling increases from 20.0 to 152.0, and A' increases from 118.0 to 269.8 **metric sabins**. Then,

$$RT_{60} = .161 (600/269.8) = 358 \text{ ms}.$$

Alphabetic Listing of Selected Equations

TERM	EQUATION	NO.
absorption coefficient	$\alpha = I_a/I_i$	10-1
absorption (total)	$A' = S_1\alpha_1 + S_2\alpha_2 + \dots + S_n\alpha_n$	10-4
acceleration	$a = \Delta c/t$	1-3
angular velocity	$\omega = 2\pi f$	2-12
bandwidth	$\Delta f = f_U - f_L$	7-1
combining sound intensity from independent sources	$dB_N = dB_1 + 10 \log_{10} N$	5-9
compliant reactance	$X_c = 1/2\pi f_c$	3-4
constant percentage		
bandwidth filter	$f_L = \text{antilog}_{10} (\log_{10} f_c - 0.3/2n)$	7-6
	$f_U = \text{antilog}_{10} (\log_{10} f_c + 0.3/2n)$	7-7
	$\Delta f = f_U - f_L$	7-8
1/1-octave filter	$\Delta f = 0.707 (f_c)$	7-2
1/2-octave filter	$\Delta f = 0.346 (f_c)$	
1/3-octave filter	$\Delta f = 0.231 (f_c)$	
1/10-octave filter	$\Delta f = 0.069 (f_c)$	
damping factor	$d_f = \ln (A_1 / A_2)$	3-1
decibels		
intensity	$dB = 10 \log_{10} (I_x / I_r)$	5-4

TERM	EQUATION	NO.
decibels (<i>continued</i>)		
intensity level	$\text{dB IL} = 10 \log_{10} (I_x / 10^{-12} \text{ N/m}^2)$	5–4
inverse square law	$\text{dB} = -20 \log_{10} (d_i / d_r)$	9–4
pressure	$\text{dB} = 20 \log_{10} (p_x / p_r)$	5–7
narrow-band filter	$\text{SPL}_{\text{nb}} = \text{SPL}_{\text{wb}} - 10 \log_{10} (\Delta f_{\text{wb}} / \Delta f_{\text{nb}})$	7–12
sound pressure level	$\text{dB SPL} = 20 \log_{10} (p_x / 20 \text{ } \mu\text{Pa})$	
wide-band filter	$\text{SPL}_{\text{wb}} = \text{SPL}_{\text{nb}} + 10 \log_{10} (\Delta f_{\text{wb}} / \Delta f_{\text{nb}})$	7–13
Doppler effect		
toward observer	$f' = f(s / s - s_s)$	9–14
away from observer	$f' = f(s / s + s_s)$	9–15
exponents		
Law I	$X^a \times X^b = X^{a+b}$	
Law II	$X^a / X^b = X^{a-b}$	
Law III	$(X^a)^b = X^{ab}$	
force		
restoring force	$F_r = -kx$	1–8
frequency		
natural	$f_{\text{nat}} = \sqrt{(s/m)}$	2–10
(and) period	$f = 1/T$	1–12
(of) vibrating string	$f = 1/2L [\sqrt{(t/m)}]$	2–11
harmonic distortion	$\frac{\sqrt{(V_2^2 + V_3^2 + \dots + V_n^2)}}{\sqrt{(V_1^2 + V_2^2 + \dots + V_n^2)}}$	8–1
Hooke's law	$F_r = -kx$	1–8
impedance	$Z = \sqrt{(R^2 + (X_m + X_c)^2)}$	3–7

TERM	EQUATION	NO.
inverse square law	$I \propto 1/(d_i / d_r)^2$	9-3
in decibels	$dB = -20 \log_{10} (d_i / d_r)$	9-4
logarithms		
Law I	$\log ab = \log a + \log b$	
Law II	$\log a/b = \log a - \log b$	
Law III	$\log a^b = b \log a$	
Law IV	$\log 1/a = -\log a$	
mass reactance	$X_m = 2\pi f m$	3-3
mean square amplitude	$\text{mean square} = A^2 / 2$	2-7
momentum	$M = mc$	1-9
newton per square meter	$1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2$	1-6
pascal	$1 \text{ Pa} = 1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2$	1-7
period		
and frequency	$T = 1/f$	1-11
of pendulum	$T = 2\pi\sqrt{L/G}$	1-13
pressure	$p = F/A$	1-5
MKS and cgs	$1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2$	1-6
pascals, newtons, and dynes	$1 \text{ Pa} = 1 \text{ N/m}^2 = 10 \text{ dynes/cm}^2$	1-7
pressure spectrum level	$L_{ps} = \text{SPL}_{wb} - 10 \log_{10} (\Delta f_{wb} / \Delta_0 f)$	7-9
	$= \text{SPL}_{wb} - 10 \log_{10} \Delta f_{wb}$	7-10
proportionality		
inverse	$A \propto 1/B$	1-14
direct	$A \propto B$	1-15

TERM	EQUATION	NO.
rectified average		
full-wave	$FW_{avg} = 2A/\pi = A (0.636)$	2-8
half-wave	$HW_{avg} = A/\pi = A (0.318)$	2-9
reverberation time	$T_{60} = k(V/A)$	10-2
root-mean-square amplitude	$rms = A / \sqrt{2}$	2-4
	$= A(0.707)$	2-6
spectral envelope		
sawtooth wave	$dB = 20 \log_{10} (1/h_i)$	6-1
	$= -20 \log_{10} h_i$	6-2
triangular wave	$dB = 20 \log_{10} (1/h_i^2)$	6-3
	$dB = -40 \log_{10} h_i$	6-4
speed	$s = d/t$	1-1
speed of sound	$s = \sqrt{\frac{E}{\rho}}$	1-16
string resonant frequencies	$f = 1/2L [\sqrt{(t/m)}]$	2-11
	$f_0 = s / 2L$	9-12
	$f_n = ns / 2L$	9-13
tube resonant frequencies		
(open-closed)	$F_1 = s / 4L$	9-6
	$F_2 = 3s / 4L$	9-7
	$F_3 = 5s / 4L$	9-8
	$F_N = [(2n) - 1]s / 4L$	9-9
(open-open) or (closed-closed)	$F_1 = s / 2L$	9-10
	$F_n = ns / 2L$	9-11

TERM	EQUATION	NO.
velocity	$c_{\text{resultant}} = \sqrt{s_1^2 + s_2^2}$	1-2
wavelength	$\lambda = s / f$	2-13
work	$W = Fd$	1-10

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